KunYuEcosystem

AI 参考包 v3.0

全加载器 · 全版本 · 全内容类型 反幻觉代码模板库

3 加载器 × 7 MC 版本 × 14 全类型组合 × 243 个模板文件

核心原则: 检索优先于生成

2026年9月

1. 概览

AI 参考包是 KunYuEcosystem 全流程 AI 制作功能的反幻觉基础设施。AI 在生成任何模组代码之前，必须先检索匹配的模板和 API 签名，基于验证过的代码出发进行生成。

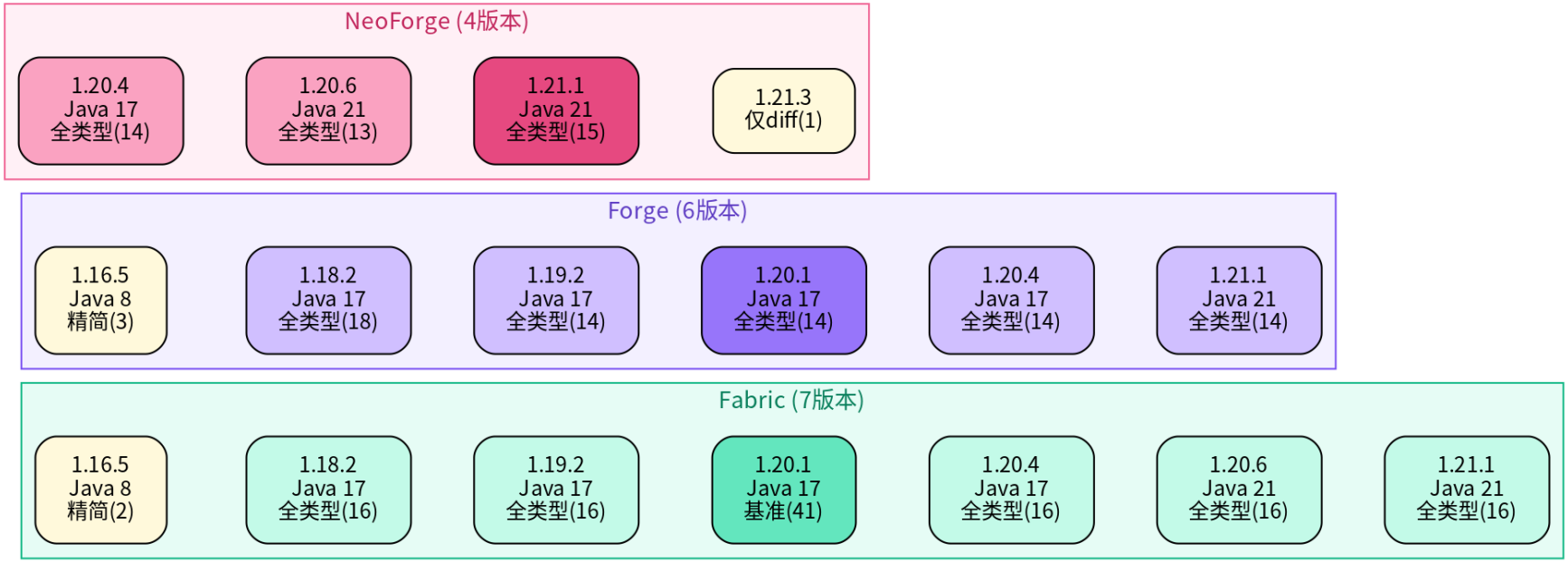


图 1: AI 参考包覆盖矩阵 - 3 加载器 × 17 版本组合

维度	数量	说明
模板文件	243	12个内容类型 × 14个全类型版本
加载器	3	Fabric / Forge / NeoForge
MC版本	7	1.16.5 ~ 1.21.3
版本组合	17	3加载器 × 各自支持的版本
全类型覆盖	14	12个内容类型均有完整模板
错误模式	50条	6大类常见编译错误+修复方案
API索引	40条	Registry + Event + Mixin
文件总数	338	模板+错误+API+Prompt

2. 三层反幻觉架构

层级	名称	作用
第一层	代码模板库 243个文件	AI从模板出发改，不从零生成
第二层	API参考索引 40条签名	防止编造不存在的方法，验证API存在性
第三层	错误模式库 50条错误	自动修复循环基础，AI修不了自己的bug

> **设计原则**
>
> 检索优先于生成 - AI 不凭空写代码，从验证过的模板出发修改。不确定时必须声明「参考包未覆盖此功能」。编译通过才交付 - AI生成后自动Gradle编译，失败则匹配错误模式库自动修复，最多重试3次。

3. 全版本方块注册代码对比

方块注册是最核心、版本间差异最大的模板。以下展示所有14个全类型版本的方块注册代码，红色标注为该版本的标志性差异点。

3.1 Fabric（6个版本）

Fabric 1.18.2 · Java 17

```
// Fabric 1.18.2 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(FabricBlockSettings.of(Material.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registry.BLOCK,
            new Identifier("exmod", name), block);
        Registry.register(Registry.ITEM,
            new Identifier("exmod", name),
            new BlockItem(block, new FabricItemSettings()
                .group(ItemGroup.BUILDING_BLOCKS)));
        return block;
    }
}
// □□: util.registry.Registry (□□)
//       new Identifier()
//       .group() □□□□□□□□
//       Material □□□
```

Fabric 1.19.2 · Java 17

```
// Fabric 1.19.2 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(FabricBlockSettings.of(Material.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registry.BLOCK,
            new Identifier("exmod", name), block);
        Registry.register(Registry.ITEM,
            new Identifier("exmod", name),
            new BlockItem(block, new FabricItemSettings()
                .group(ItemGroup.BUILDING_BLOCKS)));
        return block;
    }
}
// □□: util.registry.Registry (□□)
//       new Identifier()
//       .group() □□□
//       Material □□□□(1.19.3□□)
```

Fabric 1.20.1 · Java 17

```
// Fabric 1.20.1 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(FabricBlockSettings.copyOf(Blocks.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registries.BLOCK,
            new Identifier("exmod", name), block);
        Registry.register(Registries.ITEM,
            new Identifier("exmod", name),
            new BlockItem(block, new FabricItemSettings()));
        return block;
    }
}
// □□: registry.Registries (□□) ← □□□
//       new Identifier()
//       .group() □□□ → ItemGroupEvents
//       Material □□□□
```

Fabric 1.20.4 · Java 17

```
// Fabric 1.20.4 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(FabricBlockSettings.copyOf(Blocks.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registries.BLOCK,
            new Identifier("exmod", name), block);
        Registry.register(Registries.ITEM,
            new Identifier("exmod", name),
            new BlockItem(block, new FabricItemSettings()));
        return block;
    }
}
// □□: Registries (□□)
//       new Identifier()
//       ItemGroupEvents □□□□□
//       DrawContext □□ MatrixStack (1.20.2+)
//  □1.20.1□□□□□□□□□□□API□□
```

Fabric 1.20.6 · Java 21

```
// Fabric 1.20.6 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(FabricBlockSettings.copyOf(Blocks.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registries.BLOCK,
            Identifier.of("exmod", name), block);
        Registry.register(Registries.ITEM,
            Identifier.of("exmod", name),
            new BlockItem(block, new FabricItemSettings()));
        return block;
    }
}
// □□: Registries (□□)
//       Identifier.of() ← □□□□□□
//       Java 21 ← Java□□□□
//       Data Components □□□□
//       □□□□□□□□
```

Fabric 1.21.1 · Java 21

```
// Fabric 1.21.1 □□□□
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

public class ModBlocks {
    public static final Block EXAMPLE_BLOCK = register("example_block",
        new Block(AbstractBlock.Settings.copy(Blocks.STONE)
            .strength(3.0f)
            .requiresTool()));

    private static Block register(String name, Block block) {
        Registry.register(Registries.BLOCK,
            Identifier.of("exmod", name), block);
        Registry.register(Registries.ITEM,
            Identifier.of("exmod", name),
            new BlockItem(block, new Item.Settings()));
        return block;
    }
}
// □□: Registries (□□)
//       Identifier.of()
//       Java 21
//       Data Components □□□□NBT
//       □□□□□□□□(JSON)
//       FabricItemSettings → Item.Settings
```

Fabric 版本三大分水岭

1.19.2 → 1.20.1: Registry包路径变更 (util.registry.Registry → registry.Registries，单数→复数) 1.20.4 → 1.20.6: Identifier.of() 替代构造函数；Java 17→21；引入 Data Components 1.20.6 → 1.21.1: 附魔完全数据驱动；FabricItemSettings 回归 Item.Settings

3.2 Forge（5个版本）

Forge 1.18.2 · Java 17

```
// Forge 1.18.2 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.Material;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(ForgeRegistries.BLOCKS, "exmod");

    public static final RegistryObject<Block> EXAMPLE = BLOCKS.register("example",
        () -> new Block(BlockBehaviour.Properties
            .of(Material.STONE)
            .strength(3f)
            .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + ForgeRegistries
//       RegistryObject<T>
//       Material □□□
//       .tab(CreativeModeTab.TAB_MISC)
//       EntityType.Builder.create()
```

Forge 1.19.2 · Java 17

```
// Forge 1.19.2 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.Material;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(ForgeRegistries.BLOCKS, "exmod");

    public static final RegistryObject<Block> EXAMPLE = BLOCKS.register("example",
        () -> new Block(BlockBehaviour.Properties
            .of(Material.STONE)
            .strength(3f)
            .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + ForgeRegistries
//       Material □□□□
//       CreativeModeTab.builder() □□
//       □1.18.2□□□□
//       TileEntity→BlockEntity□□□□
```

Forge 1.20.1 · Java 17

```
// Forge 1.20.1 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.Material;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(ForgeRegistries.BLOCKS, "exmod");

    public static final RegistryObject<Block> EXAMPLE = BLOCKS.register("example",
        () -> new Block(BlockBehaviour.Properties
            .of(Material.STONE)
            .strength(3f)
            .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + ForgeRegistries
//       Material □□□□□?
//       CreativeModeTab.builder()
//       EntityType.Builder.of() (□□create)
//       net.minecraftforge.fml.common.Mod
```

Forge 1.20.4 · Java 17

```
// Forge 1.20.4 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.Blocks;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(ForgeRegistries.BLOCKS, "exmod");

    public static final RegistryObject<Block> EXAMPLE = BLOCKS.register("example",
        () -> new Block(BlockBehaviour.Properties
            .copyOf(Blocks.STONE)
            .strength(3f)
            .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + ForgeRegistries
//       BlockBehaviour.Properties.copyOf()
//       Material □□□□□□
//       □1.20.1 API□□□□
//       Forge 46.x (1.20.1□47.x)
```

Forge 1.21.1 · Java 21

```
// Forge 1.21.1 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredBlock;

public class ModBlocks {
    public static final DeferredRegister.Blocks BLOCKS =
        DeferredRegister.createBlocks("exmod");

    public static final DeferredBlock<Block> EXAMPLE =
        BLOCKS.registerSimpleBlock("example",
            BlockBehaviour.Properties.of()
                .strength(3f)
                .requiresCorrectToolForDrops());
}
// □□: DeferredRegister.createBlocks() □□
//       DeferredBlock<T> □□ RegistryObject<Block>
//       Java 21
//       Data Components □□NBT
//       □□□□□□
//       □□Tier□TagKey□□int
```

Forge 版本关键变化

1.18.2 → 1.19.2: 差异很小，均为DeferredRegister+ForgeRegistries模式 1.20.1 → 1.20.4: Forge版本47.x→46.x(注意是降序)，Material移除，API接近 1.20.4 → 1.21.1: Java 17→21，RegistryObject→DeferredBlock，Data Components

3.3 NeoForge（3个版本）

NeoForge 1.20.4 · Java 17

```
// NeoForge 1.20.4 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.bus.api.IEventBus;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.minecraft.core.registries.Registries;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(Registries.BLOCK, "exmod");

    public static final DeferredRegister.BlockEntry<Block> EXAMPLE =
        BLOCKS.register("example",
            () -> new Block(BlockBehaviour.Properties
                .copyOf(Blocks.STONE)
                .strength(3f)
                .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + Registries.BLOCK
//       BlockEntry<T> (□□RegistryObject)
//       Java 17
//       mods.toml (□□neoforge.mods.toml)
//       NeoGradle □□ (□□ModDevGradle)
```

NeoForge 1.20.6 · Java 21

```
// NeoForge 1.20.6 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.bus.api.IEventBus;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.minecraft.core.registries.Registries;

public class ModBlocks {
    public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(Registries.BLOCK, "exmod");

    public static final DeferredRegister.BlockEntry<Block> EXAMPLE =
        BLOCKS.register("example",
            () -> new Block(BlockBehaviour.Properties
                .copyOf(Blocks.STONE)
                .strength(3f)
                .requiresCorrectToolForDrops()));

    public static void register(IEventBus bus) {
        BLOCKS.register(bus);
    }
}
// □□: DeferredRegister + Registries.BLOCK
//       BlockEntry<T>
//       Java 21 ← Java□□□□
//       neoforge.mods.toml ← □□
//       Data Components □□□□
//       Mixin JAVA_21
```

NeoForge 1.21.1 · Java 21

```
// NeoForge 1.21.1 □□□□
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredBlock;

public class ModBlocks {
    public static final DeferredRegister.Blocks BLOCKS =
        DeferredRegister.createBlocks("exmod");

    public static final DeferredBlock<Block> EXAMPLE =
        BLOCKS.registerSimpleBlock("example",
            BlockBehaviour.Properties.of()
                .strength(3f)
                .requiresCorrectToolForDrops());
}
// □□: DeferredRegister.createBlocks() □□
//       DeferredBlock<T> □□ BlockEntry<T>
//       Java 21
//       ModDevGradle □□
//       Data Components □□□□
//       □□□□□□□□□
//       neoforge.mods.toml
```

NeoForge 版本演进

1.20.4: 早期稳定版，Java 17，mods.toml，NeoGradle，BlockEntry 1.20.6: Java升级到21，改名neoforge.mods.toml，Data Components引入 1.21.1: 完全体，ModDevGradle，createBlocks()工厂，DeferredBlock

4. 关键系统跨版本演进

4.1 附魔系统：类继承 → 数据驱动

1.21.x 中附魔系统从面向对象完全转向数据驱动，是 AI 幻觉的重灾区。1.20.6 为过渡版本，两种方式并存但类继承已开始废弃。

传统方式 (≤1.20.4) · Java Java

```
// □□□□ (1.20.4□□□, Fabric□□)
public class FireTouchEnchantment extends Enchantment {

    public FireTouchEnchantment() {
        super(Enchantment.Rarity.RARE,
              EnchantmentCategory.DIGGER,
              new EquipmentSlot[] { EquipmentSlot.MAINHAND });
    }

    @Override
    public int getMinPower(int level) {
        return 1 + level * 10;
    }

    @Override
    public int getMaxLevel() {
        return 3;
    }
}
// □□:
// Registry.register(Registries.ENCHANTMENT,
//     new Identifier("exmod", "fire_touch"),
//     new FireTouchEnchantment());
//
// □□: □□Enchantment□
//       □□□□□Rarity/Category/Slots
//       □□getMinPower/getMaxLevel□□□
//       EnchantmentCategory□□□□□□□□
```

数据驱动 (≥1.21) · Java JSON

```
// □□□□□□ (1.21+, JSON□□)
// □□: data/exmod/enchantment/fire_touch.json
{
  "description": { "translate": "enchantment.exmod.fire_touch" },
  "supported_items": "#exmod:fire_touch_supporters",
  "primary_items": "#exmod:fire_touch_primary",
  "max_level": 3,
  "min_cost": { "base": 1, "per_level_above_first": 10 },
  "max_cost": { "base": 31, "per_level_above_first": 10 },
  "anvil_cost": 2,
  "slots": ["mainhand"],
  "effects": {
    "tick": {
      "effect": {
        "type": "minecraft:set_fire",
        "duration": 3,
        "probability": 0.5
      }
    }
  }
}
// Java□□□□□RegistryKey□□:
// public static final RegistryKey<Enchantment> FIRE_TOUCH =
//     RegistryKey.of(Registries.ENCHANTMENT,
//         Identifier.of("exmod", "fire_touch"));
//
// □□: JSON□□□□□□□□□□
//       □□□□□(effects□□)
//       supported_items□□□□□□□
//       □□□□□□□□□□
```

4.2 Identifier / ResourceLocation 构造

版本范围	Fabric	Forge/NeoForge	说明
≤ 1.20.4	new Identifier("mod", "name")	new ResourceLocation("mod", "name")	构造函数直接创建
1.20.5 ~ 1.20.6	Identifier.of("mod", "name")	ResourceLocation.fromNamespaceAndPath()	工厂方法替代构造函数
≥ 1.21	Identifier.of("mod", "name")	ResourceLocation.fromNamespaceAndPath()	构造函数已私有

4.3 注册表包路径（Fabric）

版本	包路径	引用方式	说明
≤ 1.19.2	net.minecraft.util.registry	Registry.BLOCK	单数Registry类
≥ 1.20.1	net.minecraft.registry	Registries.BLOCK	复数Registries类
			注意: 1.19.3就改了，但通常按1.20.1作为分界线

4.4 Mixin compatibilityLevel

Java版本	compatibilityLevel	对应MC版本
Java 8	JAVA_8	1.16.5及更早
Java 17	JAVA_17	1.17.1 ~ 1.20.4
Java 21	JAVA_21	1.20.5 ~ 1.21.x

5. 错误修复样例

以下是错误模式库中最典型的幻觉错误及其修复方案。AI 生成代码编译失败后，系统自动匹配错误模式并注入正确用法。

Top 1 高频错误：编造API方法 · Java critical

```
// □ □□□□□□□□□□□□□□□□
public static Block badRegister(String name, Block block) {
    return RegistryMethods.getRegistry(Registries.BLOCK)
        .register(new Identifier("exmod", name), block);
}

// □ □□□Fabric 1.20.1 □□□□□□□
public static Block goodRegister(String name, Block block) {
    return Registry.register(Registries.BLOCK,
        new Identifier("exmod", name), block);
}

// □□: ERR_001  cannot find symbol.*getRegistry
// □□: AI□□□□□□□□RegistryMethods□
// □□: □□ Registry.register(Registries.BLOCK, ...)
```

类别	数量	典型错误
API方法不存在/编造	12	cannot find symbol, method does not exist
版本混淆	10	incompatible API version, wrong signature
Mixin错误	8	Mixin apply failed, target not found
构建配置错误	8	Could not resolve, entrypoint not found
JSON格式错误	7	JsonSyntaxException, model not found
导入/包名错误	5	package does not exist

6. 模板覆盖矩阵

加载器	版本	Java	全类型	说明
Fabric	1.16.5	8		超老版本，仅工程模板
	1.18.2	17		
	1.19.2	17		
	1.20.1	17		基准版本，最完整
	1.20.4	17		
	1.20.6	21		Data Components引入
	1.21.1	21		附魔数据驱动
Forge	1.18.2	17		
	1.19.2	17		
	1.20.1	17		基准版本
	1.20.4	17		
	1.21.1	21		
NeoForge	1.20.4	17		早期稳定版
	1.20.6	21		Java21过渡
	1.21.1	21		基准版本，最活跃

7. 使用流程

Step 1 检索模板

根据 MC 版本 + 加载器 + 内容类型，从 templates/ 中检索匹配的模板文件

Step 2 注入 Prompt

将模板代码注入 System Prompt，由 prompts/context_builder.py 自动完成

Step 3 AI 生成

AI 基于参考模板生成代码，约束在验证过的 API 范围内

Step 4 编译验证

Gradle 编译检查，提取错误信息

Step 5 错误匹配

编译失败时，从 error-patterns/ 中匹配已知错误模式，注入修复方案

Step 6 交付用户

编译通过才输出，否则进入修复重试循环（上限 3 次）

8. 版本历史

版本	日期	主要更新
v3.0	2026-09	补齐14个全类型版本 + 全版本代码对比章节 3加载器×14版本×12内容类型，243个模板文件
v2.0	2026-09	新增Fabric/Forge 1.19.2/1.20.4/1.21.1 + NeoForge 1.20.4 新增Legacy/过渡版本工程模板
v1.0	2026-09	初始发布：Fabric/Forge 1.20.1 + NeoForge 1.21.1 50条错误模式 + 40条API索引 + Prompt模板

KunYuEcosystem

AI 参考包 v3.0 · 代码附录

229 个模板文件 · 3 加载器 · 14 全类型版本

Fabric 1.18.2~1.21.1 (6版) · Forge 1.18.2~1.21.1 (5版) · NeoForge 1.20.4~1.21.1 (3版)

Fabric 模组模板

Fabric 1.18.2

advancement/advancement.json

```
{
    // ============================================================
    // □□/□□□□ - Fabric 1.18.2
    // ============================================================
    // □□□□□: data/examplemod/advancements/example_advancement.json
    //
    // □□□□□□Advancement□□ Minecraft 1.12+ □□□□□□
    // □□□□□□□□□□□□□□□□□□□□toast□□□□□□□□
    //
    // □□ 1.19.2 □□□□□□□
    //   1.18.2 □ inventory_changed □□□□□□□□□ "item"□□□□□□□
    //   1.19.2 □□□ "item"□□□□□□□□□
    //   □1.20.1 □□□ "items" □□□□□□□□
    //
    //   1.18.2 □□:
    //     "conditions": {
    //         "items": [
    //             { "item": "examplemod:example_item" }
    //         ]
    //     }
    //
    //   1.20.1 □□:
    //     "conditions": {
    //         "items": [
    //             { "items": ["examplemod:example_item"] }
    //         ]
    //     }
    //
    // □□□□:
    // - display:     □□□□□□□□□□□□□□□□□□□□□□□□
    // - parent:      □□□□□□□□□□□□□□□□
    // - criteria:    □□□□□□□□□□□□□□□□□□□□□□□□
    // - requirements: □□□□□□□□□□□□AND / OR□
    // - rewards:     □□□□□□□□□□□□□□□□□□□□
    //
    // □□□□:
    // 1. □□□□□□parent□□□□□□ display.background □□
    // 2. □1.18.2 □□□inventory_changed □□□□□ "item"□□□□□□□ "items"□□□□□
    // 3. criteria □□□□□□□□□□ trigger □□
    // 4. requirements □□□□□□□□□□□ AND□□□□□ OR
    // 5. frame □□: task(□□) / goal(□□) / challenge(□□)
    // ============================================================

    // □□□ID□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    "parent": "examplemod:example/root",

    // □□□□□
    "display": {
        // □□
        "icon": {
            "item": "examplemod:example_item"
        },

        // □□□□□□ translate □□□□□ literal □□□□□□
        "title": {
            "translate": "advancements.examplemod.example.title"
        },

        // □□
        "description": {
            "translate": "advancements.examplemod.example.description"
        },

        // □□□□□: task(□□,□□) / goal(□□,□□) / challenge(□□,□□)
        "frame": "task",
        // □□□□□□□□□□□□□
```

```
        "show_toast": true,

        // □□□□□□□□□
        "announce_to_chat": true,

        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        "hidden": false

        // □□□□□ background □□□□□□□□□□
        // "background": "minecraft:textures/block/stone.png"
    },

    // □□□□
    // □□ key □□□□□□□□□□□□□□ trigger □ conditions
    "criteria": {
        "get_example_item": {
            // □□□: □□□□□□□□□□
            "trigger": "minecraft:inventory_changed",
            "conditions": {
                // □1.18.2 □□□□□ "item"□□□□□□□□□
                // 1.20.1 □□ "items"□□□□□□□□
                "items": [
                    {
                        "item": "examplemod:example_item"
                    }
                ]
            }
        },
        "another_condition": {
            // □□□: □□□□□□□□□□□□
            "trigger": "minecraft:location",
            "conditions": {
                "player": {
                    "location": {
                        "dimension": "minecraft:overworld"
                    }
                }
            }
        }
    },

    // □□□□□□□□□
    // □□□□□ AND□□□□□□
    // □□□□□ OR□□□□□□
    // □□□□: □□ get_example_item AND another_condition
    "requirements": [
        ["get_example_item"],
        ["another_condition"]
    ],

    // □□□□
    "rewards": {
        // □□□□□
        "experience": 10,

        // □□□□□□□□□□□□
        "loot": [
            "examplemod:example_loot_table"
        ],

        // □□□□□□□□□□□□
        "function": "examplemod:grant_example_reward"
    }
}
```

block/simple_block.java

```
/*
 * simple_block.java
 * ------------------------------------------------------------------------
 * □□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□
 *   1. □□□□□□□□□□
```

```
 *      1.18.2 ～ 1.19.2 ：包路径 net.minecraft.util.registry.Registry
 *      ：字段 Registry.BLOCK（大写）
 *
 *   2. 从Material 到设置：
 *      ：方法名不变 FabricBlockSettings.of(Material.STONE)
 *      ：Material 在 1.19.3 被移除
 *
 *   3. 从ItemGroup 到设置 .group()：
 *      ：方法名不变 new Item.Settings().group(ItemGroup.BUILDING_BLOCKS)
 *      ：.group() 在 1.19.3 被移除
 *
 *   4. 从Identifier 到资源定位：
 *      ：方法名不变 new Identifier("modid", "name")
 *
 *   5. 从Text API 到文本组件：
 *      1.18.2: Text.literal() / Text.translatable()（静态工厂方法）
 *      1.19.2: new LiteralText() / new TranslatableText()（构造函数）
 *      ：在1.19.0-1.19.2 之间的过渡版本中两者都可用
 *
 * 对于 1.20.1 的迁移要点：
 *   - 1.20.1 中 net.minecraft.registry.Registries（复数）
 *   - 1.20.1 中 Material 被彻底移除 FabricBlockSettings.create()
 *   - 1.20.1 中 .group() 被彻底移除 ItemGroupEvents
 * ----------------------------------------------------------------------------
 */
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.block.Material;
import net.minecraft.item.BlockItem;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * 简单方块类，演示方块注册 (Fabric 1.18.2)
 *
 * 关键要点：
 * 1. 方块注册到 Registry.BLOCK（大写）1.18.2 中仍是 Registry。
 * 2. 方块 BlockItem 注册到物品注册表 Registry.ITEM。
 * 3. 1.18.2 中 ItemGroup 仍可用，新版本中被移除，用 .group() 将物品放入创造栏。
 * 4. Fabric 使用 DeferredRegister 和 @Mod 注解，与 Forge 不同。
 * 5. 从1.18.2 开始，Text 工厂方法从 Text.literal() 变为构造函数。
 */
public class SimpleBlock extends Block {

    public static final String MOD_ID = "examplemod";

    /**
     * 方块实例。
     * 在 static final 字段初始化时完成注册。
     */
    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * 注册方块及其对应物品。
     * 先注册 Block 本身，再注册一个 BlockItem。
     *
     * 对于 1.19.2 的说明：
     * - 方块注册到大写的 Registry.BLOCK
     * - FabricBlockSettings.of(Material.STONE) 在新版本中被移除
     * - .group(ItemGroup.BUILDING_BLOCKS) 在新版本中被移除
     * - 文本工厂从 Text API（1.18.2 ： Text.literal()，1.19.2 ： new LiteralText()）
     */
    private static Block registerBlock(String name) {
        // ---- 注册方块本身 ----
        // 在1.18.2 中仍可用 Material ：
        // 也可以用 FabricBlockSettings.copyOf(Blocks.STONE)（新版本中推荐）
```

```
        AbstractBlock.Settings settings = FabricBlockSettings.of(Material.STONE)
                .strength(3.0f)          // □□=□□=3.0
                .requiresTool()          // □□□□□□□□□□□□
                .luminance(15);           // □□□□ 15

        Block block = new SimpleBlock(settings);

        // ---- □□□□□□□ Block □ Registry.BLOCK ----
        // □1.18.2□□□ Registry.BLOCK□□□□□□□□
        // 1.20.1+ □□ Registries.BLOCK□□□□
        Registry.register(Registry.BLOCK, new Identifier(MOD_ID, name), block);

        // ---- □□□□□□□ BlockItem □ Registry.ITEM ----
        // □1.18.2 □□□□□□□□□□ .group(ItemGroup.BUILDING_BLOCKS) □□□□□□□□
        // 1.20.1 □ .group() □□□□□□□□□ ItemGroupEvents □□
        //
        // ItemGroup □ 1.18.2 □□□□□□□□□□
        //   ItemGroup.BUILDING_BLOCKS  ItemGroup.MISC  ItemGroup.COMBAT
        //   ItemGroup.TOOLS  ItemGroup.FOOD  ItemGroup.INGREDIENTS  □
        BlockItem item = new BlockItem(block,
                new FabricItemSettings().group(ItemGroup.BUILDING_BLOCKS));

        Registry.register(Registry.ITEM, new Identifier(MOD_ID, name), item);

        return block;
    }

    // ========================================================================
    // □□□□□□□□
    // ========================================================================
    //
    // □□□□□□□ ModInitializer □□□□□□□□
    //
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
    // ========================================================================
    // 1.18.2 □ .group() □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□ 1.20.1 □□□□□□□□□□□□□□□□□□
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    // ========================================================================
}
```

command/simple_command.java

```
/*
 * simple_command.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□□
 *   1. □CommandRegistrationCallback □□□□□□
 *      1.18.2 □ 1.19.2 □□□□ v1 □□□□□□2 □□□□□
 *      import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;
 *      v1 □□: (dispatcher, dedicated)
 *
 *   2. □Text API □□□□□□□□□□
 *      1.18.2: Text.literal("...") / Text.translatable("...")□□□□□□□□
 *      1.19.2: new LiteralText("...") / new TranslatableText("...")□□□□□□□
 *      □□□1.19.0-1.19.2 □□□□□□□□□□□□□□□□□1.19.3 □□□□□□□□
 *
 *   3. □CommandManager API □□□
 *      □□□□□□□ CommandManager.literal() □ CommandManager.argument()
 *
 *   4. □EntityArgumentType □□□□□
 *      □□□□□□ net.minecraft.command.argument.EntityArgumentType
 *
 * □□ 1.20.1 □□□□□□□□□□
 *   - 1.20.1 □□ v2 □□□□□3 □□□□□(dispatcher, registryAccess, environment)
```

```
 *   - 1.20.1 □□ Text.literal()□□ 1.18.2 □□□
 * ----------------------------------------------------------------------------
 */
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.CommandManager;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.text.Text;
import net.minecraft.util.Formatting;

import static net.minecraft.server.command.CommandManager.literal;
import static net.minecraft.server.command.CommandManager.argument;

/**
 * □□□□□□□□ - Fabric 1.18.2
 *
 * □□□□□□□□□□
 * 1. □□ CommandRegistrationCallback v1 □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * □□ 1.19.2 □□□□□□□
 * 1. □□□ v1 □□□□□□2 □□□□
 * 2. □□□□1.18.2 □□ Text.literal()□1.19.2 □□ new LiteralText()
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□□□ v1 □□□□□□□□ 2 □□□ (dispatcher, dedicated)
 *    1.20.1 □□ v2 □□□□□□□ 3 □□□ (dispatcher, registryAccess, environment)
 * 2. Text.literal() □□□□□□□□□□
 *
 * □□□□:
 * - 1.18.2 □□□□ v1 □□□□□□□□□□ v2□v2 □ 1.20+ □□□□
 * - CommandManager.literal() □ argument() □□□□□□□
 * - □□□□□□ int□□□□□□□□□□□□□□ 1□
 * - Feedback □□ Supplier<Text>□lambda□□□□□□
 */
public class ModCommands {

    /**
     * □□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        // □1.18.2□□□ v1 □□□□
        // v1 □□□□: (CommandDispatcher<ServerCommandSource> dispatcher, boolean dedicated)
        // dedicated: true □□□□□□□□false □□□□□□□□□□
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, dedicated) -> {
                // □□□□□
                registerExampleCommand(dispatcher);
            }
        );
    }

    /**
     * □□ /example □□□
     *
     * □□□□:
     * /example                          - □□□□□□
     * /example hello                    - □□□□□□□
     * /example hello <player>           - □□□□□□□□□
     * /example set <value>              - □□□□□
     * /example give <player> <count>    - □□□□□□□
     */
    private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
        dispatcher.register(
            literal("example")
                // □□□□□□□: □□□□
```

```
            .executes(context -> {
                // 在1.18.2及以前 Text.literal() 可以直接用
                // 1.19.2 及以后 new LiteralText() 已经移除
                context.getSource().sendFeedback(
                    () -> Text.literal("用法: /example <hello|set|give>")
                        .formatted(Formatting.YELLOW),
                    false
                );
                return 1;
            })

            // /example hello - 对自己说你好
            .then(literal("hello")
                .executes(ModCommands::sayHelloToSelf)
                // /example hello <player> - 对指定玩家说你好
                .then(argument("target", EntityArgumentType.player())
                    .executes(ModCommands::sayHelloToTarget)
                )
            )

            // /example set <value> - 设置值
            .then(literal("set")
                .then(argument("value", IntegerArgumentType.integer(0, 100))
                    .executes(ModCommands::setValue)
                )
            )

            // /example give <player> <count> - 给予物品示例
            .then(literal("give")
                .then(argument("target", EntityArgumentType.player())
                    .then(argument("count", IntegerArgumentType.integer(1, 64))
                        .executes(ModCommands::giveItem)
                    )
                )
            )
    );
}

/**
 * /example hello - 对自己说你好
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    // 在1.18.2及以前 Text.literal() 可以 new LiteralText()
    source.sendFeedback(
        () -> Text.literal("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;
}

/**
 * /example hello <player> - 对指定玩家说你好
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    // 在1.18.2及以前 Text.literal() 可以 new LiteralText()
    target.sendMessage(
        Text.literal("Hello from " + source.getName() + "!")
            .formatted(Formatting.GOLD)
    );

    source.sendFeedback(
        () -> Text.literal("已对 " + target.getName().getString() + " 说你好")
            .formatted(Formatting.GREEN),
        true
    );
```

```
        return 1;
    }

    /**
     * /example set <value> - □□□
     */
    private static int setValue(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        int value = IntegerArgumentType.getInteger(context, "value");
        ServerCommandSource source = context.getSource();

        source.sendFeedback(
            () -> Text.literal("□□□□□: " + value)
                .formatted(Formatting.AQUA),
            false
        );
        return 1;
    }

    /**
     * /example give <player> <count> - □□□□□□□□
     */
    private static int giveItem(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
        int count = IntegerArgumentType.getInteger(context, "count");
        ServerCommandSource source = context.getSource();

        // □□: □□□□□□□□
        net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
            net.minecraft.item.Items.DIAMOND, count
        );
        target.getInventory().insertStack(stack);

        source.sendFeedback(
            () -> Text.literal("□□ " + target.getName().getString()
                + " □□□ " + count + " □□□")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }
}

/*
 * ============================================================================
 * 1.19.2 □□□□□□□□□□
 * ============================================================================
 *
 * // 1.19.2 □□ new LiteralText() □□ Text.literal():
 * source.sendFeedback(
 *     () -> new LiteralText("Hello!").formatted(Formatting.GREEN),
 *     false
 * );
 *
 * ============================================================================
 * 1.20.1 □□□□□□□□□□
 * ============================================================================
 *
 * // 1.20.1 □□ v2 □□□3 □□□□:
 * import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
 *
 * CommandRegistrationCallback.EVENT.register(
 *     (dispatcher, registryAccess, environment) -> {
 *         registerExampleCommand(dispatcher);
 *     }
 * );
 *
 * // 1.20.1 □□ Text.literal()□□ 1.18.2 □□□:
 * source.sendFeedback(
 *     () -> Text.literal("Hello!").formatted(Formatting.GREEN),
 *     false
 * );
 * ============================================================================
```

```
 */
```

enchantment/custom_enchantment.java

```
/*
 * custom_enchantment.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□
 *      1.18.2 □ 1.19.2 □□□□ net.minecraft.util.registry.Registry
 *      □□□□ Registry.ENCHANTMENT□□□□□□□
 *
 *   2. □□□□□□□ API □□□
 *      □□□□□□□□□ Enchantment □
 *      □□□□ EnchantmentTarget □□□□□□□□□□□□□□
 *      □1.21+ □□□□□□□□□□□□
 *
 *   3. □Identifier □□□□□□
 *      □□□□□□□□ new Identifier("modid", "name")
 *
 *   4. □Text API □□□□□□□□
 *      1.18.2: Text.literal() / Text.translatable()
 *      1.19.2: new LiteralText() / new TranslatableText()
 *
 * □□ 1.20.1 □□□□□□□□□□□□
 *   - 1.20.1 □□ Registries.ENCHANTMENT□□□□
 *   - 1.20.1 □□□□ Enchantment □□□□□□□□□□□□ 1.21 □□□□□
 * ------------------------------------------------------------------------
 */
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.enchantment.EnchantmentTarget;
import net.minecraft.entity.EquipmentSlot;
import net.minecraft.item.ItemStack;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□□□□□□□ - Fabric 1.18.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□ Enchantment□
 * 2. □□ Registry.register □□□□ Registry.ENCHANTMENT
 * 3. □□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□
 * - □□ Registry.ENCHANTMENT□□ 1.19.2 □□□
 * - □□□□: net.minecraft.util.registry.Registry□□ 1.19.2 □□□
 * - □□□□□□□□□ 1.19.2 □□□□□
 * - □□□□□□□ Text API□1.18.2 □ Text.literal()□1.19.2 □ new LiteralText()□
 *
 * □□□□□□□:
 * - Enchantment.Rarity: □□□□□□□□□□□□□□□□□□□□
 *   COMMON      weight=10  □□□□□□□□□
 *   UNCOMMON    weight=5   □□□□□□□□□□□□□
 *   RARE        weight=2   □□□□□□□□□
 *   VERY_RARE   weight=1   □□□□□□□□□□□□□
 *
 * - EnchantmentTarget: □□□□□□□□□□□□
 *   ALL          □□□□
 *   ARMOR         □□
 *   FEET/LEGS/CHEST/HEAD  □□□□□□
 *   WEAPON        □
 *   TOOL          □□□□/□/□/□□
 *   BOW           □
 *   FISHING_ROD   □□□
 *   BREAKABLE     □□□□□
 *   TRIDENT       □□□
 *   CROSSBOW      □
 *   VANISHABLE    □□□□□□
 *
```

```
 * - EquipmentSlot[]: □□□□□□□□□□□□
 *   MAINHAND    □□
 *   OFFHAND     □□
 *   HEAD        □□
 *   CHEST       □□
 *   LEGS        □□
 *   FEET        □□
 *
 * □□□□□:
 * - □□□ mod_id □□□□□□ mod ID
 * - getMinPower □□□□□: 10 + (level - 1) * 15
 * - getMaxPower □□□□ getMinPower + 15~50
 * - getMaxLevel □□□□□□□□□□
 * - 1.18.2 □ Enchantment.canCombine() □□□□□□1.21+ □□□□□
 */
public class ModEnchantments {

    /**
     * □□□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□
     *
     * □1.18.2□□□□ Registry.ENCHANTMENT□□□□□□□□□□
     */
    public static final Enchantment EXAMPLE_ENCHANTMENT = Registry.register(
        Registry.ENCHANTMENT,
        new Identifier("examplemod", "example_enchantment"),
        new ExampleEnchantment(
            Enchantment.Rarity.RARE,           // □□□□: □□
            EnchantmentTarget.WEAPON,          // □□□□□□□□□□: □□
            EquipmentSlot.MAINHAND             // □□□□□: □□
        )
    );

    /**
     * □□□□□□□□□□□□□□□□□□□□
     */
    public static void register() {
        // □□□□□□□□□□□□□□□□□□□□
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    public static class ExampleEnchantment extends Enchantment {

        /**
         * □□□□□□
         *
         * @param rarity  □□□□□□□
         * @param target  □□□□□□□□□□
         * @param slots   □□□□□□□□□□□□□
         */
        public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
            // □□□□□□□□□□□
            super(rarity, target, slots);
        }

        /**
         * □□□□□□□□□□□□□□□
         * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□"□□□□□"□□□□□□□□□□□□□
         *
         * □□□□□□: 10 + (level - 1) * 15
         * 1□: 10, 2□: 25, 3□: 40
         *
         * @param level □□□□□
         * @return □□□□□□□□□□□□□
         */
        @Override
        public int getMinPower(int level) {
            return 10 + (level - 1) * 15;
        }

        /**
         * □□□□□□□□□□□□□□□
         * □□□□□□□□□□□□□□□□□□□□
```

```
         *
         * @param level □□□□
         * @return □□□□□□□□□□□
         */
        @Override
        public int getMaxPower(int level) {
            return super.getMinPower(level) + 50;
        }

        /**
         * □□□□□□□□□
         * □□: □□ 3 □□□□□□□□ 3 □
         *
         * @return □□□□
         */
        @Override
        public int getMaxLevel() {
            return 3;
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□□
         * □□□□□ EnchantmentTarget □□□□□□□□□□□□□□□□□□□□
         *
         * @param stack □□□□
         * @return □□□□□
         */
        @Override
        public boolean isAcceptableItem(ItemStack stack) {
            // □□□□□: □□□□□□□□□ EnchantmentTarget
            // □□□□□□□□□□□□□
            return super.isAcceptableItem(stack);
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□□
         * □□□□□ true□□□□□□□□□□□□□□□
         *
         * □□□□□1.18.2 □ 1.19.2 □□□□□□□
         * 1.21+ □□□□□□□□□□ JSON □□ exclusive_set □□
         *
         * @param other □□□□
         * @return □□□□□
         */
        @Override
        public boolean canCombine(Enchantment other) {
            // □□: □□□□□□□□
            // if (other == Enchantments.SHARPNESS) {
            //     return false;
            // }
            return super.canCombine(other);
        }
    }
}

/*
 * ============================================================================
 * □□□□□□□□□
 * ============================================================================
 * □□□□□: assets/examplemod/lang/en_us.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment",
 *     "enchantment.examplemod.example_enchantment.desc": "A custom enchantment"
 * }
 *
 * □□□□□□□: assets/examplemod/lang/zh_cn.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "□□□□□",
 *     "enchantment.examplemod.example_enchantment.desc": "□□□□□□□□□"
 * }
 *
 * ============================================================================
 * □□ ID □□□□□
 * ============================================================================
```

```
 * - □□□ Identifier □□□□□
 * - □□□□□□□□: enchantment.<mod_id>.<enchantment_name>
 * - □□□□□□□□□□ ID □□
 */
```

entity/simple_entity.java

```
/*
 * simple_entity.java
 * ------------------------------------------------------------------------
 * □□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□
 *      1.18.2 □ 1.19.2 □□□□ net.minecraft.util.registry.Registry
 *      □□□□ Registry.ENTITY_TYPE□□□□□□□
 *
 *   2. □RegistryKey / RegistryKeys □□□□□□□
 *      □□ net.minecraft.util.registry.RegistryKey / RegistryKeys
 *
 *   3. □Text API □□□□□□□□
 *      1.18.2: Text.literal() / Text.translatable()
 *      1.19.2: new LiteralText() / new TranslatableText()
 *
 *   4. □EntityType.Builder.build() □□□□□
 *      □□□□□□□ .build() □ .build(null)
 *
 *   5. □FabricDefaultAttributeRegistry □□□□□
 *      □□□□□□ net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry
 *
 * □□ 1.20.1 □□□□□□□□□□□□
 *   - 1.20.1 □□ Registries.ENTITY_TYPE□□□□
 *   - 1.20.1 □□ net.minecraft.registry.RegistryKey / RegistryKeys
 * ------------------------------------------------------------------------
 */
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;
import net.minecraft.world.World;

/**
 * □□□□□□□□□ - Fabric 1.18.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□
 * - □□ Registry.ENTITY_TYPE□□ 1.19.2 □□□□□□□□□ Registry□
 * - □□□: net.minecraft.util.registry.Registry□□ 1.19.2 □□□
 * - □□ new Identifier() □□□□□□□□ 1.19.2 □□□
 * - □□□□□□□ Text API□1.18.2 □ Text.literal()□1.19.2 □ new LiteralText()□
 *
 * □□□□□□□□:
 * - □□□□□□□□□□□□□□□□ (EntityType, World) □□
 * - EntityType.Builder □ dimensions □□□□□□"□□"□
 *   □□ (0.6f, 1.8f) □□ 0.6x1.8 □□□□
 * - createMobAttributes() □□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□ FabricDefaultAttributeRegistry
 */
public class ExampleEntity extends PathAwareEntity {

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □1.18.2□□□ Registry.ENTITY_TYPE□□□□□□□□
```

```
// 注册: mod_id 必须与你的 mod ID
public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
    Registry.register(
        Registry.ENTITY_TYPE,
        new Identifier("examplemod", "example_entity"),
        EntityType.Builder.<ExampleEntity>create(
                // 构造函数引用（实体实例化时调用）
                ExampleEntity::new,
                // 生成组（决定生物刷怪蛋颜色）
                SpawnGroup.CREATURE
            )
            // 碰撞箱尺寸（宽x高，单位：格）
            .dimensions(0.6f, 1.8f)
            // 追踪范围（玩家在多少区块内能看到实体） 5
            .trackingRange(8)
            // 是否每次移动都发送速度更新（通常 false）
            .build()
    );

// ============================================================
// 构造函数
// ============================================================
// 必须调用父类的 EntityType 构造函数！
// 参数 <? extends PathAwareEntity> 表示通用类型
public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
    super(entityType, world);
}

// ============================================================
// 属性注册表
// ============================================================
// 返回值类型是DefaultAttributeContainer，
// 必须在 createMobAttributes() 中注册所有属性！
//
// 常用属性:
// - GENERIC_MAX_HEALTH      最大生命值（默认 20 = 10 颗心）
// - GENERIC_MOVEMENT_SPEED  移动速度（默认 0.25）
// - GENERIC_ATTACK_DAMAGE   攻击伤害（默认 2）
// - GENERIC_FOLLOW_RANGE     AI 追踪范围（默认 16）
// - GENERIC_ARMOR           护甲值
// - GENERIC_KNOCKBACK_RESISTANCE 击退抗性（0.0~1.0）
public static DefaultAttributeContainer createMobAttributes() {
    return PathAwareEntity.createMobAttributes()
        .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)          // 10颗心
        .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)      // 正常速度
        .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)       // 攻击伤害
        .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)        // 追踪范围
        .add(EntityAttributes.GENERIC_ARMOR, 0.0)               // 无护甲
        .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0); // 无击退抗性
}

// ============================================================
// 每帧 tick 调用
// ============================================================
@Override
public void tick() {
    super.tick();
    // 在这里添加每 tick 逻辑
    // 提示: 客户端和服务端都会调用！
}

// ============================================================
// 初始化 AI 目标
// ============================================================
@Override
protected void initGoals() {
    // 在这里添加你的 AI 目标（Goal）
    // 常用 AI:
    // - MeleeAttackGoal        近战攻击
    // - WanderAroundGoal       随机游走
    // - LookAtEntityGoal       看向实体
    // - FollowTargetGoal       追踪目标
    // - SwimGoal               游泳
}

// ============================================================
```

```
    // □□□□□□□
    // ================================================================
    // □□□□□□□□□ModInitializer□□ onInitialize() □□□□
    //
    //   FabricDefaultAttributeRegistry.register(
    //       ExampleEntity.EXAMPLE_ENTITY,
    //       ExampleEntity.createMobAttributes()
    //   );
    //
    // FabricDefaultAttributeRegistry □□□□□□□ 1.18.2 □ 1.19.2 □□□□□
    //   import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
    // ================================================================
}
```

gui/screen_with_menu.java

```
/*
 * screen_with_menu.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□□: □□□GUI□□□□ ScreenHandler + HandledScreen □□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□
 *      1.18.2 □ 1.19.2 □□□□ net.minecraft.util.registry.Registry
 *      □□□ Registry.SCREEN_HANDLER□□□□□□□
 *
 *   2. □Text API □□□□□□□□□□□
 *      1.18.2: Text.literal() / Text.translatable()□□□□□□□□□
 *      1.19.2: new LiteralText() / new TranslatableText()□□□□□□□
 *
 *   3. □□□□ API □□□□
 *      □□□□□□□□ MatrixStack □□ GUI □□
 *      □DrawContext □ 1.20.2+ □□□□□
 *
 *   4. □ScreenHandlerType □□□□□□□□
 *      □□□□□□□□ new ScreenHandlerType<>(factory)
 *
 * □□ 1.20.1 □□□□□□□□□□□□□
 *   - 1.20.1 □□ Registries.SCREEN_HANDLER□□□□□
 *   - 1.20.1 □□□□ MatrixStack□DrawContext □ 1.20.2 □□□□□
 *   - 1.20.1 □□ Text.literal()□□ 1.18.2 □□□□
 * ------------------------------------------------------------------------
 */
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.text.Text;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.18.2
 *
 * □□□□□□□□□□□
 * 1. ExampleScreenHandler - □□□□□□□□□□□□□□□
```

```
 *
 * □□□□□:
 * - ScreenHandlerType □□□□ Registry.SCREEN_HANDLER □□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   □□□□: (int syncId, PlayerInventory)
 *   □□□□: (int syncId, PlayerInventory, Inventory)
 * - transferSlot□shift+click□□□□□□□□□□□□□ shift □□□□□
 * - canUse □□□□□ true□□□□□□□□□□□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ================================================================
    // ScreenHandlerType □□
    // ================================================================
    // □1.18.2□□□ Registry.SCREEN_HANDLER□□□□□□□□□
    // ScreenHandlerType □□□□□□□ ScreenHandlerFactory□□□□□□□□
    public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
        Registry.register(
            Registry.SCREEN_HANDLER,
            new Identifier("examplemod", "example_gui"),
            new ScreenHandlerType<>(ExampleScreenHandler::new)
        );

    // □□□□□□□□□□□□□□□□ Inventory□
    private final Inventory inventory;

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□□□□□ ScreenHandlerType □□ ScreenHandler □□□□
    // □□ syncId □ PlayerInventory□□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
        this(syncId, playerInventory, new SimpleInventory(9));
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
        // □□□□□□□□□□□□□□□ ScreenHandlerType □ syncId
        super(EXAMPLE_SCREEN_HANDLER, syncId);
        // □□□□□□□□□□□□□□
        checkSize(inventory, 9);
        this.inventory = inventory;

        // □□□□□□□□□□□□□□□ Inventory □□□□"□□"
        inventory.onOpen(playerInventory.player);

        // ================================================================
        // □□□□□□□□□□□□□□□□□□□□□□□□
        // ================================================================
        // □□: inventory, index(□□□□□), x(□□□□□), y(□□□□□)
        // □□: x, y □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                // 3x3 □□□□□□□□□ (8, 18) □□□□□□
                this.addSlot(new Slot(inventory, j + i * 3, 8 + j * 18, 18 + i * 18));
            }
        }

        // ================================================================
        // □□□□□□□□□□
        // ================================================================
        // □□□□□: 3□9□ = 27□
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 9; j++) {
                // □□□□□□□□ y=66 □□
                this.addSlot(new Slot(playerInventory, 9 + j + i * 9, 8 + j * 18, 66 + i * 18));
            }
        }

        // □□□□□□: 1□9□
        for (int j = 0; j < 9; j++) {
            // □□□□□□□□□ y=124
            this.addSlot(new Slot(playerInventory, j, 8 + j * 18, 124));
```

```
        }
    }

    // ================================================================
    // shift+click □□□□□□
    // ================================================================
    @Override
    public ItemStack transferSlot(PlayerEntity player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasStack()) {
            return ItemStack.EMPTY;
        }

        ItemStack originalStack = slot.getStack();
        ItemStack copyStack = originalStack.copy();

        // □□□□□:
        // 0~8: □□□□□□3x3□
        // 9~35: □□□□□
        // 36~44: □□□
        if (index < 9) {
            // □□□□□□□□□□□□
            if (!this.insertItem(originalStack, 9, 45, true)) {
                return ItemStack.EMPTY;
            }
        } else {
            // □□□/□□□□□□□□□
            if (!this.insertItem(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        // □□□□□□□□□□□□□□□□□□□□
        if (originalStack.isEmpty()) {
            slot.setStack(ItemStack.EMPTY);
        } else {
            slot.markDirty();
        }
        return copyStack;
    }

    // ================================================================
    // □□□□□□□□□□□□□□□□
    // ================================================================
    @Override
    public boolean canUse(PlayerEntity player) {
        // □□□□□ true□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□
        return this.inventory.canPlayerUse(player);
    }
}

/**
 * ================================================================
 * □□□ Screen □□HandledScreen□
 * ================================================================
 * HandledScreen □□□ ScreenHandler □□□□□□□
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □1.18.2□□□ MatrixStack □□□□□□□ 1.19.2 □□□
 * 1.20.2+ □□ DrawContext
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    /**
     * □□□ Screen □□□□
     * ScreenRegistry.register □□□□□□□□□
     */
    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        // □□□□□□□□□
```

```
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        // □□□□□□
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□□□
     * □□□□□: src/main/resources/assets/examplemod/textures/gui/example_gui.png
     * □□□□□□□□ 256x256□□□□□□ 176x166 □□
     *
     * □1.18.2□□□ MatrixStack □□□□□ 1.19.2 □□□□
     * 1.20.2+ □□ DrawContext □□
     */
    @Override
    protected void drawBackground(net.minecraft.client.util.math.MatrixStack matrices, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);
        net.minecraft.client.render.RenderSystem.setShaderTexture(0,
            new Identifier("examplemod", "textures/gui/example_gui.png"));

        // □□□□□: x, y, u(□□x□□), v(□□y□□), □, □
        int x = (this.width - this.backgroundWidth) / 2;
        int y = (this.height - this.backgroundHeight) / 2;
        this.drawTexture(matrices, x, y, 0, 0, this.backgroundWidth, this.backgroundHeight);
    }

    @Override
    public void render(net.minecraft.client.util.math.MatrixStack matrices, int mouseX, int mouseY, float delta) {
        // □□□□□□□□□□□□□□□□
        this.renderBackground(matrices);
        super.render(matrices, mouseX, mouseY, delta);
        // □□□□□□□□□
        this.drawMouseoverTooltip(matrices, mouseX, mouseY);
    }
}

/**
 * ================================================================
 * □□□□□□□
 * ================================================================
 * ScreenHandlerType □□□□□□□□ModInitializer□:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER □□□□□□□□□□□
 *
 * Screen □□□□□□□□ClientModInitializer□:
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
 *   );
 *
 * ================================================================
 * □□ GUI □□□□□□□□ BlockEntity □ Item □□□□
 * ================================================================
 *
 *   // □1.18.2□□□ Text.literal() □□ new LiteralText()
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       Text.literal("□□□□")
 *   ));
 *
 *   // 1.19.2 □□□□:
 *   // player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *   //     (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *   //     new LiteralText("□□□□")
 *   // ));
 */
```

item/simple_item.java

```
/*
 * simple_item.java
 * ------------------------------------------------------------------------
 * □□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
```

```
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□□
 *      1.18.2 □ 1.19.2 □□□□ net.minecraft.util.registry.Registry
 *      □□□□ Registry.ITEM□□□□□□□
 *
 *   2. □ItemGroup □□□□ .group() □□□
 *      □□□□□□□□ new Item.Settings().group(ItemGroup.MISC)
 *      ItemGroup □□□□□□□□□□ItemGroup.MISC, ItemGroup.BUILDING_BLOCKS □
 *
 *   3. □Identifier □□□□□□□□
 *      □□□□□□□□ new Identifier("modid", "name")
 *
 *   4. □FabricItemSettings □□□
 *      □□□□□□□□ FabricItemSettings□□□□□□□□
 *
 *   5. □FoodComponent □□□□□□
 *      1.18.2: net.minecraft.item.FoodComponent□□ 1.19.2 □□□
 *      1.21+ □□□□□ Food
 *
 *   6. □Text API □□□□□□□□
 *      1.18.2: Text.literal() / Text.translatable()
 *      1.19.2: new LiteralText() / new TranslatableText()
 *
 * □□ 1.20.1 □□□□□□□□□□□□
 *   - 1.20.1 □□ Registries.ITEM□□□□
 *   - 1.20.1 □ .group() □□□□□□□□ ItemGroupEvents
 *   - 1.20.1 □□ ItemGroups □□□□ ItemGroup □□
 * ------------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.minecraft.item.Item;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□□□□□□□ (Fabric 1.18.2)
 *
 * □□ 1.19.2 □□□□□□□
 * 1. □□ Registry.ITEM□□ 1.19.2 □□□□□□□□ Registry□
 * 2. ItemGroup □□□□ .group() □□□□□□□□□□□□□□□ 1.19.2 □□□
 * 3. □□□□□□□ Text API□1.18.2 □ Text.literal()□1.19.2 □ new LiteralText()□
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□ Registry.ITEM □□ Registries.ITEM□□□□□□□□
 * 2. ItemGroup □□□□ .group() □□□□□□□□□□□□□
 * 3. ItemGroup □□□□□□□□□ItemGroup.MISC□□□□ ItemGroups □
 */
public class ModItems {

    public static final String MOD_ID = "examplemod";

    /**
     * □□□□□□□□□□□□□□□□
     *
     * □1.18.2 □□□□□□□□□□□ .group(ItemGroup.MISC) □□□□□□□□□□
     * 1.20.1 □□□□□□□□□□□□□□□□ ItemGroupEvents □□□
     *
     * □□□□□□
     * - FabricItemSettings □□□ Item.Settings□□□□□ Fabric □□□□□□
     * - □□□□□ new Item.Settings()□□□□□ 1.18.2 □□□□□□
     */
    public static final Item EXAMPLE_ITEM = register(
            "example_item",
            // □1.18.2 □□□.group(ItemGroup.MISC) □□□□□□□□□□
            // 1.20.1 □□□□□□□□□□□□□□□□ ItemGroupEvents
            new Item(new FabricItemSettings().group(ItemGroup.MISC))
    );

    /**
     * □□□□□□□□□□□□□□
     * maxDamage □□□□□□□□□□□□□□□□□maxCount=1□□
```

```
     */
    public static final Item EXAMPLE_DURABLE_ITEM = register(
            "example_durable_item",
            new Item(new FabricItemSettings()
                    .maxDamage(256)
                    .group(ItemGroup.TOOLS)  // □1.18.2 □□□□□□□□
            )
    );

    /**
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *
     * □1.18.2□□□ Registry.ITEM□□□□□□□□□ Registries.ITEM
     * □□□□net.minecraft.util.registry.Registry□□□ net.minecraft.registry.Registry□
     */
    public static Item register(String name, Item item) {
        // new Identifier(namespace, path)□namespace = mod id□path = □□□
        // 1.18.2 □ 1.19.2 □□□ new Identifier() □□□□
        return Registry.register(Registry.ITEM, new Identifier(MOD_ID, name), item);
    }

    /**
     * □□□□□□□□□□□□□□□ JVM □□ ModItems □□
     */
    public static void registerModItems() {
        @SuppressWarnings("unused")
        Item ensureClassInit = EXAMPLE_ITEM;
    }

    /*
     * ============================================================================
     * □□□□□□□□□□
     * ============================================================================
     *
     * □1.18.2 □□□□□□□□□□□
     *   new Item(new FabricItemSettings().group(ItemGroup.MISC))
     *
     *   ItemGroup □□□1.18.2 □□□□□□□
     *   ItemGroup.BUILDING_BLOCKS  ItemGroup.MISC       ItemGroup.COMBAT
     *   ItemGroup.TOOLS             ItemGroup.FOOD       ItemGroup.INGREDIENTS
     *   ItemGroup.BREWING           ItemGroup.REDSTONE   ItemGroup.SEARCH
     *
     * □1.20.1 □□□□□□□□□□
     *   new Item(new FabricItemSettings())  // □□ .group()
     *
     *   // □□□□□□□□□□□□□□□□□□□□
     *   ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
     *       .register(entries -> entries.add(ModItems.EXAMPLE_ITEM));
     *
     *   ItemGroups □□□1.20.1 □□□□□
     *   ItemGroups.BUILDING_BLOCKS  ItemGroups.MISC       ItemGroups.COMBAT
     *   ItemGroups.TOOLS             ItemGroups.FOOD       ItemGroups.INGREDIENTS
     *   □□
     * ============================================================================
     */

    /*
     * ============================================================================
     * □□□□□□□
     * ============================================================================
     * 1. □□□□□assets/<modid>/lang/en_us.json
     *      { "item.examplemod.example_item": "Example Item" }
     * 2. □□□□□assets/<modid>/models/item/example_item.json
     * 3. □□□□□assets/<modid>/textures/item/example_item.png
     * ============================================================================
     */

    // □□□□□□□□□
    private ModItems() {
    }
}
```

mixin/inject_mixin.java

```
/*
 * inject_mixin.java
```

```
 * ------------------------------------------------------------------------------
 * 适用版本: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn 映射)
 * Java 版本: Java 17
 * 功能说明: 演示如何@Inject Mixin 注入到目标方法中
 *
 * 兼容 1.19.2 的说明:
 *   Mixin API 在 1.18.2 和 1.19.2 之间保持稳定
 *   注入点和回调 org.spongepowered.asm.mixin.* 不变
 *   compatibilityLevel 使用 JAVA_17 即可
 *
 * 兼容 1.20.1 的说明（重要）:
 *   Mixin 核心 API 仍然稳定
 *   1.20.1 起 compatibilityLevel 推荐 JAVA_17（1.21+ 推荐 JAVA_21）
 * ------------------------------------------------------------------------------
 */
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfoReturnable;

/**
 * @Inject Mixin 注入示例类 - Fabric 1.18.2
 *
 * 演示如何使用 @Inject 注入到目标方法的不同位置
 *
 * 兼容 1.19.2 的说明:
 * - Mixin 核心 API 保持稳定
 * - compatibilityLevel 推荐 JAVA_17
 *
 * @Inject 注入点:
 * - 可以注入到方法的开头（HEAD, RETURN, TAIL 等多个注入点）
 * - 可以注入到方法的返回前（可以修改返回值）
 * - 可以取消原方法执行（通过 CallbackInfo 或 CallbackInfoReturnable 实现）
 *
 * 常用注入点（@At 参数）:
 * - "HEAD"         方法开头
 * - "RETURN"       方法返回前（每个 return 语句前）
 * - "TAIL"         方法末尾
 * - "INVOKE"       在调用其他方法时（target 指定方法）
 * - "INVOKE_ASSIGN" 在调用方法并赋值时
 * - "FIELD_ACCESS" 访问字段时
 *
 * 使用步骤:
 * 1. 创建一个类 mixin 并添加注解
 * 2. @Mixin 指定要注入的目标类（MinecraftClient.class）
 * 3. 创建一个方法，方法名可以任意，参数为 CallbackInfo）
 * 4. 方法上添加注解，如 CallbackInfoReturnable<T> 或者 CallbackInfo
 * 5. CallbackInfo.cancel() 可以取消原方法执行（仅在 RETURN/HEAD 注入点有效）
 * 6. CallbackInfoReturnable.setReturnValue() 可以修改返回值
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {

    /**
     * 在 MinecraftClient.tick() 方法开头注入
     *
     * @param ci 回调信息对象
     *           - ci.cancel(): 取消原方法执行
     *           - 注意: HEAD 注入 cancel 会跳过整个方法体执行
     */
    @Inject(
        method = "tick",           // 目标方法名（开发环境 SRG 名）
        at = @At("HEAD"),          // 注入点: 方法开头
        cancellable = true         // 允许取消原方法执行（调用 cancel 前必须 true）
    )
    private void onTickStart(CallbackInfo ci) {
        // 这里会在 tick 方法开头执行
        // 注意: method 和 private，方法名可以任意修改

        // 示例: 条件取消原方法 tick
        // if (someCondition) {
        //     ci.cancel();  // 取消原方法执行
```

```
        // }
    }

    /**
     * 在 MinecraftClient.tick() 方法末尾注入
     * 使用 TAIL 表示在方法最后执行，不取消原逻辑
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // 这里可以做 tick 统计等
        // 注意：不要在这里做重操作
    }

    /**
     * 演示如何修改返回值
     * 使用 CallbackInfoReturnable 可以获取并修改返回值
     *
     * @param cir 回调信息，可用于设置返回值
     */
    @Inject(
        method = "getCurrentFps",   // 目标方法
        at = @At("RETURN"),         // 在方法返回前
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // 示例：翻倍返回值
        // cir.setReturnValue(cir.getReturnValue() * 2);  // 修改返回值
        // cir.cancel();  // 取消后续执行（跳过原方法）
    }
}
```

mixin/mixin_config.json

```
{
    // ============================================================
    // Mixin 配置文件模板 - Fabric 1.18.2
    // ============================================================
    // 文件位置: src/main/resources/examplemod.mixins.json
    //
    // 需要在 fabric.mod.json 中添加 "mixins" 字段引用:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // 关于 1.19.2 的说明:
    //   Mixin 配置格式与 1.18.2 和 1.19.2 完全相同。
    //   唯一区别是 compatibilityLevel: "JAVA_17"。
    //
    // 关于 1.20.1 及更高版本的说明:
    //   Mixin 配置格式同样兼容。
    //   1.20.1 起 compatibilityLevel 支持 JAVA_17、1.21+ 支持 JAVA_21。
    //
    // 字段说明:
    // - required:           是否必须（true=加载失败则崩溃，false=静默跳过）
    // - package:            Mixin 类的根包路径
    // - compatibilityLevel: Java版本兼容级别（1.18.2 至少 JAVA_17）
    // - mixins:             客户端Mixin类列表（仅在客户端加载）
    // - client:             客户端专用Mixin列
    // - server:              专用服务端Mixin类列表
    // - injectors:           注入器设置
    //   - defaultRequire:    默认要求注入数量（1=至少匹配一个，0=不检查）
    //
    // 常见问题:
    // 1. compatibilityLevel 必须是 JAVA_17（1.18.2 需要 Java 17+）
    // 2. package 必须与实际Mixin类的包路径一致
    // 3. mixins 中的类名不含 .java 后缀，只写类名
    // 4. client 中的类只在客户端加载，不要引用客户端专有类
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_17",
```

```
    // 客户端Mixin（仅在客户端加载，可选）
    "mixins": [
        "ExampleInjectMixin"
    ],

    // 客户端专用Mixin
    "client": [
        "client.ExampleClientMixin"
    ],

    // 注入配置
    "injectors": {
        // defaultRequire: 1 表示所有 @Inject 默认必须至少匹配一次
        // 防止目标方法签名变更后静默失效
        // 设为 0 则匹配失败不报错（调试时可能掩盖bug）
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ============================================================================
// build.gradle —— Fabric 1.18.2 模组构建脚本（Groovy DSL）
// ============================================================================
// 说明：
//  - 本脚本适用 Fabric 1.18.2 模组开发，使用 Loom 构建工具
//  - 所有可调参数集中在 gradle.properties 中，本脚本只读取
//  - 推荐运行环境：JDK 17 + Gradle 7.x（最低 7.6 以上）
//
// 关于 1.19.2 build.gradle 的说明：
//  1. Loom 插件：0.12-SNAPSHOT（1.19.2 用 1.0-SNAPSHOT）
//  2. Minecraft 版本：1.18.2（1.19.2 用 1.19.2）
//  3. Yarn 映射：1.18.2+build.4（1.19.2 用 1.19.2+build.28）
//  4. Fabric Loader：0.14.10（1.19.2 用 0.14.18）
//  5. Fabric API：0.77.0+1.18.2（1.19.2 用 0.77.0+1.19.2）
//  6. Java 版本：17（与 1.19.2 相同）
// ============================================================================
// 如果你坚持用1.18.2 搭配 Java 17，与 1.19.2 相同：
//  1.18.2 官方推荐使用 Java 17 ，支持从1.17+ 开始 Java 16/17。
// ============================================================================

plugins {
    // Fabric Loom 0.12：适用 Minecraft 1.18.2 的 Loom 版本
    // 1.19.2 则使用 Loom 的 1.0-SNAPSHOT（1.18.2 不能用 0.12-SNAPSHOT）
    // 如果你用Loom 0.12 构建 1.18.x 版本
    id 'fabric-loom' version '0.12-SNAPSHOT'
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
    archivesName = project.name
}

repositories {
    mavenCentral()
}

// ----------------------------------------------------------------------------
// 依赖项，版本号来自 gradle.properties
// ----------------------------------------------------------------------------
dependencies {
    // Minecraft 1.18.2 官方 jar
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn 1.18.2+build.4 映射表
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader 0.14.10
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
    // Fabric API 0.77.0+1.18.2
    // 如果你用1.18.2 的 Fabric API 而不 1.19.2 的，则使用版本号0.77.0。
    //         版本号格式+1.18.2 vs +1.19.2，必须匹配你的 Minecraft 版本。
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
}
// ----------------------------------------------------------------------------
```

```
// □□□□□□□ gradle.properties □□ mod_version □□□□ fabric.mod.json
// ------------------------------------------------------------------------
processResources {
    inputs.property "version", project.version

    filesMatching("fabric.mod.json") {
        expand "version": project.version
    }
}

// ------------------------------------------------------------------------
// □□□□□□Minecraft 1.18.2 □□ Java 17
// □□ 1.19.2 □□□□1.17+ □□ Java 17□1.16.5 □ Java 8□
// ------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}

java {
    withSourcesJar()

    sourceCompatibility = JavaVersion.VERSION_17
    targetCompatibility = JavaVersion.VERSION_17
}

// ------------------------------------------------------------------------
// jar □□
// ------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/version_diff.md

```
# Fabric 1.18.2 □ 1.19.2 □□□□□□□

> □□□□□□ Fabric 1.18.2 □□□□ 1.19.2 □ API □□□□
> 1.18.2 □ 1.19.2 □□□□ Java 17□□ Registry □□□□□□□□
> □□□□□□□□□□□□□□□□□□□□□□□□ Fabric API □□□□□□□□□ API □□□□

---

## □□□□□

1.18.2 □□ 1.18 □□□□1.18 → 1.18.1 → 1.18.2□□□□ 1.19 □□□□□□□□□□□□□□□
1.18 □□□□□□□□□□□□Caves & Cliffs□□1.19 □□□□□□□□□□□□□□□□□□□□□□□□□□□□
□□□□□□ Java 17□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

| □□□□ | 1.18.2 | 1.19.2 |
|--------|--------|--------|
| Java □□ | 17 | 17□□□□□ |
| Registry □□□□ | `net.minecraft.util.registry.Registry` | □□ |
| Registries □ | □□□□□1.19.3 □□□□□ | □□□□□□□□□ |
| □□□□□□□ | `Registry.BLOCK` / `Registry.ITEM`□□□□□□□ | □□ |
| Identifier □□ | `new Identifier(modid, name)` | □□ |
| □□□□□□□ | `.group(ItemGroup.MISC)` | □□ |
| ItemGroup □□ | `ItemGroup.BUILDING_BLOCKS`□□□□□□□ | □□ |
| Material □ | □□□□1.19.3 □□□□□ | □□□□□□□ |
| □□□□□ | `new LiteralText("...")` / `new TranslatableText("...")` | □□□1.20 □□□□ |
| □□□□□ | `command.v1` | `command.v1`□□□□1.19.3 □□ v2□ |
| □□□□□ | `BuiltinRegistries`□1.18 □□□□ | □□ |
| Loom □□ | 0.12-SNAPSHOT | 1.0-SNAPSHOT |
| Fabric Loader | 0.14.10 | 0.14.18 |
| Fabric API | 0.77.0+1.18.2 | 0.77.0+1.19.2 |

> □□□1.18.2 □ 1.19.2 □ Fabric API □□□□□□□ 0.77.0□
> □□□□□+1.18.2 vs +1.19.2□□□□□□□□□□□□ Minecraft □□□□□□

---

## □□□□□□□□□□

### 2.1 □□□□□□
1.18.2 □□□□□□□□ 1.19.2 □□□□□□□□□□ `net.minecraft.util.registry.Registry`□
```

```java
// 1.18.2 —— 至 1.19.2 通用
import net.minecraft.util.registry.Registry;
import net.minecraft.util.Identifier;

Registry.register(Registry.BLOCK, new Identifier("modid", "name"), block);
Registry.register(Registry.ITEM, new Identifier("modid", "name"), item);
```

2.2 Registry 的变化

这两个静态字段 `Registry` 的类型变成了`Registry.BLOCK`和`Registry.ITEM` 的类型
`Registries` 类型在 1.19.3 中新增，1.18.2 至 1.19.2 中不存在。

2.3 BuiltinRegistries

1.18.2 至 1.19.2 中新增 `BuiltinRegistries` 类，用于注册世界生成

```java
// 1.18.2 至 1.19.2 通用
import net.minecraft.util.registry.BuiltinRegistries;

BuiltinRegistries.add(
    BuiltinRegistries.CONFIGURED_FEATURE,
    new Identifier("modid", "example_ore"),
    configuredFeature
);
```

> 在旧版1.18 中，注册世界生成需要使用`BuiltinRegistries`类
> 在 1.16.5 中，注册世界生成需要使用 1.18.2 至 1.19.2 中的注册方式

三、命令注册的变化

3.1 命令注册

1.18.2 至 1.19.2 中新增 `command.v1` 命令注册

```java
// 1.18.2 —— 至 1.19.2 通用
import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;

CommandRegistrationCallback.EVENT.register((dispatcher, dedicated) -> {
    dispatcher.register(
        CommandManager.literal("example")
            .executes(context -> {
                context.getSource().sendFeedback(
                    new LiteralText("Hello!"),
                    false
                );
                return 1;
            })
    );
});
```

> 新增`command.v2`中的 `registryAccess` 参数，需要在 1.19.3 中使用
> 1.18.2 至 1.19.2 中新增 `command.v1`。

四、渲染的变化

4.1 渲染 API 的变化

1.18.2 至 1.19.2 中渲染相关的类在 1.17 中的渲染相关的类中的位置发生了变化

```java
// 1.18.2 至 1.19.2 通用
import net.minecraft.client.render.RenderLayer;
import net.minecraft.client.render.VertexConsumerProvider;
import net.minecraft.client.render.entity.EntityRenderer;
import net.minecraft.client.render.entity.EntityRendererFactory;
import net.minecraft.client.util.math.MatrixStack;
```

```

### 4.2 □□□□□□□□□□

1.18.2 □ 1.19.2 □□□ `EntityRendererFactory.Context`□

```java
// 1.18.2 □ 1.19.2 □□
public class MyEntityRenderer extends EntityRenderer<MyEntity> {
    public MyEntityRenderer(EntityRendererFactory.Context ctx) {
        super(ctx);
    }

    @Override
    public void render(MyEntity entity, float yaw, float tickDelta,
                       MatrixStack matrices, VertexConsumerProvider vertexConsumers, int light) {
        // □□□□
    }
}
```

> □□□1.16.5 □□□□□□□□□□□□□□□□□ 1.18.2/1.19.2 □□□□□□
> □ 1.18.2 □ 1.19.2 □□□□□□□□

4.3 □□□□□□□□□□

1.19.2 □□□□□□□□□□□□□□□□□□□□□□□□□□ `RenderLayer` □□□□
□□□□□□□□□□□□□□□□□□□□□□□□ `ClassNotFoundException`□□□□□□□□□

□□□□□□□□□□□□□□

5.1 □□□□□□

1.18.2 □ 1.19.2 □ `FabricBlockSettings`□`Item.Settings`□`Material` □□□□□□□

```java
// 1.18.2 □ 1.19.2 □□
import net.minecraft.block.Material;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;

FabricBlockSettings.of(Material.STONE).strength(1.5f);
```

```java
// 1.18.2 □ 1.19.2 □□
new Item(new Item.Settings().group(ItemGroup.MISC).maxCount(64));
```

5.2 □□□□

```java
// 1.18.2 □ 1.19.2 □□
import net.minecraft.item.FoodComponent;

new Item(new Item.Settings()
    .food(new FoodComponent.Builder()
        .hunger(6)
        .saturationModifier(0.6f)
        .meat()
        .build()
    )
);
```

> □□□1.20.2+ □□ `hunger()` □□ `nutrition()`□
> 1.18.2 □ 1.19.2 □□□ `hunger()`□

□□□□□□□□□□□□

6.1 □□□□□□

1.18.2 □ 1.19.2 □□□□□□□□□□□□□□□□
```

```java
// 1.18.2 → 1.19.2 通用
import net.minecraft.item.ItemGroup;

new Item(new Item.Settings().group(ItemGroup.MISC));
```

自定义创造模式物品栏

```java
// 1.18.2 → 1.19.2 通用
import net.fabricmc.fabric.api.itemgroup.v1.FabricItemGroup;

Registry.register(Registry.ITEM_GROUP, new Identifier("modid", "example"),
    FabricItemGroup.create()
        .icon(() -> new ItemStack(Items.STONE))
        .appendItems(stacks -> {
            stacks.add(new ItemStack(EXAMPLE_ITEM));
        })
        .build()
);
```

> 注意：1.19.3 起移除 `.group()` 方法，改用 `ItemGroupEvents`。
> 1.18.2 → 1.19.2 仍可使用 `.group()`。

七、文本与翻译

7.1 字面文本

1.18.2 → 1.19.2 通用，两个版本均可使用以下写法：

```java
// 1.18.2 → 1.19.2 通用
import net.minecraft.text.LiteralText;
import net.minecraft.text.TranslatableText;

new LiteralText("Hello World");
new TranslatableText("item.modid.example");
```

> 注意：1.20 起 `new LiteralText(...)` 改为 `Text.literal(...)`。

八、1.19 新增内容 1.18.2 中的对应处理

以下内容为 1.19 新增，在1.18.2 中不存在。

8.1 深板岩洞穴

1.19 新增深板岩洞穴生物群系 Mangrove Swamp、Deep Dark 等。
1.18.2 中无对应生物群系，如需在世界生成中使用，请改用原版已有的生物群系。

8.2 红树林相关

1.19 新增红树林木材 Mangrove 系列、Sculk 感测体等方块。
在 1.18.2 中 `Items.MANGROVE_PROPAGULE` 等常量不存在。

8.3 Allay 生物

1.19 新增的 Allay 生物在1.18.2 中不存在，无法生成。

> 如果你的模组依赖这些内容，建议在文档中明确标注 API 不兼容。
> API 不兼容，仅在 1.18.2 → 1.19.2 迁移时适用。

九、迁移检查清单（1.18.2 → 1.19.2）

1. 修改 `gradle.properties` 中的版本：
   - Loom：`0.12-SNAPSHOT` → `1.0-SNAPSHOT`
   - Minecraft：`1.18.2` → `1.19.2`
   - Yarn：`1.18.2+build.4` → `1.19.2+build.28`

   - Fabric Loader：`0.14.10` → `0.14.18`
   - Fabric API：`0.77.0+1.18.2` → `0.77.0+1.19.2`
2. 将所有从 1.19 开始新增的方块、物品、实体、生物群系等内容全部删除
3. Registry 相关代码改为使用旧版本的 `net.minecraft.util.registry.Registry`
4. 将指令系统改为旧版本的 `command.v1`
5. 将文本组件改为旧版本的 `new LiteralText()`
6. 将物品栏分组改为旧版本的 `.group(ItemGroup.XXX)`
7. 将内置注册表改为旧版本的 `BuiltinRegistries`
8. 删除所有数据包内容

常见问题排查

10.1 从 1.18.2 迁移到 1.19 时找不到/报错

```java
// ❌ 错误（1.18.2 中不存在 MANGROVE 系列）
Items.MANGROVE_PROPAGULE;
Blocks.MANGROVE_LOG;

// ✅ 正确（1.18.2 中使用以下写法）
Items.OAK_SAPLING;
Blocks.OAK_LOG;
```

10.2 从 1.18.2 迁移到 1.19.3+ 的 Registries 类

```java
// ❌ 错误（1.18.2 中不存在 Registries 类）
import net.minecraft.registry.Registries;
Registry.register(Registries.BLOCK, ...);

// ✅ 正确（1.18.2 中直接使用 Registry 的静态字段）
import net.minecraft.util.registry.Registry;
Registry.register(Registry.BLOCK, ...);
```

10.3 从 1.18.2 迁移到 1.20 时报错 API

```java
// ❌ 错误（1.18.2 中不存在 Text.literal()
Text.literal("Hello");

// ✅ 正确（1.18.2 中使用构造函数）
new LiteralText("Hello");
```

recipe/shaped_recipe.json

```
{
    // ============================================================
    // 有序合成配方示例 - Fabric 1.18.2
    // ============================================================
    // 文件路径: data/examplemod/recipes/shaped_recipe.json
    //
    // 有序合成（crafting_shaped）必须严格按照 pattern 和 key 定义
    // 物品在网格中的位置与 pattern 完全一致
    //
    // 适用 1.19.2 及以上
    //   将此 JSON 文件从 1.18.2 的 1.19.2 及以上版本中
    //   删除以下字段 type / pattern / key / result 及其内容
    //
    // 适用 1.20.1 及以上
    //   将此 JSON 文件从 1.18.2 的 1.20.1 及以上版本中删除
    //
    // 注意事项:
    // 1. pattern 数组的每个字符串长度必须相同
    // 2. pattern 中使用的每个字符必须在 key 中有定义
    // 3. key 中的字符可以是任意单个字符，推荐使用 #, X, *, @ 等
    // 4. 空格字符表示该位置为空
    // 5. result 中的 count 不能超过 64
    // 6. ingredient 中的 tag 必须以 "tag" 字段而非 "item"
    // ============================================================

    "type": "minecraft:crafting_shaped",
```

```
    // pattern: □□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/shapeless_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□ - Fabric 1.18.2
    // ============================================================
    // □□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□□□□crafting_shapeless□□□□□□□□□□□
    // □□□ ingredients □□□□□□□□□□□□□□□□□
    //
    // □□ 1.19.2 □□□□
    //   □□ JSON □□□ 1.18.2 □ 1.19.2 □□□□□□□□
    //
    // □□ 1.20.1 □□□□
    //   □□ JSON □□□ 1.18.2 □ 1.20.1 □□□□□□□□
    //
    // □□□□:
    // 1. ingredients □□□□□□□□□□□□□□□□
    // 2. □□□□□□□□□□□□□□□□□□□□□□□□
    // 3. □□□ "tag" □□□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□□
    // □□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    // ============================================================
```

```
    // □□□□□□□□ - Fabric 1.18.2
    // ============================================================
    // □□□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□ 1.19.2 □□□□□
    //   1.18.2 □□□□ JSON □□□□ 1.19.2 □□□□□□
    //   □□□□□□□□□□: data/<mod_id>/tags/blocks/
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□□□□□□□□
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□□: data/<mod_id>/tags/items/□□□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
    // □□□□□□□□ - Fabric 1.18.2
    // ============================================================
    // □□□□□□: data/examplemod/tags/items/example_item_tag.json
    //
    // □□ 1.19.2 □□□□□
    //   1.18.2 □□□□ JSON □□□□ 1.19.2 □□□□□□
    //   □□□□□□□□□□: data/<mod_id>/tags/items/
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□□□□□□□□
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□:
    //   minecraft:logs          □□
    //   minecraft:planks        □□
    //   minecraft:coals         □□□
    //   minecraft:saplings      □□
    //   minecraft:flowers       □
    //
    // □□□□□:
    // 1. □□□□□□ data/<mod_id>/tags/items/ □□□
    // 2. □□□□□□□□□□□□□□□□□□□□
    //    □□: data/<mod_id>/tags/items/
    //    □□: data/<mod_id>/tags/blocks/
    // 3. values □□□□ID□□□□□□□□□□
    // 4. replace □ true □□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
```

```
    ]
}
```

worldgen/ore_feature.java

```
/*
 * ore_feature.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.18.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * Java □□: Java 17
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□□
 *      1.18.2 □ 1.19.2 □□□□ net.minecraft.util.registry.*
 *      Registry / RegistryKey / RegistryKeys / BuiltinRegistries □□□□□□
 *
 *   2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *      □□□□□□□□ ConfiguredFeature □ PlacedFeature □□□□ BuiltinRegistries
 *      BuiltinRegistries.CONFIGURED_FEATURE
 *      BuiltinRegistries.PLACED_FEATURE
 *      □1.20.1 □□□□ Registries□
 *
 *   3. □BiomeModifications API □□□
 *      □□□□□□□□ Fabric API □ BiomeModifications
 *
 *   4. □YOffset API □□□
 *      □□□□□□□□□ YOffset.getBottom() / YOffset.fixed() □□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□
 *   - 1.20.1 □□□□□□□□□□□□ Registries□□□ BuiltinRegistries□
 *   - 1.20.1 □□ net.minecraft.registry.RegistryKey / RegistryKeys
 * ------------------------------------------------------------------------
 */
package com.example.mod.worldgen;

import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.BuiltinRegistries;
import net.minecraft.util.registry.Registry;
import net.minecraft.util.registry.RegistryKey;
import net.minecraft.util.registry.RegistryKeys;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.YOffset;
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.feature.PlacedFeature;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.PlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;

import java.util.List;

/**
 * □□□□□□□□□ - Fabric 1.18.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□ ConfiguredFeature□□□□□□
 * 2. □□□□□□ PlacedFeature□□□□□□
 * 3. □□ BiomeModifications □□□□□□□□□□□
 *
 * □□ 1.19.2 □□□□□□□
 * 1. □□□□□□□□□□□ BuiltinRegistries□□ 1.19.2 □□□
 * 2. RegistryKey / RegistryKeys □□□□□ net.minecraft.util.registry.*□□ 1.19.2 □□□
 * 3. BiomeModifications API □ 1.19.2 □□
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□ConfiguredFeature □ PlacedFeature □□□□ BuiltinRegistries
 *    □□ Registries□1.20.1 □□□ Registries□
 * 2. RegistryKey / RegistryKeys □□□□□ net.minecraft.util.registry.*
 *    □1.20.1 □□□ net.minecraft.registry.*□
 *
 * □□□□□:
```

```
 *   ConfiguredFeature -> □□□□ BuiltinRegistries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□□ BuiltinRegistries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□□
 *
 * □□□□□:
 * - □□□□□□□ ConfiguredFeature□□□□□ PlacedFeature
 * - □1.18.2 □□□□□□□ BuiltinRegistries □□ Registries
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□□□□
 * - □□□□□(size)□□□□ 64□□□□ 3~12
 */
public class ModWorldGen {

    public static final String MOD_ID = "examplemod";

    // ConfiguredFeature □ RegistryKey
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    // PlacedFeature □ RegistryKey
    public static final RegistryKey<PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    /**
     * □□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□□
     */
    public static void register() {
        // ================================================================
        // 1. □□□□□□ ConfiguredFeature
        // ================================================================
        // □□□□□□□□□□□
        // □□1: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        // □□2: □□□□□□□ BlockState
        // □□3: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            // □□□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
            // □□: ModBlocks.EXAMPLE_ORE □□□□□□□□□□□□□□□□□□□□□□□□
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),  // □□□□□□□□□□□□
            9  // □□□□□
        );

        // □□ ConfiguredFeature □□
        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,    // □□□□□□: □□
            oreConfig       // □□□□□□
        );

        // □1.18.2□□□□□ BuiltinRegistries.CONFIGURED_FEATURE
        // 1.20.1 □□ Registries.CONFIGURED_FEATURE
        Registry.register(
            BuiltinRegistries.CONFIGURED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            configuredFeature
        );

        // ================================================================
        // 2. □□□□□□ PlacedFeature
        // ================================================================
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        List<PlacementModifier> placementModifiers = List.of(
            // □□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
            CountPlacementModifier.of(20),

            // □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
            SquarePlacementModifier.of(),

            // □□□□□: □ Y=-64 □ Y=64 □□□□□□□□□□□□□□□□□□□□
```

```
        // 注意: 1.18.2 的 PlacedFeature 构造函数是 (ConfiguredFeature, List<PlacementModifier>)
        // 需要用 RegistryEntry.of(configuredFeature) 包装
        PlacedFeature placedFeature = new PlacedFeature(
            net.minecraft.util.registry.RegistryEntry.of(configuredFeature),
            placementModifiers
        );

        // 在1.18.2中使用 BuiltinRegistries.PLACED_FEATURE
        // 1.20.1 中是 Registries.PLACED_FEATURE
        Registry.register(
            BuiltinRegistries.PLACED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            placedFeature
        );

        // ================================================================
        // 3. 使用 BiomeModifications 添加到生物群系
        // ================================================================
        // BiomeModifications API 在 1.18.2 和 1.19.2 中相同
        BiomeModifications.addFeature(
            // 选择要添加到的生物群系: 这里选择主世界
            BiomeSelectors.foundInOverworld(),
            // 生成阶段: 地下矿石阶段
            GenerationStep.Feature.UNDERGROUND_ORES,
            // PlacedFeature 的 RegistryKey
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                new Identifier(MOD_ID, "ore_example")
            )
        );
    }
}

/*
 * ============================================================================
 * 生物群系选择器
 * ============================================================================
 *
 * BiomeSelectors 提供了多种选择生物群系的方法:
 *
 * BiomeSelectors.all()                  选择所有
 * BiomeSelectors.foundInOverworld()     主世界
 * BiomeSelectors.foundInNether()        下界
 * BiomeSelectors.foundInTheEnd()        末地
 * BiomeSelectors.includeByKey(keys)     通过 RegistryKey 选择
 *
 * ============================================================================
 * 生成阶段说明
 * ============================================================================
 *
 * GenerationStep.Feature枚举值包括:
 * - RAW_GENERATION          原始生成阶段
 * - LAKES                   湖泊
 * - LOCAL_MODIFICATIONS     局部修改阶段
 * - UNDERGROUND_STRUCTURES  地下结构
 * - SURFACE_STRUCTURES      地表结构
 * - STRONGHOLDS             要塞
 * - UNDERGROUND_ORES        地下矿石阶段（我们用这个）
 * - UNDERGROUND_DECORATION  地下装饰阶段
 * - FLUID_SPRINGS           流体泉
 * - VEGETAL_DECORATION      植被装饰阶段
 * - TOP_LAYER_MODIFICATION  顶层修改阶段
 */
```

Fabric 1.19.2

advancement/advancement.json

```
{
    // ============================================================
    // 进度/成就文件 - Fabric 1.19.2
    // ============================================================
    // 文件位置: data/examplemod/advancements/example_advancement.json
    //
```

```
// ■■■■■■Advancement■■ Minecraft 1.12+ ■■■■■■
// ■■■■■■■■■■■■■■■■■■■■■■■■toast■■■■■■■■■
//
// ■■ 1.20.1 ■■■■■■■■
//   1.19.2 ■ inventory_changed ■■■■■■■■■■■ "item"■■■■■■■■■
//   1.20.1 ■■ "items"■■■■■■■■■■
//
//   1.19.2 ■■:
//     "conditions": {
//         "items": [
//             { "item": "examplemod:example_item" }
//         ]
//     }
//
//   1.20.1 ■■:
//     "conditions": {
//         "items": [
//             { "items": ["examplemod:example_item"] }
//         ]
//     }
//
// ■■■■■:
// - display:     ■■■■■■■■■■■■■■■■■■■■■■■■■■■■
// - parent:      ■■■■■■■■■■■■■■■■■■■
// - criteria:    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■
// - requirements: ■■■■■■■■■■■■■■AND / OR■
// - rewards:     ■■■■■■■■■■■■■■■■■■■■■■■■■■■■
//
// ■■■■■:
// 1. ■■■■■■parent■■■■■■ display.background ■■
// 2. ■1.19.2 ■■■inventory_changed ■■■■■ "item"■■■■■■■■ "items"■■■■■
// 3. criteria ■■■■■■■■■■■■■ trigger ■■
// 4. requirements ■■■■■■■■■■■■■ AND■■■■■ OR
// 5. frame ■■: task(■■) / goal(■■) / challenge(■■)
// ============================================================

// ■■■■ID■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
"parent": "examplemod:example/root",

// ■■■■■
"display": {
    // ■■
    "icon": {
        "item": "examplemod:example_item"
    },

    // ■■■■■■ translate ■■■■■ literal ■■■■■■
    "title": {
        "translate": "advancements.examplemod.example.title"
    },

    // ■■
    "description": {
        "translate": "advancements.examplemod.example.description"
    },

    // ■■■■■: task(■■,■■) / goal(■■,■■) / challenge(■■,■■)
    "frame": "task",

    // ■■■■■■■■■■■■■■■■
    "show_toast": true,

    // ■■■■■■■■■■■■
    "announce_to_chat": true,

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    "hidden": false

    // ■■■■■■ background ■■■■■■■■■■■■■■■■
    // "background": "minecraft:textures/block/stone.png"
},

// ■■■■■
// ■■ key ■■■■■■■■■■■■■■■■■■■■ trigger ■ conditions
"criteria": {
    "get_example_item": {
```

```
            // □□□□: □□□□□□□□□□□□
            "trigger": "minecraft:inventory_changed",
            "conditions": {
                // □1.19.2 □□□□□□ "item"□□□□□□□□□□□
                // 1.20.1 □□ "items"□□□□□□□□□□
                "items": [
                    {
                        "item": "examplemod:example_item"
                    }
                ]
            }
        },
        "another_condition": {
            // □□□□: □□□□□□□□□□□□□□□□
            "trigger": "minecraft:location",
            "conditions": {
                "player": {
                    "location": {
                        "dimension": "minecraft:overworld"
                    }
                }
            }
        }
    },

    // □□□□□□□□□□□□
    // □□□□□□□ AND□□□□□□□□
    // □□□□□□□ OR□□□□□□□□
    // □□□□□□: □□ get_example_item AND another_condition
    "requirements": [
        ["get_example_item"],
        ["another_condition"]
    ],

    // □□□□□
    "rewards": {
        // □□□□□□
        "experience": 10,

        // □□□□□□□□□□□□□□□
        "loot": [
            "examplemod:example_loot_table"
        ],

        // □□□□□□□□□□□□□□□
        "function": "examplemod:grant_example_reward"
    }
}
```

block/simple_block.java

```
/*
 * simple_block.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□
 *      1.19.2: import net.minecraft.util.registry.Registry;
 *              □□ Registry.BLOCK□□□□□□□
 *      1.20.1: import net.minecraft.registry.Registry;
 *              import net.minecraft.registry.Registries;
 *              □□ Registries.BLOCK
 *
 *   2. □Material □□□□□
 *      1.19.2 □ FabricBlockSettings.of(Material.STONE) □□□Material □ 1.19.3 □□□□
 *      1.20.1 □□□□□ FabricBlockSettings.create() □ .copyOf(Blocks.STONE)
 *
 *   3. □ItemGroup □□□ .group()□
 *      1.19.2 □ new Item.Settings().group(ItemGroup.BUILDING_BLOCKS) □□
 *      1.20.1 □ .group() □□□□□□□□ ItemGroupEvents
 *
 *   4. □Identifier □□□□□□□□
 *      □□□□□□□□ new Identifier("modid", "name")
 *
```

```
 *   5. 【Registry.ITEM_GROUP 变为 Registries.ITEM_GROUP】
 *      1.19.2: Registry.ITEM_GROUP
 *      1.20.1: Registries.ITEM_GROUP
 * ------------------------------------------------------------------------
 */
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.block.Material;
import net.minecraft.item.BlockItem;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * 简单方块示例类 (Fabric 1.19.2)
 *
 * 演示如何：
 * 1. 创建一个自定义方块并注册到 Registry.BLOCK（适用于1.19.2 以及 Registry 尚未 Registries）
 * 2. 为该 BlockItem 一并注册到 Registry.ITEM。
 * 3. 1.19.2 中 ItemGroup 尚未被拆分，可以直接使用 .group() 指定创造模式标签
 * 4. Fabric 没有内置 DeferredRegister 和 @Mod 注解，这是 Forge 的特性。
 */
public class SimpleBlock extends Block {

    public static final String MOD_ID = "examplemod";

    /**
     * 方块实例
     * 在 static final 字段初始化时完成注册。
     */
    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * 注册方块及其对应的物品。
     * 将给定 Block 注册为方块，并注册一个 BlockItem。
     *
     * 适用 1.20.1 及以上
     * - 使用 Registry.BLOCK 而非 Registries.BLOCK
     * - FabricBlockSettings.of(Material.STONE) 在 1.19.2 可用，1.20.1 中 Material 已移除
     * - new BlockItem(block, new FabricItemSettings().group(ItemGroup.BUILDING_BLOCKS))
     *   在 1.19.2 可用，但在更高版本1.20.1 中 .group() 已移除
     */
    private static Block registerBlock(String name) {
        // ---- 创建方块设置（材质） ----
        // 在1.19.2 可以直接使用 Material 。
        // 1.20.1 中 Material 已经移除， FabricBlockSettings.create().mapColor(MapColor.STONE_GRAY)
        //
        // 复制石头 FabricBlockSettings.copyOf(Blocks.STONE)，也可以自定义
        AbstractBlock.Settings settings = FabricBlockSettings.of(Material.STONE)
                .strength(3.0f)          // 硬度=爆炸=3.0
                .requiresTool()          // 需要工具才能挖掘
                .luminance(15);           // 发光等级 15

        Block block = new SimpleBlock(settings);

        // ---- 注册方块到 Block 和 Registry.BLOCK ----
        // 在1.19.2 可以使用 Registry.BLOCK 而非 Registries.BLOCK
        // 1.20.1 必须使用 Registry.register(Registries.BLOCK, new Identifier(...), block)
        Registry.register(Registry.BLOCK, new Identifier(MOD_ID, name), block);

        // ---- 注册方块对应的 BlockItem 到 Registry.ITEM ----
        // 在1.19.2 可以直接使用 .group(ItemGroup.BUILDING_BLOCKS) 指定创造模式标签
        // 1.20.1 中 .group() 已经移除，需要 ItemGroupEvents 处理
        //
        // ItemGroup 在 1.19.2 中尚未被拆分
        //   ItemGroup.BUILDING_BLOCKS  ItemGroup.MISC  ItemGroup.COMBAT
        //   ItemGroup.TOOLS  ItemGroup.FOOD  ItemGroup.INGREDIENTS  等
```

```
        // 1.20.1 □□□□ ItemGroups □□
        //   ItemGroups.BUILDING_BLOCKS  ItemGroups.INGREDIENTS  □
        BlockItem item = new BlockItem(block,
                new FabricItemSettings().group(ItemGroup.BUILDING_BLOCKS));

        Registry.register(Registry.ITEM, new Identifier(MOD_ID, name), item);

        return block;
    }

    // ========================================================================
    // □□□□□□□□
    // ========================================================================
    //
    // □□□□□□□ ModInitializer □□□□□□□□
    //
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
    // ========================================================================
    // 1.19.2 □ .group() □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□ 1.20.1 □□□□□□□□□□□□□□□□
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    // ========================================================================
}
```

command/simple_command.java

```
/*
 * simple_command.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □CommandRegistrationCallback □□□□□□□□□□□□□□
 *      1.19.2: import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;
 *              v1 □□□□□□□□: (dispatcher, dedicated)
 *              - dispatcher: CommandDispatcher<ServerCommandSource>
 *              - dedicated: boolean□true=□□□□□□
 *      1.20.1: import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
 *              v2 □□□□□□□□: (dispatcher, registryAccess, environment)
 *              - dispatcher: CommandDispatcher<ServerCommandSource>
 *              - registryAccess: CommandRegistryAccess
 *              - environment: CommandManager.RegistrationEnvironment
 *
 *   2. □□□□□□□□□
 *      1.19.2: new LiteralText("...") / new TranslatableText("...")
 *      1.20.1: Text.literal("...") / Text.translatable("...")
 *
 *   3. □CommandManager API □□□
 *      □□□□□□□ CommandManager.literal() □ CommandManager.argument()
 *      □□□□□□□□□□
 *
 *   4. □EntityArgumentType □□□□□
 *      □□□□□ net.minecraft.command.argument.EntityArgumentType
 * ------------------------------------------------------------------------
 */
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.CommandManager;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.text.LiteralText;
import net.minecraft.util.Formatting;
```

```java
import static net.minecraft.server.command.CommandManager.literal;
import static net.minecraft.server.command.CommandManager.argument;

/**
 * □□□□□□□□ - Fabric 1.19.2
 *
 * □□□□□□□□□□
 * 1. □□ CommandRegistrationCallback v1 □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□□□ v1 □□□□□□□□ 2 □□□ (dispatcher, dedicated)
 *    1.20.1 □□ v2 □□□□□□□ 3 □□□ (dispatcher, registryAccess, environment)
 * 2. □□ new LiteralText() □□ Text.literal() □□□□□□□
 *
 * □□□□:
 * - 1.19.2 □□□□□ v1 □□□□□□□□□□ v2□v2 □ 1.20+ □□□□
 * - CommandManager.literal() □ argument() □□□□□□□
 * - □□□□□□□ int□□□□□□□□□□□□□□□ 1□
 * - Feedback □□ Supplier<Text>□lambda□□□□□□□
 */
public class ModCommands {

    /**
     * □□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        // □1.19.2 □□□□□□ v1 □□□□
        // v1 □□□□: (CommandDispatcher<ServerCommandSource> dispatcher, boolean dedicated)
        // dedicated: true □□□□□□□□□false □□□□□□□□□□
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, dedicated) -> {
                // □□□□□
                registerExampleCommand(dispatcher);
            }
        );
    }

    /**
     * □□ /example □□□
     *
     * □□□□:
     * /example                          - □□□□□□
     * /example hello                    - □□□□□□□
     * /example hello <player>           - □□□□□□□□□
     * /example set <value>              - □□□□□
     * /example give <player> <count>    - □□□□□□□
     */
    private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
        dispatcher.register(
            literal("example")
                // □□□□□□□: □□□□
                .executes(context -> {
                    // □1.19.2 □□□□□□ new LiteralText() □□ Text.literal()
                    context.getSource().sendFeedback(
                        () -> new LiteralText("□□: /example <hello|set|give>")
                            .formatted(Formatting.YELLOW),
                        false
                    );
                    return 1;
                })

                // /example hello - □□□□□□□□
                .then(literal("hello")
                    .executes(ModCommands::sayHelloToSelf)
                    // /example hello <player> - □□□□□□□□□
                    .then(argument("target", EntityArgumentType.player())
                        .executes(ModCommands::sayHelloToTarget)
                    )
                )

                // /example set <value> - □□□
                .then(literal("set")
```

```
                .then(argument("value", IntegerArgumentType.integer(0, 100))
                    .executes(ModCommands::setValue)
                )
            )

            // /example give <player> <count> - □□□□□□□□
            .then(literal("give")
                .then(argument("target", EntityArgumentType.player())
                    .then(argument("count", IntegerArgumentType.integer(1, 64))
                        .executes(ModCommands::giveItem)
                    )
                )
            )
    );
}

/**
 * /example hello - □□□□□□□□
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    // □1.19.2 □□□□□□ new LiteralText() □□ Text.literal()
    source.sendFeedback(
        () -> new LiteralText("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;
}

/**
 * /example hello <player> - □□□□□□□□□□
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    // □1.19.2 □□□□□□ new LiteralText() □□ Text.literal()
    target.sendMessage(
        new LiteralText("Hello from " + source.getName() + "!")
            .formatted(Formatting.GOLD)
    );

    source.sendFeedback(
        () -> new LiteralText("□□ " + target.getName().getString() + " □□□□□")
            .formatted(Formatting.GREEN),
        true
    );
    return 1;
}

/**
 * /example set <value> - □□□
 */
private static int setValue(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    int value = IntegerArgumentType.getInteger(context, "value");
    ServerCommandSource source = context.getSource();

    source.sendFeedback(
        () -> new LiteralText("□□□□□□: " + value)
            .formatted(Formatting.AQUA),
        false
    );
    return 1;
}

/**
 * /example give <player> <count> - □□□□□□□□
```

```
     */
    private static int giveItem(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
        int count = IntegerArgumentType.getInteger(context, "count");
        ServerCommandSource source = context.getSource();

        // 创建: 物品堆栈对象
        net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
            net.minecraft.item.Items.DIAMOND, count
        );
        target.getInventory().insertStack(stack);

        source.sendFeedback(
            () -> new LiteralText("给予 " + target.getName().getString()
                + " 个钻石 " + count + " 颗。")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }
}

/*
 * ============================================================================
 * 1.20.1 版本的注册方式
 * ============================================================================
 *
 * // 1.20.1 使用 v2 回调（3 个参数）:
 * import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
 *
 * CommandRegistrationCallback.EVENT.register(
 *     (dispatcher, registryAccess, environment) -> {
 *         registerExampleCommand(dispatcher);
 *     }
 * );
 *
 * // 1.20.1 使用 Text.literal() 或 new LiteralText():
 * source.sendFeedback(
 *     () -> Text.literal("Hello!").formatted(Formatting.GREEN),
 *     false
 * );
 * ============================================================================
 */
```

enchantment/custom_enchantment.java

```
/*
 * custom_enchantment.java
 * ------------------------------------------------------------------------
 * 适用版本: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn 映射)
 * 功能说明: 自定义附魔：注册一个新的附魔效果
 *
 * 迁移 1.20.1 时需要修改的地方：
 *   1. 注册表访问方式变化
 *      1.19.2: import net.minecraft.util.registry.Registry;
 *              使用 Registry.ENCHANTMENT 注册附魔
 *      1.20.1: import net.minecraft.registry.Registry;
 *              import net.minecraft.registry.Registries;
 *              使用 Registries.ENCHANTMENT
 *
 *   2. 附魔构造 API 变化
 *      1.19.2 与 1.20.1 均使用 Enchantment 的
 *      构造器 EnchantmentTarget 参数，无变化。
 *      但1.21+ 版本已移除。
 *
 *   3. 用Identifier 创建对象
 *      旧版本使用 new Identifier("modid", "name")
 * ------------------------------------------------------------------------
 */
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.enchantment.EnchantmentTarget;
import net.minecraft.entity.EquipmentSlot;
```

```java
import net.minecraft.item.ItemStack;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□□□□□□□ - Fabric 1.19.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□ Enchantment□
 * 2. □□ Registry.register □□□□ Registry.ENCHANTMENT
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□
 * - □□ Registry.ENCHANTMENT □□ Registries.ENCHANTMENT
 * - □□□□: net.minecraft.util.registry.Registry□□□□ net.minecraft.registry.Registry□
 * - □□□□□□□□□□ 1.20.1 □□□□□
 *
 * □□□□□□□□:
 * - Enchantment.Rarity: □□□□□□□□□□□□□□□□□□□□□□□□
 *   COMMON      weight=10  □□□□□□□□□□
 *   UNCOMMON    weight=5   □□□□□□□□□□□□□□□□
 *   RARE        weight=2   □□□□□□□□□□
 *   VERY_RARE   weight=1   □□□□□□□□□□□□□□□□
 *
 * - EnchantmentTarget: □□□□□□□□□□□□□
 *   ALL          □□□□□
 *   ARMOR         □□
 *   FEET/LEGS/CHEST/HEAD  □□□□□□□□
 *   WEAPON        □
 *   TOOL          □□□□/□/□/□□
 *   BOW           □
 *   FISHING_ROD   □□□
 *   BREAKABLE     □□□□□□
 *   TRIDENT       □□□
 *   CROSSBOW      □
 *   VANISHABLE    □□□□□□□
 *
 * - EquipmentSlot[]: □□□□□□□□□□□□
 *   MAINHAND    □□
 *   OFFHAND     □□
 *   HEAD        □□
 *   CHEST       □□
 *   LEGS        □□
 *   FEET        □□
 *
 * □□□□□:
 * - □□□ mod_id □□□□□□ mod ID
 * - getMinPower □□□□□: 10 + (level - 1) * 15
 * - getMaxPower □□□□ getMinPower + 15~50
 * - getMaxLevel □□□□□□□□□□
 * - 1.19.2 □ Enchantment.canCombine() □□□□□□1.21+ □□□□□
 */
public class ModEnchantments {

    /**
     * □□□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□□
     *
     * □1.19.2 □□□□□□ Registry.ENCHANTMENT □□ Registries.ENCHANTMENT
     */
    public static final Enchantment EXAMPLE_ENCHANTMENT = Registry.register(
        Registry.ENCHANTMENT,
        new Identifier("examplemod", "example_enchantment"),
        new ExampleEnchantment(
            Enchantment.Rarity.RARE,           // □□□□: □□
            EnchantmentTarget.WEAPON,          // □□□□□□□□□: □□
            EquipmentSlot.MAINHAND             // □□□□□: □□
        )
    );

    /**
     * □□□□□□□□□□□□□□□□□□□
     */
    public static void register() {
        // □□□□□□□□□□□□□□□□□□□
    }
```

```
// ================================================================
// □□□□□□□
// ================================================================
public static class ExampleEnchantment extends Enchantment {

    /**
     * □□□□□
     *
     * @param rarity  □□□□□□
     * @param target  □□□□□□□□□
     * @param slots   □□□□□□□□□□□□
     */
    public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        // □□□□□□□□□□
        super(rarity, target, slots);
    }

    /**
     * □□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□"□□□□□"□□□□□□□□□
     *
     * □□□□□□: 10 + (level - 1) * 15
     * 1□: 10, 2□: 25, 3□: 40
     *
     * @param level □□□□□
     * @return □□□□□□□□□□□□
     */
    @Override
    public int getMinPower(int level) {
        return 10 + (level - 1) * 15;
    }

    /**
     * □□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * □□□□ getMinPower + 50
     *
     * @param level □□□□□
     * @return □□□□□□□□□□□□
     */
    @Override
    public int getMaxPower(int level) {
        return super.getMinPower(level) + 50;
    }

    /**
     * □□□□□□□□□□
     * □□: □□ 3 □□□□□□□□□ 3 □
     *
     * @return □□□□□
     */
    @Override
    public int getMaxLevel() {
        return 3;
    }

    /**
     * □□□□□□□□□□□□□□□□□□□□□□
     * □□□□□ EnchantmentTarget □□□□□□□□□□□□□□□□□□□□□□□□
     *
     * @param stack □□□□□
     * @return □□□□□□
     */
    @Override
    public boolean isAcceptableItem(ItemStack stack) {
        // □□□□□: □□□□□□□□□□ EnchantmentTarget
        // □□□□□□□□□□□□□□□
        return super.isAcceptableItem(stack);
    }

    /**
     * □□□□□□□□□□□□□□□□□□□□□□
     * □□□□□ true□□□□□□□□□□□□□□□□□□□
     *
     * □□□□□1.19.2 □ 1.20.1 □□□□□□□
     * 1.21+ □□□□□□□□□□□ JSON □□ exclusive_set □□
```

```
         *
         * @param other □□□□
         * @return □□□□□□
         */
        @Override
        public boolean canCombine(Enchantment other) {
            // □□: □□□□□□□□
            // if (other == Enchantments.SHARPNESS) {
            //     return false;
            // }
            return super.canCombine(other);
        }
    }
}

/*
 * ============================================================================
 * □□□□□□□□□
 * ============================================================================
 * □□□□: assets/examplemod/lang/en_us.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment",
 *     "enchantment.examplemod.example_enchantment.desc": "A custom enchantment"
 * }
 *
 * □□□□□□: assets/examplemod/lang/zh_cn.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "□□□□",
 *     "enchantment.examplemod.example_enchantment.desc": "□□□□□□□□"
 * }
 *
 * ============================================================================
 * □□ ID □□□□
 * ============================================================================
 * - □□□ Identifier □□□□□
 * - □□□□□□□: enchantment.<mod_id>.<enchantment_name>
 * - □□□□□□□□□ ID □□
 */
```

entity/simple_entity.java

```
/*
 * simple_entity.java
 * ------------------------------------------------------------------------
 * □□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□
 *      1.19.2: import net.minecraft.util.registry.Registry;
 *              □□ Registry.ENTITY_TYPE□□□□□□
 *      1.20.1: import net.minecraft.registry.Registry;
 *              import net.minecraft.registry.Registries;
 *              □□ Registries.ENTITY_TYPE
 *
 *   2. □RegistryKey / RegistryKeys □□□□□□
 *      1.19.2: net.minecraft.util.registry.RegistryKey / RegistryKeys
 *      1.20.1: net.minecraft.registry.RegistryKey / RegistryKeys
 *
 *   3. □Text □□□□□□□
 *      1.19.2: new LiteralText("...") / new TranslatableText("...")
 *      1.20.1: Text.literal("...") / Text.translatable("...")
 *
 *   4. □EntityType.Builder.build() □□□□□
 *      □□□□□□ .build() □ .build(null)
 *
 *   5. □FabricDefaultAttributeRegistry □□□□□
 *      □□□□□ net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry
 * ------------------------------------------------------------------------
 */
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.EntityType;
```

```java
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;
import net.minecraft.world.World;

/**
 * □□□□□□□□□□ - Fabric 1.19.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□
 * - □□ Registry.ENTITY_TYPE□□□ Registries.ENTITY_TYPE□
 * - □□□□: net.minecraft.util.registry.Registry□□□ net.minecraft.registry.Registry□
 * - □□ new Identifier() □□□□□□□□ 1.20.1 □□□□
 *
 * □□□□□□□□:
 * - □□□□□□□□□□□□□□□ (EntityType, World) □□
 * - EntityType.Builder □ dimensions □□□□□□"□□□□□□"□
 *   □□ (0.6f, 1.8f) □□ 0.6x1.8 □□□□□
 * - createMobAttributes() □□□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□□ FabricDefaultAttributeRegistry
 */
public class ExampleEntity extends PathAwareEntity {

    // ==================================================================
    // □□□□□□□□
    // ==================================================================
    // □1.19.2 □□□□□□□ Registry.ENTITY_TYPE □□ Registries.ENTITY_TYPE
    // □□: mod_id □□□□□□□□□□ mod ID
    public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
        Registry.register(
            Registry.ENTITY_TYPE,
            new Identifier("examplemod", "example_entity"),
            EntityType.Builder.<ExampleEntity>create(
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    ExampleEntity::new,
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    SpawnGroup.CREATURE
                )
                // □□□□□□□□□□□□x□□□□□□□□□□
                .dimensions(0.6f, 1.8f)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ 5
                .trackingRange(8)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ false□
                .build()
        );

    // ==================================================================
    // □□□□□□
    // ==================================================================
    // □□□□□□□□□□□□ EntityType □□□□□□□
    // □□ <? extends PathAwareEntity> □□□□□□□
    public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
        super(entityType, world);
    }

    // ==================================================================
    // □□□□□□□□
    // ==================================================================
    // □□□□□□□□□□□□□□DefaultAttributeContainer□
    // □□□□ createMobAttributes() □□□□□□□□□□□□□□□□
    //
    // □□□□□□:
    // - GENERIC_MAX_HEALTH      □□□□□□□□□□ 20 = 10 □□□□
    // - GENERIC_MOVEMENT_SPEED  □□□□□□□□□ 0.25□
    // - GENERIC_ATTACK_DAMAGE   □□□□□□□□□ 2□
    // - GENERIC_FOLLOW_RANGE     AI □□□□□□□□□ 16□
    // - GENERIC_ARMOR           □□□□
    // - GENERIC_KNOCKBACK_RESISTANCE □□□□□□0.0~1.0□
    public static DefaultAttributeContainer createMobAttributes() {
```

```
        return PathAwareEntity.createMobAttributes()
            .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)          // 10□□
            .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)      // □□□□
            .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)       // □□□□
            .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)        // □□□□
            .add(EntityAttributes.GENERIC_ARMOR, 0.0)               // □□□
            .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0); // □□□□
    }

    // ================================================================
    // □□ tick □□
    // ================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□□□□□ tick □□□
        // □□: □□□□□□□□□□□□□□□□
    }

    // ================================================================
    // □□□ AI □□
    // ================================================================
    @Override
    protected void initGoals() {
        // □□□□□□□□ AI □□□Goal□
        // □□ AI:
        // - MeleeAttackGoal        □□□□□□□
        // - WanderAroundGoal       □□□□
        // - LookAtEntityGoal       □□□□
        // - FollowTargetGoal       □□□□
        // - SwimGoal               □□
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □□□□□□□□ModInitializer□□ onInitialize() □□□□
    //
    //   FabricDefaultAttributeRegistry.register(
    //       ExampleEntity.EXAMPLE_ENTITY,
    //       ExampleEntity.createMobAttributes()
    //   );
    //
    // FabricDefaultAttributeRegistry □□□□□□□ 1.19.2 □ 1.20.1 □□□□□
    //   import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
    // ================================================================
}
```

gui/screen_with_menu.java

```
/*
 * screen_with_menu.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□GUI□□□□ ScreenHandler + HandledScreen □□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□
 *      1.19.2: import net.minecraft.util.registry.Registry;
 *              □□ Registry.SCREEN_HANDLER□□□□□□□
 *      1.20.1: import net.minecraft.registry.Registry;
 *              import net.minecraft.registry.Registries;
 *              □□ Registries.SCREEN_HANDLER
 *
 *   2. □□□□□□□□□□
 *      1.19.2: new LiteralText("...") / new TranslatableText("...")
 *      1.20.1: Text.literal("...") / Text.translatable("...")
 *
 *   3. □□□ API □□□
 *      □□□□□□□ MatrixStack □□ GUI □□
 *      □DrawContext □ 1.20.2+ □□□□□
 *
 *   4. □ScreenHandlerType □□□□□□□□
 *      □□□□□□□ new ScreenHandlerType<>(factory)
 * ------------------------------------------------------------------------
 */
```

```java
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.text.LiteralText;
import net.minecraft.text.Text;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.19.2
 *
 * □□□□□□□□□□□
 * 1. ExampleScreenHandler - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 2. ExampleScreen        - □□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□
 * 1. □□ Registry.SCREEN_HANDLER □□ Registries.SCREEN_HANDLER
 * 2. □□ new LiteralText() □□ Text.literal()
 * 3. GUI □□□□□ MatrixStack□□ 1.20.1 □□□DrawContext □ 1.20.2+ □□□□□
 *
 * □□□□□:
 * - □□□□□□□ ScreenHandler □□□□ syncId
 * - □□□□□□□ Packet □ syncId □□□□□□□
 * - □□□□□□□ syncId □ ScreenHandlerType □□ Screen □□
 * - Screen □ ScreenHandler □□ syncId □□
 *
 * □□□□□:
 * - ScreenHandlerType □□□□ Registry.SCREEN_HANDLER □□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   □□□□: (int syncId, PlayerInventory)
 *   □□□□: (int syncId, PlayerInventory, Inventory)
 * - transferSlot□shift+click□□□□□□□□□□□□□ shift □□□□□
 * - canUse □□□□□ true□□□□□□□□□□□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ====================================================================
    // ScreenHandlerType □□
    // ====================================================================
    // □1.19.2 □□□□□□□ Registry.SCREEN_HANDLER □□ Registries.SCREEN_HANDLER
    // ScreenHandlerType □□□□□□□□ ScreenHandlerFactory□□□□□□□□□
    public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
        Registry.register(
            Registry.SCREEN_HANDLER,
            new Identifier("examplemod", "example_gui"),
            new ScreenHandlerType<>(ExampleScreenHandler::new)
        );

    // □□□□□□□□□□□□□□□□□□□□ Inventory□
    private final Inventory inventory;

    // ====================================================================
    // □□□□□□□□
    // ====================================================================
    // □□□□□□□ ScreenHandlerType □□ ScreenHandler □□□□
    // □□ syncId □ PlayerInventory□□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
        this(syncId, playerInventory, new SimpleInventory(9));
    }

    // ====================================================================
    // □□□□□□□□
    // ====================================================================
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
        // □□□□□□□□□□□□□□□□ ScreenHandlerType □ syncId
        super(EXAMPLE_SCREEN_HANDLER, syncId);
        // □□□□□□□□□□□□□□□
        checkSize(inventory, 9);
```

```
        this.inventory = inventory;

        // □□□□□□□□□□□□□ Inventory □□□"□□"
        inventory.onOpen(playerInventory.player);

        // ================================================================
        // □□□□□□□□□□□□□□□□□□□□
        // ================================================================
        // □□: inventory, index(□□□□□), x(□□□□□), y(□□□□□)
        // □□: x, y □□□□□□□□□□□□□□□□□□□□□□□□
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                // 3x3 □□□□□□□□□ (8, 18) □□□□□□
                this.addSlot(new Slot(inventory, j + i * 3, 8 + j * 18, 18 + i * 18));
            }
        }

        // ================================================================
        // □□□□□□□□□
        // ================================================================
        // □□□□□: 3□9□ = 27□
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 9; j++) {
                // □□□□□□□□ y=66 □□
                this.addSlot(new Slot(playerInventory, 9 + j + i * 9, 8 + j * 18, 66 + i * 18));
            }
        }

        // □□□□□□: 1□9□
        for (int j = 0; j < 9; j++) {
            // □□□□□□□□ y=124
            this.addSlot(new Slot(playerInventory, j, 8 + j * 18, 124));
        }
    }

    // ================================================================
    // shift+click □□□□□□□
    // ================================================================
    @Override
    public ItemStack transferSlot(PlayerEntity player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasStack()) {
            return ItemStack.EMPTY;
        }

        ItemStack originalStack = slot.getStack();
        ItemStack copyStack = originalStack.copy();

        // □□□□□:
        // 0~8: □□□□□□3x3□
        // 9~35: □□□□□
        // 36~44: □□□□
        if (index < 9) {
            // □□□□□□□□□□□□□
            if (!this.insertItem(originalStack, 9, 45, true)) {
                return ItemStack.EMPTY;
            }
        } else {
            // □□□□/□□□□□□□□□□
            if (!this.insertItem(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        // □□□□□□□□□□□□□□□□□□□□□□□□
        if (originalStack.isEmpty()) {
            slot.setStack(ItemStack.EMPTY);
        } else {
            slot.markDirty();
        }
        return copyStack;
    }

    // ================================================================
    // □□□□□□□□□□□□□□□□□□□□
    // ================================================================
```

```
    @Override
    public boolean canUse(PlayerEntity player) {
        // □□□□ true□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□
        return this.inventory.canPlayerUse(player);
    }
}

/**
 * ================================================================
 * □□□ Screen □□HandledScreen□
 * ================================================================
 * HandledScreen □□□ ScreenHandler □□□□□□□
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □1.19.2 □□□□□□ MatrixStack □□□□□□□ 1.20.1 □□□
 * 1.20.2+ □□ DrawContext
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    /**
     * □□□ Screen □□□□
     * ScreenRegistry.register □□□□□□□□□
     */
    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        // □□□□□□□□□
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        // □□□□□□
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□□
     * □□□□□: src/main/resources/assets/examplemod/textures/gui/example_gui.png
     * □□□□□□□ 256x256□□□□□□ 176x166 □□
     *
     * □1.19.2 □□□□□□ MatrixStack □□□□□ 1.20.1 □□□
     * 1.20.2+ □□ DrawContext □□
     */
    @Override
    protected void drawBackground(net.minecraft.client.util.math.MatrixStack matrices, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);
        net.minecraft.client.render.RenderSystem.setShaderTexture(0,
            new Identifier("examplemod", "textures/gui/example_gui.png"));

        // □□□□□: x, y, u(□□x□□), v(□□y□□), □, □
        int x = (this.width - this.backgroundWidth) / 2;
        int y = (this.height - this.backgroundHeight) / 2;
        this.drawTexture(matrices, x, y, 0, 0, this.backgroundWidth, this.backgroundHeight);
    }

    @Override
    public void render(net.minecraft.client.util.math.MatrixStack matrices, int mouseX, int mouseY, float delta) {
        // □□□□□□□□□□□□□□□□
        this.renderBackground(matrices);
        super.render(matrices, mouseX, mouseY, delta);
        // □□□□□□□□□
        this.drawMouseoverTooltip(matrices, mouseX, mouseY);
    }
}

/**
 * ================================================================
 * □□□□□□□
 * ================================================================
 * ScreenHandlerType □□□□□□□□ModInitializer□:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER □□□□□□□□□□□
 *
 * Screen □□□□□□□□ClientModInitializer□:
```

```
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
 *   );
 *
 * ================================================================
 * □□ GUI □□□□□□□□ BlockEntity □ Item □□□□
 * ================================================================
 *
 *   // □1.19.2 □□□□□□ new LiteralText() □□ Text.literal()
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       new LiteralText("□□□□□")
 *   ));
 *
 *   // 1.20.1 □□□□□:
 *   // player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *   //     (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *   //     Text.literal("□□□□□")
 *   // ));
 */
```

item/simple_item.java

```
/*
 * simple_item.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□□
 *      1.19.2: import net.minecraft.util.registry.Registry;
 *              □□ Registry.ITEM□□□□□□□
 *      1.20.1: import net.minecraft.registry.Registry;
 *              import net.minecraft.registry.Registries;
 *              □□ Registries.ITEM
 *
 *   2. □ItemGroup □□□□ .group() □□□□□□□□□□
 *      1.19.2: new Item.Settings().group(ItemGroup.MISC) □□
 *              ItemGroup □□□□□□□□□□ItemGroup.MISC, ItemGroup.BUILDING_BLOCKS □
 *      1.20.1: .group() □□□□□1.19.3 □□□□□
 *              □□□□□ ItemGroupEvents.modifyEntriesEvent(ItemGroups.MISC)
 *              ItemGroups □□□□□□ItemGroups.MISC, ItemGroups.BUILDING_BLOCKS □
 *
 *   3. □Identifier □□□□□□□□
 *      □□□□□□□□□ new Identifier("modid", "name")
 *
 *   4. □FabricItemSettings □□□
 *      □□□□□□□□□ FabricItemSettings□□□□□□□□
 *
 *   5. □FoodComponent □□□□□□
 *      1.19.2: net.minecraft.item.FoodComponent□□ 1.20.1 □□□
 *      1.21+ □□□□□ Food
 * ------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.minecraft.item.Item;
import net.minecraft.item.ItemGroup;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.Registry;

/**
 * □□□□□□□□ (Fabric 1.19.2)
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□ Registry.ITEM □□ Registries.ITEM□□□□□□□□
 * 2. ItemGroup □□□□ .group() □□□□□□□□□□□□
 * 3. ItemGroup □□□□□□□□□□ItemGroup.MISC□□□□ ItemGroups □
 */
public class ModItems {
```

```
public static final String MOD_ID = "examplemod";

/**
 * □□□□□□□□□□□□□□□□
 *
 * □1.19.2 □□□□□□□□□□ .group(ItemGroup.MISC) □□□□□□□□□
 * 1.20.1 □□□□□□□□□□□□□□□□ ItemGroupEvents □□□
 *
 * □□□□□□
 * - FabricItemSettings □□□ Item.Settings□□□□□ Fabric □□□□□□
 * - □□□□ new Item.Settings()□□□□ 1.19.2 □□□□□□
 */
public static final Item EXAMPLE_ITEM = register(
        "example_item",
        // □1.19.2 □□□.group(ItemGroup.MISC) □□□□□□□□□□
        // 1.20.1 □□□□□□□□□□□□□□□□ ItemGroupEvents
        new Item(new FabricItemSettings().group(ItemGroup.MISC))
);

/**
 * □□□□□□□□□□□□□□
 * maxDamage □□□□□□□□□□□□□□□□□maxCount=1□□
 */
public static final Item EXAMPLE_DURABLE_ITEM = register(
        "example_durable_item",
        new Item(new FabricItemSettings()
                .maxDamage(256)
                .group(ItemGroup.TOOLS)  // □1.19.2 □□□□□□□□
        )
);

/**
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □1.19.2 □□□□□□ Registry.ITEM□□□□□□□□ Registries.ITEM
 * □□□□net.minecraft.util.registry.Registry□□□ net.minecraft.registry.Registry□
 */
public static Item register(String name, Item item) {
    // new Identifier(namespace, path)□namespace = mod id□path = □□□
    // 1.19.2 □ 1.20.1 □□□ new Identifier() □□□□
    return Registry.register(Registry.ITEM, new Identifier(MOD_ID, name), item);
}

/**
 * □□□□□□□□□□□□□□□□ JVM □□ ModItems □□
 */
public static void registerModItems() {
    @SuppressWarnings("unused")
    Item ensureClassInit = EXAMPLE_ITEM;
}

/*
 * ============================================================================
 * □□□□□□□□□□
 * ============================================================================
 *
 * □1.19.2 □□□□□□□□□□□□
 *   new Item(new FabricItemSettings().group(ItemGroup.MISC))
 *
 *   ItemGroup □□□1.19.2 □□□□□□□
 *   ItemGroup.BUILDING_BLOCKS  ItemGroup.MISC       ItemGroup.COMBAT
 *   ItemGroup.TOOLS             ItemGroup.FOOD       ItemGroup.INGREDIENTS
 *   ItemGroup.BREWING           ItemGroup.REDSTONE   ItemGroup.SEARCH
 *
 * □1.20.1 □□□□□□□□□□□□
 *   new Item(new FabricItemSettings())  // □□ .group()
 *
 *   // □□□□□□□□□□□□□□□□□□□□□□
 *   ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
 *       .register(entries -> entries.add(ModItems.EXAMPLE_ITEM));
 *
 *   ItemGroups □□□1.20.1 □□□□□□
 *   ItemGroups.BUILDING_BLOCKS  ItemGroups.MISC       ItemGroups.COMBAT
 *   ItemGroups.TOOLS             ItemGroups.FOOD       ItemGroups.INGREDIENTS
 *   □□
 * ============================================================================
```

```
     */

    /*
     * ============================================================================
     * □□□□□□□□
     * ============================================================================
     * 1. □□□□□□assets/<modid>/lang/en_us.json
     *      { "item.examplemod.example_item": "Example Item" }
     * 2. □□□□□□assets/<modid>/models/item/example_item.json
     * 3. □□□□□□assets/<modid>/textures/item/example_item.png
     * ============================================================================
     */

    // □□□□□□□□□□
    private ModItems() {
    }
}
```

mixin/inject_mixin.java

```
/*
 * inject_mixin.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□@Inject Mixin □□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□
 *   Mixin API □ 1.19.2 □ 1.20.1 □□□□□□□□
 *   □□□□□□□□ org.spongepowered.asm.mixin.* □□□
 *   compatibilityLevel □□ JAVA_17□
 * ------------------------------------------------------------------------------
 */
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfoReturnable;

/**
 * @Inject Mixin □□□□□□ - Fabric 1.19.2
 *
 * □□□□□□□□□□ @Inject □□□□□□□□□□□□□□
 *
 * @Inject □□□□:
 * - □□□□□□□□□□□□HEAD, RETURN, TAIL □□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□ CallbackInfo □ CallbackInfoReturnable □□
 *
 * □□□□□□□□@At □□:
 * - "HEAD"        □□□□
 * - "RETURN"      □□□□□□□□ return □□□□□
 * - "TAIL"        □□□□
 * - "INVOKE"      □□□□□□□□□□□□ target □□□□□
 * - "INVOKE_ASSIGN" □□□□□□□□□□□
 * - "FIELD_ACCESS" □□□□□□□
 *
 * □□□□:
 * 1. □□□□□□□ mixin □□□□□□□□□□□
 * 2. @Mixin □□□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ CallbackInfo□
 * 4. □□□□□□□□□□□□□□□□ CallbackInfoReturnable<T> □□ CallbackInfo
 * 5. CallbackInfo.cancel() □□□□□□□□□□□□□ RETURN/HEAD □□□□□□□
 * 6. CallbackInfoReturnable.setReturnValue() □□□□□□□□
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     *
     * @param ci □□□□□□□
     *           - ci.cancel(): □□□□□□□□
     *           - □□: HEAD □□ cancel □□□□□□□□□□□□
```

```
     */
    @Inject(
        method = "tick",           // □□□□□□□□□□□□□□□ SRG □□
        at = @At("HEAD"),          // □□□□□: □□□□□
        cancellable = true         // □□□□□□□□□□□□□□□ cancel □□□ true□
    )
    private void onTickStart(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□□□□□□
        // □□: method □ private□□□□□□□□□□□□□

        // □□: □□□□□□□□□□ tick
        // if (someCondition) {
        //     ci.cancel();  // □□□□□□□□
        // }
    }

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     * □□ TAIL □□□□□□□□□□□□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□
        // □□□□□□□□□□□□□□□□□□□
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□ CallbackInfoReturnable □□□□□□□□□□□□
     *
     * @param cir □□□□□□□□□□□□□□□
     */
    @Inject(
        method = "getCurrentFps",   // □□□□□□
        at = @At("RETURN"),         // □□□□□□□□
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // □□□□□□□□□:
        // cir.setReturnValue(cir.getReturnValue() * 2);  // □□□□□□
        // cir.cancel();  // □□□□□□□□□□□□□□□□□□□
    }
}
```

mixin/mixin_config.json

```
{
    // ============================================================
    // Mixin □□□□□□□ - Fabric 1.19.2
    // ============================================================
    // □□□□□: src/main/resources/examplemod.mixins.json
    //
    // □□□□□ fabric.mod.json □□□□ "mixins" □□□□□:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // □□ 1.20.1 □□□□□
    //   Mixin □□□□□□□ 1.19.2 □ 1.20.1 □□□□□□□□□
    //   □□□□□□ compatibilityLevel: "JAVA_17"□
    //
    // □□□□□□:
    // - required:           □□□□□□true=□□□□□□□false=□□□
    // - package:            Mixin □□□□□□□□□
    // - compatibilityLevel: Java□□□□□□1.19.2 □□ JAVA_17□
    // - mixins:             □□Mixin□□□□□□□□□□□□□□□□□
    // - client:             □□□□□□□□□Mixin□
    // - server:              □□□□□□□□Mixin□□□□□□
    // - injectors:           □□□□□□
    //   - defaultRequire:    □□□□□□□□□1=□□□□□□□□□□0=□□□
    //
    // □□□□□□:
```

```
    // 1. compatibilityLevel □□□□ JAVA_17□1.19.2 □□ Java 17+□
    // 2. package □□□□□□Mixin□□□□□□□□
    // 3. mixins □□□□□□□ .java □□□□□□□□□
    // 4. client □□□□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_17",

    // □□Mixin□□□□□□□□□□□□□□□□□
    "mixins": [
        "ExampleInjectMixin"
    ],

    // □□□□□□□□Mixin□
    "client": [
        "client.ExampleClientMixin"
    ],

    // □□□□□□
    "injectors": {
        // defaultRequire: 1 □□□□□ @Inject □□□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□
        // □□ 0 □□□□□□□□□□□□□□□□□□□□□bug□
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ============================================================================
// build.gradle —— Fabric 1.19.2 □□□□□□□□□□Groovy DSL□
// ============================================================================
// □□□□
//  - □□□□□□ Fabric 1.19.2 □□□□□□□ Loom □□□□□□
//  - □□□□□□□□□□ gradle.properties □□□□□□□□□□
//  - □□□□□□□□□JDK 17 + Gradle 7.x□□□ 7.6 □□□□□□
//
// □□ 1.20.1 build.gradle □□□□□
//  1. Loom □□□□1.0-SNAPSHOT□1.20.1 □ 1.6-SNAPSHOT□
//  2. Minecraft □□□□1.19.2
//  3. Yarn □□□□1.19.2+build.28
//  4. Fabric Loader□0.14.18□1.20.1 □ 0.16.x□
//  5. Fabric API□0.77.0+1.19.2
//  6. Java □□□17□□ 1.20.1 □□□□1.21+ □□□□ 21□
// ============================================================================

plugins {
    // Fabric Loom 1.0□□□ Minecraft 1.19.2 □ Loom □□□
    // 1.20.1 □□□□ Loom □ 1.6-SNAPSHOT□1.19.2 □□□□□ 1.0-SNAPSHOT□
    id 'fabric-loom' version '1.0-SNAPSHOT'
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
    archivesName = project.name
}

repositories {
    mavenCentral()
}

// ---------------------------------------------------------------------------
// □□□□□□□□□□□ gradle.properties
// ---------------------------------------------------------------------------
dependencies {
    // Minecraft 1.19.2 □□□□ jar
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn 1.19.2+build.28 □□□□□
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader 0.14.18
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
```

```
    // Fabric API 0.77.0+1.19.2
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
}

// ------------------------------------------------------------------------
// □□□□□□□ gradle.properties □□ mod_version □□□□ fabric.mod.json
// ------------------------------------------------------------------------
processResources {
    inputs.property "version", project.version

    filesMatching("fabric.mod.json") {
        expand "version": project.version
    }
}

// ------------------------------------------------------------------------
// □□□□□□Minecraft 1.19.2 □□ Java 17
// ------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}

java {
    withSourcesJar()

    sourceCompatibility = JavaVersion.VERSION_17
    targetCompatibility = JavaVersion.VERSION_17
}

// ------------------------------------------------------------------------
// jar □□
// ------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/version_diff.md

```
# Fabric 1.19.2 □ 1.20.1 □□□□□□□

> □□□□□□ Fabric 1.19.2 □□□□ 1.20.1□□□□□□□□ API □□□□
> □□ AI □□□□ 1.19.2 □□□□□□□□□□ API□

---

## □□□□□

1.19.2 □ 1.19 □□□□□□□□□□□□□□□□□□□□□ 1.19.3 □□□□□□□□□□
1.19.3 □□□□□□□□□□□□□□□□`Registries` □□□□□□□□□□
□□ **1.19.2 □□□□ API □□□□ 1.20.1 □□□□□□**□

| □□□□ | 1.19.2 | 1.20.1 |
|--------|--------|--------|
| Registry □□□□ | `net.minecraft.util.registry.Registry` | `net.minecraft.registry.Registry` |
| □□□□□□□□ | `Registry.BLOCK` / `Registry.ITEM`□□□□□□□ | `Registries.BLOCK` / `Registries.ITEM` |
| Registries □ | **□□□□** | □□□`net.minecraft.registry.Registries`□ |
| Identifier □□ | `new Identifier(modid, name)` | `new Identifier(modid, name)`□□□□□ |
| □□□□□□□□□ | `.group(ItemGroup.MISC)` □□ | `.group()` □□□□□□□ `ItemGroupEvents` |
| ItemGroup □□ | `ItemGroup.BUILDING_BLOCKS`□□□□□□□ | `ItemGroups.BUILDING_BLOCKS`□□□□□□ |
| Material □ | □□□`net.minecraft.block.Material`□ | □□□□□□ `MapColor` □□□□ |
| FabricBlockSettings | `.of(Material.STONE)` | `.create()` / `.copyOf(Blocks.STONE)` |
| Enchantment □□ | □□ `Enchantment` □ | □□ `Enchantment` □□□□□□ |
| EnchantmentTarget | `EnchantmentTarget.WEAPON` □□ | □□□□1.21+ □□□□□ |
| Java □□ | 17 | 17 |
| Loom □□ | 1.0-SNAPSHOT | 1.6-SNAPSHOT |
| Fabric Loader | 0.14.18 | 0.16.x |
| Fabric API | 0.77.0+1.19.2 | 0.83.0+1.20.1 |

---

## □□□□□□□□□□□□□□□□□□□□

### 2.1 □□□□□□
1.19.2 □□□□□□□□□ `net.minecraft.util.registry` □□□
```

```java
// 1.19.2 —— 旧版
import net.minecraft.util.registry.Registry;

Registry.register(Registry.BLOCK, new Identifier("modid", "name"), block);
Registry.register(Registry.ITEM, new Identifier("modid", "name"), item);
```

1.20.1 起，原版注册表统一移入 `net.minecraft.registry` 包，并新增了 `Registries` 类。

```java
// 1.20.1 —— 新版
import net.minecraft.registry.Registry;
import net.minecraft.registry.Registries;

Registry.register(Registries.BLOCK, new Identifier("modid", "name"), block);
Registry.register(Registries.ITEM, new Identifier("modid", "name"), item);
```

迁移提示：若 1.19.2 代码中 `import net.minecraft.registry.Registries;` 报错找不到，
说明你正在/将从 1.19.2 迁移过来。

2.2 Registry 常量名

1.19.2 中注册表常量直接挂在 `Registry` 类上，名称如下：

```java
// 1.19.2
Registry.BLOCK      // 方块注册表
Registry.ITEM       // 物品注册表
Registry.ENCHANTMENT // 附魔注册表
Registry.SOUND_EVENT // 音效事件注册表
Registry.STATUS_EFFECT // 状态效果注册表
// ... 等等
```

1.20.1 起，这些常量统一移至 `Registries` 类，`Registry` 仅保留注册表本身的操作方法。

2.3 Identifier 构造

两个版本的构造方式完全相同：

```java
// 1.19.2 与 1.20.1 相同
new Identifier("modid", "name")
```

> 提示：1.21+ 将 `new Identifier()` 改为私有，改用 `Identifier.of()`。

三、物品创造栏（Creative Tab）注册方式变化

3.1 1.19.2：通过 `.group()` 设置

1.19.2 中，`Item.Settings` 直接调用 `.group(ItemGroup.XXX)` 将物品放入对应创造栏：

```java
// 1.19.2 —— 旧版
import net.minecraft.item.ItemGroup;

new Item(new Item.Settings().group(ItemGroup.MISC));
new Item(new Item.Settings().group(ItemGroup.BUILDING_BLOCKS));
```

常用的原版创造栏 `ItemGroup` 常量如下：

```java
ItemGroup.BUILDING_BLOCKS
ItemGroup.MISC
ItemGroup.COMBAT
ItemGroup.TOOLS
ItemGroup.REDSTONE
ItemGroup.FOOD
ItemGroup.INGREDIENTS
ItemGroup.BREWING
```

```
ItemGroup.SEARCH
```

3.2 1.20.1□□□□□ ItemGroupEvents

1.19.3+ □□□ `Item.Settings.group()` □□□1.20.1 □□□□□ Fabric API □□□□

```java
// 1.20.1 —— □□
import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
import net.minecraft.item.ItemGroups;

ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
    .register(entries -> entries.add(myItem));
```

□□□□□□□□□□□□□□ `ItemGroups` □□□□□□□□

```java
ItemGroups.BUILDING_BLOCKS
ItemGroups.MISC
ItemGroups.COMBAT
// ... □□
```

3.3 □□□□□□□□

1.19.2 □□□□□□ `FabricItemGroup` □□□□□□□□□

```java
// 1.19.2
import net.fabricmc.fabric.api.itemgroup.v1.FabricItemGroup;
import net.minecraft.item.ItemStack;

Registry.register(Registry.ITEM_GROUP, new Identifier("modid", "example"),
    FabricItemGroup.create()
        .icon(() -> new ItemStack(Items.STONE))
        .displayName(new LiteralText("Example Tab"))
        .appendItems(stacks -> {
            stacks.add(new ItemStack(EXAMPLE_ITEM));
        })
        .build()
);
```

1.20.1 □□□ `FabricItemGroup.builder()`□API □□□□

```java
// 1.20.1
import net.fabricmc.fabric.api.itemgroup.v1.FabricItemGroup;

Registry.register(Registries.ITEM_GROUP, new Identifier("modid", "example"),
    FabricItemGroup.builder()
        .displayName(Text.literal("Example Tab"))
        .icon(() -> new ItemStack(Items.STONE))
        .entries((context, entries) -> {
            entries.add(EXAMPLE_ITEM);
        })
        .build()
);
```

> □□□1.19.2 □□ `new LiteralText("...")`□1.20.1 □□ `Text.literal("...")`□

□□□□□□□□

4.1 Material □

1.19.2 □ `Material` □□□□□□□□□□ `FabricBlockSettings.of(Material.STONE)`□

```java
// 1.19.2 —— □□
import net.minecraft.block.Material;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
```

```
FabricBlockSettings.of(Material.STONE).strength(1.5f);
FabricBlockSettings.of(Material.METAL).strength(3.0f);
FabricBlockSettings.of(Material.WOOD).strength(2.0f);
```

1.19.3+ □□□ `Material` □□1.20.1 □□□ `MapColor` □□□□□

```java
// 1.20.1 —— □□
import net.minecraft.block.MapColor;
import net.minecraft.sound.BlockSoundGroup;

FabricBlockSettings.create()
    .mapColor(MapColor.STONE_GRAY)
    .sounds(BlockSoundGroup.STONE)
    .strength(1.5f);
```

4.2 FabricBlockSettings.copyOf

□□□□□□□ `FabricBlockSettings.copyOf(Blocks.STONE)`□

```java
// 1.19.2 □ 1.20.1 □□
FabricBlockSettings.copyOf(Blocks.STONE).strength(3.0f).requiresTool();
```

□□□□□□□□□

5.1 Item.Settings

1.19.2 □ `Item.Settings` □ 1.20.1 □□□□□□□□□

```java
// 1.19.2 □ 1.20.1 □□□□□□□
new Item.Settings()
    .maxCount(64)
    .maxDamage(256)
    .rarity(Rarity.UNCOMMON)
    .fireproof()  // FabricItemSettings □□
```

□□□□□□ `.group()` □□□□□□□□□□□□

5.2 FabricItemSettings

□□□□□□□ `FabricItemSettings`□□□□□□□□

```java
import net.fabricmc.fabric.api.item.v1.FabricItemSettings;

// 1.19.2 □ 1.20.1 □□
new FabricItemSettings().maxCount(64);
```

5.3 □□□□

1.19.2 □□□ `FoodComponent`□

```java
// 1.19.2
import net.minecraft.item.FoodComponent;

new Item(new Item.Settings()
    .food(new FoodComponent.Builder()
        .hunger(6)        // □□□□
        .saturationModifier(0.6f) // □□□□
        .meat()           // □□□
        .build()
    )
);
```

1.20.1 □□□□□□`FoodComponent`□□□ 1.21+ □□□ `Food`□

□□□□□□□□□□

□□□	1.19.2	1.20.1
□□□□□	`new LiteralText("hello")`	`Text.literal("hello")`
□□□□□□	`new TranslatableText("key")`	`Text.translatable("key")`

1.19.2 □□□ `new LiteralText(...)` / `new TranslatableText(...)` □□□□□
1.20.1 □□□□□□□□□□□□□□□□ `Text.literal(...)` / `Text.translatable(...)` □□□□□□□□

□□□□□□□□□ 1.19.2 □□□□□□ API

□□ API □ 1.19.2 □□□□□□□□□ 1.19.3+ □□□□□□□□
□□□□□□□□□□ 1.19.2 □□□□□□□□□□□□"□"API□

7.1 `ItemGroup` □□□□□

```java
// 1.19.2 □ API□□ 1.19.2 □□□□□□□□
ItemGroup.BUILDING_BLOCKS  // □□□□
ItemGroup.MISC
```

7.2 `Item.Settings.group()`

```java
// 1.19.2 □ API□□ 1.19.2 □□□□□□□□
new Item.Settings().group(ItemGroup.MISC)
```

7.3 `Material` □

```java
// 1.19.2 □ API□□ 1.19.2 □□□□□□□□
import net.minecraft.block.Material;
FabricBlockSettings.of(Material.STONE)
```

7.4 `new LiteralText()` / `new TranslatableText()`

```java
// 1.19.2 □ API□□ 1.19.2 □□□□□□□□
import net.minecraft.text.LiteralText;
import net.minecraft.text.TranslatableText;

new LiteralText("Hello");
new TranslatableText("item.modid.example");
```

7.5 `Registry.BLOCK` □□□□□

```java
// 1.19.2 □ API□□ 1.19.2 □□□□□□□□
Registry.register(Registry.BLOCK, new Identifier("modid", "name"), block);
```

□□□□□□□

1.19.2 □□□□□□□ 1.20.1 □□□□□□□□□□ `Enchantment` □□

```java
// 1.19.2 □ 1.20.1 □□
public class MyEnchantment extends Enchantment {
    public MyEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        super(rarity, target, slots);
    }
    @Override
    public int getMinPower(int level) { return 10 + (level - 1) * 15; }
    @Override
    public int getMaxLevel() { return 3; }
```

```
}

// □□
Registry.register(Registry.ENCHANTMENT, new Identifier("modid", "my_enchant"), new MyEnchantment(...));
```

> □□□1.21+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ 1.21.1 □ version_diff.md□

□□□□□□□□1.19.2 → 1.20.1□

□□□□□ 1.19.2 □□□□ 1.20.1□□□□□□□□□□□

1. `import net.minecraft.util.registry.Registry` → `import net.minecraft.registry.Registry`
2. □□ `import net.minecraft.registry.Registries;`
3. `Registry.BLOCK` → `Registries.BLOCK`□□□□□□□□□□
4. `.group(ItemGroup.MISC)` → □□ `ItemGroupEvents.modifyEntriesEvent(ItemGroups.MISC)`
5. `ItemGroup.BUILDING_BLOCKS` → `ItemGroups.BUILDING_BLOCKS`
6. `FabricBlockSettings.of(Material.STONE)` → `FabricBlockSettings.create().mapColor(MapColor.STONE_GRAY)`
7. `new LiteralText("...")` → `Text.literal("...")`
8. `new TranslatableText("...")` → `Text.translatable("...")`
9. `FabricItemGroup.create()` → `FabricItemGroup.builder()`
10. Loom □□□□□□□□`1.0-SNAPSHOT` → `1.6-SNAPSHOT`
11. Fabric Loader / Fabric API □□□□□

recipe/shaped_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.19.2
    // ============================================================
    // □□□□□: data/examplemod/recipes/shaped_recipe.json
    //
    // □□□□□□crafting_shaped□□□□□□ pattern □ key □□
    // □□□□□□□□□□□□□□ pattern □□□□□□
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□□ 1.19.2 □ 1.20.1 □□□□□□□□□
    //   □□□□□□□□ type / pattern / key / result □□□□
    //
    // □□□□□:
    // 1. pattern □□□□□□□□□□□□□□
    // 2. pattern □□□□□□□□□□□□□□ key □□□□□□□□
    // 3. key □□□□□□□□□□□□□□□□□□□□□□□□ #, X, *, @ □□
    // 4. □□□□□□□□□□□□□□□□
    // 5. result □□ count □□□□□ 64
    // 6. ingredient □□ tag □□□□ "tag" □□□□□ "item"
    // ============================================================

    "type": "minecraft:crafting_shaped",

    // pattern: □□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
```

```
}
```

recipe/shapeless_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.19.2
    // ============================================================
    // □□□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□□□□□crafting_shapeless□□□□□□□□□□□□□
    // □□□□ ingredients □□□□□□□□□□□□□□□□□□□□
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□□ 1.19.2 □ 1.20.1 □□□□□□□□□
    //
    // □□□□□:
    // 1. ingredients □□□□□□□□□□□□□□□□□□□□
    // 2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // 3. □□□□ "tag" □□□□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    // ============================================================
    // □□□□□□□□ - Fabric 1.19.2
    // ============================================================
    // □□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□ 1.20.1 □□□□□
    //   1.19.2 □□□ JSON □□□□ 1.20.1 □□□□□□
    //   □□□□□□□□: data/<mod_id>/tags/blocks/
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□□: data/<mod_id>/tags/items/□□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,
```

```
    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.19.2
    // ============================================================
    // □□□□□: data/examplemod/tags/items/example_item_tag.json
    //
    // □□ 1.20.1 □□□□□
    //   1.19.2 □□□□ JSON □□□ 1.20.1 □□□□□□
    //   □□□□□□□□: data/<mod_id>/tags/items/
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□:
    //   minecraft:logs          □□
    //   minecraft:planks        □□
    //   minecraft:coals         □□□
    //   minecraft:saplings      □□
    //   minecraft:flowers       □
    //
    // □□□□□:
    // 1. □□□□□□ data/<mod_id>/tags/items/ □□□
    // 2. □□□□□□□□□□□□□□□□□□□□
    //    □□: data/<mod_id>/tags/items/
    //    □□: data/<mod_id>/tags/blocks/
    // 3. values □□□□ID□□□□□□□□□□
    // 4. replace □ true □□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
    ]
}
```

worldgen/ore_feature.java

```
/*
 * ore_feature.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.19.2 (Fabric Loader 0.14.18 + Fabric API 0.77.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□□
 *   1. □□□□□□□□□□□□□
 *      1.19.2: net.minecraft.util.registry.Registry
 *              net.minecraft.util.registry.RegistryKey
 *              net.minecraft.util.registry.RegistryKeys
 *              net.minecraft.util.registry.BuiltinRegistries
 *      1.20.1: net.minecraft.registry.Registry
 *              net.minecraft.registry.RegistryKey
 *              net.minecraft.registry.RegistryKeys
 *              net.minecraft.registry.Registries
 *
 *   2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *      1.19.2: ConfiguredFeature □ PlacedFeature □□□ BuiltinRegistries
 *              BuiltinRegistries.CONFIGURED_FEATURE
 *              BuiltinRegistries.PLACED_FEATURE
 *      1.20.1: □□□ Registries
 *              Registries.CONFIGURED_FEATURE
 *              Registries.PLACED_FEATURE
 *
 *   3. □BiomeModifications API □□□
 *      □□□□□□□ Fabric API □ BiomeModifications
 * ------------------------------------------------------------------------------
 */
package com.example.mod.worldgen;
```

```java
import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.util.Identifier;
import net.minecraft.util.registry.BuiltinRegistries;
import net.minecraft.util.registry.Registry;
import net.minecraft.util.registry.RegistryKey;
import net.minecraft.util.registry.RegistryKeys;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.YOffset;
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.feature.PlacedFeature;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.PlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;

import java.util.List;

/**
 * □□□□□□□□□□ - Fabric 1.19.2
 *
 * □□□□□□□□□□
 * 1. □□□□□□ ConfiguredFeature□□□□□□
 * 2. □□□□□□ PlacedFeature□□□□□□□
 * 3. □□ BiomeModifications □□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□ConfiguredFeature □ PlacedFeature □□□ BuiltinRegistries
 *    □□ Registries□1.19.2 □ Registries □□□□□
 * 2. RegistryKey / RegistryKeys □□□□□□ net.minecraft.util.registry.*
 * 3. BiomeModifications API □ 1.20.1 □□
 *
 * □□□□:
 *   ConfiguredFeature -> □□□ BuiltinRegistries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□ BuiltinRegistries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□
 *
 * □□□□:
 * - □□□□□□ ConfiguredFeature□□□□ PlacedFeature
 * - □1.19.2 □□□□□□ BuiltinRegistries □□ Registries
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□
 * - □□□□(size)□□□ 64□□□ 3~12
 */
public class ModWorldGen {

    public static final String MOD_ID = "examplemod";

    // ConfiguredFeature □ RegistryKey
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    // PlacedFeature □ RegistryKey
    public static final RegistryKey<PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    /**
     * □□□□□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        // ================================================================
        // 1. □□□□□□ ConfiguredFeature
        // ================================================================
        // □□□□□□□□□
        // □□1: □□□□□□□□□□□□□□□□□□□□□□□□
        // □□2: □□□□□□ BlockState
        // □□3: □□□□□□□□□□□□□□□□□□□□□□□□
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            // □□□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
            // □□: ModBlocks.EXAMPLE_ORE □□□□□□□□□□□□□□□□
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),  // □□□□□□□□□□
            9  // □□□□□
```

```
        );

        // 创建 ConfiguredFeature 实例
        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,    // 特征类型: 矿石
            oreConfig       // 特征配置
        );

        // 在1.19.2 及之前版本使用 BuiltinRegistries.CONFIGURED_FEATURE
        // 1.20.1 使用 Registries.CONFIGURED_FEATURE
        Registry.register(
            BuiltinRegistries.CONFIGURED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            configuredFeature
        );

        // ================================================================
        // 2. 注册放置的 PlacedFeature
        // ================================================================
        // 定义矿石在世界中的放置规则（生成多少、在哪个高度范围）
        List<PlacementModifier> placementModifiers = List.of(
            // 数量: 每个区块尝试生成多少次
            CountPlacementModifier.of(20),

            // 分布方式: 在区块内均匀随机分布
            SquarePlacementModifier.of(),

            // 高度范围: 从 Y=-64 到 Y=64 均匀随机分布
            HeightRangePlacementModifier.uniform(
                YOffset.getBottom(),  // Y=-64（世界底部）
                YOffset.fixed(64)     // Y=64
            )
        );

        // 创建 PlacedFeature 实例
        // 注意: 1.19.2 中 PlacedFeature 构造函数为 (ConfiguredFeature, List<PlacementModifier>)
        // 需要将其包装为 RegistryEntry.of(configuredFeature) 形式
        PlacedFeature placedFeature = new PlacedFeature(
            net.minecraft.registry.entry.RegistryEntry.of(configuredFeature),
            placementModifiers
        );

        // 在1.19.2 及之前版本使用 BuiltinRegistries.PLACED_FEATURE
        // 1.20.1 使用 Registries.PLACED_FEATURE
        Registry.register(
            BuiltinRegistries.PLACED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            placedFeature
        );

        // ================================================================
        // 3. 使用 BiomeModifications 将矿石添加到生物群系
        // ================================================================
        // BiomeModifications API 在 1.19.2 和 1.20.1 中相同
        BiomeModifications.addFeature(
            // 生物群系选择器: 选择哪些生物群系
            BiomeSelectors.foundInOverworld(),
            // 生成阶段: 在地下矿石阶段生成
            GenerationStep.Feature.UNDERGROUND_ORES,
            // PlacedFeature 的 RegistryKey
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                new Identifier(MOD_ID, "ore_example")
            )
        );
    }
}

/*
 * ================================================================================
 * 生物群系选择器参考
 * ================================================================================
 *
 * BiomeSelectors 提供了多种选择生物群系的方式:
 *
 * BiomeSelectors.all()                  选择所有生
```

```
 * BiomeSelectors.foundInOverworld()      □□□□□□□□
 * BiomeSelectors.foundInNether()         □□□□□□□
 * BiomeSelectors.foundInTheEnd()         □□□□□□□
 * BiomeSelectors.includeByKey(keys)      □□□ RegistryKey □□
 *
 * ============================================================================
 * □□□□□□□
 * ============================================================================
 *
 * GenerationStep.Feature□□□□□□□:
 * - RAW_GENERATION          □□□□□□□□□□
 * - LAKES                    □□
 * - LOCAL_MODIFICATIONS      □□□□□□□□□□
 * - UNDERGROUND_STRUCTURES   □□□□
 * - SURFACE_STRUCTURES       □□□□
 * - STRONGHOLDS              □□
 * - UNDERGROUND_ORES        □□□□□□□□□□□□□□□
 * - UNDERGROUND_DECORATION   □□□□□□□□□□
 * - FLUID_SPRINGS            □□□
 * - VEGETAL_DECORATION       □□□□□□□□□□
 * - TOP_LAYER_MODIFICATION   □□□□□□□□□
 */
```

Fabric 1.20.1

advancement/advancement.json

```
{
    // ============================================================
    // □□/□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□: data/examplemod/advancements/example_advancement.json
    //
    // □□□□□Advancement□□ Minecraft 1.12+ □□□□□
    // □□□□□□□□□□□□□□□□□□□□toast□□□□□□□
    //
    // □□□□:
    // - display:     □□□□□□□□□□□□□□□□□□□□□□□□
    // - parent:      □□□□□□□□□□□□□□□□
    // - criteria:    □□□□□□□□□□□□□□□□□□□□□□□□
    // - requirements: □□□□□□□□□□□□AND / OR□
    // - rewards:     □□□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□:
    // 1. □□□□□parent□□□□□ display.background □□
    // 2. 1.20.1 □ inventory_changed □□□□□□□□ "items" □□□□□□ "item"□
    // 3. criteria □□□□□□□□□□ trigger □□
    // 4. requirements □□□□□□□□□□□□ AND□□□□ OR
    // 5. frame □□: task(□□) / goal(□□) / challenge(□□)
    // ============================================================

    // □□□ID□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    "parent": "examplemod:example/root",

    // □□□□□
    "display": {
        // □□
        "icon": {
            "item": "examplemod:example_item"
        },

        // □□□□□□ translate □□□□□ literal □□□□□□
        "title": {
            "translate": "advancements.examplemod.example.title"
        },

        // □□
        "description": {
            "translate": "advancements.examplemod.example.description"
        },

        // □□□□□: task(□□,□□) / goal(□□,□□) / challenge(□□,□□)
        "frame": "task",
        // □□□□□□□□□□□□□□
```

```
        "show_toast": true,

        // □□□□□□□□□
        "announce_to_chat": true,

        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        "hidden": false

        // □□□□□□ background □□□□□□□□□□□
        // "background": "minecraft:textures/block/stone.png"
    },

    // □□□□□
    // □□ key □□□□□□□□□□□□□□□ trigger □ conditions
    "criteria": {
        "get_example_item": {
            // □□□: □□□□□□□□□□
            "trigger": "minecraft:inventory_changed",
            "conditions": {
                // □□□□1.20.1 □□ "items"□□□□□□□□
                // □□□□ "item"□□□□□□□□
                "items": [
                    {
                        "items": [
                            "examplemod:example_item"
                        ]
                    }
                ]
            }
        },
        "another_condition": {
            // □□□: □□□□□□□□□□□□□
            "trigger": "minecraft:location",
            "conditions": {
                "player": {
                    "location": {
                        "dimension": "minecraft:overworld"
                    }
                }
            }
        }
    },

    // □□□□□□□□□
    // □□□□□ AND□□□□□□
    // □□□□□ OR□□□□□□
    // □□□□: □□ get_example_item AND another_condition
    "requirements": [
        ["get_example_item"],
        ["another_condition"]
    ],

    // □□□□
    "rewards": {
        // □□□□□
        "experience": 10,

        // □□□□□□□□□□□
        "loot": [
            "examplemod:example_loot_table"
        ],

        // □□□□□□□□□□□
        "function": "examplemod:grant_example_reward"
    }
}
```

block/block_entity.java

```
package com.example.mod.block.entity;

import net.fabricmc.fabric.api.object.builder.v1.block.entity.FabricBlockEntityTypeBuilder;
import net.minecraft.block.BlockState;
import net.minecraft.block.entity.BlockEntity;
import net.minecraft.block.entity.BlockEntityType;
import net.minecraft.nbt.NbtCompound;
```

```java
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.util.math.BlockPos;

/**
 * □□□□□□□□ (Fabric 1.20.1)
 * <p>
 * □□□□□□BlockEntity□□□□□□□
```

```
/**
 * □ NBT □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * @param nbt □□□□□□□□□ NbtCompound □□
 *
 * □□□□□□
 * - 1.20.1 □ readNbt □□ void□□□□1.20.5+ □□ API □□□□□□
 * - □□□□□□ super.readNbt()□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   □□ writeNbt □□□writeNbt □□□□□□□ super□□
 * - □□□□□□□□□□□□□□□ key □□
 * - □□□□□□readNbt □ writeNbt □ key □□□□□□□□□□□□□□□
 *
 * NbtCompound □□□□□□□
 *   nbt.getInt("key")       ── □□ int□□□ 0□
 *   nbt.getString("key")    ── □□ String□□□ ""□
 *   nbt.getBoolean("key")   ── □□ boolean□□□ false□
 *   nbt.contains("key")     ── □□ key □□□□□
 */
@Override
public void readNbt(NbtCompound nbt) {
    super.readNbt(nbt);  // 1.20.1 □□□□ super □□□□□□□□□□□□□□□□

    // □□□□□□□□□
    this.counter = nbt.getInt("counter");
}

// ========================================================================
// BlockEntityType □□
// ========================================================================

/**
 * BlockEntityType □□□
 * <p>
 * □□□□□□□□□
 * 1. □□□□□□□□□□SimpleBlock.EXAMPLE_BLOCK□——□□□□□□ BlockEntityType □□□□□□
 * 2. □□ FabricBlockEntityTypeBuilder.create(...) □□□□□□□□□□□
 *    - BlockEntity □□□□□□□□□□□ ExampleBlockEntity::new□
 *    - □□□□ Block □□□□□□□□□□□
 * 3. □□ .build() □□ BlockEntityType□
 * 4. □□ Registry.register(Registries.BLOCK_ENTITY_TYPE, ...) □□□
 * <p>
 * □□□□□□□□□□□
 * - Fabric □□□□□□□□□
 *     FabricBlockEntityTypeBuilder.create(ExampleBlockEntity::new, BLOCK).build()
 *   FabricBlockEntityTypeBuilder □□□□□□ BlockEntityType.Builder□
 *   □□□□□□ BlockEntityType □□□□□□□□□□□□□□□
 * <p>
 * - □□□□□□□□□□□□□
 *     BlockEntityType.Builder.create(ExampleBlockEntity::new, BLOCK).build(null)
 *   □□ BlockEntityType.Builder.build(Type<?> type) □□□□□□□ Type<?> □□□
 *   □□□□□□□ Type □□□□□□□ null□□ Registry.register □□□□□□
 *   Fabric □□□□□□□ null□□□□□□
 * <p>
 * □□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□ BLOCK □□□□□□□□□□ EXAMPLE_BLOCK_ENTITY□
 *   □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 */
public static final BlockEntityType<ExampleBlockEntity> EXAMPLE_BLOCK_ENTITY =
        Registry.register(
                Registries.BLOCK_ENTITY_TYPE,
                new Identifier("examplemod", "example_block_entity"),
                // Fabric □□□□□
                FabricBlockEntityTypeBuilder.create(
                        ExampleBlockEntity::new,
                        com.example.mod.block.SimpleBlock.EXAMPLE_BLOCK
                ).build()
        );

// ========================================================================
// □□□□□□□□□□□□□□□□
// ========================================================================
//
// □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
// □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
//
```

```
    //   @Override
    //   public NbtCompound toUpdatePacket() {
    //       NbtCompound nbt = new NbtCompound();
    //       this.writeNbt(nbt);
    //       return nbt;
    //   }
    //
    //   @Override
    //   public void readUpdatePacket(NbtCompound nbt) {
    //       this.readNbt(nbt);
    //   }
    //
    // □□□□□□□□□□□□ world.updateListeners(pos, state, state, flags) □□□□
    // □□□□□□ BlockEntity □□□□toUpdatePacket -> □□□□readUpdatePacket -> □□□□□
    //
    // □□□□□□□□□□□□ markDirty() □□□□□□□□□□□
    //
    //   public void incrementCounter() {
    //       this.counter++;
    //       markDirty();  // □□□□□□□□□□□□□□□□□□□□
    //   }
    //
    // markDirty() □□□□□□□□□□□□□□□□□□□□□

    // ============================================================================
    // □□□□□□□□□□□□□□□□
    // ============================================================================
    //
    // □□□□□□□□ Block □□□□□□ BlockEntityProvider □□□
    //
    //   import net.minecraft.block.BlockWithEntity;
    //   import net.minecraft.block.entity.BlockEntity;
    //
    //   public class ExampleBlock extends BlockWithEntity {
    //
    //       // □□□□ createBlockEntity□□□□□□ BlockEntity □□
    //       @Override
    //       public BlockEntity createBlockEntity(BlockPos pos, BlockState state) {
    //           return new ExampleBlockEntity(pos, state);
    //       }
    //
    //       // □□□□□□□□□□□□□□□□ getRenderType
    //       @Override
    //       public BlockRenderType getRenderType(BlockState state) {
    //           // BLOCK □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //           // □□□□□□□□□□□□□□□ ENTITYBLOCK_ANIMATED
    //           return BlockRenderType.MODEL;
    //       }
    //   }
}
```

block/blockstate.json

```
{
  // ================================================================
  // □□□□ JSON □□ (Fabric 1.20.1)
  // ================================================================
  //
  // □□□□□src/main/resources/assets/examplemod/blockstates/<block_name>.json
  // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ mod_id □□□□
  //
  // blockstate JSON □□□□
  //   □□ BlockState □□□□□□□□ facing=north, powered=false□
  //   □□□□□□□□□□□□□□model□□
  //
  // □□□□□□□□
  //   1. "variants"□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //   2. "multipart"□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //      □□□□□□ + □□□□□□□
  //   □□□□□□ variants □□□□□□□□□
  //
  // □□□Minecraft □ GSON □□□□□□□ JSON □□□
  //       □□□□□□□□□□□□□ // □□□□□□□□
  // ================================================================

  // "variants"□□□□□□□□□
```

```
  // □□□□ BlockState □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  "variants": {

    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ""
    // □□□□□□□□□"examplemod:block/<block_name>"□□□□ .json □□□
    //
    // "model" □□□□□ assets/examplemod/models/block/<model_name>.json
    //   □□□□□□□□□ "block/" □□□□□□□□□□□□□□□□
    //
    // □□□□□□□
    //   "x": 0/90/180    ── □□□□ X □□□□□□□□□□□□□□□□□□□□
    //   "y": 0/90/180/270 ── □□□□ Y □□□□□□□□□□□□□□□□□□□□□□□□
    //   "uvlock": true/false ── □□□□□ UV□□□□□□□□□□□□□□□□□□
    //      □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ true□
    //
    // □□□□□□□
    //   - model □□□□□□ "block/" □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //   - □□□□□□□□□□□□□□□□□□□□□□□□ blockstate □□□□□□□□□□□□
    //   - □□□□□□□□□□/□□ Java □□□□□ BlockState □□□□□□□
    //     □□□□□ "facing=north" □□□□□□□□□ "facing"□□□□□□□□□□□□□
    "": {
      "model": "examplemod:block/example_block"
    }

    // ---- □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ----
    //
    // □□□□□□□ facing □□□□□□□□□□□□□□□□□□□□
```

```
{
  // "rolls"□□□
```

```
      //       "name": "examplemod:example_block",
      //       "conditions": [
      //         {
      //           "condition": "minecraft:silk_touch",
      //           "enchantment": "minecraft:silk_touch"
      //         }
      //       ]
      //     },
      //     {
      //       "type": "minecraft:item",
      //       "name": "minecraft:cobblestone"
      //     }
      //   ]
      //
      // □□□minecraft:silk_touch □□□□ 1.20.1 □□□□□□
      // □□□□□□□ minecraft:match_tool + predicate □□□□□□□□□□

      // "functions"□□□□□□□□□□□□
      // □□ functions□
      //   minecraft:set_count ── □□□□□□□
      //   minecraft:set_damage ── □□□□□□
      //   minecraft:set_nbt   ── □□ NBT □□
      //   minecraft:furnace_smelt ── □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
      //
      // □□□□□□□□□□ 1 □□□□□□□□□□□□
      "functions": [
        {
          "function": "minecraft:set_count",
          "count": 1
          // □□□□□□□□
          //   "count": { "min": 1, "max": 3 }
        }
      ]
    }
  ]
}
```

block/model.json

```
{
  // ================================================================
  // □□□□ JSON □□ (Fabric 1.20.1)
  // ================================================================
  //
  // □□□□□src/main/resources/assets/examplemod/models/block/<model_name>.json
  // □□□□□□ blockstate.json □ "model" □□□□□□□□□□□
  //
  // □□□□ JSON □□□□
  //   □□□□□□□□□□□□□□□/□□□□□□□□□□□
  //   □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //
  // □□ parent □□□□□ block/ □□□□□□□□
  //   block/cube_all        ── □□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //   block/cube            ── □□□□□□□□□□□□□□□□□□□□
  //   block/cube_column     ── □□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //   block/cube_bottom_top── □/□/□□□□□□□□□□
  //   block/cross           ── □□□□□□□□□□□□□□□
  //   block/tinted_cross    ── □□□□□□□□□□□□
  //   block/cube_directional ── □□□□□□□□
  //
  // □□□Minecraft □ GSON □□□□□□□ JSON □□□
  //       □□□□□□□□□□□□□□□ // □□□□□□□□□
  // ================================================================

  // "parent"□□□□□□□□
  // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ textures□□
  // □□ block/cube_all □□□□□□□□□□□□□□□□□□□□ "all" □□□□
  // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //
  // □□□□□□□
  //   - parent □□□□□□□□□ "cube_all" □□ "block/cube_all"
  //     □□□□ "block/" □□□□□□□□□□□□□□□□
  //   - □□□□□□□□□□□□□□□□□cube_all □□□□ "all"□cube □□ "up/down/north/..."□
  "parent": "block/cube_all",

  // "textures"□□□□□□
```

```
  // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
  //
  // □□ block/cube_all□□□□□□□□□□ "all" □□□
  // □□□□□□□□□□□□□□□□□□□□ "all" □□□□□□□□
  //
  // □□□□□□□□"examplemod:block/<texture_name>"
  //   □□□□□□□□assets/examplemod/textures/block/<texture_name>.png
  //   □□□□□□□□ .png □□□□□□□□ "block/" □□
  //
  // □□□□□□
  //   - □□□□□□□□ "examplemod:example_block" □□ "examplemod:block/example_block"
  //   - .png □□□□□□ textures/ □□□□□□ textures/block/ □□□
  //   - □□□□□□□□□□□□□□□□□□□□
  "textures": {
    "all": "examplemod:block/example_block"
  }

  // ---- □□□□□□□□□□□□□□□ block/cube□ ----
  //
  // □□□□□□□□□□□□□□□□□□□ "block/cube" □□□□□
  //
  //   "parent": "block/cube",
  //   "textures": {
  //     "up":    "examplemod:block/example_block_top",
  //     "down":  "examplemod:block/example_block_bottom",
  //     "north": "examplemod:block/example_block_side",
  //     "south": "examplemod:block/example_block_side",
  //     "east":  "examplemod:block/example_block_side",
  //     "west":  "examplemod:block/example_block_side"
  //   }
  //
  // □□□□□□
  //   north = □□□Z □□□□□□□□□□□□□□□□
  //   south = □□□Z □□□□□□
  //   east  = □□□X □□□□□□
  //   west  = □□□X □□□□□□
  //   up    = □□□Y □□□□□□
  //   down  = □□□Y □□□□□□

  // ---- □□□□□□□□□□□□□□□□□□□□ ----
  //
  // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ parent□
  // □□□□□□□□ "elements" □□□□□□□□□□□□□□□□□□□□
  //
  //   {
  //     "elements": [
  //       {
  //         "from": [0, 0, 0],      // □□□□□□□□□□ 0~16 □□□
  //         "to":   [16, 16, 16],   // □□□□□□
  //         "faces": {
  //           "up":    { "uv": [0, 0, 16, 16], "texture": "#all" },
  //           "down":  { "uv": [0, 0, 16, 16], "texture": "#all" },
  //           "north": { "uv": [0, 0, 16, 16], "texture": "#all" },
  //           "south": { "uv": [0, 0, 16, 16], "texture": "#all" },
  //           "east":  { "uv": [0, 0, 16, 16], "texture": "#all" },
  //           "west":  { "uv": [0, 0, 16, 16], "texture": "#all" }
  //         }
  //       }
  //     ],
  //     "textures": {
  //       "all": "examplemod:block/example_block"
  //     }
  //   }
  //
  // □□□□□□□□□□□□□□□ 16x16x16 □□□□□□□□
  //   "from" □ "to" □□□□□□□□□□ 0~16□
  //   [0,0,0] = □□□□□□□□□□[16,16,16] = □□□□□□□□□□
  //
  // UV □□□□
  //   "uv": [u0, v0, u1, v1] □□□□□□□□□□□□□□□□
  //   □□ 0~16 □□ 16x16 □□□□
  //   "#all" □□ # □□□□□□□□ textures □□□□□□□□□□
}
```

block/simple_block.java

```java
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□□□□□ (Fabric 1.20.1)
 * <p>
 * □□□□□□□□□ Fabric 1.20.1 □□□□□□□□□□□□□□□□□□
 * □□□□□□□
 * 1. □□□□□□□□□ Registries.BLOCK□
 * 2. □□□□□□□BlockItem□□□□□□□□ Registries.ITEM□□□□□□□□□□□□□□□□□□□□□□□
 * 3. Fabric □□□ @Mod □□□□□□□ Forge □□□□□□□□□ Registry.register() □□□□□
 * 4. □□□□□□□□□□□□□□□□□□□□□□□ ModInitializer □□□
 * <p>
 * □□□□□□
 * - □□□□ BLOCK □□□□ ITEM□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□ new Item.Settings() □□ FabricItemSettings□□□□□□□□
 * - Identifier □□ mod_id□□□□□□□□blockstate/model□□□□□□
 */
public class SimpleBlock extends Block {

    /**
     * □□□□□□
     * □ static final □□□□□□□□□□□□□□
     * □□□□□□□□□□□□□□BLOCK □□□□ ITEM □□□□□□ BlockItem □□□□ block □□□
     */
    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    /**
     * □□□□□□
     * □□ AbstractBlock.Settings□FabricBlockSettings □□□□□□□□□□
     * □□□□□□□□ Block □□□□□□□□□□
     *
     * @param settings □□□□□□□□□ FabricBlockSettings □□
     */
    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * □□□□□□□□□□□□□□□
     * □□□□□□□□ Block □□□□□□□□□□□□ BlockItem□□□□□□□□□
     *
     * @param name □□□□□□□□□□ mod_id □□□□□□□□□□□□ "examplemod:example_block"
     * @return □□□□□□ Block □□
     */
    private static Block registerBlock(String name) {
        // ---- □□□□□□□□□□□□ ----
        // FabricBlockSettings.copyOf(Blocks.STONE) □□□□□□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //
        // □□□□□□□□□
        //   .strength(3.0f)          —— □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //   .strength(3.0f, 6.0f)    —— □□□□□□□□□□□□□□□□□□□□□□□□□□□
        //   .resistance(6.0f)        —— □□□□□□□□□□□
        //   .requiresTool()          —— □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //   .luminance(15)           —— □□□□□ 0~15
        //   .nonOpaque()             —— □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //   .mapColor(MapColor.STONE_GRAY) —— □□□□□
        //   .sounds(BlockSoundGroup.STONE) —— □□/□□/□□□□
        //
        // □□□□□□□
        //   - □ 1.20.1 □□□□□□ AbstractBlock.Settings.copy() + of()□□□□ FabricBlockSettings.copyOf()□
        //   - 1.20.1 □ strength □□□□□□□ float□
        AbstractBlock.Settings settings = FabricBlockSettings.copyOf(Blocks.STONE)
                .strength(3.0f)          // □□=□□=3.0□□□□□□□
                .requiresTool()           // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                .luminance(15);           // □□□□□ 15□□□□
```

```
        // ---- □□□□□□□□□□□□ ----
        // □□ new SimpleBlock(settings) □□□
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ new Block(settings)□
        Block block = new SimpleBlock(settings);

        // ---- □□□□□□□ Block □ Registries.BLOCK ----
        // □□ Fabric 1.20.1 □□□□□□□□□□
        //   Registry.register(Registries.BLOCK, new Identifier("mod_id", "name"), block)
        //
        // □□□□□□
        //   - Registries.BLOCK□□□□□□□□1.20.1 □□□ Registries□□□□ Registry.BLOCK □□□□
        //   - new Identifier("examplemod", name)□□□□□□□□□□□□ mod_id□□□□□□□□
        //   - block□□□□□□
        //
        // □□□□□□
        //   - □□□□ DeferredRegister □ @ObjectHolder□□□ Forge □□□□
        //   - mod_id □□□□□□□□□□□□□□□□□
        //   - Identifier □□□□□□□□ fabric.mod.json □□ id □□□□□□
        Registry.register(Registries.BLOCK, new Identifier("examplemod", name), block);

        // ---- □□□□□□□ BlockItem □ Registries.ITEM ----
        // □□□□□□□□□"□□□□□□□□"□□□□□□□□□□□□□□□/□□□□□□□□□ BlockItem□
        //
        // FabricItemSettings □□□□
        //   import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
        //
        // □ 1.20.1 □□□□□□□ FabricItemSettings □□ new Item.Settings()□
        // □□ FabricItemSettings □□□ Fabric □□□□□□□□ food□fuel □□□
        //
        // □□□□□□
        //   .maxCount(64)             —— □□□□□□
        //   .maxDamage(256)           —— □□□□□□
        //   .fireproof()              —— □□□□□□□□ FabricItemSettings □□□
        //   .group(ItemGroup.BUILDING_BLOCKS) —— 1.20.1 □ creative tab □□□□□□□□□
        //                                       □□□ ItemGroupEvents □□□□□□□□□□□□□□
        BlockItem item = new BlockItem(block, new FabricItemSettings());

        Registry.register(Registries.ITEM, new Identifier("examplemod", name), item);

        return block;
    }

    // ========================================================================
    // □□□□□□□□
    // ========================================================================
    //
    // □□□□□□□□ ModInitializer □□□□□□□□□□□□□□□
    //
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
    //           // □□ SimpleBlock □□□□□□□□□□□□□□□□□
    //           // □□□ SimpleBlock.class □□□□□□□□□□□□□ EXAMPLE_BLOCK □□
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
    // □□□□□□□□□□□ onInitialize() □□□□□□□□/□□□□□□□□□□□
    // □□□□□□ onInitializeClient□□□□□□□□□□□□□□□□□□
    //
    // ========================================================================
    // □□□□□□□□□□1.20.1 □□□□□
    // ========================================================================
    //
    // 1.20.1 □□□□□ .group(ItemGroup.XXX) □□□□□□□□□□□□□□□□□□□□□□
    //
    //   // □□□□□□□□□□ onInitialize □□
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    //
    // □□□□
    //   import net.fabricmc.fabric.api.event.player.UseBlockCallback;
    //   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
    //   import net.minecraft.item.ItemGroups;
}
```

command/simple_command.java

```
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.arguments.StringArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.server.command.CommandManager;
import net.minecraft.text.Text;
import net.minecraft.util.Formatting;

import static net.minecraft.server.command.CommandManager.literal;
import static net.minecraft.server.command.CommandManager.argument;

/**
 * □□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□
 * 1. □□ CommandRegistrationCallback □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * □□□□Fabric 1.20.1 □□ CommandRegistrationCallback v2 □□
 * v2 □□□□□□□□□□□□□□:
 *   (CommandDispatcher<ServerCommandSource> dispatcher,
 *    CommandRegistryAccess registryAccess,
 *    CommandManager.RegistrationEnvironment environment)
 *
 * environment □□□□:
 *   DEDICATED - □□□□□
 *   INTEGRATED - □□/□□□□□□□□□□□□□□
 *
 * □□□□:
 * - 1.20.1 □□□□ v2 □□□□□□□□□□□□ v1
 * - CommandManager.literal() □ argument() □□□□□□
 * - □□□□□□ int□□□□□□□□□□□□□□□ 1□
 * - Feedback □□ Supplier<Text>□lambda□□□□□□
 * - □□□□□□□□□□□□□□ sendFeedback □□□□□ true
 */
public class ModCommands {

    /**
     * □□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        // □□□□□□□□□
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, registryAccess, environment) -> {
                // □□□□□
                registerExampleCommand(dispatcher);
            }
        );
    }

    /**
     * □□ /example □□□
     *
     * □□□□:
     * /example                          - □□□□□□
     * /example hello                    - □□□□□□□□
     * /example hello <player>           - □□□□□□□□□□
     * /example set <value>              - □□□□□
     * /example give <player> <count>    - □□□□□□□
     */
    private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
        dispatcher.register(
            literal("example")
                // □□□□□□□: □□□□
```

```
            .executes(context -> {
                context.getSource().sendFeedback(
                    () -> Text.literal("□□: /example <hello|set|give>")
                        .formatted(Formatting.YELLOW),
                    false
                );
                return 1;
            })

            // /example hello - □□□□□□□□
            .then(literal("hello")
                .executes(ModCommands::sayHelloToSelf)
                // /example hello <player> - □□□□□□□□□□
                .then(argument("target", EntityArgumentType.player())
                    .executes(ModCommands::sayHelloToTarget)
                )
            )

            // /example set <value> - □□□
            .then(literal("set")
                .then(argument("value", IntegerArgumentType.integer(0, 100))
                    .executes(ModCommands::setValue)
                )
            )

            // /example give <player> <count> - □□□□□□□
            .then(literal("give")
                .then(argument("target", EntityArgumentType.player())
                    .then(argument("count", IntegerArgumentType.integer(1, 64))
                        .executes(ModCommands::giveItem)
                    )
                )
            )
    );
}

/**
 * /example hello - □□□□□□□□
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    // □□□□□□□□□□□□
    // sendFeedback □□: (Supplier<Text> feedback, boolean broadcastToOps)
    // broadcastToOps=true □□□□□□□□□□OP
    source.sendFeedback(
        () -> Text.literal("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;  // □□□□□□
}

/**
 * /example hello <player> - □□□□□□□□□□
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    // □□ EntityArgumentType □□□□□□
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    // □□□□□□□□□□
    target.sendMessage(
        Text.literal("Hello from " + source.getName() + "!")
            .formatted(Formatting.GOLD)
    );

    // □□□□□□□□□□□□
    source.sendFeedback(
        () -> Text.literal("□□ " + target.getName().getString() + " □□□□")
            .formatted(Formatting.GREEN),
```

```
            true  // □□□OP
        );
        return 1;
    }

    /**
     * /example set <value> - □□□
     * □□□□□□□
     */
    private static int setValue(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        // □□ IntegerArgumentType □□□□□
        int value = IntegerArgumentType.getInteger(context, "value");
        ServerCommandSource source = context.getSource();

        source.sendFeedback(
            () -> Text.literal("□□□□□: " + value)
                .formatted(Formatting.AQUA),
            false
        );
        return 1;
    }

    /**
     * /example give <player> <count> - □□□□□□□
     * □□□□□□□□□□□
     */
    private static int giveItem(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
        int count = IntegerArgumentType.getInteger(context, "count");
        ServerCommandSource source = context.getSource();

        // □□: □□□□□□□
        // □□□□□□□□□□□□□
        net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
            net.minecraft.item.Items.DIAMOND, count
        );
        target.getInventory().insertStack(stack);

        source.sendFeedback(
            () -> Text.literal("□□ " + target.getName().getString()
                + " □□□ " + count + " □□□")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }
}
```

enchantment/custom_enchantment.java

```
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.enchantment.EnchantmentTarget;
import net.minecraft.enchantment.Enchantments;
import net.minecraft.entity.EquipmentSlot;
import net.minecraft.entity.EquipmentSlotGroup;
import net.minecraft.item.ItemStack;
import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□
 * 1. □□□□□□□□□□□□ Enchantment□
 * 2. □□ Registry.register □□□ Registries.ENCHANTMENT
 * 3. □□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□:
```

```
 * - Enchantment.Rarity: □□□□□□□□□□□□□□□□□□□□□□□□
 *   COMMON      weight=10  □□□□□□□□□□
 *   UNCOMMON    weight=5   □□□□□□□□□□□□□□□□
 *   RARE        weight=2   □□□□□□□□□□
 *   VERY_RARE   weight=1   □□□□□□□□□□□□□□□□
 *
 * - EnchantmentTarget: □□□□□□□□□□□□□□
 *   ALL          □□□□□
 *   ARMOR         □□
 *   FEET/LEGS/CHEST/HEAD  □□□□□□□
 *   WEAPON        □
 *   TOOL          □□□□□/□/□/□□
 *   BOW           □
 *   FISHING_ROD   □□□
 *   BREAKABLE     □□□□□□
 *   TRIDENT       □□□
 *   CROSSBOW      □
 *   VANISHABLE    □□□□□□□
 *
 * - EquipmentSlot[]: □□□□□□□□□□□□
 *   MAINHAND    □□
 *   OFFHAND     □□
 *   HEAD        □□
 *   CHEST       □□
 *   LEGS        □□
 *   FEET        □□
 *
 * □□□□□:
 * - □□□ mod_id □□□□□□ mod ID
 * - getMinPower □□□□□: 10 + (level - 1) * 15
 * - getMaxPower □□□ getMinPower + 15~50
 * - getMaxLevel □□□□□□□□□□
 * - canCombine □□ false □□□□□□□□□□□□
 */
public class ModEnchantments {

    /**
     * □□□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static final Enchantment EXAMPLE_ENCHANTMENT = Registry.register(
        Registries.ENCHANTMENT,
        new Identifier("examplemod", "example_enchantment"),
        new ExampleEnchantment(
            Enchantment.Rarity.RARE,           // □□□: □□
            EnchantmentTarget.WEAPON,          // □□□□□□□□□: □□
            EquipmentSlot.MAINHAND             // □□□□□: □□
        )
    );

    /**
     * □□□□□□□□□□□□□□□□□
     */
    public static void register() {
        // □□□□□□□□□□□□□□□□
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    public static class ExampleEnchantment extends Enchantment {

        /**
         * □□□□□
         *
         * @param rarity  □□□□□□
         * @param target  □□□□□□□□□
         * @param slots   □□□□□□□□□□□□
         */
        public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
            // □□□□□□□□□□
            super(rarity, target, slots);
        }

        /**
         * □□□□□□□□□□□□□
```

```
         * □□□□□□□□□□□□□□□□□□□□□□□□□□□"□□□□□"□□□□□□□□□□
         *
         * □□□□□: 10 + (level - 1) * 15
         * 1□: 10, 2□: 25, 3□: 40
         *
         * @param level □□□□□
         * @return □□□□□□□□□□□□
         */
        @Override
        public int getMinPower(int level) {
            return 10 + (level - 1) * 15;
        }

        /**
         * □□□□□□□□□□□□
         * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
         * □□□□ getMinPower + 50
         *
         * @param level □□□□□
         * @return □□□□□□□□□□□□
         */
        @Override
        public int getMaxPower(int level) {
            return super.getMinPower(level) + 50;
        }

        /**
         * □□□□□□□□□□
         * □□: □□ 3 □□□□□□□□□ 3 □
         *
         * @return □□□□□
         */
        @Override
        public int getMaxLevel() {
            return 3;
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□□□
         * □□□□□ EnchantmentTarget □□□□□□□□□□□□□□□□□□□□□□□□
         *
         * @param stack □□□□□
         * @return □□□□□□
         */
        @Override
        public boolean isAcceptableItem(ItemStack stack) {
            // □□□□□: □□□□□□□□□□ EnchantmentTarget
            // □□□□□□□□□□□□□□□□
            return super.isAcceptableItem(stack);
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□□□
         * □□□□□ true□□□□□□□□□□□□□□□□□□□□
         *
         * @param other □□□□□
         * @return □□□□□□
         */
        @Override
        public boolean canCombine(Enchantment other) {
            // □□: □□□□□□□□□
            // if (other == Enchantments.SHARPNESS) {
            //     return false;
            // }
            return super.canCombine(other);
        }
    }
}

/**
 * ================================================================
 * □□□□□□□□□□
 * ================================================================
 * □□□□□: assets/examplemod/lang/en_us.json
 *
 * {
```

```
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment",
 *     "enchantment.examplemod.example_enchantment.desc": "A custom enchantment"
 * }
 *
 * □□□□□□□□: assets/examplemod/lang/zh_cn.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "□□□□□",
 *     "enchantment.examplemod.example_enchantment.desc": "□□□□□□□□□"
 * }
 *
 * ================================================================
 * □□ ID □□□□□
 * ================================================================
 * - □□□□ Identifier □□□□□□
 * - □□□□□□□□□: enchantment.<mod_id>.<enchantment_name>
 * - □□□□□□□□□□□ ID □□
 */
```

entity/entity_attribute.java

```
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;

/**
 * □□□□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□
 * 1. □□ FabricDefaultAttributeRegistry □□□□□□□□Fabric □□□□□□
 * 2. □□ DefaultAttributeRegistry□Vanilla □□□□□□
 *
 * □□□□□Fabric 1.20.1 □□□□□□□□□□□□□□□□□
 *
 * □□□□□□□□□: FabricDefaultAttributeRegistry
 *   - Fabric API □□□□□□□□□
 *   - □□□□□□□□□□□□□
 *   - □□□□□□□□□□□□□□□□
 *
 * □□□□: DefaultAttributeRegistry□Vanilla □□□□
 *   - Minecraft □□□□□□□□□
 *   - □□□□□□□□□□□□□□□□
 *   - □□□□□□□□□□□□□ Fabric □□□□□□□□
 *
 * □□□□□:
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□
 * - createMobAttributes() □□□□□□□□□□□□□□□□□□□□
 */
public class ModEntityAttributes {

    /**
     * □□□□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□□
     */
    public static void register() {

        // ================================================================
        // □□□□: FabricDefaultAttributeRegistry□□□□
        // ================================================================
        // □□1: EntityType□□□□□□□
        // □□2: DefaultAttributeContainer□□□□□□□□□□□□□ createMobAttributes() □□□□
        FabricDefaultAttributeRegistry.register(
            ExampleEntity.EXAMPLE_ENTITY,
            ExampleEntity.createMobAttributes()
        );

        // ================================================================
        // □□□□: DefaultAttributeRegistry□Vanilla □□□□□□□□□□
        // ================================================================
        // DefaultAttributeRegistry.register(
        //     ExampleEntity.EXAMPLE_ENTITY,
        //     ExampleEntity.createMobAttributes()
```

```
        // );
        // □□□□□□□□□□□□□□□ FabricDefaultAttributeRegistry □□□
    }
}

/**
 * ================================================================
 * □□□□□□□□□□
 * ================================================================
 *
 * □□□□□                          □□□    □□
 * ----------------------------------------------------------------
 * GENERIC_MAX_HEALTH              20.0     □□□□□□□20 = 10□□□
 * GENERIC_KNOCKBACK_RESISTANCE     0.0     □□□□□0.0~1.0□
 * GENERIC_MOVEMENT_SPEED           0.25    □□□□□
 * GENERIC_FLYING_SPEED             0.0     □□□□□□□□□□□□□□
 * GENERIC_ATTACK_DAMAGE            2.0     □□□□□
 * GENERIC_ATTACK_KNOCKBACK         0.0     □□□□□□□
 * GENERIC_ATTACK_SPEED             4.0     □□□□□□□□□□□□□□□□
 * GENERIC_ARMOR                    0.0     □□□□□0~20□
 * GENERIC_ARMOR_TOUGHNESS          0.0     □□□□□
 * GENERIC_FOLLOW_RANGE            16.0     AI □□/□□□□
 * GENERIC_LUCK                     0.0     □□□□□□□□□□
 * GENERIC_MAX_ABSORPTION           0.0     □□□□□□□□□□□
 * GENERIC_JUMP_STRENGTH            0.42     □□□□
 * GENERIC_GRAVITY                  0.08    □□□1.20.5+ □□□
 *
 * □□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 */
```

entity/entity_renderer.java

```
package com.example.mod.client.render;

import com.example.mod.entity.ExampleEntity;
import net.minecraft.client.render.entity.EntityRenderer;
import net.minecraft.client.render.entity.EntityRendererFactory;
import net.minecraft.client.render.entity.model.EntityModel;
import net.minecraft.client.render.entity.model.EntityModelLayer;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□ EntityRenderer□
 * 2. □□ EntityRendererRegistry □□□□□□
 * 3. □□□□□□□□EntityModel□
 *
 * □□□□□□□:
 * - EntityRenderer □□□□□□□□□□□□□□□ClientModInitializer□
 * - EntityRendererFactory.Context □□□□□□□□□□□□□□□□
 * - getTexture() □□□□□□□□□□□□□
 * - □□□□□□□□ EntityModelLayerRegistry □□
 * - □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□□□□□□□□□□□□□□□□□□:
 *   EntityRendererRegistry.register(ExampleEntity.EXAMPLE_ENTITY, ExampleEntityRenderer::new);
 *
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□:
 *   EntityModelLayerRegistry.registerModelLayer(ExampleEntityModel.LAYER_LOCATION, ExampleEntityModel::getTexturedModelData);
 */
public class ExampleEntityRenderer extends EntityRenderer<ExampleEntity> {

    // □□□□□□□
    // □□□□□□□□ EntityModel □□□□ ExampleEntityModel□
    private final EntityModel<ExampleEntity> model;

    /**
     * □□□□
     * EntityRendererFactory.Context □ Fabric □□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *
     * @param ctx □□□□□□□□□
     */
    public ExampleEntityRenderer(EntityRendererFactory.Context ctx) {
```

```
        super(ctx);
        // □□ ctx.getPart() □□□□□□□□□□
        // □□□□□ EntityModelLayerRegistry □□□□□
        this.model = new ExampleEntityModel<>(ctx.getPart(ExampleEntityModel.LAYER_LOCATION));
        // □□□□□□□□□□0.0~1.0□0.5 □□□□□□□
        this.shadowRadius = 0.5f;
    }

    /**
     * □□□□□□□
     * □□□□□□□: mod_id:textures/entity/xxx.png
     * □□□□□□□: src/main/resources/assets/examplemod/textures/entity/example_entity.png
     *
     * @param entity □□□□
     * @return □□□□□
     */
    @Override
    public Identifier getTexture(ExampleEntity entity) {
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        return new Identifier("examplemod", "textures/entity/example_entity.png");
    }
}

/**
 * ================================================================
 * □□□□□□□□□□□□□□□□ ClientModInitializer □□□□
 * ================================================================
 *
 * import net.fabricmc.fabric.api.client.rendering.v1.EntityRendererRegistry;
 * import net.fabricmc.fabric.api.client.rendering.v1.EntityModelLayerRegistry;
 *
 * public class ExampleModClient implements ClientModInitializer {
 *
 *     @Override
 *     public void onInitializeClient() {
 *         // □□□□□□□□
 *         EntityRendererRegistry.register(
 *             ExampleEntity.EXAMPLE_ENTITY,
 *             ExampleEntityRenderer::new
 *         );
 *
 *         // □□□□□□□□
 *         EntityModelLayerRegistry.registerModelLayer(
 *             ExampleEntityModel.LAYER_LOCATION,
 *             ExampleEntityModel::getTexturedModelData
 *         );
 *     }
 * }
 */
```

entity/simple_entity.java

```
package com.example.mod.entity;

import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.SpawnRestriction;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.world.Heightmap;
import net.minecraft.world.World;
import net.minecraft.world.entity.SpawnLocation;

/**
 * □□□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□
 * 1. □□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□
 * 4. □□□□□□□□SpawnRestriction□
 *
```

```
 * □□□□□□□□:
 * - □□□□□□□□□□□□□□□□ (EntityType, World) □□
 * - EntityType.Builder □ dimensions □□□□□□ "□□□□□□"□
 *   □□ (0.6f, 1.8f) □□ 0.6x1.8 □□□□□
 * - EntityType □□□□□□□□□□ build() □□□ DataFixer□□□□ null□
 * - createMobAttributes() □□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□ FabricDefaultAttributeRegistry□□ entity_attribute.java□
 * - SpawnRestriction □□□□□□□□□□□□□□
 */
public class ExampleEntity extends PathAwareEntity {

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□ Registry.register □□□□ Registries.ENTITY_TYPE
    // □□: mod_id □□□□□□□□□□ mod ID
    public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
        Registry.register(
            Registries.ENTITY_TYPE,
            new Identifier("examplemod", "example_entity"),
            EntityType.Builder.<ExampleEntity>create(
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    ExampleEntity::new,
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    SpawnGroup.CREATURE
                )
                // □□□□□□□□□□□□x□□□□□□□□□
                .dimensions(0.6f, 1.8f)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ 5
                .trackingRange(8)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ false□
                .build(null)
        );

    // ================================================================
    // □□□□□
    // ================================================================
    // □□□□□□□□□□□□ EntityType □□□□□□□□
    // □□ <? extends PathAwareEntity> □□□□□□□□
    public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
        super(entityType, world);
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□□□□□□□□□□□□□DefaultAttributeContainer□
    // □□□□ createMobAttributes() □□□□□□□□□□□□□□□□
    //
    // □□□□□□:
    // - GENERIC_MAX_HEALTH      □□□□□□□□□□ 20 = 10 □□□□
    // - GENERIC_MOVEMENT_SPEED  □□□□□□□□□□ 0.25□
    // - GENERIC_ATTACK_DAMAGE   □□□□□□□□□□ 2□
    // - GENERIC_FOLLOW_RANGE     AI □□□□□□□□□□ 16□
    // - GENERIC_ARMOR           □□□□
    // - GENERIC_ARMOR_TOUGHNESS □□□□□
    // - GENERIC_KNOCKBACK_RESISTANCE □□□□□□0.0~1.0□
    public static DefaultAttributeContainer createMobAttributes() {
        return PathAwareEntity.createMobAttributes()
            .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)          // 10□□
            .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)      // □□□□□
            .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)        // □□□□□
            .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)        // □□□□□
            .add(EntityAttributes.GENERIC_ARMOR, 0.0)                // □□□□
            .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0); // □□□□□
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // SpawnRestriction □□□□□□□□□□□□□□
    // □□□□ EntityType □□□□□□□□
    public static void registerSpawnRestrictions() {
        // □□□□□□:
        // 1. □□□□□
        // 2. □□□□□□□ON_GROUND □□ / IN_WATER □□ / ANY □□□
```

```
        // 3. □□□□□□□MOTION_BLOCKING □□
        // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        SpawnRestriction.register(
            EXAMPLE_ENTITY,
            SpawnRestriction.Location.ON_GROUND,
            Heightmap.Type.MOTION_BLOCKING,
            SpawnRestriction.Method.ON_GROUND,
            (entityType, worldAccess, spawnReason, blockPos, random) -> {
                // □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                return worldAccess.getBlockState(blockPos.down()).isSolidBlock(
                    worldAccess, blockPos.down()
                );
            }
        );
    }

    // ================================================================
    // □□ tick □□
    // ================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□□□□□ tick □□□
        // □□: □□□□□□□□□□□□□□□□□
    }

    // ================================================================
    // □□□ AI □□
    // ================================================================
    @Override
    protected void initGoals() {
        // □□□□□□□□ AI □□□Goal□
        // □□ AI:
        // - MeleeAttackGoal        □□□□□□□
        // - WanderAroundGoal       □□□□
        // - LookAtEntityGoal       □□□□
        // - FollowTargetGoal       □□□□
        // - SwimGoal               □□
    }
}
```

entity/spawn_egg_item.java

```
package com.example.mod.item;

import com.example.mod.entity.ExampleEntity;
import net.fabricmc.fabric.api.object.builder.v1.spawn_egg.FabricSpawnEggItem;
import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.item.Item;
import net.minecraft.item.ItemGroup;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.util.Rarity;

/**
 * □□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□
 * 1. □□ FabricSpawnEggItem □□□□□
 * 2. □□ Registry.register □□□ Registries.ITEM
 *
 * □□□□Fabric 1.20.1 □ FabricSpawnEggItem □□□□□□□□
 * Supplier<EntityType<?>> □□□□□□□□□□ EntityType
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * □□ Supplier □□□□□□□□□□□□□□□□
 *
 * □□□□□:
 * - □□□□□□ new SpawnEggItem()□□□□□ FabricSpawnEggItem
 * - Supplier □□□□□□□□□□ EntityType□□□□□□□□□□□□□□□
 * - □□□□□□ RGB □□□□□0xRRGGBB □□□
 * - Item.Settings □□□□□□□□□□□□□□ new Item.Settings()
 * - □□□□□□□□□□□□ ItemGroup□□□□□□□□□□□
 */
public class ModSpawnEggItems {
```

```
    /**
     * □□□□□□□□□
     * □□□□□□□□:
     * - primaryColor:   □□□□□□□□□□0xRRGGBB □□□
     * - secondaryColor: □□□□□□□□□□0xRRGGBB □□□
     */
    public static final Item EXAMPLE_SPAWN_EGG = registerSpawnEgg(
        // □□ Supplier □□□□□□□□□
        () -> ExampleEntity.EXAMPLE_ENTITY,
        0xFFFFFF,   // □□□□: □□
        0x000000,   // □□□□: □□
        new Item.Settings()
            .rarity(Rarity.UNCOMMON)
            // □□□□□□□□□□□□□1.20.1 □□ ItemGroups□
            // □□: 1.20.1 □ ItemGroup □□□□□□□□□
            // □□□□□ ItemGroupEvents.modifyEntriesEvent □□□□□
    );

    /**
     * □□□□□□□□□□□□
     *
     * @param entityTypeSupplier □□□□ Supplier□□□□□□
     * @param primaryColor      □□□□□□0xRRGGBB□
     * @param secondaryColor    □□□□□□0xRRGGBB□
     * @param settings          □□□□□□□
     * @return □□□□□ Item □□
     */
    private static Item registerSpawnEgg(
        java.util.function.Supplier<EntityType<?>> entityTypeSupplier,
        int primaryColor,
        int secondaryColor,
        Item.Settings settings
    ) {
        // □□ FabricSpawnEggItem □□
        // FabricSpawnEggItem □□□□□□□□□□□□□□□□□□□□□
        Item spawnEgg = new FabricSpawnEggItem(
            entityTypeSupplier,
            primaryColor,
            secondaryColor,
            settings
        );

        // □□□□ Registries.ITEM
        // □□: mod_id □□□□□□ mod ID
        return Registry.register(
            Registries.ITEM,
            new Identifier("examplemod", "example_spawn_egg"),
            spawnEgg
        );
    }

    /**
     * □□□□□□□□
     * □□□□□□□□□□ModInitializer□□ onInitialize() □□□□
     */
    public static void register() {
        // □□□□□□□□□□□□□□□□□□□□
```

gui/screen_with_menu.java

```
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.util.Identifier;

/**
 * ============================================================
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.20.1
 * ============================================================
 *
 * □□□□□□□□□□:
 * 1. ExampleScreenHandler - □□□□□□□□□□□□□□□□□□□□□□□□
 * 2. ExampleScreen        - □□□□□□□□□□□□□□□□
 *
 * □□□□□:
 * - □□□□□□ ScreenHandler □□□□ syncId
 * - □□□□□□ Packet □ syncId □□□□□□□
 * - □□□□□□ syncId □ ScreenHandlerType □□ Screen □□
 * - Screen □ ScreenHandler □□ syncId □□
 *
 * □□□□□:
 * - ScreenHandlerType □□□□ Registries.SCREEN_HANDLER □□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   □□□□: (int syncId, PlayerInventory)
 *   □□□□: (int syncId, PlayerInventory, Inventory)
 * - transferSlot□shift+click□□□□□□□□□□□□ shift □□□□
 * - canUse □□□□□ true□□□□□□□□□□
 * - Slot □□□□□□□ 0 □□□□□□□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ============================================================
    // ScreenHandlerType □□
    // ============================================================
    // □□ Registry.register □□□□ Registries.SCREEN_HANDLER
    // ScreenHandlerType □□□□□□□ ScreenHandlerFactory□□□□□□□□
    public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
        Registry.register(
            Registries.SCREEN_HANDLER,
            new Identifier("examplemod", "example_gui"),
            new ScreenHandlerType<>(ExampleScreenHandler::new)
        );

    // □□□□□□□□□□□□□□□□ Inventory□
    private final Inventory inventory;

    // ============================================================
    // □□□□□□□
    // ============================================================
    // □□□□□□ ScreenHandlerType □□ ScreenHandler □□□
    // □□ syncId □ PlayerInventory□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
        this(syncId, playerInventory, new SimpleInventory(9));
    }

    // ============================================================
    // □□□□□□□
    // ============================================================
    // □□□□□□□□□□□□□□□□□□□□ Inventory
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
        // □□□□□□□□□□□□□ ScreenHandlerType □ syncId
        super(EXAMPLE_SCREEN_HANDLER, syncId);
        // □□□□□□□□□□□□
        checkSize(inventory, 9);
        this.inventory = inventory;
```

```
    // □□□□□□□□□□□□□□ Inventory □□□"□□"
    inventory.onOpen(playerInventory.player);

    // ================================================================
    // □□□□□□□□□□□□□□□□□□□□□□
    // ================================================================
    // □□: inventory, index(□□□□), x(□□□□), y(□□□□)
    // □□: x, y □□□□□□□□□□□□□□□□□□□□□□□□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            // 3x3 □□□□□□□□□ (8, 18) □□□□□□
            this.addSlot(new Slot(inventory, j + i * 3, 8 + j * 18, 18 + i * 18));
        }
    }

    // ================================================================
    // □□□□□□□□□□
    // ================================================================
    // □□□□: 3□9□ = 27□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 9; j++) {
            // □□□□□□□□ y=66 □□
            this.addSlot(new Slot(playerInventory, 9 + j + i * 9, 8 + j * 18, 66 + i * 18));
        }
    }

    // □□□□□: 1□9□
    for (int j = 0; j < 9; j++) {
        // □□□□□□□□ y=124
        this.addSlot(new Slot(playerInventory, j, 8 + j * 18, 124));
    }
}

// ================================================================
// shift+click □□□□□□
// ================================================================
// □□□ shift+□□□□□□□
// □□□□□□□□□□□□□□ ItemStack
@Override
public ItemStack transferSlot(PlayerEntity player, int index) {
    Slot slot = this.slots.get(index);
    if (!slot.hasStack()) {
        return ItemStack.EMPTY;
    }

    ItemStack originalStack = slot.getStack();
    ItemStack copyStack = originalStack.copy();

    // □□□□□:
    // 0~8: □□□□□3x3□
    // 9~35: □□□□
    // 36~44: □□□
    if (index < 9) {
        // □□□□□□□□□□□□
        if (!this.insertItem(originalStack, 9, 45, true)) {
            return ItemStack.EMPTY;
        }
    } else {
        // □□□/□□□□□□□□
        if (!this.insertItem(originalStack, 0, 9, false)) {
            return ItemStack.EMPTY;
        }
    }

    // □□□□□□□□□□□□□□□□□□□□
    if (originalStack.isEmpty()) {
        slot.setStack(ItemStack.EMPTY);
    } else {
        slot.markDirty();
    }
    return copyStack;
}

// ================================================================
// □□□□□□□□□□□□□□□□
// ================================================================
```

```
    @Override
    public boolean canUse(PlayerEntity player) {
        // □□□□ true□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□
        return this.inventory.canPlayerUse(player);
    }
}

/**
 * ================================================================
 * □□□ Screen □□HandledScreen□
 * ================================================================
 * HandledScreen □□□ ScreenHandler □□□□□□
 * □□□□□□□□□□□□□□□□□□□□□□□□
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    /**
     * □□□ Screen □□□□
     * ScreenRegistry.register □□□□□□□□
     */
    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, net.minecraft.text.Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        // □□□□□□□□
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        // □□□□□
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□
     * □□□□: src/main/resources/assets/examplemod/textures/gui/example_gui.png
     * □□□□□□ 256x256□□□□□ 176x166 □□
     */
    @Override
    protected void drawBackground(net.minecraft.client.util.math.MatrixStack matrices, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);
        net.minecraft.client.render.RenderSystem.setShaderTexture(0,
            new Identifier("examplemod", "textures/gui/example_gui.png"));

        // □□□□: x, y, u(□□x□□), v(□□y□□), □, □
        int x = (this.width - this.backgroundWidth) / 2;
        int y = (this.height - this.backgroundHeight) / 2;
        this.drawTexture(matrices, x, y, 0, 0, this.backgroundWidth, this.backgroundHeight);
    }

    @Override
    public void render(net.minecraft.client.util.math.MatrixStack matrices, int mouseX, int mouseY, float delta) {
        // □□□□□□□□□□□□□□□
        this.renderBackground(matrices);
        super.render(matrices, mouseX, mouseY, delta);
        // □□□□□□□□
        this.drawMouseoverTooltip(matrices, mouseX, mouseY);
    }
}

/**
 * ================================================================
 * □□□□□□
 * ================================================================
 * ScreenHandlerType □□□□□□□□ModInitializer□:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER □□□□□□□□□□
 *
 * Screen □□□□□□□□ClientModInitializer□:
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
```

```
 *   );
 *
 * ================================================================
 * □□ GUI □□□□□□□□ BlockEntity □ Item □□□□
 * ================================================================
 *
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       Text.literal("□□□□□")
 *   ));
 */
```

gui/simple_screen.java

```
package com.example.mod.gui;

import net.minecraft.client.gui.screen.Screen;
import net.minecraft.client.gui.widget.ButtonWidget;
import net.minecraft.client.gui.widget.TextFieldWidget;
import net.minecraft.client.util.math.MatrixStack;
import net.minecraft.text.Text;

/**
 * □□□□□ Screen □□ - Fabric 1.20.1
 *
 * □□□□□□□□□□□□□□□□□□□□□□□□ Screen □□
 * □ HandledScreen □□□Screen □□□□ ScreenHandler□
 * □□□□□□□□/□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□:
 * - □□□□□
 * - □□□□□□□□
 * - □□□□□□□/□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□:
 * - init() □□□□□□□□□□□□ this.width / this.height □□□□□□
 * - □□□□□□□□□ addDrawableChild() □□ addDrawable()
 *   □addDrawable □□□□□□□□□□□□
 * - □□□□□□ addSelectableChild() □□□□□□□□□□
 * - shouldPause() □□ false □□□□□□□□□
 * - render() □□□□□□□□ renderBackground() □□□□□
 *   □□□□ super.render() □□□□□□
 */
public class ExampleScreen extends Screen {

    // □□□□□□
    private TextFieldWidget textField;

    /**
     * □□□□□
     * □□□□□□□□□Text □□□
     */
    public ExampleScreen() {
        // □□ Text.literal() □□□□□□1.20.1 □□□□□□□□ String□
        super(Text.literal("□□□□□□"));
    }

    /**
     * □□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * □□: □□□□□□□□□□□□□□□□□□□□ init()
     */
    @Override
    protected void init() {
        // ================================================================
        // □□□□□
        // ================================================================
        // 1.20.1 □□ ButtonWidget.builder() □□ API
        this.addDrawableChild(
            ButtonWidget.builder(
                Text.literal("□□"),  // □□□□□
                button -> {
                    // □□□□□□□□
                    // □□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    // □□□□□□□□□□□□□□□□ ClientPlayNetworking □□
```

```
                System.out.println("□□□□□□□□□□□□□□: " + textField.getText());
                // □□□□
                this.client.setScreen(null);
            }
        )
        // □□□□□□□□□□: x, y, width, height
        .dimensions(this.width / 2 - 100, this.height / 2 - 10, 200, 20)
        .build()
    );

    // ================================================================
    // □□□□□□□□
    // ================================================================
    this.addDrawableChild(
        ButtonWidget.builder(
            Text.literal("□□"),
            button -> this.client.setScreen(null)  // □□□□
        )
        .dimensions(this.width / 2 - 100, this.height / 2 + 20, 200, 20)
        .build()
    );

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□: textRenderer, x, y, width, height, □□□□
    this.textField = new TextFieldWidget(
        this.textRenderer,
        this.width / 2 - 100,  // x □□
        this.height / 2 - 50,  // y □□
        200,                    // □□
        20,                     // □□
        Text.literal("□□□□...")  // □□□□□□□
    );

    // □□□□□□□□
    this.textField.setMaxLength(50);          // □□□□□□
    this.textField.setDrawsBackground(true);   // □□□□
    // □□□□□□□□□□□□□□□□□□□□□□□□
    this.textField.active = true;

    // □□: □□□□□□□□ addSelectableChild □□□□□□□□□
    // addSelectableChild □□□□□□□□□□□□□□□□□□□□□□□□
    this.addSelectableChild(this.textField);
    // □□□□□□□□□□ addDrawableChild
    this.addDrawableChild(this.textField);
}

/**
 * □□□□
 * □□□□□□□□□□□□□□□□□□□□
 *
 * @param matrices □□□□□□□□□□□□□□□□□□
 * @param mouseX   □□ x □□
 * @param mouseY   □□ y □□
 * @param delta    □□□□□□□□□□□□□□
 */
@Override
public void render(MatrixStack matrices, int mouseX, int mouseY, float delta) {
    // 1. □□□□□□□□□□□□□□□□□□□□
    this.renderBackground(matrices);

    // 2. □□ super.render() □□□□□□□□□□□□□
    //    □□□□□□□□□□□□ addDrawableChild/addSelectableChild □□□□□□□
    super.render(matrices, mouseX, mouseY, delta);

    // 3. □□□□□□□
    // drawCenteredText □□: matrices, text, x(□□), y, color
    this.textRenderer.drawCentered(
        matrices,
        this.title,
        this.width / 2,
        20,
        0xFFFFFF   // □□
    );
    // 4. □□□□□□□
```

```
        this.textRenderer.draw(
            matrices,
            Text.literal("□□□□□□□□□□:"),
            this.width / 2 - 100,
            this.height / 2 - 65,
            0xAAAAAA   // □□
        );
    }

    /**
     * □□□□□□□□□□□□□□□□□□□□
     * - true: □□□□□□□□□□□□□□□□□□□□□□□□
     * - false: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     */
    @Override
    public boolean shouldPause() {
        return false;
    }

    /**
     * □□□□□□□□
     * □□□□□□□□□□□□□□ Esc □□□□□□
     */
    @Override
    public boolean keyPressed(int keyCode, int scanCode, int modifiers) {
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        if (this.textField.isActive() && this.textField.keyPressed(keyCode, scanCode, modifiers)) {
            return true;
        }
        // □□□□□□□□
        return super.keyPressed(keyCode, scanCode, modifiers);
    }
}

/**
 * ================================================================
 * □□ Screen □□□□□□
 * ================================================================
 * □□□□□□□□□□□□:
 *
 *   MinecraftClient.getInstance().setScreen(new ExampleScreen());
 *
 * □□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 */
```

item/armor_item.java

```
/*
 * armor_item.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.1 (Fabric Loader + Fabric API + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ API □□□□ Fabric 1.20.1 □□ Yarn □□□□□□□yarn-1.20.1+build.10□□
 *
 * ================================================================================
 * □□□□ ArmorMaterial.Builder □□□□□□□ —— □□□□□□□
 * ================================================================================
 *   □ 1.20.1 □□□□□□□□ArmorMaterial.Builder□
 *   - 1.20.1 □ ArmorMaterial □□□□□□□□□□net.minecraft.item.ArmorMaterial□□
 *     □□□□□□□□□□□□□□□□□□□□□□□□□□implements ArmorMaterial □□□□□□□□□□□□□□□□
 *   - ArmorMaterial.Builder□□□ record □□□EquipmentType□RegistryEntry<SoundEvent>□
 *     □□□1.20.5 □□□□□□1.21 □□/□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *     ArmorMaterial.builder()...build()□□□□□□□□□□□□□□□□□□□□□□□□□□□□ 1.20.1□
 *
 * □1.20.1 □ ArmorMaterial □□□□□□□□□□□□□□□□ javadoc□□
 *   - int          getDurability(ArmorItem.Type type)   // □□□□□□□□
 *   - int          getProtection(ArmorItem.Type type)    // □□□□□□□□
 *   - int          getEnchantability()                  // □□□□□
 *   - SoundEvent   getEquipSound()                       // □□□□□□□□□□□□ SoundEvent□□ RegistryEntry□
 *   - Ingredient   getRepairIngredient()                // □□□□□
 *   - String       getName()                              // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   - float        getToughness()                         // □□□□□□ 2.0F□□□□□□ 4.0F□□□□□□ 0□
 *   - float        getKnockbackResistance()               // □□□□□□□□□□□□ 0.1F□□□□□□ 0□
 *
```

```
 *   □□□□□□□1.20.1 □ getDurability / getProtection / getEnchantability□
 *   □ 1.20.4 □□□□□□ Yarn □□□□ getDurabilityForType / getDefenseForType / getEnchantmentValue□
 *   1.20.5+ □□□□□□□ record + Builder □□□□□□□□□□□ 1.20.1□
 *
 * □ArmorItem □□□□□□1.20.1□□
 *   public ArmorItem(ArmorMaterial material, ArmorItem.Type type, Item.Settings settings)
 *   □□ ArmorItem.Type □ ArmorItem □□□□□□□HELMET / CHESTPLATE / LEGGINGS / BOOTS
 *   □□□□ BODY □□□□/□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   □□□□□□□□□ = □□□□□□□□ × □□□□□□□□
 *     HELMET=11, CHESTPLATE=16, LEGGINGS=15, BOOTS=13
 *   □□□□□□□□ 15□□□□ 11×15=165□□□□ 16×15=240□□□□ 15×15=225□□□□ 13×15=195□
 *
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *   boots=3, leggings=6, chestplate=8, helmet=3□□□□□ 1/2/3/1□□
 * ----------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.minecraft.item.ArmorItem;
import net.minecraft.item.ArmorMaterial;
import net.minecraft.item.Item;
import net.minecraft.recipe.Ingredient;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.sound.SoundEvent;
import net.minecraft.sound.SoundEvents;
import net.minecraft.util.Identifier;

import java.util.function.Supplier;

/**
 * □□□□□□□□□□□□□□
 */
public class ModArmorItems {

    public static final String EXAMPLE_MOD_ID = "examplemod";

    // ============================================================================
    // □□□□□□□□□□□□□□□□ ArmorMaterial □□
    // ============================================================================

    /**
     * □□□□□□□□ “example_armor”□
     *
     * □□□□□
     *  - 1.20.1 □□□□□□□□□ 8 □□□□□□□□ Builder □□□□
     *  - getRepairIngredient() □□ Ingredient□□□□□□□□□□□□□□□□□□□
```

```
            default -> 0; // BODY□□/□□□□□□□□□□□□□□□□□□□□ 0
        };
    }

    @Override
    public int getProtection(ArmorItem.Type type) {
        // □□□□□□□□□□□□□boots 3 / leggings 6 / chestplate 8 / helmet 3
        return switch (type) {
            case BOOTS, HELMET -> 3;
            case LEGGINGS -> 6;
            case CHESTPLATE -> 8;
            default -> 0;
        };
    }

    @Override
    public int getEnchantability() {
        // □□ 10□□ 9□□ 22□□□□□ 10□
        return 10;
    }

    @Override
    public SoundEvent getEquipSound() {
        // 1.20.1 □□□□□□□□□□□□ ITEM_ □□□□□ ITEM_ARMOR_EQUIP_IRON□□
        // □□□1.20.5+ □□□□□□□□□□□□□□□□□□□□ ARMOR_EQUIP_IRON□□□□□□□□
        return SoundEvents.ITEM_ARMOR_EQUIP_IRON;
    }

    @Override
    public Ingredient getRepairIngredient() {
        return repairIngredientSupplier.get();
    }

    @Override
    public String getName() {
        return NAME;
    }

    @Override
    public float getToughness() {
        // □□□□□□ 2.0F□□□□□□ 4.0F□□□□□□ 0□□□□□ 2.0F□
        return 2.0F;
    }

    @Override
    public float getKnockbackResistance() {
        // □□□□□□□□□□□□□ 0.1F□□□□ 0□□□□□ 0□
        return 0.0F;
    }
}

// ============================================================================
// □□□□□□□□□□□□□□□□□□□/□□/□□/□□□
// ============================================================================
// ArmorItem □□□new ArmorItem(□□, □□□□□, Item.Settings)
// □□□ 1.20.5+ □□□1.20.1 □□□□□□ ArmorItem.Type□□□ EquipmentType□□
public static final Item EXAMPLE_HELMET = register(
        "example_helmet",
        new ArmorItem(ExampleArmorMaterial.INSTANCE, ArmorItem.Type.HELMET, new FabricItemSettings())
);
public static final Item EXAMPLE_CHESTPLATE = register(
        "example_chestplate",
        new ArmorItem(ExampleArmorMaterial.INSTANCE, ArmorItem.Type.CHESTPLATE, new FabricItemSettings())
);
public static final Item EXAMPLE_LEGGINGS = register(
        "example_leggings",
        new ArmorItem(ExampleArmorMaterial.INSTANCE, ArmorItem.Type.LEGGINGS, new FabricItemSettings())
);
public static final Item EXAMPLE_BOOTS = register(
        "example_boots",
        new ArmorItem(ExampleArmorMaterial.INSTANCE, ArmorItem.Type.BOOTS, new FabricItemSettings())
);

// ============================================================================
// □□□□□□□
// ============================================================================
```

```
    public static Item register(String name, Item item) {
        return Registry.register(Registries.ITEM, new Identifier(EXAMPLE_MOD_ID, name), item);
    }

    /**
     * □□□□□□□ onInitialize() □□□□□□□□□□□□□□□□□□□□□□□□
     */
    public static void registerArmorItems() {
        @SuppressWarnings("unused")
        Item ensureClassInit = EXAMPLE_HELMET;
    }

    /*
     * ============================================================================
     * □□□□□□□□□□□□□□□□□□□□□□□□
     * ============================================================================
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *   assets/minecraft/textures/models/armor/example_armor_layer_1.png   □□□+□□□
     *   assets/minecraft/textures/models/armor/example_armor_layer_2.png   □□□+□□□
     * □□□ example_armor □□□□□□ getName() □□□□□□□□
     *
     * □□□□□□□□□□□□□□□□□□□□□□□□
     *   □□ assets/<modid>/models/item/example_helmet.json □□□□□□□ model.json □□□□
     *   □□ assets/<modid>/textures/item/example_helmet.png
     * □□□□□□□□□□□□□□□□□□□□□□□□
     * ============================================================================
     */

    /*
     * ============================================================================
     * □□□□□□□□□□□□
     * ============================================================================
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *   new DyeableArmorItem(ExampleArmorMaterial.INSTANCE, ArmorItem.Type.CHESTPLATE, new FabricItemSettings())
     * □□□□□□□□ JSON □ textures □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * ============================================================================
     */

    private ModArmorItems() {
    }
}
```

item/food_item.java

```
/*
 * food_item.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.1 (Fabric Loader + Fabric API + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ API □□□□ Fabric 1.20.1 □□ Yarn □□□□□□□yarn-1.20.1+build.10□□□
 *
 * □1.20.1 □□ API □□□□□□□□□□□□□□□□□□□□
```

```
 *      Item.Settings.recipeRemainder(Item) 可以用于/可以用于。
 *   4. 自定义 food(...) 可以在食物上挂“效果”，在use/finishUsing里也可以手动处理。
 * ------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.minecraft.entity.effect.StatusEffectInstance;
import net.minecraft.entity.effect.StatusEffects;
import net.minecraft.item.FoodComponent;
import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * 模组的食物物品集合。
 */
public class ModFoodItems {

    public static final String EXAMPLE_MOD_ID = "examplemod";

    /**
     * 示例食物 1：普通熟食
     * hunger=6，3 饱食度，saturationModifier=0.6F（常见值）
     */
    public static final Item EXAMPLE_FOOD = register(
            "example_food",
            new Item(new FabricItemSettings().food(
                    new FoodComponent.Builder()
                            .hunger(6)
                            .saturationModifier(0.6F)
                            .meat()
                            .build()
            ))
    );

    /**
     * 示例食物 2：特殊食物——即时吃 + 给自己一个短暂的速度效果（演示用）
     */
    public static final Item EXAMPLE_SPECIAL_FOOD = register(
            "example_special_food",
            new Item(new FabricItemSettings().food(
                    new FoodComponent.Builder()
                            .hunger(4)
                            .saturationModifier(0.8F)
                            .alwaysEdible()           // 饱食时也能吃
                            .snack()                    // 标记为小食
                            // 给自己 100% 概率 5 秒的 I 级 chance 概率 0.0F~1.0F）
                            .statusEffect(
                                    new StatusEffectInstance(StatusEffects.SPEED, 5 * 20, 0),
                                    1.0F
                            )
                            // 再给一个小概率的再生效果 30% 概率触发（演示）
                            .statusEffect(
                                    new StatusEffectInstance(StatusEffects.REGENERATION, 3 * 20, 1),
                                    0.3F
                            )
                            .build()
            ))
            // 如果想让这个食物堆叠数量小一些，可以覆盖 64，例如 .maxCount(16)
    );

    // ========================================================================
    // 注册工具方法
    // ========================================================================
    public static Item register(String name, Item item) {
        return Registry.register(Registries.ITEM, new Identifier(EXAMPLE_MOD_ID, name), item);
    }

    /**
     * 在模组初始化 onInitialize() 中调用此方法，触发类加载。
     */
    public static void registerFoodItems() {
        @SuppressWarnings("unused")
        Item ensureClassInit = EXAMPLE_FOOD;
```

```
    }

    /*
     * ============================================================================
     * □□□□□□□□□□□□□□□□
     * ============================================================================
     *   □□□□□item.<modid>.<name>
     *   □□□□□□assets/<modid>/models/item/example_food.json □□□□□□ model.json □□□
     *   □□□□□□assets/<modid>/textures/item/example_food.png
     *
     * □□□□□□□□□□□“□□”□□□□□□□ FabricItemGroupEvents□
     *   ItemGroupEvents.modifyEntriesEvent(ItemGroups.FOOD_AND_DRINKS)
     *       .register(entries -> entries.add(ModFoodItems.EXAMPLE_FOOD));
     * ============================================================================
     */

    private ModFoodItems() {
    }
}
```

item/model.json

```
{
  "//□□": [
    "model.json — □□□□□ JSON □□□Fabric 1.20.1□",
    "□□□□□□assets/<modid>/models/item/<item_name>.json",
    "□□□□□□□□□ example_item□□□□□□ examplemod□",
    "□□□□□□□□□assets/<modid>/textures/item/<item_name>.png",
    "",
    "parent □□□□□□",
    "  - item/generated      □□□□□□□□/□□/□□/□□□□□□□□□□□□□□□□",
    "  - item/handheld       □□□□□□□□□/□/□□□□□□□□□□□□□□□□□□□□□□□□□□□□",
    "  - item/handheld_rods  □□/□/□□",
    "  - builtin/entity      □□ 3D/□□□□□□□□□□□□□□□□□□□
```

```java
import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□□
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 */
public class ModItems {

    // TODO: □ EXAMPLE_MOD_ID □□□□ fabric.mod.json □□□□□ mod id□
    public static final String EXAMPLE_MOD_ID = "examplemod";

    /**
     * □□□□□□□□□□□□□□□□
     *
     * □□ A□□□□□□new Item(new FabricItemSettings())
     * □□ B□□□□□□new Item(new Item.Settings())   // Item.Settings = net.minecraft.item.Item$Settings
     *
     * □□□□□ FabricItemSettings□□□□□□□□□□□ maxCount/maxDamage/rarity/fireproof □□
     */
    public static final Item EXAMPLE_ITEM = register(
            "example_item",
            new Item(new FabricItemSettings())
            // □□□□□□□□□□□□□□□□□□□□
            // .maxCount(16)        // □□□□□□□□□□ 64□□□□□□□□ maxDamage□maxCount □□□□□□ 1
            // .rarity(Rarity.UNCOMMON)
            // .fireproof()         // □□□□□□□□□□
    );

    /**
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * maxDamage □□□□□□□□“□□□□□”□□□□□□□□□□□□□□ 1□
     */
    public static final Item EXAMPLE_DURABLE_ITEM = register(
            "example_durable_item",
            new Item(new FabricItemSettings().maxDamage(256))
    );

    /**
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Registries.ITEM□
     *
     * Registry.register □□□□□□□□ item □□□□□□□□□□□□□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□onInitialize □□□□□□□□□□
     * □□□□□□□ static final □□□□□□□□□□ ModItems □□□□□□□□□□□□□□□□
     */
    public static Item register(String name, Item item) {
        // new Identifier(namespace, path)□namespace = mod id□path = □□□
        return Registry.register(Registries.ITEM, new Identifier(EXAMPLE_MOD_ID, name), item);
    }

    /**
     * □□□□□□□□□□□□□□□□□□
     *   public class ExampleMod implements ModInitializer {
     *       @Override
     *       public void onInitialize() {
     *           ModItems.registerModItems();
     *           ...
     *       }
     *   }
     *
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ JVM □□ ModItems □□□□
     * □□□□□□□□□□ static final □□□□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□ ModItems□□□□□□□□□□□□□□□
     */
    public static void registerModItems() {
        // □□□□□□□□□□□□□□□□□□□□
```

```
     * ========================================================================
     * 1.20.1 □□□□□ Item.Settings.group(ItemGroup)□□□□□ Fabric API □□□
     *
     * import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
     * import net.minecraft.item.ItemGroups;
     *
     * ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS).register(
     *         entries -> entries.add(ModItems.EXAMPLE_ITEM));
     *
     * □□□□□□□□□□□□□□/□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * ========================================================================
     */

    /*
     * ========================================================================
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *   1. □□□□□□assets/<modid>/lang/en_us.json □ zh_cn.json
     *       { "item.examplemod.example_item": "Example Item" }
     *      □□□□□□□□item.<modid>.<name>□□□□□□□□□
     *   2. □□□□□□□□□□
     *      □□□□assets/<modid>/models/item/example_item.json   □□□□□□ model.json □□□
     *      □□□□assets/<modid>/textures/item/example_item.png
     * ========================================================================
     */

    // □□□□□□□□□□
    private ModItems() {
    }
}
```

item/tool_item.java

```
/*
 * tool_item.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.1 (Fabric Loader + Fabric API + Yarn □□)
 * □□□□□: □□□□□□/□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ API □□□□ Fabric 1.20.1 □□ Yarn □□□□□□□yarn-1.20.1+build.10□□
 *
 * □1.20.1 □□□□□□□□□□□□□□□□□□□□ javadoc□□
 *   - public  SwordItem (ToolMaterial, int   attackDamage, float attackSpeed, Item.Settings)
 *   - protected PickaxeItem(ToolMaterial, int   attackDamage, float attackSpeed, Item.Settings)
 *   - protected AxeItem    (ToolMaterial, float attackDamage, float attackSpeed, Item.Settings)
 *   □□□□□□□□/□□ attackDamage □ int□□□ attackDamage □ float□□□□□□□□□□
 *
 * □□□□□□□□□protected □□□□□□
 *   PickaxeItem □ AxeItem □□□□□□□□ 1.20.1 □□ protected □□□□□□ net.minecraft.item
 *   □□□□□□□□□□□□□□□□□ super □□□□□□□□□□□□□□□□ com.example.mod.item□□
 *   net.minecraft.item □□□□□□□□□□□□□□ new PickaxeItem(...)/new AxeItem(...) □□□□□□□□
 *   □Java □□□□□ JLS 6.6.2.1□protected □□□□□□□□□□□□□ new □□□□□
 *   => □□□□□□□□□□/□□□□□□□□□□□□□□□□□□□□□□□□□□□□ super(...) □□□□□ protected □□□□□
 *      □□□□□□□□□□□□□□ public□□□□□ new CustomPickaxe(...) □□□□□
 *   SwordItem □□□□□□□□ public□□□□□ new□
 *
 *   □□□□□□□□□□/□□□□□□□□□□ new PickaxeItem(...)□□□□□□□□□□□□□□□□□□□□□□□□□□□□ public□
 *   □□□□□□□□□□□□□□□ 1.20.1 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□
 *   □□□□□□□□□□□□□□ ToolMaterial□□ ToolMaterials.IRON □□□□□□□ 250□□
 *   □□□□□ FabricItemSettings().maxDamage(n) □□□□□□□□□□ maxDamage □□□□□□□□□□ 1□□
 *
 * □attackSpeed □□□□
 *   attackSpeed □□□□□□□□□□□□□□□□□□□□ -2.8F□□□□□□□□□□□□□4.0F □□□□□□□□□□□
 *
 * □□□□□□□□□□□
 *   - 1.20.2~1.20.4 □□ Yarn □□□□□□□□□□□□□□□□□□□□
```

```
import net.minecraft.item.Item;
import net.minecraft.item.PickaxeItem;
import net.minecraft.item.SwordItem;
import net.minecraft.item.ToolMaterial;
import net.minecraft.item.ToolMaterials;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□/□□□□□□□□□
 */
public class ModToolItems {

    public static final String EXAMPLE_MOD_ID = "examplemod";

    // ============================================================================
    // □□□□SwordItem □□□□□□ public□□□□□□□□
    // ============================================================================
    // □□ IRON_SWORD □□□□□□new SwordItem(ToolMaterials.IRON, 3, -2.4F, new Item.Settings())
    // attackDamage □ int□□□□□□ = □□□□□(1) + attackDamage + □□□□□□□□□□ 3 + IRON □□□□□□□□□
    public static final Item EXAMPLE_SWORD = register(
            "example_sword",
            new SwordItem(
                    ToolMaterials.IRON, // □□□□□□WOOD/STONE/IRON/DIAMOND/GOLD/NETHERITE
                    3,                  // attackDamage□int □□□□
                    -2.4F,              // attackSpeed□float□
                    new FabricItemSettings()
                    // □□□□□□□□□ IRON □□□□250□□□□□□□□□□□□.maxDamage(512)
            )
    );

    // ============================================================================
    // □□□□PickaxeItem □□□□□□ protected□□□□□□□□
    // ============================================================================
    // □□ IRON_PICKAXE □□□□□□(ToolMaterials.IRON, 1, -2.8F, settings)
    public static final Item EXAMPLE_PICKAXE = register(
            "example_pickaxe",
            new CustomPickaxeItem(
                    ToolMaterials.IRON,
                    1,        // attackDamage□int □□□□
                    -2.8F,    // attackSpeed
                    new FabricItemSettings()
            )
    );

    // ============================================================================
    // □□□□AxeItem □□□□□□ protected□□□□□□□□
    // ============================================================================
    // □□ IRON_AXE □□□□□□(ToolMaterials.IRON, 6.0F, -3.0F, settings)
    // □□□□□□ attackDamage □ float □□□□□□□/□□ int □□□□ 6□int□□□□□□ float□
    // □□□□□□□□□□ float □□□□□□□ 6.0F □□□□□□
    public static final Item EXAMPLE_AXE = register(
            "example_axe",
            new CustomAxeItem(
                    ToolMaterials.IRON,
                    6.0F,     // attackDamage□float □□□□
                    -3.0F,    // attackSpeed
                    new FabricItemSettings()
            )
    );

    // ============================================================================
    // □□□□□□ public □□□□□□□□ super(...) □□ protected □□□□□□□□
    // ============================================================================

    /**
     * □□□□□□□□ PickaxeItem□□ protected □□□□“□□□”□
     * □□□□□□□□□□□□□(ToolMaterial, int attackDamage, float attackSpeed, Item.Settings)
     */
    public static class CustomPickaxeItem extends PickaxeItem {
        public CustomPickaxeItem(ToolMaterial material, int attackDamage, float attackSpeed, Settings settings) {
            // super □□ protected PickaxeItem(...) —— □□□□□□□□ super □□□□□□□
            super(material, attackDamage, attackSpeed, settings);
        }
    }
```

```
    /**
     * □□□□□□□□ AxeItem□□ protected □□□□□“□□□”□
     * □□□□□□□□□□□□(ToolMaterial, float attackDamage, float attackSpeed, Item.Settings)
     * □□ attackDamage □ float□□□□ int □□□□□□□□□
     */
    public static class CustomAxeItem extends AxeItem {
        public CustomAxeItem(ToolMaterial material, float attackDamage, float attackSpeed, Settings settings) {
            super(material, attackDamage, attackSpeed, settings);
        }
    }

    /**
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□/□□□□□□□□□
     *
     *   public static class CustomSwordItem extends SwordItem {
     *       public CustomSwordItem(ToolMaterial m, int attackDamage, float attackSpeed, Settings s) {
     *           super(m, attackDamage, attackSpeed, s);
     *       }
     *   }
     *
     * □□□ SwordItem □□□□□□□□ public□□□ new SwordItem(...) □□□□□□□□□□□□
     */

    // ============================================================================
    // □□□□□□□
    // ============================================================================
    public static Item register(String name, Item item) {
        return Registry.register(Registries.ITEM, new Identifier(EXAMPLE_MOD_ID, name), item);
    }

    /**
     * □□□□□□□ onInitialize() □□□□□□□□□□□□□□□□□□□□□□
     */
    public static void registerToolItems() {
        @SuppressWarnings("unused")
        Item ensureClassInit = EXAMPLE_SWORD;
    }

    /*
     * ============================================================================
     * □□□□□□□ ToolMaterial□□□ ToolMaterial □□□
     * ============================================================================
     * □ ToolMaterials □□□□□□□□□□□□□□□□□□□□
     *
     *   public class ExampleToolMaterial implements ToolMaterial {
     *       public static final ExampleToolMaterial INSTANCE = new ExampleToolMaterial();
     *
     *       @Override public int getDurability()             { return 512; }          // □□□
     *       @Override public float getMiningSpeedMultiplier() { return 6.0F; }         // □□□□□□□
     *       @Override public float getAttackDamage()          { return 2.0F; }         // □□□□□□□□□□□□□□□□
     *       @Override public int getMiningLevel()             { return 2; }          // □□□□□□□□□□/□□□□□□□□
     *       @Override public int getEnchantability()          { return 14; }          // □□□□
     *       @Override public Ingredient getRepairIngredient() { return Ingredient.of(Items.IRON_INGOT); }
     *   }
     *
     *   □□□□□□□ ToolMaterials.IRON □□□ ExampleToolMaterial.INSTANCE □□□
     * □□□1.20.1 □ ToolMaterial □□□□□□□□□□ Yarn □□□□□getDurability / getMiningLevel □□□
     * ============================================================================
     */

    private ModToolItems() {
    }
}
```

mixin/inject_mixin.java

```
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;

/**
 * @Inject Mixin □□□□□□ - Fabric 1.20.1
```

```
 *
 * □□□□□□□□□□ @Inject □□□□□□□□□□□□□□□
 *
 * @Inject □□□□□:
 * - □□□□□□□□□□□□□HEAD, RETURN, TAIL □□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□ CallbackInfo □ CallbackInfoReturnable □□
 *
 * □□□□□□□□@At □□:
 * - "HEAD"         □□□□□
 * - "RETURN"       □□□□□□□□□ return □□□□□□
 * - "TAIL"         □□□□□
 * - "INVOKE"       □□□□□□□□□□□□□ target □□□□□□
 * - "INVOKE_ASSIGN" □□□□□□□□□□□□
 * - "FIELD_ACCESS" □□□□□□□
 *
 * □□□□□:
 * 1. □□□□□□□ mixin □□□□□□□□□□□
 * 2. @Mixin □□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ CallbackInfo□
 * 4. □□□□□□□□□□□□□□□□ CallbackInfoReturnable<T> □□ CallbackInfo
 * 5. CallbackInfo.cancel() □□□□□□□□□□□□□□ RETURN/HEAD □□□□□□□
 * 6. CallbackInfoReturnable.setReturnValue() □□□□□□□□
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     *
     * @param ci □□□□□□□
     *           - ci.cancel(): □□□□□□□□
     *           - □□: HEAD □□ cancel □□□□□□□□□□□□
     */
    @Inject(
        method = "tick",           // □□□□□□□□□□□□□□ SRG □□
        at = @At("HEAD"),          // □□□□: □□□□□
        cancellable = true         // □□□□□□□□□□□□□□ cancel □□□ true□
    )
    private void onTickStart(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□□□□□□
        // □□: method □ private□□□□□□□□□□□□□

        // □□: □□□□□□□□□□ tick
        // if (someCondition) {
        //     ci.cancel();  // □□□□□□□□
        // }
    }

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     * □□ TAIL □□□□□□□□□□□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□
        // □□□□□□□□□□□□□□□□□□
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□ CallbackInfoReturnable □□□□□□□□□□□□
     *
     * @param cir □□□□□□□□□□□□□□□□
     */
    @Inject(
        method = "getCurrentFps",   // □□□□□□
        at = @At("RETURN"),         // □□□□□□□□
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // □□□□□□□□:
```

```
        // cir.setReturnValue(cir.getReturnValue() * 2);  // □□□□□
        // cir.cancel();  // □□□□□□□□□□□□□□□□□□□
    }
}
```

mixin/mixin_config.json

```
{
    // ============================================================
    // Mixin □□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□: src/main/resources/examplemod.mixins.json
    //
    // □□□□ fabric.mod.json □□□ "mixins" □□□□:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // □□□□:
    // - required:           □□□□□true=□□□□□□false=□□□
    // - package:            Mixin □□□□□□□
    // - compatibilityLevel: Java□□□□□1.20.1 □□ JAVA_17□
    // - mixins:             □□Mixin□□□□□□□□□□□□□□□
    // - client:             □□□□□□□Mixin□
    // - server:              □□□□□□□Mixin□□□□□
    // - injectors:           □□□□□
    //   - defaultRequire:    □□□□□□□1=□□□□□□□0=□□□
    //
    // □□□□:
    // 1. compatibilityLevel □□□ JAVA_17□1.20.1 □□ Java 17+□
    // 2. package □□□□□Mixin□□□□□□
    // 3. mixins □□□□□□ .java □□□□□□□
    // 4. client □□□□□□□□□□□□□□□□□
    // 5. □□□□ Mixin □□□□□□□□□□□ fabric.mod.json □□
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_17",

    // □□Mixin□□□□□□□□□□□□□
    "mixins": [
        "ExampleInjectMixin",
        "ExampleModifyArgMixin"
    ],

    // □□□□□□□Mixin□
    "client": [
        "client.ExampleClientMixin"
    ],

    // □□□□□□□Mixin□□□□□□□□□□□□□
    // "server": [
    //     "server.ExampleServerMixin"
    // ],

    // □□□□□
    "injectors": {
        // defaultRequire: 1 □□□□ @Inject □□□□□□□□□
        // □□□□□□□□□□□□□□□□
        // □□ 0 □□□□□□□□□□□□□□□□□□bug□
        "defaultRequire": 1
    }
}
```

mixin/modify_arg_mixin.java

```
package com.example.mod.mixin;

import net.minecraft.client.gui.screen.TitleScreen;
import net.minecraft.client.util.math.MatrixStack;
import net.minecraft.text.Text;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.ModifyArg;

/**
```

```
 * @ModifyArg Mixin □□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□ @ModifyArg □□□□□□□□□□□□□□□□□□□□
 *
 * @ModifyArg □□□□□:
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □ @Inject □□□@ModifyArg □□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * @ModifyArg □□□□□:
 * - method:   □□□□□□□□□ @ModifyArg □□□□□
 * - at:        □□□□□□□□□□ INVOKE□□□□□□□□□□□□□□□
 *   - value:   "INVOKE" □□□□□□□□□
 *   - target:  □□□□□□□□□□□□□ JVM □□□□□□□
 * - index:     □□□□□□□□□□□ 0 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - name:      □□□□□□□□□□□□
 *
 * JVM □□□□□□□□□□:
 *   Lnet/minecraft/client/font/TextRenderer;drawWithShadow(
 *       Lnet/minecraft/client/util/math/MatrixStack;  <- □□0
 *       Lnet/minecraft/text/Text;                      <- □□1
 *       F                                              <- □□2 (float)
 *       F                                              <- □□3 (float)
 *       I                                              <- □□4 (int, □□)
 *   )I                                                 <- □□□□ int
 *
 * □□□□□:
 * 1. target □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 4. index □ 0 □□□□□□□□□□□□□□□□□□□□□□□□
 * 5. □□□□□□□□□□□□□□□□□□□□□□□□ index □ target □□□□□
 */
@Mixin(TitleScreen.class)
public class ExampleModifyArgMixin {

    /**
     * □□1: □□ TitleScreen □□□□□□□□ x □□
     *
     * □□: TitleScreen.render() □□□□□□□□□ TextRenderer.drawWithShadow()
     * □□: □ 3 □□□□□□index=2□□ x □□□□float □□□□
     */
    @ModifyArg(
        method = "render",   // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        at = @At(
            value = "INVOKE",
            // target □□: "L□□□□□;□□□□(□□□□□□□)□□□□□□□□"
            target = "Lnet/minecraft/client/font/TextRenderer;drawWithShadow(Lnet/minecraft/client/util/math/MatrixStack;Lnet/minecraft/text/Text;FFI)I"
        ),
        index = 2   // □□□□ 3 □□□□□□x □□□□float □□□□□□□□□0□□□□
    )
    private float modifyTextX(float originalX) {
        // □□: originalX □□□□□ x □□□□
        // □□: □□□□□ x □□□□
        return originalX + 10.0f;  // □□□□□ 10 □□
    }

    /**
     * □□2: □□□□□□□□□□□□□□□□
     *
     * □□ TextRenderer.drawWithShadow() □□□□□□□□int □□□□
     */
    @ModifyArg(
        method = "render",
        at = @At(
            value = "INVOKE",
            target = "Lnet/minecraft/client/font/TextRenderer;drawWithShadow(Lnet/minecraft/client/util/math/MatrixStack;Lnet/minecraft/text/Text;FFI)I"
        ),
        index = 4   // □□□□ 5 □□□□□□□□□□□□int □□□□
    )
    private int modifyTextColor(int originalColor) {
        // □□: originalColor □□□□□□□□□ARGB □□□□
        // □□: □□□□□□□□□□
        return 0xFF0000;  // □□□□□□□
    }
    /**
```

```
     * 方式3: 修改传入的 Text 对象内容
     *
     * 适用于 TextRenderer.draw() 方法调用
     */
    @ModifyArg(
        method = "render",
        at = @At(
            value = "INVOKE",
            // 目标: draw 方法的 drawWithShadow 重载版本
            target = "Lnet/minecraft/client/font/TextRenderer;draw(Lnet/minecraft/client/util/math/MatrixStack;Lnet/minecraft/text/Text;FFI)I"
        ),
        index = 1   // 第二个 (索引 2) 参数是Text 对象
    )
    private Text modifyTextContent(Text originalText) {
        // 参数: originalText 是原始的文本对象
        // 返回: 修改后的文本对象
        return Text.literal("[修改后的内容]");
    }
}
```

project/README.md

```
# Fabric 1.20.1 模组开发项目

这是一个使用 **Fabric 1.20.1 官方模板**，由 AI 辅助生成的模组开发项目。
支持 API 调用，适用于 Minecraft 1.20.1 + Fabric 环境。

## 项目结构

```
project/
├── build.gradle                                # Gradle 构建脚本（Groovy DSL）
├── settings.gradle                             # Gradle 设置文件（插件 + 项目名）
├── gradle.properties                           # 模组版本与依赖版本
├── LICENSE                                     # 开源协议（默认 MIT，build.gradle 会打入 jar）
├── README.md                                   # 项目说明
└── src/main/
    ├── resources/
    │   └── fabric.mod.json                     # 模组元数据（入口点、依赖、版本等）
    └── java/com/example/mod/
        ├── ModMain.java                        # 通用（main）初始化入口
        ├── ModClient.java                      # 客户端（client）
        └── ModServer.java                      # 专用服务端（server）
```

文件说明

文件	说明
`build.gradle`	配置 Loom 插件、Minecraft/Yarn/Loader/Fabric API 依赖、Java 17 编译选项，将 `mod_version` 写入 `fabric.mod.json` 的 `${version}` 占位符
`settings.gradle`	配置 `pluginManagement` 插件仓库（Fabric Maven 与 `fabric-loom`）及项目 `rootProject.name`（影响输出 jar 文件名）
`gradle.properties`	集中管理 `mod_version`、`maven_group`、`minecraft_version`、`yarn_mappings`、`loader_version`、`fabric_version` 等版本号
`fabric.mod.json`	模组元数据（id、版本、名称、入口点 entrypoints（main/client/server）、依赖（fabricloader / fabric-api / minecraft / java）等）
`ModMain.java`	实现 `ModInitializer` 接口，在 `onInitialize()` 中输出日志，作为模组通用初始化入口
`ModClient.java`	实现 `ClientModInitializer` 接口，在 `onInitializeClient()` 中输出日志，仅在客户端环境加载
`ModServer.java`	实现 `DedicatedServerModInitializer` 接口，在 `onInitializeServer()` 中输出日志，仅在专用服务端加载
`LICENSE`	开源协议文本，默认 MIT，构建时会被打入 jar 包

默认版本（对应 1.20.1）

项目	版本
Minecraft	`1.20.1`
Yarn Mappings	`1.20.1+build.10`
Fabric Loader	`0.16.9`
Fabric API	`0.83.0+1.20.1`
Loom（Gradle 插件）	`1.6-SNAPSHOT`
Java 编译版本	`17`
最低 Gradle 版本	`8.6` 及以上

构建项目

> 本项目未包含 Gradle Wrapper，请先在本机安装 Gradle 并生成 wrapper。

```bash
cd project
```
```

```
gradle wrapper --gradle-version 8.6      # □□ ./gradlew□□□□□□□□□ Gradle□
./gradlew build                          # □□□□□□
./gradlew runClient                      # □□□□□□□□
./gradlew runServer                      # □□□□□□□□
```

□□□□□□□□**JDK 17** □□□□□ Fabric Maven / Mojang □□□□□□

□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□`examplemod` / `Example Mod` / `com.example.mod` □□□

1. **mod_id / □□ id**
   - `settings.gradle` → `rootProject.name = 'examplemod'`
   - `src/main/resources/fabric.mod.json` → `"id": "examplemod"`
   - `src/main/java/com/example/mod/ModMain.java` → `MOD_ID = "examplemod"`

2. **□□ / package**
   - □ `src/main/java/com/example/mod/` □□□□□□□□□□□□□□□□ `com/yourname/yourmod/`□
   - □□ Java □□□□□ `package com.example.mod;` □□□□□
   - `gradle.properties` → `maven_group = com.example.mod`
   - `fabric.mod.json` □□□ entrypoint □□□□□□□□□□□□

3. **□□□□□□□□**
   - `fabric.mod.json` → `"name"`□`"authors"`□`"description"`□`"contact"`

4. **□□□**
   - `gradle.properties` → `mod_version`□□□□□□□□ `fabric.mod.json` □ `"version": "${version}"`□

5. **□□□**
   - `fabric.mod.json` → `"license"`□□□□□ `LICENSE` □□□□

6. **□□□□□□□**
   - □ `src/main/resources/assets/<mod_id>/icon.png` □□□□□□□□ 128×128 □ 512×512□
   - □ `fabric.mod.json` □□□□□ `"icon": "assets/<mod_id>/icon.png"`

□□ settings.gradle □□ `fabricgeorge`

□□□□□□ “□□□□□ fabric-loom □ fabricgeorge”□□□□□□

- `fabric-loom` □ Fabric □□ Gradle □□□□□□□ Fabric Maven□□□□□□□□□□□
- `fabricgeorge` □ Fabric □□□□□ Gradle □□□□□□□□□□□□□□□□□□□□□□/□□□□□□
  □□□□□□ **100% □□□□**□settings.gradle □□□□□□□□□□□□Fabric Maven / Maven Central / Gradle Plugin Portal□□□□□□□ `fabricgeorge` □□□□□□□——□□□□□□□□□□/□□□ `fabricgeorge` □□□□□□□□□□□□□□□ URL □□□

□□ Mixin

□□□□□□□□□ Mixin□`fabric.mod.json` □□□ `mixins` □□□□□□□□□□□□□□□□□□□□□□□□
□□□□ Mixin□□□□□
1. □ `fabric.mod.json` □□ `"mixins": ["<mod_id>.mixins.json"]`
2. □□ `<mod_id>.mixins.json` □□□□
3. □□ Mixin □□□□□□□□□□Loom □□□□□ refmap□

project/build.gradle

```
// ============================================================================
// build.gradle —— Fabric 1.20.1 □□□□□□□□□□Groovy DSL□
// ============================================================================
// □□□
//  - □□□□□□□ Fabric □□ example mod □□□□□□□□□□ 1.20.1 □□□□□□
//  - □□□□□□□□□ gradle.properties □□□□□□□□□□
//  - □□□□□□□□JDK 17 + Gradle 8.x□□□ 8.6 □□□□□□
// ============================================================================

plugins {
    // Fabric Loom□Fabric □□□□□ Gradle □□□□□□□□ Minecraft □□□
    // Yarn □□□□□□□□□runClient/runServer□□remap □□□□□□□□□□□
    // 1.6-SNAPSHOT □□□ settings.gradle □□□□ Fabric Maven □□□
    id 'fabric-loom' version '1.6-SNAPSHOT'
    // maven-publish□□□□□□□□□□□□ Maven □□□□□□□□□□□□□ publishing □□□
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
```

```
    // □□□□□ jar □□□□□□□□□□ Gradle □ project.name□□□□□□□□
    // project.name □ settings.gradle □ rootProject.name □□□□□□□□ examplemod□
    archivesName = project.name
}

repositories {
    // Loom □□□□□□ Fabric□Mojang □ Maven Central □□□□□□
    // □□□□□□□□□□□/□□□□□ Mod Menu□Ribbon □□□□□□□□□□□□□ maven □□□□□□□
    // maven { url = 'https://maven.terraformersmc.com/' }
    mavenCentral()
}

// ------------------------------------------------------------------------------
// □□□□□□□□□□□ gradle.properties
// ------------------------------------------------------------------------------
dependencies {
    // Minecraft □□□ jar
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn □□□□□□□ :v2 □□□□□□□□□□□□□□/□□□□
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader□□□□□□□□
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
    // Fabric API□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
}

// ------------------------------------------------------------------------------
// □□□□□□□ gradle.properties □□ mod_version □□□ fabric.mod.json
// ------------------------------------------------------------------------------
processResources {
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    inputs.property "version", project.version

    // □ fabric.mod.json □□ ${version} □□□□□□□□□□□□□
    filesMatching("fabric.mod.json") {
        expand "version": project.version
    }
}

// ------------------------------------------------------------------------------
// □□□□□□Minecraft 1.20.1 □□ Java 17
// ------------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}

java {
    // □□ sources jar□□□□□□□□□
    withSourcesJar()

    sourceCompatibility = JavaVersion.VERSION_17
    targetCompatibility = JavaVersion.VERSION_17
}

// ------------------------------------------------------------------------------
// jar □□□□□ LICENSE □□□□□ jar □□□□□□□□□□□
// ------------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/settings.gradle

```
// ==============================================================================
// settings.gradle —— Gradle □□□□□
// ==============================================================================
// □□□□□□□□□□□□□□□□
// □□□□□□□□□□□□□□ settings.gradle □ pluginManagement □□□□□
//       □□ Gradle □□□□□ fabric-loom □□□□□□
// ==============================================================================

pluginManagement {
    repositories {
        // Fabric □□ Maven □□□fabric-loom □□□□□□ SNAPSHOT □□□□□□□□□□□□□□□
```

```
        maven {
            name = 'Fabric'
            url = 'https://maven.fabricmc.net/'
        }
        // Maven Central□□□□□□□
        mavenCentral()
        // Gradle □□□□□□□□□□ maven-publish □□□□□□
        gradlePluginPortal()

        // ------------------------------------------------------------------------
        // □□ fabricgeorge□
        // □□□□"fabricgeorge" □□ Fabric □□□□□□□□□ Gradle □□□□□□ Maven □□
        // □□ Fabric □□□□□□ Gradle □□□□□□□□□□□□□□□□
        // □□□□□□□□□□/□□□ fabricgeorge □□□□□□□□□□□□□□□□□□□□□□ URL□
        // maven {
        //     name = 'fabricgeorge'
        //     url = '<your-fabricgeorge-repo-url>'
        // }
        // ------------------------------------------------------------------------
    }
}

// □□□□□□□□□□□□□ jar □□examplemod-<version>.jar□
// □□ mod_id □□□□□□□□□□ rootProject.name □ fabric.mod.json □ id □□
rootProject.name = 'examplemod'
```

recipe/shaped_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□: data/examplemod/recipes/shaped_recipe.json
    //
    // □□□□□□crafting_shaped□□□□□□ pattern □ key □□
    // □□□□□□□□□□□□□ pattern □□□□□□
    //
    // □□□□□:
    // 1. pattern □□□□□□□□□□□□
    // 2. pattern □□□□□□□□□□□ key □□□□□□□
    // 3. key □□□□□□□□□□□□□□□□□□□□□□ #, X, *, @ □□
    // 4. □□□□□□□□□□□□□□
    // 5. result □□ count □□□□□ 64
    // 6. ingredient □□ tag □□□□ "tag" □□□□□ "item"
    // ============================================================

    "type": "minecraft:crafting_shaped",

    // pattern: □□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/shapeless_recipe.json

```
{
```

```
    // ============================================================
    // □□□□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□□□□□crafting_shapeless□□□□□□□□□□□□
    // □□□ ingredients □□□□□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. ingredients □□□□□□□□□□□□□□□□□
    // 2. □□□□□□□□□□□□□□□□□□□□□□□□□□
    // 3. □□□ "tag" □□□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/smelting_recipe.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/recipes/smelting_recipe.json
    //
    // □□□□□□smelting□□□□□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. cookingtime □□□□ tick□20 tick = 1 □□
    // 2. □□□□□□□ cookingtime □ 200□10□□
    // 3. experience □□□□□□□□□□□□□□□□□□□□□
    // 4. ingredient □□□□□□□□□□□ tag □□□□□□□
    // 5. result □□□□□□□□□□□□□□□□□□□□□□ID
    // ============================================================

    "type": "minecraft:smelting",

    // ingredient: □□□□
    // □□□□ tag □□□□□□□
    "ingredient": {
       "item": "minecraft:raw_iron"
    },

    // result: □□□□□□□□□□□□□□□□□□□□□□□□
    "result": "minecraft:iron_ingot",

    // experience: □□□□□□□□□
    "experience": 0.7,

    // cookingtime: □□□□□□□□tick□□200 = 10□
    "cookingtime": 200
}
```

recipe/smithing_recipe.json

```
{
```

```
    // ============================================================
    // □□□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/recipes/smithing_recipe.json
    //
    // □□□□□1.20.1 □□□□□□□□□□□□□□□□□□□□□
    //
    // □□□1.19 □□□□□□□□ "minecraft:smithing" □□□
    // 1.20+ □□□□□□□□□□□□□□□□□□□□□□□□□
    //
    // 1. minecraft:smithing_transform - □□□□□□□□□□->□□□□□
    // 2. minecraft:smithing_trim       - □□□□□□□□□□□□
    //
    // smithing_transform □□□□□□□□:
    //   - template:  □□□□□□□ netherite_upgrade_smithing_template□
    //   - base:      □□□□□□□□□□□□
    //   - addition:  □□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. 1.20+ □□□□□□□□□ "minecraft:smithing" □□
    // 2. template □□□□□□□□1.20 □□□
    // 3. result □□□□□□□□□□□
    // ============================================================

    "type": "minecraft:smithing_transform",

    // template: □□□□□
    // 1.20 □□□□□□□□□□□
    "template": {
        "item": "minecraft:netherite_upgrade_smithing_template"
    },

    // base: □□□□□□□□□□□□□□□□
    "base": {
        "item": "minecraft:diamond_sword"
    },

    // addition: □□□□□
    "addition": {
        "item": "minecraft:netherite_ingot"
    },

    // result: □□□□□
    "result": "minecraft:netherite_sword"
}
```

recipe/stonecutting_recipe.json

```
{
    // ============================================================
    // □□□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/recipes/stonecutting_recipe.json
    //
    // □□□□□□□□stonecutting□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□□□ ingredient□□□□□□□□
    // 2. ingredient □□□□ item □ tag
    // 3. count □□□□□1□□□□□□□□□□□□□□□□□□□□
    // 4. result □□□□□□□□
    // 5. □□□□□□□ cookingtime □ experience
    // ============================================================

    "type": "minecraft:stonecutting",

    // ingredient: □□□□□
    "ingredient": {
        "item": "minecraft:stone"
    },

    // result: □□□□□
    "result": "minecraft:stone_bricks",

    // count: □□□□□□□□□□□□
```

```
    "count": 1
}
```

tag/block_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□
    //   minecraft:overworld_natural_logs □□□□□□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□: data/<mod_id>/tags/items/□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/entity_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/tags/entity_types/example_entity_tag.json
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:skeletons     □□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:raiders       □□□□
    //   minecraft:beehives      □□□□□□□
    //   minecraft:arrows         □□□□□
    //   minecraft:impact_projectiles □□□□□□□
    //   minecraft:fall_damage_immune  □□□□□□□
    //   minecraft:freezable_hurt_type □□□□□□□□□□
    //   minecraft:freeze_immune_entity_type □□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□: data/<mod_id>/tags/entity_types/
    // 2. □□ID□□□□□□□□□□
    // 3. □□□□□□□□□□□ EntityType.getTag() □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:zombie",
        "minecraft:skeleton",
        "examplemod:example_entity"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
```

```
    // □□□□□□□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/tags/items/example_item_tag.json
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:logs          □□
    //   minecraft:planks        □□
    //   minecraft:coals         □□□
    //   minecraft:saplings      □□
    //   minecraft:flowers       □
    //
    // □□□□□:
    // 1. □□□□□□ data/<mod_id>/tags/items/ □□□
    // 2. □□□□□□□□□□□□□□□□□□
    //    □□: data/<mod_id>/tags/items/
    //    □□: data/<mod_id>/tags/blocks/
    // 3. values □□□□ID□□□□□□□□□
    // 4. replace □ true □□□□□□□□□□□□□□□□□□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
    ]
}
```

worldgen/biome_modifier.json

```
{
    // ============================================================
    // □□□□□□□/□□□□□ JSON □□ - Fabric 1.20.1
    // ============================================================
    // □□□□□: data/examplemod/worldgen/placed_feature/ore_example.json
    //
    // □□□□Fabric 1.20.1 □□□□□□□□□□□□□□□ API
    // □BiomeModifications.addFeature□□□□□□ JSON □□□
    // □ JSON □□□□□□□□□□ PlacedFeature □□□
    // □□□ ore_feature.java □□□□□□□□□□□
    //
    // □□□□□□□□ vs □□□□□:
    // - □□□□□: □□ Registry.register + BiomeModifications API
    // - □□□□□: □□ data □□□□ JSON □□□□□
    //   □□ mixin □ Loom □ datagen □□
    //
    // PlacedFeature JSON □□:
    // - feature:    □□□ ConfiguredFeature ID
    // - placement:  □□□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. feature □□□□□□ ConfiguredFeature □□□□□□
    //    □□ data/<mod_id>/worldgen/configured_feature/ □□□□□
    // 2. placement □□ type □□□□□□ minecraft: □□
    // 3. □□□□□□□□□□□□□□□□ type □□□
    // 4. □ JSON □□□□□□□□□□□ datagen □ Mixin □□
    // ============================================================

    // □□□ ConfiguredFeature ID
    "feature": "examplemod:ore_example",

    // □□□□□□□□□□□□□□□□□□
    "placement": [
        // 1. □□□□□□: □□□□□□□□□□□□
        {
            "type": "minecraft:count",
            "count": 20
        },

        // 2. □□□□□□: □□□□□□□□□□□□
        {
            "type": "minecraft:in_square"
        },
        // 3. □□□□□: □□□□□□□□□□
```

```
        {
            "type": "minecraft:height_range",
            "height": {
                // □□□□
                "type": "minecraft:uniform",
                // □□ Y □□
                "min_inclusive": {
                    "absolute": -64
                },
                // □□ Y □□
                "max_inclusive": {
                    "absolute": 64
                }
            }
        },

        // 4. □□□□□□□: □□□□□□□□□□□□□□
        {
            "type": "minecraft:biome"
        }
    ]
}

// ================================================================
// □□: ConfiguredFeature JSON □□
// ================================================================
// □□□□: data/examplemod/worldgen/configured_feature/ore_example.json
//
// {
//     "type": "minecraft:ore",
//     "config": {
//         "size": 9,
//         "targets": [
//             {
//                 "target": {
//                     "predicate_type": "minecraft:tag_match",
//                     "tag": "minecraft:stone_ore_replaceables"
//                 },
//                 "state": {
//                     "Name": "examplemod:example_ore"
//                 }
//             },
//             {
//                 "target": {
//                     "predicate_type": "minecraft:tag_match",
//                     "tag": "minecraft:deepslate_ore_replaceables"
//                 },
//                 "state": {
//                     "Name": "examplemod:deepslate_example_ore"
//                 }
//             }
//         ]
//     }
// }
// ================================================================
```

worldgen/ore_feature.java

```
package com.example.mod.worldgen;

import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.registry.entry.RegistryEntry;
import net.minecraft.util.Identifier;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;
import net.minecraft.world.gen.YOffset;
```

```java
import java.util.List;

/**
 * □□□□□□□□□□ - Fabric 1.20.1
 *
 * □□□□□□□□□□
 * 1. □□□□□□ ConfiguredFeature□□□□□□□
 * 2. □□□□□□ PlacedFeature□□□□□□□
 * 3. □□ BiomeModifications □□□□□□□□□□□□
 *
 * □□□□□1.20.1 □□□□□□□□□□□□□
 *   - ConfiguredFeature: □□"□□□□□"□□□□□□□□□□□□□□□
 *   - PlacedFeature: □□"□□□□□□"□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□:
 *   ConfiguredFeature -> □□□□ Registries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□□ Registries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□□
 *
 * □□□□□:
 * - □□□□□□ ConfiguredFeature□□□□□ PlacedFeature
 * - PlacedFeature □□ RegistryEntry<ConfiguredFeature> □□□□□□□□□□ ConfiguredFeature
 * - BiomeModifications.addFeature □□ RegistryKey<PlacedFeature> □□ PlacedFeature □□
 * - OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES □□□□□□□□□
 *   □□□□□ NETHERRACK□□□□□□ ENDSTONE
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□□□
 * - □□□□□(size)□□□□ 64□□□ 3~12
 */
public class ModWorldGen {

    // ConfiguredFeature □ RegistryKey
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, new Identifier("examplemod", "ore_example"));

    // PlacedFeature □ RegistryKey
    public static final RegistryKey<net.minecraft.world.gen.feature.PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, new Identifier("examplemod", "ore_example"));

    /**
     * □□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□□
     */
    public static void register() {
        // ================================================================
        // 1. □□□□□□ ConfiguredFeature
        // ================================================================
        // □□□□□□□□□□
        // □□1: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        // □□2: □□□□□□□ BlockState
        // □□3: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            // □□□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
            // □□: ModBlocks.EXAMPLE_ORE □□□□□□□□□□□□□□□□□□□□
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),  // □□□□□□□□□□
            9  // □□□□□□
        );

        // □□ ConfiguredFeature □□
        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,    // □□□□□□: □□
            oreConfig        // □□□□□□
        );

        // □□□□ Registries.CONFIGURED_FEATURE
        Registry.register(
            Registries.CONFIGURED_FEATURE,
            new Identifier("examplemod", "ore_example"),
            configuredFeature
        );

        // ================================================================
        // 2. □□□□□□ PlacedFeature
        // ================================================================
        // PlacedFeature □□ RegistryEntry<ConfiguredFeature> □□
```

```
        // 获取 RegistryEntry.of() 需要的是 ConfiguredFeature
        RegistryEntry<ConfiguredFeature<?, ?>> cfEntry = RegistryEntry.of(configuredFeature);

        // 创建放置修饰器列表（控制生成频率和位置）
        List<net.minecraft.world.gen.placementmodifier.PlacementModifier> placementModifiers = List.of(
            // 数量: 每个区块尝试生成次数
            CountPlacementModifier.of(20),

            // 平方分布: 在区块内随机水平分布
            SquarePlacementModifier.of(),

            // 高度范围: 从 Y=-64 到 Y=64 均匀分布
            HeightRangePlacementModifier.uniform(
                YOffset.getBottom(),  // Y=-64（世界底部）
                YOffset.fixed(64)     // Y=64
            )
        );

        // 创建 PlacedFeature 实例
        net.minecraft.world.gen.feature.PlacedFeature placedFeature =
            new net.minecraft.world.gen.feature.PlacedFeature(
                cfEntry,            // ConfiguredFeature 条目
                placementModifiers  // 放置修饰器列表
            );

        // 注册到 Registries.PLACED_FEATURE
        Registry.register(
            Registries.PLACED_FEATURE,
            new Identifier("examplemod", "ore_example"),
            placedFeature
        );

        // ================================================================
        // 3. 使用 BiomeModifications 将特征添加到生物群系
        // ================================================================
        BiomeModifications.addFeature(
            // 生物群系选择器: 选择哪些生物群系
            BiomeSelectors.foundInOverworld(),
            // 生成步骤: 在什么阶段生成
            GenerationStep.Feature.UNDERGROUND_ORES,
            // PlacedFeature 的 RegistryKey
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                new Identifier("examplemod", "ore_example")
            )
        );
    }
}

/**
 * ================================================================
 * 生物群系选择器
 * ================================================================
 *
 * BiomeSelectors 提供了多种选择生物群系的方法:
 *
 * BiomeSelectors.all()                  选择所有
 * BiomeSelectors.foundInOverworld()      主世界生物群系
 * BiomeSelectors.foundInNether()         下界生物群系
 * BiomeSelectors.foundInTheEnd()         末地生物群系
 * BiomeSelectors.includeByKey(keys)      通过 RegistryKey 选择
 * BiomeSelectors.tag(tag)                通过标签选择
 *
 * 可以组合使用:
 * BiomeSelectors.foundInOverworld().and(context -> ...)
 * BiomeSelectors.foundInOverworld().or(BiomeSelectors.foundInNether())
 *
 * ================================================================
 * 生成步骤说明
 * ================================================================
 *
 * GenerationStep.Feature 枚举值包括:
 * - RAW_GENERATION          原始地形生成阶段
 * - LAKES                   湖泊
 * - LOCAL_MODIFICATIONS     地形局部修改阶段
 * - UNDERGROUND_STRUCTURES  地下结构
```

```
 * - SURFACE_STRUCTURES       □□□□
 * - STRONGHOLDS              □□
 * - UNDERGROUND_ORES         □□□□□□□□□□□□□□□
 * - UNDERGROUND_DECORATION   □□□□□□□□□□
 * - FLUID_SPRINGS            □□□
 * - VEGETAL_DECORATION       □□□□□□□□□□□
 * - TOP_LAYER_MODIFICATION   □□□□□□□□□
 */
```

Fabric 1.20.4

advancement/advancement.json

```
{
    // ============================================================
    // □□/□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: data/examplemod/advancements/example_advancement.json
    //
    // □□ 1.20.1 □□□□
    //   □□ JSON □□□ 1.20.4 □ 1.20.1 □□□□□□□□
    //   inventory_changed □□□□□□ "items"□□□□□□□□□□
    //
    // □□□□□□Advancement□□ Minecraft 1.12+ □□□□□□
    // □□□□□□□□□□□□□□□□□□□□□toast□□□□□□□□
    //
    // □□□□□:
    // - display:     □□□□□□□□□□□□□□□□□□□□□□□□□□
    // - parent:      □□□□□□□□□□□□□□□□□
    // - criteria:    □□□□□□□□□□□□□□□□□□□□□□□□□
    // - requirements: □□□□□□□□□□□□AND / OR□
    // - rewards:     □□□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□parent□□□□□□ display.background □□
    // 2. 1.20.4 □ inventory_changed □□□□□□□□ "items" □□□□□ 1.20.1 □□□
    // 3. criteria □□□□□□□□□□□ trigger □□
    // 4. requirements □□□□□□□□□□□ AND□□□□ OR
    // 5. frame □□: task(□□) / goal(□□) / challenge(□□)
    // ============================================================

    // □□□ID□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    "parent": "examplemod:example/root",

    // □□□□□
    "display": {
        // □□
        "icon": {
            "item": "examplemod:example_item"
        },

        // □□□□□□ translate □□□□□ literal □□□□□□
        "title": {
            "translate": "advancements.examplemod.example.title"
        },

        // □□
        "description": {
            "translate": "advancements.examplemod.example.description"
        },

        // □□□□□: task(□□,□□) / goal(□□,□□) / challenge(□□,□□)
        "frame": "task",

        // □□□□□□□□□□□□□□
        "show_toast": true,

        // □□□□□□□□□□
        "announce_to_chat": true,

        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        "hidden": false

        // □□□□□□ background □□□□□□□□□□□□
```

```
        // "background": "minecraft:textures/block/stone.png"
    },

    // □□□□□
    // □□ key □□□□□□□□□□□□□□□ trigger □ conditions
    "criteria": {
        "get_example_item": {
            // □□□: □□□□□□□□□□□
            "trigger": "minecraft:inventory_changed",
            "conditions": {
                // 1.20.4 □□ "items"□□□□□□□□□□□□ 1.20.1 □□
                // 1.19.2 □□ "item"□□□□□□□
                "items": [
                    {
                        "items": [
                            "examplemod:example_item"
                        ]
                    }
                ]
            }
        },
        "another_condition": {
            // □□□: □□□□□□□□□□□□□□□
            "trigger": "minecraft:location",
            "conditions": {
                "player": {
                    "location": {
                        "dimension": "minecraft:overworld"
                    }
                }
            }
        }
    },

    // □□□□□□□□□□□
    // □□□□□□ AND□□□□□□□
    // □□□□□□ OR□□□□□□□
    // □□□□: □□ get_example_item AND another_condition
    "requirements": [
        ["get_example_item"],
        ["another_condition"]
    ],

    // □□□□
    "rewards": {
        // □□□□□
        "experience": 10,

        // □□□□□□□□□□□□□
        "loot": [
            "examplemod:example_loot_table"
        ],

        // □□□□□□□□□□□□□
        "function": "examplemod:grant_example_reward"
    }
}
```

block/simple_block.java

```
/*
 * simple_block.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□
 *   1. □□□□□□□□□□Registry API □ 1.20.1 □□□□□□
 *   2. FabricBlockSettings API □□□□□□□
 *   3. FabricItemSettings □□□□□□ Item.Settings □□□□ Fabric □□□□□□
 *   4. □□□□□□□□□□ItemGroupEvents + ItemGroups□□ 1.20.1 □□□
 *   5. □□□□□□□□FoodComponent.Builder □ hunger() □□□□□□ nutrition()□
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □□ net.minecraft.util.registry.Registry□1.20.4 □□ net.minecraft.registry.Registry
 *   - 1.19.2 □□ .group()□1.20.4 □□□ ItemGroupEvents
```

```
 *   - 1.19.2 □ Material □□1.20.4 □□□
 * ------------------------------------------------------------------------
 */
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□□□□□□□ (Fabric 1.20.4)
 *
 * □□□□□□□□
 * 1. □□□□□□ 1.20.1 □□□□□□□□□ Registries.BLOCK / Registries.ITEM□
 * 2. FabricBlockSettings API □ 1.20.1 □□□□□□□□
 * 3. □□□□□□□□□ ItemGroupEvents □□□□□ 1.20.1 □□□□□□
 */
public class SimpleBlock extends Block {

    public static final String MOD_ID = "examplemod";

    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□□□ Block □□□□□□□□□□□□ BlockItem□
     *
     * □□ 1.20.1 □□□□□□□□□□□□□□API □□□□□□□□□
     */
    private static Block registerBlock(String name) {
        // ---- □□□□□□□□□□□□□□ ----
        // □ 1.20.1 □□□□□□ FabricBlockSettings □□
        AbstractBlock.Settings settings = FabricBlockSettings.copyOf(Blocks.STONE)
                .strength(3.0f)
                .requiresTool()
                .luminance(15);

        Block block = new SimpleBlock(settings);

        // ---- □□□□□□□ Block □ Registries.BLOCK ----
        // □ 1.20.1 □□□□□
        Registry.register(Registries.BLOCK, new Identifier(MOD_ID, name), block);

        // ---- □□□□□□□ BlockItem □ Registries.ITEM ----
        // □ 1.20.1 □□□□□
        // □□□□□□□□□ .group()□1.20.4 □□□□□□□□□□1.19.3 □□□□□
        BlockItem item = new BlockItem(block, new FabricItemSettings());

        Registry.register(Registries.ITEM, new Identifier(MOD_ID, name), item);

        return block;
    }

    // ========================================================================
    // □□□□□□□□□
    // ========================================================================
    //
    // □□□□□□□□ ModInitializer □□□□□□□□□
    //
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
```

```
    // ========================================================================
    // □□□□□□□□□□ 1.20.1 □□□□□□□
    // ========================================================================
    //
    //   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
    //   import net.minecraft.item.ItemGroups;
    //
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    //
    // ========================================================================
}
```

command/simple_command.java

```
/*
 * simple_command.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□
 *   1. CommandRegistrationCallback API □ 1.20.1 □□□□□□v2 □□□□
 *   2. CommandManager API □□
 *   3. Text API □□□Text.literal()□
 *   4. EntityArgumentType □□□□
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □□ v1 □□□□2 □□□□□□1.20.4 □□ v2 □□□3 □□□□□
 *   - 1.19.2 □ new LiteralText()□1.20.4 □ Text.literal()
 * ------------------------------------------------------------------------
 */
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.CommandManager;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.text.Text;
import net.minecraft.util.Formatting;

import static net.minecraft.server.command.CommandManager.literal;
import static net.minecraft.server.command.CommandManager.argument;

/**
 * □□□□□□□□ - Fabric 1.20.4
 *
 * □□ 1.20.1 □□□□□□□□□□□□API □□□□□□□□
 *
 * □□□□□□□□□
 * 1. □□ CommandRegistrationCallback v2 □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * Fabric 1.20.4 □□ CommandRegistrationCallback v2 □□
 * v2 □□□□□□□□□□□□□□:
 *   (CommandDispatcher<ServerCommandSource> dispatcher,
 *    CommandRegistryAccess registryAccess,
 *    CommandManager.RegistrationEnvironment environment)
 */
public class ModCommands {

    /**
     * □□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, registryAccess, environment) -> {
                registerExampleCommand(dispatcher);
```

```
        }
    );
}

/**
 * □□ /example □□□
 *
 * □□□□:
 * /example                          - □□□□□□
 * /example hello                    - □□□□□□□□
 * /example hello <player>           - □□□□□□□□□□
 * /example set <value>              - □□□□□
 * /example give <player> <count>    - □□□□□□□□
 */
private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
    dispatcher.register(
        literal("example")
            .executes(context -> {
                context.getSource().sendFeedback(
                    () -> Text.literal("□□: /example <hello|set|give>")
                        .formatted(Formatting.YELLOW),
                    false
                );
                return 1;
            })

            .then(literal("hello")
                .executes(ModCommands::sayHelloToSelf)
                .then(argument("target", EntityArgumentType.player())
                    .executes(ModCommands::sayHelloToTarget)
                )
            )

            .then(literal("set")
                .then(argument("value", IntegerArgumentType.integer(0, 100))
                    .executes(ModCommands::setValue)
                )
            )

            .then(literal("give")
                .then(argument("target", EntityArgumentType.player())
                    .then(argument("count", IntegerArgumentType.integer(1, 64))
                        .executes(ModCommands::giveItem)
                    )
                )
            )
    );
}

/**
 * /example hello - □□□□□□□
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    source.sendFeedback(
        () -> Text.literal("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;
}

/**
 * /example hello <player> - □□□□□□□□□
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    target.sendMessage(
```

```
            Text.literal("Hello from " + source.getName() + "!")
                .formatted(Formatting.GOLD)
        );

        source.sendFeedback(
            () -> Text.literal("□□ " + target.getName().getString() + " □□□□□")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }

    /**
     * /example set <value> - □□□
     */
    private static int setValue(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        int value = IntegerArgumentType.getInteger(context, "value");
        ServerCommandSource source = context.getSource();

        source.sendFeedback(
            () -> Text.literal("□□□□□□: " + value)
                .formatted(Formatting.AQUA),
            false
        );
        return 1;
    }

    /**
     * /example give <player> <count> - □□□□□□□□
     */
    private static int giveItem(CommandContext<ServerCommandSource> context)
            throws CommandSyntaxException {

        PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
        int count = IntegerArgumentType.getInteger(context, "count");
        ServerCommandSource source = context.getSource();

        net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
            net.minecraft.item.Items.DIAMOND, count
        );
        target.getInventory().insertStack(stack);

        source.sendFeedback(
            () -> Text.literal("□□ " + target.getName().getString()
                + " □□□ " + count + " □□□")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }
}
```

enchantment/custom_enchantment.java

```
/*
 * custom_enchantment.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□
 *   1. □□□□□ API □ 1.20.1 □□□□□
 *   2. □□□□ Enchantment □
 *   3. □□□□ EnchantmentTarget □□
 *   4. □□□□□□□□Registries.ENCHANTMENT□
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □ Registry.ENCHANTMENT□1.20.4 □ Registries.ENCHANTMENT
 *   - 1.19.2 □ net.minecraft.util.registry.Registry□1.20.4 □ net.minecraft.registry.Registry
 *
 * □□ 1.21.1 □□□□□□□□□□□
 *   - 1.21.1 □□□□□□□□□□□□□□□JSON□□□□□□□ Enchantment □
 *   - 1.20.4 □□□□□ Enchantment □□□□□□□□
 * ------------------------------------------------------------------------
```

```
 */
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.enchantment.EnchantmentTarget;
import net.minecraft.entity.EquipmentSlot;
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□ - Fabric 1.20.4
 *
 * □□ 1.20.1 □□□□□□□□□□□□API □□□□□□□□
 *
 * □□□□□□□□□
 * 1. □□□□□□□□□□□□ Enchantment□
 * 2. □□ Registry.register □□□ Registries.ENCHANTMENT
 * 3. □□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□:
 * - Enchantment.Rarity: □□□□□□
 *   COMMON(10) / UNCOMMON(5) / RARE(2) / VERY_RARE(1)
 *
 * - EnchantmentTarget: □□□□□□□□□□□
 *   ALL / ARMOR / FEET / LEGS / CHEST / HEAD / WEAPON / TOOL
 *   BOW / FISHING_ROD / BREAKABLE / TRIDENT / CROSSBOW / VANISHABLE
 *
 * - EquipmentSlot[]: □□□□□□□□□
 *   MAINHAND / OFFHAND / HEAD / CHEST / LEGS / FEET
 *
 * □□□□:
 * - □□□ mod_id □□□□□ mod ID
 * - getMinPower □□□□: 10 + (level - 1) * 15
 * - getMaxPower □□□ getMinPower + 15~50
 * - 1.20.4 □ canCombine() □□□□□1.21+ □□□□
 */
public class ModEnchantments {

    /**
     * □□□□□□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static final Enchantment EXAMPLE_ENCHANTMENT = Registry.register(
        Registries.ENCHANTMENT,
        new Identifier("examplemod", "example_enchantment"),
        new ExampleEnchantment(
            Enchantment.Rarity.RARE,
            EnchantmentTarget.WEAPON,
            EquipmentSlot.MAINHAND
        )
    );

    /**
     * □□□□□□□□□□□□□□□
     */
    public static void register() {
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    public static class ExampleEnchantment extends Enchantment {

        public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
            super(rarity, target, slots);
        }

        @Override
        public int getMinPower(int level) {
            return 10 + (level - 1) * 15;
        }

        @Override
        public int getMaxPower(int level) {
```

```
            return super.getMinPower(level) + 50;
        }

        @Override
        public int getMaxLevel() {
            return 3;
        }

        @Override
        public boolean isAcceptableItem(ItemStack stack) {
            return super.isAcceptableItem(stack);
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□
         * 1.20.4 □□□□□□□□□1.21+ □□□□□□□ JSON □ exclusive_set □□□
         */
        @Override
        public boolean canCombine(Enchantment other) {
            return super.canCombine(other);
        }
    }
}

/*
 * ============================================================================
 * □□□□□□□□□□
 * ============================================================================
 * □□□□□: assets/examplemod/lang/en_us.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment",
 *     "enchantment.examplemod.example_enchantment.desc": "A custom enchantment"
 * }
 *
 * ============================================================================
 * □□ ID □□□□
 * ============================================================================
 * - □□□ Identifier □□□□□
 * - □□□□□□□□: enchantment.<mod_id>.<enchantment_name>
 */
```

entity/simple_entity.java

```
/*
 * simple_entity.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□
 *   1. □□□□□□□□□□Registry API □ 1.20.1 □□□□□□
 *   2. EntityType.Builder □□□□□□
 *   3. FabricDefaultAttributeRegistry □□□□□□
 *   4. □□□□□□ API □□□□□□□ 1.20.1 □□□
 *
 * □□ 1.19.2 □□□□□□□□□□
 *   - 1.19.2 □ Registry.ENTITY_TYPE□1.20.4 □ Registries.ENTITY_TYPE
 *   - 1.19.2 □ net.minecraft.util.registry.Registry□1.20.4 □ net.minecraft.registry.Registry
 * ------------------------------------------------------------------------
 */
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.world.World;

/**
 * □□□□□□□□□ - Fabric 1.20.4
```

```
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□API □□□□□□□□
 *
 * □□□□□□□□:
 * - □□□□□□□□□□□□□□□□ (EntityType, World) □□
 * - EntityType.Builder □ dimensions □□□□□□"□□□□□□"□
 *   □□ (0.6f, 1.8f) □□ 0.6x1.8 □□□□□
 * - createMobAttributes() □□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□□ FabricDefaultAttributeRegistry
 */
public class ExampleEntity extends PathAwareEntity {

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□ Registry.register □□□□ Registries.ENTITY_TYPE
    // □□: mod_id □□□□□□□□□□ mod ID
    public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
        Registry.register(
            Registries.ENTITY_TYPE,
            new Identifier("examplemod", "example_entity"),
            EntityType.Builder.<ExampleEntity>create(
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    ExampleEntity::new,
                    // □□□□□□□□□□□□□□□□□□□□□□□□
                    SpawnGroup.CREATURE
                )
                // □□□□□□□□□□□□x□□□□□□□□□
                .dimensions(0.6f, 1.8f)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ 5
                .trackingRange(8)
                // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ false□
                .build(null)
        );

    // ================================================================
    // □□□□□
    // ================================================================
    public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
        super(entityType, world);
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□□□□□:
    // - GENERIC_MAX_HEALTH      □□□□□□□□□□ 20 = 10 □□□□
    // - GENERIC_MOVEMENT_SPEED  □□□□□□□□□□ 0.25□
    // - GENERIC_ATTACK_DAMAGE   □□□□□□□□□□ 2□
    // - GENERIC_FOLLOW_RANGE     AI □□□□□□□□□□ 16□
    // - GENERIC_ARMOR           □□□□
    // - GENERIC_KNOCKBACK_RESISTANCE □□□□□□0.0~1.0□
    public static DefaultAttributeContainer createMobAttributes() {
        return PathAwareEntity.createMobAttributes()
            .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)
            .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)
            .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)
            .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)
            .add(EntityAttributes.GENERIC_ARMOR, 0.0)
            .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0);
    }

    // ================================================================
    // □□ tick □□
    // ================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□□□□□ tick □□□□
        // □□: □□□□□□□□□□□□□□□□□□□□
    }
```

```
    // ================================================================
    // □□□□ AI □□
    // ================================================================
    @Override
    protected void initGoals() {
        // □□□□□□□□□ AI □□□□Goal□
        // □□ AI:
        // - MeleeAttackGoal        □□□□□□□
        // - WanderAroundGoal       □□□□□
        // - LookAtEntityGoal       □□□□□
        // - FollowTargetGoal       □□□□□
        // - SwimGoal               □□
    }

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □□□□□□□□□ModInitializer□□ onInitialize() □□□□□
    //
    //   FabricDefaultAttributeRegistry.register(
    //       ExampleEntity.EXAMPLE_ENTITY,
    //       ExampleEntity.createMobAttributes()
    //   );
    // ================================================================
}
```

gui/screen_with_menu.java

```
/*
 * screen_with_menu.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□GUI□□□□□ ScreenHandler + HandledScreen □□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □□□□ API □□□□□□□□□
 *      1.20.1: □□ MatrixStack □□ GUI □□
 *              drawBackground(MatrixStack matrices, ...)
 *              render(MatrixStack matrices, ...)
 *      1.20.4: □□ DrawContext □□ GUI □□□□1.20.2 □□□□
 *              drawBackground(DrawContext context, ...)
 *              render(DrawContext context, ...)
 *
 *   2. □□□□□□□□□□
 *      □□□□□□□□ Registries.SCREEN_HANDLER □□ ScreenHandlerType
 *
 *   3. □□□□□□□□□□
 *      □□□□□□□□ Text.literal() / Text.translatable()
 *
 *   4. □ScreenHandlerType □□□□□□□□
 *      □□□□□□□□ new ScreenHandlerType<>(factory)
 * ------------------------------------------------------------------------------
 */
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.text.Text;
import net.minecraft.util.Identifier;

/**
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.20.4
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□GUI □□□□□ DrawContext □□ MatrixStack
 *    - drawBackground □□□□□: DrawContext□□□ MatrixStack□
 *    - render □□□□□: DrawContext□□□ MatrixStack□
 *    - drawMouseoverTooltip □□□□□: DrawContext□□□ MatrixStack□
```

```
 *    - renderBackground □□□□: DrawContext□□□ MatrixStack□
 * 2. ScreenHandler □□□□□□ 1.20.1 □□□□□□Registries.SCREEN_HANDLER□
 * 3. Text API □ 1.20.1 □□□□□□Text.literal()□
 *
 * □□□□:
 * - □□□□□□□ ScreenHandler □□□□ syncId
 * - □□□□□□□ Packet □ syncId □□□□□□□
 * - □□□□□□□ syncId □ ScreenHandlerType □□ Screen □□
 * - Screen □ ScreenHandler □□ syncId □□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ====================================================================
    // ScreenHandlerType □□
    // ====================================================================
    // □□□□□□ 1.20.1 □□□□□
    public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
        Registry.register(
            Registries.SCREEN_HANDLER,
            new Identifier("examplemod", "example_gui"),
            new ScreenHandlerType<>(ExampleScreenHandler::new)
        );

    // □□□□□
    private final Inventory inventory;

    // ====================================================================
    // □□□□□□□□
    // ====================================================================
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
        this(syncId, playerInventory, new SimpleInventory(9));
    }

    // ====================================================================
    // □□□□□□□□
    // ====================================================================
    public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
        super(EXAMPLE_SCREEN_HANDLER, syncId);
        checkSize(inventory, 9);
        this.inventory = inventory;
        inventory.onOpen(playerInventory.player);

        // □□□□□□□□
```

```
            if (!this.insertItem(originalStack, 9, 45, true)) {
                return ItemStack.EMPTY;
            }
        } else {
            if (!this.insertItem(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        if (originalStack.isEmpty()) {
            slot.setStack(ItemStack.EMPTY);
        } else {
            slot.markDirty();
        }
        return copyStack;
    }

    // ================================================================
    // □□□□□□□□□□□□□□□□
    // ================================================================
    @Override
    public boolean canUse(PlayerEntity player) {
        return this.inventory.canPlayerUse(player);
    }
}

/**
 * ================================================================
 * □□□ Screen □□HandledScreen□
 * ================================================================
 * □1.20.4 □□□□□□□□ DrawContext □□ MatrixStack □□□□
 * 1.20.1 □□ MatrixStack□1.20.2+ □□ DrawContext
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□□
     *
     * □1.20.4 □□□□□□□□□□□ MatrixStack □□ DrawContext
     * 1.20.1: drawBackground(MatrixStack matrices, float delta, int mouseX, int mouseY)
     * 1.20.4: drawBackground(DrawContext context, float delta, int mouseX, int mouseY)
     */
    @Override
    protected void drawBackground(net.minecraft.client.gui.DrawContext context, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);
        net.minecraft.client.render.RenderSystem.setShaderTexture(0,
            new Identifier("examplemod", "textures/gui/example_gui.png"));

        // □1.20.4 □□□□□□ context □□ matrices
        // DrawContext □ drawTexture □□□□ MatrixStack □□□□□□□
        int x = (this.width - this.backgroundWidth) / 2;
        int y = (this.height - this.backgroundHeight) / 2;
        context.drawTexture(new Identifier("examplemod", "textures/gui/example_gui.png"),
            x, y, 0, 0, this.backgroundWidth, this.backgroundHeight);
    }

    /**
     * □□□□□
     *
     * □1.20.4 □□□□□□□□□□□ MatrixStack □□ DrawContext
     * 1.20.1: render(MatrixStack matrices, int mouseX, int mouseY, float delta)
     * 1.20.4: render(DrawContext context, int mouseX, int mouseY, float delta)
     */
    @Override
```

```
    public void render(net.minecraft.client.gui.DrawContext context, int mouseX, int mouseY, float delta) {
        // 在1.20.4 中需要renderBackground 需要传入 MatrixStack 而是 DrawContext
        this.renderBackground(context);
        super.render(context, mouseX, mouseY, delta);
        // 在1.20.4 中需要drawMouseoverTooltip 需要传入 MatrixStack 而是 DrawContext
        this.drawMouseoverTooltip(context, mouseX, mouseY);
    }
}

/**
 * ================================================================
 * 使用说明：
 * ================================================================
 * ScreenHandlerType 需要在主类（ModInitializer）:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER 需要在主类中注册
 *
 * Screen 需要在客户端类（ClientModInitializer）:
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
 *   );
 *
 * ================================================================
 * 打开 GUI 的方式（在 BlockEntity 或 Item 中）：
 * ================================================================
 *
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       Text.literal("容器名称")
 *   ));
 */
```

item/simple_item.java

```
/*
 * simple_item.java
 * ------------------------------------------------------------------------------
 * 版本号: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn 映射)
 * 文件名: 简单物品注册（食物、普通物品）
 *
 * 适用 1.20.1 的写法请注意：
 *   1. 物品注册方式与 1.20.1 基本相同
 *   2. FabricItemSettings 已经被 Item.Settings 取代，Fabric 专用类
 *      在 1.20.4 中直接用 new Item.Settings()。
 *   3. FoodComponent.Builder 的 hunger() 方法名改为 nutrition()（1.20.2+ 变更）
 *   4. 创造模式标签用 ItemGroupEvents + ItemGroups，与 1.20.1 相同
 *   5. Identifier 构造 new Identifier() 在部分版本1.21+ 改为 Identifier.of()。
 *
 * 适用 1.19.2 的写法请注意：
 *   - 1.19.2 中是 Registry.ITEM，1.20.4 中是 Registries.ITEM
 *   - 1.19.2 中是 .group(ItemGroup.MISC)，1.20.4 中改为 ItemGroupEvents
 *   - 1.19.2 的 ItemGroup.MISC，1.20.4 改为 ItemGroups.MISC
 * ------------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * 简单物品注册类 (Fabric 1.20.4)
 *
 * 适用 1.20.1 的写法
 * 1. API 基本相同，无需改动
 * 2. 唯一区别 new Item.Settings() 而非 new FabricItemSettings()。
 *    （在 1.20.2+ 中 Item.Settings 已经被 Fabric 专用类
 *    （FabricItemSettings 取代，无需改动）
 */
public class ModItems {

    public static final String MOD_ID = "examplemod";
```

```
/**
 * □□□□□□□□□□□□□□□□
 *
 * □1.20.4 □□□□□□□□ new Item.Settings() □□ FabricItemSettings□
 * □ 1.20.1 □□□□ FabricItemSettings□□ 1.20.4 □□□□□□□Item.Settings □□□□
 *
 * □□□□□□□□□ .group()□□□□□□ 1.19.3+ □□□□□
 * □□□□□□□□□□□□□□□□□ ItemGroupEvents□□□□□□□
 */
public static final Item EXAMPLE_ITEM = register(
        "example_item",
        new Item(new Item.Settings())
        // □□□□□□□□□□□□□□□□
        // .maxCount(16)
        // .maxDamage(256)
        // .rarity(Rarity.UNCOMMON)
        // .fireproof()
);

/**
 * □□□□□□□□□□□□□□□□
 */
public static final Item EXAMPLE_DURABLE_ITEM = register(
        "example_durable_item",
        new Item(new Item.Settings().maxDamage(256))
);

/**
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Registries.ITEM□
 *
 * □ 1.20.1 □□□□□□
 * - □□ Registries.ITEM□□□ 1.19.2 □ Registry.ITEM□
 * - □□ new Identifier() □□□□□□□□ 1.21+ □ Identifier.of()□
 */
public static Item register(String name, Item item) {
    return Registry.register(Registries.ITEM, new Identifier(MOD_ID, name), item);
}

/**
 * □□□□□□□□□□□□□□□□ JVM □□ ModItems □□
 */
public static void registerModItems() {
    @SuppressWarnings("unused")
    Item ensureClassInit = EXAMPLE_ITEM;
}

/*
 * ============================================================================
 * □□□□□□□□□□□□□□□□ 1.20.1 □□□□□□□□
 * ============================================================================
 *
 *   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
 *   import net.minecraft.item.ItemGroups;
 *
 *   ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
 *       .register(entries -> entries.add(ModItems.EXAMPLE_ITEM));
 *
 * ItemGroups □□□1.20.4 □ 1.20.1 □□□□□
 *   ItemGroups.BUILDING_BLOCKS  ItemGroups.MISC       ItemGroups.COMBAT
 *   ItemGroups.TOOLS             ItemGroups.FOOD       ItemGroups.INGREDIENTS
 *   □□
 * ============================================================================
 */

/*
 * ============================================================================
 * □□□□□□□□□□
 * ============================================================================
 * 1.20.1: new FoodComponent.Builder().hunger(6).saturationModifier(0.6f).build()
 * 1.20.4: new FoodComponent.Builder().nutrition(6).saturationModifier(0.6f).build()
 *         ↑ hunger() □□□□ nutrition()□1.20.2+ □□□
 * ============================================================================
 */

/*
 * ============================================================================
```

```
     * □□□□□□□□
     * ============================================================================
     * 1. □□□□□□assets/<modid>/lang/en_us.json
     *       { "item.examplemod.example_item": "Example Item" }
     * 2. □□□□□□assets/<modid>/models/item/example_item.json
     * 3. □□□□□□assets/<modid>/textures/item/example_item.png
     * ============================================================================
     */

    // □□□□□□□□□□
    private ModItems() {
    }
}
```

mixin/inject_mixin.java

```
/*
 * inject_mixin.java
 * ----------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn □□)
 * □□□□□: □□□□@Inject Mixin □□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□
 *   Mixin API □ 1.20.4 □ 1.20.1 □□□□□□□□□
 *   □□□□□□□□□ org.spongepowered.asm.mixin.* □□□□
 *   compatibilityLevel □□ JAVA_17□
 *
 * □□ 1.19.2 □□□□□□□□□□□□
 *   Mixin API □□□□□□□□1.19.2 □□□□□ JAVA_17□
 * ----------------------------------------------------------------------------
 */
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfoReturnable;

/**
 * @Inject Mixin □□□□□□□ - Fabric 1.20.4
 *
 * □□□□□□□□□□□ @Inject □□□□□□□□□□□□□□□
 *
 * @Inject □□□□□:
 * - □□□□□□□□□□□□□□□HEAD, RETURN, TAIL □□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□ CallbackInfo □ CallbackInfoReturnable □□
 *
 * □□□□□□□□□@At □□:
 * - "HEAD"        □□□□□
 * - "RETURN"      □□□□□□□□□□ return □□□□□□
 * - "TAIL"        □□□□□
 * - "INVOKE"      □□□□□□□□□□□□□ target □□□□□□
 * - "INVOKE_ASSIGN" □□□□□□□□□□□□□
 * - "FIELD_ACCESS" □□□□□□□□
 *
 * □□□□□:
 * 1. □□□□□□□ mixin □□□□□□□□□□□□□
 * 2. @Mixin □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ CallbackInfo□
 * 4. □□□□□□□□□□□□□□□□ CallbackInfoReturnable<T> □□ CallbackInfo
 * 5. CallbackInfo.cancel() □□□□□□□□□□□□□□□ RETURN/HEAD □□□□□□□
 * 6. CallbackInfoReturnable.setReturnValue() □□□□□□□□□
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("HEAD"),
        cancellable = true
```

```
    )
    private void onTickStart(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□□□□□□□
        
        // □□: □□□□□□□□□□□□ tick
        // if (someCondition) {
        //     ci.cancel();
        // }
    }

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□ CallbackInfoReturnable □□□□□□□□□□□□
     */
    @Inject(
        method = "getCurrentFps",
        at = @At("RETURN"),
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // □□□□□□□□:
        // cir.setReturnValue(cir.getReturnValue() * 2);
        // cir.cancel();
    }
}
```

mixin/mixin_config.json

```
{
    // ============================================================
    // Mixin □□□□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: src/main/resources/examplemod.mixins.json
    //
    // □□□□□ fabric.mod.json □□□ "mixins" □□□□□:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // □□ 1.20.1 □□□□
    //   Mixin □□□□□□ 1.20.4 □ 1.20.1 □□□□□□□□
    //   □□□□□□ compatibilityLevel: "JAVA_17"□
    //
    // □□□□□:
    // - required:           □□□□□□true=□□□□□□□false=□□□
    // - package:            Mixin □□□□□□□□□
    // - compatibilityLevel: Java□□□□□□1.20.4 □□ JAVA_17□
    // - mixins:             □□Mixin□□□□□□□□□□□□□□□□
    // - client:             □□□□□□□□Mixin□
    // - server:              □□□□□□□□Mixin□□□□□□
    // - injectors:           □□□□□□
    //   - defaultRequire:    □□□□□□□□1=□□□□□□□□□0=□□□
    //
    // □□□□□:
    // 1. compatibilityLevel □□□ JAVA_17□1.20.4 □□ Java 17+□
    // 2. package □□□□□□Mixin□□□□□□□□
    // 3. mixins □□□□□□□ .java □□□□□□□□□
    // 4. client □□□□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_17",
    // □□Mixin□□□□□□□□□□□□□□□
```

```
    "mixins": [
        "ExampleInjectMixin"
    ],

    // □□□□□□□□Mixin□
    "client": [
        "client.ExampleClientMixin"
    ],

    // □□□□□□
    "injectors": {
        // defaultRequire: 1 □□□□□ @Inject □□□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□□□
        // □□ 0 □□□□□□□□□□□□□□□□□□□□□□□□bug□
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ============================================================================
// build.gradle —— Fabric 1.20.4 □□□□□□□□□□□Groovy DSL□
// ============================================================================
// □□□□
//  - □□□□□□ Fabric 1.20.4 □□□□□□□□ Loom □□□□□□
//  - □□□□□□□□□□ gradle.properties □□□□□□□□□□
//  - □□□□□□□□□JDK 17 + Gradle 8.x□□□ 8.6 □□□□□□
//
// □□ 1.20.1 build.gradle □□□□□
//  1. Loom □□□□1.6-SNAPSHOT□□ 1.20.1 □□□□ Loom □□□□□□□ 1.20.4□
//  2. Minecraft □□□□1.20.4□1.20.1 □ 1.20.1□
//  3. Yarn □□□□1.20.4+build.3□1.20.1 □ 1.20.1+build.10□
//  4. Fabric Loader□0.15.1□1.20.1 □ 0.16.x□1.20.4 □□ 0.15.x□
//  5. Fabric API□0.96.11+1.20.4□1.20.1 □ 0.83.0+1.20.1□
//  6. Java □□□17□□ 1.20.1 □□□□1.21+ □□□□ 21□
// ============================================================================

plugins {
    // Fabric Loom 1.6□□□ Minecraft 1.20.4□
    // □ 1.20.1 □□□□□□ Loom □□□□□
    id 'fabric-loom' version '1.6-SNAPSHOT'
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
    archivesName = project.name
}

repositories {
    mavenCentral()
}

// ----------------------------------------------------------------------------
// □□□□□□□□□□□ gradle.properties
// ----------------------------------------------------------------------------
dependencies {
    // Minecraft 1.20.4 □□□□ jar
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn 1.20.4+build.3 □□□□□
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader 0.15.1
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
    // Fabric API 0.96.11+1.20.4
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
}

// ----------------------------------------------------------------------------
// □□□□□□□ gradle.properties □□ mod_version □□□□ fabric.mod.json
// ----------------------------------------------------------------------------
processResources {
    inputs.property "version", project.version

    filesMatching("fabric.mod.json") {
```

```
        expand "version": project.version
    }
}

// --------------------------------------------------------------------------
// □□□□□Minecraft 1.20.4 □□ Java 17
// --------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}

java {
    withSourcesJar()

    sourceCompatibility = JavaVersion.VERSION_17
    targetCompatibility = JavaVersion.VERSION_17
}

// --------------------------------------------------------------------------
// jar □□
// --------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/version_diff.md

```
# Fabric 1.20.4 □ 1.20.1 □□□□□□□

> □□□□□□ Fabric 1.20.4 □□□□ 1.20.1□□□□□□□□ API □□□□
> 1.20.4 □ 1.20.1 □□□□□□□□□□□□□□ Fabric API □□□□□□□□□ API □□□□

---

## □□□□□

1.20.4 □□ 1.20.x □□□□□□□□□□1.20.1 → 1.20.2 → 1.20.3 → 1.20.4□□
□□□□□□ API □□□□□□□□□□□□□□
```

□ `Registries.ITEM` □ `Registries.BLOCK` □ 3 □□□□□`registry, id, value`□□□□□□

□□□□ 1.20.4 □□□□□□□□□□□□□□□□□□□□□□□□ `RegistryKey` □□□

```java
// 1.20.4 □□□□□□□□ RegistryKey□
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;

RegistryKey<Item> key = RegistryKey.of(RegistryKeys.ITEM, new Identifier("modid", "name"));
Registry.register(Registries.ITEM, key, item);
```

> □□□□□□□□□□□□□□□□ `Identifier` □ 3 □□□□□□□□

□□FabricItemSettings □□

3.1 FabricItemSettings □□□□□□□□□

1.20.2+ □ Fabric API □ `FabricItemSettings` □□□□□□□□
- □□ `Item.Settings` □□□□□□□□□ Fabric □□□□□
- `FabricItemSettings` □□□□□□□□□ Fabric API □□□□□□□□□□□□□ `Item.Settings`□

```java
// 1.20.4 —— □□□□□□□
import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.minecraft.item.Item;

// □□ A□FabricItemSettings□□□□□
new Item(new FabricItemSettings().maxCount(64));

// □□ B□Item.Settings□□□□
new Item(new Item.Settings().maxCount(64));
```

3.2 □□□□

1.20.4 □□□□□□□□□□ `FoodComponent`□□ 1.20.1 □□□□

```java
// 1.20.4 —— □ 1.20.1 □□
import net.minecraft.item.FoodComponent;

new Item(new Item.Settings()
    .food(new FoodComponent.Builder()
        .nutrition(6)
        .saturationModifier(0.6f)
        .meat()
        .build()
    )
);
```

> □□□`hunger()` □□□ 1.20.2+ □□□□□ `nutrition()`□1.20.1 □□□ `hunger()`□

□□□□□□□□□

4.1 FabricBlockSettings □□□□□

1.20.4 □ `FabricBlockSettings` API □ 1.20.1 □□□□□

```java
// 1.20.4 —— □ 1.20.1 □□
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.Blocks;

FabricBlockSettings.copyOf(Blocks.STONE)
    .strength(3.0f)
    .requiresTool()
    .luminance(15);

// □□
```

```
FabricBlockSettings.create()
    .strength(3.0f)
    .requiresTool();
```

4.2 AbstractBlock.Settings

1.20.4 □ `AbstractBlock.Settings` □□□□□□□□□ 1.20.1 □□□□

```java
import net.minecraft.block.AbstractBlock;

AbstractBlock.Settings.create()
    .strength(3.0f)
    .requiresTool();
```

□□□□□□□□□Creative Tab□□□

5.1 □□□□□

1.20.4 □□□□□□□□□□ 1.20.1 □□□□□

```java
// 1.20.4 ── □ 1.20.1 □□
import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
import net.minecraft.item.ItemGroups;

ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
    .register(entries -> entries.add(myItem));
```

5.2 □□□□□□□□□

```java
// 1.20.4 ── □ 1.20.1 □□
import net.fabricmc.fabric.api.itemgroup.v1.FabricItemGroup;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

Registry.register(Registries.ITEM_GROUP, new Identifier("modid", "example"),
    FabricItemGroup.builder()
        .displayName(Text.literal("Example Tab"))
        .icon(() -> new ItemStack(Items.STONE))
        .entries((context, entries) -> {
            entries.add(myItem);
        })
        .build()
);
```

□□1.20.2+ □□□□□□ API □□

□□□□□□□□ 1.20.2/1.20.3 □□1.20.4 □□□□□□□□□
□□□□□ 1.20.1 □□□□ 1.20.4□□□□□□□□□□

6.1 FoodComponent.Builder □□□□□

```java
// 1.20.1
new FoodComponent.Builder().hunger(6).saturationModifier(0.6f).build();

// 1.20.4□hunger → nutrition□
new FoodComponent.Builder().nutrition(6).saturationModifier(0.6f).build();
```

6.2 □□□□□□□□

□□□□□□□□ 1.20.2+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□
□□□□□□ `ClassNotFoundException`□□□□□□□□□□□□□□□□□ 1.20.1 □□□
6.3 ScreenHandlerType □□

1.20.2+ □ `ScreenHandlerType` □□□□□□□□□□□□ `ExtendedScreenHandlerType` □□□
□□□□□□□□□

□□□□□□□□1.20.1 → 1.20.4□

1. □□ `gradle.properties` □□□□□□MC□Yarn□Loader□Fabric API□□
2. □□□□□□ `FoodComponent.Builder().hunger()`□□□ `.nutrition()`□
3. □□□□ `Registry.register` □□□□□□□□ `RegistryKey` □□□
4. □□□□□□□□□□□□□□□□□□ API □□□ 1.20.1 □□□

recipe/shaped_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: data/examplemod/recipes/shaped_recipe.json
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□ 1.20.4 □ 1.20.1 □□□□□□□□
    //   □□□□□□□ type / pattern / key / result □□□
    //
    // □□□□□□crafting_shaped□□□□□ pattern □ key □□
    // □□□□□□□□□□□□□ pattern □□□□□□
    //
    // □□□□□:
    // 1. pattern □□□□□□□□□□□□
    // 2. pattern □□□□□□□□□□ key □□□□□□□
    // 3. key □□□□□□□□□□□□□□□□□□□□ #, X, *, @ □□
    // 4. □□□□□□□□□□□□□
    // 5. result □□ count □□□□ 64
    // 6. ingredient □□ tag □□□ "tag" □□□□ "item"
    // ============================================================

    "type": "minecraft:crafting_shaped",

    // pattern: □□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/shapeless_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□ 1.20.1 □□□□□
    //   □□ JSON □□□ 1.20.4 □ 1.20.1 □□□□□□□□
    //
    // □□□□□□crafting_shapeless□□□□□□□□□□□□
    // □□□ ingredients □□□□□□□□□□□□□□□□□□
```

```
    //
    // □□□□□:
    // 1. ingredients □□□□□□□□□□□□□□□□□□□□
    // 2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // 3. □□□□ "tag" □□□□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□ 1.20.1 □□□□□
    //   1.20.4 □□□ JSON □□□ 1.20.1 □□□□□
    //   □□□□□□□□: data/<mod_id>/tags/blocks/
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□: data/<mod_id>/tags/items/□□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.4
    // ============================================================
    // □□□□□: data/examplemod/tags/items/example_item_tag.json
```

```
    //
    // 适用 1.20.1 及以后
    //   1.20.4 及以后 JSON 格式与 1.20.1 完全一致
    //   文件存放路径: data/<mod_id>/tags/items/
    //
    // 原版常用物品标签（可直接引用，无需自己定义）:
    //   minecraft:logs          原木
    //   minecraft:planks        木板
    //   minecraft:coals         煤炭类
    //   minecraft:saplings      树苗
    //   minecraft:flowers       花
    //
    // 使用说明:
    // 1. 将此文件放入 data/<mod_id>/tags/items/ 目录下
    // 2. 文件名即为标签名（不含扩展名）
    //    物品: data/<mod_id>/tags/items/
    //    方块: data/<mod_id>/tags/blocks/
    // 3. values 中可填ID或其他标签引用
    // 4. replace 为 true 时会覆盖其他模组对同名标签的定义
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
    ]
}
```

worldgen/ore_feature.java

```
/*
 * ore_feature.java
 * ------------------------------------------------------------------------------
 * 适用版本: Minecraft 1.20.4 (Fabric Loader 0.15.1 + Fabric API 0.96.11 + Yarn 映射)
 * 文件说明: 自定义矿石世界生成特性（Feature）
 *
 * 适用 1.20.1 及以后版本（本文件）
 *   1. 使用新 API 自 1.20.1 起可用
 *   2. Registries.CONFIGURED_FEATURE / Registries.PLACED_FEATURE 注册表
 *   3. BiomeModifications API 稳定
 *   4. 放置修饰器 (PlacementModifier) API 稳定
 *
 * 适用 1.19.2 及以前版本（旧版）
 *   - 1.19.2 用 BuiltinRegistries 手动注册特性
 *   - 1.20.4 用 Registries 注册表
 *   - 1.19.2 用 RegistryKey/RegistryKeys 在 net.minecraft.util.registry 包
 *   - 1.20.4 用 RegistryKey/RegistryKeys 在 net.minecraft.registry 包
 * ------------------------------------------------------------------------------
 */
package com.example.mod.worldgen;

import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.registry.entry.RegistryEntry;
import net.minecraft.util.Identifier;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.YOffset;
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.feature.PlacedFeature;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.PlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;

import java.util.List;

/**
```

```
 * □□□□□□□□□□ - Fabric 1.20.4
 *
 * □□ 1.20.1 □□□□□□□□□□□□API □□□□□□□□
 *
 * □□□□□□□□□□
 * 1. □□□□□□ ConfiguredFeature□□□□□□□
 * 2. □□□□□□ PlacedFeature□□□□□□□
 * 3. □□ BiomeModifications □□□□□□□□□□□□
 *
 * □□□□□:
 *   ConfiguredFeature -> □□□ Registries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□ Registries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□□
 *
 * □□□□□:
 * - □□□□□□ ConfiguredFeature□□□□ PlacedFeature
 * - PlacedFeature □□ RegistryEntry<ConfiguredFeature> □□
 * - BiomeModifications.addFeature □□ RegistryKey<PlacedFeature>
 * - OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES □□□□□□□□
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□
 */
public class ModWorldGen {

    public static final String MOD_ID = "examplemod";

    // ConfiguredFeature □ RegistryKey
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    // PlacedFeature □ RegistryKey
    public static final RegistryKey<PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, new Identifier(MOD_ID, "ore_example"));

    /**
     * □□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□□
     */
    public static void register() {
        // ================================================================
        // 1. □□□□□□ ConfiguredFeature
        // ================================================================
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),
            9
        );

        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,
            oreConfig
        );

        // □□□□ Registries.CONFIGURED_FEATURE□□ 1.20.1 □□□□
        Registry.register(
            Registries.CONFIGURED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            configuredFeature
        );

        // ================================================================
        // 2. □□□□□□ PlacedFeature
        // ================================================================
        RegistryEntry<ConfiguredFeature<?, ?>> cfEntry = RegistryEntry.of(configuredFeature);

        List<PlacementModifier> placementModifiers = List.of(
            CountPlacementModifier.of(20),
            SquarePlacementModifier.of(),
            HeightRangePlacementModifier.uniform(
                YOffset.getBottom(),
                YOffset.fixed(64)
            )
        );

        PlacedFeature placedFeature = new PlacedFeature(
            cfEntry,
            placementModifiers
```

```
        );

        // □□□ Registries.PLACED_FEATURE□□ 1.20.1 □□□
        Registry.register(
            Registries.PLACED_FEATURE,
            new Identifier(MOD_ID, "ore_example"),
            placedFeature
        );

        // ================================================================
        // 3. □□ BiomeModifications □□□□□□□□□□
        // ================================================================
        BiomeModifications.addFeature(
            BiomeSelectors.foundInOverworld(),
            GenerationStep.Feature.UNDERGROUND_ORES,
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                new Identifier(MOD_ID, "ore_example")
            )
        );
    }
}

/*
 * ================================================================================
 * □□□□□□□□□□
 * ================================================================================
 *
 * BiomeSelectors.all()                  □□□□□□□
 * BiomeSelectors.foundInOverworld()      □□□□□□□□
 * BiomeSelectors.foundInNether()         □□□□□□□
 * BiomeSelectors.foundInTheEnd()         □□□□□□□
 *
 * ================================================================================
 * □□□□□□□
 * ================================================================================
 *
 * GenerationStep.Feature□□□□□□□:
 * - RAW_GENERATION          □□□□□□□□□□
 * - LAKES                    □□
 * - LOCAL_MODIFICATIONS      □□□□
 * - UNDERGROUND_STRUCTURES   □□□□
 * - SURFACE_STRUCTURES       □□□□
 * - STRONGHOLDS              □□
 * - UNDERGROUND_ORES        □□□□□□□□□□
 * - UNDERGROUND_DECORATION   □□□□
 * - FLUID_SPRINGS            □□□
 * - VEGETAL_DECORATION       □□□□
 * - TOP_LAYER_MODIFICATION   □□□□
 */
```

Fabric 1.20.6

advancement/advancement.json

```
{
    // ============================================================
    // □□/□□□□ - Fabric 1.20.6
    // ============================================================
    // □□□□: data/examplemod/advancements/example_advancement.json
    //
    // □□ 1.21.1 □□□□
    //   □□ JSON □□□ 1.20.6 □ 1.21.1 □□□□□□□□
    //   inventory_changed □□□□□ "items"□□□□□□□□□
    //   □□: 1.21.2+ □□□□□□□□□□□□□□□
    //   □ 1.20.6 □ 1.21.1 □□□□□□□□□□
    //
    // □□□□□Advancement□□ Minecraft 1.12+ □□□□□
    // □□□□□□□□□□□□□□□□□□□□toast□□□□□□□
    //
    // □□□□:
    // - display:     □□□□□□□□□□□□□□□□□□□□□□□□
    // - parent:      □□□□□□□□□□□□□□□□
    // - criteria:    □□□□□□□□□□□□□□□□□□□□□□□□
```

```
// - requirements: □□□□□□□□□□□□AND / OR□
// - rewards:     □□□□□□□□□□□□□□□□□□□□□□□□
//
// □□□□□:
// 1. □□□□□□parent□□□□□□ display.background □□
// 2. 1.20.6 □ inventory_changed □□□□□□□□ "items" □□□□□ 1.21.1 □□□
//    □1.19.2 □□ "item" □□□□□□
// 3. criteria □□□□□□□□□□□ trigger □□
// 4. requirements □□□□□□□□□□□ AND□□□□□ OR
// 5. frame □□: task(□□) / goal(□□) / challenge(□□)
// ============================================================

// □□□□ID□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
"parent": "examplemod:example/root",

// □□□□□
"display": {
    // □□
    "icon": {
        "item": "examplemod:example_item"
    },

    // □□□□□□ translate □□□□□ literal □□□□□□
    "title": {
        "translate": "advancements.examplemod.example.title"
    },

    // □□
    "description": {
        "translate": "advancements.examplemod.example.description"
    },

    // □□□□□: task(□□,□□) / goal(□□,□□) / challenge(□□,□□)
    "frame": "task",

    // □□□□□□□□□□□□□□
    "show_toast": true,

    // □□□□□□□□□□
    "announce_to_chat": true,

    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    "hidden": false

    // □□□□□□ background □□□□□□□□□□□□□
    // "background": "minecraft:textures/block/stone.png"
},

// □□□□□
// □□ key □□□□□□□□□□□□□□□□ trigger □ conditions
"criteria": {
    "get_example_item": {
        // □□□□: □□□□□□□□□□□□
        "trigger": "minecraft:inventory_changed",
        "conditions": {
            // 1.20.6 □□ "items"□□□□□□□□□□□□ 1.21.1 □□
            // 1.19.2 □□ "item"□□□□□□□□
            "items": [
                {
                    "items": [
                        "examplemod:example_item"
                    ]
                }
            ]
        }
    },
    "another_condition": {
        // □□□□: □□□□□□□□□□□□□□□□
        "trigger": "minecraft:location",
        "conditions": {
            "player": {
                "location": {
                    "dimension": "minecraft:overworld"
                }
            }
        }
```

```
        }
    },

    // □□□□□□□□□□
    // □□□□□□ AND□□□□□□□
    // □□□□□□ OR□□□□□□□
    // □□□□□: □□ get_example_item AND another_condition
    "requirements": [
        ["get_example_item"],
        ["another_condition"]
    ],

    // □□□□□
    "rewards": {
        // □□□□□□
        "experience": 10,

        // □□□□□□□□□□□□□□
        "loot": [
            "examplemod:example_loot_table"
        ],

        // □□□□□□□□□□□□□□□
        "function": "examplemod:grant_example_reward"
    }
}
```

block/simple_block.java

```
/*
 * simple_block.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□□ Identifier.of("modid", "name")□□□□□□□□□
 *      □1.20.5 □□□□□ Identifier.of()□□□□□□□□□□□□□□□□
 *
 *   2. □Registries □□□□
 *      □□□□□□□□ net.minecraft.registry.Registries□□□□
 *      □□□ Registries.BLOCK / Registries.ITEM
 *
 *   3. □Data Components □□□□□□□□□□□□□□□□
 *      1.20.6: Data Components □□□□□□□1.20.5 □□□□□
 *              □□□□□□□□□□□□□□□□□□□ API □□□□□□
 *              Item.Settings □□□□□□□ .component() □□
 *      1.21.1: Data Components □□□□□□
 *              □□□□□□□□□ Registries.DATA_COMPONENT_TYPE
 *              Item.Settings □ .component() □□
 *
 *   4. □BlockEntity □□/□□□□□□□□□□
 *      1.20.6 □ saveAdditional / loadAdditional □□ HolderLookup.Provider □□
 *      1.21.1 □□□ HolderLookup.Provider □□
 *
 *   5. □FabricBlockSettings API □□□□□□□
 *      .copyOf(), .create(), .strength(), .requiresTool() □□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□
 *   - 1.20.1 □□ new Identifier() □□□□□□1.20.5 □□□□ Identifier.of()□
 *   - 1.20.1 □□ Data Components □□
 *   - 1.20.1 □□ Java 17□1.20.6 □□ Java 21
 *   - 1.20.1 □□ MatrixStack □□□1.20.6 □□ DrawContext
 * ------------------------------------------------------------------------
 */
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
```

```java
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□□□□□□□□ (Fabric 1.20.6)
 *
 * □□□□□□□□
 * 1. □□□□□□□ Registries.BLOCK / Registries.ITEM□□ 1.21.1 □□□□□
 * 2. □1.20.5+ □□□Identifier □□ Identifier.of() □□ new Identifier()□
 * 3. FabricBlockSettings API □ 1.21.1 □□□□□□
 * 4. □□□□□□□□ ItemGroupEvents □□□□□ 1.21.1 □□□□□
 * 5. □□□□□□□□□□Data Components □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 */
public class SimpleBlock extends Block {

    public static final String MOD_ID = "examplemod";

    /**
     * □□□□□□□
     * □ static final □□□□□□□□□□□□□□□□□
     */
    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * □□□□□□□□□□□□□□□□□
     * □□□□ Block □□□□□□□□□□□□ BlockItem□
     *
     * □□ 1.21.1 □□□□□
     * - □□ Identifier.of()□□ 1.21.1 □□□□
     * - Registries.BLOCK / Registries.ITEM □ 1.21.1 □□
     * - FabricBlockSettings □□□□ 1.21.1 □□
     * - □□□□□1.20.6 □ Data Components □□□□□□□□□□
     *   BlockItem □ Item.Settings □□□□□□ .component() □□
     *   □1.21.1 □ .component() □□□□□□□□
     */
    private static Block registerBlock(String name) {
        // ---- □□□□□□□□□□□□□□□ ----
        // □ 1.21.1 □□□□□□□ FabricBlockSettings □□
        AbstractBlock.Settings settings = FabricBlockSettings.copyOf(Blocks.STONE)
                .strength(3.0f)
                .requiresTool()
                .luminance(15);

        Block block = new SimpleBlock(settings);

        // ---- □□□□□□□□ Block □ Registries.BLOCK ----
        // □1.20.6□□□□ Identifier.of() □□ new Identifier()
        // 1.20.4 □□□□: Registry.register(Registries.BLOCK, new Identifier(MOD_ID, name), block);
        // 1.20.5+: Registry.register(Registries.BLOCK, Identifier.of(MOD_ID, name), block);
        Registry.register(Registries.BLOCK, Identifier.of(MOD_ID, name), block);

        // ---- □□□□□□□□ BlockItem □ Registries.ITEM ----
        // □□□□□ Identifier.of()
        // □□□□□□□□□ .group()□□□□□□ 1.19.3+ □□□□
        // □1.20.6 □□□□Data Components □□□□□□□□
        // Item.Settings □ .component() □□□□□□□□□□□□□□□□
        BlockItem item = new BlockItem(block, new FabricItemSettings());

        Registry.register(Registries.ITEM, Identifier.of(MOD_ID, name), item);

        return block;
    }

    // ========================================================================
    // □□□□□□□□□
    // ========================================================================
    //
    // □□□□□□□□ ModInitializer □□□□□□□□□
    //
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
```

```
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
    // ========================================================================
    // □□□□□□□□□□ 1.21.1 □□□□□□□
    // ========================================================================
    //
    //   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
    //   import net.minecraft.item.ItemGroups;
    //
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    //
    // ========================================================================
    // BlockEntity □□/□□□□□□□□□□□□□□□□□□
    // ========================================================================
    //
    // □1.20.6□saveAdditional □ loadAdditional □□ HolderLookup.Provider □□□
    //
    //   @Override
    //   public void saveAdditional(CompoundTag tag) {
    //       super.saveAdditional(tag);
    //       tag.putInt("value", this.value);
    //   }
    //
    //   @Override
    //   public void loadAdditional(CompoundTag tag) {
    //       super.loadAdditional(tag);
    //       this.value = tag.getInt("value");
    //   }
    //
    // 1.21.1 □□□□□□□□□ HolderLookup.Provider □□□
    // ========================================================================
}
```

command/simple_command.java

```
/*
 * simple_command.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□
 *   1. CommandRegistrationCallback API □ 1.21.1 □□□□□□v2 □□□
 *   2. CommandManager API □□
 *   3. Text API □□□Text.literal()□
 *   4. EntityArgumentType □□□□
 *   5. Java 21 □□□□□□□□□ record □□□□□switch □□□□□□□
 *
 * □□ 1.20.4 □□□□□□□□□□
 *   - 1.20.4 □□□□ v2 □□□□3 □□□□□□API □□□□
 *   - 1.20.4 □□ Java 17□1.20.6 □□ Java 21
 *   - 1.20.4 □□ new Identifier()□1.20.6 □□ Identifier.of()
 *
 * □□ 1.19.2 □□□□□□□□□□
 *   - 1.19.2 □□ v1 □□□□2 □□□□□□1.20.6 □□ v2 □□□3 □□□□□
 *   - 1.19.2 □ new LiteralText()□1.20.6 □ Text.literal()
 * ------------------------------------------------------------------------
 */
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.CommandManager;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.text.Text;
import net.minecraft.util.Formatting;
import static net.minecraft.server.command.CommandManager.literal;
```

```java
import static net.minecraft.server.command.CommandManager.argument;

/**
 * □□□□□□□□ - Fabric 1.20.6
 *
 * □□ 1.21.1 □□□□□□□□□□□□API □□□□□□□□
 * □□□□ Java 21 □□□□□□□□ switch □□□□□□record □□□
 *
 * □□□□□□□□□
 * 1. □□ CommandRegistrationCallback v2 □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * Fabric 1.20.6 □□ CommandRegistrationCallback v2 □□
 * v2 □□□□□□□□□□□□□□:
 *   (CommandDispatcher<ServerCommandSource> dispatcher,
 *    CommandRegistryAccess registryAccess,
 *    CommandManager.RegistrationEnvironment environment)
 */
public class ModCommands {

    /**
     * □□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, registryAccess, environment) -> {
                registerExampleCommand(dispatcher);
            }
        );
    }

    /**
     * □□ /example □□□
     *
     * □□□□:
     * /example                          - □□□□□□
     * /example hello                    - □□□□□□□□
     * /example hello <player>           - □□□□□□□□□□
     * /example set <value>              - □□□□□□
     * /example give <player> <count>    - □□□□□□□□
     */
    private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
        dispatcher.register(
            literal("example")
                .executes(context -> {
                    context.getSource().sendFeedback(
                        () -> Text.literal("□□: /example <hello|set|give>")
                            .formatted(Formatting.YELLOW),
                        false
                    );
                    return 1;
                })

                .then(literal("hello")
                    .executes(ModCommands::sayHelloToSelf)
                    .then(argument("target", EntityArgumentType.player())
                        .executes(ModCommands::sayHelloToTarget)
                    )
                )

                .then(literal("set")
                    .then(argument("value", IntegerArgumentType.integer(0, 100))
                        .executes(ModCommands::setValue)
                    )
                )

                .then(literal("give")
                    .then(argument("target", EntityArgumentType.player())
                        .then(argument("count", IntegerArgumentType.integer(1, 64))
                            .executes(ModCommands::giveItem)
                        )
                    )
                )
```

```
    );
}

/**
 * /example hello - □□□□□□□□
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    source.sendFeedback(
        () -> Text.literal("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;
}

/**
 * /example hello <player> - □□□□□□□□□□
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    target.sendMessage(
        Text.literal("Hello from " + source.getName() + "!")
            .formatted(Formatting.GOLD)
    );

    source.sendFeedback(
        () -> Text.literal("□□ " + target.getName().getString() + " □□□□")
            .formatted(Formatting.GREEN),
        true
    );
    return 1;
}

/**
 * /example set <value> - □□□
 */
private static int setValue(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    int value = IntegerArgumentType.getInteger(context, "value");
    ServerCommandSource source = context.getSource();

    source.sendFeedback(
        () -> Text.literal("□□□□□: " + value)
            .formatted(Formatting.AQUA),
        false
    );
    return 1;
}

/**
 * /example give <player> <count> - □□□□□□□□
 */
private static int giveItem(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
    int count = IntegerArgumentType.getInteger(context, "count");
    ServerCommandSource source = context.getSource();

    net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
        net.minecraft.item.Items.DIAMOND, count
    );
    target.getInventory().insertStack(stack);

    source.sendFeedback(
        () -> Text.literal("□□ " + target.getName().getString()
```

```
                + " □□□ " + count + " □□□")
                .formatted(Formatting.GREEN),
            true
        );
        return 1;
    }
}

/*
 * ============================================================================
 * 1.19.2 □□□□□□□□□□v1 □□□□□
 * ============================================================================
 *
 * // 1.19.2 □□ v1 □□□2 □□□□:
 * import net.fabricmc.fabric.api.command.v1.CommandRegistrationCallback;
 *
 * CommandRegistrationCallback.EVENT.register(
 *     (dispatcher, dedicated) -> {
 *         registerExampleCommand(dispatcher);
 *     }
 * );
 *
 * // 1.19.2 □□ new LiteralText() □□ Text.literal():
 * source.sendFeedback(
 *     () -> new LiteralText("Hello!").formatted(Formatting.GREEN),
 *     false
 * );
 * ============================================================================
 */
```

enchantment/custom_enchantment.java

```
/*
 * custom_enchantment.java
 * ------------------------------------------------------------------------
 * □□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□□
 *
 *   1. □□□□□□□□□□□□□□□□□□□□□□□□□□
 *      1.20.6: □□□□□□□□□□ Enchantment □□□□□□□□□□□□
 *              □□□□□□□□□□JSON □□□□□□□□□□□□□□□
 *              Registry.register(Registries.ENCHANTMENT, Identifier.of(...), enchantment)
 *
 *      1.21.1: Enchantment □ record□□□□□□□□□□□□ JSON □□□□□□□
 *              Java □□□□□□ RegistryKey □□
 *              □□□□□□□□ data/<modid>/enchantment/ □□□ JSON □□□
 *
 *   2. □EnchantmentTarget □□□□□□□
 *      1.20.6: EnchantmentTarget.WEAPON / ARMOR / TOOL □□□□□□□
 *      1.21.1: □□□□□□□ (supported_items) □□□□□□□□
 *
 *   3. □□□□□□□□□□□□□□□□
 *      1.20.6: getMinPower() / getMaxPower() / getMaxLevel() / canCombine() □□□□
 *      1.21.1: JSON □□ min_cost / max_cost / max_level / exclusive_set
 *
 *   4. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□□ Identifier.of("modid", "name")
 *
 * □□ 1.20.4 □□□□□□□□□□□
 *   - 1.20.4 □□ new Identifier()□1.20.6 □□ Identifier.of()
 *   - 1.20.4 □□ Java 17□1.20.6 □□ Java 21
 *   - □□□□ API □□□□□□□□□ Enchantment □□
 *
 * □□□□□□□
 * 1.20.x □□□□□□□□□□□□□□□□□□□□1.21 □□□□□□□□□□□□□
 * □ 1.20.6 □□□□□□□□□□□□□ Enchantment □□□□□□□
 * ------------------------------------------------------------------------
 */
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.enchantment.EnchantmentTarget;
import net.minecraft.entity.EquipmentSlot;
```

```java
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□ - Fabric 1.20.6
 *
 * □□□□□□□□□□1.20.6 □□□□□□□□□ Java □□□□□□□□
 * □□ 1.20.5+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ 1.21 □□□□□
 * □ 1.20.6 □□□□□□□□□ Enchantment □□□□□□□□□□□□□
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□ Enchantment□
 * 2. □□ Registry.register □□□ Registries.ENCHANTMENT
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□
 * - □□□□□□□1.20.6 □□□ Enchantment □□□□□□□□□
 *   1.21.1 □ Enchantment □ record□□□□□ JSON □□
 * - □□ Registries.ENCHANTMENT□□ 1.21.1 □□□□□□□□□□□□□□□□□□□□□□
 * - EnchantmentTarget □□□□□□□1.21.1 □□□□□□□□□□□□□
 * - canCombine() □□□□□□□1.21.1 □□ JSON □ exclusive_set□
 *
 * □□□□□□□□:
 * - Enchantment.Rarity: □□□□□□□□□□□□□□□□□□□□
 *   COMMON      weight=10  □□□□□□□□□
 *   UNCOMMON    weight=5   □□□□□□□□□□□□□□□
 *   RARE        weight=2   □□□□□□□□□
 *   VERY_RARE   weight=1   □□□□□□□□□□□□□□□
 *
 * - EnchantmentTarget: □□□□□□□□□□□□
 *   ALL          □□□□
 *   ARMOR         □□
 *   FEET/LEGS/CHEST/HEAD  □□□□□□
 *   WEAPON        □
 *   TOOL          □□□□/□/□/□□
 *   BOW           □
 *   FISHING_ROD   □□□
 *   BREAKABLE     □□□□□□
 *   TRIDENT       □□□
 *   CROSSBOW      □
 *   VANISHABLE    □□□□□□□
 *
 * - EquipmentSlot[]: □□□□□□□□□□
 *   MAINHAND    □□
 *   OFFHAND     □□
 *   HEAD        □□
 *   CHEST       □□
 *   LEGS        □□
 *   FEET        □□
 *
 * □□□□:
 * - □□□ mod_id □□□□□□ mod ID
 * - getMinPower □□□□□: 10 + (level - 1) * 15
 * - getMaxPower □□□ getMinPower + 15~50
 * - getMaxLevel □□□□□□□□□
 * - □□□□□1.21+ □ Enchantment.canCombine() □□□□□□
 *   1.20.6 □□□□□□□□
 */
public class ModEnchantments {

    /**
     * □□□□□□□□
     * □□□□□□□□□ModInitializer□□ onInitialize() □□□
     *
     * □1.20.6□□□ Registries.ENCHANTMENT□□□□
     * □□ Identifier.of() □□ new Identifier()
     *
     * □□ 1.21.1 □□□□□
     * 1.20.6 □□□□□ Registry.register() □□□□□□□
     * 1.21.1 □□□□□□□□□ JSON □□□□□□□ Java □□
     */
    public static final Enchantment EXAMPLE_ENCHANTMENT = Registry.register(
        Registries.ENCHANTMENT,
        Identifier.of("examplemod", "example_enchantment"),  // □1.20.6□Identifier.of()
```

```
    new ExampleEnchantment(
        Enchantment.Rarity.RARE,           // □□□: □□
        EnchantmentTarget.WEAPON,          // □□□□□□□□: □□
        EquipmentSlot.MAINHAND             // □□□□□: □□
    )
);

/**
 * □□□□□□□□□□□□□□□□
 */
public static void register() {
    // □□□□□□□□□□□□□□□□
}

// ================================================================
// □□□□□□□□
// ================================================================
// □1.20.6 □□□□□□□□ Enchantment □
// 1.21.1 □ Enchantment □ record□□□□□□
public static class ExampleEnchantment extends Enchantment {

    /**
     * □□□□□
     *
     * @param rarity  □□□□□□
     * @param target  □□□□□□□□
     * @param slots   □□□□□□□□□□□
     */
    public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        // □□□□□□□□□□
        super(rarity, target, slots);
    }

    /**
     * □□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□"□□□□□"□□□□□□□□□
     *
     * □□□□□□: 10 + (level - 1) * 15
     * 1□: 10, 2□: 25, 3□: 40
     *
     * @param level □□□□□
     * @return □□□□□□□□□□□
     */
    @Override
    public int getMinPower(int level) {
        return 10 + (level - 1) * 15;
    }

    /**
     * □□□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     * □□□□ getMinPower + 50
     *
     * @param level □□□□□
     * @return □□□□□□□□□□□
     */
    @Override
    public int getMaxPower(int level) {
        return super.getMinPower(level) + 50;
    }

    /**
     * □□□□□□□□□□
     * □□: □□ 3 □□□□□□□□ 3 □
     *
     * @return □□□□□
     */
    @Override
    public int getMaxLevel() {
        return 3;
    }

    /**
     * □□□□□□□□□□□□□□□□□□□□□□
     * □□□□□ EnchantmentTarget □□□□□□□□□□□□□□□□□□□□□□□□
     *
```

```
         * @param stack □□□□
         * @return □□□□□
         */
        @Override
        public boolean isAcceptableItem(ItemStack stack) {
            // □□□□: □□□□□□□□ EnchantmentTarget
            // □□□□□□□□□□□□
            return super.isAcceptableItem(stack);
        }

        /**
         * □□□□□□□□□□□□□□□□□□□□
         * □□□□ true□□□□□□□□□□□□□□□□
         *
         * □□□□1.20.6 □□□□□□□□
         * 1.21+ □□□□□□□□□□ JSON □□ exclusive_set □□
         *
         * @param other □□□□
         * @return □□□□□
         */
        @Override
        public boolean canCombine(Enchantment other) {
            // □□: □□□□□□□□
            // if (other == Enchantments.SHARPNESS) {
            //     return false;
            // }
            return super.canCombine(other);
        }
    }
}

/*
 * ============================================================================
 * □□□□□□□□□
 * ============================================================================
 * □□□□□: assets/examplemod/lang/en_us.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment",
 *     "enchantment.examplemod.example_enchantment.desc": "A custom enchantment"
 * }
 *
 * □□□□□□□: assets/examplemod/lang/zh_cn.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "□□□□□",
 *     "enchantment.examplemod.example_enchantment.desc": "□□□□□□□□□"
 * }
 *
 * ============================================================================
 * □□ ID □□□□□
 * ============================================================================
 * - □□□ Identifier □□□□□□
 * - □□□□□□□□□: enchantment.<mod_id>.<enchantment_name>
 * - □□□□□□□□□□□ ID □□
 *
 * ============================================================================
 * 1.21.1 □□□□□□□□□□□□□□□□□□□□□ 1.20.6□
 * ============================================================================
 *
 * // 1.21.1: □□□ RegistryKey □□□□□□□□ JSON □□
 * public static final RegistryKey<Enchantment> EXAMPLE_ENCHANTMENT =
 *     RegistryKey.of(
 *         RegistryKeys.ENCHANTMENT,
 *         Identifier.of("examplemod", "example_enchantment")
 *     );
 *
 * // JSON □□: data/examplemod/enchantment/example_enchantment.json
 * // {
 * //     "description": { "translate": "enchantment.examplemod.example_enchantment" },
 * //     "supported_items": "#examplemod:enchantable/example_weapon",
 * //     "max_level": 3,
 * //     ...
 * // }
 * ============================================================================
 */
```

entity/simple_entity.java

```java
/*
 * simple_entity.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□□ Identifier.of("modid", "name")□□□□□□□□□
 *      □1.20.5 □□□□□□□□□□□□□□□□
 *
 *   2. □Java 21 □□□
 *      1.20.5+ □□□□□ Java 21□□ 1.21.1 □□□
 *
 *   3. □□□□□ API □□□□□□
 *      1.20.6 □□□□ Registries.ENTITY_TYPE□□ 1.21.1 □□□□
 *
 *   4. □EntityAttribute □□□□□□□□
 *      FabricDefaultAttributeRegistry API □□□□□□□□□□
 *
 *   5. □Entity □□□□□□□□□□□□□□
 *      1.21+ □□□□ NBT □□□□□□□□□□□□□□
 *      1.20.6 □□□□□□□□ NBT □□
 *
 * □□ 1.20.4 □□□□□□□□□□□□
 *   - 1.20.4 □□ new Identifier() □□□□□
 *   - 1.20.4 □□ Java 17□1.20.6 □□ Java 21
 *   - □□□□□ API □□□□□
 * ------------------------------------------------------------------------
 */
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.world.World;

/**
 * □□□□□□□□□□ - Fabric 1.20.6
 *
 * □□□□□□□□□□
 * 1. □□□□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□
 * 1. □□ Identifier.of()□□ 1.21.1 □□□□
 * 2. Registries.ENTITY_TYPE □ 1.21.1 □□
 * 3. FabricDefaultAttributeRegistry API □ 1.21.1 □□
 * 4. Java 21 □□□□□□□□□□ record □□□□□□□
 * 5. □□□□□□□□□□□□□□□□1.21 □□□□□□□□□□□□□□
 *
 * □□□□□□□□:
 * - □1.20.6□□□□□□ Identifier.of() □□ new Identifier()
 * - □□□□□□□□□□□□□□□ (EntityType, World) □□
 * - createMobAttributes() □□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□□ FabricDefaultAttributeRegistry
 */
public class ExampleEntity extends PathAwareEntity {

    // ================================================================
    // □□□□□□□□
    // ================================================================
    // □1.20.6□□□□ Identifier.of() □□ new Identifier()
    // Registries.ENTITY_TYPE □ 1.21.1 □□
    public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
        Registry.register(
```

```
            Registries.ENTITY_TYPE,
            Identifier.of("examplemod", "example_entity"),  // □1.20.6□Identifier.of()
            EntityType.Builder.<ExampleEntity>create(
                    ExampleEntity::new,
                    SpawnGroup.CREATURE
                )
                .dimensions(0.6f, 1.8f)
                .trackingRange(8)
                .build()
        );

    // ================================================================
    // □□□□□
    // ================================================================
    public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
        super(entityType, world);
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □□□□□:
    // - GENERIC_MAX_HEALTH      □□□□□□□□□ 20 = 10 □□□
    // - GENERIC_MOVEMENT_SPEED  □□□□□□□□ 0.25□
    // - GENERIC_ATTACK_DAMAGE   □□□□□□□□ 2□
    // - GENERIC_FOLLOW_RANGE     AI □□□□□□□□ 16□
    // - GENERIC_ARMOR           □□□
    // - GENERIC_KNOCKBACK_RESISTANCE □□□□□0.0~1.0□
    public static DefaultAttributeContainer createMobAttributes() {
        return PathAwareEntity.createMobAttributes()
            .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)
            .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)
            .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)
            .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)
            .add(EntityAttributes.GENERIC_ARMOR, 0.0)
            .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0);
    }

    // ================================================================
    // □□ tick □□
    // ================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□□□□□ tick □□□
    }

    // ================================================================
    // □□□ AI □□
    // ================================================================
    @Override
    protected void initGoals() {
        // □□□□□□□□ AI □□□Goal□
        // □□ AI:
        // - MeleeAttackGoal        □□□□□□□
        // - WanderAroundGoal       □□□□
        // - LookAtEntityGoal       □□□□
        // - FollowTargetGoal       □□□□
        // - SwimGoal               □□
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □□□□□□□□ModInitializer□□ onInitialize() □□□□
    //
    //   FabricDefaultAttributeRegistry.register(
    //       ExampleEntity.EXAMPLE_ENTITY,
    //       ExampleEntity.createMobAttributes()
    //   );
    // ================================================================
}
```

gui/screen_with_menu.java

```
/*
```

```
 * screen_with_menu.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□GUI□□□□□ ScreenHandler + HandledScreen □□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□□ Identifier.of("modid", "name")
 *
 *   2. □□□□ API □□□□
 *      1.20.6 □ 1.21.1 □□□□ DrawContext □□ GUI □□
 *      □DrawContext □ 1.20.2 □□□□1.20.6 □□□□□□□□
 *
 *   3. □ScreenHandler □□□□□□□□□□□□□
 *      1.21+ □ ScreenHandler □□□□□□□□□□□□□□□□□□□□
 *      1.20.6 □□□□□□ 1.20.x □□□□
 *
 *   4. □Java 21 □□□□
 *      1.20.5+ □□□□□□ Java 21□□ 1.21.1 □□□□
 *
 * □□ 1.20.4 □□□□□□□□□□□□□
 *   - 1.20.4 □□□□ DrawContext□1.20.2 □□□□□
 *   - 1.20.4 □□ new Identifier()□1.20.6 □□ Identifier.of()
 *   - 1.20.4 □□ Java 17□1.20.6 □□ Java 21
 *
 * □□ 1.19.2 □□□□□□□□□□□□□
 *   - 1.19.2 □□ Registry.SCREEN_HANDLER□□□□□
 *   - 1.19.2 □□ MatrixStack □□
 *   - 1.19.2 □□ new LiteralText()
 * ------------------------------------------------------------------------
 */
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.text.Text;
import net.minecraft.util.Identifier;

/**
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.20.6
 *
 * □□ 1.21.1 □□□□□□□
 * 1. □□ Identifier.of()□□ 1.21.1 □□□□
 * 2. □□ DrawContext □□□□□□□ 1.21.1 □□□□
 * 3. ScreenHandler □□□□□□ 1.21.1 □□□□□□Registries.SCREEN_HANDLER□
 * 4. Text API □ 1.21.1 □□□Text.literal()□
 *
 * □□□□□:
 * - □□□□□□□ ScreenHandler □□□□ syncId
 * - □□□□□□□ Packet □ syncId □□□□□□□
 * - □□□□□□□ syncId □ ScreenHandlerType □□ Screen □□
 * - Screen □ ScreenHandler □□ syncId □□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ====================================================================
    // ScreenHandlerType □□
    // ====================================================================
    // □1.20.6□□□□ Identifier.of() □□ new Identifier()
    // Registries.SCREEN_HANDLER □ 1.21.1 □□
    public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
        Registry.register(
            Registries.SCREEN_HANDLER,
            Identifier.of("examplemod", "example_gui"),  // □1.20.6□Identifier.of()
            new ScreenHandlerType<>(ExampleScreenHandler::new)
        );
    // □□□□□
```

```
private final Inventory inventory;

// ================================================================
// □□□□□□□
// ================================================================
public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
    this(syncId, playerInventory, new SimpleInventory(9));
}

// ================================================================
// □□□□□□□
// ================================================================
public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
    super(EXAMPLE_SCREEN_HANDLER, syncId);
    checkSize(inventory, 9);
    this.inventory = inventory;
    inventory.onOpen(playerInventory.player);

    // □□□□□□□□□□□□□□□□□□□□□□□□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            this.addSlot(new Slot(inventory, j + i * 3, 8 + j * 18, 18 + i * 18));
        }
    }

    // □□□□□□□□□□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 9; j++) {
            this.addSlot(new Slot(playerInventory, 9 + j + i * 9, 8 + j * 18, 66 + i * 18));
        }
    }

    // □□□□□□
    for (int j = 0; j < 9; j++) {
        this.addSlot(new Slot(playerInventory, j, 8 + j * 18, 124));
    }
}

// ================================================================
// shift+click □□□□□□□
// ================================================================
@Override
public ItemStack transferSlot(PlayerEntity player, int index) {
    Slot slot = this.slots.get(index);
    if (!slot.hasStack()) {
        return ItemStack.EMPTY;
    }

    ItemStack originalStack = slot.getStack();
    ItemStack copyStack = originalStack.copy();

    if (index < 9) {
        if (!this.insertItem(originalStack, 9, 45, true)) {
            return ItemStack.EMPTY;
        }
    } else {
        if (!this.insertItem(originalStack, 0, 9, false)) {
            return ItemStack.EMPTY;
        }
    }

    if (originalStack.isEmpty()) {
        slot.setStack(ItemStack.EMPTY);
    } else {
        slot.markDirty();
    }
    return copyStack;
}

// ================================================================
// □□□□□□□□□□□□□□□□
// ================================================================
@Override
public boolean canUse(PlayerEntity player) {
    return this.inventory.canPlayerUse(player);
}
```

```
}

/**
 * ============================================================
 * □□□ Screen □□HandledScreen□
 * ============================================================
 * □1.20.6□□□ DrawContext □□□□□□ 1.21.1 □□□
 * □1.20.6□Identifier □□ Identifier.of()
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□□
     *
     * □1.20.6□□□ DrawContext □□□□ 1.21.1 □□□
     * 1.20.1 □□□: drawBackground(MatrixStack matrices, ...)
     * 1.20.2+: drawBackground(DrawContext context, ...)
     *
     * □1.20.6□□□ Identifier □□ Identifier.of()
     */
    @Override
    protected void drawBackground(net.minecraft.client.gui.DrawContext context, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);

        // □1.20.6□□□ Identifier.of() □□ new Identifier()
        context.drawTexture(
            Identifier.of("examplemod", "textures/gui/example_gui.png"),
            (this.width - this.backgroundWidth) / 2,
            (this.height - this.backgroundHeight) / 2,
            0, 0, this.backgroundWidth, this.backgroundHeight
        );
    }

    /**
     * □□□□
     *
     * □1.20.6□□□ DrawContext □□
     */
    @Override
    public void render(net.minecraft.client.gui.DrawContext context, int mouseX, int mouseY, float delta) {
        // □1.20.6□renderBackground □□ DrawContext □□
        this.renderBackground(context, mouseX, mouseY, delta);
        super.render(context, mouseX, mouseY, delta);
        this.drawMouseoverTooltip(context, mouseX, mouseY);
    }
}

/**
 * ============================================================
 * □□□□□□
 * ============================================================
 * ScreenHandlerType □□□□□□□ModInitializer□:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER □□□□□□□□□□
 *
 * Screen □□□□□□□ClientModInitializer□:
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
 *   );
 *
 * ============================================================
```

```
 * □□ GUI □□□□□□□□ BlockEntity □ Item □□□□
 * ================================================================
 *
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       Text.literal("□□□□")
 *   ));
 *
 * ================================================================
 * ScreenHandler □□□□□
 * ================================================================
 * 1.20.6 □□ ScreenHandler □□□□□□ 1.21.1 □□□□□□
 * □ 1.21+ □□□ readNbt / writeNbt □□□□□□□□
 * RegistryWrapper.WrapperLookup □□□
 * □ 1.20.6 □□□□□□□□□□□□□□□□□□□ NBT □□□
 * ================================================================
 */
```

item/simple_item.java

```
/*
 * simple_item.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□ Data Components □□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□ Identifier.of("modid", "name")
 *      □1.20.5 □□□□□□□□□□□□□□□
 *
 *   2. □Data Components □□□□□□□□□□□□□□□
 *      1.20.6: Data Components □□□□□□□1.20.5 □□□□
 *              □□□□□□□ FOOD, MAX_DAMAGE □□□□□□□
 *              □□□□□□□□□ API □□□□□□□□□□□
 *              Item.Settings □ .component() □□□□□□□□
 *              FoodComponent □□□□□□□ Food □□□ component.type □
 *      1.21.1: Data Components □□□□□
 *              □□□□□□□□□ Registries.DATA_COMPONENT_TYPE
 *              Item.Settings □□□□ .component() □□
 *
 *   3. □FoodComponent / Food □□□□□□□□□
 *      1.20.4 □□□: net.minecraft.item.FoodComponent
 *      1.20.5+: □□□□□□□ Food □□□ component.type □
 *              □□□□□□□ 1.20.5 □□□□□□□□□
 *      1.21.1: net.minecraft.component.type.Food
 *
 *   4. □Java 21 □□□
 *      1.20.5+ □□□□□ Java 21□1.20.4 □□□□□ Java 17□
 *
 * □□ 1.20.4 □□□□□□□□□□□
 *   - 1.20.4 □□ new Identifier() □□□□
 *   - 1.20.4 □□ Data Components □□
 *   - 1.20.4 □□ Java 17
 *   - 1.20.4 □□ MatrixStack □□
 * ------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□ Data Components □□□□□□□ (Fabric 1.20.6)
 *
 * □□□□□□□□
 * 1. □1.20.5+ □□□□□□ Identifier.of() □□ new Identifier()□
 * 2. □□□□□□□□Data Components □□□□□□□□□□□□□□□□□□□□
```

```
 * - 使用 Data Component Types（如 FOOD, MAX_DAMAGE, CUSTOM_DATA）替代
 * - 使用 NBT 的物品改为 Data Components 组件
 * - 物品注册方式 API 也有较大变化，详见 1.21+ 版本文档
 * - ItemStack 的数据读写 set(DataComponentType, value) 等方法
 * - 在 Item.Settings 中 .component() 方法添加自定义组件数据
 */
public class ModItems {

    public static final String MOD_ID = "examplemod";

    // ========================================================================
    // 自定义组件类型 Data Component 注册（1.20.6 及以上）
    // ========================================================================
    // 1.20.5+ 引入了 Data Components 系统，自定义组件类型在 1.20.6 中
    // 的注册方式与 API 在后续版本（如 1.21+）有较大变化，详见文档。
    //
    // 1.21.1 中自定义组件类型（1.20.6 中同样适用）:
    //
    //   import net.minecraft.component.DataComponentType;
    //   import com.mojang.serialization.Codec;
    //
    //   public static final DataComponentType<Integer> CHARGE =
    //       Registry.register(
    //           Registries.DATA_COMPONENT_TYPE,
    //           Identifier.of(MOD_ID, "charge"),
    //           DataComponentType.<Integer>builder()
    //               .codec(Codec.INT)
    //               .build()
    //       );
    //
    // ========================================================================

    // ========================================================================
    // 物品注册
    // ========================================================================

    /**
     * 普通物品（不可堆叠）
     *
     * Item.Settings 常用方法在 1.21.1 中：
     *   .maxCount(64)     最大堆叠
     *   .maxDamage(256)   最大耐久
     *   .rarity(Rarity.UNCOMMON)  稀有度
     *   .fireproof()      防火
     *
     * 在1.20.6 中可通过Item.Settings 的 .component() 方法添加组件
     * 但在后续 1.21 版本中该方法已被移除，请注意
     * 自定义组件请使用 Fabric API 的附件系统
     */
    public static final Item EXAMPLE_ITEM = register(
            "example_item",
            new Item(new Item.Settings()
                    .maxCount(64)
                    .rarity(net.minecraft.item.Rarity.UNCOMMON)
            )
    );

    /**
     * 耐久物品（可损坏）
     * maxDamage 设置后物品自动变为不可堆叠（maxCount=1）
     */
    public static final Item EXAMPLE_DURABLE_ITEM = register(
            "example_durable_item",
            new Item(new Item.Settings()
                    .maxDamage(256)
                    .rarity(net.minecraft.item.Rarity.UNCOMMON)
            )
    );

    /**
     * 食物物品（1.20.6 及以上）
     *
     * 在1.20.6 中用FoodComponent 类，在后续版本 Food 类已被移除
     *   1.20.4 及以下: import net.minecraft.item.FoodComponent;
     *   1.20.5+: 已被移除 import net.minecraft.component.type.Food;
     *            食物组件现在通过数据组件系统注册
```

```
 *
 * □□□□□□□□□□ 1.20.5+ □□□□□□ Data Components □□□
 * □ Food.Builder □□□□□□□□□□□□□□□□□□□□
 */
// public static final Item EXAMPLE_FOOD = register(
//         "example_food",
//         new Item(new Item.Settings()
//                 .food(new net.minecraft.item.FoodComponent.Builder()
//                         .hunger(6)
//                         .saturationModifier(0.6f)
//                         .meat()
//                         .build()
//                 )
//         )
// );

/**
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Registries.ITEM□
 *
 * □1.20.6□□□ Identifier.of() □□ new Identifier()
 * □1.20.5 □□□□□□□□□□□□□□□□□□□□□□□□
 */
public static Item register(String name, Item item) {
    // □1.20.4 □□□□new Identifier(MOD_ID, name)
    // □1.20.5+□Identifier.of(MOD_ID, name) ── □□□□□□□□
    return Registry.register(Registries.ITEM, Identifier.of(MOD_ID, name), item);
}

/**
 * □□□□□□□□□□□□□□□□ JVM □□ ModItems □□
 */
public static void registerModItems() {
    @SuppressWarnings("unused")
    Item ensureClassInit = EXAMPLE_ITEM;
}

// ============================================================================
// Data Components □□□□□□□□1.20.6 □□□□□□
// ============================================================================
//
// 1.20.5+ □□□ Data Components □□□□□□□□□□□□ ItemStack NBT□
// □ 1.20.6 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
//
// □□□□□□□□□□□□□□□□□□□□□□:
//
//   import net.minecraft.component.DataComponentTypes;
//
//   ItemStack stack = new ItemStack(ModItems.EXAMPLE_ITEM);
//
//   // □□□□□□□□□□□□□□□□□□□□
//   // Text customName = stack.get(DataComponentTypes.CUSTOM_NAME);
//
//   // □□□□□□□□
//   // stack.set(DataComponentTypes.CUSTOM_NAME, Text.literal("□□□□□"));
//
// □□□□□□□ NBT □□□1.20.4 □□□□□□
//   stack.getOrCreateTag().putString("CustomName", "□□□□□");
//
// □□□□□1.20.6 □□□□□□□□Data Components API □□□□□□□□
// □□□□ 1.21+ □□□□□□□□□□□□ Data Components□
// ============================================================================

// ============================================================================
// □□□□□□□□□□ 1.21.1 □□□□□□□□
// ============================================================================
//
//   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
//   import net.minecraft.item.ItemGroups;
//
//   ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
//       .register(entries -> entries.add(ModItems.EXAMPLE_ITEM));
// ============================================================================

// ============================================================================
// □□□□□□□□
// ============================================================================
```

```
    // 1. □□□□□assets/<modid>/lang/en_us.json
    //      { "item.examplemod.example_item": "Example Item" }
    // 2. □□□□□assets/<modid>/models/item/example_item.json
    // 3. □□□□□assets/<modid>/textures/item/example_item.png
    // ========================================================================

    // □□□□□□□□□□
    private ModItems() {
    }
}
```

mixin/inject_mixin.java

```
/*
 * inject_mixin.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□@Inject Mixin □□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□
 *   Mixin □□ API □ 1.20.6 □ 1.21.1 □□□□□□□□
 *   □□□□□□ org.spongepowered.asm.mixin.* □□□
 *   □□□□□: □□□□□□ JAVA_21 □ compatibilityLevel□
 *
 * □□ 1.20.4 □□□□□□□□□□□
 *   Mixin □□ API □□□□□□
 *   1.20.4 □□ JAVA_17□1.20.6 □□ JAVA_21□1.20.5+ □□ Java 21□□
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   Mixin API □□□□□□□1.19.2 □□ JAVA_17□
 * ------------------------------------------------------------------------
 */
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfoReturnable;

/**
 * @Inject Mixin □□□□□□□ - Fabric 1.20.6
 *
 * □□□□□□□□□□□ @Inject □□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□
 * - Mixin □□ API □□□□□
 * - compatibilityLevel □□ JAVA_21
 * - □□□□ Java 21 □□□□□□□□□ record □□□□□□switch □□□□□□
 *
 * @Inject □□□□□:
 * - □□□□□□□□□□□□□□HEAD, RETURN, TAIL □□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□ CallbackInfo □ CallbackInfoReturnable □□
 *
 * □□□□□□□□@At □□:
 * - "HEAD"         □□□□□
 * - "RETURN"       □□□□□□□□□ return □□□□□□
 * - "TAIL"         □□□□□
 * - "INVOKE"       □□□□□□□□□□□□□ target □□□□□□
 * - "INVOKE_ASSIGN" □□□□□□□□□□□□
 * - "FIELD_ACCESS" □□□□□□□
 *
 * □□□□□:
 * 1. □□□□□□□ mixin □□□□□□□□□□□
 * 2. @Mixin □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ CallbackInfo□
 * 4. □□□□□□□□□□□□□□□□ CallbackInfoReturnable<T> □□ CallbackInfo
 * 5. CallbackInfo.cancel() □□□□□□□□□□□□□□ RETURN/HEAD □□□□□□□
 * 6. CallbackInfoReturnable.setReturnValue() □□□□□□□□
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {
    /**
```

```
     * □ MinecraftClient.tick() □□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("HEAD"),
        cancellable = true
    )
    private void onTickStart(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□□□□□□□
        
        // □□: □□□□□□□□□□□□ tick
        // if (someCondition) {
        //     ci.cancel();
        // }
    }

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□ CallbackInfoReturnable □□□□□□□□□□□□
     */
    @Inject(
        method = "getCurrentFps",
        at = @At("RETURN"),
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // □□□□□□□□:
        // cir.setReturnValue(cir.getReturnValue() * 2);
        // cir.cancel();
    }
}
```

mixin/mixin_config.json

```
{
    // ==========================================================
    // Mixin □□□□□□□ - Fabric 1.20.6
    // ==========================================================
    // □□□□□: src/main/resources/examplemod.mixins.json
    //
    // □□□□□ fabric.mod.json □□□ "mixins" □□□□□:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // □□ 1.21.1 □□□□□□□
    //   1. □compatibilityLevel □□□
    //      1.20.6 □ 1.21.1 □□□□ JAVA_21
    //      □1.20.5+ □□□□□ Java 21□
    //
    //   2. Mixin □□ API □□□□□□
    //      □□□ @Mixin, @Inject, @At □□□
    //
    // □□ 1.20.4 □□□□□□□□□□□
    //   1.20.4 □□ JAVA_17□1.20.6 □□ JAVA_21
    //   □1.20.5 □ Java □□□□□□□
    //
    // □□□□□:
    // - required:           □□□□□□true=□□□□□□□false=□□□
    // - package:            Mixin □□□□□□□□□
    // - compatibilityLevel: Java□□□□□□1.20.6 □□ JAVA_21□
    // - mixins:             □□Mixin□□□□□□□□□□□□□□□□□
    // - client:             □□□□□□□□Mixin□
    // - server:              □□□□□□□□Mixin□□□□□□
```

```
    // - injectors:           □□□□□□
    //   - defaultRequire:    □□□□□□□□1=□□□□□□□□□0=□□□
    //
    // □□□□□:
    // 1. □1.20.6 □□□compatibilityLevel □□□ JAVA_21□1.20.5+ □□ Java 21□
    // 2. package □□□□□□Mixin□□□□□□□□
    // 3. mixins □□□□□□□ .java □□□□□□□□□
    // 4. client □□□□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_21",

    // □□Mixin□□□□□□□□□□□□□□□
    "mixins": [
        "ExampleInjectMixin"
    ],

    // □□□□□□□□Mixin□
    "client": [
        "client.ExampleClientMixin"
    ],

    // □□□□□□
    "injectors": {
        // defaultRequire: 1 □□□□□ @Inject □□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□
        // □□ 0 □□□□□□□□□□□□□□□□□□□□□□bug□
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ================================================================================
// build.gradle —— Fabric 1.20.6 □□□□□□□□□□Groovy DSL□
// ================================================================================
// □□□
//  - □□□□□□ Fabric 1.20.6 □□□□□□□□ Loom □□□□□□
//  - □□□□□□□□□□ gradle.properties □□□□□□□□□□
//  - □□□□□□□□□JDK 21 + Gradle 8.x□□□ 8.7 □□□□□□
//
// □□ 1.20.4 build.gradle □□□□□
//  1. Loom □□□□1.6-SNAPSHOT□□ 1.20.4 □□□□ Loom □□□□□□□ 1.20.6□
//  2. Minecraft □□□□1.20.6□1.20.4 □ 1.20.4□
//  3. Yarn □□□□1.20.6+build.1□1.20.4 □ 1.20.4+build.3□
//  4. Fabric Loader□0.15.11□1.20.4 □ 0.15.1□
//  5. Fabric API□0.100.7+1.20.6□1.20.4 □ 0.96.11+1.20.4□
//  6. Java □□□□21□□□□□□□□1.20.4 □ 17□1.20.5+ □□□□□□ Java 21!□
// ================================================================================
// □□□□□□□1.20.5+ □□□□□ Java 21□
//  - Mojang □ 1.20.5 □□□□□□□□□□□□ Java 21
//  - □□□□□□□□□□ Java 21 □□
//  - □□□□□ JDK 21□□ Temurin 21 / Microsoft OpenJDK 21□
//  - 1.20.6 □□□□ Data Components □□□□1.20.5 □□□□1.20.6 □□□□
// ================================================================================

plugins {
    // Fabric Loom 1.6□□□□ Minecraft 1.20.6□
    // □ 1.20.4 □□□□□□□ Loom □□□□□
    id 'fabric-loom' version '1.6-SNAPSHOT'
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
    archivesName = project.name
}

repositories {
    mavenCentral()
}
// --------------------------------------------------------------------------------
```

```
// □□□□□□□□□□□ gradle.properties
// ------------------------------------------------------------------------
dependencies {
    // Minecraft 1.20.6 □□□□
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn 1.20.6+build.1 □□□□
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader 0.15.11
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
    // Fabric API 0.100.7+1.20.6
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
}

// ------------------------------------------------------------------------
// □□□□□□□ gradle.properties □□ mod_version □□□□ fabric.mod.json
// ------------------------------------------------------------------------
processResources {
    inputs.property "version", project.version

    filesMatching("fabric.mod.json") {
        expand "version": project.version
    }
}

// ------------------------------------------------------------------------
// □□□□□□Minecraft 1.20.6 □□ Java 21
// □□□□□□□1.20.5+ □□□□□ Java 21□1.20.4 □ Java 17□
// □□□□□ 21 □ JDK □□□□□□□□□□
// ------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 21
}

java {
    withSourcesJar()

    // □□□□□1.20.5+ □□□□□ Java 21
    sourceCompatibility = JavaVersion.VERSION_21
    targetCompatibility = JavaVersion.VERSION_21
}

// ------------------------------------------------------------------------
// jar □□
// ------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/version_diff.md

```
# Fabric 1.20.6 □ 1.20.4 □□□□□□□□

> □□□□□□ Fabric 1.20.6 □□□□ 1.20.4 □□□□ API □□□□
> 1.20.5/1.20.6 □□□□□□□□□□□□□Java □□□□ 17 □□□□ 21□
> □□□□□ Data Components □□□□□□□□□□□ ItemStack NBT □□□□□□□□□
> AI □□□□ 1.20.6 □□□□□□□□□□□□□□□□□

---

## □□□□□

1.20.6 □□ 1.20.x □□□□□□□□□□1.20.4 → 1.20.5 → 1.20.6□□
□□ MC □□□□□□□□□□□□ 1.20.5 □□□□□□□□□□□□□
□□□□ Data Components □□□□ Java □□□□□□

| □□□□ | 1.20.4 | 1.20.6 |
|--------|--------|--------|
| **Java □□** | **17** | **21□□□□□□** |
| Registry □□□□ | `net.minecraft.registry.Registry` | □□ |
| Registries □ | `net.minecraft.registry.Registries` | □□ |
| Registry □□□□ | □□□□□□□□ | **□□□** |
| Identifier □□ | `new Identifier(modid, name)` | **`Identifier.of(modid, name)`** |
| □□□□□□□ | NBT (`ItemStack.getOrCreateTag()`) | **Data Components** (`stack.get/set()`) |
| FoodComponent | `net.minecraft.item.FoodComponent` | `net.minecraft.component.type.Food` |
```

FoodComponent.Builder.hunger()	`hunger(6)`	`nutrition(6)`
□□□□□	□□ `Enchantment` □	□□ `Enchantment` □□□□□□□□1.21 □□□□□□□
EnchantmentTarget □□	□□	□□□1.21 □□□□
□□□□□□	`ItemGroupEvents` + `ItemGroups`	□□
FabricBlockSettings	`.create()` / `.copyOf()`	□□
Material □	□□□	□□□□□□□□
Loom □□	1.6-SNAPSHOT	1.6-SNAPSHOT□□□□
Fabric Loader	0.15.1	0.15.11
Fabric API	0.96.11+1.20.4	0.100.7+1.20.6

□□Java □□□□□□□□□□

2.1 Java 21 □□□□

1.20.5+ **□□□□ Java 21**□1.20.4 □□□ Java 17□

```gradle
// 1.20.4 build.gradle
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}
java {
    sourceCompatibility = JavaVersion.VERSION_17
    targetCompatibility = JavaVersion.VERSION_17
}
```

```gradle
// 1.20.6 build.gradle
tasks.withType(JavaCompile).configureEach {
    it.options.release = 21
}
java {
    sourceCompatibility = JavaVersion.VERSION_21
    targetCompatibility = JavaVersion.VERSION_21
}
```

□□□□□□□□□ JDK 21□□□ Eclipse Temurin 21 □ Microsoft OpenJDK 21□□

2.2 □□□ Java 21 □□

1.20.6 □□□□□□□□ Java 21 □□□□□□
- Record □□Java 16 □□□
- Switch □□□□□□Java 21 □□□
- □□□ `"""..."""`□Java 15+ □□□15 □□□
- □□□□□□Virtual Threads□Java 21□
- `sealed` □□□□Java 17 □□□

□□Identifier □□□□□□□□□□

3.1 □□□□□□□□

1.20.5+ □□□ `new Identifier()` □□□□□□□□□□□□□□□□ `Identifier.of()`□

```java
// 1.20.4 —— □□□□□1.20.6 □□□□□□
new Identifier("modid", "name")

// 1.20.6 —— □□□□□
Identifier.of("modid", "name")
```

3.2 □□□□□□□□□□□

```java
// 1.20.4
Registry.register(Registries.ITEM, new Identifier("modid", "name"), item);

// 1.20.6
Registry.register(Registries.ITEM, Identifier.of("modid", "name"), item);
```

> `new Identifier()` □ 1.20.6 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□Data Components □□□□□□□□□

4.1 □□

1.20.5+ □□□□ **Data Components □□**□□□□□□□□ `ItemStack` NBT □□□□□□□□□
□□□□□□□□□□□□□□□□ Data Components □□□□

□□□□	1.20.4 (NBT)	1.20.6 (Data Components)
□□□□□□	`stack.getOrCreateTag().getInt("value")`	`stack.get(MY_COMPONENT)`
□□□□□□	`stack.getOrCreateTag().putInt("value", 100)`	`stack.set(MY_COMPONENT, 100)`
□□□□□□	`stack.getOrCreateTag().contains("value")`	`stack.contains(MY_COMPONENT)`
□□□□□□	NBT tag □□	`DataComponentTypes.XXX`

4.2 □□ Data Component □□

1.20.6 □□□□□□□□□□□□□□□`DataComponentTypes` □□□□

```java
import net.minecraft.component.DataComponentTypes;

DataComponentTypes.MAX_STACK_SIZE   // □□□□□□
DataComponentTypes.MAX_DAMAGE         // □□□□□□□
DataComponentTypes.DAMAGE             // □□□□□□□
DataComponentTypes.RARITY             // □□□
DataComponentTypes.FOOD               // □□□□□
DataComponentTypes.CUSTOM_NAME        // □□□□□□□
DataComponentTypes.LORE               // □□□□□
DataComponentTypes.ENCHANTMENTS       // □□
DataComponentTypes.REPAIR_COST        // □□□□□
DataComponentTypes.CREATIVE_SLOT_LOCK // □□□□□□□□
```

4.3 □□ ItemStack □□□□

```java
import net.minecraft.component.DataComponentTypes;
import net.minecraft.item.ItemStack;

// □□
int maxDamage = stack.get(DataComponentTypes.MAX_DAMAGE);

// □□
stack.set(DataComponentTypes.MAX_DAMAGE, 500);

// □□
boolean hasFood = stack.contains(DataComponentTypes.FOOD);
```

4.4 □□□□□ Data Component

```java
import net.minecraft.component.DataComponentType;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import com.mojang.serialization.Codec;

public class ModComponents {
    public static final String MOD_ID = "examplemod";

    // □□□□□ Integer □□□□□□□□□
    public static final DataComponentType<Integer> CHARGE =
        Registry.register(
            Registries.DATA_COMPONENT_TYPE,
            Identifier.of(MOD_ID, "charge"),
            DataComponentType.<Integer>builder()
                .codec(Codec.INT)
                .build()
        );

    public static void initialize() {
```

```
        // □□□□□□
    }
}
```

4.5 □ Item.Settings □□□□□□□□

```java
import net.minecraft.component.DataComponentTypes;

// 1.20.6 —— □ Item.Settings □□□□□□□□□
new Item(new Item.Settings()
    .maxCount(64)                    // □□□□□□□□□
    .maxDamage(256)                  // □□□□□□□□□
    .component(ModComponents.CHARGE, 0)  // □□□□□□□□□□□□□□
);
```

> □□□□□□`.maxCount()`, `.maxDamage()` □□□□□□□□□□□□□ Data Components□
> □□□□□□□□□ API□□□□□□□□□□□ `.component()` □□□□□□□□□

□□Food □□□□□□□□

5.1 □□□□□□□□□□

`FoodComponent` □□□□□□ `Food` □□□□□□□

```java
// 1.20.4
import net.minecraft.item.FoodComponent;
new FoodComponent.Builder().nutrition(6).saturationModifier(0.6f).build();

// 1.20.6
import net.minecraft.component.type.Food;
new Food.Builder().nutrition(6).saturationModifier(0.6f).build();
```

5.2 □□□□□□□

```java
// 1.20.6
new Item(new Item.Settings()
    .food(new Food.Builder()
        .nutrition(6)
        .saturationModifier(0.6f)
        .meat()
        .build()
    )
);
```

> □□□1.20.6 □ `FoodComponent` □□□□□□□□□□□ `net.minecraft.component.type.Food`□
> □□□□□□□□□□ Data Components □□□□`DataComponentTypes.FOOD`□

□□□□□□□□□

6.1 Registry □□□□□□

1.20.5+ □□□□□□ `Registry` □□□□□□□□□□□ `Registry.BLOCK`, `Registry.ITEM`□□
□□□□□ `Registries` □□

```java
// 1.20.4 —— Registry.BLOCK □□□□□□□□□□□□□□□
Registry.register(Registry.BLOCK, new Identifier("modid", "name"), block);

// 1.20.6 —— Registry.BLOCK □□□□□□□□ Registries.BLOCK
Registry.register(Registries.BLOCK, Identifier.of("modid", "name"), block);
```

6.2 □□ DATA_COMPONENT_TYPE □□□

```java
```

```
// 1.20.6 □□
Registries.DATA_COMPONENT_TYPE  // Data Component □□□□□
```

6.3 RegistryKey □□□

1.20.5+ □□□ `RegistryKey` □□□□

```java
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;

// □□ RegistryKey
RegistryKey<Item> key = RegistryKey.of(RegistryKeys.ITEM, Identifier.of("modid", "name"));
Registry.register(Registries.ITEM, key, item);
```

> □□□□□ `Registry.register(registry, id, value)` □□□□□□□□□ `Identifier`□

□□□□□□□□□□□□□□□□

7.1 □□□□

1.20.6 □□□□□□□□□□ `Enchantment` □□□ 1.20.4 □□□□
□ 1.20.5+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```java
// 1.20.6 —— □□□□□□ Enchantment
public class MyEnchantment extends Enchantment {
    public MyEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        super(rarity, target, slots);
    }
    @Override
    public int getMinPower(int level) { return 10 + (level - 1) * 15; }
    @Override
    public int getMaxLevel() { return 3; }
}
```

> □□□1.20.6 □□□□□□□□□ 1.20.4 □□□□□□□□□□□□□□□ 1.21+ □□□□
> 1.20.6 □□□□□□□□□□□□ `EnchantmentTarget` □□□1.21 □□□□□□□□□□

□□□□□□□

8.1 ItemStack NBT □□□□

1.20.5+ □□□□□□ `ItemStack.getOrCreateTag()` □ NBT □□□
□□□□□□□□□ Data Components □□□

```java
// 1.20.4 —— NBT □□
NbtCompound tag = stack.getOrCreateTag();
tag.putInt("value", 100);
int value = tag.getInt("value");

// 1.20.6 —— Data Components □□
stack.set(ModComponents.CHARGE, 100);
Integer charge = stack.get(ModComponents.CHARGE);
```

8.2 BlockEntity □□/□□□□

1.20.5+ □ `BlockEntity` □ `saveAdditional` □ `loadAdditional` □□□□□ `HolderLookup.Provider` □□□

```java
// 1.20.6
@Override
public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.saveAdditional(tag, registries);
    tag.putInt("value", this.value);
}
@Override
```

```
public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.loadAdditional(tag, registries);
    this.value = tag.getInt("value");
}
```

8.3 ScreenHandler □□

1.20.5+ □ `ScreenHandler` □□□□□□□□□□□□
□□□□□□□□□□ `RegistryWrapper.WrapperLookup` □□□

□□□□□□□□1.20.4 → 1.20.6□

1. **□□ JDK □ 21**□□□ build.gradle □□ Java □□□17 → 21□□
2. `new Identifier("modid", "name")` → `Identifier.of("modid", "name")`□□□□□□□
3. `Registry.BLOCK` □□□□□ → `Registries.BLOCK`□□□□□□□□□□□□
4. ItemStack NBT □□ → Data Components □□□□□□□□□
5. `FoodComponent` → `Food`□□□□□□□□□□□□
6. `FoodComponent.Builder().nutrition()` → `Food.Builder().nutrition()`□Builder □□□□□□
7. `BlockEntity.saveAdditional(CompoundTag)` → `saveAdditional(CompoundTag, HolderLookup.Provider)`□
8. □□ gradle.properties □□□□□□□□
   - Loom□`1.6-SNAPSHOT`□□□
   - Minecraft□`1.20.4` → `1.20.6`
   - Yarn□`1.20.4+build.3` → `1.20.6+build.1`
   - Fabric Loader□`0.15.1` → `0.15.11`
   - Fabric API□`0.96.11+1.20.4` → `0.100.7+1.20.6`

□□□□□□□

10.1 □ 1.20.6 □□□ Java 17

```java
// □ □□□1.20.6 □□ Java 21□build.gradle □□□□ release = 17
tasks.withType(JavaCompile).configureEach {
    it.options.release = 17
}

// □ □□□□□ Java 21
tasks.withType(JavaCompile).configureEach {
    it.options.release = 21
}
```

10.2 □ 1.20.6 □□□□ NBT API

```java
// □ □□□1.20.6 □□□ getOrCreateTag()
NbtCompound tag = stack.getOrCreateTag();

// □ □□□□□ Data Components
stack.set(ModComponents.CHARGE, 100);
```

10.3 □ 1.20.6 □□□□□ FoodComponent

```java
// □ □□□1.20.6 □ FoodComponent □□□□□□□□
import net.minecraft.item.FoodComponent;
new FoodComponent.Builder().nutrition(6).build();

// □ □□□□□□□□ Food □
import net.minecraft.component.type.Food;
new Food.Builder().nutrition(6).build();
```

10.4 □ 1.20.6 □□□ new Identifier()

```java
// □ □□□1.20.6 □ new Identifier() □□□
new Identifier("modid", "name");

// □ □□□□□□□□□□□□□
```

```
Identifier.of("modid", "name");
```

recipe/shaped_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.20.6
    // ============================================================
    // □□□□□: data/examplemod/recipes/shaped_recipe.json
    //
    // □□□□□□crafting_shaped□□□□□□ pattern □ key □□
    // □□□□□□□□□□□□□□ pattern □□□□□□
    //
    // □□ 1.21.1 □□□□□
    //   □□ JSON □□□□ 1.20.6 □ 1.21.1 □□□□□□□□□
    //   □□: 1.21.2+ □□□□□□□□□□□□□□□
    //   □ 1.20.6 □ 1.21.1 □□□□□□□□□□□□□□
    //
    // □□ 1.19.2 □□□□□
    //   □□ JSON □□□□□□□□□□
    //
    // □□□□□:
    // 1. pattern □□□□□□□□□□□□□
    // 2. pattern □□□□□□□□□□□ key □□□□□□□
    // 3. key □□□□□□□□□□□□□□□□□□□□□□□ #, X, *, @ □□
    // 4. □□□□□□□□□□□□□□
    // 5. result □□ count □□□□□ 64
    // 6. ingredient □□ tag □□□□ "tag" □□□□□ "item"
    // ============================================================

    "type": "minecraft:crafting_shaped",

    // pattern: □□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/shapeless_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□□ - Fabric 1.20.6
    // ============================================================
    // □□□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□□□□□crafting_shapeless□□□□□□□□□□□□
    // □□□□ ingredients □□□□□□□□□□□□□□□□□□□□
    //
    // □□ 1.21.1 □□□□□
    //   □□ JSON □□□□ 1.20.6 □ 1.21.1 □□□□□□□□□
    //   □□: 1.21.2+ □□□□□□□□□□□□□□□
    //   □ 1.20.6 □ 1.21.1 □□□□□□□□□□□□□□
    //
    // □□□□□:
    // 1. ingredients □□□□□□□□□□□□□□□□□
```

```
    // 2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // 3. □□□□ "tag" □□□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.6
    // ============================================================
    // □□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□ 1.21.1 □□□□□
    //   1.20.6 □□□ JSON □□□ 1.21.1 □□□□□
    //   □□□□□□□□: data/<mod_id>/tags/blocks/
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□
    //
    // □□□□:
    // 1. □□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□: data/<mod_id>/tags/items/□□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.20.6
    // ============================================================
    // □□□□□: data/examplemod/tags/items/example_item_tag.json
    //
    // □□ 1.21.1 □□□□□
    //   1.20.6 □□□ JSON □□□ 1.21.1 □□□□□
```

```
    //   □□□□□□□□: data/<mod_id>/tags/items/
    //
    // □1.21.1 □□□□□□
    //   □□□□□□ supported_items □□□□□□□□□□□□□□□□□□□□
    //   □□□ 1.20.x □ EnchantmentTarget □□
    //   □1.20.6 □ EnchantmentTarget □□□□
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□:
    //   minecraft:logs          □□
    //   minecraft:planks        □□
    //   minecraft:coals         □□□
    //   minecraft:saplings      □□
    //   minecraft:flowers       □
    //
    // □□□□□:
    // 1. □□□□□□ data/<mod_id>/tags/items/ □□□
    // 2. □□□□□□□□□□□□□□□□□□□□
    // 3. values □□□□ID□□□□□□□□□□
    // 4. replace □ true □□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
    ]
}
```

worldgen/ore_feature.java

```
/*
 * ore_feature.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.20.6 (Fabric Loader 0.16.0 + Fabric API 0.100.7 + Yarn □□)
 * Java □□: Java 21
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.21.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.6 □ 1.21.1 □□□ Identifier.of("modid", "name")
 *
 *   2. □□□□□ API □□□□□□
 *      1.20.6 □□□ Registries.CONFIGURED_FEATURE / Registries.PLACED_FEATURE
 *      □□ 1.21.1 □□□□□□□ Identifier □□ Identifier.of()□
 *
 *   3. □BiomeModifications API □□□
 *      Fabric API □ BiomeModifications □□□ 1.21.1 □□
 *
 *   4. □Java 21 □□□
 *      1.20.5+ □□□□□ Java 21□□ 1.21.1 □□□
 *
 *   5. □□□□□□ JSON □□□□□□□□□□□□
 *      1.21+ □□□□□□□□ JSON □□□□□□□
 *      1.20.6 □□□□□ 1.20.x □□□
 *
 * □□ 1.19.2 □□□□□□□□□□
 *   - 1.19.2 □ BuiltinRegistries □□□□□□□□□
 *   - 1.20.6 □ Registries □□□□□□□□□
 *   - 1.19.2 □ new Identifier()□1.20.6 □ Identifier.of()
 * ------------------------------------------------------------------------------
 */
package com.example.mod.worldgen;

import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.registry.entry.RegistryEntry;
import net.minecraft.util.Identifier;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.YOffset;
```

```java
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.feature.PlacedFeature;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.PlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;

import java.util.List;

/**
 * □□□□□□□□□□ - Fabric 1.20.6
 *
 * □□ 1.21.1 □□□□□□□
 * 1. □□ Identifier.of()□□ 1.21.1 □□□
 * 2. Registries.CONFIGURED_FEATURE / PLACED_FEATURE □ 1.21.1 □□
 * 3. BiomeModifications API □ 1.21.1 □□
 * 4. □□□□□□□□□□□ API □ 1.20.6 □ 1.21.1 □□□□□□□
 *
 * □□□□:
 *   ConfiguredFeature -> □□□ Registries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□ Registries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□
 *
 * □□□□:
 * - □1.20.6□□□□□ Identifier.of() □□ new Identifier()
 * - □□□□□□ ConfiguredFeature□□□□ PlacedFeature
 * - PlacedFeature □□ RegistryEntry<ConfiguredFeature> □□
 * - BiomeModifications.addFeature □□ RegistryKey<PlacedFeature>
 * - OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES □□□□□□□□
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□
 */
public class ModWorldGen {

    public static final String MOD_ID = "examplemod";

    // ConfiguredFeature □ RegistryKey
    // □1.20.6□□□ Identifier.of() □□ new Identifier()
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, Identifier.of(MOD_ID, "ore_example"));

    // PlacedFeature □ RegistryKey
    public static final RegistryKey<PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, Identifier.of(MOD_ID, "ore_example"));

    /**
     * □□□□□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        // ================================================================
        // 1. □□□□□ ConfiguredFeature
        // ================================================================
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),
            9
        );

        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,
            oreConfig
        );

        // □1.20.6□□□ Identifier.of() □□ new Identifier()
        // □□□ Registries.CONFIGURED_FEATURE
        // □1.19.2 □□□□□ BuiltinRegistries.CONFIGURED_FEATURE□
        Registry.register(
            Registries.CONFIGURED_FEATURE,
            Identifier.of(MOD_ID, "ore_example"),
            configuredFeature
        );

        // ================================================================
        // 2. □□□□□ PlacedFeature
```

```
        // ================================================================
        RegistryEntry<ConfiguredFeature<?, ?>> cfEntry = RegistryEntry.of(configuredFeature);

        List<PlacementModifier> placementModifiers = List.of(
            CountPlacementModifier.of(20),
            SquarePlacementModifier.of(),
            HeightRangePlacementModifier.uniform(
                YOffset.getBottom(),
                YOffset.fixed(64)
            )
        );

        PlacedFeature placedFeature = new PlacedFeature(
            cfEntry,
            placementModifiers
        );

        // □1.20.6□□□ Identifier.of() □□ new Identifier()
        // □□□ Registries.PLACED_FEATURE
        // □1.19.2 □□□□□ BuiltinRegistries.PLACED_FEATURE□
        Registry.register(
            Registries.PLACED_FEATURE,
            Identifier.of(MOD_ID, "ore_example"),
            placedFeature
        );

        // ================================================================
        // 3. □□ BiomeModifications □□□□□□□□□□
        // ================================================================
        BiomeModifications.addFeature(
            BiomeSelectors.foundInOverworld(),
            GenerationStep.Feature.UNDERGROUND_ORES,
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                Identifier.of(MOD_ID, "ore_example")  // □1.20.6□Identifier.of()
            )
        );
    }
}

/*
 * ================================================================================
 * □□□□□□□□□□
 * ================================================================================
 *
 * BiomeSelectors.all()                  □□□□□□□
 * BiomeSelectors.foundInOverworld()      □□□□□□□□
 * BiomeSelectors.foundInNether()         □□□□□□□
 * BiomeSelectors.foundInTheEnd()         □□□□□□□
 *
 * ================================================================================
 * □□□□□□□
 * ================================================================================
 *
 * GenerationStep.Feature□□□□□□□:
 * - RAW_GENERATION          □□□□□□□□□□
 * - LAKES                    □□
 * - LOCAL_MODIFICATIONS      □□□□
 * - UNDERGROUND_STRUCTURES   □□□□
 * - SURFACE_STRUCTURES       □□□□
 * - STRONGHOLDS              □□
 * - UNDERGROUND_ORES        □□□□□□□□□□
 * - UNDERGROUND_DECORATION   □□□□
 * - FLUID_SPRINGS            □□□
 * - VEGETAL_DECORATION       □□□□
 * - TOP_LAYER_MODIFICATION   □□□□
 */
```

Fabric 1.21.1

advancement/advancement.json

```
{
    // ============================================================
```

```
// 进度/成就配置 - Fabric 1.21.1
// ============================================================
// 文件路径: data/examplemod/advancements/example_advancement.json
//
// 适配 1.20.1 的要点：
//   整个 JSON 结构与 1.21.1 和 1.20.1 完全一致
//   inventory_changed 触发器的 "items"条件格式也一致
//   注意: 1.21.2+ 的数据组件格式在这里不适用
//   从 1.21.1 迁移到 1.20.1 无需修改
//
// 本文件演示Advancement的从 Minecraft 1.12+ 的所有用法
// 包括父子关系、显示设置、多种触发器、奖励、toast通知等完整功能
//
// 文件结构:
// - display:     显示设置（图标、标题、描述、背景、框架）
// - parent:      父成就（构建成就树）
// - criteria:    触发条件（定义如何完成成就）
// - requirements: 完成要求（多个条件时的AND / OR）
// - rewards:     奖励（经验、物品、战利品表、函数）
//
// 使用步骤:
// 1. 将此文件放入parent指定的路径下 display.background 可选
// 2. 1.21.1 与 inventory_changed 的物品条件格式 "items" 数组与 1.20.1 完全一致
// 3. criteria 中的键名可以自定义 trigger 必填
// 4. requirements 决定多个条件之间 AND还是 OR
// 5. frame 可选: task(默认) / goal(目标) / challenge(挑战)
// ============================================================

// 父成就ID，指向本成就在成就树中的上一级成就
"parent": "examplemod:example/root",

// 显示设置
"display": {
    // 图标
    "icon": {
        "item": "examplemod:example_item"
    },

    // 标题，使用 translate 翻译键或 literal 直接文本
    "title": {
        "translate": "advancements.examplemod.example.title"
    },

    // 描述
    "description": {
        "translate": "advancements.examplemod.example.description"
    },

    // 框架类型: task(圆形,默认) / goal(圆形,目标) / challenge(六边形,挑战)
    "frame": "task",

    // 是否显示右上角通知
    "show_toast": true,

    // 是否在聊天栏公告
    "announce_to_chat": true,

    // 是否隐藏（隐藏成就在完成前不可见）
    "hidden": false

    // 背景图，仅 background 在根成就上有效
    // "background": "minecraft:textures/block/stone.png"
},

// 触发条件
// 每个 key 是自定义名称，对应一个 trigger 和 conditions
"criteria": {
    "get_example_item": {
        // 触发器: 物品栏变化时触发
        "trigger": "minecraft:inventory_changed",
        "conditions": {
            // 1.21.1 格式 "items"数组，每个元素与 1.20.1 相同
            // 1.19.2 以前 "item"单个对象
            "items": [
                {
                    "items": [
```

```
                        "examplemod:example_item"
                    ]
                }
            ]
        }
    },
    "another_condition": {
        // 条件2: プレイヤーがオーバーワールドにいる
        "trigger": "minecraft:location",
        "conditions": {
            "player": {
                "location": {
                    "dimension": "minecraft:overworld"
                }
            }
        }
    }
},

// 複数条件の設定
// 配列内は AND（すべて必要）
// 配列外は OR（どれか一つ）
// 今回は: 条件1 get_example_item AND another_condition
"requirements": [
    ["get_example_item"],
    ["another_condition"]
],

// 報酬の設定
"rewards": {
    // 経験値報酬
    "experience": 10,

    // ルートテーブル報酬（アイテム）
    "loot": [
        "examplemod:example_loot_table"
    ],

    // 関数報酬（カスタム処理）
    "function": "examplemod:grant_example_reward"
}
}
```

block/simple_block.java

```
/*
 * simple_block.java
 * --------------------------------------------------------------------------
 * 対象環境: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn 推奨)
 * 主な内容: ブロック登録・設定・アイテム化の最小サンプル
 *
 * 注意 1.20.1 からの主な変更点:
 *   1. 「Identifier の生成方法」
 *      1.20.1: new Identifier("modid", "name")（コンストラクタ）
 *      1.21.1: Identifier.of("modid", "name")（ファクトリメソッド）に変更
 *
 *   2. 「Registry 参照の変更」
 *      1.20.1: Registry.BLOCK のように直接参照
 *      1.21.1: Registry.BLOCK は廃止 → Registries.BLOCK
 *
 *   3. 「Java 21 必須」
 *      1.21+ では最低 Java 21（1.20.x は Java 17）
 *
 *   4. 「BlockEntity 保存/読込のシグネチャ」
 *      1.21+ は saveAdditional / loadAdditional に HolderLookup.Provider 引数
 *
 *   5. 「FabricBlockSettings API の変更点」
 *      .copyOf(), .create(), .strength(), .requiresTool() などは 1.20.1 と同じ
 *
 * 注意 1.19.2 からの主な変更点:
 *   - 1.19.2 の net.minecraft.util.registry.Registry → 1.21.1 は net.minecraft.registry.Registry
 *   - 1.19.2 の Material 指定 → 1.21.1 は廃止
 *   - 1.19.2 の .group() → 1.21.1 は廃止 ItemGroupEvents
 *   - 1.19.2 の new Identifier() → 1.21.1 は Identifier.of()
 * --------------------------------------------------------------------------
 */
```

```java
package com.example.mod.block;

import net.fabricmc.fabric.api.item.v1.FabricItemSettings;
import net.fabricmc.fabric.api.object.builder.v1.block.FabricBlockSettings;
import net.minecraft.block.AbstractBlock;
import net.minecraft.block.Block;
import net.minecraft.block.Blocks;
import net.minecraft.item.BlockItem;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□□□□□□ (Fabric 1.21.1)
 *
 * □□□□□□□□
 * 1. □□□□□□□ Registries.BLOCK / Registries.ITEM□□ 1.20.1 □□□□□
 * 2. □□□□□□□Identifier □□ Identifier.of() □□ new Identifier()□
 * 3. FabricBlockSettings API □ 1.20.1 □□□□□□
 * 4. □□□□□□□□ ItemGroupEvents □□□□□ 1.20.1 □□□□□
 */
public class SimpleBlock extends Block {

    public static final String MOD_ID = "examplemod";

    /**
     * □□□□□□
     * □ static final □□□□□□□□□□□□□□□□
     */
    public static final Block EXAMPLE_BLOCK = registerBlock("example_block");

    public SimpleBlock(AbstractBlock.Settings settings) {
        super(settings);
    }

    /**
     * □□□□□□□□□□□□□□□□
     * □□□ Block □□□□□□□□□□□ BlockItem□
     *
     * □□ 1.20.1 □□□□□
     * - □□ Identifier.of() □□ new Identifier()□□□□□□□
     * - Registries.BLOCK / Registries.ITEM □ 1.20.1 □□
     * - FabricBlockSettings □□□ 1.20.1 □□
     */
    private static Block registerBlock(String name) {
        // ---- □□□□□□□□□□□□ ----
        // □ 1.20.1 □□□□□□ FabricBlockSettings □□
        AbstractBlock.Settings settings = FabricBlockSettings.copyOf(Blocks.STONE)
                .strength(3.0f)
                .requiresTool()
                .luminance(15);

        Block block = new SimpleBlock(settings);

        // ---- □□□□□□□ Block □ Registries.BLOCK ----
        // □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
        // 1.20.1: Registry.register(Registries.BLOCK, new Identifier(MOD_ID, name), block);
        // 1.21.1: Registry.register(Registries.BLOCK, Identifier.of(MOD_ID, name), block);
        Registry.register(Registries.BLOCK, Identifier.of(MOD_ID, name), block);

        // ---- □□□□□□□ BlockItem □ Registries.ITEM ----
        // □□□□ Identifier.of()
        // □□□□□□□□ .group()□□□□□ 1.19.3+ □□□
        BlockItem item = new BlockItem(block, new FabricItemSettings());

        Registry.register(Registries.ITEM, Identifier.of(MOD_ID, name), item);

        return block;
    }

    // ===========================================================================
    // □□□□□□□□
    // ===========================================================================
    //
    // □□□□□□□ ModInitializer □□□□□□□□
    //
```

```
    //   public class ExampleMod implements ModInitializer {
    //       @Override
    //       public void onInitialize() {
    //           Block block = SimpleBlock.EXAMPLE_BLOCK;
    //       }
    //   }
    //
    // ========================================================================
    // □□□□□□□□□□ 1.20.1 □□□□□□□
    // ========================================================================
    //
    //   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
    //   import net.minecraft.item.ItemGroups;
    //
    //   ItemGroupEvents.modifyEntriesEvent(ItemGroups.BUILDING_BLOCKS)
    //       .register(entries -> entries.add(SimpleBlock.EXAMPLE_BLOCK));
    //
    // ========================================================================
    // BlockEntity □□/□□□□□□□□□□□□□□□□□□□□
    // ========================================================================
    //
    // □1.21.1 □□□saveAdditional □ loadAdditional □□ HolderLookup.Provider □□□
    //
    //   @Override
    //   public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    //       super.saveAdditional(tag, registries);
    //       tag.putInt("value", this.value);
    //   }
    //
    //   @Override
    //   public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    //       super.loadAdditional(tag, registries);
    //       this.value = tag.getInt("value");
    //   }
    //
    // 1.20.1 □□□□□□□□ HolderLookup.Provider □□□
    // ========================================================================
}
```

command/simple_command.java

```
/*
 * simple_command.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□
 *   1. CommandRegistrationCallback API □ 1.20.1 □□□□□□v2 □□□
 *   2. CommandManager API □□
 *   3. Text API □□□Text.literal()□
 *   4. EntityArgumentType □□□□
 *   5. Java 21 □□□□□□□□□ record □□□□□□switch □□□□□□□
 *
 * □□ 1.19.2 □□□□□□□□□□
 *   - 1.19.2 □□ v1 □□□2 □□□□□□1.21.1 □□ v2 □□□3 □□□□
 *   - 1.19.2 □ new LiteralText()□1.21.1 □ Text.literal()
 * ------------------------------------------------------------------------------
 */
package com.example.mod.command;

import com.mojang.brigadier.CommandDispatcher;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.fabricmc.fabric.api.command.v2.CommandRegistrationCallback;
import net.minecraft.command.argument.EntityArgumentType;
import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.server.command.CommandManager;
import net.minecraft.server.command.ServerCommandSource;
import net.minecraft.text.Text;
import net.minecraft.util.Formatting;

import static net.minecraft.server.command.CommandManager.literal;
import static net.minecraft.server.command.CommandManager.argument;
/**
```

```
 * □□□□□□□□□ - Fabric 1.21.1
 *
 * □□ 1.20.1 □□□□□□□□□□□□API □□□□□□□□
 * □□□□ Java 21 □□□□□□□□ switch □□□□□□record □□□
 *
 * □□□□□□□□□□
 * 1. □□ CommandRegistrationCallback v2 □□□□
 * 2. □□ CommandDispatcher □□□□□□□
 * 3. □□□□□□argument□□□□
 * 4. □□□□□□□
 *
 * Fabric 1.21.1 □□ CommandRegistrationCallback v2 □□
 * v2 □□□□□□□□□□□□□□:
 *   (CommandDispatcher<ServerCommandSource> dispatcher,
 *    CommandRegistryAccess registryAccess,
 *    CommandManager.RegistrationEnvironment environment)
 */
public class ModCommands {

    /**
     * □□□□□
     * □□□□□□□□ModInitializer□□ onInitialize() □□□
     */
    public static void register() {
        CommandRegistrationCallback.EVENT.register(
            (dispatcher, registryAccess, environment) -> {
                registerExampleCommand(dispatcher);
            }
        );
    }

    /**
     * □□ /example □□□
     *
     * □□□□□:
     * /example                          - □□□□□□□
     * /example hello                    - □□□□□□□□
     * /example hello <player>           - □□□□□□□□□□
     * /example set <value>              - □□□□□□
     * /example give <player> <count>    - □□□□□□□□
     */
    private static void registerExampleCommand(CommandDispatcher<ServerCommandSource> dispatcher) {
        dispatcher.register(
            literal("example")
                .executes(context -> {
                    context.getSource().sendFeedback(
                        () -> Text.literal("□□: /example <hello|set|give>")
                            .formatted(Formatting.YELLOW),
                        false
                    );
                    return 1;
                })

                .then(literal("hello")
                    .executes(ModCommands::sayHelloToSelf)
                    .then(argument("target", EntityArgumentType.player())
                        .executes(ModCommands::sayHelloToTarget)
                    )
                )

                .then(literal("set")
                    .then(argument("value", IntegerArgumentType.integer(0, 100))
                        .executes(ModCommands::setValue)
                    )
                )

                .then(literal("give")
                    .then(argument("target", EntityArgumentType.player())
                        .then(argument("count", IntegerArgumentType.integer(1, 64))
                            .executes(ModCommands::giveItem)
                        )
                    )
                )
        );
    }
    /**
```

```
 * /example hello - □□□□□□□□
 */
private static int sayHelloToSelf(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity player = source.getPlayer();

    source.sendFeedback(
        () -> Text.literal("Hello, " + player.getName().getString() + "!")
            .formatted(Formatting.GREEN),
        false
    );
    return 1;
}

/**
 * /example hello <player> - □□□□□□□□□□
 */
private static int sayHelloToTarget(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    ServerCommandSource source = context.getSource();
    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");

    target.sendMessage(
        Text.literal("Hello from " + source.getName() + "!")
            .formatted(Formatting.GOLD)
    );

    source.sendFeedback(
        () -> Text.literal("□□ " + target.getName().getString() + " □□□□")
            .formatted(Formatting.GREEN),
        true
    );
    return 1;
}

/**
 * /example set <value> - □□□
 */
private static int setValue(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    int value = IntegerArgumentType.getInteger(context, "value");
    ServerCommandSource source = context.getSource();

    source.sendFeedback(
        () -> Text.literal("□□□□□: " + value)
            .formatted(Formatting.AQUA),
        false
    );
    return 1;
}

/**
 * /example give <player> <count> - □□□□□□□
 */
private static int giveItem(CommandContext<ServerCommandSource> context)
        throws CommandSyntaxException {

    PlayerEntity target = EntityArgumentType.getPlayer(context, "target");
    int count = IntegerArgumentType.getInteger(context, "count");
    ServerCommandSource source = context.getSource();

    net.minecraft.item.ItemStack stack = new net.minecraft.item.ItemStack(
        net.minecraft.item.Items.DIAMOND, count
    );
    target.getInventory().insertStack(stack);

    source.sendFeedback(
        () -> Text.literal("□□ " + target.getName().getString()
            + " □□□ " + count + " □□□")
            .formatted(Formatting.GREEN),
        true
    );
```

```
        return 1;
    }
}
```

enchantment/custom_enchantment.java

```
/*
 * custom_enchantment.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□
 *
 *   1. □□□□□□□□□□□□□□□□□□□□□□□□
 *      1.20.1: □□ Enchantment □□□□□ Java □□□□□
 *              new ExampleEnchantment(Rarity.RARE, EnchantmentTarget.WEAPON, EquipmentSlot.MAINHAND)
 *              Registry.register(Registries.ENCHANTMENT, new Identifier(...), enchantment)
 *
 *      1.21.1: Enchantment □ record□□□□□□□□□□ JSON □□□□□□□
 *              Java □□□□□□ RegistryKey □□
 *              □□□□□□□□ data/<modid>/enchantment/ □□□ JSON □□□
 *
 *   2. □EnchantmentTarget □□□□□□□
 *      1.20.1: EnchantmentTarget.WEAPON / ARMOR / TOOL □□□
 *      1.21.1: □□□□□□□ (supported_items) □□□□□□□□
 *
 *   3. □□□□□□□□ JSON □□□
 *      1.20.1: getMinPower() / getMaxPower() / getMaxLevel() / canCombine() □□□□
 *      1.21.1: JSON □□ min_cost / max_cost / max_level / exclusive_set
 *
 *   4. □□□□□□□□□□□□□□□
 *      1.20.1: □□□□□□□□□□□□
 *      1.21.1: JSON □□ effects □□□□□□□□□□□□
 *
 *   5. □Identifier □□□□□□□□
 *      1.20.1: new Identifier("modid", "name")
 *      1.21.1: Identifier.of("modid", "name")
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □ 1.20.1 □□□□ Enchantment □□□□□□□□
 *   - 1.21.1 □□□□□□□□□□
 *   - 1.19.2 □ Registry.ENCHANTMENT□1.21.1 □ Registries.ENCHANTMENT
 * ------------------------------------------------------------------------
 */
package com.example.mod.enchantment;

import net.minecraft.enchantment.Enchantment;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.util.Identifier;

/**
 * □□□□□□□□ - Fabric 1.21.1
 *
 * □□□□□□□1.21.1 □□□□□□□□□□□□□□□□
 * □□□□□□□ Enchantment □□□□□□□□
 * □□□□□□□□□□□□ JSON □□□□□□Java □□□□□□□□□□□□
 *
 * □□□□□□□□□
 * 1. □ Java □□□□□□□ RegistryKey □□
 * 2. □□□□ JSON □□□□□□□□
 * 3. □□□□□□□□□□□□□□□□
 *
 * □□□□□:
 *   1. □□ JSON □□□□□□□ -> data/<modid>/enchantment/<name>.json
 *   2. □□□□□□□□□□□□□□□□ -> data/<modid>/tags/items/enchantable/<tag>.json
 *   3. □ Java □□□□□□ RegistryKey □□□□□□□□
 *   4. □□ RegistryKey □□□□□□□□□□□□
 */
public class ModEnchantments {

    public static final String MOD_ID = "examplemod";

    // ====================================================================
    // □□ RegistryKey□Java □□□□□
```

```
// ================================================================
// 1.21.1 
// 
// RegistryKey :
// - 
// - /
// - 
//
// 1.21.1 Identifier.of() new Identifier()
public static final RegistryKey<Enchantment> EXAMPLE_ENCHANTMENT =
    RegistryKey.of(
        RegistryKeys.ENCHANTMENT,
        Identifier.of(MOD_ID, "example_enchantment")  // 1.21.1Identifier.of()
    );

/**
 * 
 * ModInitializer onInitialize()
 *
 * 1.21.1 Registry.register()
 * JSON 
 * Java RegistryKey 
 */
public static void register() {
    // RegistryKey 
    @SuppressWarnings("unused")
    RegistryKey<Enchantment> key = EXAMPLE_ENCHANTMENT;
}

// ================================================================
// JSON 
// ================================================================
// : data/examplemod/enchantment/example_enchantment.json
//
// 1.21.1 JSON
//
// {
//     "description": {
//         "translate": "enchantment.examplemod.example_enchantment"
//     },
//     "supported_items": "#examplemod:enchantable/example_weapon",
//     "weight": 2,
//     "max_level": 3,
//     "min_cost": {
//         "base": 10,
//         "per_level_above_first": 15
//     },
//     "max_cost": {
//         "base": 60,
//         "per_level_above_first": 15
//     },
//     "anvil_cost": 4,
//     "slots": ["mainhand"],
//     "effects": {
//         "minecraft:damage": [
//             {
//                 "effect": {
//                     "type": "minecraft:add_value",
//                     "value": {
//                         "type": "minecraft:linear",
//                         "base": 2.0,
//                         "per_level_above_first": 1.5
//                     }
//                 }
//             }
//         ]
//     }
// }
//
// :
// - description:       
// - supported_items:    # 
// - weight:             1=VERY_RARE, 2=RARE, 5=UNCOMMON, 10=COMMON
// - max_level:          getMaxLevel()
// - min_cost:           getMinPower()
//   - base:             
```

```
    //   - per_level_above_first: □□□□□□□
    // - max_cost:           □□□□□□□□□□ getMaxPower()□
    // - anvil_cost:         □□□□□□□
    // - slots:              □□□□□□□□mainhand/offhand/head/chest/legs/feet□
    // - effects:            □□□□□□□□□□□□□□□□

    // ================================================================
    // □□□□□□□□□□
    // ================================================================
    // □□□□□: data/examplemod/tags/items/enchantable/example_weapon.json
    //
    // □1.21.1 □□□□□□□□□□□□□□□□□□□□□□□□□□□□ EnchantmentTarget □□
    //
    // {
    //     "replace": false,
    //     "values": [
    //         "minecraft:iron_sword",
    //         "minecraft:diamond_sword",
    //         "minecraft:netherite_sword",
    //         "examplemod:example_sword"
    //     ]
    // }
    //
    // □□ JSON □□ supported_items □□□□□□□:
    //   "supported_items": "#examplemod:enchantable/example_weapon"
}

/*
 * ================================================================================
 * □□□□□□□□ RegistryKey □□□□□
 * ================================================================================
 *
 * 1. □□□□□□□□□□□□:
 *
 *   // □□□□□□□□□□□□□□□□
 *   Enchantment enchantment = world.getRegistryManager()
 *       .getOrThrow(RegistryKeys.ENCHANTMENT)
 *       .getOrThrow(ModEnchantments.EXAMPLE_ENCHANTMENT);
 *
 *   // □□□□□□□□□□□□
 *   int level = stack.getEnchantments().getLevel(enchantment);
 *   if (level > 0) {
 *       // □□□□□□□□□□□□ level
 *   }
 *
 * 2. □□□□□□□□□:
 *   /enchant @s examplemod:example_enchantment 3
 *
 * ================================================================================
 * 1.20.1 □□□□□□□□□□□□□□□□□□□□ 1.21.1□
 * ================================================================================
 *
 * // 1.20.1: □□ Enchantment □
 * public class ExampleEnchantment extends Enchantment {
 *     public ExampleEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
 *         super(rarity, target, slots);
 *     }
 *     @Override
 *     public int getMinPower(int level) { return 10 + (level - 1) * 15; }
 *     @Override
 *     public int getMaxLevel() { return 3; }
 * }
 *
 * // 1.20.1: □□ Registry.register □□
 * Registry.register(Registries.ENCHANTMENT,
 *     new Identifier("examplemod", "example_enchantment"),
 *     new ExampleEnchantment(Enchantment.Rarity.RARE,
 *         EnchantmentTarget.WEAPON, EquipmentSlot.MAINHAND));
 *
 * ================================================================================
 * □□□□□□□□□□
 * ================================================================================
 * □□□□□: assets/examplemod/lang/en_us.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "Example Enchantment"
```

```
 * }
 *
 * □□□□□□□: assets/examplemod/lang/zh_cn.json
 *
 * {
 *     "enchantment.examplemod.example_enchantment": "□□□□□"
 * }
 *
 * ============================================================================
 * □□ ID □□□□□
 * ============================================================================
 * - □□□ Identifier □□□□□□
 * - □□□□□□□□□: enchantment.<mod_id>.<enchantment_name>
 * - □□ JSON □□□□□: data/<modid>/enchantment/<name>.json
 * - □□□□□□□□□□: data/<modid>/tags/items/enchantable/<tag>.json
 */
```

entity/simple_entity.java

```
/*
 * simple_entity.java
 * ----------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.1: new Identifier("modid", "name")
 *      1.21.1: Identifier.of("modid", "name")□□□□□□□□□□
 *
 *   2. □Java 21 □□□
 *      1.21+ □□□□□ Java 21□1.20.x □ Java 17□
 *
 *   3. □□□□□ API □□□□□□
 *      1.21.1 □□□ Registries.ENTITY_TYPE□□ 1.20.1 □□□
 *      □ Registry □□□□□□□□□□□□□□
 *
 *   4. □EntityType.Builder.build() □□□□□□□□
 *      1.21+ □□□□□□□ RegistryKey<EntityType<?>> □□
 *
 * □□ 1.19.2 □□□□□□□□□□□□
 *   - 1.19.2 □ Registry.ENTITY_TYPE□1.21.1 □ Registries.ENTITY_TYPE
 *   - 1.19.2 □ net.minecraft.util.registry.Registry□1.21.1 □ net.minecraft.registry.Registry
 *   - 1.19.2 □ new Identifier()□1.21.1 □ Identifier.of()
 * ----------------------------------------------------------------------------
 */
package com.example.mod.entity;

import net.fabricmc.fabric.api.object.builder.v1.entity.FabricDefaultAttributeRegistry;
import net.minecraft.entity.EntityType;
import net.minecraft.entity.SpawnGroup;
import net.minecraft.entity.attribute.DefaultAttributeContainer;
import net.minecraft.entity.attribute.EntityAttributes;
import net.minecraft.entity.mob.PathAwareEntity;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import net.minecraft.world.World;

/**
 * □□□□□□□□□ - Fabric 1.21.1
 *
 * □□□□□□□□□
 * 1. □□□□□□□□□□□□□□
 * 2. □□ Registry.register □□ EntityType
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□□□ Identifier.of() □□ new Identifier()
 * 2. Registries.ENTITY_TYPE □ 1.20.1 □□
 * 3. FabricDefaultAttributeRegistry API □ 1.20.1 □□
 * 4. Java 21 □□□□□□□□□ record □□□□□□□
 *
 * □□□□□□□:
 * - □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
 * - □□□□□□□□□□□□□□□ (EntityType, World) □□
```

```
 * - createMobAttributes() □□□□□□□□□ DefaultAttributeContainer
 * - □□□□□□□□□ FabricDefaultAttributeRegistry
 */
public class ExampleEntity extends PathAwareEntity {

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
    // Registries.ENTITY_TYPE □ 1.20.1 □□
    public static final EntityType<ExampleEntity> EXAMPLE_ENTITY =
        Registry.register(
            Registries.ENTITY_TYPE,
            Identifier.of("examplemod", "example_entity"),  // □1.21.1□Identifier.of()
            EntityType.Builder.<ExampleEntity>create(
                    ExampleEntity::new,
                    SpawnGroup.CREATURE
                )
                .dimensions(0.6f, 1.8f)
                .trackingRange(8)
                .build()
        );

    // ================================================================
    // □□□□□
    // ================================================================
    public ExampleEntity(EntityType<? extends PathAwareEntity> entityType, World world) {
        super(entityType, world);
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
    // □□□□□:
    // - GENERIC_MAX_HEALTH      □□□□□□□□□ 20 = 10 □□□
    // - GENERIC_MOVEMENT_SPEED  □□□□□□□□ 0.25□
    // - GENERIC_ATTACK_DAMAGE   □□□□□□□□ 2□
    // - GENERIC_FOLLOW_RANGE     AI □□□□□□□□ 16□
    // - GENERIC_ARMOR           □□□
    // - GENERIC_KNOCKBACK_RESISTANCE □□□□□0.0~1.0□
    public static DefaultAttributeContainer createMobAttributes() {
        return PathAwareEntity.createMobAttributes()
            .add(EntityAttributes.GENERIC_MAX_HEALTH, 20.0)
            .add(EntityAttributes.GENERIC_MOVEMENT_SPEED, 0.25)
            .add(EntityAttributes.GENERIC_ATTACK_DAMAGE, 3.0)
            .add(EntityAttributes.GENERIC_FOLLOW_RANGE, 16.0)
            .add(EntityAttributes.GENERIC_ARMOR, 0.0)
            .add(EntityAttributes.GENERIC_KNOCKBACK_RESISTANCE, 0.0);
    }

    // ================================================================
    // □□ tick □□
    // ================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□□□□□ tick □□□
    }

    // ================================================================
    // □□□ AI □□
    // ================================================================
    @Override
    protected void initGoals() {
        // □□□□□□□□ AI □□□Goal□
        // □□ AI:
        // - MeleeAttackGoal        □□□□□□□
        // - WanderAroundGoal       □□□□
        // - LookAtEntityGoal       □□□□
        // - FollowTargetGoal       □□□□
        // - SwimGoal               □□
    }

    // ================================================================
    // □□□□□□□
    // ================================================================
```

```
    // □□□□□□□□ModInitializer□□ onInitialize() □□□□
    //
    //   FabricDefaultAttributeRegistry.register(
    //       ExampleEntity.EXAMPLE_ENTITY,
    //       ExampleEntity.createMobAttributes()
    //   );
    // ================================================================
}
```

gui/screen_with_menu.java

```
/*
 * screen_with_menu.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□GUI□□□□□ ScreenHandler + HandledScreen □□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.1: new Identifier("modid", "name")
 *      1.21.1: Identifier.of("modid", "name")
 *
 *   2. □□□ API □□□
 *      1.20.1: □□ MatrixStack □□ GUI □□
 *      1.21.1: □□ DrawContext □□ GUI □□□1.20.2 □□□
 *
 *   3. □ScreenHandler □□□□□□□□□
 *      1.21+ □ ScreenHandler □□□□□□□□□□□□
 *      □□□□□□□□□□□□□□□ RegistryWrapper.WrapperLookup □□□
 *
 *   4. □Java 21 □□□
 *      1.21+ □□□□□ Java 21
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □ Registry.SCREEN_HANDLER□1.21.1 □ Registries.SCREEN_HANDLER
 *   - 1.19.2 □ new LiteralText()□1.21.1 □ Text.literal()
 *   - 1.19.2 □ MatrixStack□1.21.1 □ DrawContext
 *   - 1.19.2 □ new Identifier()□1.21.1 □ Identifier.of()
 * ------------------------------------------------------------------------------
 */
package com.example.mod.gui;

import net.minecraft.entity.player.PlayerEntity;
import net.minecraft.entity.player.PlayerInventory;
import net.minecraft.inventory.Inventory;
import net.minecraft.inventory.SimpleInventory;
import net.minecraft.item.ItemStack;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.screen.ScreenHandler;
import net.minecraft.screen.ScreenHandlerType;
import net.minecraft.screen.slot.Slot;
import net.minecraft.text.Text;
import net.minecraft.util.Identifier;

/**
 * □□GUI□□□ScreenHandler + HandledScreen□- Fabric 1.21.1
 *
 * □□ 1.20.1 □□□□□□
 * 1. □□□□□□□□□ Identifier.of() □□ new Identifier()
 * 2. □□□□□□□□□ DrawContext □□ MatrixStack□□ 1.20.4 □□□
 * 3. ScreenHandler □□□□□ 1.20.1 □□□□□□Registries.SCREEN_HANDLER□
 * 4. Text API □ 1.20.1 □□□Text.literal()□
 *
 * □□□□□:
 * - □□□□□□ ScreenHandler □□□ syncId
 * - □□□□□□ Packet □ syncId □□□□□□
 * - □□□□□□ syncId □ ScreenHandlerType □□ Screen □□
 */
public class ExampleScreenHandler extends ScreenHandler {

    // ================================================================
    // ScreenHandlerType □□
    // ================================================================
    // □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
    // Registries.SCREEN_HANDLER □ 1.20.1 □□
```

```
public static final ScreenHandlerType<ExampleScreenHandler> EXAMPLE_SCREEN_HANDLER =
    Registry.register(
        Registries.SCREEN_HANDLER,
        Identifier.of("examplemod", "example_gui"),  // □1.21.1□Identifier.of()
        new ScreenHandlerType<>(ExampleScreenHandler::new)
    );

// □□□□□
private final Inventory inventory;

// ================================================================
// □□□□□□□
// ================================================================
public ExampleScreenHandler(int syncId, PlayerInventory playerInventory) {
    this(syncId, playerInventory, new SimpleInventory(9));
}

// ================================================================
// □□□□□□□
// ================================================================
public ExampleScreenHandler(int syncId, PlayerInventory playerInventory, Inventory inventory) {
    super(EXAMPLE_SCREEN_HANDLER, syncId);
    checkSize(inventory, 9);
    this.inventory = inventory;
    inventory.onOpen(playerInventory.player);

    // □□□□□□□□□□□□□□□□□□□□□□□□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            this.addSlot(new Slot(inventory, j + i * 3, 8 + j * 18, 18 + i * 18));
        }
    }

    // □□□□□□□□□□
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 9; j++) {
            this.addSlot(new Slot(playerInventory, 9 + j + i * 9, 8 + j * 18, 66 + i * 18));
        }
    }

    // □□□□□□
    for (int j = 0; j < 9; j++) {
        this.addSlot(new Slot(playerInventory, j, 8 + j * 18, 124));
    }
}

// ================================================================
// shift+click □□□□□□□
// ================================================================
@Override
public ItemStack transferSlot(PlayerEntity player, int index) {
    Slot slot = this.slots.get(index);
    if (!slot.hasStack()) {
        return ItemStack.EMPTY;
    }

    ItemStack originalStack = slot.getStack();
    ItemStack copyStack = originalStack.copy();

    if (index < 9) {
        if (!this.insertItem(originalStack, 9, 45, true)) {
            return ItemStack.EMPTY;
        }
    } else {
        if (!this.insertItem(originalStack, 0, 9, false)) {
            return ItemStack.EMPTY;
        }
    }

    if (originalStack.isEmpty()) {
        slot.setStack(ItemStack.EMPTY);
    } else {
        slot.markDirty();
    }
    return copyStack;
}
```

```
    // ================================================================
    // □□□□□□□□□□□□□□□□
    // ================================================================
    @Override
    public boolean canUse(PlayerEntity player) {
        return this.inventory.canPlayerUse(player);
    }
}

/**
 * ================================================================
 * □□□ Screen □□HandledScreen□
 * ================================================================
 * □1.21.1 □□□□□ DrawContext □□□□□□ 1.20.4 □□□1.20.1 □ MatrixStack□
 * □1.21.1 □□□Identifier □□ Identifier.of()
 */
class ExampleScreen extends net.minecraft.client.gui.screen.ingame.HandledScreen<ExampleScreenHandler> {

    public ExampleScreen(ExampleScreenHandler handler, PlayerInventory inventory, Text title) {
        super(handler, inventory, title);
    }

    @Override
    protected void init() {
        super.init();
        this.backgroundWidth = 176;
        this.backgroundHeight = 166;
        this.titleX = (this.backgroundWidth - this.textRenderer.getWidth(this.title)) / 2;
    }

    /**
     * □□□□□□□
     *
     * □1.21.1 □□□□□ DrawContext □□□□ 1.20.4 □□□
     * 1.20.1: drawBackground(MatrixStack matrices, ...)
     * 1.21.1: drawBackground(DrawContext context, ...)
     *
     * □1.21.1 □□□□□ Identifier □□ Identifier.of()
     */
    @Override
    protected void drawBackground(net.minecraft.client.gui.DrawContext context, float delta, int mouseX, int mouseY) {
        net.minecraft.client.render.RenderSystem.setShaderColor(1.0f, 1.0f, 1.0f, 1.0f);

        // □1.21.1□□□ Identifier.of() □□ new Identifier()
        context.drawTexture(
            Identifier.of("examplemod", "textures/gui/example_gui.png"),
            (this.width - this.backgroundWidth) / 2,
            (this.height - this.backgroundHeight) / 2,
            0, 0, this.backgroundWidth, this.backgroundHeight
        );
    }

    /**
     * □□□□
     *
     * □1.21.1 □□□□□ DrawContext □□
     */
    @Override
    public void render(net.minecraft.client.gui.DrawContext context, int mouseX, int mouseY, float delta) {
        // □1.21.1 □□□renderBackground □□ DrawContext □□
        this.renderBackground(context, mouseX, mouseY, delta);
        super.render(context, mouseX, mouseY, delta);
        this.drawMouseoverTooltip(context, mouseX, mouseY);
    }
}

/**
 * ================================================================
 * □□□□□□□
 * ================================================================
 * ScreenHandlerType □□□□□□□□ModInitializer□:
 *   // ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER □□□□□□□□□□□□
 *
 * Screen □□□□□□□□ClientModInitializer□:
 *
 *   import net.fabricmc.fabric.api.client.screen.v1.ScreenRegistry;
```

```
 *
 *   ScreenRegistry.register(
 *       ExampleScreenHandler.EXAMPLE_SCREEN_HANDLER,
 *       ExampleScreen::new
 *   );
 *
 * ================================================================
 * □□ GUI □□□□□□□□ BlockEntity □ Item □□□□
 * ================================================================
 *
 *   player.openHandledScreen(new SimpleNamedScreenHandlerFactory(
 *       (syncId, playerInventory, player) -> new ExampleScreenHandler(syncId, playerInventory, blockInventory),
 *       Text.literal("□□□□")
 *   ));
 *
 * ================================================================
 * 1.21.1 ScreenHandler □□□□□□□□
 * ================================================================
 * 1.21+ □ ScreenHandler □□□□□□□□□□□□:
 * - □□ readNbt / writeNbt □□□□□□□ RegistryWrapper.WrapperLookup □□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ API
 * - BlockEntity □ saveAdditional / loadAdditional □□□□
 *   HolderLookup.Provider □□
 * ================================================================
 */
```

item/simple_item.java

```
/*
 * simple_item.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□□□□ Data Components □□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.1: new Identifier("modid", "name")
 *      1.21.1: Identifier.of("modid", "name")□□□□□□□□□□
 *
 *   2. □Data Components □□□□□ NBT□□□□□□□□
 *      1.20.1: ItemStack NBT (stack.getOrCreateTag().putInt("value", 100))
 *      1.21.1: Data Components (stack.set(MY_COMPONENT, 100))
 *      □□□□□□□□□□ Registries.DATA_COMPONENT_TYPE
 *
 *   3. □Item.Settings □□ .component() □□□
 *      1.21.1 □□□ Item.Settings □□□□□□ Data Component □□
 *      new Item(new Item.Settings().component(MY_COMPONENT, 0))
 *
 *   4. □Food □□□□□□
 *      1.20.1: net.minecraft.item.FoodComponent
 *      1.21.1: net.minecraft.component.type.Food□□□□□□□□□□□□
 *
 *   5. □Java 21 □□□
 *      1.21+ □□□□□ Java 21□1.20.x □ Java 17□
 *
 * □□ 1.19.2 □□□□□□□□□□□□
 *   - 1.19.2 □ Registry.ITEM□1.21.1 □ Registries.ITEM
 *   - 1.19.2 □□ .group()□1.21.1 □□□ ItemGroupEvents
 *   - 1.19.2 □ new Identifier()□1.21.1 □ Identifier.of()
 *   - 1.19.2 □ Data Components □□
 * ------------------------------------------------------------------------------
 */
package com.example.mod.item;

import net.minecraft.component.DataComponentType;
import net.minecraft.component.DataComponentTypes;
import net.minecraft.item.Item;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import com.mojang.serialization.Codec;

/**
 * □□□□□ Data Components □□ (Fabric 1.21.1)
 *
 * □□□□□□□
```

```
 * 1. □□□□□□□□□ Identifier.of() □□ new Identifier()□
 * 2. □□□□Data Components □□□□□ NBT□□□□□□□□□□□□□
 * 3. Item.Settings □□□□□□□maxCount, maxDamage □□□ 1.20.1 □□□
 * 4. □□ .component() □□□□□□ Item.Settings □□□□□□□□□□
 */
public class ModItems {

    public static final String MOD_ID = "examplemod";

    // ============================================================================
    // □□□□□□□□ Data Component □□
    // ============================================================================
    // 1.21+ □□□□□□□□□□□□□□ Data Components □□□□□□□□ NBT□
    // □□□□□□□□□□□ Registries.DATA_COMPONENT_TYPE□
    // □□□□□□□□□□□□ Codec □□□□□□□□□□□□□□□

    /**
     * □□□□□□□ "charge"□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
     *
     * □□□□□□
     *   1. □□ DataComponentType.Builder
     *   2. □□ Codec□□□□□□□□
     *   3. □□□ Registries.DATA_COMPONENT_TYPE
     */
    public static final DataComponentType<Integer> CHARGE =
        Registry.register(
            Registries.DATA_COMPONENT_TYPE,
            Identifier.of(MOD_ID, "charge"),    // □1.21.1□Identifier.of()
            DataComponentType.<Integer>builder()
                .codec(Codec.INT)               // □□ Codec.INT □□□
                .build()
        );

    // ============================================================================
    // □□□□□
    // ============================================================================

    /**
     * □□□□□□□□□□□□□□□□
     *
     * Item.Settings □□□□□□□ 1.20.1 □□□
     *   .maxCount(64)     □□□□□□
     *   .maxDamage(256)   □□□□□□
     *   .rarity(Rarity.UNCOMMON)  □□□
     *   .fireproof()      □□
     *
     * □1.21.1 □□□.component() □□□□□□□□□□□□
     *   .component(ModItems.CHARGE, 0)  □□ CHARGE □□□□□□□□ 0
     */
    public static final Item EXAMPLE_ITEM = register(
            "example_item",
            new Item(new Item.Settings()
                    .maxCount(64)
                    .rarity(Rarity.UNCOMMON)
                    .component(ModItems.CHARGE, 0)  // □1.21.1 □□□□□□□□□□□□
            )
    );

    /**
     * □□□□□□□□□□□□□□□□
     * maxDamage □□□□□□□□□□□□□□□□□maxCount=1□□
     */
    public static final Item EXAMPLE_DURABLE_ITEM = register(
            "example_durable_item",
            new Item(new Item.Settings()
                    .maxDamage(256)
                    .rarity(Rarity.UNCOMMON)
            )
    );

    /**
     * □□□□□□□□□□
     *
     * □1.21.1 □□□FoodComponent □□□ Food□□□□□□□
     *   1.20.1: import net.minecraft.item.FoodComponent;
```

```
 *   1.21.1: import net.minecraft.component.type.Food;
 *
 * Food.Builder □□□□□
 *   1.20.1: .hunger(6)         □1.20.2+ □□□
 *   1.21.1: .nutrition(6)
 */
// public static final Item EXAMPLE_FOOD = register(
//         "example_food",
//         new Item(new Item.Settings()
//                 .food(new Food.Builder()
//                         .nutrition(6)
//                         .saturationModifier(0.6f)
//                         .meat()
//                         .build()
//                 )
//         )
// );

/**
 * □□□□□□□□□□□□□□□□□□□□□□□□□□□□ Registries.ITEM□
 *
 * □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
 */
public static Item register(String name, Item item) {
    // □1.20.1□new Identifier(MOD_ID, name)
    // □1.21.1□Identifier.of(MOD_ID, name) —— □□□□□□□□
    return Registry.register(Registries.ITEM, Identifier.of(MOD_ID, name), item);
}

/**
 * □□□□□□□□□□□□□□□ JVM □□ ModItems □□
 */
public static void registerModItems() {
    @SuppressWarnings("unused")
    Item ensureClassInit = EXAMPLE_ITEM;
}

// ============================================================================
// Data Components □□□□
// ============================================================================
//
// □□□□□□□ ItemStack □□□□□□
//
//   ItemStack stack = new ItemStack(ModItems.EXAMPLE_ITEM);
//
//   // □□□□□□□□□
//   stack.set(ModItems.CHARGE, 42);
//
//   // □□□□□□□□□□□□□ Integer □ null□
//   Integer charge = stack.get(ModItems.CHARGE);
//   if (charge != null && charge > 0) {
//       // □□□□
//   }
//
//   // □□□□□□□
//   int maxDamage = stack.get(DataComponentTypes.MAX_DAMAGE);  // □□□ null
//
//   // □□□□□□□□□
//   boolean hasCharge = stack.contains(ModItems.CHARGE);
//
// □□□ 1.20.1 □ NBT □□□□□□□□□
//   stack.getOrCreateTag().putInt("charge", 42);
//   int charge = stack.getOrCreateTag().getInt("charge");
// ============================================================================

// ============================================================================
// □□□□□□□□ 1.20.1 □□□□□□□
// ============================================================================
//
//   import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
//   import net.minecraft.item.ItemGroups;
//
//   ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
//       .register(entries -> entries.add(ModItems.EXAMPLE_ITEM));
// ============================================================================
// ============================================================================
```

```
    // □□□□□□□□
    // ============================================================================
    // 1. □□□□□□assets/<modid>/lang/en_us.json
    //      { "item.examplemod.example_item": "Example Item" }
    // 2. □□□□□□assets/<modid>/models/item/example_item.json
    // 3. □□□□□□assets/<modid>/textures/item/example_item.png
    // ============================================================================

    // □□□□□□□□□□
    private ModItems() {
    }
}
```

mixin/inject_mixin.java

```
/*
 * inject_mixin.java
 * ------------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□@Inject Mixin □□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□
 *   Mixin □□ API □ 1.21.1 □ 1.20.1 □□□□□□□□
 *   □□□□□□ org.spongepowered.asm.mixin.* □□□
 *   □□□□□: compatibilityLevel □ JAVA_17 □□ JAVA_21□
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   Mixin API □□□□□□□1.19.2 □□ JAVA_17□1.21.1 □□ JAVA_21□
 * ------------------------------------------------------------------------------
 */
package com.example.mod.mixin;

import net.minecraft.client.MinecraftClient;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfoReturnable;

/**
 * @Inject Mixin □□□□□□□ - Fabric 1.21.1
 *
 * □□□□□□□□□□□ @Inject □□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□
 * - Mixin □□ API □□□□□
 * - compatibilityLevel □ JAVA_17 □□ JAVA_21□□ mixin_config.json □□□□□
 * - □□□ Java 21 □□□□□□□□□ record □□□□□□switch □□□□□□
 *
 * @Inject □□□□□:
 * - □□□□□□□□□□□□□□HEAD, RETURN, TAIL □□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□ CallbackInfo □ CallbackInfoReturnable □□
 *
 * □□□□□□□□□@At □□:
 * - "HEAD"         □□□□□
 * - "RETURN"       □□□□□□□□□ return □□□□□□
 * - "TAIL"         □□□□□
 * - "INVOKE"       □□□□□□□□□□□□□ target □□□□□□
 * - "INVOKE_ASSIGN" □□□□□□□□□□□□□
 * - "FIELD_ACCESS" □□□□□□□□
 *
 * □□□□□:
 * 1. □□□□□□□ mixin □□□□□□□□□□□□
 * 2. @Mixin □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ CallbackInfo□
 * 4. □□□□□□□□□□□□□□□□□ CallbackInfoReturnable<T> □□ CallbackInfo
 * 5. CallbackInfo.cancel() □□□□□□□□□□□□□□ RETURN/HEAD □□□□□□□
 * 6. CallbackInfoReturnable.setReturnValue() □□□□□□□□
 */
@Mixin(MinecraftClient.class)
public class ExampleInjectMixin {

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     */
```

```
    @Inject(
        method = "tick",
        at = @At("HEAD"),
        cancellable = true
    )
    private void onTickStart(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□□□□□□□□

        // □□: □□□□□□□□□□□ tick
        // if (someCondition) {
        //     ci.cancel();
        // }
    }

    /**
     * □ MinecraftClient.tick() □□□□□□□□□
     */
    @Inject(
        method = "tick",
        at = @At("TAIL"),
        cancellable = false
    )
    private void onTickEnd(CallbackInfo ci) {
        // □□□□□□ tick □□□□□□
    }

    /**
     * □□□□□□□□□□□□□□□□□□
     * □□ CallbackInfoReturnable □□□□□□□□□□□□
     */
    @Inject(
        method = "getCurrentFps",
        at = @At("RETURN"),
        cancellable = true
    )
    private void onGetCurrentFps(CallbackInfoReturnable<Integer> cir) {
        // □□□□□□□□:
        // cir.setReturnValue(cir.getReturnValue() * 2);
        // cir.cancel();
    }
}
```

mixin/mixin_config.json

```
{
    // ============================================================
    // Mixin □□□□□□□ - Fabric 1.21.1
    // ============================================================
    // □□□□□: src/main/resources/examplemod.mixins.json
    //
    // □□□□□ fabric.mod.json □□□□ "mixins" □□□□□:
    //   "mixins": [
    //     "examplemod.mixins.json"
    //   ]
    //
    // □□ 1.20.1 □□□□□□□
    //   1. □compatibilityLevel □□□□
    //      1.20.1: JAVA_17
    //      1.21.1: JAVA_21□1.21+ □□□□□ Java 21□
    //
    //   2. Mixin □□ API □□□□□□
    //      □□□□ @Mixin, @Inject, @At □□□□
    //
    // □□□□□:
    // - required:           □□□□□□true=□□□□□□□false=□□□
    // - package:            Mixin □□□□□□□□□
    // - compatibilityLevel: Java□□□□□□1.21.1 □□ JAVA_21□
    // - mixins:             □□Mixin□□□□□□□□□□□□□□□□□□
    // - client:             □□□□□□□□Mixin□
    // - server:              □□□□□□□□Mixin□□□□□□
    // - injectors:           □□□□□□
    //   - defaultRequire:    □□□□□□□□□1=□□□□□□□□□0=□□□
    //
    // □□□□□:
    // 1. □1.21.1 □□□compatibilityLevel □□□□ JAVA_21□1.20.x □ JAVA_17□
    // 2. package □□□□□□Mixin□□□□□□□□
```

```
    // 3. mixins □□□□□□□ .java □□□□□□□□□
    // 4. client □□□□□□□□□□□□□□□□□□□□□□□□
    // ============================================================

    "required": true,
    "package": "com.example.mod.mixin",
    "compatibilityLevel": "JAVA_21",

    // □□Mixin□□□□□□□□□□□□□□□□
    "mixins": [
        "ExampleInjectMixin"
    ],

    // □□□□□□□□Mixin□
    "client": [
        "client.ExampleClientMixin"
    ],

    // □□□□□□
    "injectors": {
        // defaultRequire: 1 □□□□□ @Inject □□□□□□□□□□□□
        // □□□□□□□□□□□□□□□□□□□□□
        // □□ 0 □□□□□□□□□□□□□□□□□□□□□□□□bug□
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ================================================================================
// build.gradle —— Fabric 1.21.1 □□□□□□□□□□Groovy DSL□
// ================================================================================
// □□□
//  - □□□□□□ Fabric 1.21.1 □□□□□□□ Loom □□□□□
//  - □□□□□□□□□ gradle.properties □□□□□□□□□
//  - □□□□□□□□JDK 21 + Gradle 8.x□□□ 8.7 □□□□□□
//
// □□ 1.20.1 build.gradle □□□□
//  1. Loom □□□1.7-SNAPSHOT□1.20.1 □ 1.6-SNAPSHOT□
//  2. Minecraft □□□1.21.1□1.20.1 □ 1.20.1□
//  3. Yarn □□□1.21.1+build.3□1.20.1 □ 1.20.1+build.10□
//  4. Fabric Loader□0.16.0□1.20.1 □ 0.16.x□□□□□□□□□□
//  5. Fabric API□0.102.0+1.21.1□1.20.1 □ 0.83.0+1.20.1□
//  6. Java □□□21□1.20.1 □ 17□□□□□□□□1.21+ □□□□□ Java 21□
// ================================================================================

plugins {
    // Fabric Loom 1.7□□□ Minecraft 1.21.1□
    // 1.21+ □□ Loom 1.7+□□□□ Loom □□□□□ 1.21 □□□□
    id 'fabric-loom' version '1.7-SNAPSHOT'
    id 'maven-publish'
}

version = project.mod_version
group = project.maven_group

base {
    archivesName = project.name
}

repositories {
    mavenCentral()
}

// --------------------------------------------------------------------------------
// □□□□□□□□□□ gradle.properties
// --------------------------------------------------------------------------------
dependencies {
    // Minecraft 1.21.1 □□□ jar
    minecraft "com.mojang:minecraft:${project.minecraft_version}"
    // Yarn 1.21.1+build.3 □□□□
    mappings "net.fabricmc:yarn:${project.yarn_mappings}:v2"
    // Fabric Loader 0.16.0
    modImplementation "net.fabricmc:fabric-loader:${project.loader_version}"
    // Fabric API 0.102.0+1.21.1
    modImplementation "net.fabricmc.fabric-api:fabric-api:${project.fabric_version}"
```

```
}

// --------------------------------------------------------------------------
// □□□□□□□ gradle.properties □□ mod_version □□□□ fabric.mod.json
// --------------------------------------------------------------------------
processResources {
    inputs.property "version", project.version

    filesMatching("fabric.mod.json") {
        expand "version": project.version
    }
}

// --------------------------------------------------------------------------
// □□□□□□Minecraft 1.21.1 □□ Java 21
// □□□□□□□1.21+ □□□□□ Java 21□1.20.x □ Java 17□
// □□□□□ 21 □ JDK □□□□□□□□□□
// --------------------------------------------------------------------------
tasks.withType(JavaCompile).configureEach {
    it.options.release = 21
}

java {
    withSourcesJar()

    // □□□□□1.21+ □□□□□ Java 21
    sourceCompatibility = JavaVersion.VERSION_21
    targetCompatibility = JavaVersion.VERSION_21
}

// --------------------------------------------------------------------------
// jar □□
// --------------------------------------------------------------------------
jar {
    from("LICENSE") {
        rename { "${it}_${project.base.archivesName.get()}" }
    }
}
```

project/version_diff.md

```
# Fabric 1.21.1 □ 1.20.1 □□□□□□□

> □□□□□□ Fabric 1.21.1 □□□□ 1.20.1□□□□□□□□□□□ API □□□□
> 1.21.x □□□□□□□□□□□□□□□□ Data Components □□□□□□□□□□□□□
> Java □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

---

## □□□□□

| □□□□ | 1.20.1 | 1.21.1 |
|--------|--------|--------|
| Java □□ | 17 | **21□□□□□□□** |
| Loom □□ | 1.6-SNAPSHOT | 1.7-SNAPSHOT |
| Fabric Loader | 0.16.x | 0.16.0 |
| Fabric API | 0.83.0+1.20.1 | 0.102.0+1.21.1 |
| Registry □□□□ | `net.minecraft.registry.Registry` | □□ |
| Registries □ | `net.minecraft.registry.Registries` | □□ |
| Registry.BLOCK □□□□□ | □□□□□□□□□ | **□□□** |
| Identifier □□ | `new Identifier(modid, name)` | **`Identifier.of(modid, name)`** |
| □□□□□□□□ | NBT (`ItemStack.getOrCreateTag()`) | **Data Components** (`stack.get/set()`) |
| □□□□□ | □□ `Enchantment` □ | **□□□□□□JSON/Record□** |
| EnchantmentTarget □□ | □□ | **□□□□□□□□□□□** |
| FoodComponent | `net.minecraft.item.FoodComponent` | `net.minecraft.component.type.Food` |
| FoodComponent.Builder.hunger() | `hunger(6)` | `nutrition(6)` |
| BlockBehaviour/AbstractBlock | `AbstractBlock.Settings` | □□□□□□ `AbstractBlock.Settings`□ |
| □□□□□□ | `ItemGroupEvents` + `ItemGroups` | □□ |
| FabricItemSettings | □□ | □□□□□□□□□□ Item.Settings |

---

## □□Java □□□□□□□□□□□□

### 2.1 Java 21 □□□□

1.21+ **□□□□□ Java 21**□□□ Java 17 □□□□□□□□□□
```

```gradle
// build.gradle
tasks.withType(JavaCompile).configureEach {
    it.options.release = 21
}

java {
    sourceCompatibility = JavaVersion.VERSION_21
    targetCompatibility = JavaVersion.VERSION_21
}
```

□□□□□□□□□ JDK 21□□□ Eclipse Temurin 21 □ Oracle JDK 21□□

2.2 Java 21 □□□□

1.21.x □□□□□□□□ Java 21 □□□□□□
- Record □□□□□
- Switch □□□□□
- □□□□□Text Blocks□Java 15+ □□□21 □□□
- □□□□□□Virtual Threads□

□□Identifier □□□□□□□□□□

3.1 □□□□□□□

1.21+ □□□ `new Identifier()` □□□□□□□□□□□□□□□ `Identifier.of()`□

```java
// 1.20.1 —— □□□□□1.21 □□□□□□
new Identifier("modid", "name")

// 1.21.1 —— □□□□
Identifier.of("modid", "name")
```

3.2 □□□□□□□□□□

□□□□ `Identifier` □□□□□□□□□□□□

```java
// 1.20.1
Registry.register(Registries.ITEM, new Identifier("modid", "name"), item);

// 1.21.1
Registry.register(Registries.ITEM, Identifier.of("modid", "name"), item);
```

> `new Identifier()` □ 1.21 □□□□□□□□□□□□□□□

```
DataComponentTypes.MAX_STACK_SIZE   // □□□□□
DataComponentTypes.MAX_DAMAGE         // □□□□□
DataComponentTypes.DAMAGE             // □□□□□
DataComponentTypes.RARITY             // □□□
DataComponentTypes.FOOD               // □□□□
DataComponentTypes.CUSTOM_NAME        // □□□□□
DataComponentTypes.LORE               // □□□□
DataComponentTypes.ENCHANTMENTS       // □□
DataComponentTypes.REPAIR_COST        // □□□□
DataComponentTypes.CREATIVE_SLOT_LOCK // □□□□□□
```

4.3 □□ ItemStack □□□□

```java
import net.minecraft.component.DataComponentTypes;
import net.minecraft.item.ItemStack;

// □□
int maxDamage = stack.get(DataComponentTypes.MAX_DAMAGE);

// □□
stack.set(DataComponentTypes.MAX_DAMAGE, 500);

// □□
boolean hasFood = stack.contains(DataComponentTypes.FOOD);
```

4.4 □□□□□ Data Component

```java
import net.minecraft.component.DataComponentType;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.util.Identifier;
import com.mojang.serialization.Codec;

public class ModComponents {
    public static final String MOD_ID = "examplemod";

    // □□□□ Integer □□□□□□□□□
    public static final DataComponentType<Integer> CHARGE =
        Registry.register(
            Registries.DATA_COMPONENT_TYPE,
            Identifier.of(MOD_ID, "charge"),
            DataComponentType.<Integer>builder()
                .codec(Codec.INT)
                .build()
        );

    // □□□□□□□□□□□□□□□□□□□□ Codec □□□□
    // public static final DataComponentType<MyData> MY_DATA =
    //     Registry.register(
    //         Registries.DATA_COMPONENT_TYPE,
    //         Identifier.of(MOD_ID, "my_data"),
    //         DataComponentType.<MyData>builder()
    //             .codec(MyData.CODEC)
    //             .build()
    //     );

    public static void initialize() {
        // □□□□□□
    }
}
```

4.5 □ Item.Settings □□□□□□□

```java
import net.minecraft.component.DataComponentTypes;

// 1.21.1 —— □ Item.Settings □□□□□□□□
new Item(new Item.Settings()
    .maxCount(64)                    // □□□□□□□□
    .maxDamage(256)                  // □□□□□□□□
    .component(ModComponents.CHARGE, 0)  // □□□□□□□□□□□□
```

);
```

> □□□□□`.maxCount()`, `.maxDamage()` □□□□□□□□□□□□□ Data Components□
> □□□□□□□□□□ API□□□□□□□□□□□ `.component()` □□□□□□□□□

### 4.6 □ ItemStack □□□□□□□□

```java
ItemStack stack = new ItemStack(ModItems.EXAMPLE_ITEM);

// □□□□□□□□□
stack.set(ModComponents.CHARGE, 42);

// □□□□□□□□□
int charge = stack.get(ModComponents.CHARGE);  // □□ 42
// □□□□□□□□□ null□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

// □□□□□
Integer charge = stack.get(ModComponents.CHARGE);
if (charge != null) {
    // □□ charge
}
```

□□□□□□□□□□□□□□□□□□□□□□

5.1 □□

1.21+ □□□□□□□□□□□□□□□□□`Enchantment` □□□□□□□□□□□□□□□
□□□□□ **record□□□□□□□□**□□□□□□□□□□□JSON□□□□□□□□□

□□□□	1.20.1	1.21.1
Enchantment □	□□□□□□□	**record□□□□□□□□**
□□□□□□□□□□	`extends Enchantment` + `Registry.register`	**JSON □□□□□** □ `Enchantment.Builder`
□□□□□□	`EnchantmentTarget` □□	**□□□□□□** (`supported_items`)
□□□□□□	`EquipmentSlot[]`	`EquipmentSlotGroup`
□□/□□□□□□□□	`getMinPower()` / `getMaxPower()`	**JSON □□□** `min_cost` / `max_cost`
□□□□□□□□□	`getMaxLevel()`	**JSON □□□** `max_level`
□□□□□□	`canCombine()`	**JSON □□□** `exclusive_set`
□□□□□□	□□□□□□	**□□□□□□□□** (`effects`)

5.2 1.20.1 □□□□□□□□□□□□□□□ 1.21.1□

```java
// 1.20.1 —— □□ Enchantment □
public class MyEnchantment extends Enchantment {
    public MyEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        super(rarity, target, slots);
    }
    @Override
    public int getMinPower(int level) { return 10 + (level - 1) * 15; }
    @Override
    public int getMaxLevel() { return 3; }
}
```

5.3 1.21.1 □□□□□□□□□□□□□□

1.21.1 □□□□□□ JSON □□□□□□□□□□□ `data/<modid>/enchantment/` □□□□□

```json
// data/examplemod/enchantment/example_enchantment.json
{
    "description": {
        "translate": "enchantment.examplemod.example_enchantment"
    },
    "supported_items": "#examplemod:enchantable/example_weapon",
    "weight": 2,
    "max_level": 3,
    "min_cost": {
        "base": 10,
        "per_level_above_first": 15
```
```

```
    },
    "max_cost": {
        "base": 60,
        "per_level_above_first": 15
    },
    "anvil_cost": 4,
    "slots": ["mainhand"],
    "effects": {
        "minecraft:damage": [
            {
                "effect": {
                    "type": "minecraft:add_value",
                    "value": {
                        "type": "minecraft:linear",
                        "base": 2.0,
                        "per_level_above_first": 1.5
                    }
                }
            }
        ]
    }
}
```

5.4 □□□□□□□

1.21.1 □□□□□□□□□□□□ **□□□□□** □□□□□□□□□ `EnchantmentTarget` □□□

```json
// data/examplemod/tags/item/enchantable/example_weapon.json
{
    "replace": false,
    "values": [
        "minecraft:iron_sword",
        "minecraft:diamond_sword",
        "minecraft:netherite_sword",
        "examplemod:example_sword"
    ]
}
```

5.5 □□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□ JSON□□□□□□□ `Enchantment` □□□□□□□

```java
// 1.21.1 —— □□□□□□□□□□□□□□□□
import net.minecraft.enchantment.Enchantment;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.util.Identifier;

public static final RegistryKey<Enchantment> EXAMPLE_ENCHANTMENT =
    RegistryKey.of(RegistryKeys.ENCHANTMENT, Identifier.of("examplemod", "example_enchantment"));

// □□□□ 1.21.x □□□□□□□□ JSON □□□□□□□□□□□□□□□□□□□□
// □□□□□□□□□□□□□□ JSON □□□□□□□□□□□□□□□□ RegistryKey □□□
```

□□□□□□□□

6.1 Registry □□□□□□

1.21+ □□□□□ `Registry` □□□□□□□□□□ `Registry.BLOCK`, `Registry.ITEM`□□
□□□□ `Registries` □□

```java
// 1.20.1 —— Registry.BLOCK □□□□□□□□□□□□
Registry.register(Registry.BLOCK, new Identifier("modid", "name"), block);

// 1.21.1 —— Registry.BLOCK □□□□□□□ Registries.BLOCK
Registry.register(Registries.BLOCK, Identifier.of("modid", "name"), block);
```

```

### 6.2 □□□□□□□□□

| □□□ | 1.20.1 | 1.21.1 |
|---|---|---|
| □□ | `Registries.BLOCK` | `Registries.BLOCK`□□□□ |
| □□ | `Registries.ITEM` | `Registries.ITEM`□□□□ |
| □□ | `Registries.ENCHANTMENT` | `Registries.ENCHANTMENT`□□□□ |
| Data Component | □□□ | **`Registries.DATA_COMPONENT_TYPE`□□□□** |

### 6.3 RegistryKey □□□

1.21+ □□□ `RegistryKey` □□□□□□□□□□□□□□□□□ key□

```java
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;

// □□ RegistryKey
RegistryKey<Item> key = RegistryKey.of(RegistryKeys.ITEM, Identifier.of("modid", "name"));

// □□ key □□□□□□□□
Registry.register(Registries.ITEM, key, item);
```

> □□□□□□ `Registry.register(registry, id, value)` □□□□□□□□□□ `Identifier`□

□□□□□□□□□

7.1 □□□□□

`FabricBlockSettings` □ `AbstractBlock.Settings` □ API □ 1.20.1 □□□□□

```java
// 1.21.1 ── □ 1.20.1 □□
FabricBlockSettings.copyOf(Blocks.STONE)
    .strength(3.0f)
    .requiresTool()
    .luminance(15);

// □□
AbstractBlock.Settings.create()
    .strength(3.0f)
    .requiresTool();
```

7.2 MapColor

`MapColor` □□□□□ 1.20.1 □□□`Material` □ 1.19.3 □□□□□

```java
import net.minecraft.block.MapColor;

FabricBlockSettings.create()
    .mapColor(MapColor.STONE_GRAY)
    .strength(3.0f);
```

□□□□□□□□□

8.1 Item.Settings □□□□

`Item.Settings` □□□□□□ 1.20.1 □□□

```java
// 1.21.1 ── □ 1.20.1 □□
new Item(new Item.Settings()
    .maxCount(64)
    .maxDamage(256)
    .rarity(Rarity.UNCOMMON)
    .fireproof()
```
```

);
```

### 8.2 □□ .component() □□

1.21+ □□□ `.component()` □□□□□□□ `Item.Settings` □□□□□□ Data Component□

```java
// 1.21.1 —— □□
new Item(new Item.Settings()
    .maxCount(64)
    .component(ModComponents.CHARGE, 0)  // □□□□□□□□□□□□
);
```

8.3 Food □□□□

`FoodComponent` □□□□□□ `Food` □□□□□□□

```java
// 1.20.1
import net.minecraft.item.FoodComponent;
new FoodComponent.Builder().nutrition(6).saturationModifier(0.6f).build();

// 1.21.1
import net.minecraft.component.type.Food;
new Food.Builder().nutrition(6).saturationModifier(0.6f).build();
```

□□ `Item.Settings.food()` □□□□□□□□

```java
// 1.21.1
new Item(new Item.Settings()
    .food(new Food.Builder()
        .nutrition(6)
        .saturationModifier(0.6f)
        .build()
    )
);
```

8.4 FabricItemSettings

`FabricItemSettings` □ 1.21.1 □□□□□□□ `Item.Settings` □□□□□□□ Fabric □□□□□□
□□□□□□□ `new Item.Settings()`□

□□□□□□□□□

1.21.1 □□□□□□□□□□□ 1.20.1 □□□□□□

```java
// 1.21.1 —— □ 1.20.1 □□
import net.fabricmc.fabric.api.itemgroup.v1.ItemGroupEvents;
import net.minecraft.item.ItemGroups;

ItemGroupEvents.modifyEntriesEvent(ItemGroups.INGREDIENTS)
    .register(entries -> entries.add(myItem));
```

□□□□□□□□□□

```java
// 1.21.1 —— Identifier.of() □□ new Identifier()
Registry.register(Registries.ITEM_GROUP,
    Identifier.of("modid", "example"),
    FabricItemGroup.builder()
        .displayName(Text.literal("Example Tab"))
        .icon(() -> new ItemStack(Items.STONE))
        .entries((context, entries) -> {
            entries.add(myItem);
        })
        .build()
);
```
```

```

---

## □□□□□□□

### 10.1 ItemStack NBT □□□□

1.21+ □□□□□□ `ItemStack.getOrCreateTag()` □ NBT □□□
□□□□□□□□□ Data Components □□□

### 10.2 ScreenHandler □ PacketByteBuf □□

1.21+ □ `ScreenHandler` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ `RegistryWrapper.WrapperLookup` □□□□

### 10.3 BlockEntity □□/□□

1.21+ □ `BlockEntity` □ `saveAdditional` □ `loadAdditional` □□□□□□ `HolderLookup.Provider` □□□

```java
// 1.21.1
@Override
public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.saveAdditional(tag, registries);
    tag.putInt("value", this.value);
}

@Override
public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.loadAdditional(tag, registries);
    this.value = tag.getInt("value");
}
```

□□□□□□□□□□1.20.1 → 1.21.1□

1. **□□ JDK □ 21**□□□ build.gradle □□ Java □□□
2. `new Identifier("modid", "name")` → `Identifier.of("modid", "name")`□□□□□□□
3. `Registry.BLOCK` □□□□□ → `Registries.BLOCK`□□□□□□□□□□□□
4. ItemStack NBT □□ → Data Components □□□□□□□□□
5. □□□□□□□□ `Enchantment` □ → JSON □□□□□
6. `EnchantmentTarget` □□ → □□□□□
7. `FoodComponent` → `Food`□□□□□□□□□□□
8. `FoodComponent.Builder().hunger()` → `Food.Builder().nutrition()`□
9. `BlockEntity.saveAdditional(CompoundTag)` → `saveAdditional(CompoundTag, HolderLookup.Provider)`□
10. Loom □□□□□□□`1.6-SNAPSHOT` → `1.7-SNAPSHOT`□
11. □□ gradle.properties □□□□□□□□

recipe/shaped_recipe.json

```
{
    // ================================================================
    // □□□□□□□□□ - Fabric 1.21.1
    // ================================================================
    // □□□□: data/examplemod/recipes/shaped_recipe.json
    //
    // □□ 1.20.1 □□□□
    //   □□ JSON □□□ 1.21.1 □ 1.20.1 □□□□□□□
    //   □□: 1.21.2+ □□□□□□□□□□□□
    //   □ 1.21.1 □□□□ 1.20.1 □□□□□□
    //
    // □□□□□crafting_shaped□□□□□ pattern □ key □□
    // □□□□□□□□□□□□ pattern □□□□□
    //
    // □□□□:
    // 1. pattern □□□□□□□□□□□
    // 2. pattern □□□□□□□□□□ key □□□□□□
    // 3. key □□□□□□□□□□□□□□□□□□□□ #, X, *, @ □□
    // 4. □□□□□□□□□□□□
    // 5. result □□ count □□□□ 64
    // 6. ingredient □□ tag □□□ "tag" □□□□ "item"
    // ================================================================

    "type": "minecraft:crafting_shaped",
```
```

```
    // pattern: □□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□
    "pattern": [
        "###",
        "#X#",
        "###"
    ],

    // key: □ pattern □□□□□□□□□□□□
    "key": {
        "#": {
            "item": "minecraft:diamond"
        },
        "X": {
            "item": "minecraft:nether_star"
        }
    },

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

recipe/shapeless_recipe.json

```
{
    // ============================================================
    // □□□□□□□□□ - Fabric 1.21.1
    // ============================================================
    // □□□□: data/examplemod/recipes/shapeless_recipe.json
    //
    // □□ 1.20.1 □□□□
    //   □□ JSON □□□ 1.21.1 □ 1.20.1 □□□□□□□
    //   □□: 1.21.2+ □□□□□□□□□□□□
    //   □ 1.21.1 □□□□ 1.20.1 □□□□□□
    //
    // □□□□□crafting_shapeless□□□□□□□□□□
    // □□□ ingredients □□□□□□□□□□□□□□□□
    //
    // □□□□:
    // 1. ingredients □□□□□□□□□□□□□□□
    // 2. □□□□□□□□□□□□□□□□□□□□
    // 3. □□□ "tag" □□□□□□ "item"
    // 4. □ pattern □ key □□
    // ============================================================

    "type": "minecraft:crafting_shapeless",

    // ingredients: □□□□□□□□□
    // □□□□□□□□□□□□□□□
    "ingredients": [
        {
            "item": "minecraft:diamond"
        },
        {
            "item": "minecraft:emerald"
        },
        {
            "tag": "minecraft:logs"
        }
    ],

    // result: □□□□
    "result": {
        "item": "examplemod:example_item",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    // ============================================================
    // □□□□□□ - Fabric 1.21.1
```

```
    // ============================================================
    // □□□□□: data/examplemod/tags/blocks/example_block_tag.json
    //
    // □□ 1.20.1 □□□□□
    //   1.21.1 □□□ JSON □□□ 1.20.1 □□□□□
    //   □□□□□□□□: data/<mod_id>/tags/blocks/
    //
    // □□□□□□□□□□□□□□□□□□□□
    // □□□□□□□:
    //   minecraft:logs          □□□□□□□□□□□□□□□□□□□□□□□□
    //   minecraft:mineable/axe  □□□□□□□□□
    //   minecraft:mineable/pickaxe □□□□□□□□
    //   minecraft:mineable/shovel □□□□□□□□
    //   minecraft:needs_iron_tool □□□□□□□
    //   minecraft:needs_diamond_tool □□□□□□□□
    //   minecraft:base_stone_overworld □□□□□□□□
    //   minecraft:base_stone_nether □□□□□□□
    //
    // □□□□□:
    // 1. □□□□□□□: data/<mod_id>/tags/blocks/
    // 2. □□□□□□□: data/<mod_id>/tags/items/□□□□□□
    // 3. □□□□□□□□□□□□□□□□□□□□□□□□□□□ mineable/* □□
    // 4. □□□□□□□□□□□□□□□□□□□□□□□□□□□ needs_*_tool □□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond_block",
        "minecraft:emerald_block",
        "examplemod:example_block"
    ]
}
```

tag/item_tag.json

```
{
    // ============================================================
    // □□□□□□□ - Fabric 1.21.1
    // ============================================================
    // □□□□□: data/examplemod/tags/items/example_item_tag.json
    //
    // □□ 1.20.1 □□□□□
    //   1.21.1 □□□ JSON □□□ 1.20.1 □□□□□
    //   □□□□□□□□: data/<mod_id>/tags/items/
    //
    // □1.21.1 □□□□□□
    //   □□□□□□ supported_items □□□□□□□□□□□□□□□□□□□
    //   □□□ 1.20.x □ EnchantmentTarget □□
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□□□:
    //   minecraft:logs          □□
    //   minecraft:planks        □□
    //   minecraft:coals         □□□
    //   minecraft:saplings      □□
    //   minecraft:flowers       □
    //
    // □□□□□:
    // 1. □□□□□□ data/<mod_id>/tags/items/ □□□
    // 2. □□□□□□□□□□□□□□□□□□
    // 3. values □□□□ID□□□□□□□□□□
    // 4. replace □ true □□□□□□□□□□□□□□□□□□
    // ============================================================

    "replace": false,

    "values": [
        "minecraft:diamond",
        "minecraft:emerald",
        "examplemod:example_item"
    ]
}
```

worldgen/ore_feature.java

```
/*
```

```
 * ore_feature.java
 * ------------------------------------------------------------------------
 * □□□□□: Minecraft 1.21.1 (Fabric Loader 0.16.0 + Fabric API 0.102.0 + Yarn □□)
 * □□□□□: □□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□ 1.20.1 □□□□□□□□□□□□□□□□
 *   1. □Identifier □□□□□□□□
 *      1.20.1: new Identifier("modid", "name")
 *      1.21.1: Identifier.of("modid", "name")
 *
 *   2. □□□□□ API □□□□□□
 *      1.21.1 □□□ Registries.CONFIGURED_FEATURE / Registries.PLACED_FEATURE
 *      □□ 1.20.1 □□□□□□□ Identifier □□ Identifier.of()□
 *
 *   3. □BiomeModifications API □□□
 *      Fabric API □ BiomeModifications □□□□ 1.20.1 □□
 *
 *   4. □Java 21 □□□
 *      1.21+ □□□□□ Java 21
 *
 * □□ 1.19.2 □□□□□□□□□□□
 *   - 1.19.2 □ BuiltinRegistries □□□□□□□□□□
 *   - 1.21.1 □ Registries □□□□□□□□□□
 *   - 1.19.2 □ new Identifier()□1.21.1 □ Identifier.of()
 * ------------------------------------------------------------------------
 */
package com.example.mod.worldgen;

import net.fabricmc.fabric.api.biome.v1.BiomeModifications;
import net.fabricmc.fabric.api.biome.v1.BiomeSelectors;
import net.minecraft.registry.Registries;
import net.minecraft.registry.Registry;
import net.minecraft.registry.RegistryKey;
import net.minecraft.registry.RegistryKeys;
import net.minecraft.registry.entry.RegistryEntry;
import net.minecraft.util.Identifier;
import net.minecraft.world.gen.GenerationStep;
import net.minecraft.world.gen.YOffset;
import net.minecraft.world.gen.feature.ConfiguredFeature;
import net.minecraft.world.gen.feature.Feature;
import net.minecraft.world.gen.feature.OreFeatureConfig;
import net.minecraft.world.gen.feature.PlacedFeature;
import net.minecraft.world.gen.placementmodifier.CountPlacementModifier;
import net.minecraft.world.gen.placementmodifier.HeightRangePlacementModifier;
import net.minecraft.world.gen.placementmodifier.PlacementModifier;
import net.minecraft.world.gen.placementmodifier.SquarePlacementModifier;

import java.util.List;

/**
 * □□□□□□□□□ - Fabric 1.21.1
 *
 * □□ 1.20.1 □□□□□□□
 * 1. □□□□□□□□ Identifier.of() □□ new Identifier()
 * 2. Registries.CONFIGURED_FEATURE / PLACED_FEATURE □ 1.20.1 □□
 * 3. BiomeModifications API □ 1.20.1 □□
 *
 * □□□□□:
 *   ConfiguredFeature -> □□□ Registries.CONFIGURED_FEATURE
 *   PlacedFeature     -> □□□ Registries.PLACED_FEATURE
 *   BiomeModifications -> □ PlacedFeature □□□□□□□
 *
 * □□□□□:
 * - □1.21.1 □□□□□□□□ Identifier.of() □□ new Identifier()
 * - □□□□□□ ConfiguredFeature□□□□ PlacedFeature
 * - PlacedFeature □□ RegistryEntry<ConfiguredFeature> □□
 * - BiomeModifications.addFeature □□ RegistryKey<PlacedFeature>
 * - OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES □□□□□□□□
 * - GenerationStep.Feature.UNDERGROUND_ORES □□□□□□□□□□□□
 */
public class ModWorldGen {

    public static final String MOD_ID = "examplemod";

    // ConfiguredFeature □ RegistryKey
    // □1.21.1□□□ Identifier.of() □□ new Identifier()
```

```
    public static final RegistryKey<ConfiguredFeature<?, ?>> ORE_EXAMPLE_CF =
        RegistryKey.of(RegistryKeys.CONFIGURED_FEATURE, Identifier.of(MOD_ID, "ore_example"));

    // PlacedFeature 的 RegistryKey
    public static final RegistryKey<PlacedFeature> ORE_EXAMPLE_PF =
        RegistryKey.of(RegistryKeys.PLACED_FEATURE, Identifier.of(MOD_ID, "ore_example"));

    /**
     * 注册所有内容
     * 在你的主类的ModInitializer中的 onInitialize() 中调用
     */
    public static void register() {
        // ========================================================================
        // 1. 注册配置 ConfiguredFeature
        // ========================================================================
        OreFeatureConfig oreConfig = new OreFeatureConfig(
            OreFeatureConfig.Rules.STONE_ORE_REPLACEABLES,
            net.minecraft.block.Blocks.IRON_BLOCK.getDefaultState(),
            9
        );

        ConfiguredFeature<?, ?> configuredFeature = new ConfiguredFeature<>(
            Feature.ORE,
            oreConfig
        );

        // 在1.21.1中使用 Identifier.of() 而非 new Identifier()
        Registry.register(
            Registries.CONFIGURED_FEATURE,
            Identifier.of(MOD_ID, "ore_example"),
            configuredFeature
        );

        // ========================================================================
        // 2. 注册放置 PlacedFeature
        // ========================================================================
        RegistryEntry<ConfiguredFeature<?, ?>> cfEntry = RegistryEntry.of(configuredFeature);

        List<PlacementModifier> placementModifiers = List.of(
            CountPlacementModifier.of(20),
            SquarePlacementModifier.of(),
            HeightRangePlacementModifier.uniform(
                YOffset.getBottom(),
                YOffset.fixed(64)
            )
        );

        PlacedFeature placedFeature = new PlacedFeature(
            cfEntry,
            placementModifiers
        );

        // 在1.21.1中使用 Identifier.of() 而非 new Identifier()
        Registry.register(
            Registries.PLACED_FEATURE,
            Identifier.of(MOD_ID, "ore_example"),
            placedFeature
        );

        // ========================================================================
        // 3. 使用 BiomeModifications 将矿物添加到生物群系
        // ========================================================================
        BiomeModifications.addFeature(
            BiomeSelectors.foundInOverworld(),
            GenerationStep.Feature.UNDERGROUND_ORES,
            RegistryKey.of(
                RegistryKeys.PLACED_FEATURE,
                Identifier.of(MOD_ID, "ore_example")  // 在1.21.1中Identifier.of()
            )
        );
    }
}

/*
 * ============================================================================
 * 其他可用的生成步骤
```

```
* ============================================================================
*
* BiomeSelectors.all()                  □□□□□□□
* BiomeSelectors.foundInOverworld()      □□□□□□□□
* BiomeSelectors.foundInNether()         □□□□□□□
* BiomeSelectors.foundInTheEnd()         □□□□□□□
*
* ============================================================================
* □□□□□□□
* ============================================================================
*
* GenerationStep.Feature□□□□□□□:
* - RAW_GENERATION          □□□□□□□□□□
* - LAKES                    □□
* - LOCAL_MODIFICATIONS      □□□□
* - UNDERGROUND_STRUCTURES   □□□□
* - SURFACE_STRUCTURES       □□□□
* - STRONGHOLDS              □□
* - UNDERGROUND_ORES        □□□□□□□□□□
* - UNDERGROUND_DECORATION   □□□□
* - FLUID_SPRINGS            □□□
* - VEGETAL_DECORATION       □□□□
* - TOP_LAYER_MODIFICATION   □□□□
*/
```

Forge 模组模板

Forge 1.18.2

advancement/advancement.json

```
{
  "_comment_advancement": "================================================================================",
  "_comment_version": "Forge 1.18.2 □□□□ - advancement.json",
  "_comment_mc": "MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0",
  "_comment_diff": "□□ Forge 1.19.2 □□□□□□ JSON □□□ 1.18+ □□□□",
  "_comment_diff_1": "1. 1.18.2 □□□□□□ 1.19.2 □□□1.18+ □□□□□□□",
  "_comment_diff_2": "2. □□□□□□□□□□□□□□□□",
  "_comment_diff_old": "□□ Forge 1.16.5 □□□□",
  "_comment_diff_old_1": "1. 1.16.5 □□□□□□□□ JSON □□□1.12 □□□□□□□□",
  "_comment_diff_old_2": "2. 1.18.2 □□□□□□□□□□□□□□□□□□□□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/advancements/example_advancement.json",
  "_comment_end": "================================================================================",

  "_comment_structure": "□□□□□□□□display + criteria + requirements □□□□",

  "display": {
    "_comment_display": "□□□□□□□□□□□□□□□□",

    "icon": {
      "item": "examplemod:example_item",
      "_comment_icon": "□□□□□□□□□□ ID"
    },

    "title": {
      "translate": "advancement.examplemod.example_advancement.title",
      "_comment_title": "□□□□□□□□□□□□□□□□□□"
    },

    "description": {
      "translate": "advancement.examplemod.example_advancement.description",
      "_comment_description": "□□□□□□□□□□□□"
    },

    "frame": "task",
    "_comment_frame": "□□□□□□□□task□□□□/ goal□□□□/ challenge□□□□",

    "show_toast": true,
    "_comment_toast": "□□□□□□□□□□□□□□□□",

    "announce_to_chat": true,
    "_comment_announce": "□□□□□□□□□□□□□□",

    "hidden": false,
    "_comment_hidden": "□□□□□□□□□□□",

    "background": "minecraft:textures/gui/advancements/backgrounds/stone.png",
    "_comment_background": "□□□□□□□□□□□□□□□□□□□□□□"
  },

  "parent": "minecraft:story/root",
  "_comment_parent": "□□□□ ID□□□□□□□□□□□□□□□",

  "criteria": {
    "_comment_criteria": "□□□□□□□□□□□□□□□□□□",

    "get_example_item": {
      "trigger": "minecraft:inventory_changed",
      "_comment_trigger": "□□□□□□□inventory_changed □□□□□□□□□□□□",
      "conditions": {
        "items": [
          {
            "item": "examplemod:example_item",
            "count": {
              "min": 1
            }
          }
```

```
        ]
      }
    },

    "kill_zombie": {
      "trigger": "minecraft:player_killed_entity",
      "_comment_trigger_2": "□□□□□□□□□□□",
      "conditions": {
        "entity": {
          "type": "minecraft:zombie"
        }
      }
    }
  },

  "requirements": [
    [
      "get_example_item"
    ],
    [
      "kill_zombie"
    ]
  ],
  "_comment_requirements": "□□□□□□□□□□",
  "_comment_req_1": "□□□□□□ AND □□□□□□□□□□□□□□□□",
  "_comment_req_2": "□□□□□□ OR □□□□□□□□□□□□□□□□",
  "_comment_req_3": "□□□□□□□□□□□□□ get_example_item □ kill_zombie",

  "rewards": {
    "_comment_rewards": "□□□□□□□□□□□□□□□□",
    "experience": 50,
    "_comment_exp": "□□□□□□□",
    "loot": [
      "minecraft:chests/simple_dungeon"
    ],
    "_comment_loot": "□□□□□□□□",
    "recipes": [
      "examplemod:example_recipe"
    ],
    "_comment_recipes": "□□□□□□"
  }
}
```

block/simple_block.java

```
// ================================================================================
// Forge 1.18.2 □□□□□□□ - simple_block.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ================================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. 1.18.2 □□ AbstractBlock.Properties□1.19.x □□□□1.20.1 □□ BlockBehaviour.Properties□
// 2. 1.18.2 □ Material □□1.19.3+ □□□□□□□ .of(Material.STONE)
// 3. 1.18.2 □ BlockEntity □□□□□□ ForgeRegistries.BLOCK_ENTITIES
//    □1.17+ □ TILE_ENTITIES □□□1.19.2 □□□□□
// 4. DeferredRegister + ForgeRegistries.BLOCKS □□□□ 1.19.2 □□
// □□ Forge 1.16.5 □□□□□□□
// 5. TileEntity □□□□ BlockEntity□1.17+ □□□□
// 6. □□□□□□ RegistryEvent □□ DeferredRegister□1.18+ □□□□
// 7. Material □□□□□□□□□□□□□□□□□□□□□□
// ================================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.material.Material;
import net.minecraft.world.level.material.MaterialColor;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
```

```
 *
 * □□□□□□□□ ModMain □□□□□□□□□□
 *   ModBlocks.register(modEventBus);
 *
 * □1.18.2 □□□□□□
 * - 1.18.2 □□□□□□ TileEntity □□□□□□□ BlockEntity□1.17 □□□
 * - □□□□□□□ AbstractBlock.Properties.of(Material.XXX) □□
 * - Material □□ 1.18.2 □□□□□□□1.19.3 □□□□□□□□□□□
 */
public class ModBlocks {

    // ===== □□□□□ DeferredRegister =====
    // □□□□□Forge □ ForgeRegistries.BLOCKS
    //         1.16.5 □ RegistryEvent.Register<Block> □□
    public static final DeferredRegister<Block> BLOCKS =
            DeferredRegister.create(ForgeRegistries.BLOCKS, ExampleMod.MODID);

    // □□□□□□□□□□□
    // □□□□□□□1.18.2 □□□□□ BLOCK_ENTITIES□1.17+ □ TILE_ENTITIES □□□
    //            1.16.5 □ TILE_ENTITIES
    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITIES =
            DeferredRegister.create(ForgeRegistries.BLOCK_ENTITIES, ExampleMod.MODID);

    // ===== □□□□□ =====
    // □□□□□□□1.18.2 □□ AbstractBlock.Properties.of(Material.STONE)
    //            1.19.x □□ BlockBehaviour.Properties□AbstractBlock □□□□
    //            1.20.1 □□□□ BlockBehaviour.Properties
    //
    // □□ 1.16.5 □□□□
    // 1.16.5: Block.Properties.create(Material.STONE)
    // 1.18.2: BlockBehaviour.Properties.of(Material.STONE)
    //         □□□□ AbstractBlock.Properties□□□□□ BlockBehaviour.Properties□
    public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register("example_block",
            () -> new Block(BlockBehaviour.Properties.of(Material.STONE)
                    .color(MaterialColor.STONE)
                    .strength(3.0F, 6.0F)         // □□ / □□□□
                    .requiresCorrectToolForDrops()  // □□□□□□□□□□□
                    .sound(net.minecraft.world.level.block.SoundType.STONE)
            ));

    // ===== □□ BlockEntityType =====
    // BlockEntityType.Builder.of(factory, blocks...).build(null)
    // □□□□□1.16.5 □ TileEntityType.Builder.create()
    //        1.18.2 □ BlockEntityType.Builder.of()
    //
    // public static final RegistryObject<BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
    //         BLOCK_ENTITIES.register("example_block_entity",
    //                 () -> BlockEntityType.Builder
    //                         .of(ExampleBlockEntity::new, EXAMPLE_BLOCK.get())
    //                         .build(null));

    // ===== □□□□□□□□□□□□ =====
    // □□□□□1.16.5 □□ TileEntity
    //        1.18.2 □□ BlockEntity
    //
    // public static class ExampleBlockEntity extends BlockEntity {
    //     public ExampleBlockEntity(BlockPos pos, BlockState state) {
    //         super(EXAMPLE_BLOCK_ENTITY.get(), pos, state);
    //     }
    // }

    // ===== □□□□□□□ =====
    public static void register(IEventBus modEventBus) {
        BLOCKS.register(modEventBus);
        BLOCK_ENTITIES.register(modEventBus);
    }
}
```

command/simple_command.java

```
// ================================================================================
// Forge 1.18.2 □□□□□□□ - simple_command.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ================================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. □□□□□ API □□□□□□RegisterCommandsEvent + Brigadier
```

```
// 2. sendSuccess() / sendFailure() □□□□□□□
// 3. □□□□□□1.18.2 □□ Component.literal() □ new TextComponent()
//    □1.19+ □□□□ Component.literal()□1.18.2 □□□□□□□□
// □□ Forge 1.16.5 □□□□□□□
// 4. 1.16.5 □ CommandRegistrationEvent□□□□□□ RegisterCommandsEvent
// 5. 1.16.5 □□□□□□ net.minecraft.util.text □□
// 6. 1.16.5 □ sendMessage □□□□□□□
// ================================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.network.chat.Component;
import net.minecraft.server.level.ServerPlayer;
import net.minecraftforge.event.RegisterCommandsEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * Forge □□□□□ - □□ RegisterCommandsEvent □□
 *
 * □1.18.2 □□□□□□
 * - □□□□□□□ RegisterCommandsEvent □□
 * - □□□□□ FORGE □□□□□□□□□bus = Bus.FORGE□
 * - □□□□□ Brigadier □□□□□
 * - □□□□□□□□□ Component.literal()□1.18.2 □□□□□
 *   □□□□□ new TextComponent("text") □□□□
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        // □□□□□□□□□
        var dispatcher = event.getDispatcher();

        // □□□□□□/examplemod give <target> <amount>
        dispatcher.register(Commands.literal(ExampleMod.MODID)
                .requires(source -> source.hasPermission(2))  // □□ OP □□ 2
                .then(Commands.literal("give")
                        .then(Commands.argument("target", EntityArgument.player())
                                .then(Commands.argument("amount", IntegerArgumentType.integer(1, 64))
                                        .executes(ModCommands::executeGive)
                                )
                        )
                )
                .executes(ModCommands::executeInfo)
        );
    }

    /**
     * □□ give □□□
     * /examplemod give <target> <amount>
     */
    private static int executeGive(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        ServerPlayer target = EntityArgument.getPlayer(context, "target");
        int amount = IntegerArgumentType.getInteger(context, "amount");

        CommandSourceStack source = context.getSource();

        // □□□□sendSuccess() □□ Supplier<Component>
        // 1.18.2 □□ Component.literal() □□□□□□□□□□□□
        // □□□□□ new TextComponent("...")□□ API□□□□□□□
        source.sendSuccess(
                () -> Component.literal("□□□ " + amount + " □□□□ " + target.getName().getString()),
                true  // true = □□□□□ OP□false = □□□□□□□□□□
        );
        return Command.SINGLE_SUCCESS;
```

```
    }

    /**
     * □□□□□□□□□□□□□□□□
     * /examplemod
     */
    private static int executeInfo(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();

        // □1.18.2 □□□□□□□□□
        // □□□□□□Component.literal("text")
        // □□□□□new TextComponent("text")  —— □□□□□□□□□□□
        // □□□□□□Component.translatable("key") □ new TranslatableComponent("key")
        source.sendSuccess(
                () -> Component.literal("Example Mod v1.0.0 - Forge 1.18.2"),
                false
        );

        return Command.SINGLE_SUCCESS;
    }

    // ============================================================================
    // □□ 1.16.5 □□□□ API □□□□
    //
    // 1. □□□□□
    //    1.16.5: RegisterCommandsEvent □□□□□
    //
    // 2. □□□□□□□□□□
    //    1.16.5: import net.minecraft.util.text.StringTextComponent;
    //            new StringTextComponent("text")
    //    1.18.2: import net.minecraft.network.chat.Component;
    //            Component.literal("text")
    //
    // 3. sendFeedback □□□□
    //    1.16.5: source.sendFeedback(text, allowLogging)
    //    1.18.2: source.sendSuccess(componentSupplier, allowLogging)
    //            □□□□□□□□□□□□ ITextComponent □□ Supplier<Component>□
    // ============================================================================
}
```

enchantment/custom_enchantment.java

```
// ============================================================================
// Forge 1.18.2 □□□□□□□□ - custom_enchantment.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ============================================================================
// □□ Forge 1.19.2 □□□□□□□□
// 1. □□□□□□ 1.19.2 □□□□□□□□□ Enchantment □
// 2. □□□□□□□□□Enchantment(Rarity, EnchantmentCategory, EquipmentSlot[])
// 3. DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) □□
// 4. □□□1.21+ □□□□□□□□□□□□□JSON□□1.18.2 □□□□□□□□
// □□ Forge 1.16.5 □□□□□□□□
// 5. 1.16.5 □□□□□□ RegistryEvent□1.18.2 □ DeferredRegister
// 6. 1.16.5 □□□□□□□□□□□□□□Enchantment □□□□□□□□□□
// 7. 1.18.2 □ EquipmentSlot □□□□□□□
// ============================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □1.18.2 □□□□□□
 * - □□□□□□□ Enchantment □□□
 * - □□ DeferredRegister + ForgeRegistries.ENCHANTMENTS □□
 * - □□□□□□□□□□□□□□1.21+ □□□□□□□□ JSON□
 */
```

```
public class ModEnchantments {

    // ===== □□□□□□ =====
    // □□ 1.16.5 □□□□□
    // 1.16.5: □ RegistryEvent.Register<Enchantment> □□
    // 1.18.2: □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID)
    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, ExampleMod.MODID);

    // ===== □□□□□ =====
    // Enchantment □□□□□□□ 1.19.2 □□□□□
    //   - Rarity rarity: □□□□□COMMON/UNCOMMON/RARE/VERY_RARE□
    //   - EnchantmentCategory category: □□□□□□□□
    //   - EquipmentSlot[] slots: □□□□□□□□
    //
    // EnchantmentCategory □□□□□
    //   ALL / ARMOR / ARMOR_FEET / ARMOR_LEGS / ARMOR_CHEST / ARMOR_HEAD
    //   WEAPON / FISHING_ROD / BREAKABLE / DIGGER / BOW / CROSSBOW / TRIDENT / VANISHABLE
    public static final RegistryObject<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND));

    // ===== □□□□□□□ =====
    // □□ Enchantment□□□□□□□□□□□□□□□
    public static class ExampleEnchantment extends Enchantment {

        public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                                  EquipmentSlot... slots) {
            super(rarity, category, slots);
        }

        // □□□□□□□□□□□□□□□□
        @Override
        public int getMinCost(int level) {
            return 10 * level;
        }

        // □□□□□□□□□□□□□□□□
        @Override
        public int getMaxCost(int level) {
            return getMinCost(level) + 15;
        }

        // □□□□□□□
        @Override
        public int getMaxLevel() {
            return 3;
        }

        // □□□□□□□□□□□□□□
        @Override
        protected boolean checkCompatibility(Enchantment other) {
            return super.checkCompatibility(other)
                    && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
        }

        // □□□□□□□□□□□□□□□□□
        @Override
        public boolean canEnchant(net.minecraft.world.item.ItemStack stack) {
            return super.canEnchant(stack);
        }

        // □□□□□□□□
        @Override
        public boolean isCurse() {
            return false;
        }

        // □□□□□□□□□□□□□□□□
        @Override
        public boolean isTradeable() {
            return true;
```

```
        }

        // □□□□□□□□□□□□□□□□□□□□□□
        @Override
        public boolean isDiscoverable() {
            return true;
        }

        // □□□□□□□□□□□□□□□□□□□□□□
        @Override
        public boolean isAllowedOnBooks() {
            return true;
        }

        // ===== □□□□□□□□ =====
        // □□□□□□□□□□□□□□□□□□□□□□
        @Override
        public void doPostAttack(
                net.minecraft.world.entity.LivingEntity attacker,
                net.minecraft.world.entity.Entity target,
                int level) {
            if (target instanceof net.minecraft.world.entity.LivingEntity livingTarget) {
                // □□□□□□□□□□
                livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                        net.minecraft.world.effect.MobEffects.POISON,
                        60,           // □□□□□tick□
                        level - 1));  // □□□□
            }
            super.doPostAttack(attacker, target, level);
        }

        // ===== □□□□□□□□□□ =====
        // □□□□□□□□□□□□□□□□□□
        // @Override
        // public void doPostHurt(
        //         LivingEntity user,
        //         Entity attacker,
        //         int level) {
        //     // □□□□□...
        //     super.doPostHurt(user, attacker, level);
        // }
    }

    // ===== □□□□□□□ =====
    public static void register(IEventBus modEventBus) {
        ENCHANTMENTS.register(modEventBus);
    }

    // ============================================================================
    // □□□□□□□ - Forge □□□
    // Forge □ EnchantmentLevelSetEvent □□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□
    //
    // □□ Fabric □□□□Fabric □□□□□□□□ Mixin □□
    // ============================================================================
    // @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.FORGE)
    // public static class ForgeEvents {
    //     @SubscribeEvent
    //     public static void onEnchantLevel(EnchantmentLevelSetEvent event) {
    //         // □□□□□□□□□□□□□□□□□□□□□□
    //         // event.getEnchantments().add(new EnchantmentInstance(EXAMPLE_ENCHANTMENT.get(), 1));
    //     }
    // }
}
```

entity/simple_entity.java

```
// ============================================================================
// Forge 1.18.2 □□□□□□□ - simple_entity.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ============================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. □□□□□□□□□1.18.2 □ ForgeRegistries.ENTITIES
//    □1.19.x □□□□□□□ ENTITY_TYPES□1.18.2 □□ ENTITIES□
// 2. EntityType.Builder.create() □ 1.18.2 □□□1.20.1 □□□ .of()□
// 3. EntityAttributeCreationEvent □□□□□ 1.19.2 □□
```

```
// 4. ForgeSpawnEggItem □ 1.18.2 □□□
// □□ Forge 1.16.5 □□□□□□□
// 5. 1.16.5 □ EntityType.Builder.create()□□□□□□□□
// 6. 1.16.5 □□□□□□ RegistryEvent□1.18.2 □ DeferredRegister
// 7. 1.16.5 □□□□□□ GlobalEntityTypeAttributes.put()
//    1.18.2 □ EntityAttributeCreationEvent □□
// ============================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.entity.ai.attributes.AttributeSupplier;
import net.minecraft.world.entity.ai.attributes.Attributes;
import net.minecraft.world.entity.monster.Monster;
import net.minecraftforge.event.entity.EntityAttributeCreationEvent;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □1.18.2 □□□□□□
 * - □□□□□□□□□□ ForgeRegistries.ENTITIES□□□□□ ENTITY_TYPES□
 * - □□□□□□□□ EntityType.Builder.create()
 * - □□□□□□□ EntityAttributeCreationEvent □□□□
 */
public class ModEntities {

    // ===== □□□□□□ =====
    // □□□□□□□1.18.2 □□□□ ENTITIES□□□ ENTITY_TYPES□
    //            1.19.x □□□□□□□□ ENTITY_TYPES
    //            1.16.5 □□ ENTITIES
    public static final DeferredRegister<EntityType<?>> ENTITIES =
            DeferredRegister.create(ForgeRegistries.ENTITIES, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // □□□□□□□1.18.2 □ EntityType.Builder.create()
    //            1.20.1 □□ EntityType.Builder.of()
    //
    // □□ 1.16.5 □□□□
    // 1.16.5: □ RegistryEvent.Register<EntityType<?>> □□□□
    // 1.18.2: □ DeferredRegister □□□□
    public static final RegistryObject<EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITIES.register("example_entity",
                    () -> EntityType.Builder
                            // □□□□1.18.2: .create()   1.20.1: .of()
                            .create(ExampleEntity::new, MobCategory.CREATURE)
                            .sized(1.0F, 1.0F)            // □□□□□□□□/□□
                            .clientTrackingRange(10)        // □□□□□□□□
                            .build("example_entity"));

    // ============================================================================
    // □□□□□□□ - Forge □□□□□□□
    // □□ 1.16.5 □□□□
    // 1.16.5: □ FMLCommonSetupEvent □□□
    //         DeferredWorkQueue.runLater(() ->
    //             GlobalEntityTypeAttributes.put(type, builder.build()))
    // 1.18.2: □□ EntityAttributeCreationEvent□□□ event.put(type, builder.build())
    //
    // □□ Fabric □□□□
    // Fabric: FabricDefaultAttributeRegistry.register(type, builder)
    // Forge: EntityAttributeCreationEvent □□
    public static void registerAttributes(EntityAttributeCreationEvent event) {
        AttributeSupplier.Builder builder = Monster.createMonsterAttributes();
        builder.add(Attributes.MAX_HEALTH, 20.0D);
        builder.add(Attributes.MOVEMENT_SPEED, 0.25D);
        builder.add(Attributes.ATTACK_DAMAGE, 5.0D);
        builder.add(Attributes.FOLLOW_RANGE, 32.0D);

        event.put(EXAMPLE_ENTITY.get(), builder.build());
```

```
    }

    // □ @SubscribeEvent + □□□□□□□□□□□□
    // bus = Mod.EventBusSubscriber.Bus.MOD □□□□ MOD □□□□
    @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.MOD)
    public static class ModBusEvents {
        @SubscribeEvent
        public static void onAttributeCreate(EntityAttributeCreationEvent event) {
            registerAttributes(event);
        }
    }

    // ===== □□□□□□□□□ =====
    // □□□□□ EntityRenderersEvent.RegisterRenderers □□□□□
    // □□□□□□□□□□□□□ value = Dist.CLIENT
    //
    // @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.MOD, value = Dist.CLIENT)
    // public static class ClientBusEvents {
    //     @SubscribeEvent
    //     public static void onRegisterRenderers(EntityRenderersEvent.RegisterRenderers event) {
    //         event.registerEntityRenderer(EXAMPLE_ENTITY.get(), ExampleRenderer::new);
    //     }
    // }

    // ===== □□□□□□□□□□ =====
    // Forge □□ ForgeSpawnEggItem □
    // □□□ ModItems.ITEMS □□□□
    // public static final RegistryObject<Item> EXAMPLE_SPAWN_EGG = ITEMS.register("example_spawn_egg",
    //         () -> new ForgeSpawnEggItem(() -> EXAMPLE_ENTITY.get(),
    //                 0xFFFFFF, 0x000000,
    //                 new Item.Properties().tab(CreativeModeTab.TAB_MISC)));

    public static void register(IEventBus modEventBus) {
        ENTITIES.register(modEventBus);
    }

    // ============================================================================
    // □□□□□
    // □□ Monster □ PathfinderMob □
    // ============================================================================
    public static class ExampleEntity extends Monster {

        public ExampleEntity(EntityType<? extends Monster> type,
                             net.minecraft.world.level.Level level) {
            super(type, level);
        }

        // □□□□□ AI □□
        @Override
        protected void registerGoals() {
            this.goalSelector.addGoal(1,
                    new net.minecraft.world.entity.ai.goal.MeleeAttackGoal(this, 1.0D, false));
            this.goalSelector.addGoal(2,
                    new net.minecraft.world.entity.ai.goal.WaterAvoidingRandomStrollGoal(this, 0.8D));
            this.targetSelector.addGoal(1,
                    new net.minecraft.world.entity.ai.goal.target.HurtByTargetGoal(this));
        }
    }
}
```

gui/container_screen.java

```
// ============================================================================
// Forge 1.18.2 GUI / □□□□□□□ - container_screen.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ============================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. MenuType □□□: ForgeRegistries.CONTAINERS□□ 1.19.2 □□□
// 2. IForgeMenuType.create() □ 1.18.2 □□□
// 3. AbstractContainerScreen □ AbstractContainerMenu API □□
// 4. 1.18.2 □□ GuiGraphics □□1.20.1 □□□□
//    □ PoseStack + RenderSystem □□ GUI □□
// □□ Forge 1.16.5 □□□□□□□
// 5. 1.16.5 □ Container □□1.18.2 □ AbstractContainerMenu□□□□□
// 6. 1.16.5 □ ContainerType□1.18.2 □ MenuType□□□□□
// 7. 1.16.5 □□□ MatrixStack□1.17+ □□□ PoseStack□
```

```
// ================================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.client.gui.screens.MenuScreens;
import net.minecraft.client.gui.screens.inventory.AbstractContainerScreen;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraftforge.common.extensions.IForgeMenuType;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 菜单/容器注册 - Forge DeferredRegister + MenuScreens
 *
 * 与1.18.2 相同结构
 * - 注册表改为使用 ForgeRegistries.CONTAINERS
 * - 菜单类继承 AbstractContainerMenu
 * - 渲染使用 PoseStack，改为 GuiGraphics（GuiGraphics 在 1.20.1 引入）
 * - MenuScreens.register() 在 FMLClientSetupEvent 中调用
 */
public class ModMenus {

    // ===== 注册菜单类型 =====
    // 注册表：ForgeRegistries.CONTAINERS
    //        1.16.5 中为 CONTAINERS，名称不同 ContainerType）
    public static final DeferredRegister<MenuType<?>> MENUS =
            DeferredRegister.create(ForgeRegistries.CONTAINERS, ExampleMod.MODID);

    // ===== 创建 MenuType =====
    // IForgeMenuType.create() 的工厂参数：接收服务端传来的数据，创建 MenuType
    // 对比 1.16.5 的变化：
    // 1.16.5: ContainerType.create(factory)
    // 1.18.2: IForgeMenuType.create((windowId, inv, data) -> new Menu(...))
    public static final RegistryObject<MenuType<ExampleMenu>> EXAMPLE_MENU = MENUS.register("example_menu",
            () -> IForgeMenuType.create((containerId, inventory, data) -> {
                // 从数据中读取 extraData 的示例
                // data 是服务端写入的缓冲区（FriendlyByteBuf）
                return new ExampleMenu(containerId, inventory);
            }));

    // ================================================================================
    // AbstractContainerMenu：服务端逻辑
    // 对比 1.16.5 的变化：
    // 1.16.5: 名为 Container
    // 1.18.2: 名为 AbstractContainerMenu
    // 构造函数参数和注册表绑定方式不同
    public static class ExampleMenu extends AbstractContainerMenu {

        public ExampleMenu(int containerId, Inventory playerInventory) {
            super(EXAMPLE_MENU.get(), containerId);

            // 玩家背包槽位（3x9 = 27个）
            for (int row = 0; row < 3; row++) {
                for (int col = 0; col < 9; col++) {
                    this.addSlot(new Slot(playerInventory, 9 + row * 9 + col,
                            8 + col * 18, 84 + row * 18));
                }
            }

            // 玩家快捷栏（9个）
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
            }
        }

        // 快速移动（Shift+点击）
        @Override
```

```
    public ItemStack quickMoveStack(Player player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasItem()) return ItemStack.EMPTY;

        ItemStack original = slot.getItem().copy();
        ItemStack stack = slot.getItem();

        // □□□□□□□□
        if (!this.moveItemStackTo(stack, 0, this.slots.size(), true)) {
            return ItemStack.EMPTY;
        }

        slot.setChanged();
        return original;
    }

    // □□□□□□□□□□□□□□□□
    @Override
    public boolean stillValid(Player player) {
        return true;
    }
}

// ========================================================================
// AbstractContainerScreen□□□□□□
// □□□□□□□1.18.2 □ PoseStack □□□□ GuiGraphics□
//            1.20.1 □□ GuiGraphics □□□□
//
// □□ 1.16.5 □□□□
// 1.16.5: □□□□□□ MatrixStack
// 1.18.2: □□□□□□ PoseStack□1.17 □□□
//         □□□□□□□□□□□
public static class ExampleScreen extends AbstractContainerScreen<ExampleMenu> {

    // □□□□□□□□□□□□
    // private static final net.minecraft.resources.ResourceLocation TEXTURE =
    //         new net.minecraft.resources.ResourceLocation(ExampleMod.MODID,
    //                 "textures/gui/example_gui.png");

    public ExampleScreen(ExampleMenu menu, Inventory inventory, Component title) {
        super(menu, inventory, title);
        this.imageWidth = 176;      // GUI □□
        this.imageHeight = 166;     // GUI □□
        this.inventoryLabelY = this.imageHeight - 94; // □□□□□□ Y □□
    }

    // ===== □□□□ ===== 
    // □□□□□□□1.18.2 □□□□□renderBg(PoseStack poseStack, float partialTick, int x, int y)
    //            1.20.1 □□□□□renderBg(GuiGraphics graphics, float partialTick, int x, int y)
    @Override
    protected void renderBg(com.mojang.blaze3d.vertex.PoseStack poseStack,
                            float partialTick, int mouseX, int mouseY) {
        // 1.18.2 □□□□□□□ RenderSystem + blit□□
        //
        // // □□□□□□□□□
        // RenderSystem.setShader(GameRenderer::getPositionTexShader);
        // RenderSystem.setShaderColor(1.0F, 1.0F, 1.0F, 1.0F);
        // RenderSystem.setShaderTexture(0, TEXTURE);
        //
        // // □□ GUI □□□□□
        // int x = (this.width - this.imageWidth) / 2;
        // int y = (this.height - this.imageHeight) / 2;
        //
        // // □□□□□□□
        // blit(poseStack, x, y, 0, 0, this.imageWidth, this.imageHeight);
    }

    // ===== □□□□□□□ ===== 
    // □□□□□1.18.2 □□□render(PoseStack poseStack, int mouseX, int mouseY, float partialTick)
    //        1.20.1 □□□render(GuiGraphics graphics, int mouseX, int mouseY, float partialTick)
    @Override
    public void render(com.mojang.blaze3d.vertex.PoseStack poseStack,
                       int mouseX, int mouseY, float partialTick) {
        this.renderBackground(poseStack);          // □□□□□□□□□□□□□□□□
        super.render(poseStack, mouseX, mouseY, partialTick);
        this.renderTooltip(poseStack, mouseX, mouseY); // □□□□□□□
```

```
        }
    }

    // ===== Screen 注册 =====
    // 在客户端MenuScreens.register(type, factory)
    // 在 FMLClientSetupEvent 中调用，例如：
    //
    // 在客户端事件 ModMain 中添加以下内容：
    // @SubscribeEvent
    // public static void onClientSetup(FMLClientSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         MenuScreens.register(EXAMPLE_MENU.get(), ExampleScreen::new);
    //     });
    // }

    // ===== 打开菜单辅助方法 =====
    // 服务端调用 NetworkHooks.openScreen() 打开菜单
    // 注意 1.16.5 不同：
    // 1.16.5: player.openContainer(container) 或 NetworkHooks.openGui()
    // 1.18.2: NetworkHooks.openScreen(player, menuProvider, bufferWriter)
    //
    // 示例：
    // NetworkHooks.openScreen((ServerPlayer) player,
    //     new SimpleMenuProvider((id, inv, p) -> new ExampleMenu(id, inv), title),
    //     buf -> { /* 写入额外数据 */ });

    public static void register(IEventBus modEventBus) {
        MENUS.register(modEventBus);
    }
}
```

item/simple_item.java

```
// ============================================================================
// Forge 1.18.2 物品注册模板 - simple_item.java
// MC版本: 1.18.2 | Java版本: 17 | Forge版本: 40.2.0
// ============================================================================
// 适用 Forge 1.19.2 物品注册模板
// 1. 注册使用 API 为官方的DeferredRegister + ForgeRegistries.ITEMS
// 2. CreativeModeTab 创造模式物品栏.tab(CreativeModeTab.TAB_MISC)
//    在1.19.3+ 版本为 CreativeModeTab.builder()在1.18.2 版本为
// 3. 1.18.2 版本为 Material 在新版本中被移除
// 适用 Forge 1.16.5 物品注册模板
// 4. 注册事件1.16.5 为 RegistryEvent在1.18.2 为 DeferredRegister
// 5. 物品栏类在1.16.5 为 ItemGroup在1.18.2 为 CreativeModeTab创造模式物品栏
// 6. 文本组件相关类在1.16.5 为 net.minecraft.util.text
//    1.18.2 为 net.minecraft.network.chat
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 物品注册类 - Forge DeferredRegister 模式
 *
 * 在主类构造函数中调用 ModMain 中添加以下内容：
 *   ModItems.register(modEventBus);
 *
 * 在1.18.2 版本中：
 * - 1.18.2 中 CreativeModeTab 创造模式物品栏为TAB_BUILDING_BLOCKS 等
 * - 使用 CreativeModeTab.builder() 方法在1.19.3+ 版本中
 * - 自定义创造模式物品栏需要在 CreativeModeTab 类中添加 Forge 支持
 */
public class ModItems {

    // ===== 物品注册 =====
    // 注册在1.16.5 为 RegistryEvent.Register<Item> 监听
    //        1.18.2 为 DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
    public static final DeferredRegister<Item> ITEMS =
```

```
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // □□□□□□□1.18.2 □ .tab(CreativeModeTab.TAB_MISC) □□□□□□□□
    //            1.20.1 □ CreativeModeTab.builder() □ BuildCreativeModeTabContentsEvent
    //
    // □□ 1.16.5 □□□□
    // 1.16.5: .group(ItemGroup.MISC)
    // 1.18.2: .tab(CreativeModeTab.TAB_MISC)
    //         □□□□ ItemGroup □□ CreativeModeTab□□□□ group □□ tab□
    public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register("example_item",
            () -> new Item(new Item.Properties()
                    .stacksTo(64)                       // □□□□
                    .tab(CreativeModeTab.TAB_MISC)      // □1.18.2 □□□□□□□□□□□□□□□
                    // .rarity(Rarity.RARE)             // □□□
                    // .fireResistant()                 // □□
            ));

    // ===== □□□□□□□□BlockItem□ =====
    // □□□□□□□Forge □□□□ BlockItem □□□□□□ ITEMS □□□□□□
    // □Fabric □ BlockItem □□□□□□□□□□
    // public static final RegistryObject<Item> EXAMPLE_BLOCK_ITEM = ITEMS.register("example_block",
    //         () -> new net.minecraft.world.item.BlockItem(
    //                 com.example.examplemod.block.ModBlocks.EXAMPLE_BLOCK.get(),
    //                 new Item.Properties().tab(CreativeModeTab.TAB_BUILDING_BLOCKS)));

    // ========================================================================
    // □1.18.2 □□□□□□□□□□□□□□
    // 1.18.2 □□ CreativeModeTab.builder()□□□□□□□□□□□□□□□□ CreativeModeTab □□
    //
    // public static final CreativeModeTab EXAMPLE_TAB = new CreativeModeTab("examplemod.example_tab") {
    //     @Override
    //     public ItemStack makeIcon() {
    //         return new ItemStack(EXAMPLE_ITEM.get());
    //     }
    // };
    //
    // □□□□□□□□□□□□□□.tab(EXAMPLE_TAB)
    //
    // □□□□1.20.1 □□□ DeferredRegister<CreativeModeTab> □□□□□□ Tab
    //         1.18.2 □□□□□□□□□CreativeModeTab □□□□□□□□
    // ========================================================================

    // ===== □□□□□□□ =====
    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

item/tool_item.java

```
// ========================================================================
// Forge 1.18.2 □□/□□□□□□□ - tool_item.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ========================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. Tier □□□□□□ 1.19.2 □□□□□□getUses/getSpeed/getLevel □□
// 2. □□□□□□□□□□□ 1.19.2 □□□PickaxeItem(tier, damage, speed, props)□
// 3. TierSortingRegistry □ 1.18.2 □□□□Forge □□□□□□
// 4. 1.18.2 □ CreativeModeTab □ TAB_TOOLS / TAB_COMBAT □□
// □□ Forge 1.16.5 □□□□□□□
// 5. 1.16.5 □ IItemTier □□□1.18.2 □ Tier □□□□□□□□
// 6. 1.16.5 □□□□□□□□□□□□□□ PickaxeItem □□□□□□□□□□□□
// ========================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraftforge.eventbus.api.IEventBus;
```

```
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□□ - Forge Tier □□
 *
 * □1.18.2 □□□□□□
 * - Tier □□□□□□ net.minecraft.world.item.Tier
 * - □□□□□□□□ Tiers□WOOD/STONE/IRON/DIAMOND/GOLD/NETHERITE□
 * - Forge □□ TierSortingRegistry □□□□□□□□□□□□□□
 */
public class ModTools {

    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ============================================================================
    // □□□□□□□□□Tier□
    // □□ Forge □ Tier □□
    // □□ 1.16.5 □□□□
    //   1.16.5: IItemTier □□□□□□□□□□□□
    //   1.18.2: Tier □□
    //     - getUses()             □□□
    //     - getSpeed()            □□□□
    //     - getAttackDamageBonus() □□□□□□□
    //     - getLevel()            □□□□
    //     - getEnchantmentValue() □□□□
    //     - getRepairIngredient() □□□□
    // ============================================================================

    public enum ModTiers implements Tier {
        EXAMPLE_TIER(
            1561,    // getUses()□□□□□□□□1561□
            8.0F,    // getSpeed()□□□□□□
            3.0F,    // getAttackDamageBonus()□□□□□□□□
            3,       // getLevel()□□□□□□□□0/□1/□2/□3/□□□□4□
            10       // getEnchantmentValue()□□□□□□
        );

        private final int uses;
        private final float speed;
        private final float attackDamageBonus;
        private final int level;
        private final int enchantmentValue;
        private final Ingredient repairIngredient;

        ModTiers(int uses, float speed, float attackDamageBonus,
                 int level, int enchantmentValue) {
            this.uses = uses;
            this.speed = speed;
            this.attackDamageBonus = attackDamageBonus;
            this.level = level;
            this.enchantmentValue = enchantmentValue;
            this.repairIngredient = Ingredient.of(net.minecraft.world.item.Items.DIAMOND);
        }

        @Override
        public int getUses() {
            return uses;
        }

        @Override
        public float getSpeed() {
            return speed;
        }

        @Override
        public float getAttackDamageBonus() {
            return attackDamageBonus;
        }

        @Override
        public int getLevel() {
            return level;
        }
```

```
        @Override
        public int getEnchantmentValue() {
            return enchantmentValue;
        }

        @Override
        public Ingredient getRepairIngredient() {
            return repairIngredient;
        }
    }

    // ===== □□□□□□□ =====
    // □□□□□□new PickaxeItem(tier, attackDamage, attackSpeed, properties)
    // □□ 1.16.5 □□□□
    // 1.16.5 □ PickaxeItem □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // 1.18.2 □□□□ attackDamageBonus □ attackSpeedModifier
    public static final RegistryObject<Item> EXAMPLE_PICKAXE = ITEMS.register("example_pickaxe",
            () -> new PickaxeItem(ModTiers.EXAMPLE_TIER,
                    1,      // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    -2.8F,  // □□□□□□□□□
                    new Item.Properties().tab(CreativeModeTab.TAB_TOOLS)));

    public static final RegistryObject<Item> EXAMPLE_SWORD = ITEMS.register("example_sword",
            () -> new SwordItem(ModTiers.EXAMPLE_TIER,
                    3,      // □□□□□□□
                    -2.4F,  // □□□□□□□□□
                    new Item.Properties().tab(CreativeModeTab.TAB_COMBAT)));

    // ===== Tier □□□□□Forge □□□ =====
    // TierSortingRegistry □□□□□ Tier□□□□□□□□□□□□□□□
    // □□□ FMLCommonSetupEvent □□□
    //
    // TierSortingRegistry.registerTier(
    //     ModTiers.EXAMPLE_TIER,
    //     new ResourceLocation(ExampleMod.MODID, "example_tier"),
    //     List.of(Tiers.DIAMOND),    // □□□□□□□
    //     List.of()                  // □□□□□□□
    // );

    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

mixin/inject_mixin.java

```
// ================================================================================
// Forge 1.18.2 Mixin □□□□ - inject_mixin.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ================================================================================
// □□ Forge 1.19.2 □□□□□□
// 1. Mixin □□□□□□□□□□@Mixin / @Inject / @Shadow □□
// 2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
// 3. 1.18.2 □□□□□□□□/□□□□□□ 1.19.2 □□
// □□ Forge 1.16.5 □□□□□□□
// 4. 1.16.5 □ Java 8□1.18.2 □ Java 17□□□□□□□□□□
// 5. □ Mixin □□□□□□□□□□□□□□□□
// ================================================================================

package com.example.examplemod.mixin;

import net.minecraft.server.MinecraftServer;
import org.spongepowered.asm.mixin.Mixin;
import org.spongepowered.asm.mixin.Shadow;
import org.spongepowered.asm.mixin.injection.At;
import org.spongepowered.asm.mixin.injection.Inject;
import org.spongepowered.asm.mixin.injection.callback.CallbackInfo;

/**
 * Mixin □□□□□ MinecraftServer □
 *
 * □1.18.2 □□□□□
 * - □□ official □□□□Mojang □□□□□
 * - Mixin □□□□□□□□□ refmap □□□□□□□□□
 * - □□□□□ target = "Lnet/minecraft/server/MinecraftServer;..." □□
 *
```

```
 * □□□ Mixin □□□
 * @Mixin: □□□□□□□ Mixin □□□□□□□□
 * @Inject: □□□□□□□□□□□□□□□□□□
 * @Shadow: □□□□□□□□□□□□□□
 * @Overwrite: □□□□□□□□□□□□□□□
 * @ModifyVariable: □□□□□□□□□□□□
 * @Redirect: □□□□□□□□
 */
@Mixin(MinecraftServer.class)
public abstract class ExampleServerMixin {

    // ===== Shadow □□/□□ =====
    // @Shadow □□□□□□□□□□□□□□□□□□□□
    // □□□□□□ abstract□□□□□□□□□□□□□ abstract □□□
    //
    // @Shadow
    // private int tickCount;
    //
    // @Shadow
    // public abstract boolean isRunning();

    // ===== Inject □□□□□ =====
    // □ MinecraftServer.runServer() □□□□□□□□□□
    //
    // □□□□□□□□□
    // method = "runServer" : □□□□□□
    // at = @At("HEAD")     : □□□□□□□□
    // at = @At("RETURN")   : □□□□□□□□□
    // at = @At(value = "INVOKE", target = "...") : □□□□□□□□□□□□
    //
    // □□ 1.16.5 □□□□
    // □□□□□□□□□□□□□□□□□□
    // 1.16.5 (MCP □□): func_71217_p() □ SRG □
    // 1.18.2 (official □□): runServer() □ Mojang □□□
    @Inject(method = "runServer", at = @At("HEAD"))
    private void onServerStart(CallbackInfo ci) {
        // □□□□□□□□□□□□□□
        // System.out.println("[ExampleMod] □□□□□□□□□□");

        // □□□□
        // 1. □□□□□□□□□ private + □□□□□□
        // 2. CallbackInfo □□□□□□□□ void □□□
        //    □□□□□□□□□□□□□□ CallbackInfoReturnable<T>
        // 3. □□□□ this □□□□□□□□□□□□□□□ @Shadow□
    }

    // ===== □□□□□□□□ =====
    // □□ @ModifyReturnValue □□□□□□□□□
    //
    // @ModifyReturnValue(method = "getServerModName", at = @At("RETURN"))
    // private String modifyServerModName(String original) {
    //     return original + " + ExampleMod";
    // }

    // ===== Redirect □□□□□ =====
    // □□□□□□□□□□□□□□□□□□□□
    //
    // @Redirect(method = "runServer",
    //           at = @At(value = "INVOKE",
    //                    target = "Lnet/minecraft/server/MinecraftServer;initServer()Z"))
    // private boolean redirectInitServer(MinecraftServer server) {
    //     // □□□□□□...
    //     boolean result = server.initServer(); // □□□□□
    //     // □□□□□...
    //     return result;
    // }
}

// ============================================================================
// □□□□ Mixin □□□
// □□□ Mixin □□□□ mixin □□□ client □□□
// ============================================================================
//
// @Mixin(net.minecraft.client.Minecraft.class)
// public class ExampleClientMixin {
//
```

```
//     @Inject(method = "run", at = @At("HEAD"))
//     private void onClientStart(CallbackInfo ci) {
//         // □□□□□□□□□
//     }
// }

// ================================================================================
// □□ 1.16.5 Mixin □□□□□□
//
// 1. □□□□□
//    1.16.5: □□ MCP/SRG □□□(func_xxxx、field_xxxx)
//    1.18.2: □□ Mojang □□□□□□□□□
//
// 2. Java □□□
//    1.16.5: Java 8□□□□ var、switch □□□□□□□□
//    1.18.2: Java 17□□□□ Java 17 □□□□□
//
// 3. □□□□□□□□□
//    □□□□□□□ net.minecraft.util.text □□ net.minecraft.network.chat
//    TileEntity □□□ BlockEntity □
//
// 4. □□□□□□□□□
//    □□ render □□□□□□ MatrixStack □□ PoseStack
// ================================================================================
```

mixin/mixin_config.json

```
{
  "_comment_mixin_config": "================================================================================",
  "_comment_version": "Forge 1.18.2 Mixin □□ - mixin_config.json",
  "_comment_mc": "MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0",
  "_comment_diff": "□□ Forge 1.19.2 □□□□□□□□□□□□□□□□□□ JAVA_17",
  "_comment_diff_old": "□□ Forge 1.16.5 □□□□",
  "_comment_diff_old_1": "1. 1.16.5 □ JAVA_8 □□□□□□1.18.2 □ JAVA_17",
  "_comment_diff_old_2": "2. 1.16.5 □ Mixin □□□□□□□□□□□□□□□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□",
  "_comment_diff_1": "1. Forge □□ Mixin、SpongePowered Mixin□□□□□□□□",
  "_comment_diff_2": "2. Forge □□ jar □ MANIFEST.MF □□□ MixinConfigs",
  "   _comment_manifest": "   build.gradle: attributes([\"MixinConfigs\": \"examplemod.mixins.json\"])",
  "_comment_diff_3": "3. Fabric □ fabric.mod.json □ mixins □□□□",
  "   _comment_fabric": "   \"mixins\": [\"examplemod.mixins.json\"]",
  "_comment_diff_4": "4. Forge □ official □□□Fabric □ yarn/parchment",
  "   _comment_mapping": "   Mixin □□□□/□□□□□□□□□□□",
  "_comment_location": "□□□□: src/main/resources/examplemod.mixins.json",
  "_comment_end": "================================================================================",

  "required": true,
  "_comment_required": "required: true □□□ Mixin □□□□□□□□□□□□□□□□",

  "minVersion": "0.8",
  "_comment_minVersion": "Mixin □□□□□□□□",

  "package": "com.example.examplemod.mixin",
  "_comment_package": "Mixin □□□□ Java □□□",

  "compatibilityLevel": "JAVA_17",
  "_comment_compatibilityLevel": "Java □□□□□1.18.2 □□ JAVA_17、1.16.5 □ JAVA_8□",

  "refmap": "examplemod.refmap.json",
  "_comment_refmap": "□□□□□□□□□ForgeGradle □□□□□□□□□□□□□□□□□□□□□□□□",

  "verbose": false,
  "showMessageTypes": false,

  "mixins": [
    {
      "_comment": "□□ Mixin□□□□+□□□□□□□",
      "name": "MinecraftServerMixin"
    }
  ],

  "client": [
    {
      "_comment": "□□□□□ Mixin□□□□□□□□□",
      "name": "ClientLevelMixin"
    },
```

```
    {
      "name": "GuiMixin"
    }
  ],

  "server": [
    {
      "_comment": "□□□□□□ Mixin□□□□□□□□□□",
      "name": "DedicatedServerMixin"
    }
  ],

  "injectors": {
    "defaultRequire": 1,
    "_comment_defaultRequire": "Mixin □□□□□□□□□1 = □□□□□□□□□□1□□□□□□□□□□□"
  },

  "_plugin_comment": "□□□□□□ Mixin □□□□□□□□□□□□□□□□□□□ Mixin",
  "plugin": "com.example.examplemod.mixin.ExampleMixinPlugin"
}
```

project/ModMain.java

```
// ============================================================================
// Forge 1.18.2 □□□□□□□□ - ModMain.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ============================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. API □□□□□□@Mod □□ + DeferredRegister + ForgeRegistries
// 2. 1.18.2 □□□□□□□□ Component.literal() □ TextComponent □□□□
// 3. 1.18.2 □ CreativeModeTab □□ builder() □□□□□□□□□□□□
// 4. 1.18.2 □□□□□□□□□ ForgeRegistries.ENTITIES□□□ ENTITY_TYPES□
// 5. 1.18.2 □□ AbstractBlock.Properties □ BlockBehaviour.Properties
// □□ Forge 1.16.5 □□□□□□□
// 6. 1.16.5 □ @Mod □□ + RegistryEvent □□
//    1.18.2 □ @Mod □□ + DeferredRegister □□□□□□
// 7. 1.16.5 TileEntity → 1.18.2 BlockEntity□1.17 □□□
// 8. 1.16.5 □□□□□ net.minecraft.util.text □
//    1.18.2 □ net.minecraft.network.chat □
// ============================================================================

package com.example.examplemod;

import net.minecraftforge.common.MinecraftForge;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext;

/**
 * □□□□□□□ - Forge 1.18.2
 *
 * □□□ @Mod □□□□□□□□
 * - □□□ modId□□□□ mods.toml □□ modId □□
 * - Forge □□□□□□□□□□□□□□□□□□□□□□□□□□
 *
 * □□□□□□□
 * 1. □□ modEventBus□□□□□□□□□□□
 * 2. □□□□□□□□ register(modEventBus) □□□□ DeferredRegister
 * 3. □□□□□□□□□□□commonSetup, clientSetup □□
 * 4. □□□ FORGE □□□□□□□□□□□□□□□
 */
@Mod(ExampleMod.MODID)
public class ExampleMod {

    public static final String MODID = "examplemod";

    // □□□□□□□□Forge □□□□□□□□□□□□□□□□
    // □□□□□□ DeferredRegister□□□□□□□□□□□□□□□□
    public ExampleMod() {
        // □□□□□□□□□□□□□modEventBus□
        // □□□□□1.18.2 □ 1.19.2 □□□ FMLJavaModLoadingContext
        //         1.16.5 □□□□□□□□□□□□□□□□
        IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();

        // ===== □□ DeferredRegister =====
        // □□□□□□□□□ register(modEventBus)
```

```
        //
        // □□□
        // ModBlocks.register(modEventBus);
        // ModItems.register(modEventBus);
        // ModTools.register(modEventBus);
        // ModEntities.register(modEventBus);
        // ModMenus.register(modEventBus);
        // ModEnchantments.register(modEventBus);

        // ===== □□□□□□□□□ =====
        // □□□□□□□□□+□□□□□□□□
        modEventBus.addListener(this::commonSetup);

        // □□□□□□□□□□□□□□□□
        // modEventBus.addListener(this::clientSetup);

        // □□□□□ FORGE □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
        MinecraftForge.EVENT_BUS.register(this);
    }

    /**
     * □□□□□□□
     * □□□□□□□ common setup □□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□
     */
    private void commonSetup(final net.minecraftforge.fml.event.lifecycle.FMLCommonSetupEvent event) {
        // □□□□□□□□□□□□□□□□□□□□□□□
        // event.enqueueWork(() -> {
        //     // □□□□□□□□□□□□□...
        // });
    }

    /**
     * □□□□□□□□
     * □□□□□□□□□□□□
     * □□□□□□□□□□□□□□□□□□□□□□□□□□□
     *
     * □□ @OnlyIn(Dist.CLIENT) □□□□□□□□□
     */
    // private void clientSetup(final net.minecraftforge.fml.event.lifecycle.FMLClientSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         // □□□□
    //         // MenuScreens.register(ModMenus.EXAMPLE_MENU.get(), ModMenus.ExampleScreen::new);
    //
    //         // □□□□□□□□
    //         // EntityRenderers.register(ModEntities.EXAMPLE_ENTITY.get(), ExampleRenderer::new);
    //     });
    // }

    // ===== □□□□□□□□□ =====
    // □□ FORGE □□□□□□□□□
    // □ @SubscribeEvent □□□□□□□□□□□
    //
    // @SubscribeEvent
    // public void onServerTick(net.minecraftforge.event.TickEvent.ServerTickEvent event) {
    //     if (event.phase == TickEvent.Phase.END) {
    //         // □□□ tick □□□□□...
    //     }
    // }
    //
    // @SubscribeEvent
    // public void onPlayerJoin(net.minecraftforge.event.entity.player.PlayerEvent.PlayerLoggedInEvent event) {
    //     Player player = event.getPlayer();
    //     player.sendMessage(Component.literal("□□□□ Example Mod□"), player.getUUID());
    // }
}
```

project/build.gradle

```
// ================================================================================
// Forge 1.18.2 □□□□□□□ - build.gradle
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ================================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. ForgeGradle □□: 1.18.2 □ 5.1.x□□ 1.19.2 □□□ ForgeGradle □□□□
// 2. Forge □□: 40.2.0□1.19.2 □ 43.3.0□
```

```
// 3. 映射表: 1.18.2 用 official 1.18.2 即可
// 4. Java 版本: 17（与 1.19.2 相同，1.18+ 统一 Java 17）
// 5. 1.18.2 新增生物群系：与 1.19.2 相同，需要在1.18 的世界中测试
// 6. 1.18.2 没有 1.19 的深暗之域 Mangrove、Sculk、Allay 等
// 如果 Forge 1.16.5 不支持这些
// 7. Java 版本: 1.18.2 用 17，1.16.5 用 8。
// 8. ForgeGradle: 1.18.2 用 5.1.x 的 plugins DSL，1.16.5 用 3.1 的 buildscript。
// 9. TileEntity 改名为 BlockEntity（1.17+ 改名）
// 10. 聊天组件包路径变化: net.minecraft.util.text → net.minecraft.network.chat
// ============================================================================

// 使用 ForgeGradle 插件（1.18.2 用 5.1 版本）
// 注意：1.16.5 用 buildscript classpath 方式，ForgeGradle 3.1 不支持 plugins DSL。
//         1.18.2 支持 plugins DSL，ForgeGradle 5.1 支持。
plugins {
    id 'eclipse'
    id 'idea'
    id 'maven-publish'
    id 'net.minecraftforge.gradle' version '5.1.+'
}

// ----------------------------------------------------------------------------
// 项目基本信息配置
// ----------------------------------------------------------------------------
version = '1.0.0'
group = 'com.example.examplemod'  // 包名，对应你的 ModMain.java 所在的包路径

// Java 版本：1.18.2 必须使用 Java 17
// 注意：1.16.5 使用 Java 8，1.18.2 必须用 Java 17
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// ----------------------------------------------------------------------------
// Minecraft / Forge 配置块
// ----------------------------------------------------------------------------
// 格式：通常为Forge 版本 "MC版本-Forge版本" 或简写。
// 1.18.2 推荐 Forge 40.2.0
minecraft {
    // 映射表：1.18.2 推荐官方映射
    // 可选用 Parchment（mappings channel: 'parchment', version: '2022.11.27-1.18.2'
    mappings channel: 'official', version: '1.18.2'

    // 运行配置
    runs {
        // 客户端运行配置
        client {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }

        // 服务端运行配置
        server {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }

        // 数据生成器配置
        data {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            args '--mod', 'examplemod', '--all', '--output', file('src/generated/resources/'),
                    '--existing', file('src/main/resources/')
            mods {
                examplemod {
```

```
                    source sourceSets.main
                }
            }
        }
    }
}

// ------------------------------------------------------------------------
// □□□□□
// ------------------------------------------------------------------------
repositories {
    maven { url = 'https://maven.minecraftforge.net/' }
    maven { url = 'https://maven.parchmentmc.org' }
    maven { url = 'https://maven.blamejared.com' }
    mavenCentral()
}

// ------------------------------------------------------------------------
// □□□□□
// ------------------------------------------------------------------------
dependencies {
    // Minecraft + Forge □□□□□□1.18.2-40.2.0
    minecraft 'net.minecraftforge:forge:1.18.2-40.2.0'

    // ===== □□□□□□□□□□□□□□□ =====
    // JEI□1.18.2 □□□
    // compileOnly(fg.deobf("mezz.jei:jei-1.18.2-common-api:9.7.+.+:api"))
    // runtimeOnly(fg.deobf("mezz.jei:jei-1.18.2-forge:9.7.+.+"))
}

// ------------------------------------------------------------------------
// □□□□□□□□□□□□□□□□□
// ------------------------------------------------------------------------
sourceSets.main.resources { srcDir 'src/generated/resources' }

// ------------------------------------------------------------------------
// Jar □□□□□
// ------------------------------------------------------------------------
jar {
    manifest {
        attributes([
            "Specification-Title"     : "examplemod",
            "Specification-Vendor"     : "examplemod",
            "Specification-Version"   : "1",
            "Implementation-Title"    : project.name,
            "Implementation-Version"  : project.jar.archiveVersion,
            "Implementation-Vendor"    : "examplemod",
            "MixinConfigs"            : "examplemod.mixins.json"
        ])
    }
}

// ------------------------------------------------------------------------
// □□□□□□ build □□□ mods.toml □□□□□
// ------------------------------------------------------------------------
tasks.named('processResources', ProcessResources).configure {
    var replaceProperties = [
            minecraft_version   : "1.18.2",
            minecraft_version_range: "[1.18.2,1.19)",
            forge_version       : "40.2.0",
            forge_version_range : "[40,)",
            loader_version_range: "[40,)",
            mod_id              : "examplemod",
            mod_name            : "Example Mod",
            mod_license         : "MIT",
            mod_version         : "1.0.0",
            mod_authors         : "Author",
            mod_description     : "Example mod for Forge 1.18.2"
    ]
    inputs.properties replaceProperties

    filesMatching(['META-INF/mods.toml', 'pack.mcmeta']) {
        expand replaceProperties + [project: project]
    }
}
```

project/mods.toml

```
# ======================================================================================
# Forge 1.18.2 □□□□□□□□ - mods.toml
# MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
# ======================================================================================
# □□ Forge 1.19.2 □□□□□□□
# 1. 1.18.2 □ mods.toml □□□ 1.19.2 □□□□
# 2. □□□□□□□□1.18.2 □ [1.18.2,1.19) □□ [1.19.2,1.19.3)
# 3. Forge □□□□□1.18.2 □ [40,) □□ [43,)
# □□ Forge 1.16.5 □□□□□□□
# 4. 1.16.5 □ mods.toml □□□□□□□□□□□□□□□
# 5. 1.18.2 □□□□□□□□□□ displayTest □□
# 6. □□□□□□ src/main/resources/META-INF/mods.toml
# ======================================================================================

# □□□□□□□□□javafmloader □ Forge Java □□□□□□
modLoader="javafmloader"

# □□□□□□□□□
# 1.18.2 □□ Forge 40.x □□
loaderVersion="${loader_version_range}"

# □□□□□□□□□□□
license="${mod_license}"

# ----------------------------------------------------------------------------------
# displayTest
# □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
# - "MATCH_VERSION"□□□□□□□□□□□□□□□□
# - "IGNORE_ALL"□□□□□□□□□□□□
# - "IGNORE_SERVER_VERSION"□□□□□□□□□□□
# ----------------------------------------------------------------------------------
#displayTest="MATCH_VERSION"

# ----------------------------------------------------------------------------------
# □□□□□
# □□□□□□ [[mods]] □□□□□□□□□□□□
# ----------------------------------------------------------------------------------
[[mods]]
modId="${mod_id}"
version="${mod_version}"
displayName="${mod_name}"
authors="${mod_authors}"
description='''
${mod_description}
'''

# □□□□□□□□□□
# □□□□□□src/main/resources/assets/${mod_id}/textures/item/icon.png
#logoFile="examplemod.png"

# □□□□□ URL□□□□
#displayURL="https://example.com/"

# ----------------------------------------------------------------------------------
# □□□□□□
# □□□□□□□ [[dependencies.mod_id]] □
# ----------------------------------------------------------------------------------
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

[[dependencies.${mod_id}]]
    modId="forge"
    type="required"
    versionRange="${forge_version_range}"
    ordering="NONE"
    side="BOTH"

# □□□□□□□
# [[dependencies.${mod_id}]]
#     modId="jei"
```

```
#     type="optional"
#     versionRange="[9.7.0,)"
#     ordering="NONE"
#     side="BOTH"
```

project/version_diff.md

```
# Forge 1.18.2 □ 1.19.2 □□□□□□□

> □□□□□□ Forge 1.18.2 □□□□ 1.19.2 □ API □□□□
> 1.18.2 □ 1.19.2 □□□□ Java 17 □ ForgeGradle 5.1+□
> □□□□□□□□□ Forge □□□□□□□□□□□□□□□□□□□□□□□□□ API □□□□

---

## □□□□□

1.18.2 □□ 1.18 □□□□1.18 → 1.18.1 → 1.18.2□□□□ 1.19 □□□□□□□□□□□□□□
1.18 □□□□□□□□□□□□Caves & Cliffs□□1.19 □□□□□□□□□□□□□□□□□□□□□□□□
□□□□□□ Java 17□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

| □□□□ | 1.18.2 | 1.19.2 |
|--------|--------|--------|
| Java □□ | 17 | 17□□□□□ |
| Forge □□ | 40.2.0 | 43.3.0 |
| ForgeGradle | 5.1.+ | 5.1.+□□□□ |
| □□□□□ | DeferredRegister | DeferredRegister□□□□ |
| mods.toml | □□ | □□□□□□□□□ |
| □□□□□□ | `net.minecraft.network.chat` | □□ |
| □□□□□ | `BlockEntity` | `BlockEntity`□□□□ |
| □□□□□ | □□□□□□□□□1.18 □□□□□ | □□ |
| BiomeModifier | □□□□1.18.2 □□□□ | □□□□□□ |
| Forge Event Bus | `@SubscribeEvent` + `IEventBus` | □□ |

---

## □□ForgeGradle □□

### 2.1 □□□□□□

1.18.2 □ 1.19.2 □□□□ ForgeGradle 5.1+□□□□ plugins DSL□

```gradle
// 1.18.2 □ 1.19.2 □□
plugins {
    id 'net.minecraftforge.gradle' version '5.1.+'
}
```

2.2 Forge □□□□

Minecraft □□	Forge □□	ForgeGradle □□
1.16.5	36.2.42	3.1.+
1.18.2	40.2.0	5.1.+
1.19.2	43.3.0	5.1.+

□□□□□□□□□

3.1 DeferredRegister□□□□□□

1.18.2 □ 1.19.2 □□□□□□□□□□□□

```java
// 1.18.2 □ 1.19.2 □□
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;

public static final DeferredRegister<Item> ITEMS =
    DeferredRegister.create(ForgeRegistries.ITEMS, "examplemod");

public static final RegistryObject<Item> EXAMPLE_ITEM =
    ITEMS.register("example_item", () -> new Item(new Item.Properties()));
```
```

3.2 ForgeRegistries

```java
// 1.18.2  1.19.2 
ForgeRegistries.ITEMS
ForgeRegistries.BLOCKS
ForgeRegistries.ENTITIES
ForgeRegistries.BLOCK_ENTITIES  // 1.17+  BLOCK_ENTITIES
```

> 1.16.5 `TILE_ENTITIES`1.17+ 1.18.2 1.19.2 `BLOCK_ENTITIES`

4.1

1.18.2 1.19.2 `net.minecraft.network.chat`

```java
// 1.18.2  1.19.2 
import net.minecraft.network.chat.TextComponent;
import net.minecraft.network.chat.TranslatableComponent;

new TextComponent("Hello World");
new TranslatableComponent("item.modid.example");
```

> 1.16.5 `net.minecraft.util.text` `StringTextComponent`
> 1.18.2 1.19.2

5.1

1.18.2 1.19.2 1.18
`BiomeModifier`

```java
// 1.18.2  1.19.2  —— BiomeModifier JSON
//  data/examplemod/forge/biome_modifier/example_ore.json
```

5.2 ConfiguredFeature

```java
// 1.18.2  1.19.2  —— ConfiguredFeature 
ConfiguredFeature<?, ?> feature = Feature.ORE
    .configured(new OreConfiguration(...))
    .range(64)
    .squared()
    .count(20);
```

> 1.16.5 `withConfiguration()` `square()`
> 1.18+ `configured()` `squared()`

Forge Event

6.1

1.18.2 1.19.2 Forge

```java
// 1.18.2  1.19.2 
@SubscribeEvent
public static void onRegisterCommand(RegisterCommandsEvent event) {
    event.getDispatcher().register(
        Commands.literal("example")
            .executes(context -> {
                context.getSource().sendSuccess(
```

```
                    new TextComponent("Hello!"),
                    false
                );
                return 1;
            })
    );
}
```

□□□□□□□□□□

7.1 □□□□

1.18.2 □ 1.19.2 □□□□ `BlockEntity`□1.17+ □ `TileEntity` □□□□

```java
// 1.18.2 □ 1.19.2 □□
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraft.world.level.block.entity.BlockEntityType;

public class MyBlockEntity extends BlockEntity { ... }
```

□□1.19 □□□ 1.18.2 □□□□□

8.1 □□□□□

1.19 □□□□□□□□□□□ Mangrove Swamp□Deep Dark □□□
1.18.2 □□□□□□□□□□

8.2 □□□□□□□

1.19 □□□□□□□□ Mangrove □□□Sculk □□□□□□□□□□
□ 1.18.2 □□□□□□□□□□

8.3 Allay □□

1.19 □□□ Allay □□□1.18.2 □□□□□□□□□□□

> □□□□□□□□□□□□□□□□□□□□□□□□□□□□ API □□□
> API □□□□□□□ 1.18.2 □ 1.19.2 □□□□□□□□

□□□□□□□□1.18.2 → 1.19.2□

1. □□ build.gradle □□□□□□
   - Forge□`1.18.2-40.2.0` → `1.19.2-43.3.0`
   - □□□□`official 1.18.2` → `official 1.19.2`
   - □□□□□□`[1.18.2,1.19)` → `[1.19.2,1.19.3)`□`[40,)` → `[43,)`
2. □□□□□ 1.19 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
3. DeferredRegister □□□□□□□□□□
4. □□□□□□□□□□□□□ `net.minecraft.network.chat` □□□
5. BlockEntity □□□□□□□□□ `BlockEntity` □□□□
6. □□□□□□□□□□□□□□□□□□□□ `BiomeModifier` □ `ConfiguredFeature`□□
7. Forge Event □□□□□□□□

□□□□□□□

10.1 □ 1.18.2 □□□ 1.19 □□□□□

```java
// □ □□□1.18.2 □□□ Mangrove □□
Items.MANGROVE_PROPAGULE;
Blocks.MANGROVE_LOG;
Blocks.SCULK;

// □ □□□□□ 1.18.2 □□□□□□□□□
Items.OAK_SAPLING;
Blocks.OAK_LOG;
```

```

### 10.2 □ 1.18.2 □□□□ 1.16.5 □□ API

```java
// □ □□□1.18.2 □□ TileEntity □□□□ BlockEntity
import net.minecraft.tileentity.TileEntity;

// □ □□□1.18.2 □□ BlockEntity
import net.minecraft.world.level.block.entity.BlockEntity;
```

10.3 □ 1.18.2 □□□□ 1.16.5 □□□□□□□□□

```java
// □ □□□1.18.2 □□□□□□□□□□□□□
import net.minecraft.util.text.StringTextComponent;

// □ □□□1.18.2 □□□□□□□□
import net.minecraft.network.chat.TextComponent;
```

recipe/shaped_recipe.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.18.2 □□□□□□□□ - shaped_recipe.json",
  "_comment_mc": "MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0",
  "_comment_diff_1": "□□ Forge 1.19.2 □□□□□",
  "_comment_diff_2": "1. JSON □□□□ 1.19.2 □□□□□□1.18+ □□□□□□□□",
  "_comment_diff_3": "2. Forge 1.18.2 □□ result □□ nbt □□",
  "_comment_diff_4": "3. Forge □□□□□□□□forge:mod_loaded, forge:item_exists □□",
  "_comment_diff_5": "4. □□□□□ data/<modid>/recipes/ □",
  "_comment_diff_old": "□□ Forge 1.16.5 □□□□□",
  "_comment_diff_old_1": "5. 1.16.5 □□□□□□□□ JSON □□□□□□□□□□ 1.12 □□□□□□",
  "_comment_diff_old_2": "6. 1.18.2 □□□□□□□□□□□□□□□□ shaped/shapeless □□□□",
  "_comment_location": "□□□□□: src/main/resources/data/examplemod/recipes/example_recipe.json",
  "_comment_end": "================================================================================",

  "type": "minecraft:crafting_shaped",

  "pattern": [
    "III",
    "I I",
    "III"
  ],

  "key": {
    "I": {
      "_comment": "□□ Forge □ forge:ingots/iron □□",
      "tag": "forge:ingots/iron"
    }
  },

  "result": {
    "item": "examplemod:example_item",
    "count": 1,
    "_comment_nbt": "□□□□□Forge □□ nbt □□□□□□□□ NBT □□",
    "nbt": "{display:{Name:\"{\\\"text\\\":\\\"Example Item\\\",\\\"color\\\":\\\"gold\\\",\\\"italic\\\":false}\\\"}}"
  },

  "_comment_conditions": "Forge □□□□□□□□□□□□□□□□Forge □□□□□□",
  "conditions": [
    {
      "type": "forge:mod_loaded",
      "modid": "examplemod"
    }
  ]
}
```

tag/block_tag.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.18.2 □□□□□□□□ - block_tag.json",
  "_comment_mc": "MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0",
  "_comment_diff": "□□ Forge 1.19.2 □□□□□□□□□□□□□□□□Forge □□□□□□ 1.18+ □□",
```
```

```
  "_comment_forge_tags": "Forge □□□□□ forge: □□□□□□□□□□□□□/□□□□□□□",
  "_comment_forge_needs": "  - forge:needs_wood_tool / forge:needs_stone_tool / forge:needs_iron_tool □",
  "_comment_forge_ores": "  - forge:ores, forge:ores/iron: □□□□□□□",
  "_comment_forge_ingots": "  - forge:ingots, forge:ingots/gold: □□□□□□",
  "_comment_diff_old": "□□ Forge 1.16.5 □□□□□",
  "_comment_diff_old_1": "1.16.5 □□□□□□□□□□□□□□□□□□□□□□□□ 1.13 □□□□□□",
  "_comment_diff_old_2": "1.18.2 □□□□□□□□□□□□□□□□□□□□□□□□□□□□",
  "_comment_location": "□□□□□: src/main/resources/data/<tag_namespace>/tags/blocks/<tag_name>.json",
  "_comment_end": "================================================================================",

  "replace": false,
  "_comment_replace": "replace: false □□□□□□□□□□□□□true □□□□□□□□",

  "values": [
    {
      "id": "examplemod:example_block",
      "required": true,
      "_comment": "required: true □□□□□□□□□□□□□□□□□□□□□□□□□□□□"
    },
    {
      "id": "minecraft:stone",
      "required": true
    },
    {
      "#": "minecraft:mineable/pickaxe",
      "required": false,
      "_comment": "# □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□"
    },
    {
      "id": "othermod:other_block",
      "required": false,
      "_comment": "required: false □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□"
    }
  ]
}
```

worldgen/biome_loading_event.java

```
// ================================================================================
// Forge 1.18.2 □□□□□□□□□□□□ - biome_loading_event.java
// MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0
// ================================================================================
// □□ Forge 1.19.2 □□□□□□□
// 1. 1.18.2 □□□□□□□□□□□□□□□□□□□□
//    - □□□□□□BiomeLoadingEvent□□□□□□□□□□□□
//    - □□□□□□biome_modifier JSON□□□□□□□□□□
// 2. BiomeLoadingEvent □ 1.18.2 □ 1.19.2 □□□□
// 3. 1.19+ □□□□□ BiomeLoadingEvent□□□□□□□□ biome_modifier JSON
// □□ Forge 1.16.5 □□□□□□□
// 4. 1.16.5 □□ BiomeLoadingEvent □□□□□
// 5. 1.18.2 □□□□□□□□□□□□□□□1.18 □□□□□□□□□□□□□□□
// 6. 1.16.5 □ BiomeGenerationSettings.Builder□1.18.2 □□□□□□□□□
// ================================================================================

package com.example.examplemod.worldgen;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.level.biome.MobSpawnSettings;
import net.minecraft.world.level.levelgen.GenerationStep;
import net.minecraftforge.event.world.BiomeLoadingEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * □□□□□□□□□□ - BiomeLoadingEvent
 *
 * □1.18.2 □□□□□□
 * - BiomeLoadingEvent □ Forge □□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - 1.18.2 □□□□□□□□□□ biome_modifier JSON□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□ JSON □□□□□□
 *
 * □□□□□
 * - □□□□□ FORGE □□□□□□□□
```

```
 * - □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
 * - □□□□□□□□□□□□□□□□/□□□□□□□□ biome_modifier JSON
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModBiomeEvents {

    /**
     * □□□□□□□□□□
     * □□□□□□□□□□□□□□
     * - □□□□□□□feature generation□
     * - □□□□□□□mob spawning□
     * - □□□□□□□□□climate, effects □□
     */
    @SubscribeEvent
    public static void onBiomeLoading(BiomeLoadingEvent event) {
        // ===== □□□□□□□□□□ =====
        // event.getName() □□□□□□□□□ ResourceLocation
        // □□□□ null□□□□□□□□□□□□
        var biomeName = event.getName();
        if (biomeName == null) return;

        // ===== □□□□□□□□□□□□□□□□□□□□□□□□□□□□ =====
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□ biome_modifier JSON
        //
        // if (biomeName.getNamespace().equals("minecraft")) {
        //     // □□□□□□□□□□□□□□□□□□
        //     event.getGeneration().addFeature(
        //         GenerationStep.Decoration.UNDERGROUND_ORES,
        //         ModConfiguredFeatures.EXAMPLE_ORE_PLACED
        //     );
        // }

        // ===== □□□□□□□□□□□□□□ =====
        // □□□□□□□□□□□□□□□□□□□□□□
        //
        // String biomePath = biomeName.getPath();
        // if (biomePath.equals("plains") || biomePath.equals("forest")) {
        //     event.getSpawns().addSpawn(
        //         MobCategory.CREATURE,
        //         new MobSpawnSettings.SpawnerData(
        //             EntityType.SHEEP,    // □□□□□
        //             10,                  // □□□□□
        //             2,                   // □□□□□□□□
        //             6                    // □□□□□□□□
        //         )
        //     );
        // }

        // ===== □□□□□□□□□□□□□□□□ =====
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //
        // event.getClimate().setTemperature(0.5F);
        // event.getEffects().setSkyColor(0x7F7FE5);
    }

    // ============================================================================
    // □BiomeLoadingEvent vs biome_modifier JSON □□□
    //
    // □□                  | BiomeLoadingEvent    | biome_modifier JSON
    // --------------------|----------------------|---------------------
    // □□□□□□              | □□□□□□               | JSON □□□□□
    // □□                  | □□□□□□□□□□           | □□□□□□□□□□□
    // □□□□                | □□□□□□□□□□□□□□□□     | □□□□□□□□□□
    // □□□□□□□□□           | □□□□                 | □□□□□□□□□□□□
    // □□□□□□              | □□□□□□□□□□           | □□□□□/□□
    //
    // □□□□□□□□
    // 1. □□□□□□□□□□/□□/□□ → □ biome_modifier JSON
    // 2. □□□□□□□□□/□□□□□□□□□□ → □ BiomeLoadingEvent
    // 3. □□□□□□□□□□□
    // ============================================================================

    // ============================================================================
    // □□ 1.16.5 □□□□□
    //
    // 1. □□□□□□□□□BiomeLoadingEvent□□□□□□□□□□□□□□
```

```
    // 2. 1.16.5 □□□□□□□□□□□□ getGeneration().getFeatures().add()
    //    1.18.2 □ getGeneration().addFeature(step, placedFeature)
    // 3. 1.16.5 □□□□□□□□□□Feature + ConfiguredFeature □□□□□
    //    1.18.2 □□□□□□□□□□ConfiguredFeature + PlacedFeature□
    // ========================================================================
}
```

worldgen/biome_modifier.json

```
{
  "_comment_biome_modifier": "================================================================================",
  "_comment_version": "Forge 1.18.2 □□□□□□□□□□ - biome_modifier.json",
  "_comment_mc": "MC□□: 1.18.2 | Java□□: 17 | Forge□□: 40.2.0",
  "_comment_diff": "□□ Forge 1.19.2 □□□□□",
  "_comment_diff_1": "1. 1.18.2 □□□□□□□□□□ BiomeModifier□Forge □□□",
  "_comment_diff_2": "2. JSON □□□ 1.19.2 □□□□□",
  "_comment_diff_3": "3. □□□□□□ BiomeLoadingEvent □□□□□□□□□□ biome_loading_event.java□",
  "_comment_diff_old": "□□ Forge 1.16.5 □□□□□",
  "_comment_diff_old_1": "1. 1.16.5 □□ BiomeLoadingEvent □□□□□",
  "_comment_diff_old_2": "2. 1.18.2 □□□□□□□□□□□□□□□Caves & Cliffs □□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□□",
  "_comment_diff_1": "1. Forge □ biome_modifier JSON □□□Forge □□□□□□□□□□",
  "_comment_diff_2": "2. Fabric □ BiomeModification API□□□□□□□",
  "_comment_location": "□□□□□: src/main/resources/data/examplemod/forge/biome_modifier/example_modifier.json",
  "_comment_end": "================================================================================",

  "type": "forge:add_features",
  "_comment_type": "biome_modifier □□□□",
  "_comment_type_1": "  forge:add_features     - □□□□□□□□□/□□□□□",
  "_comment_type_2": "  forge:add_spawns       - □□□□□□□□",
  "_comment_type_3": "  forge:remove_features  - □□□□□",
  "_comment_type_4": "  forge:remove_spawns    - □□□□□□□□",
  "_comment_type_5": "  forge:modify_properties - □□□□□□□□□□",

  "biomes": "#minecraft:is_overworld",
  "_comment_biomes": "□□□□□□□□□□□□□□",
  "_comment_biomes_1": "  - □□□□□□\"#minecraft:is_overworld\"",
  "_comment_biomes_2": "  - □□ ID□\"minecraft:plains\"",
  "_comment_biomes_3": "  - ID □□□[\"minecraft:plains\", \"minecraft:forest\"]",

  "features": [
    "examplemod:example_ore_placed"
  ],
  "_comment_features": "□□□□□ placed_feature □□□□□□□□□□□□ placed_feature ID□",

  "step": "underground_ores",
  "_comment_step": "□□□□□□□□□Decoration □□□□□",
  "_comment_step_1": "  - raw_generation          □□□□□□□□□□□",
  "_comment_step_2": "  - lakes                   □□□",
  "_comment_step_3": "  - local_modifications     □□□□□",
  "_comment_step_4": "  - underground_structures  □□□□□",
  "_comment_step_5": "  - surface_structures      □□□□□",
  "_comment_step_6": "  - strongholds             □□□",
  "_comment_step_7": "  - underground_ores        □□□□□□□□□□□□",
  "_comment_step_8": "  - underground_decoration  □□□□□",
  "_comment_step_9": "  - fluid_springs           □□□",
  "_comment_step_10": " - vegetal_decoration      □□□□□",
  "_comment_step_11": " - top_layer_modification  □□□□"
}
```

Forge 1.19.2

block/simple_block.java

```
// ========================================================================
// Forge 1.19.2 □□□□□□□ - simple_block.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ========================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. 1.19.2 □□ AbstractBlock.Properties□1.20.1 □□ BlockBehaviour.Properties□
// 2. 1.19.2 □□ Material □□1.19.3+ □□□□□□□ .of(Material.STONE)
// 3. 1.19.2 □ SoundType □□□□□□□
// 4. DeferredRegister + ForgeRegistries.BLOCKS □□□ 1.20.1 □□
// □□ Fabric □□□□□□□
```

```
// 5. Fabric □ Registry.register(Registries.BLOCK, id, block) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.BLOCKS, MODID) □□□□
// 6. Forge □ RegistryObject<Block> □□□□□□□□□□□
//    Fabric □□□□ Block □□
// ============================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.material.Material;
import net.minecraft.world.level.material.MapColor;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□ ModMain □□□□□□□□□
 *   ModBlocks.BLOCKS.register(modEventBus);
 */
public class ModBlocks {

    // ===== □□□□ DeferredRegister =====
    // □□□□Forge □ ForgeRegistries.BLOCKS
    //         Fabric 1.19.2 □ Registry.BLOCK□1.19.2 □ net.minecraft.util.registry □□□
    public static final DeferredRegister<Block> BLOCKS =
            DeferredRegister.create(ForgeRegistries.BLOCKS, ExampleMod.MODID);

    // □□□□□□□□□□
    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITIES =
            DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□ =====
    // □□□□□□1.19.2 □□ AbstractBlock.Properties□1.20.1 □□ BlockBehaviour.Properties□
    // 1.19.2 □□□ .of(Material.STONE) □□□□□1.19.3+ Material □□□□
    //
    // □□□□Fabric 1.19.2:
    //   public static final Block EXAMPLE_BLOCK = Registry.register(
    //       Registry.BLOCK, new ResourceLocation("examplemod", "example_block"),
    //       new Block(FabricBlockSettings.of(Material.STONE).strength(3.0f)));
    // Forge:
    //   public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register(
    //       "example_block", () -> new Block(AbstractBlock.Properties.of(Material.STONE)...));
    public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register("example_block",
            () -> new Block(BlockBehaviour.Properties.of(Material.STONE)
                    .mapColor(MapColor.STONE)
                    .strength(3.0F, 6.0F)         // □□ / □□□□
                    .requiresCorrectToolForDrops()  // □□□□□□□□□□□
                    .sound(net.minecraft.world.level.block.SoundType.STONE)
            ));

    // ===== □□ BlockEntityType =====
    // BlockEntityType.Builder.of(factory, blocks...).build(null)
    // □□□□Fabric: BlockEntityType.Builder.create() + Registry.register
    // Forge: □□□□ Builder□□□□ DeferredRegister □□
    // public static final RegistryObject<BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
    //         BLOCK_ENTITIES.register("example_block_entity",
    //                 () -> BlockEntityType.Builder
    //                         .of(ExampleBlockEntity::new, EXAMPLE_BLOCK.get())
    //                         .build(null));

    // ===== □□□□□□ =====
    public static void register(IEventBus modEventBus) {
        BLOCKS.register(modEventBus);
        BLOCK_ENTITIES.register(modEventBus);
    }
}
```

command/simple_command.java

```
// ============================================================================
```

```
// Forge 1.19.2 □□□□□□□ - simple_command.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□ API □□□□□□RegisterCommandsEvent + Brigadier
// 2. sendSuccess() □□□□□□ 1.20.1 □□□□□□ Supplier<Component>□
// 3. Component □□ 1.19.2 □□□□□1.19.1 □□□□
// □□ Fabric □□□□□□□
// 4. Fabric □ CommandRegistrationCallback.EVENT □□□□□
//    Forge □ RegisterCommandsEvent □□□□□ FORGE EVENT_BUS □□□□
// 5. □□□□□□□ Brigadier □□□□□□API □□
// 6. Forge □ @Mod.EventBusSubscriber □□□□□□□□
// ================================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.network.chat.Component;
import net.minecraft.server.level.ServerPlayer;
import net.minecraftforge.event.RegisterCommandsEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * Forge □□□□□ - □□ RegisterCommandsEvent □□
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        CommandDispatcher<CommandSourceStack> dispatcher = event.getDispatcher();

        // □□□□□□/examplemod give <target> <amount>
        dispatcher.register(Commands.literal(ExampleMod.MODID)
                .requires(source -> source.hasPermission(2))
                .then(Commands.literal("give")
                        .then(Commands.argument("target", EntityArgument.player())
                                .then(Commands.argument("amount", IntegerArgumentType.integer(1, 64))
                                        .executes(ModCommands::executeGive)
                                )
                        )
                )
                .executes(ModCommands::executeInfo)
        );
    }

    private static int executeGive(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        ServerPlayer target = EntityArgument.getPlayer(context, "target");
        int amount = IntegerArgumentType.getInteger(context, "amount");

        CommandSourceStack source = context.getSource();
        // □□□□Fabric □ source.sendFeedback(Text.of(...), false)
        // Forge □ source.sendSuccess(() -> Component, false)
        source.sendSuccess(() -> Component.literal(
                "Gave " + amount + " items to " + target.getName().getString()), true);

        return Command.SINGLE_SUCCESS;
    }

    private static int executeInfo(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal("Example Mod v1.0.0 - Forge 1.19.2"), false);
        return Command.SINGLE_SUCCESS;
    }
}
```

enchantment/custom_enchantment.java

```
// ========================================================================
// Forge 1.19.2 □□□□□□□ - custom_enchantment.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ========================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□ 1.20.1 □□□□□□□□ Enchantment □
// 2. □□□□□□□□Enchantment(Rarity, EnchantmentCategory, EquipmentSlot[])
// 3. DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) □□
// 4. □□□1.21+ □□□□□□□□□□□□□JSON□□1.19.2 □□□□□□□
// □□ Fabric □□□□□□□
// 5. Fabric □ Registry.register(Registries.ENCHANTMENT) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS) □□□□
// 6. □□□ Enchantment API □□□□□□□□ MC □□□
// ========================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 */
public class ModEnchantments {

    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, ExampleMod.MODID);

    // ===== □□□□□ =====
    // Enchantment □□□□□□□ 1.20.1 □□□□□□ API□□
    //   - Rarity rarity: □□□□□COMMON/UNCOMMON/RARE/VERY_RARE□
    //   - EnchantmentCategory category: □□□□□□□□
    //   - EquipmentSlot[] slots: □□□□□□□□
    //
    // EnchantmentCategory □□□□□
    //   ALL / ARMOR / ARMOR_FEET / ARMOR_LEGS / ARMOR_CHEST / ARMOR_HEAD
    //   WEAPON / FISHING_ROD / BREAKABLE / DIGGER / BOW / CROSSBOW / TRIDENT / VANISHABLE
    public static final RegistryObject<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND));

    // ===== □□□□□□□ =====
    public static class ExampleEnchantment extends Enchantment {

        public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                                  EquipmentSlot... slots) {
            super(rarity, category, slots);
        }

        @Override
        public int getMinCost(int level) {
            return 10 * level;
        }

        @Override
        public int getMaxCost(int level) {
            return getMinCost(level) + 15;
        }

        @Override
        public int getMaxLevel() {
            return 3;
        }
        @Override
```

```
        protected boolean checkCompatibility(Enchantment other) {
            return super.checkCompatibility(other)
                    && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
        }

        @Override
        public boolean canEnchant(net.minecraft.world.item.ItemStack stack) {
            return super.canEnchant(stack);
        }

        @Override
        public boolean isCurse() {
            return false;
        }

        @Override
        public boolean isTradeable() {
            return true;
        }

        @Override
        public boolean isDiscoverable() {
            return true;
        }

        @Override
        public boolean isAllowedOnBooks() {
            return true;
        }

        // ===== □□□□□□ =====
        @Override
        public void doPostAttack(
                net.minecraft.world.entity.LivingEntity attacker,
                net.minecraft.world.entity.Entity target,
                int level) {
            if (target instanceof net.minecraft.world.entity.LivingEntity livingTarget) {
                livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                        net.minecraft.world.effect.MobEffects.POISON, 60, level - 1));
            }
            super.doPostAttack(attacker, target, level);
        }
    }

    public static void register(IEventBus modEventBus) {
        ENCHANTMENTS.register(modEventBus);
    }

    // ===== □□□□□□ - Forge □□ =====
    // Forge □ EnchantmentLevelSetEvent □□□□□□□□□□
    // □□□□Fabric □□□□□□□□ Mixin □□
    // @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.FORGE)
    // public static class ForgeEvents {
    //     @SubscribeEvent
    //     public static void onEnchantLevel(EnchantmentLevelSetEvent event) {
    //         // event.getEnchantments().put(EXAMPLE_ENCHANTMENT.get(), 1);
    //     }
    // }
}
```

entity/simple_entity.java

```
// ================================================================================
// Forge 1.19.2 □□□□□□□ - simple_entity.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□ API □□□□□□DeferredRegister + ForgeRegistries.ENTITY_TYPES
// 2. EntityType.Builder.create() □ 1.19.2 □□□1.20.1 □□ .of()□
//    □□□1.19.2 □ Builder.create()□1.20.1 □ Builder.of()
// 3. EntityAttributeCreationEvent □□□□□ 1.20.1 □□
// 4. ForgeSpawnEggItem □ 1.19.2 □□□
// □□ Fabric □□□□□□□
// 5. Fabric □ Registry.register(Registries.ENTITY_TYPE) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ENTITY_TYPES) □□□□
// 6. □□□□□Fabric □ FabricDefaultAttributeRegistry.register()
```

```
//    Forge 用 EntityAttributeCreationEvent 事件
// 7. 注册渲染器：Fabric 用 EntityRendererRegistry.register()
//    Forge 用 EntityRenderersEvent.RegisterRenderers 事件
// ============================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.entity.ai.attributes.AttributeSupplier;
import net.minecraft.world.entity.ai.attributes.Attributes;
import net.minecraft.world.entity.monster.Monster;
import net.minecraftforge.event.entity.EntityAttributeCreationEvent;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 实体注册类 - Forge DeferredRegister 模式
 */
public class ModEntities {

    public static final DeferredRegister<EntityType<?>> ENTITIES =
            DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, ExampleMod.MODID);

    // ===== 实体类型注册 =====
    // 注意：1.19.2 用 EntityType.Builder.create()，1.20.1 改用 .of()！
    // 对比Fabric: EntityType.Builder.create() + .build() + Registry.register
    // Forge: 用 RegistryObject 延迟supplier lambda 注册模式
    public static final RegistryObject<EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITIES.register("example_entity",
                    () -> EntityType.Builder
                            // 注意：1.19.2: .create()   1.20.1: .of()
                            .create(ExampleEntity::new, MobCategory.CREATURE)
                            .sized(1.0F, 1.0F)
                            .clientTrackingRange(10)
                            .build("example_entity"));

    // ============================================================================
    // 实体属性注册 - Forge 事件驱动模式
    // 对比Fabric: FabricDefaultAttributeRegistry.register(type, builder)
    // Forge: 监听 EntityAttributeCreationEvent，调用 event.put(type, builder.build())
    public static void registerAttributes(EntityAttributeCreationEvent event) {
        AttributeSupplier.Builder builder = Monster.createMonsterAttributes();
        builder.add(Attributes.MAX_HEALTH, 20.0D);
        builder.add(Attributes.MOVEMENT_SPEED, 0.25D);
        builder.add(Attributes.ATTACK_DAMAGE, 5.0D);
        builder.add(Attributes.FOLLOW_RANGE, 32.0D);

        event.put(EXAMPLE_ENTITY.get(), builder.build());
    }

    // 用 @SubscribeEvent + 事件总线注册
    @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.MOD)
    public static class ModBusEvents {
        @SubscribeEvent
        public static void onAttributeCreate(EntityAttributeCreationEvent event) {
            registerAttributes(event);
        }
    }

    // ===== 实体渲染器注册 =====
    // 对比Fabric: EntityRendererRegistry.register(type, renderer)
    // Forge: EntityRenderersEvent.RegisterRenderers 事件
    // @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.MOD, value = Dist.CLIENT)
    // public static class ClientBusEvents {
    //     @SubscribeEvent
    //     public static void onRegisterRenderers(EntityRenderersEvent.RegisterRenderers event) {
    //         event.registerEntityRenderer(EXAMPLE_ENTITY.get(), ExampleRenderer::new);
    //     }
    // }
    public static void register(IEventBus modEventBus) {
```

```
        ENTITIES.register(modEventBus);
    }
}
```

gui/container_screen.java

```
// ================================================================================
// Forge 1.19.2 GUI / □□□□□□□ - container_screen.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. MenuType □□□: ForgeRegistries.CONTAINERS□□ 1.20.1 □□□
// 2. IForgeMenuType.create() □ 1.19.2 □□□
// 3. AbstractContainerScreen □ AbstractContainerMenu API □□
// 4. 1.19.2 □ GuiGraphics □□□□□ PoseStack + RenderSystem □□
//    □1.20.1 □□□ GuiGraphics □□□□□
// □□ Fabric □□□□□□□
// 5. Fabric □ ScreenHandlerType + Registry.register(Registries.SCREEN_HANDLER)
//    Forge □ DeferredRegister.create(ForgeRegistries.CONTAINERS) □□ MenuType
// 6. Forge □ MenuType = Fabric □ ScreenHandlerType
//    Forge □ AbstractContainerMenu = Fabric □ ScreenHandler
// 7. Forge □□□□□ NetworkHooks.openScreen()
//    Fabric □ player.openHandledScreen()
// ================================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.client.gui.screens.MenuScreens;
import net.minecraft.client.gui.screens.inventory.AbstractContainerScreen;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraftforge.common.extensions.IForgeMenuType;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□ - Forge DeferredRegister + MenuScreens
 */
public class ModMenus {

    // ===== □□□□□□□□ =====
    // □□□□Forge: ForgeRegistries.CONTAINERS
    //         Fabric: Registries.SCREEN_HANDLER
    public static final DeferredRegister<MenuType<?>> MENUS =
            DeferredRegister.create(ForgeRegistries.CONTAINERS, ExampleMod.MODID);

    // ===== □□ MenuType =====
    // IForgeMenuType.create() □□□□□□□□□□□□□ MenuType
    // □□□□Fabric □ new ScreenHandlerType<>(factory)
    public static final RegistryObject<MenuType<ExampleMenu>> EXAMPLE_MENU = MENUS.register("example_menu",
            () -> IForgeMenuType.create((containerId, inventory, data) -> {
                // □□□□□□ extraData □□
                return new ExampleMenu(containerId, inventory);
            }));

    // ================================================================================
    // AbstractContainerMenu□□□□□□□□
    // □□□□Forge □ AbstractContainerMenu = Fabric □ ScreenHandler
    public static class ExampleMenu extends AbstractContainerMenu {

        public ExampleMenu(int containerId, Inventory playerInventory) {
            super(EXAMPLE_MENU.get(), containerId);

            // □□□□□□□□□3x9□
            for (int row = 0; row < 3; row++) {
                for (int col = 0; col < 9; col++) {
                    this.addSlot(new Slot(playerInventory, 9 + row * 9 + col,
                            8 + col * 18, 84 + row * 18));
```

```
                }
            }

            // □□□□□□□□□9□□
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
            }
        }

        @Override
        public ItemStack quickMoveStack(Player player, int index) {
            Slot slot = this.slots.get(index);
            if (!slot.hasItem()) return ItemStack.EMPTY;

            ItemStack original = slot.getItem().copy();
            ItemStack stack = slot.getItem();

            if (!this.moveItemStackTo(stack, 0, this.slots.size(), true)) {
                return ItemStack.EMPTY;
            }

            slot.setChanged();
            return original;
        }

        @Override
        public boolean stillValid(Player player) {
            return true;
        }
    }

    // ============================================================================
    // AbstractContainerScreen□□□□□□
    // □□□□□□□1.19.2 □ PoseStack + RenderSystem □□□□ GuiGraphics□
    // 1.20.1 □ GuiGraphics □□□□
    //
    // □□□□Forge □ AbstractContainerScreen = Fabric □ HandledScreen
    public static class ExampleScreen extends AbstractContainerScreen<ExampleMenu> {
        public ExampleScreen(ExampleMenu menu, Inventory inventory, Component title) {
            super(menu, inventory, title);
            this.imageWidth = 176;
            this.imageHeight = 166;
            this.inventoryLabelY = this.imageHeight - 94;
        }

        // □□□□1.19.2 □ PoseStack □□□1.20.1 □ GuiGraphics□
        @Override
        protected void renderBg(net.minecraft.client.renderer.MultiBufferSource.BufferSource
                                // □□□1.19.2 □ renderBg □□□ 1.20.1 □□
                                // 1.19.2: renderBg(PoseStack poseStack, float partialTick, int x, int y)
                                // 1.20.1: renderBg(GuiGraphics graphics, float partialTick, int x, int y)
                                ignored,
                                float partialTick, int mouseX, int mouseY) {
            // 1.19.2 □□□□□□□ RenderSystem + blit□□
            // RenderSystem.setShader(GameRenderer::getPositionTexShader);
            // RenderSystem.setShaderTexture(0, TEXTURE);
            // int x = (this.width - this.imageWidth) / 2;
            // int y = (this.height - this.imageHeight) / 2;
            // blit(poseStack, x, y, 0, 0, this.imageWidth, this.imageHeight);
        }
    }

    // ===== Screen □□ =====
    // □□□□Fabric: HandledScreens.register(type, factory)
    // Forge: MenuScreens.register(type, factory)
    // □ FMLClientSetupEvent □□□□
    // event.enqueueWork(() -> {
    //     MenuScreens.register(EXAMPLE_MENU.get(), ExampleScreen::new);
    // });

    public static void register(IEventBus modEventBus) {
        MENUS.register(modEventBus);
    }
}
```

item/simple_item.java

```
// ========================================================================
// Forge 1.19.2 □□□□□□□ - simple_item.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ========================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□□1.19.2 □ CreativeModeTab □□ builder() □□□
//    1.19.2 □□□□ .tab(CreativeModeTab.TAB_MISC) □□□□□□□□
//    1.19.3+ □□□□ CreativeModeTab.builder() □ DeferredRegister<CreativeModeTab>
// 2. 1.19.2 □ CreativeModeTab □□□□□□□TAB_BUILDING_BLOCKS □□
//    1.20.1 □□ CreativeModeTabs □□□□□□
// 3. □□□□□□DeferredRegister + ForgeRegistries.ITEMS□□ 1.20.1 □□
// □□ Fabric □□□□□□□
// 4. Fabric □ Registry.register(Registries.ITEM, id, item) □□□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ITEMS, MODID) □□□□□
// 5. Fabric 1.19.2 □□ .group(ItemGroup.MISC) □□□□□□□□□□ API□
// ========================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□□ ModMain □□□□□□□□□□
 *   ModItems.ITEMS.register(modEventBus);
 */
public class ModItems {

    // ===== □□□□□□ =====
    // □□□□□Forge: ForgeRegistries.ITEMS
    //         Fabric 1.19.2: Registry.ITEM□net.minecraft.util.registry.Registry□
    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // □□□□□□□1.19.2 □ .tab(CreativeModeTab.TAB_MISC) □□□□□□□□
    // 1.20.1 □ CreativeModeTab.builder() □ BuildCreativeModeTabContentsEvent
    //
    // □□□□□Fabric 1.19.2:
    //   new Item(new Item.Settings().group(ItemGroup.MISC));
    // Forge 1.19.2:
    //   new Item(new Item.Properties().tab(CreativeModeTab.TAB_MISC));
    public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register("example_item",
            () -> new Item(new Item.Properties()
                    .stacksTo(64)                          // □□□□
                    .tab(CreativeModeTab.TAB_MISC)         // □1.19.2 □□□□□□□□□□□□□□
                    // .rarity(Rarity.RARE)                // □□□
                    // .fireResistant()                    // □□
            ));

    // ===== □□□□□□□□BlockItem□ =====
    // □□□□□□□Forge □□□□ BlockItem □□□□□□ ITEMS □□□□□□
    // public static final RegistryObject<Item> EXAMPLE_BLOCK_ITEM = ITEMS.register("example_block",
    //         () -> new net.minecraft.world.item.BlockItem(
    //                 com.example.examplemod.block.ModBlocks.EXAMPLE_BLOCK.get(),
    //                 new Item.Properties().tab(CreativeModeTab.TAB_MISC)));

    // ========================================================================
    // □1.19.2 □□□□□□□□□□□□□□
    // 1.19.2 □□CreativeModeTab.builder()□□□□□□□□□□□□ Forge □
    // CreativeModeTabRegistry □□□ CreativeModeTab □□□□□
    //
    // □□□□□1.20.1 □□□ DeferredRegister<CreativeModeTab> □□□□□□ Tab
    //         1.19.2 □□□□□□□□□CreativeModeTab □□□□□□□□
    // ========================================================================
    // Forge 1.19.2 □□□ Tab □□□□□□ CreativeModeTabRegistry□□
    //
```

```
    // public static final CreativeModeTab EXAMPLE_TAB =
    //         CreativeModeTabRegistry.create(new ResourceLocation(ExampleMod.MODID, "example_tab"),
    //                 builder -> builder
    //                         .title(Component.translatable("itemGroup.examplemod.example_tab"))
    //                         .icon(() -> new ItemStack(EXAMPLE_ITEM.get()))
    //                         .displayItems((params, output) -> {
    //                             output.accept(EXAMPLE_ITEM.get());
    //                         })
    //         );

    // ===== □□□□□□□ =====
    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

item/tool_item.java

```
// ================================================================================
// Forge 1.19.2 □□/□□□□□□ - tool_item.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. Tier □□□□□□ 1.20.1 □□□□□□getUses/getSpeed/getLevel □□
// 2. □□□□□□□□□□ 1.20.1 □□□PickaxeItem(tier, damage, speed, props)□
// 3. TierSortingRegistry □ 1.19.2 □□□
// □□ Fabric □□□□□□□
// 4. Forge □ Tier □□□net.minecraft.world.item.Tier□
//    Fabric 1.19.2 □ ToolMaterial□net.minecraft.item.ToolMaterial□
//    □□□□□□□Forge getUses() vs Fabric getDurability() □
// 5. Forge □ DeferredRegister □□□□□□□□Fabric □ Registry.register
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□ - Forge Tier □□
 */
public class ModTools {

    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ================================================================================
    // □□□□□□□□□Tier□
    // □□ Forge □ Tier □□
    // □□□□Fabric 1.19.2 □ ToolMaterial □□□□□□
    //   Forge: getUses()        Fabric: getDurability()
    //   Forge: getSpeed()      Fabric: getMiningSpeedMultiplier()
    //   Forge: getLevel()      Fabric: getMiningLevel()
    //   Forge: getEnchantmentValue()  Fabric: getEnchantability()
    // ================================================================================

    public enum ModTiers implements Tier {
        EXAMPLE_TIER(
            1561,    // getUses()□□□□□□□□1561□
            8.0F,    // getSpeed()□□□□□□
            3.0F,    // getAttackDamageBonus()□□□□□□□□
            3,       // getLevel()□□□□□□□□0/□1/□2/□3/□□□□4□
            10,      // getEnchantmentValue()□□□□□□
        );

        private final int uses;
        private final float speed;
```

```
        private final float attackDamageBonus;
        private final int level;
        private final int enchantmentValue;
        private final Ingredient repairIngredient;

        ModTiers(int uses, float speed, float attackDamageBonus,
                 int level, int enchantmentValue) {
            this.uses = uses;
            this.speed = speed;
            this.attackDamageBonus = attackDamageBonus;
            this.level = level;
            this.enchantmentValue = enchantmentValue;
            this.repairIngredient = Ingredient.of(net.minecraft.world.item.Items.DIAMOND);
        }

        @Override
        public int getUses() {
            return uses;                        // Fabric: getDurability()
        }

        @Override
        public float getSpeed() {
            return speed;                       // Fabric: getMiningSpeedMultiplier()
        }

        @Override
        public float getAttackDamageBonus() {
            return attackDamageBonus;           // Fabric: getAttackDamage()
        }

        @Override
        public int getLevel() {
            return level;                       // Fabric: getMiningLevel()
        }

        @Override
        public int getEnchantmentValue() {
            return enchantmentValue;            // Fabric: getEnchantability()
        }

        @Override
        public Ingredient getRepairIngredient() {
            return repairIngredient;
        }
    }

    // ===== □□□□□□□ =====
    // □□□□□ 1.20.1 □□□new PickaxeItem(tier, attackDamage, attackSpeed, properties)
    // □□□□Fabric 1.19.2 □□□□□□□□□ tier □□□□□□□
    public static final RegistryObject<Item> EXAMPLE_PICKAXE = ITEMS.register("example_pickaxe",
            () -> new PickaxeItem(ModTiers.EXAMPLE_TIER,
                    1,      // □□□□□□□
                    -2.8F,  // □□□□
                    new Item.Properties().tab(CreativeModeTab.TAB_TOOLS)));

    public static final RegistryObject<Item> EXAMPLE_SWORD = ITEMS.register("example_sword",
            () -> new SwordItem(ModTiers.EXAMPLE_TIER,
                    3,      // □□□□□□□
                    -2.4F,  // □□□□
                    new Item.Properties().tab(CreativeModeTab.TAB_COMBAT)));

    // ===== Tier □□□□□Forge □□□ =====
    // TierSortingRegistry □□□□□ Tier□□□□□□□□
    // TierSortingRegistry.registerTier(
    //     ModTiers.EXAMPLE_TIER,
    //     new ResourceLocation(ExampleMod.MODID, "example_tier"),
    //     List.of(Tiers.DIAMOND),
    //     List.of()
    // );

    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

mixin/mixin_config.json

```
{
  "_comment_mixin_config": "================================================================================",
  "_comment_version": "Forge 1.19.2 Mixin □□ - mixin_config.json",
  "_comment_mc": "MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□□□□□□□□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□□",
  "_comment_diff_1": "1. Forge □□ Mixin□SpongePowered Mixin□□□□□□□□□",
  "_comment_diff_2": "2. Forge □□ jar □ MANIFEST.MF □□□ MixinConfigs",
  "   _comment_manifest": "   build.gradle: attributes([\"MixinConfigs\": \"examplemod.mixins.json\"])",
  "_comment_diff_3": "3. Fabric □ fabric.mod.json □ mixins □□□□",
  "   _comment_fabric": "   \"mixins\": [\"examplemod.mixins.json\"]",
  "_comment_diff_4": "4. Forge □ official □□□Fabric □ yarn/parchment",
  "   _comment_mapping": "   Mixin □□□□/□□□□□□□□□□□",
  "_comment_location": "□□□□□: src/main/resources/examplemod.mixins.json",
  "_comment_end": "================================================================================",

  "required": true,
  "_comment_required": "required: true □□□ Mixin □□□□□□□",

  "minVersion": "0.8",
  "_comment_minVersion": "Mixin □□□□□□",

  "package": "com.example.examplemod.mixin",
  "_comment_package": "Mixin □□□□□ Java □",

  "compatibilityLevel": "JAVA_17",
  "_comment_compatibilityLevel": "Java □□□□□□1.19.2 □□ JAVA_17",

  "refmap": "examplemod.refmap.json",
  "_comment_refmap": "□□□□□□□□□ForgeGradle □□□□□□",

  "verbose": false,
  "showMessageTypes": false,

  "mixins": [
    {
      "_comment": "□□ Mixin□□□□□+□□□□□□□□",
      "name": "MinecraftServerMixin"
    }
  ],

  "client": [
    {
      "_comment": "□□□□□□ Mixin",
      "name": "ClientLevelMixin"
    },
    {
      "name": "GuiMixin"
    }
  ],

  "server": [
    {
      "_comment": "□□□□□□ Mixin",
      "name": "DedicatedServerMixin"
    }
  ],

  "injectors": {
    "defaultRequire": 1,
    "_comment_defaultRequire": "Mixin □□□□□□□□□1 = □□□□□□□□1□□□□□□"
  }
}
```

project/ModMain.java

```
// ================================================================================
// Forge 1.19.2 □□□□□□□□ - ModMain.java
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. API □□□□□@Mod □□ + DeferredRegister + ForgeRegistries
// 2. 1.19.2 □ Component □□□□□1.19.1 □□□□□□□□ TextComponent□
// 3. 1.19.2 □ CreativeModeTab □□ builder() □□□1.19.3+ □□□□
```

```
// 4. 1.19.2 □□ AbstractBlock.Properties□1.20.1 □□ BlockBehaviour.Properties□
// □□ Fabric □□□□□□□
// 5. Fabric □□ ModInitializer □□□□ onInitialize() □□
//    Forge □□ @Mod □□□□□□□□□□□□□□□ DeferredRegister
// 6. Fabric □ Registry.register() □□□□□ Registries
//    Forge □ DeferredRegister.create(ForgeRegistries.XXX, MODID) □□□□
// 7. Fabric □□□□□□□□□□□□□□□□
//    Forge □ IEventBus□modEventBus□□□□□□□□□ addListener / @SubscribeEvent
// ============================================================================

package com.example.examplemod;

import net.minecraftforge.common.MinecraftForge;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext;

// □□□□□□□□□ @Mod □□□□□□□ modId
@Mod(ExampleMod.MODID)
public class ExampleMod {

    public static final String MODID = "examplemod";

    // □□□□□□□□Forge □□□□□□□□□□□□□□□□
    // □□□□□□ DeferredRegister□□□□□□□□□□□□□□□□□□□
    public ExampleMod() {
        // □□□□□□□□□□□□□modEventBus□
        // □□□□□1.19.2 □ 1.20.1 □□□□ FMLJavaModLoadingContext
        //         □1.21+ □ NeoForge □□□□□□□□
        IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();

        // ===== □□ DeferredRegister =====
        // □□□□□□□□□□ register(modEventBus)
        // □□
        //   ModBlocks.BLOCKS.register(modEventBus);
        //   ModItems.ITEMS.register(modEventBus);
        //   ModEntities.ENTITIES.register(modEventBus);
        //   ModMenus.MENUS.register(modEventBus);
        //   ModEnchantments.ENCHANTMENTS.register(modEventBus);

        // □□□□□□□□□□
        modEventBus.addListener(this::commonSetup);

        // □□□□□□ FORGE □□□□□□□□□□□□□□□□
        MinecraftForge.EVENT_BUS.register(this);
    }

    // □□□□□□□
    private void commonSetup(final net.minecraftforge.fml.event.lifecycle.FMLCommonSetupEvent event) {
        // □□□□□□□□□□
        // event.enqueueWork(() -> { ... });
    }

    // ===== □□□□□□□□ =====
    // @SubscribeEvent
    // public void onServerTick(net.minecraftforge.event.TickEvent.ServerTickEvent event) {
    //     if (event.phase == TickEvent.Phase.END) {
    //         // □□...
    //     }
    // }
}
```

project/build.gradle

```
// ============================================================================
// Forge 1.19.2 □□□□□□□ - build.gradle
// MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
// ============================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. ForgeGradle □□: 1.19.2 □ 5.1.x□1.20.1 □ 6.0.x□
// 2. Forge □□: 43.3.0□1.20.1 □ 47.3.0□
// 3. □□□: 1.19.2 □ official 1.19.2 □□
// 4. mods.toml □□□□□□□1.19.2 □ displayTest □□□□
// 5. 1.19.2 □ creative tab □□□□□□□ builder□□□□ .tab() □□□
// □□ Fabric □□□□□□□
// 6. Forge □ ForgeGradle □□□□ Loom□□□□ minecraftForge □□□□□□□
```

```
// 7. Forge 用 fg.deobf() 处理混淆依赖
// ============================================================================

// 使用 ForgeGradle 最新（1.19.2 用 5.1 系列）
// 如果你1.20.1 用 'net.minecraftforge.gradle' version '6.0.24'
//         1.19.2 用 'net.minecraftforge.gradle' version '5.1.+'
plugins {
    id 'eclipse'
    id 'idea'
    id 'maven-publish'
    id 'net.minecraftforge.gradle' version '5.1.+'
}

// ----------------------------------------------------------------------------
// 项目基本信息
// ----------------------------------------------------------------------------
version = '1.0.0'
group = 'com.example.examplemod'  // 改成你自己的 ModMain.java 所在的包路径

// Java 版本设置1.19.2 需要至少 Java 17
// 如果你Fabric 1.19.2 也需要 Java 17或更高版
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// ----------------------------------------------------------------------------
// Minecraft / Forge 配置块
// ----------------------------------------------------------------------------
// 这个块告诉Forge 你的 "MC版本-Forge版本" 是什么组
// 1.19.2 推荐 Forge 43.3.0
minecraft {
    // 映射版本1.19.2 官方映射即可
    // 如果你 Parchment（mappings channel: 'parchment', version: '2022.11.27-1.19.2'
    mappings channel: 'official', version: '1.19.2'

    // 运行配置
    runs {
        // 客户端运行配置
        client {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            // 1.19.2 的 mod 加载配置
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }

        // 服务端运行配置
        server {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }

        // 数据生成器配置
        data {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            args '--mod', 'examplemod', '--all', '--output', file('src/generated/resources/'),
                 '--existing', file('src/main/resources/')
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }
    }
}
// ----------------------------------------------------------------------------
```

```
// □□□□
// ------------------------------------------------------------------------
repositories {
    maven { url = 'https://maven.minecraftforge.net/' }
    maven { url = 'https://maven.parchmentmc.org' }
    maven { url = 'https://maven.blamejared.com' }
    mavenCentral()
}

// ------------------------------------------------------------------------
// □□□□
// ------------------------------------------------------------------------
dependencies {
    // Minecraft + Forge □□□□□1.19.2-43.3.0
    minecraft 'net.minecraftforge:forge:1.19.2-43.3.0'

    // ===== □□□□□□□□□□□□□□□ =====
    // JEI□1.19.2 □□□
    // compileOnly(fg.deobf("mezz.jei:jei-1.19.2-common-api:11.x.x.x:api"))
    // runtimeOnly(fg.deobf("mezz.jei:jei-1.19.2-forge:11.x.x.x"))
}

// ------------------------------------------------------------------------
// □□□□□□□□□□□□□□□□
// ------------------------------------------------------------------------
sourceSets.main.resources { srcDir 'src/generated/resources' }

// ------------------------------------------------------------------------
// Jar □□□□
// ------------------------------------------------------------------------
jar {
    manifest {
        attributes([
            "Specification-Title"     : "examplemod",
            "Specification-Vendor"     : "examplemod",
            "Specification-Version"   : "1",
            "Implementation-Title"    : project.name,
            "Implementation-Version"  : project.jar.archiveVersion,
            "Implementation-Vendor"    : "examplemod",
            "MixinConfigs"            : "examplemod.mixins.json"
        ])
    }
}

// ------------------------------------------------------------------------
// □□□□□ build □□□ mods.toml □□□□□
// ------------------------------------------------------------------------
tasks.named('processResources', ProcessResources).configure {
    var replaceProperties = [
            minecraft_version   : "1.19.2",
            minecraft_version_range: "[1.19.2,1.19.3)",
            forge_version       : "43.3.0",
            forge_version_range : "[43,)",
            loader_version_range: "[43,)",
            mod_id              : "examplemod",
            mod_name            : "Example Mod",
            mod_license         : "MIT",
            mod_version         : "1.0.0",
            mod_authors         : "Author",
            mod_description     : "Example mod for Forge 1.19.2"
    ]
    inputs.properties replaceProperties

    filesMatching(['META-INF/mods.toml', 'pack.mcmeta']) {
        expand replaceProperties + [project: project]
    }
}
```

project/mods.toml

```
# ======================================================================
# Forge 1.19.2 □□□□□□□□ - mods.toml
# MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0
# ======================================================================
# □□ Forge 1.20.1 □□□□□□□
# 1. 1.19.2 □ mods.toml □□□ 1.20.1 □□□□
```

```
# 2. 1.19.2 □ displayTest □□□1.20.1 □□□□□□□□□□□□
# 3. □□□□□□□□1.19.2 □ [1.19.2,1.19.3) □□ [1.20.1,1.20.2)
# □□ Fabric □□□□□□□
# 4. Forge □ mods.toml□TOML □□□□Fabric □ fabric.mod.json□JSON□
# 5. Forge □ modLoader □□□ "javafmloader"
# 6. □□□□□□ src/main/resources/META-INF/mods.toml
# ============================================================================

# □□□□□□□□□javafmloader □ Forge Java □□□□□
modLoader="javafmloader"

# □□□□□□□□
loaderVersion="${loader_version_range}"

# □□□□□□□□□□
license="${mod_license}"

# ----------------------------------------------------------------------------
# displayTest□1.19.2 □□□
# □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
# - "MATCH_VERSION"□□□□□□□□□□□□□□□□
# - "IGNORE_ALL"□□□□□□□□□□□
# - "IGNORE_SERVER_VERSION"□□□□□□□□□□
# ----------------------------------------------------------------------------
#displayTest="MATCH_VERSION"

# ----------------------------------------------------------------------------
# □□□□□
# ----------------------------------------------------------------------------
[[mods]]
modId="${mod_id}"
version="${mod_version}"
displayName="${mod_name}"
authors="${mod_authors}"
description='''
${mod_description}
'''


# ----------------------------------------------------------------------------
# □□□□□□
# ----------------------------------------------------------------------------
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

[[dependencies.${mod_id}]]
    modId="forge"
    type="required"
    versionRange="${forge_version_range}"
    ordering="NONE"
    side="BOTH"

# □□□□□□□
# [[dependencies.${mod_id}]]
#     modId="jei"
#     type="optional"
#     versionRange="[11.0.0,)"
#     ordering="NONE"
#     side="BOTH"
```

recipe/shaped_recipe.json

```
{
  "_comment": "============================================================================",
  "_comment_version": "Forge 1.19.2 □□□□□□□ - shaped_recipe.json",
  "_comment_mc": "MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0",
  "_comment_diff_1": "□□ Forge 1.20.1 □□□□□",
  "_comment_diff_2": "1. JSON □□□ 1.20.1 □□□□□",
  "_comment_diff_3": "2. Forge 1.19.2 □□ result □□ nbt □□□□ 1.20.1 □□□",
  "_comment_diff_4": "3. Forge □□□□□□□□forge:mod_loaded, forge:item_exists □□",
  "_comment_diff_5": "4. □□□□ data/<modid>/recipes/ □",
  "_comment_fabric_diff": "□□ Fabric □□□□□",
  "_comment_fabric_diff_1": "5. Forge □□ nbt □□□forge: ingredient □□□forge □□",
```

```
  "_comment_fabric_diff_2": "6. Fabric □□ conditions □□□□□□□□□fabric:□",
  "_comment_location": "□□□□□: src/main/resources/data/examplemod/recipes/example_recipe.json",
  "_comment_end": "================================================================================",

  "type": "minecraft:crafting_shaped",

  "pattern": [
    "III",
    "I I",
    "III"
  ],

  "key": {
    "I": {
      "_comment": "□□ Forge □ forge:ingots/iron □□",
      "tag": "forge:ingots/iron"
    }
  },

  "result": {
    "item": "examplemod:example_item",
    "count": 1,
    "_comment_nbt": "□□□□□Forge □□ nbt □□□□□□□□ NBT",
    "nbt": "{display:{Name:\"{\\\"text\\\":\\\"Example Item\\\",\\\"color\\\":\\\"gold\\\",\\\"italic\\\":false}\\\"}}"
  },

  "_comment_conditions": "Forge □□□□□□□□□□□□□□□□",
  "conditions": [
    {
      "type": "forge:mod_loaded",
      "modid": "examplemod"
    }
  ]
}
```

tag/block_tag.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.19.2 □□□□□□□ - block_tag.json",
  "_comment_mc": "MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□□□□□□□□□□Forge □□□□□□□",
  "_comment_forge_tags": "Forge □□□□□ forge: □□□□□□□□",
  "_comment_forge_needs": "  - forge:needs_wood_tool / forge:needs_stone_tool / forge:needs_iron_tool □",
  "_comment_forge_ores": "  - forge:ores, forge:ores/iron: □□□□□□□",
  "_comment_fabric_diff": "□□ Fabric □□□□□Fabric □ forge: □□□□□□□□□ c: □□",
  "_comment_location": "□□□□□: src/main/resources/data/<tag_namespace>/tags/blocks/<tag_name>.json",
  "_comment_end": "================================================================================",

  "replace": false,

  "values": [
    {
      "id": "examplemod:example_block",
      "required": true
    },
    {
      "id": "minecraft:stone",
      "required": true
    },
    {
      "#": "minecraft:mineable/pickaxe",
      "required": false
    },
    {
      "id": "othermod:other_block",
      "required": false
    }
  ]
}
```

worldgen/biome_modifier.json

```
{
  "_comment_biome_modifier": "================================================================================",
  "_comment_version": "Forge 1.19.2 □□□□□□□□□□ - biome_modifier.json",
  "_comment_mc": "MC□□: 1.19.2 | Java□□: 17 | Forge□□: 43.3.0",
```

```
  "_comment_forge_vs_fabric": "□□ Fabric □□□□",
  "_comment_diff_1": "1. Forge □ biome_modifier JSON □□□Forge □□□□□□□□□",
  "_comment_diff_2": "2. Fabric □ BiomeModification API□□□□□□BiomeModifications.addFeature□",
  "_comment_diff_3": "3. Forge □ forge: □□□NeoForge 1.21+ □□ neoforge:□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/forge/biome_modifier/example_modifier.json",
  "_comment_end": "================================================================================",

  "type": "forge:add_features",
  "_comment_type": "biome_modifier □□□forge:add_features / forge:add_spawns / forge:remove_features □",

  "biomes": "#minecraft:is_overworld",
  "_comment_biomes": "□□□□□□□□□□□□□□□ #minecraft:is_overworld □□□ biome ID □□□",

  "features": [
    "examplemod:example_ore_placed"
  ],
  "_comment_features": "□□□□ placed_feature □□□□□□□□□□ placed_feature ID□",

  "step": "underground_ores",
  "_comment_step": "□□□□□□□□underground_ores□□□□□□□/ vegetal_decoration□□□□□□□□"
}
```

Forge 1.20.1

block/simple_block.java

```
// ================================================================================
// Forge 1.20.1 □□□□□□□ - simple_block.java
// ================================================================================
// □Forge □ Fabric □□□□□□□
// 1. Fabric □ Registry.register(Registries.BLOCK, id, block) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.BLOCKS, MODID) □□□□
// 2. Fabric □□□□□□□□□□□□ BlockItem □□□
//    Forge □ BLOCKS □ ITEMS □□□□□□□□□□□ DeferredRegister □□
// 3. Forge □ BlockEntityType □ ForgeRegistries.BLOCK_ENTITY_TYPES □□
//    Fabric □ BlockEntityType.Builder.create() + Registry.register(Registries.BLOCK_ENTITY_TYPE)
// 4. Forge □ RegistryObject<Block> □□□□□□□□□□□□□□
//    Fabric □□□□ Block □□
// 5. Forge □□□□□□□□ BlockBehaviour.Properties□□ Fabric □□□ MC API□
//    □□□□□□□□□□
// ================================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.Blocks;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.block.entity.BlockEntityType.Builder;
import net.minecraft.world.level.material.MapColor;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□ ModMain □□□□□□□□□
 *   ModBlocks.BLOCKS.register(modEventBus);
 *
 * □□□□□□□
 * Fabric:
 *   public static final Block EXAMPLE_BLOCK = Registry.register(
 *       Registries.BLOCK, new ResourceLocation("examplemod", "example_block"),
 *       new Block(AbstractBlock.Settings.create()...));
 * Forge:
 *   public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register(
 *       "example_block", () -> new Block(BlockBehaviour.Properties.of()...));
 */
public class ModBlocks {
    // ===== □□□□ DeferredRegister =====
```

```
// □□1□ForgeRegistries.BLOCKS□Forge □□□□□□□□□□ Fabric □ Registries.BLOCK□
// □□2□MODID□□□ID□□□□□ ResourceLocation □□□□□
// □□□□□□□Forge □ ForgeRegistries□Fabric □ net.minecraft.registry.Registries
public static final DeferredRegister<Block> BLOCKS =
        DeferredRegister.create(ForgeRegistries.BLOCKS, ExampleMod.MODID);

// □□□□□□□□□□□□□□□ DeferredRegister□
public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITIES =
        DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, ExampleMod.MODID);

// ===== □□□□□□RegistryObject □□ =====
// RegistryObject □ Forge □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
// BLOCKS.register(name, supplier) □□ RegistryObject<Block>
// supplier □□□□ () -> new Block(...) lambda□□□□□□□□□□
//
// □□□□□Fabric □□□□□ Block □□□Forge □□ RegistryObject □□
// □□□□□□□□□□□□□□□□□Forge □ Mod loading □□□□□□□□□□□□
public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register("example_block",
        () -> new Block(BlockBehaviour.Properties.of()
                // □□□□□□□□ Fabric □ AbstractBlock.Settings □□□□□□ MC □□ API□
                .mapColor(MapColor.STONE)
                .strength(3.0F, 6.0F)         // □□ / □□□□□
                .requiresCorrectToolForDrops()  // □□□□□□□□□□□□□
                .sound(net.minecraft.world.level.block.SoundType.STONE)
        ));

// ===== □□□□ BlockEntity □□□□ =====
public static final RegistryObject<Block> EXAMPLE_ENTITY_BLOCK = BLOCKS.register("example_entity_block",
        () -> new net.minecraft.world.level.block.EntityBlock() {
            // □□ EntityBlock □□□□□□ BlockEntity
            // □□□□□□□□□□□□□□□□□□□□ BaseEntityBlock □□□□□□□□□

            @Override
            public BlockEntity newBlockEntity(net.minecraft.core.BlockPos pos,
                                              net.minecraft.world.level.block.state.BlockState state) {
                // □□□□□□ BlockEntity □□
                return ModBlocks.EXAMPLE_BLOCK_ENTITY.get().create(pos, state);
            }

            @Override
            public net.minecraft.world.level.block.RenderShape getRenderShape(
                     net.minecraft.world.level.block.state.BlockState state) {
                // □□□□□□□□ INVISIBLE □□□□□ BlockEntityRenderer □□
                return net.minecraft.world.level.block.RenderShape.MODEL;
            }
        } instanceof Block ? (Block) null : null);
// □□□□□□□□□□□□□□□□□□□□□□□□□□□□ BaseEntityBlock □□□□□□□□□

/*
public static final RegistryObject<Block> EXAMPLE_ENTITY_BLOCK = BLOCKS.register("example_entity_block",
        () -> new BaseEntityBlock(BlockBehaviour.Properties.of().strength(3.0F)) {
            @Override
            public BlockEntity newBlockEntity(BlockPos pos, BlockState state) {
                return EXAMPLE_BLOCK_ENTITY.get().create(pos, state);
            }
            @Override
            public RenderShape getRenderShape(BlockState state) {
                return RenderShape.MODEL;
            }
        });
*/

// ===== □□ BlockEntityType =====
// □□□□□Fabric: BlockEntityType.Builder.create(ExampleBlockEntity::new, BLOCK).build(null)
//               □□ Registry.register(Registries.BLOCK_ENTITY_TYPE, id, type)
// Forge: □□□□ BlockEntityType.Builder□□□□ DeferredRegister □□
//
// BlockEntityType.Builder.of(factory, blocks...) □ blocks □□□□□□□□□□
// □ RegistryObject.get() □□□□□□□□□□□□□□□□
public static final RegistryObject<BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
        BLOCK_ENTITIES.register("example_block_entity",
                () -> BlockEntityType.Builder
                        .of(ExampleBlockEntity::new, EXAMPLE_ENTITY_BLOCK.get())
                        // .build(null)□Type? □□□□ null□Forge □□□□ DataFixerType□
                        .build(null));
// ===== □□□□□□□ =====
```

```
    public static void register(IEventBus modEventBus) {
        BLOCKS.register(modEventBus);
        BLOCK_ENTITIES.register(modEventBus);
    }
}
```

command/simple_command.java

```
// ================================================================================
// Forge 1.20.1 □□□□□□□ - simple_command.java
// ================================================================================
// □Forge □ Fabric □□□□□□ - □□□
// 1. Fabric □ CommandRegistrationCallback.EVENT □□□□
//    Forge □ RegisterCommandsEvent □□□□□ FORGE EVENT_BUS □□□
// 2. Fabric □□□□□□□□□ CommandDispatcher<CommandSourceStack>
//    Forge □□□□□□□ dispatcher□event.getDispatcher()
// 3. □□□□□□□ Brigadier□com.mojang.brigadier□□□□□□□API □□
//    □□□□□□□□□□□□□□□□
// 4. Forge □□□□□□□□□/□□□□□□□□
// 5. Fabric □□□□□ fabric.mod.json □□ entrypoint □□□□□
//    Forge □ @Mod.EventBusSubscriber □□□□□□□□
// ================================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.network.chat.Component;
import net.minecraft.server.level.ServerPlayer;
import net.minecraftforge.event.RegisterCommandsEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * Forge □□□□□ - □□ RegisterCommandsEvent □□
 *
 * □□□□□□□□ @Mod.EventBusSubscriber □□□□□□ FORGE □□□□□
 *   bus = Mod.EventBusSubscriber.Bus.FORGE□FORGE □□□□□ MOD □□□
 *   □□ RegisterCommandsEvent □ FORGE □□□□□
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    // ===== □□ RegisterCommandsEvent =====
    // □□□□Fabric: CommandRegistrationCallback.EVENT.register((dispatcher, registryAccess, env) -> {...})
    // Forge:  @SubscribeEvent void onRegisterCommands(RegisterCommandsEvent event)
    //
    // event.getDispatcher() □□ CommandDispatcher<CommandSourceStack>
    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        // □□□□□□□□
        CommandDispatcher<CommandSourceStack> dispatcher = event.getDispatcher();

        // □□□□□□/examplemod <target> <amount>
        // □□ Fabric □□□□□□□ Brigadier API□
        //   Commands.literal() / argument()
        dispatcher.register(Commands.literal(ExampleMod.MODID)
                // □□□□□□□□2 = □□□□□□□□4 = OP□
                .requires(source -> source.hasPermission(2))
                // □□□□give <target> <amount>
                .then(Commands.literal("give")
                        .then(Commands.argument("target", EntityArgument.player())
                                .then(Commands.argument("amount", IntegerArgumentType.integer(1, 64))
                                        // □□□□
                                        .executes(ModCommands::executeGive)
                                )
                        )
                )
                // □□□□info□□□□□□□□□□
                .executes(ModCommands::executeInfo)
```

```
        );
    }

    // ===== □□□□□□□ =====
    // CommandContext<CommandSourceStack> □□□□□□□□□□□□□
    private static int executeGive(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        // □□□□□□□□□□
        ServerPlayer target = EntityArgument.getPlayer(context, "target");
        // □□□□□□□
        int amount = IntegerArgumentType.getInteger(context, "amount");

        // □□□□□□□□□□□□□□□
        // target.addItem(new ItemStack(ModItems.EXAMPLE_ITEM.get(), amount));

        // □□□□□□□
        // □□□□□Fabric □ source.sendFeedback(Text.of(...), false)
        // Forge □ source.sendSuccess(() -> Component, false)
        // □□□sendSuccess □□ Supplier<Component>□□□□□□□□□
        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal("Gave " + amount + " items to " + target.getName().getString()), true);

        // □□□□□□□□□1 = □□□
        // Brigadier □□□□□□□□□□□□□
        return Command.SINGLE_SUCCESS;
    }

    private static int executeInfo(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal("Example Mod v1.0.0 - by Author"), false);
        return Command.SINGLE_SUCCESS;
    }

    // ============================================================================
    // □□□□□□□Client Commands□- □□
    // ============================================================================
    // □□□□□Fabric □ ClientCommandRegistrationCallback □□□□□□□□
    // Forge □□□□□□□□□□□□□□□□□□□□□□□□□□□ ClientCommands mod□
    //
    // □□□□□□□□□□□□□□
    // 1. □ Forge □ ClientCommandSourceStack□□ ClientCommands □□□□
    // 2. □□□□□□□□□□□ Forge 1.20.1 □□□□□□□□□□□□

    // ============================================================================
    // □□□□□□□ - Forge □□
    // ============================================================================
    // Forge □□□□□□□□Permission Node□□□□□□□□□□□
    // .requires(source -> source.hasPermission(2))
    // □□□ forge:permission □□□□□□□□□□□□□□□□
    //
    // □□□□□Fabric □□□□□□□□□ FabricPermissionsApi□□□□□□□□
}
```

enchantment/custom_enchantment.java

```
// ============================================================================
// Forge 1.20.1 □□□□□□□ - custom_enchantment.java
// ============================================================================
// □Forge □ Fabric □□□□□□ - □□□
// 1. Forge 1.20.1 □□□□□ Enchantment □□□□□ Enchantment□
//    Fabric 1.20.1 □□ Enchantment □□□□□□□□ RegistryEntry□
//    □□□MC 1.20.5+ □□ Enchantment □□ data-driven□□ JSON □□□
// 2. Fabric □ Registry.register(Registries.ENCHANTMENT, id, enchantment)
//    Forge □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID)
// 3. Forge 1.20.1 □ Enchantment □□□□□□□ Rarity + EnchantmentCategory + EquipmentSlot[]
//    □□□ API□□□□□□□
// 4. Forge □□□□□□□□□□□□□□□□□□□□□ EnchantmentLevelSetEvent□
// 5. Forge □ EnchantmentCategory □□□□□□□□□□□□□□□□□
// ============================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
```

```
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 */
public class ModEnchantments {

    // ===== □□□□□□ =====
    // □□□□Forge: ForgeRegistries.ENCHANTMENTS
    //         Fabric: Registries.ENCHANTMENT
    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, ExampleMod.MODID);

    // ===== □□□□□ =====
    // Enchantment □□□□□□MC 1.20.1 □□ API□□
    //   - Rarity rarity: □□□□□COMMON=10 / UNCOMMON=5 / RARE=2 / VERY_RARE=1□
    //   - EnchantmentCategory category: □□□□□□□□□□□□□□□□□□□□
    //   - EquipmentSlot[] slots: □□□□□□□□□
    //
    // □□□□Fabric 1.20.1 □□□□□□□□□□MC □□ API□
    // □□ Enchantment API □□□□□□□□□□□□□□
    //
    // EnchantmentCategory □□□□□
    //   - ALL: □□□□□
    //   - ARMOR: □□/□□/□□/□□
    //   - ARMOR_FEET / ARMOR_LEGS / ARMOR_CHEST / ARMOR_HEAD: □□□□□
    //   - WEAPON: □□□□□□□□
    //   - FISHING_ROD: □□□
    //   - BREAKABLE: □□□□□□□□□
    //   - DIGGER: □/□/□/□
    //   - BOW: □
    //   - CROSSBOW: □
    //   - TRIDENT: □□□
    //   - VANISHABLE: □□□□□
    public static final RegistryObject<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            // □□□□□□□□□□□□□□□□□□□□
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND));

    // ===== □□□□□□□ =====
    // □□ Enchantment □□□□□□□□□□□□□
    public static class ExampleEnchantment extends Enchantment {

        public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                                  EquipmentSlot... slots) {
            super(rarity, category, slots);
        }

        // ===== □□□□□□□□□□□□□□□□□□□□□□□□□□□□ =====
        @Override
        public int getMinCost(int level) {
            // □□□□□□□□□□□□□□□□
            return 10 * level;
        }

        // ===== □□□□□□□□□□ =====
        @Override
        public int getMaxCost(int level) {
            return getMinCost(level) + 15;
        }

        // ===== □□□□□□□ =====
        @Override
        public int getMaxLevel() {
            return 3;  // 3□□□
        }

        // ===== □□□□□□□□□□□□ =====
        @Override
```

```
    protected boolean checkCompatibility(Enchantment other) {
        // □□□□□□□□□□□□□□□
        return super.checkCompatibility(other)
                && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
    }

    // ===== □□□□□□□□□□□ =====
    @Override
    public boolean canEnchant(net.minecraft.world.item.ItemStack stack) {
        // □□□□□□□□□□□□□□□□□□□□□□□□
        return super.canEnchant(stack);
    }

    // ===== □□□□□□□□□□□□□□□□ =====
    @Override
    public boolean isCurse() {
        return false;
    }

    // ===== □□□□□□□□□□□□ =====
    @Override
    public boolean isTradeable() {
        return true;
    }

    // ===== □□□□□□□□□□ =====
    @Override
    public boolean isDiscoverable() {
        return true;
    }

    // ===== □□□□□□□□ =====
    @Override
    public boolean isAllowedOnBooks() {
        return true;
    }

    // ===== □□□□□□□□□□□□□□ =====
    @Override
    public void doPostAttack(
            net.minecraft.world.entity.LivingEntity attacker,
            net.minecraft.world.entity.Entity target,
            int level) {
        // □□□□□□□□□□□□
        if (target instanceof net.minecraft.world.entity.LivingEntity livingTarget) {
            livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                    net.minecraft.world.effect.MobEffects.POISON, 60, level - 1));
        }
        super.doPostAttack(attacker, target, level);
    }
}

// ===== □□□□□ =====
public static void register(IEventBus modEventBus) {
    ENCHANTMENTS.register(modEventBus);
}

// ============================================================================
// □□□□□□ - Forge □□
// ============================================================================
// Forge □ EnchantmentLevelSetEvent □□□□□□□□□□
// □□□□□□□□□□□□□□□□□□□□□□□□□□□□
//
// □□□□Fabric □□□□□□□□ Mixin □□□□□□□□□
//
// @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.FORGE)
// public static class ForgeEvents {
//     @SubscribeEvent
//     public static void onEnchantLevel(EnchantmentLevelSetEvent event) {
//         // event.getLevel() □□□□□□□
//         // event.getEnchantments() □□□□ Map
//         // event.getPower() □□□□□□□□□□
//
//         // □□□□□□□□□□□□
//         // event.getEnchantments().put(EXAMPLE_ENCHANTMENT.get(), 1);
//     }
```

```
    // }
    //
    // □□□□□□□Fabric □□ EnchantmentLevelAddedEvent□Fabric □□□□□ Mixin □□
    // Forge □ EnchantmentLevelSetEvent □□□□□□□□□□□□
}
```

entity/simple_entity.java

```
// ================================================================================
// Forge 1.20.1 □□□□□□□ - simple_entity.java
// ================================================================================
// □Forge □ Fabric □□□□□□ - □□□
// 1. Fabric □ Registry.register(Registries.ENTITY_TYPE, id, type) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, MODID) □□□□
// 2. □□□□□Attribute□□□□
//    Fabric □ EntityAttributeRegistry□FabricRegistries.ENTITY_ATTRIBUTE / □□□
//    Forge □ EntityAttributeCreationEvent □□□□ modEventBus □□□
// 3. Fabric □□□□□□□□ EntityRendererRegistry□□□□□□□□□□□
//    Forge □ EntityRenderersEvent.RegisterRenderers □□
// 4. Forge □ RegistryObject<EntityType<?>> □□□□
// ================================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.entity.ai.attributes.AttributeSupplier;
import net.minecraft.world.entity.ai.attributes.Attributes;
import net.minecraft.world.entity.monster.Monster;
import net.minecraftforge.event.entity.EntityAttributeCreationEvent;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□ ModMain □□□□□□□□□
 *   ModEntities.ENTITIES.register(modEventBus);
 *   modEventBus.addListener(ModEntities::registerAttributes);
 *   modEventBus.addListener(ModEntities::registerRenderers);  // □□□
 */
public class ModEntities {

    // ===== □□□□□□□□ =====
    // □□□□Forge: ForgeRegistries.ENTITY_TYPES
    //         Fabric: Registries.ENTITY_TYPE
    public static final DeferredRegister<EntityType<?>> ENTITIES =
            DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // EntityType.Builder.create(factory, category)
    //   - factory: EntityType.EntityFactory<T>□□□□□□□□□
    //   - category: MobCategory□□□□□□□□□□□□□□□□□□□□
    //
    // □□□□Fabric □ EntityType.Builder.create() + .build() □ Registry.register
    // Forge □ RegistryObject □□□supplier lambda □□□□
    public static final RegistryObject<EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITIES.register("example_entity",
                    () -> EntityType.Builder
                            .of(ExampleEntity::new, MobCategory.CREATURE)
                            // □□□□□□□□ x □□□□□□□□
                            .sized(1.0F, 1.0F)
                            // □□□□□□□□□□□□□ tick□□□□ 80
                            .clientTrackingRange(10)
                            // fireImmune□□□□□□□□□□□
                            // .fireImmune()
                            // .canSpawnFarFromPlayer()
                            .build("example_entity"));
                            // □□□build(null) □□□□□□□□1.20.1 □ build(id) □□□□□

    // ================================================================================
```

```
    // □□□□□□□ - Forge □□□□□□□
    // ============================================================================
    // □□□□□□□
    // Fabric: □ FabricDefaultAttributeRegistry.register(type, builder) □□□□□/□□ init □□
    //   □□:
    //   FabricDefaultAttributeRegistry.register(EXAMPLE_ENTITY,
    //       Mob.createMobAttributes().add(Attributes.MAX_HEALTH, 20.0)...);
    //
    // Forge: □□ EntityAttributeCreationEvent□□□ event.put(type, builder.build())
    //   - □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //   - □□□□□ modEventBus □□□□□□□□□□□□
    
    // □□1□□□□□□□□□□□□ ModMain □□ addListener □□□
    public static void registerAttributes(EntityAttributeCreationEvent event) {
        // □□ AttributeSupplier□□□□□□□
        // Mob.createMobAttributes() □□□□□□□□□□□□□□□□□□□□□
        AttributeSupplier.Builder builder = Monster.createMonsterAttributes();
        builder.add(Attributes.MAX_HEALTH, 20.0D);
        builder.add(Attributes.MOVEMENT_SPEED, 0.25D);
        builder.add(Attributes.ATTACK_DAMAGE, 5.0D);
        builder.add(Attributes.FOLLOW_RANGE, 32.0D);

        // □□□□□□□□□□□□□□□□
        // □□□□□□□□ RegistryObject.get() □□□□ EntityType
        event.put(EXAMPLE_ENTITY.get(), builder.build());
    }

    // □□2□□ @SubscribeEvent + □□□□□□ ModEventBus □□□
    // □□□ ModMain □□modEventBus.register(ModEntities.class);
    @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.MOD)
    public static class ModBusEvents {
        @SubscribeEvent
        public static void onAttributeCreate(EntityAttributeCreationEvent event) {
            // □□□□□□□ registerAttributes
            registerAttributes(event);
        }
    }

    // ============================================================================
    // □□□□□□□□□ - Forge □□□□□
    // ============================================================================
    // □□□□Fabric: □ EntityRendererRegistry.register(type, context -> renderer)
    // Forge: □□ EntityRenderersEvent.RegisterRenderers □□
    //
    // □□□□□□□□□□□□ @OnlyIn □ DistExecutor □□□
    // @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Bus.MOD, value = Dist.CLIENT)
    // public static class ClientBusEvents {
    //     @SubscribeEvent
    //     public static void onRegisterRenderers(EntityRenderersEvent.RegisterRenderers event) {
    //         event.registerEntityRenderer(EXAMPLE_ENTITY.get(), ExampleRenderer::new);
    //     }
    // }

    // ============================================================================
    // □□□□Spawn Egg□□□ - Forge □□
    // ============================================================================
    // Forge □ ForgeSpawnEggItem □ Forge □□□ Spawn Egg
    // □□□□Fabric □ net.minecraft.item.SpawnEggItem + Registry.register
    // Forge □ ForgeSpawnEggItem.builder() □□□□□□□□□□□□□□□
    //
    // public static final RegistryObject<Item> EXAMPLE_SPAWN_EGG = ModItems.ITEMS.register(
    //     "example_spawn_egg",
    //     () -> ForgeSpawnEggItem.builder()
    //         .of(EXAMPLE_ENTITY, 0xFFFFFF, 0x000000)  // □□ + □□
    //         .build(new Item.Properties()));

    // ===== □□□□□ =====
    public static void register(IEventBus modEventBus) {
        ENTITIES.register(modEventBus);
    }
}
```

gui/container_screen.java

```
// ============================================================================
// Forge 1.20.1 GUI / □□□□□□□ - container_screen.java
```

```
// ================================================================================
// 【Forge 与 Fabric 对比说明 - GUI】
// 1. Fabric 的 ScreenHandlerType + Registry.register(Registries.SCREEN_HANDLER)
//    Forge 的 DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID) 注册 MenuType
//    【关键区别】Fabric 用 Registries.SCREEN_HANDLER vs Forge 用 ForgeRegistries.CONTAINERS
// 2. Screen注册方式不同
//    Fabric 用 HandledScreens.register(type, factory)，通常在 init 中调用
//    Forge 用 MenuScreens.register(type, factory)，通常在客户端 FMLClientSetupEvent 中调用
// 3. Forge 的 MenuType构造方式= Fabric 的 ScreenHandlerType构造方式（参数顺序）
//    Forge 的 AbstractContainerMenu = Fabric 的 ScreenHandler
//    Forge 的 AbstractContainerScreen = Fabric 的 ScreenHandlerScreen
// 4. Forge 额外有 ContainerData辅助类，简化 Container数据同步
// ================================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.client.gui.screens.MenuScreens;
import net.minecraft.client.gui.screens.Screen;
import net.minecraft.client.gui.screens.inventory.AbstractContainerScreen;
import net.minecraft.network.FriendlyByteBuf;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraftforge.common.extensions.IForgeMenuType;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 菜单/容器注册 - Forge DeferredRegister + MenuScreens
 */
public class ModMenus {

    // ===== 注册延迟注册器 =====
    // 【对比】Forge: ForgeRegistries.CONTAINERS（旧名 CONTAINERS，新名 MENU_TYPES）
    //         Fabric: Registries.SCREEN_HANDLER
    public static final DeferredRegister<MenuType<?>> MENUS =
            DeferredRegister.create(ForgeRegistries.CONTAINERS, ExampleMod.MODID);

    // ===== 注册 MenuType =====
    // Forge 用 IForgeMenuType.create() 可以从网络数据创建 MenuType
    // 【对比】Fabric 用 new ScreenHandlerType<>(factory)
    //
    // IForgeMenuType.create() 的 factory 参数 (containerId, playerInventory, extraData)：
    //   - extraData: FriendlyByteBuf，可以读取 BlockEntity 的位置等
    //   - 这是客户端创建 Container时从服务端接收数据的关键 BlockEntity。
    //
    // 注意 Forge 和 Fabric 的参数顺序不同，容易搞混！
    public static final RegistryObject<MenuType<ExampleMenu>> EXAMPLE_MENU = MENUS.register("example_menu",
            () -> IForgeMenuType.create((containerId, inventory, data) -> {
                // 从网络数据 extraData 读取 BlockEntity 位置
                // data 是一个 BlockPos
                // BlockEntity blockEntity = inventory.player.level.getBlockEntity(data.readBlockPos());
                // return new ExampleMenu(containerId, inventory, blockEntity);

                // 简单版本：不需要额外数据
                return new ExampleMenu(containerId, inventory);
            }));

    // ================================================================================
    // AbstractContainerMenu（容器/ScreenHandler）
    // ================================================================================
    // 【对比】Forge 的 AbstractContainerMenu = Fabric 的 ScreenHandler
    // 两者 API 几乎一样，都是 MC 原版的设计，移植很简单

    public static class ExampleMenu extends AbstractContainerMenu {
        // 容器关联的TileEntity 的容器引用
        private final net.minecraft.world.Container container;

        // 容器1的槽位数量（ BlockEntity 的）
```

```
    public ExampleMenu(int containerId, Inventory playerInventory, BlockEntity blockEntity) {
        super(EXAMPLE_MENU.get(), containerId);
        // □□ blockEntity □□□□ Container □□
        this.container = (net.minecraft.world.Container) blockEntity;

        // □□□□□□□□□□
        for (int row = 0; row < 3; row++) {
            for (int col = 0; col < 3; col++) {
                // Slot(container, index, x, y)
                this.addSlot(new Slot(container, row * 3 + col, 8 + col * 18, 18 + row * 18));
            }
        }

        // □□□□□□□□□□3x9□
        for (int row = 0; row < 3; row++) {
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory, 9 + row * 9 + col,
                        8 + col * 18, 84 + row * 18));
            }
        }

        // □□□□□□□□□9□□
        for (int col = 0; col < 9; col++) {
            this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
        }
    }

    // □□□2□□□□□□□□ extraData □□□
    public ExampleMenu(int containerId, Inventory playerInventory) {
        this(containerId, playerInventory, null);  // □□
    }

    // ===== Shift □□□□□□□□□ =====
    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasItem()) return ItemStack.EMPTY;

        ItemStack original = slot.getItem().copy();
        ItemStack stack = slot.getItem();

        // □□□□□□□□□□□□ <-> □□□□□
        if (index < 9) {
            // □□□□ -> □□□□
            if (!this.moveItemStackTo(stack, 9, this.slots.size(), true)) {
                return ItemStack.EMPTY;
            }
        } else {
            // □□□□□ -> □□□
            if (!this.moveItemStackTo(stack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        slot.setChanged();
        return original;
    }

    // ===== □□□□□□□□□□ =====
    @Override
    public boolean stillValid(Player player) {
        return container.stillValid(player);
    }
}

// ============================================================================
// AbstractContainerScreen□□□/ScreenHandlerScreen□□□
// ============================================================================
// □□□□Forge □ AbstractContainerScreen = Fabric □ HandledScreen
// □□□□□□□□□□□□□□□ GUI □□□□□□□□

// □□□ @OnlyIn(Dist.CLIENT) □□□□□□□□□□□
// @OnlyIn(Dist.CLIENT)
public static class ExampleScreen extends AbstractContainerScreen<ExampleMenu> {
    public ExampleScreen(ExampleMenu menu, Inventory inventory, net.minecraft.network.chat.Component title) {
        super(menu, inventory, title);
```

```
            // GUI □□
            this.imageWidth = 176;
            this.imageHeight = 166;
            this.inventoryLabelY = this.imageHeight - 94;  // "□□"□□ Y □□
        }

        @Override
        protected void renderBg(net.minecraft.client.gui.GuiGraphics graphics,
                                float partialTick, int mouseX, int mouseY) {
            // □□ GUI □□
            // □□□□Forge 1.20.1 □ GuiGraphics□1.20 □□□□□Fabric 1.20.1 □□
            // □□□ RenderSystem + blit
            int x = (this.width - this.imageWidth) / 2;
            int y = (this.height - this.imageHeight) / 2;
            graphics.blit(new net.minecraft.resources.ResourceLocation(ExampleMod.MODID,
                    "textures/gui/example_container.png"), x, y, 0, 0,
                    this.imageWidth, this.imageHeight);
        }

        @Override
        public void render(net.minecraft.client.gui.GuiGraphics graphics,
                           int mouseX, int mouseY, float partialTick) {
            // □□□□□□□□□□
            this.renderBackground(graphics);
            // □□□□□□
            super.render(graphics, mouseX, mouseY, partialTick);
            // □□ tooltip
            this.renderTooltip(graphics, mouseX, mouseY);
        }
    }

    // ============================================================================
    // Screen□□□□□□□□ - □□□
    // ============================================================================
    // □□□□□□Fabric □ HandledScreens.register(type, factory)
    // Forge □ MenuScreens.register(type, factory)
    // □□□□□□□□□□□FMLClientSetupEvent □□

    // □ ModMain □□□□□□□□□□□
    //   modEventBus.addListener(this::clientSetup);
    //
    // private void clientSetup(FMLClientSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         MenuScreens.register(EXAMPLE_MENU.get(), ExampleScreen::new);
    //     });
    // }
    //
    // □□□□□□□□ enqueueWork □□□□□□□□□□ Screen □□□□□□□□□□
    //
    // □□□□Fabric:
    //   HandledScreens.register(EXAMPLE_MENU, ExampleScreen::new);
    //   □ ClientModInitializer.onInitializeClient() □□□□□

    // ===== □□□□ =====
    public static void register(IEventBus modEventBus) {
        MENUS.register(modEventBus);
    }

    // ===== □□□□□□□□□□ =====
    //
    // □□1□□□ MenuProvider□□□□□□□
    // player.openMenu(menuProvider);  // □□ NetworkHooks.openScreen□Forge□
    //   - Fabric □ player.openHandledScreen(menuProvider)
    //
    // □□2□□ BlockEntity □□ MenuProvider
    //   @Override
    //   public AbstractContainerMenu createMenu(int id, Inventory inv, Player player) {
    //       return new ExampleMenu(id, inv, this);
    //   }
    //
    // □□□□Forge □□□□□ NetworkHooks.openScreen(serverPlayer, provider, buf)
    //         Fabric □ player.openHandledScreen(provider)
}
```

item/simple_item.java

```
// ================================================================================
// Forge 1.20.1 □□□□□□□ - simple_item.java
// ================================================================================
// □Forge □ Fabric □□□□□□□
// 1. Fabric □ Registry.register(Registries.ITEM, id, item) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ITEMS, MODID) □□□□
// 2. Forge □□□□□□□□□□□□ ITEMS□BLOCKS □ ITEMS □□□□□□□□□
//    Fabric □□□□□□□ BlockItem□□ Forge □ RegistryObject □□
// 3. □□□□□□□□□
//    Fabric □ ItemGroupEvents.modifyEntriesEvent() □□□□
//    Forge □ CreativeModeTabRegistry + CreativeModeTabEvent □□□□□□□
//    □□□□ DeferredRegister □□□□□□ Tab
// 4. Forge □ RegistryObject<Item> □□□□
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Items;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□ ModMain □□□□□□□□□
 *   ModItems.ITEMS.register(modEventBus);
 *   ModItems.CREATIVE_TABS.register(modEventBus);
 */
public class ModItems {

    // ===== □□□□□□ =====
    // □□□□Forge: ForgeRegistries.ITEMS
    //         Fabric: Registries.ITEM
    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ===== □□□□□□□□□□□□ =====
    // Forge 1.20.1 □□ DeferredRegister □□ CreativeModeTab
    // □□□□Fabric □ ItemGroupEvents.modifyEntriesEvent □□□□□□□□□
    //         □□ FabricItemSettings □□□ group()
    // Forge □□□□□□□□□□□□ Tab
    public static final DeferredRegister<CreativeModeTab> CREATIVE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // RegistryObject<Item> □□□□□□□□□ .get() □□□□ Item □□
    // supplier □ lambda □□□□
    //
    // □□□□Fabric:
    //   public static final Item EXAMPLE_ITEM = Registry.register(
    //       Registries.ITEM, new ResourceLocation("examplemod", "example_item"),
    //       new Item(new FabricItemSettings()));
    // Forge:
    //   public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register(
    //       "example_item", () -> new Item(new Item.Properties()));
    public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register("example_item",
            () -> new Item(new Item.Properties()
                    // □□□□□□□ Fabric □ FabricItemSettings □□□□□□ MC □□ Item.Properties□
                    .stacksTo(64)              // □□□□□
                    // .rarity(Rarity.RARE)    // □□□
                    // .fireResistant()        // □□□□□□□□□□□□
            ));

    // ===== □□□□□□□□BlockItem□ =====
    // □□□□□□□□Forge □□□□□ BlockItem □□□□□□ ITEMS □□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□ ModBlocks □□ RegistryObject<Block>
    //
```

```
    // □□□□Fabric □□□□□□□□□ BlockItem□
    //   Registry.register(Registries.ITEM, blockId, new BlockItem(block, settings))
    // Forge □ RegistryObject □□□
    //   ITEMS.register("example_block", () -> new BlockItem(ModBlocks.EXAMPLE_BLOCK.get(), new Item.Properties()))
    public static final RegistryObject<Item> EXAMPLE_BLOCK_ITEM = ITEMS.register("example_block",
            () -> new net.minecraft.world.item.BlockItem(
                    com.example.examplemod.block.ModBlocks.EXAMPLE_BLOCK.get(),
                    new Item.Properties()));

    // ===== □□□□□□□□□□□□□□□ =====
    // Forge 1.20.1 □ CreativeModeTab.builder() □□
    // □□□□Fabric □ ItemGroupRegistry □□□□□□□
    // Forge □□□□□□ Tab□□□□ .icon() □□□□□.displayItems() □□□□□
    public static final RegistryObject<CreativeModeTab> EXAMPLE_TAB = CREATIVE_TABS.register("example_tab",
            () -> CreativeModeTab.builder()
                    // □□□□□□□□ Component.translatable □□□□
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    // □□□□□□
                    .icon(() -> new ItemStack(EXAMPLE_ITEM.get()))
                    // □□□□□□□□□□□□ CreativeModeTab.ItemDisplay □□□
                    .displayItems((parameters, output) -> {
                        // □□□□□□□□□□□□□ Tab
                        // □□□□□Fabric □ output.accept(item) □ ItemGroupEvents □□□□□□
                        output.accept(EXAMPLE_ITEM.get());
                        output.accept(EXAMPLE_BLOCK_ITEM.get());
                        // □□□□□□□□□□□□
                        // output.accept(Items.DIAMOND);
                    })
                    // □□□□Forge 1.20.1 □ builder □□□
                    .build());

    // ===== □□□□□□□□ =====
    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
        CREATIVE_TABS.register(modEventBus);
    }

    // ===== □□□□□□□□□□□□□□□□□□□ =====
    // □□□□□□□□□ Tab□□□□□ BuildCreativeModeTabContentsEvent □□□□□□□
    // □□□□□Fabric □ ItemGroupEvents.modifyEntriesEvent(INGREDIENTS).register(...)
    //
    // □ ModMain □□□□
    //   modEventBus.addListener(ModItems::addToExistingTabs);
    //
    // public static void addToExistingTabs(
    //         net.minecraftforge.event.BuildCreativeModeTabContentsEvent event) {
    //     if (event.getTabKey() == CreativeModeTabs.INGREDIENTS) {
    //         event.accept(EXAMPLE_ITEM);
    //     }
    // }
}
```

item/tool_item.java

```
// ================================================================================
// Forge 1.20.1 □□/□□□□□□□ - tool_item.java
// ================================================================================
// □Forge □ Fabric □□□□□□ - □□□□ Tier□
// 1. Fabric 1.20.1 □ net.minecraft.item.ToolMaterial□□□ ToolMaterial □□□□□□□
//    Forge 1.20.1 □ net.minecraftforge.common.Tier□□□ Tier □□□
//    □□□□□□□□□□Fabric □ ToolMaterial vs Forge □ Tier
// 2. Forge 1.20.1 □□□□□□ Tier □□□getUses()□getSpeed() □□□□
//    □ Fabric 1.20.1 □□□□□□ ToolMaterial□ingredient() □□ Ingredient□
//    □□□MC 1.20.5+ □□□□□□ Tier □□□1.20.1 □□□□□□
// 3. Forge □ TierSortingRegistry □□ Tier □□□□□□
//    Fabric □□□□ Tag □□□□□□
// 4. Forge □ DeferredRegister □□□□□□□□□□□□□□□□□□□
//    Fabric □ Registry.register(Registries.ITEM)
// 5. Forge □ Tier □□□□□□
//    - int getUses()              □□
//    - float getSpeed()           □□□□
//    - float getAttackDamageBonus() □□□□□□□
//    - int getLevel()             □□□□□□□0□□1□□2□□3□
//    - int getEnchantmentValue()  □□□□
//    - Ingredient getRepairIngredient() □□□□
//    □□□□Fabric 1.20.1 □ ToolMaterial□
```

```
//    - int getDurability()
//    - float getMiningSpeedMultiplier()
//    - float getAttackDamage()
//    - int getMiningLevel()
//    - int getEnchantability()
//    - Ingredient getRepairIngredient()（Fabric 1.20.1 用的是 TagKey 而 Supplier）
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Items;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 工具/武器物品 - Forge Tier 版本
 */
public class ModTools {

    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ============================================================================
    // 自定义工具等级（Tier）
    // ============================================================================
    // 注意 Forge 的 Tier 接口与原版 Forge 的不同于 Fabric 的 ToolMaterial：
    // 主要区别是Forge 的 Tier 使用的是 Fabric 使用的是 Forge 的 getLevel() 而非 getMiningLevel()

    public enum ModTiers implements Tier {
        // 示例：自定义等级EXAMPLE_TIER
        // 参数：耐久、速度、攻击加成、挖掘等级、附魔能力、修复材料
        EXAMPLE_TIER(
            1561,    // getUses()：耐久（钻石是1561）
            8.0F,    // getSpeed()：挖掘速度（钻石是8.0）
            3.0F,    // getAttackDamageBonus()：攻击加成
            3,       // getLevel()：挖掘等级（0/木1/石2/铁3/钻石4）
            10,      // getEnchantmentValue()：附魔能力（金是10）
            // 修复材料：用 Ingredients.fromValues() 或 fromItems()
            // 注意：Fabric 1.20.1 用的是 Supplier<TagKey> 而 Ingredient
            // Forge 1.20.1 直接 Ingredient
        );

        private final int uses;
        private final float speed;
        private final float attackDamageBonus;
        private final int level;
        private final int enchantmentValue;
        private final Ingredient repairIngredient;

        ModTiers(int uses, float speed, float attackDamageBonus,
                 int level, int enchantmentValue) {
            this.uses = uses;
            this.speed = speed;
            this.attackDamageBonus = attackDamageBonus;
            this.level = level;
            this.enchantmentValue = enchantmentValue;
            // 修复材料：这里用钻石锭作为修复材料（可以改成自己的物品）
            // 注意：Fabric 1.20.1 用的是 () -> Ingredient.of(tag) 而非 Supplier
            // Forge 1.20.1 的 getRepairIngredient() 直接返回 Ingredient
            this.repairIngredient = Ingredient.of(Items.DIAMOND);
        }

        // ===== Forge Tier 接口实现 =====
        // 注意：这里和 Fabric 不同！
        @Override
        public int getUses() {
```

```
            return uses;                        // Fabric: getDurability()
        }

        @Override
        public float getSpeed() {
            return speed;                       // Fabric: getMiningSpeedMultiplier()
        }

        @Override
        public float getAttackDamageBonus() {
            return attackDamageBonus;           // Fabric: getAttackDamage()
        }

        @Override
        public int getLevel() {
            return level;                       // Fabric: getMiningLevel()
        }

        @Override
        public int getEnchantmentValue() {
            return enchantmentValue;            // Fabric: getEnchantability()
        }

        @Override
        public Ingredient getRepairIngredient() {
            return repairIngredient;            // □□□□□□□□
        }
    }

    // ========================================================================
    // □□□□□□□
    // ========================================================================
    // PickaxeItem / SwordItem / AxeItem / ShovelItem / HoeItem
    // □□□□□□new PickaxeItem(tier, attackDamage, attackSpeed, properties)
    //   - tier: Tier□□□□□□
    //   - attackDamage: □□□□□□□□□□ = tier.getAttackDamageBonus() + □□ + 1□
    //   - attackSpeed: □□□□□□□□4.0□□□□
    //   - properties: Item.Properties
    //
    // □□□□□Fabric 1.20.1 □□□□□□□□□□□ MC □□ API□□
    //         □ tier □□□□□□□□Forge □ Tier vs Fabric □ ToolMaterial
    public static final RegistryObject<Item> EXAMPLE_PICKAXE = ITEMS.register("example_pickaxe",
            () -> new PickaxeItem(ModTiers.EXAMPLE_TIER,
                    1,   // □□□□□□□
                    -2.8F, // □□□□□□□□□□□□
                    new Item.Properties()));

    public static final RegistryObject<Item> EXAMPLE_SWORD = ITEMS.register("example_sword",
            () -> new SwordItem(ModTiers.EXAMPLE_TIER,
                    3,    // □□□□□□□
                    -2.4F, // □□□□
                    new Item.Properties()));

    // ========================================================================
    // Tier □□□□□□Forge □□□
    // ========================================================================
    // Forge □ TierSortingRegistry □□□□□ Tier□□□□□□□□□□□□
    // □□□ Forge □□□□□□ Tier □□□Fabric □□□□□□□
    //
    // □□□□□□□□□□□□□
    // TierSortingRegistry.registerTier(
    //     ModTiers.EXAMPLE_TIER,                          // tier □□
    //     new ResourceLocation(ExampleMod.MODID, "example_tier"),  // tier ID
    //     List.of(Tiers.DIAMOND),                         // □□□ tier □□□□□□□
    //     List.of()                                        // □□□ tier □□□□□□□
    // );

    // ========================================================================
    // □□□□
    // ========================================================================
    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

mixin/mixin_config.json

```
{
  "_comment_mixin_config": "================================================================================",
  "_comment_forge_vs_fabric": "□Forge 1.20.1 Mixin □□ - □ Fabric □□□□",
  "_comment_diff_1": "1. Forge 1.20.1 □□□ Mixin□□□ SpongePowered Mixin□Forge □□□",
  "_comment_diff_2": "2. □□□□□□□□□□□□Mixin □□□□□",
  "_comment_diff_3": "3. Forge □□ jar □ MANIFEST.MF □□□ MixinConfigs □□□□□",
  "    _comment_manifest": "   build.gradle □□attributes([\"MixinConfigs\": \"examplemod.mixins.json\"])",
  "_comment_diff_4": "4. Fabric □ fabric.mod.json □ mixins □□□□",
  "    _comment_fabric": "   \"mixins\": [\"examplemod.mixins.json\"]",
  "_comment_diff_5": "5. Forge □ refmap□□□□□□□□ ForgeGradle □□",
  "    _comment_refmap": "   refmap.json □□□ jar □□□□Forge □□□□□□□□ Mixin",
  "    _comment_fabric_refmap": "   Fabric □ loom □□ refmap□□ examplemod-refmap.json□",
  "_comment_diff_6": "6. Forge □ MCPConfig/official □□□Fabric □ yarn/parchment",
  "    _comment_mapping": "   □□ Mixin □□□□□/□□□□□□□□□□",
  "_comment_diff_7": "7. Forge □ mixin client/server □□□ side □□",
  "    _comment_fabric_side": "   Fabric □□ side □□□□□□□□",
  "_comment_location": "□□□□: src/main/resources/examplemod.mixins.json",
  "_comment_==========================================================================": "============================================================================================",

  "required": true,
  "_comment_required": "required: true □□□ Mixin □□□□□□□□□□□□□□□□false □□□",

  "minVersion": "0.8",
  "_comment_minVersion": "Mixin □□□□□□0.8 □□ 1.20.x □□□",

  "package": "com.example.examplemod.mixin",
  "_comment_package": "Mixin □□□□ Java □□□□ mixin □□□□□□□□",

  "compatibilityLevel": "JAVA_17",
  "_comment_compatibilityLevel": "Java □□□□□1.20.1 □□ JAVA_17",

  "refmap": "examplemod.refmap.json",
  "_comment_refmap": "□□□□□□□□ForgeGradle □□□□□",
  "_comment_refmap_forge": "Forge □ refmap □ ForgeGradle □□□□□□ jar",
  "_comment_refmap_fabric": "Fabric □ refmap □ Loom □□□□□□□□ <modid>-refmap.json",

  "verbose": false,
  "showMessageTypes": false,

  "_comment_mixins_server_client": "Forge □ side □□ mixin□client / server / common□",
  "mixins": [
    {
      "_comment": "□□ Mixin□□□□+□□□□□□□",
      "_comment_example": "MinecraftServerMixin□□□□□□□□",
      "name": "MinecraftServerMixin"
    }
  ],

  "client": [
    {
      "_comment": "□□□□□ Mixin□□□□□□□□",
      "_comment_example": "□□□□□□□□GUI □□□",
      "_comment_fabric_diff": "Fabric □□ client □□□□□□□",
      "name": "ClientLevelMixin"
    },
    {
      "name": "GuiMixin"
    }
  ],

  "server": [
    {
      "_comment": "□□□□□ Mixin□□□□□□□□□",
      "_comment_name": "Server □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□",
      "name": "DedicatedServerMixin"
    }
  ],

  "injectors": {
    "defaultRequire": 1,
    "_comment_defaultRequire": "Mixin □□□□□□□□1 = □□□□□□□1□□□□□",
    "_comment_default_require_strict": "□□□□□□□0 □□□□□□"
  },
  "_comment_overrides": "Mixin □□□□□ overrides □□□□□□□□□□□□□□□□",
```

```
  "_comment_plugin": "□□□ mixin plugin□□□ IMixinConfigPlugin□□□□□ mixin □□",
  "_comment_plugin_fabric": "□□□□□ IMixinConfigPlugin□API □□"
}
```

project/ModMain.java

```
// ============================================================================
// Forge 1.20.1 □□□□□□□ - ModMain.java
// ============================================================================
// □Forge □ Fabric □□□□□ - □□□□
// 1. Fabric □□ ModInitializer / ClientModInitializer □□□□ onInitialize() □□
//    Forge □□ @Mod □□□□□□□□□□□□□□□ DeferredRegister
// 2. Fabric □ Registry.register() □□□□□ net.minecraft.registry.Registries
//    Forge □ DeferredRegister.create(ForgeRegistries.XXX, MODID) □□□□
// 3. Fabric □□□□□□□□□□□□□□□□□ ServerTickEvents□
//    Forge □ IEventBus□modEventBus□□□□□□□□ addListener / @SubscribeEvent
// 4. Forge □□□□□□□□□□□□□□□□□□□□□□□□□□□□ @Mod □□
// ============================================================================

package com.example.examplemod;

import net.minecraftforge.common.MinecraftForge;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext;

// □□□□□□□□ @Mod □□□□□□ modId□□□□□□□ mods.toml □□□
// □□□□Fabric □□□□□□□□ fabric.mod.json □ entrypoint □□□
// @Mod □□ Forge □□□□□□□□□□□□□□□□□□□□
@Mod(ExampleMod.MODID)
public class ExampleMod {

    // □□ ID□□□□□□□□□□□□□□□
    public static final String MODID = "examplemod";

    // □□□□□□□Forge □□□□□□□ModLoadingContext□□□□□□□
    // □□□□□ DeferredRegister□□□□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□□□□□□□□□ Level□Player□□□□□□□□□□□
    public ExampleMod() {
        // □□□□□□□□□□□modEventBus□
        // □□□□Fabric □□□□□□□□□□□□ onInitialize □□ Registry.register
        // IEventBus □□□□□□□□□□□□□
        //   - RegisterEvent□□□□□□□ DeferredRegister □□□□□
        //   - FMLCommonSetupEvent□□□□□□
        //   - FMLClientSetupEvent□□□□□□
        //   - FMLDedicatedServerSetupEvent□□□□□□
        IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();

        // ===== □□ DeferredRegister =====
        // DeferredRegister □ Forge □□□□□□□
        // □□□□□□□□□□□□□□□ RegisterEvent □□□□□□□□□□□
        // □□□□Fabric □□ onInitialize □□□□□ Registry.register()
        // Forge □□□ DeferredRegister□□□□□□□□□□□□□□
        //
        // □□□□□□□□□□□□□□□ DeferredRegister □□□□ register(modEventBus)
        // □□
        //   ModBlocks.REGISTER.register(modEventBus);
        //   ModItems.REGISTER.register(modEventBus);
        //   ModEntities.REGISTER.register(modEventBus);
        //   ModMenus.REGISTER.register(modEventBus);
        //   ModEnchantments.REGISTER.register(modEventBus);

        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        // □□□□Fabric □ onInitialize □□□□□ setup □□
        // Forge □□□□□□□□□□□□□□□□□□□□□□□□□□
        modEventBus.addListener(this::commonSetup);

        // □□□□□ FORGE □□□□□□□□□□□□□□□□□ tick□□□□□□
        // □□□□Fabric □□□□□□□□□□□□□□
        // MinecraftForge.EVENT_BUS □□□ FORGE □□□□ modEventBus □□□
        //   - modEventBus□□□□□□□□□□□□□□□□□□□□
        //   - EVENT_BUS□□□□□□□□□□tick□□□□□□□□□
        MinecraftForge.EVENT_BUS.register(this);
    }

    // □□□□□□□□ FMLCommonSetupEvent □□□□□
```

```
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // □□□□□Fabric □ onInitialize() □□□□□□□
    private void commonSetup(final net.minecraftforge.fml.event.lifecycle.FMLCommonSetupEvent event) {
        // □□□□□□□□□□□□□□□□□□□□□□□□
        // event.enqueueWork(() -> { ... });
    }

    // ===== □□□□□□□□□□@SubscribeEvent =====
    // □□□□□Fabric □ @SubscribeEvent□□ XxxEvents.EVENT.register(callback)
    // Forge □□□□ EVENT_BUS □□□□□□□□ @SubscribeEvent □□□□□□□□

    // □□□□□□□□□ tick □□
    // @SubscribeEvent
    // public void onServerTick(net.minecraftforge.event.TickEvent.ServerTickEvent event) {
    //     if (event.phase == TickEvent.Phase.END) {
    //         // □□...
    //     }
    // }

    // □□□□□□□□□□□□□□□
    // @SubscribeEvent
    // public void onPlayerLogin(net.minecraftforge.event.entity.player.PlayerEvent.PlayerLoggedInEvent event) {
    //     // □□...
    // }
}
```

project/build.gradle

```
// ================================================================================
// Forge 1.20.1 □□□□□□□ - build.gradle
// ================================================================================
// □Forge □ Fabric □□□□□□□
// 1. Forge □□ ForgeGradle □□□□□□ Fabric □ Loom □□□
//    - Fabric: plugins { id 'fabric-loom' version '1.x' }
//    - Forge:  plugins { id 'net.minecraftforge.gradle' version '6.0.x' }
// 2. Forge □□ MC□□+Forge□□ □□□□□□□ 1.20.1-47.3.0□□□□ Fabric □ Yarn mapping □□
// 3. Forge □□□□□□□MCPConfig / official mappings□□□□□□□□ mapping □□
// 4. Forge □□ jarJar□□□□□□□□□□□□□□□ Fabric □ include/unpick □□
// ================================================================================

// □□ ForgeGradle □□□□□□□□ settings.gradle □ pluginManagement □□□□□□□
plugins {
    id 'eclipse'
    id 'idea'
    id 'maven-publish'
    id 'net.minecraftforge.gradle' version '6.0.24'
}

// --------------------------------------------------------------------------
// □□□□□□□□
// --------------------------------------------------------------------------
version = '1.0.0'
group = 'com.example.examplemod'  // □□□□□□□ ModMain.java □□□□□□□□

// Java □□□□□1.20.1 □□□□□ Java 17
// □□□□□Fabric 1.20.1 □□□□ Java 17□□□□□□
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// --------------------------------------------------------------------------
// Minecraft / Forge □□□□□
// --------------------------------------------------------------------------
// □□□□□□□□Forge □□ "MC□□-Forge□□" □□□□□□
// Fabric □□ parchment/yarn mapping + loader + API □□□□□
// Forge 47.3.0 □□ Minecraft 1.20.1 □□□□□□□□□
minecraft {
    // □□□□□Forge □□□□□□□□official□□ Parchment
    // "official" □□ Mojang □□□□□□□mcp □□□□□□
    // Parchment □□□□□□□□□□□□□□□□□□□□□□□□
    //   mappings channel: 'parchment', version: '2023.09.03-1.20.1'
    mappings channel: 'official', version: '1.20.1'

    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    runs {
        // □□□□□□□□□
        client {
            workingDirectory project.file('run')
```

```
            property 'forge.logging.markers', 'REGISTRIES'  // □□□□□□□
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod {  // modId□□□□ mods.toml / @Mod □□□□
                    source sourceSets.main
                }
            }
        }

        // □□□□□□□□
        server {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }

        // □□□□□□□□□□□□□□□□ JSON□□□□□□□□□□□□□□□□□□□□
        // □□□□Fabric □□□□□□□FabricDataGen□□□□□□□□□□
        // Forge □□ data run + GatherDataEvent □□□□
        data {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            args '--mod', 'examplemod', '--all', '--output', file('src/generated/resources/'),
                    '--existing', file('src/main/resources/')
            mods {
                examplemod {
                    source sourceSets.main
                }
            }
        }
    }
}

// ------------------------------------------------------------------------------
// □□□□□
// ------------------------------------------------------------------------------
// □□□□Fabric □□□□ maven.fabricmc.net □ modmaven
// Forge □□ maven.minecraftforge.net
repositories {
    // Forge □□□□□□ForgeGradle □□□□□□□□□□□□□□□□□□□□
    maven { url = 'https://maven.minecraftforge.net/' }
    // Parchment □□□□□□□□□ parchment mapping□
    maven { url = 'https://maven.parchmentmc.org' }
    // Minecraft □□□□□□□□□
    maven { url = 'https://maven.blamejared.com' }  // CTM / Bookshelf □
    maven { url = 'https://cursemaven.com' }        // CurseForge □□
    maven { url = 'https://dvs1.progwml6.com/files/maven/' }  // JEI
    mavenCentral()
    maven { url = 'https://maven.parchmentmc.org' }
}

// ------------------------------------------------------------------------------
// □□□□□
// ------------------------------------------------------------------------------
// □□□□□□□Fabric □□ "minecraft", "mappings", "loader", "fabric-api" □□□□
// Forge □□□□ "minecraftForge" □□□Forge □□□□□□□ Loader □ API
dependencies {
    // Minecraft + Forge □□□□□□□□□□□□ "MC□□-Forge□□"
    // □□□□□ 1.20.1 □□□ Forge 47.3.0
    minecraft 'net.minecraftforge:forge:1.20.1-47.3.0'

    // ===== □□□□□□□□□□□□□□□ =====

    // JEI□□□□□□□□□□- Forge □□ jei-api □□□Fabric □ fabric-jei-api-v1□
    // compileOnly(fg.deobf("mezz.jei:jei-1.20.1-common-api:15.x.x.x:api"))
    // runtimeOnly(fg.deobf("mezz.jei:jei-1.20.1-forge:15.x.x.x"))

    // Forge □ fg.deobf() □□□□□□□□□□□□□Fabric □□□□

    // JarJar □□□□□Forge □□□□□□□□□□□□□□□□□ jar
```

```
    // jarJar(group: 'com.example', name: 'somelib', version: '[1.0,2.0)')
}

// ------------------------------------------------------------------------
// □□□□□□□□□□□□□□□□
// ------------------------------------------------------------------------
sourceSets.main.resources { srcDir 'src/generated/resources' }

// ------------------------------------------------------------------------
// Jar □□□□
// ------------------------------------------------------------------------
// □□□□Fabric □□ Loom □ loom { } □□□□□□ include □□
// Forge □□ jarJar □□□□□□□ jarJar □□
jar {
    manifest {
        attributes([
            "Specification-Title"     : "examplemod",
            "Specification-Vendor"     : "examplemod",
            "Specification-Version"   : "1",
            "Implementation-Title"    : project.name,
            "Implementation-Version"  : project.jar.archiveVersion,
            "Implementation-Vendor"    : "examplemod",
            // Mixin □□□Forge □□□ Mixin□
            "MixinConfigs"            : "examplemod.mixins.json"
        ])
    }

    // □ JarJar □□□□□□□
    // from { configurations.jarJar.collect { it.isDirectory() ? it : zipTree(it) } }
}

// ------------------------------------------------------------------------
// □□□□□□□□□
// ------------------------------------------------------------------------
publishing {
    publications {
        mavenJava(MavenPublication) {
            artifact jar
        }
    }
    repositories {
        maven {
            url "file://${project.projectDir}/mcmodsrepo"
        }
    }
}

// ------------------------------------------------------------------------
// □□□□□ build □□□□□□□□□
// ------------------------------------------------------------------------
tasks.named('processResources', ProcessResources).configure {
    var replaceProperties = [
            minecraft_version   : "1.20.1",
            minecraft_version_range: "[1.20.1,1.20.2)",
            forge_version       : "47.3.0",
            forge_version_range : "[47.3,)",
            loader_version_range: "[47,)",
            mod_id              : "examplemod",
            mod_name            : "Example Mod",
            mod_license         : "MIT",
            mod_version         : "1.0.0",
            mod_authors         : "Author",
            mod_description     : "Example mod for Forge 1.20.1"
    ]
    inputs.properties replaceProperties

    // □□ mods.toml □□□□□ ${mod_id} □
    filesMatching(['META-INF/mods.toml', 'pack.mcmeta']) {
        expand replaceProperties + [project: project]
    }
}
```

project/mods.toml

```
# ======================================================================
# Forge 1.20.1 □□□□□□□□ - mods.toml
```

```
# ================================================================================
# 【Forge 与 Fabric 的主要区别】
# 1. Forge 使用 mods.toml（TOML 格式），而不是 fabric.mod.json（JSON 格式）
# 2. Forge 的 modLoader 通常为 "javafmloader"，而 Fabric 为 "fabric"
# 3. Forge 使用 dependency 来声明 Minecraft/Forge 版本范围，
# 4. Forge 的 [x,y) 区间表示法与 Fabric 的 >=x <y 类似，但语法不同
# 5. 文件位置：src/main/resources/META-INF/mods.toml
# ================================================================================

# 【必填】加载器类型，javafmloader 对应 Forge Java 模组加载器
# 【说明】Fabric 对应的是 loader（fabric.mod.json 的 schemaVersion 字段）
modLoader="javafmloader"

# Forge 加载器版本范围，对应 loaderVersion。
# 【说明】 META-INF/jarjar/metadata.json 中的 loaderVersion 字段
loaderVersion="${loader_version_range}"

# 【必填】模组许可证，Forge 官方模板默认使用 All Rights Reserved 。
# 【说明】MIT, Apache-2.0, GPL-3.0, All Rights Reserved 等
license="${mod_license}"

# 【可选】问题追踪地址，issueTrackerURL 即 issue 提交地址
# issueTrackerURL="https://github.com/example/examplemod/issues"

# --------------------------------------------------------------------------------
# 【必填】模组列表 [[mods]] 数组，可以同时打包多个 jar 的模组信息。
# --------------------------------------------------------------------------------
[[mods]]
# 【必填】 ID，模组的唯一标识符，仅允许 a-z0-9_。
# 【说明】 @Mod("examplemod") 的参数。
modId="${mod_id}"

# 【必填】模组版本，支持占位符，${file.jarVersion} 会从 build.gradle 读取。
version="${mod_version}"

# 【可选】
displayName="${mod_name}"

# 【可选】TOML 的特殊写法
authors="${mod_authors}"

# 【】
description='''
${mod_description}
'''

# --------------------------------------------------------------------------------
# 【依赖】
# 【】 [[dependencies.${mod_id}]] 数组，可以为空。
# Forge 会自动添加一个 minecraft, forge，不需要手动 modId。
# mandatory: true 表示必须存在，否则为 false 则可选。
# 【版本范围】[1.20.1,1.20.2) 表示 >=1.20.1 且 <1.20.2
# --------------------------------------------------------------------------------

# 对 Minecraft 的依赖。
# 【说明】Fabric 为 "minecraft": ">=1.20.1 <1.20.2"
# Forge 为 versionRange 格式
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"   # required【必须】| optional【可选】| incompatible【不兼容】
    versionRange="${minecraft_version_range}"
    ordering="NONE"    # BEFORE【在其之前加载】| AFTER【在其之后加载】| NONE【无】
    side="BOTH"       # BOTH【客户端+服务端】| CLIENT【仅客户端】| SERVER【仅服务端】

# 对 Forge 的依赖。
[[dependencies.${mod_id}]]
    modId="forge"
    type="required"
    versionRange="${forge_version_range}"
    ordering="NONE"
    side="BOTH"

# 可选依赖，例如 JEI。
# [[dependencies.${mod_id}]]
```

```
#     modId="jei"
#     type="optional"
#     versionRange="[15.0.0,)"
#     ordering="NONE"
#     side="BOTH"
```

recipe/shaped_recipe.json

```
{
  "_comment_shaped_recipe": "================================================================================",
  "_comment_forge_vs_fabric": "□Forge 1.20.1 □□□□□□□ - □ Fabric □□□□□",
  "_comment_diff_1": "1. JSON □□□□□□□□□□□□□□ minecraft:crafting_shaped □□",
  "_comment_diff_2": "2. Forge □□□□□ NBT □□□□□□□□□□□□□□ nbt □□□□□□□ NBT",
  "_comment_diff_3": "3. Forge □□□□□ Forge □ ingredient □□□□□ forge:partial_nbt, forge:intersection□",
  "_comment_diff_4": "4. Forge □□□□□□□ ingredient□fabric □□□□□□ forge □□□□□□□□□□",
  "_comment_diff_5": "5. Forge 1.20.1 □□□□□□□ data/<modid>/recipes/ □□□□□□□□□□□□",
  "_comment_diff_6": "6. Forge □□□□conditions□□□□□□□□□□ forge:mod_loaded, item_exists □",
  "_comment_diff_7": "   Fabric □□ conditions□□ Forge □□□□□□□□□□forge:tag_empty □□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/recipes/example_recipe.json",
  "_comment_================================================================================": "============================================================================================",

  "type": "minecraft:crafting_shaped",

  "_comment_pattern": "□□□□□□□□=□□□□□□□□ ingredient □□□□□",
  "pattern": [
    "III",
    "I I",
    "III"
  ],

  "_comment_key": "□□□□□pattern □□□□ -> ingredient",
  "_comment_key_forge": "Forge □□ ingredient □□□□□",
  "_comment_key_forge_item": "  - item: □□□ {\"item\": \"minecraft:iron_ingot\"}",
  "_comment_key_forge_tag": "  - tag: □□ {\"tag\": \"forge:ingots/iron\"}□Forge □□□ forge: □□□□□□□□",
  "_comment_key_forge_nbt": "  - Forge □□□partial_nbt, intersection, compound □□□ ingredient",
  "key": {
    "I": {
      "_comment": "□□ Forge □ forge:ingots/iron □□□Forge □□□□□□□□□□□",
      "tag": "forge:ingots/iron"
    }
  },

  "_comment_result": "□□□□",
  "_comment_result_forge_nbt": "Forge □□□□□□□□□□ nbt□□□□□□ NBT □□",
  "_comment_result_forge_nbt_example": "□□□\"nbt\": \"{display:{Name:\\\"{\\\"text\\\":\\\"Custom\\\"}\\\"}}\"",
  "result": {
    "item": "examplemod:example_item",
    "count": 1,
    "_comment_nbt_optional": "□□□□Forge □□ nbt □□□Fabric 1.20.1 □□□□□□",
    "nbt": "{display:{Name:\"\\\"{\\\"text\\\":\\\"Example Item\\\",\\\"color\\\":\\\"gold\\\",\\\"italic\\\":false}\\\"\"}}"
  },

  "_comment_conditions_forge": "Forge □□□□□□□□□□□□□□□",
  "_comment_conditions_forge_types": "□□□□□□□□",
  "_comment_conditions_forge_mod_loaded": "  - forge:mod_loaded: □□□□□□□□□□□□□□□",
  "_comment_conditions_forge_item_exists": "  - forge:item_exists: □□□□□□□□□□□□□□□□□□□□",
  "_comment_conditions_forge_tag_empty": "  - forge:tag_empty: □□□□□□□□□□",
  "_comment_conditions_fabric_diff": "Fabric □□ conditions□□□□□□□□ fabric:□□ fabric:all_loaded, fabric:item_exists□",
  "conditions": [
    {
      "type": "forge:mod_loaded",
      "modid": "examplemod"
    }
  ]
}
```

tag/block_tag.json

```
{
  "_comment_block_tag": "================================================================================",
  "_comment_forge_vs_fabric": "□Forge 1.20.1 □□□□□□□ - □ Fabric □□□□□",
  "_comment_diff_1": "1. JSON □□□□□□□□□□□□□□ Tag JSON □□",
  "_comment_diff_2": "2. Forge 1.20.1 □□□□□□□□□□entries -> values □1.21+□",
  "_comment_diff_3": "3. Forge □□□□□ forge: □□□□□□□□□□□□□□",
  "_comment_forge_tags": "   - forge:needs_wood_tool: □□□□□□□□□□□□",
  "_comment_forge_needs_stone": "   - forge:needs_stone_tool: □□□□□□□",
  "_comment_forge_needs_iron": "   - forge:needs_iron_tool: □□□□□□□",
```

```
  "_comment_forge_needs_diamond": "   - forge:needs_diamond_tool: □□□□□□□",
  "_comment_forge_needs_netherite": "   - forge:needs_netherite_tool: □□□□□□□□□",
  "_comment_forge_infiniburn": "   - forge:infiniburn_*: □□□□□□□□/□□□□□□□□□",
  "_comment_forge_ores": "   - forge:ores, forge:ores/iron: □□□□□□□□□□□□□□□□",
  "_comment_forge_storage_blocks": "   - forge:storage_blocks, forge:storage_blocks/iron: □□□□□□□□□",
  "_comment_diff_4": "4. Fabric □□□□□□□□ forge: □□□□□□□□□ c: □□□□ c:ores□",
  "_comment_diff_5": "5. Forge □ needs_*_tool □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□",
  "_comment_location": "□□□□□: src/main/resources/data/<tag_namespace>/tags/blocks/<tag_name>.json",
  "_comment_path_example": "□□: src/main/resources/data/forge/tags/blocks/needs_iron_tool.json",
  "_comment_path_example2": "□□: src/main/resources/data/examplemod/tags/blocks/example_tag.json",
  "_comment_=============================================================================": "============================================================================================",

  "replace": false,
  "_comment_replace": "replace: true □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□false □□□",

  "values": [
    {
      "_comment": "□□□□□□□□",
      "id": "examplemod:example_block",
      "required": true
    },
    {
      "_comment": "□□□□□□□□",
      "id": "minecraft:stone",
      "required": true
    },
    {
      "_comment": "□□□□□□□□□□□□□□□□□□# □□□",
      "#": "minecraft:mineable/pickaxe",
      "required": false
    },
    {
      "_comment": "required: false □□□□□□□□□□□□□□□□□□□□",
      "_comment_required_false": "□□□□□□□□/□□□□□□□□□□□□□□□ false□□□□□□□□□",
      "id": "othermod:other_block",
      "required": false
    }
  ]
}
```

worldgen/biome_modifier.json

```
{
  "_comment_biome_modifier": "============================================================================",
  "_comment_forge_vs_fabric": "□Forge 1.20.1 □□□□□□□□ - □ Fabric □□□□□",
  "_comment_diff_1": "1. Forge 1.20.1 □ biome_modifier JSON □□□□□□□□□□Forge □□□□□□□□□□□",
  "_comment_diff_2": "2. Fabric □ BiomeModification API□□□□□□□BiomeModifications.addFeature □□",
  "    _comment_fabric_code": "   Fabric: BiomeModifications.addFeature(BiomeSelectors.foundInOverworld(), ...)",
  "_comment_diff_3": "3. Forge □ biome_modifier JSON □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□",
  "_comment_diff_4": "4. Forge □□□□□ DatapackBuiltinRegistriesProvider □□□□□□□□□□□",
  "    _comment_datapack_provider": "   □□□ feature/placed_feature □ DeferredRegister + DatapackBuiltinRegistriesProvider",
  "_comment_diff_5": "5. Fabric □ DataGen + BiomeModification □□□□□",
  "_comment_location": "□□□□□: src/main/resources/data/examplemod/forge/biome_modifier/example_modifier.json",
  "_comment_=============================================================================": "============================================================================================",

  "type": "forge:add_features",
  "_comment_type": "type: biome_modifier □□",
  "_comment_types": "Forge □□□ biome_modifier □□□□",
  "  _comment_add_features": "  - forge:add_features: □□□□□□□□□□□□□□□□",
  "  _comment_add_spawns": "  - forge:add_spawns: □□□□□/□□□□□",
  "  _comment_remove_features": "  - forge:remove_features: □□□□□",
  "  _comment_remove_spawns": "  - forge:remove_spawns: □□□□□",
  "  _comment_add_carvers": "  - forge:add_carvers: □□□□□□□□",
  "  _comment_modify_b...": "  - forge:modify_b....",

  "biomes": "#minecraft:is_overworld",
  "_comment_biomes": "biomes: □□□□□□□□",
  "_comment_biomes_types": "□□□□□□□□",
  "  _comment_tag": "  - #namespace:tag□□□□□□□□ #minecraft:is_overworld□",
  "  _comment_single": "  - namespace:biome_id□□□□□□□□□□",
  "  _comment_array": "  - □□ [\"minecraft:plains\", \"minecraft:forest\"]",

  "features": [
    "examplemod:example_ore_placed"
  ],
  "_comment_features": "features: □□□□□ placed_feature □□□□□□□□□□□□ placed_feature ID□",
```

```
  "_comment_features_path": "placed_feature □□□□□data/<modid>/worldgen/placed_feature/<name>.json",

  "step": "underground_ores",
  "_comment_step": "step: □□□□□□□□□□□□□□□□□",
  "_comment_steps": "Forge/MC □□□□□□□□□GenerationStep.Decoration□□",
  "  _comment_raw_generation": "  - raw_generation□□□□□□□□□",
  "  _comment_lakes": "  - lakes□□□□",
  "  _comment_local_modifications": "  - local_modifications□□□□□□",
  "  _comment_underground_structures": "  - underground_structures□□□□□□",
  "  _comment_surface_structures": "  - surface_structures□□□□□□",
  "  _comment_strongholds": "  - strongholds□□□□",
  "  _comment_underground_ores": "  - underground_ores□□□□□□-- □□□□□□□",
  "  _comment_underground_decoration": "  - underground_decoration□□□□□□",
  "  _comment_fluid_springs": "  - fluid_springs□□□□□",
  "  _comment_vegetal_decoration": "  - vegetal_decoration□□□□□□-- □/□□□□",
  "  _comment_top_layer_modification": "  - top_layer_modification□□□□□□□□□"
}
```

Forge 1.20.4

block/simple_block.java

```
// ================================================================================
// Forge 1.20.4 □□□□□□□ - simple_block.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. API □□□□□DeferredRegister + ForgeRegistries.BLOCKS
// 2. BlockBehaviour.Properties□□ 1.20.1 □□□
// 3. Material □□□□□□□□ 1.20.1 □□□□ MapColor □□□
// 4. 1.20.4 □ BlockEntityType.Builder.of() □□□ 1.20.1 □□
// □□ Fabric □□□□□□□
// 5. Fabric □ Registry.register(Registries.BLOCK, id, block) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.BLOCKS, MODID) □□□□
// 6. Forge □ RegistryObject<Block> □□□□□□□□□□□
//    Fabric □□□□ Block □□
// ================================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.material.MapColor;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 *
 * □□□□□□□□ ModMain □□□□□□□□□
 *   ModBlocks.BLOCKS.register(modEventBus);
 */
public class ModBlocks {

    public static final DeferredRegister<Block> BLOCKS =
            DeferredRegister.create(ForgeRegistries.BLOCKS, ExampleMod.MODID);

    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITIES =
            DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□□ =====
    // 1.20.4 □□ BlockBehaviour.Properties□□ 1.20.1 □□□
    // □□□□□Fabric □ AbstractBlock.Settings / FabricBlockSettings
    public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register("example_block",
            () -> new Block(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.STONE)
                    .strength(3.0F, 6.0F)
                    .requiresCorrectToolForDrops()
                    .sound(net.minecraft.world.level.block.SoundType.STONE)
            ));
```

```
    // ===== □□ BlockEntityType =====
    // public static final RegistryObject<BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
    //         BLOCK_ENTITIES.register("example_block_entity",
    //                 () -> BlockEntityType.Builder
    //                         .of(ExampleBlockEntity::new, EXAMPLE_BLOCK.get())
    //                         .build(null));

    public static void register(IEventBus modEventBus) {
        BLOCKS.register(modEventBus);
        BLOCK_ENTITIES.register(modEventBus);
    }
}
```

command/simple_command.java

```
// ================================================================================
// Forge 1.20.4 □□□□□□□ - simple_command.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□ API □□□□□□RegisterCommandsEvent + Brigadier
// 2. sendSuccess() □□□□□□ 1.20.1 □□
// □□ Fabric □□□□□□□
// 3. Fabric □ CommandRegistrationCallback.EVENT □□□□
//    Forge □ RegisterCommandsEvent □□□□ FORGE EVENT_BUS □□□
// 4. Forge □ @Mod.EventBusSubscriber □□□□□□□□
// 5. □□□□□□□ Brigadier □□□□□□API □□
// ================================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.network.chat.Component;
import net.minecraft.server.level.ServerPlayer;
import net.minecraftforge.event.RegisterCommandsEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * Forge □□□□ - □□ RegisterCommandsEvent □□
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        CommandDispatcher<CommandSourceStack> dispatcher = event.getDispatcher();

        dispatcher.register(Commands.literal(ExampleMod.MODID)
                .requires(source -> source.hasPermission(2))
                .then(Commands.literal("give")
                        .then(Commands.argument("target", EntityArgument.player())
                                .then(Commands.argument("amount", IntegerArgumentType.integer(1, 64))
                                        .executes(ModCommands::executeGive)
                                )
                        )
                )
                .executes(ModCommands::executeInfo)
        );
    }

    private static int executeGive(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        ServerPlayer target = EntityArgument.getPlayer(context, "target");
        int amount = IntegerArgumentType.getInteger(context, "amount");

        CommandSourceStack source = context.getSource();
        // □□□□Fabric □ source.sendFeedback(Text.of(...), false)
        // Forge □ source.sendSuccess(() -> Component, false)
```

```
        source.sendSuccess(() -> Component.literal(
                "Gave " + amount + " items to " + target.getName().getString()), true);

        return Command.SINGLE_SUCCESS;
    }

    private static int executeInfo(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal("Example Mod v1.0.0 - Forge 1.20.4"), false);
        return Command.SINGLE_SUCCESS;
    }
}
```

enchantment/custom_enchantment.java

```
// ============================================================================
// Forge 1.20.4 □□□□□□□ - custom_enchantment.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ============================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□ 1.20.1 □□□□□□□□ Enchantment □
// 2. □□□□□□□□Enchantment(Rarity, EnchantmentCategory, EquipmentSlot[])
// 3. DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) □□
// 4. □□□1.21+ □□□□□□□□□□□□□JSON□□1.20.4 □□□□□□□
// □□ Fabric □□□□□□□
// 5. Fabric □ Registry.register(Registries.ENCHANTMENT) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS) □□□□
// 6. □□□ Enchantment API □□□□□□□□ MC □□□
// ============================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□ - Forge DeferredRegister □□
 */
public class ModEnchantments {

    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, ExampleMod.MODID);

    // ===== □□□□ =====
    // Enchantment □□□□□□ 1.20.1 □□□□□ API□
    // □□□□1.21+ □□□□□□ Enchantment □□□□□□□□
    public static final RegistryObject<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND));

    // ===== □□□□□□ =====
    public static class ExampleEnchantment extends Enchantment {

        public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                                  EquipmentSlot... slots) {
            super(rarity, category, slots);
        }

        @Override
        public int getMinCost(int level) {
            return 10 * level;
        }

        @Override
        public int getMaxCost(int level) {
            return getMinCost(level) + 15;
```

```
        }

        @Override
        public int getMaxLevel() {
            return 3;
        }

        @Override
        protected boolean checkCompatibility(Enchantment other) {
            return super.checkCompatibility(other)
                    && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
        }

        @Override
        public boolean canEnchant(net.minecraft.world.item.ItemStack stack) {
            return super.canEnchant(stack);
        }

        @Override
        public boolean isCurse() { return false; }
        @Override
        public boolean isTradeable() { return true; }
        @Override
        public boolean isDiscoverable() { return true; }
        @Override
        public boolean isAllowedOnBooks() { return true; }

        // ===== □□□□□ =====
        @Override
        public void doPostAttack(
                net.minecraft.world.entity.LivingEntity attacker,
                net.minecraft.world.entity.Entity target,
                int level) {
            if (target instanceof net.minecraft.world.entity.LivingEntity livingTarget) {
                livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                        net.minecraft.world.effect.MobEffects.POISON, 60, level - 1));
            }
            super.doPostAttack(attacker, target, level);
        }
    }

    public static void register(IEventBus modEventBus) {
        ENCHANTMENTS.register(modEventBus);
    }
}
```

entity/simple_entity.java

```
// ================================================================================
// Forge 1.20.4 □□□□□□□ - simple_entity.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□ API □□□□□□DeferredRegister + ForgeRegistries.ENTITY_TYPES
// 2. EntityType.Builder.of() □□□ 1.20.1 □□
// 3. EntityAttributeCreationEvent □□□□□ 1.20.1 □□
// 4. ForgeSpawnEggItem □ 1.20.4 □□□
// □□ Fabric □□□□□□□
// 5. Fabric □ Registry.register(Registries.ENTITY_TYPE) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ENTITY_TYPES) □□□□
// 6. □□□□□Fabric □ FabricDefaultAttributeRegistry.register()
//    Forge □ EntityAttributeCreationEvent □□
// ================================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.entity.ai.attributes.AttributeSupplier;
import net.minecraft.world.entity.ai.attributes.Attributes;
import net.minecraft.world.entity.monster.Monster;
import net.minecraftforge.event.entity.EntityAttributeCreationEvent;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;
```

```
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□□ - Forge DeferredRegister □□
 */
public class ModEntities {

    public static final DeferredRegister<EntityType<?>> ENTITIES =
            DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    // EntityType.Builder.of(factory, category)□□ 1.20.1 □□□
    // □□□□Fabric: EntityType.Builder.create() + .build() + Registry.register
    // Forge: RegistryObject □□□supplier lambda □□□□
    public static final RegistryObject<EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITIES.register("example_entity",
                    () -> EntityType.Builder
                            .of(ExampleEntity::new, MobCategory.CREATURE)
                            .sized(1.0F, 1.0F)
                            .clientTrackingRange(10)
                            .build("example_entity"));

    // ===== □□□□□□□ =====
    // □□□□Fabric: FabricDefaultAttributeRegistry.register(type, builder)
    // Forge: EntityAttributeCreationEvent □□
    public static void registerAttributes(EntityAttributeCreationEvent event) {
        AttributeSupplier.Builder builder = Monster.createMonsterAttributes();
        builder.add(Attributes.MAX_HEALTH, 20.0D);
        builder.add(Attributes.MOVEMENT_SPEED, 0.25D);
        builder.add(Attributes.ATTACK_DAMAGE, 5.0D);
        builder.add(Attributes.FOLLOW_RANGE, 32.0D);

        event.put(EXAMPLE_ENTITY.get(), builder.build());
    }

    @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.MOD)
    public static class ModBusEvents {
        @SubscribeEvent
        public static void onAttributeCreate(EntityAttributeCreationEvent event) {
            registerAttributes(event);
        }
    }

    public static void register(IEventBus modEventBus) {
        ENTITIES.register(modEventBus);
    }
}
```

gui/container_screen.java

```
// ================================================================================
// Forge 1.20.4 GUI / □□□□□□□ - container_screen.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. MenuType □□□: ForgeRegistries.CONTAINERS□□ 1.20.1 □□□
// 2. IForgeMenuType.create() □□□ 1.20.1 □□
// 3. GuiGraphics □□□ 1.20.1 □□□1.20 □□□
// 4. AbstractContainerScreen □ AbstractContainerMenu API □□
// □□ Fabric □□□□□□□
// 5. Fabric □ ScreenHandlerType + Registry.register(Registries.SCREEN_HANDLER)
//    Forge □ DeferredRegister.create(ForgeRegistries.CONTAINERS) □□ MenuType
// 6. Forge □ MenuType = Fabric □ ScreenHandlerType
// 7. Forge □□□□□ NetworkHooks.openScreen()
//    Fabric □ player.openHandledScreen()
// ================================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.client.gui.screens.MenuScreens;
import net.minecraft.client.gui.screens.inventory.AbstractContainerScreen;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
```

```
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraftforge.common.extensions.IForgeMenuType;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□ - Forge DeferredRegister + MenuScreens
 */
public class ModMenus {

    public static final DeferredRegister<MenuType<?>> MENUS =
            DeferredRegister.create(ForgeRegistries.CONTAINERS, ExampleMod.MODID);

    // ===== □□ MenuType =====
    // □□□□Fabric □ new ScreenHandlerType<>(factory)
    // Forge □ IForgeMenuType.create()
    public static final RegistryObject<MenuType<ExampleMenu>> EXAMPLE_MENU = MENUS.register("example_menu",
            () -> IForgeMenuType.create((containerId, inventory, data) -> {
                return new ExampleMenu(containerId, inventory);
            }));

    // ===== AbstractContainerMenu □□ =====
    // □□□□Forge □ AbstractContainerMenu = Fabric □ ScreenHandler
    public static class ExampleMenu extends AbstractContainerMenu {

        public ExampleMenu(int containerId, Inventory playerInventory) {
            super(EXAMPLE_MENU.get(), containerId);

            // □□□□□□□□□3x9□
            for (int row = 0; row < 3; row++) {
                for (int col = 0; col < 9; col++) {
                    this.addSlot(new Slot(playerInventory, 9 + row * 9 + col,
                            8 + col * 18, 84 + row * 18));
                }
            }

            // □□□□□□□□□9□□
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
            }
        }

        @Override
        public ItemStack quickMoveStack(Player player, int index) {
            Slot slot = this.slots.get(index);
            if (!slot.hasItem()) return ItemStack.EMPTY;

            ItemStack original = slot.getItem().copy();
            ItemStack stack = slot.getItem();

            if (!this.moveItemStackTo(stack, 0, this.slots.size(), true)) {
                return ItemStack.EMPTY;
            }

            slot.setChanged();
            return original;
        }

        @Override
        public boolean stillValid(Player player) {
            return true;
        }
    }

    // ===== AbstractContainerScreen □□ =====
    // □□□□Forge □ AbstractContainerScreen = Fabric □ HandledScreen
    // 1.20.4 □□ GuiGraphics□□ 1.20.1 □□□
    public static class ExampleScreen extends AbstractContainerScreen<ExampleMenu> {
        public ExampleScreen(ExampleMenu menu, Inventory inventory, Component title) {
            super(menu, inventory, title);
```

```
            this.imageWidth = 176;
            this.imageHeight = 166;
            this.inventoryLabelY = this.imageHeight - 94;
        }

        // 1.20.4 □ GuiGraphics□□ 1.20.1 □□□1.19.2 □ PoseStack□
        @Override
        protected void renderBg(net.minecraft.client.gui.GuiGraphics graphics,
                                float partialTick, int mouseX, int mouseY) {
            int x = (this.width - this.imageWidth) / 2;
            int y = (this.height - this.imageHeight) / 2;
            graphics.blit(new net.minecraft.resources.ResourceLocation(ExampleMod.MODID,
                    "textures/gui/example_container.png"), x, y, 0, 0,
                    this.imageWidth, this.imageHeight);
        }

        @Override
        public void render(net.minecraft.client.gui.GuiGraphics graphics,
                           int mouseX, int mouseY, float partialTick) {
            this.renderBackground(graphics);
            super.render(graphics, mouseX, mouseY, partialTick);
            this.renderTooltip(graphics, mouseX, mouseY);
        }
    }

    // ===== Screen □□ =====
    // □□□□Fabric: HandledScreens.register(type, factory)
    // Forge: MenuScreens.register(type, factory)
    // □ FMLClientSetupEvent □□□□
    // event.enqueueWork(() -> {
    //     MenuScreens.register(EXAMPLE_MENU.get(), ExampleScreen::new);
    // });

    public static void register(IEventBus modEventBus) {
        MENUS.register(modEventBus);
    }
}
```

item/simple_item.java

```
// ================================================================================
// Forge 1.20.4 □□□□□□ - simple_item.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□
// 1. API □□□□□DeferredRegister + ForgeRegistries.ITEMS
// 2. CreativeModeTab.builder() □□□□ 1.20.1 □□□
// 3. BuildCreativeModeTabContentsEvent □□□□ 1.20.1 □□□
// 4. Item.Properties □□□□ 1.20.1 □□
// □□ Fabric □□□□□□
// 5. Fabric □ Registry.register(Registries.ITEM, id, item) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ITEMS, MODID) □□□□
// 6. □□□□□□Fabric □ ItemGroupEvents.modifyEntriesEvent()
//    Forge □ CreativeModeTab.builder() □ BuildCreativeModeTabContentsEvent
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□ - Forge DeferredRegister □□
 */
public class ModItems {

    // □□□□□
    public static final DeferredRegister<Item> ITEMS =
```

```
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // □□□□□□□□□□□□□□□ 1.20.1 □□□
    public static final DeferredRegister<CreativeModeTab> CREATIVE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MODID);

    // ===== □□□□□□□ =====
    public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register("example_item",
            () -> new Item(new Item.Properties()
                    .stacksTo(64)
                    // .rarity(Rarity.RARE)
                    // .fireResistant()
            ));

    // ===== □□□□□□□ =====
    // public static final RegistryObject<Item> EXAMPLE_BLOCK_ITEM = ITEMS.register("example_block",
    //         () -> new net.minecraft.world.item.BlockItem(
    //                 com.example.examplemod.block.ModBlocks.EXAMPLE_BLOCK.get(),
    //                 new Item.Properties()));

    // ===== □□□□□□□□□□□□□□□ =====
    // 1.20.4 □ CreativeModeTab.builder()□□ 1.20.1 □□□
    // □□□□Fabric □ FabricItemGroup.builder() □ ItemGroupEvents
    public static final RegistryObject<CreativeModeTab> EXAMPLE_TAB = CREATIVE_TABS.register("example_tab",
            () -> CreativeModeTab.builder()
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    .icon(() -> new ItemStack(EXAMPLE_ITEM.get()))
                    .displayItems((parameters, output) -> {
                        output.accept(EXAMPLE_ITEM.get());
                        // output.accept(EXAMPLE_BLOCK_ITEM.get());
                    })
                    .build());

    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
        CREATIVE_TABS.register(modEventBus);
    }

    // ===== □□□□□□□□□□□□□□□□ =====
    // □□□□Fabric □ ItemGroupEvents.modifyEntriesEvent(INGREDIENTS).register(...)
    // Forge □ BuildCreativeModeTabContentsEvent
    //
    // modEventBus.addListener(ModItems::addToExistingTabs);
    // public static void addToExistingTabs(
    //         net.minecraftforge.event.BuildCreativeModeTabContentsEvent event) {
    //     if (event.getTabKey() == CreativeModeTabs.INGREDIENTS) {
    //         event.accept(EXAMPLE_ITEM);
    //     }
    // }
}
```

item/tool_item.java

```
// ================================================================================
// Forge 1.20.4 □□/□□□□□□ - tool_item.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. Tier □□□□□□ 1.20.1 □□□□□□getUses/getSpeed/getLevel □□
// 2. □□□□□□□□□□ 1.20.1 □□
// 3. TierSortingRegistry □ 1.20.4 □□□
// 4. □□□1.21+ □□ Tier □□□□□ TagKey<Block> □□ getLevel()
// □□ Fabric □□□□□□□
// 5. Forge □ Tier □□□net.minecraft.world.item.Tier□
//    Fabric 1.20.4 □ ToolMaterial □□□□□□□□□□
// 6. Forge □ DeferredRegister □□□Fabric □ Registry.register
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
```

```
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□ - Forge Tier □□
 */
public class ModTools {

    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ===== □□□□□□□□Tier□ =====
    // □□□□Forge Tier □□□ vs Fabric ToolMaterial□
    //   Forge: getUses()             Fabric: getDurability()
    //   Forge: getSpeed()            Fabric: getMiningSpeedMultiplier()
    //   Forge: getLevel()            Fabric: getMiningLevel()
    //   Forge: getEnchantmentValue() Fabric: getEnchantability()
    public enum ModTiers implements Tier {
        EXAMPLE_TIER(
            1561,    // getUses()□□□
            8.0F,    // getSpeed()□□□□□
            3.0F,    // getAttackDamageBonus()□□□□□□□□
            3,       // getLevel()□□□□□
            10,      // getEnchantmentValue()□□□□□
        );

        private final int uses;
        private final float speed;
        private final float attackDamageBonus;
        private final int level;
        private final int enchantmentValue;
        private final Ingredient repairIngredient;

        ModTiers(int uses, float speed, float attackDamageBonus,
                 int level, int enchantmentValue) {
            this.uses = uses;
            this.speed = speed;
            this.attackDamageBonus = attackDamageBonus;
            this.level = level;
            this.enchantmentValue = enchantmentValue;
            this.repairIngredient = Ingredient.of(net.minecraft.world.item.Items.DIAMOND);
        }

        @Override
        public int getUses() { return uses; }
        @Override
        public float getSpeed() { return speed; }
        @Override
        public float getAttackDamageBonus() { return attackDamageBonus; }
        @Override
        public int getLevel() { return level; }
        @Override
        public int getEnchantmentValue() { return enchantmentValue; }
        @Override
        public Ingredient getRepairIngredient() { return repairIngredient; }
    }

    // ===== □□□□□□ =====
    // □□□□□ 1.20.1 □□
    // □□□□1.21+ □□□□ Item.Properties.attributes() □□□□□□
    public static final RegistryObject<Item> EXAMPLE_PICKAXE = ITEMS.register("example_pickaxe",
            () -> new PickaxeItem(ModTiers.EXAMPLE_TIER, 1, -2.8F, new Item.Properties()));

    public static final RegistryObject<Item> EXAMPLE_SWORD = ITEMS.register("example_sword",
            () -> new SwordItem(ModTiers.EXAMPLE_TIER, 3, -2.4F, new Item.Properties()));

    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
    }
}
```

mixin/mixin_config.json

```
{
  "_comment_mixin_config": "================================================================================",
  "_comment_version": "Forge 1.20.4 Mixin □□ - mixin_config.json",
  "_comment_mc": "MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□□□□□□□□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□□",
  "_comment_diff_1": "1. Forge □□ Mixin□□□□□□□□",
  "_comment_diff_2": "2. Forge □□ jar □ MANIFEST.MF □□□ MixinConfigs",
  "   _comment_manifest": "   build.gradle: attributes([\"MixinConfigs\": \"examplemod.mixins.json\"])",
  "_comment_diff_3": "3. Fabric □ fabric.mod.json □ mixins □□□□",
  "_comment_diff_4": "4. Forge □ official □□□Fabric □ yarn/parchment",
  "_comment_location": "□□□□: src/main/resources/examplemod.mixins.json",
  "_comment_end": "================================================================================",

  "required": true,
  "minVersion": "0.8",
  "package": "com.example.examplemod.mixin",
  "compatibilityLevel": "JAVA_17",
  "refmap": "examplemod.refmap.json",
  "verbose": false,
  "showMessageTypes": false,

  "mixins": [
    {
      "name": "MinecraftServerMixin"
    }
  ],

  "client": [
    {
      "name": "ClientLevelMixin"
    },
    {
      "name": "GuiMixin"
    }
  ],

  "server": [
    {
      "name": "DedicatedServerMixin"
    }
  ],

  "injectors": {
    "defaultRequire": 1
  }
}
```

project/ModMain.java

```
// ================================================================================
// Forge 1.20.4 □□□□□□□□ - ModMain.java
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. API □□□□□□@Mod □□ + DeferredRegister + ForgeRegistries
// 2. 1.20.4 □ BlockBehaviour.Properties□□ 1.20.1 □□□
// 3. 1.20.4 □ CreativeModeTab.builder()□□ 1.20.1 □□□
// 4. 1.20.2+ □□□□□□ API □□□□□□□□□□□□□□□□
// 5. 1.20.4 □ FMLJavaModLoadingContext □□□□□□1.21+ NeoForge □□□□□
// □□ Fabric □□□□□□□
// 6. Fabric □□ ModInitializer □□□□ onInitialize() □□
//    Forge □□ @Mod □□□□□□□□□□□□□□□□ DeferredRegister
// 7. Fabric □ Registry.register() □□□□□□ Registries
//    Forge □ DeferredRegister.create(ForgeRegistries.XXX, MODID) □□□□
// 8. Forge □ IEventBus □□□□□□□□□ addListener / @SubscribeEvent
// ================================================================================

package com.example.examplemod;

import net.minecraftforge.common.MinecraftForge;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext;

// □□□□□□□□□ @Mod □□
```

```
@Mod(ExampleMod.MODID)
public class ExampleMod {

    public static final String MODID = "examplemod";

    // □□□□□□□
    public ExampleMod() {
        IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();

        // ===== □□ DeferredRegister =====
        // ModBlocks.BLOCKS.register(modEventBus);
        // ModItems.ITEMS.register(modEventBus);
        // ModItems.CREATIVE_TABS.register(modEventBus);
        // ModEntities.ENTITIES.register(modEventBus);
        // ModMenus.MENUS.register(modEventBus);
        // ModEnchantments.ENCHANTMENTS.register(modEventBus);

        modEventBus.addListener(this::commonSetup);
        MinecraftForge.EVENT_BUS.register(this);
    }

    private void commonSetup(final net.minecraftforge.fml.event.lifecycle.FMLCommonSetupEvent event) {
        // event.enqueueWork(() -> { ... });
    }
}
```

project/build.gradle

```
// ============================================================================
// Forge 1.20.4 □□□□□□□ - build.gradle
// MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
// ============================================================================
// □□ Forge 1.20.1 □□□□□□□
// 1. Forge □□: 1.20.4 □ 46.x□1.20.1 □ 47.x□
// 2. ForgeGradle □□: □□ 6.0.x □□□□□□□□□□□□
// 3. □□□□□: 1.20.4 □ official □□□ Parchment □□□□
// 4. API □□□□□□□□□□□□□□□□
// 5. 1.20.2+ □□□□□□ API □□□□□□□□□□□□□□□□
// □□ Fabric □□□□□□□
// 6. Forge □ ForgeGradle □□□□ Loom□□□□ minecraftForge □□□□□□
// 7. Forge □ fg.deobf() □□□□□□□□
// ============================================================================

plugins {
    id 'eclipse'
    id 'idea'
    id 'maven-publish'
    id 'net.minecraftforge.gradle' version '6.0.24'
}

// ----------------------------------------------------------------------------
// □□□□□□□□
// ----------------------------------------------------------------------------
version = '1.0.0'
group = 'com.example.examplemod'

// Java □□□□1.20.4 □□□□ Java 17
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// ----------------------------------------------------------------------------
// Minecraft / Forge □□□□
// ----------------------------------------------------------------------------
// □□□□1.20.4 □□ Forge 46.0.x□1.20.1 □ 47.3.0□
minecraft {
    mappings channel: 'official', version: '1.20.4'
    // □□□□ Parchment□
    // mappings channel: 'parchment', version: '2023.12.31-1.20.4'

    runs {
        client {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod { source sourceSets.main }
            }
```

```
        }

        server {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod { source sourceSets.main }
            }
        }

        data {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            args '--mod', 'examplemod', '--all', '--output', file('src/generated/resources/'),
                    '--existing', file('src/main/resources/')
            mods {
                examplemod { source sourceSets.main }
            }
        }
    }
}

repositories {
    maven { url = 'https://maven.minecraftforge.net/' }
    maven { url = 'https://maven.parchmentmc.org' }
    maven { url = 'https://maven.blamejared.com' }
    mavenCentral()
}

dependencies {
    // Minecraft + Forge □□□□□1.20.4-46.0.41
    minecraft 'net.minecraftforge:forge:1.20.4-46.0.41'

    // JEI□1.20.4 □□□
    // compileOnly(fg.deobf("mezz.jei:jei-1.20.4-common-api:16.x.x.x:api"))
    // runtimeOnly(fg.deobf("mezz.jei:jei-1.20.4-forge:16.x.x.x"))
}

sourceSets.main.resources { srcDir 'src/generated/resources' }

jar {
    manifest {
        attributes([
            "Specification-Title"     : "examplemod",
            "Specification-Vendor"     : "examplemod",
            "Specification-Version"   : "1",
            "Implementation-Title"    : project.name,
            "Implementation-Version"  : project.jar.archiveVersion,
            "Implementation-Vendor"    : "examplemod",
            "MixinConfigs"            : "examplemod.mixins.json"
        ])
    }
}

tasks.named('processResources', ProcessResources).configure {
    var replaceProperties = [
            minecraft_version   : "1.20.4",
            minecraft_version_range: "[1.20.4,1.20.5)",
            forge_version       : "46.0.41",
            forge_version_range : "[46,)",
            loader_version_range: "[46,)",
            mod_id              : "examplemod",
            mod_name            : "Example Mod",
            mod_license         : "MIT",
            mod_version         : "1.0.0",
            mod_authors         : "Author",
            mod_description     : "Example mod for Forge 1.20.4"
    ]
    inputs.properties replaceProperties

    filesMatching(['META-INF/mods.toml', 'pack.mcmeta']) {
        expand replaceProperties + [project: project]
    }
}
```

project/mods.toml

```
# ============================================================================
# Forge 1.20.4 □□□□□□□□ - mods.toml
# MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x
# ============================================================================
# □□ Forge 1.20.1 □□□□□□□
# 1. □□□□ 1.20.1 □□□□□
# 2. □□□□□□□□□1.20.4 □ [1.20.4,1.20.5)
# 3. 1.20.4 □ Forge □□□□□□ [46,)
# □□ Fabric □□□□□□□
# 4. Forge □ mods.toml□TOML □□□□□Fabric □ fabric.mod.json□JSON□
# 5. □□□□□□ src/main/resources/META-INF/mods.toml
# ============================================================================

modLoader="javafmloader"
loaderVersion="${loader_version_range}"
license="${mod_license}"

# ----------------------------------------------------------------------------
# □□□□□
# ----------------------------------------------------------------------------
[[mods]]
modId="${mod_id}"
version="${mod_version}"
displayName="${mod_name}"
authors="${mod_authors}"
description='''
${mod_description}
'''

# ----------------------------------------------------------------------------
# □□□□□□
# ----------------------------------------------------------------------------
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

[[dependencies.${mod_id}]]
    modId="forge"
    type="required"
    versionRange="${forge_version_range}"
    ordering="NONE"
    side="BOTH"

# □□□□□□□
# [[dependencies.${mod_id}]]
#     modId="jei"
#     type="optional"
#     versionRange="[16.0.0,)"
#     ordering="NONE"
#     side="BOTH"
```

recipe/shaped_recipe.json

```
{
  "_comment": "============================================================================",
  "_comment_version": "Forge 1.20.4 □□□□□□□□ - shaped_recipe.json",
  "_comment_mc": "MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□□□□□□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□□",
  "_comment_diff_1": "1. Forge □□ result □□ nbt □□□forge: ingredient □□□forge □□",
  "_comment_diff_2": "2. Fabric □□ conditions □□□□□□□□□fabric:□",
  "_comment_location": "□□□□□: src/main/resources/data/examplemod/recipes/example_recipe.json",
  "_comment_end": "============================================================================",

  "type": "minecraft:crafting_shaped",

  "pattern": [
    "III",
    "I I",
    "III"
  ],
```

```
  "key": {
    "I": {
      "tag": "forge:ingots/iron"
    }
  },

  "result": {
    "item": "examplemod:example_item",
    "count": 1,
    "nbt": "{display:{Name:\"{\\\"text\\\":\\\"Example Item\\\",\\\"color\\\":\\\"gold\\\",\\\"italic\\\":false}\\\"}}"
  },

  "conditions": [
    {
      "type": "forge:mod_loaded",
      "modid": "examplemod"
    }
  ]
}
```

tag/block_tag.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.20.4 □□□□□□□ - block_tag.json",
  "_comment_mc": "MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□□□□□□□□",
  "_comment_forge_tags": "Forge □□□□ forge: □□□□□□□□",
  "_comment_forge_needs": "  - forge:needs_wood_tool / forge:needs_stone_tool / forge:needs_iron_tool □",
  "_comment_forge_ores": "  - forge:ores, forge:ores/iron: □□□□□□□",
  "_comment_fabric_diff": "□□ Fabric □□□□Fabric □ forge: □□□□□□□□ c: □□",
  "_comment_location": "□□□□: src/main/resources/data/<tag_namespace>/tags/blocks/<tag_name>.json",
  "_comment_end": "================================================================================",

  "replace": false,

  "values": [
    {
      "id": "examplemod:example_block",
      "required": true
    },
    {
      "id": "minecraft:stone",
      "required": true
    },
    {
      "#": "minecraft:mineable/pickaxe",
      "required": false
    },
    {
      "id": "othermod:other_block",
      "required": false
    }
  ]
}
```

worldgen/biome_modifier.json

```
{
  "_comment_biome_modifier": "================================================================================",
  "_comment_version": "Forge 1.20.4 □□□□□□□□□□ - biome_modifier.json",
  "_comment_mc": "MC□□: 1.20.4 | Java□□: 17 | Forge□□: 46.0.x",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□",
  "_comment_diff_1": "1. Forge □ biome_modifier JSON □□□Forge □□□□□□□□□",
  "_comment_diff_2": "2. Fabric □ BiomeModification API□□□□□□",
  "_comment_diff_3": "3. Forge □ forge: □□□NeoForge 1.21+ □□ neoforge:□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/forge/biome_modifier/example_modifier.json",
  "_comment_end": "================================================================================",

  "type": "forge:add_features",

  "biomes": "#minecraft:is_overworld",

  "features": [
    "examplemod:example_ore_placed"
  ],
  "step": "underground_ores"
```

```
}
```

Forge 1.21.1

block/simple_block.java

```
// ========================================================================
// Forge 1.21.1 □□□□□□ - simple_block.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ========================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□!□
// 1. BlockBehaviour.Properties□□ 1.20.1 □□□
// 2. □□□1.21+ BlockEntity □ saveAdditional/loadAdditional □□
//    HolderLookup.Provider registries □□
// 3. □□□1.21+ □□ MapCodec □□□□□□□□□□□ codec()□
// 4. □□□ResourceLocation.fromNamespaceAndPath() □□ new ResourceLocation()
// 5. DeferredRegister + ForgeRegistries.BLOCKS □□□
// □□ NeoForge □□□□□□
// 6. Forge □ ForgeRegistries.BLOCKS□NeoForge □ DeferredRegister.createBlocks()
// 7. Forge □ net.minecraftforge.registries.*□NeoForge □ net.neoforged.neoforge.registries.*
// □□ Fabric □□□□□□
// 8. Fabric □ Registry.register(Registries.BLOCK) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.BLOCKS) □□□□
// ========================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.HolderLookup;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.EntityBlock;
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.block.state.BlockState;
import net.minecraft.world.level.material.MapColor;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□ - Forge DeferredRegister □□
 */
public class ModBlocks {

    // ===== □□□□□ =====
    // Forge 1.21.1 □□ ForgeRegistries.BLOCKS
    // □□□□NeoForge □ DeferredRegister.createBlocks(MODID)
    public static final DeferredRegister<Block> BLOCKS =
            DeferredRegister.create(ForgeRegistries.BLOCKS, ExampleMod.MODID);

    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITIES =
            DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□ =====
    public static final RegistryObject<Block> EXAMPLE_BLOCK = BLOCKS.register("example_block",
            () -> new Block(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.STONE)
                    .strength(3.0F, 6.0F)
                    .requiresCorrectToolForDrops()
                    .sound(net.minecraft.world.level.block.SoundType.STONE)
            ));

    // ===== □□□ BlockEntity □□□ =====
    // public static final RegistryObject<Block> EXAMPLE_ENTITY_BLOCK = BLOCKS.register(
    //         "example_entity_block",
    //         () -> new ExampleEntityBlock(BlockBehaviour.Properties.of()
    //                 .mapColor(MapColor.METAL)
    //                 .strength(3.0F)
    //         )
    // );
```

```
    // ===== □□ BlockEntityType =====
    // public static final RegistryObject<BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
    //         BLOCK_ENTITIES.register("example_block_entity",
    //                 () -> BlockEntityType.Builder
    //                         .of(ExampleBlockEntity::new, EXAMPLE_ENTITY_BLOCK.get())
    //                         .build(null));

    public static void register(IEventBus modEventBus) {
        BLOCKS.register(modEventBus);
        BLOCK_ENTITIES.register(modEventBus);
    }
}

// ============================================================================
// □1.21+ □□□□□BlockEntity □□/□□□□□□ HolderLookup.Provider □□
//
// Forge 1.20.1:
//   public void saveAdditional(CompoundTag tag)
//   public void loadAdditional(CompoundTag tag)
//
// Forge 1.21.1:
//   public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
//   public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
// ============================================================================
class ExampleBlockEntity extends BlockEntity {

    private int value = 0;

    public ExampleBlockEntity(BlockEntityType<?> type, BlockPos pos, BlockState state) {
        super(type, pos, state);
    }

    // □1.21+ □□□saveAdditional □□ HolderLookup.Provider □□
    @Override
    public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.saveAdditional(tag, registries);
        tag.putInt("value", this.value);
    }

    // □1.21+ □□□loadAdditional □□ HolderLookup.Provider □□
    @Override
    public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.loadAdditional(tag, registries);
        this.value = tag.getInt("value");
    }
}
```

command/simple_command.java

```
// ============================================================================
// Forge 1.21.1 □□□□□□ - simple_command.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ============================================================================
// □□ Forge 1.20.1 □□□□□□
// 1. □□□□ API □□□□□RegisterCommandsEvent + Brigadier
// 2. 1.21+ □□□ RegisterCommandsEvent□Forge □□□□□
// 3. sendSuccess() □□□□□ 1.20.1 □□
// 4. Java 21 □□□□□ record□switch □□□□□
// □□ NeoForge □□□□□□
// 5. Forge: net.minecraftforge.event.RegisterCommandsEvent
//    NeoForge: net.neoforged.neoforge.event.RegisterCommandsEvent
// □□ Fabric □□□□□□
// 6. Fabric □ CommandRegistrationCallback.EVENT
//    Forge □ RegisterCommandsEvent □□
// 7. Forge □ @Mod.EventBusSubscriber □□□□
// ============================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
```

```
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.network.chat.Component;
import net.minecraft.server.level.ServerPlayer;
import net.minecraftforge.event.RegisterCommandsEvent;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;

/**
 * Forge □□□□□ - □□ RegisterCommandsEvent □□
 */
@Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        CommandDispatcher<CommandSourceStack> dispatcher = event.getDispatcher();

        dispatcher.register(Commands.literal(ExampleMod.MODID)
                .requires(source -> source.hasPermission(2))
                .then(Commands.literal("give")
                        .then(Commands.argument("target", EntityArgument.player())
                                .then(Commands.argument("amount", IntegerArgumentType.integer(1, 64))
                                        .executes(ModCommands::executeGive)
                                )
                        )
                )
                .executes(ModCommands::executeInfo)
        );
    }

    private static int executeGive(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        ServerPlayer target = EntityArgument.getPlayer(context, "target");
        int amount = IntegerArgumentType.getInteger(context, "amount");

        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal(
                "Gave " + amount + " items to " + target.getName().getString()), true);

        return Command.SINGLE_SUCCESS;
    }

    private static int executeInfo(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();
        source.sendSuccess(() -> Component.literal("Example Mod v1.0.0 - Forge 1.21.1"), false);
        return Command.SINGLE_SUCCESS;
    }
}
```

enchantment/custom_enchantment.java

```
// ========================================================================
// Forge 1.21.1 □□□□□□□ - custom_enchantment.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ========================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□□□□□□!□
//
// 1. □□□□□□□□□□□□□□□□!□
//    1.20.1: □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) □□
//            □□ Enchantment □□□□□□□□□
//    1.21.1: □□□□□□□□□□□datapack registry□□□□ JSON □□□□
//            □□□□ RegistrySetBuilder □□□□□□□□□
//
// 2. □□□□□□□□□□ Enchantment □□□!□
//    1.20.1: □□□□□□□□□ extends Enchantment□□□□□
//            - getMaxLevel() / getMinCost() / getMaxCost()
//            - checkCompatibility() / canEnchant()
//            - doPostAttack() □
//    1.21.1: Enchantment □ record□□□□□□□□□□□
//            - description (Component): □□□□
//            - definition (EnchantmentDefinition): □□□□□□
//            - exclusiveSet (HolderSet<Enchantment>): □□□□
//            - effects (DataComponentMap): □□□□□□
//
// 3. □□□□□□□□□□ (Enchantment Effect Components)□
//    1.21+ □□□□□□□□□□□□
```

```
//    □□□□
```

```
                () -> DataComponentType.<Increment>builder()
                        .persistent(Increment.CODEC)
                        .build()
        );

// ============================================================================
// □□□□□□□ ResourceKey □□□□□ DeferredRegister!□
//
// 1.20.1: □ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) □□
//         RegistryObject<Enchantment> EXAMPLE = ENCHANTMENTS.register("name", () -> new MyEnchant(...))
//
// 1.21.1: □□□□□□□□□□□□□□□ DeferredRegister □□!
//         □ ResourceKey + RegistrySetBuilder □□□□□□□□□
// ============================================================================
public static final ResourceKey<Enchantment> EXAMPLE_ENCHANTMENT = ResourceKey.create(
        Registries.ENCHANTMENT,
        ResourceLocation.fromNamespaceAndPath(ExampleMod.MODID, "example_enchantment")
);

// ============================================================================
// □□□□□ - □□ RegistrySetBuilder □□□□
// □□ builder □□□□ DatapackBuiltinEntriesProvider□□ GatherDataEvent □□
//
// 1.20.1 □□□□□□□ - □□□□□□ Java □□□□□□
// 1.21.1 □□□□□□□□□□□□□□□□□ JSON □□□
// ============================================================================
public static final RegistrySetBuilder BUILDER = new RegistrySetBuilder();

static {
    BUILDER.add(Registries.ENCHANTMENT, ModEnchantments::bootstrapEnchantments);
}

// ===== □□□□□□□□□□□□□□□□□□□□ =====
private static void bootstrapEnchantments(BootstrapContext<Enchantment> context) {
    context.register(EXAMPLE_ENCHANTMENT, new Enchantment(
            // 1. □□□□□□□□□□□□□□□
            Component.translatable("enchantment.examplemod.example_enchantment"),

            // 2. □□□□□EnchantmentDefinition□
            new EnchantmentDefinition(
                    // supportedItems: □□□□□□□HolderSet□
                    // □□□□□□□□HolderSet.empty() □□□□□□□□
                    HolderSet.direct(),
                    // primaryItems: □□□□□□Optional□□□□□□□□□□
                    Optional.empty(),
                    // weight: □□□□□□□□ [1, 1024]
                    10,
                    // maxLevel: □□□□ [1, 255]
                    3,
                    // minCost: □□□□□□□
                    new Cost(5, 8),     // base=5, perLevelAboveFirst=8
                    // maxCost: □□□□□□□
                    new Cost(25, 9),    // base=25, perLevelAboveFirst=9
                    // anvilCost: □□□□ [1, inf)
                    2,
                    // slots: □□□□□□□□□□□□
                    List.of(EquipmentSlotGroup.MAINHAND)
            ),

            // 3. exclusiveSet: □□□□□□HolderSet<Enchantment>□
            HolderSet.empty(),

            // 4. effects: □□□□□□□□DataComponentMap□
            // □□□□□□□□□□ DataComponentMap □□□□□□□□□□□□□□!
            //
            // 1.20.1 □□□□□□□□□□
            //   @Override public float getDamageBonus(int level, ...) { return level * 2.5f; }
            //
            // 1.21.1 □□□□□□□□□
            //   .set(EnchantmentEffectComponents.DAMAGE, ...)
            //   .set(MY_CUSTOM_COMPONENT, new MyData())
            net.minecraft.util.DataComponentMap.builder()
                    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
                    // .set(EnchantmentEffectComponents.DAMAGE, ...)
                    // □□□□□□□□□□□□
                    .set(INCREMENT.get(), new Increment(2))
```

```
                        .build()
        ));
    }

    // ===== 在 GatherDataEvent 中生成数据 =====
    // @SubscribeEvent
    // public static void gatherData(GatherDataEvent event) {
    //     DataGenerator generator = event.getGenerator();
    //     PackOutput packOutput = generator.getPackOutput();
    //     CompletableFuture<HolderLookup.Provider> lookupProvider = event.getLookupProvider();
    //
    //     generator.addProvider(event.includeServer(),
    //         new DatapackBuiltinEntriesProvider(packOutput, lookupProvider, BUILDER,
    //             Set.of(ExampleMod.MOD_ID))
    //     );
    // }

    // ===== 对应 JSON 数据文件（放在 data/examplemod/enchantment/example_enchantment.json） =====
    //
    // {
    //     "description": { "translate": "enchantment.examplemod.example_enchantment" },
    //     "supported_items": "#examplemod:enchantable/example",
    //     "weight": 10,
    //     "max_level": 3,
    //     "min_cost": { "base": 5, "per_level_above_first": 8 },
    //     "max_cost": { "base": 25, "per_level_above_first": 9 },
    //     "anvil_cost": 2,
    //     "slots": ["mainhand"],
    //     "effects": {
    //         "examplemod:increment": [
    //             { "effect": { "value": 2 } }
    //         ]
    //     }
    // }

    public static void register(IEventBus modEventBus) {
        ENCHANTMENT_COMPONENT_TYPES.register(modEventBus);
    }

    // ================================================================================
    // 附魔系统变化对比 1.20.1 vs 1.21.1 总结
    //
    // ┌──────────────────┬──────────────────────────┬──────────────────────────────────┐
    // │ 维度             │ Forge 1.20.1             │ Forge 1.21.1                     │
    // ├──────────────────┼──────────────────────────┼──────────────────────────────────┤
    // │ 注册方式         │ 延迟注册(DeferredRegister)│ 数据驱动注册(RegistrySetBuilder)│
    // │ 附魔本体         │ 继承 Enchantment         │ Enchantment record(纯数据)       │
    // │ 附魔数据         │ 硬编码                   │ 数据包(DataComponentType)        │
    // │ 物品存储         │ ItemStack NBT            │ DataComponents.ENCHANTMENTS      │
    // │ 附魔引用         │ Enchantment              │ Holder<Enchantment>              │
    // │ Enchantments.X   │ Enchantment 实例         │ ResourceKey<Enchantment>         │
    // │ 获取等级         │ getEnchantmentLevel()    │ ItemEnchantments.getLevel()      │
    // └──────────────────┴──────────────────────────┴──────────────────────────────────┘
    // ================================================================================
}
```

entity/simple_entity.java

```
// ================================================================================
// Forge 1.21.1 实体注册示例 - simple_entity.java
// MC版本: 1.21.1 | Java版本: 21 | Forge版本: 51.0.x
// ================================================================================
// 相比 Forge 1.20.1 的变化总结 - 非常大!：
// 1. 从build() 的参数从1.21+ build() 的参数 ResourceKey
//    1.20.1: .build("example_entity")字符串名
//    1.21.1: .build(ResourceKey.create(Registries.ENTITY_TYPE, ResourceLocation))
// 2. 伤害方法名从hurt() 改成了 final不可重写 hurtServer()/hurtClient()
//    1.20.1: public boolean hurt(DamageSource, float)
//    1.21.1: public boolean hurtServer(ServerLevel, DamageSource, float)
// 3. ResourceLocation.fromNamespaceAndPath() 取代 new ResourceLocation()
// 4. readAdditionalSaveData/addAdditionalSaveData 改用 CompoundTag（1.21.1）
// 5. defineSynchedData() 必须调用1.21.5+ 新增了 Builder 模式
// 不同 NeoForge 的变化更大
// 6. Forge 用 ForgeRegistries.ENTITY_TYPES，NeoForge 用 DeferredRegister.createEntities()
// 不同 Fabric 的变化更大
// 7. Fabric 用 Registry.register(Registries.ENTITY_TYPE) 静态注册
```

```
//    Forge □ DeferredRegister.create(ForgeRegistries.ENTITY_TYPES) □□□□
// ============================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.syncher.EntityDataAccessor;
import net.minecraft.network.syncher.EntityDataSerializers;
import net.minecraft.network.syncher.SynchedEntityData;
import net.minecraft.resources.ResourceKey;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.server.level.ServerLevel;
import net.minecraft.world.damagesource.DamageSource;
import net.minecraft.world.entity.Entity;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.level.Level;
import net.minecraftforge.event.entity.EntityAttributeCreationEvent;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.eventbus.api.SubscribeEvent;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□□□□ - Forge 1.21.1
 */
public class ModEntities {

    public static final DeferredRegister<EntityType<?>> ENTITIES =
            DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, ExampleMod.MODID);

    // ===== □□□□□□ =====
    // □□□□□□1.21+ build() □□□□ ResourceKey
    // 1.20.1: .build("example_entity")
    // 1.21.1: .build(ResourceKey.create(Registries.ENTITY_TYPE, ResourceLocation...))
    public static final RegistryObject<EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITIES.register("example_entity",
                    () -> EntityType.Builder
                            .<ExampleEntity>of(ExampleEntity::new, MobCategory.CREATURE)
                            .sized(1.0F, 1.0F)
                            .clientTrackingRange(10)
                            // □□□□build() □□□□ ResourceKey
                            .build(ResourceKey.create(
                                    Registries.ENTITY_TYPE,
                                    ResourceLocation.fromNamespaceAndPath(
                                            ExampleMod.MODID, "example_entity")
                            )));

    // ===== □□□□□□ =====
    // □□□□Fabric: FabricDefaultAttributeRegistry.register()
    // Forge: EntityAttributeCreationEvent □□
    public static void registerAttributes(EntityAttributeCreationEvent event) {
        var builder = net.minecraft.world.entity.ai.attributes.AttributeSupplier.builder();
        builder.add(net.minecraft.world.entity.ai.attributes.Attributes.MAX_HEALTH, 20.0D);
        builder.add(net.minecraft.world.entity.ai.attributes.Attributes.MOVEMENT_SPEED, 0.25D);
        builder.add(net.minecraft.world.entity.ai.attributes.Attributes.ATTACK_DAMAGE, 5.0D);
        event.put(EXAMPLE_ENTITY.get(), builder.build());
    }

    @Mod.EventBusSubscriber(modid = ExampleMod.MODID, bus = Mod.EventBusSubscriber.Bus.MOD)
    public static class ModBusEvents {
        @SubscribeEvent
        public static void onAttributeCreate(EntityAttributeCreationEvent event) {
            registerAttributes(event);
        }
    }

    public static void register(IEventBus modEventBus) {
        ENTITIES.register(modEventBus);
    }
}
// ============================================================================
```

```
// □□□□□□□
// □1.21+ □□□□□□□□□□□□□□□□
// 1.20.1: public boolean hurt(DamageSource, float)
// 1.21.1: hurt() □□□□□ final□□□□ hurtServer() □ hurtClient()
// ================================================================================
class ExampleEntity extends Entity {

    private static final EntityDataAccessor<Integer> DATA_VALUE =
            SynchedEntityData.defineId(ExampleEntity.class, EntityDataSerializers.INT);

    public ExampleEntity(EntityType<? extends ExampleEntity> type, Level level) {
        super(type, level);
    }

    // ===== □1.21+ □□□□□□□□ =====
    // hurt() □□□□□ final□□□□ hurtServer()
    // 1.20.1: public boolean hurt(DamageSource source, float amount)
    // 1.21.1: public boolean hurtServer(ServerLevel level, DamageSource source, float amount)
    @Override
    public boolean hurtServer(ServerLevel level, DamageSource source, float amount) {
        if (this.isInvulnerableTo(source)) {
            return false;
        }
        return super.hurtServer(level, source, amount);
    }

    // ===== □ NBT □□□1.21.1 □□ CompoundTag□ =====
    @Override
    public void readAdditionalSaveData(CompoundTag tag) {
        super.readAdditionalSaveData(tag);
        this.entityData.set(DATA_VALUE, tag.getInt("value"));
    }

    // ===== □□ NBT□1.21.1 □□ CompoundTag□ =====
    @Override
    public void addAdditionalSaveData(CompoundTag tag) {
        super.addAdditionalSaveData(tag);
        tag.putInt("value", this.entityData.get(DATA_VALUE));
    }

    // ===== □□□□□□□□1.21.1 □□□□□□1.21.5+ □□□□ =====
    @Override
    protected void defineSynchedData() {
        this.entityData.define(DATA_VALUE, 0);
    }

    @Override
    public void tick() {
        super.tick();
        // □ tick □□
    }

    public int getValue() { return this.entityData.get(DATA_VALUE); }
    public void setValue(int value) { this.entityData.set(DATA_VALUE, value); }
}
```

gui/container_screen.java

```
// ================================================================================
// Forge 1.21.1 GUI / □□□□□□□ - container_screen.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□□!□
// 1. □□□□□□□□□□ForgeRegistries.CONTAINERS -> Registries.MENU
//    1.20.1: DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID)
//    1.21.1: DeferredRegister.create(Registries.MENU, MODID)
// 2. □MenuType □□□□□□IForgeMenuType □□□
//    □□□□□ new MenuType<>(factory, FeatureFlags.DEFAULT_FLAGS)
// 3. □□□□□□□1.21+ □□ RegistryFriendlyByteBuf□□□ FriendlyByteBuf□
// 4. GuiGraphics □□□ 1.20.1 □□
// □□ NeoForge □□□□□□□
// 5. Forge □ Registries.MENU + DeferredRegister.create()
//    NeoForge □ Registries.MENU + DeferredRegister.create()
//    □□□□□□□ API □□□
// □□ Fabric □□□□□□□
// 6. Fabric □ ScreenHandlerType□Forge □ MenuType
```

```
// 7. Fabric 用 HandledScreens.register()，Forge 用 MenuScreens.register()
// ============================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraftforge.common.extensions.IForgeMenuType;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * 菜单/容器注册 - Forge 1.21.1
 */
public class ModMenus {

    // ===== 创建注册器 =====
    // 注意：1.21+ 用 Registries.MENU，旧版用 ForgeRegistries.CONTAINERS！
    // 1.20.1: ForgeRegistries.CONTAINERS
    // 1.21.1: Registries.MENU
    public static final DeferredRegister<MenuType<?>> MENU_TYPES =
            DeferredRegister.create(Registries.MENU, ExampleMod.MODID);

    // ===== 注册 MenuType =====
    // IForgeMenuType.create() 可以传递额外数据给客户端
    // 对比Fabric 用 new ScreenHandlerType<>(factory)
    public static final RegistryObject<MenuType<ExampleMenu>> EXAMPLE_MENU = MENU_TYPES.register(
            "example_menu",
            () -> IForgeMenuType.create((containerId, inventory, data) -> {
                return new ExampleMenu(containerId, inventory);
            }));

    // ===== AbstractContainerMenu 子类 =====
    public static class ExampleMenu extends AbstractContainerMenu {

        public ExampleMenu(int containerId, Inventory playerInventory) {
            super(EXAMPLE_MENU.get(), containerId);

            // 添加玩家背包（3x9）
            for (int row = 0; row < 3; row++) {
                for (int col = 0; col < 9; col++) {
                    this.addSlot(new Slot(playerInventory, 9 + row * 9 + col,
                            8 + col * 18, 84 + row * 18));
                }
            }

            // 添加玩家快捷栏（9格）
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
            }
        }

        @Override
        public ItemStack quickMoveStack(Player player, int index) {
            Slot slot = this.slots.get(index);
            if (!slot.hasItem()) return ItemStack.EMPTY;

            ItemStack original = slot.getItem().copy();
            ItemStack stack = slot.getItem();

            if (!this.moveItemStackTo(stack, 0, this.slots.size(), true)) {
                return ItemStack.EMPTY;
            }

            slot.setChanged();
            return original;
        }
```

```
        @Override
        public boolean stillValid(Player player) {
            return true;
        }
    }

    // ===== AbstractContainerScreen □□ =====
    // 1.21.1 □□ GuiGraphics □□□□ 1.20.1 □□□
    public static class ExampleScreen extends
            net.minecraft.client.gui.screens.inventory.AbstractContainerScreen<ExampleMenu> {

        public ExampleScreen(ExampleMenu menu, Inventory inventory, Component title) {
            super(menu, inventory, title);
            this.imageWidth = 176;
            this.imageHeight = 166;
            this.inventoryLabelY = this.imageHeight - 94;
        }

        @Override
        protected void renderBg(net.minecraft.client.gui.GuiGraphics graphics,
                                float partialTick, int mouseX, int mouseY) {
            int x = (this.width - this.imageWidth) / 2;
            int y = (this.height - this.imageHeight) / 2;
            graphics.blit(
                    net.minecraft.resources.ResourceLocation.fromNamespaceAndPath(
                            ExampleMod.MODID, "textures/gui/example_container.png"),
                    x, y, 0, 0, this.imageWidth, this.imageHeight);
        }

        @Override
        public void render(net.minecraft.client.gui.GuiGraphics graphics,
                           int mouseX, int mouseY, float partialTick) {
            this.renderBackground(graphics, mouseX, mouseY, partialTick);
            super.render(graphics, mouseX, mouseY, partialTick);
            this.renderTooltip(graphics, mouseX, mouseY);
        }
    }

    // ===== Screen □□ =====
    // □□□□Fabric: HandledScreens.register(type, factory)
    // Forge: MenuScreens.register(type, factory)
    // □□□□□□□□□□□
    // event.enqueueWork(() -> {
    //     MenuScreens.register(EXAMPLE_MENU.get(), ExampleScreen::new);
    // });

    public static void register(IEventBus modEventBus) {
        MENU_TYPES.register(modEventBus);
    }
}
```

item/simple_item.java

```
// ================================================================================
// Forge 1.21.1 □□□□□□ - simple_item.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□!□
// 1. □Data Components □□□1.20.5+ □□□□□□□□□□□ NBT
//    - stacksTo -> □□□□□□ DataComponents.MAX_STACK_SIZE
//    - durability -> □□□□□□ DataComponents.MAX_DAMAGE
//    - rarity -> □□□□□□ DataComponents.RARITY
//    - food -> □□□□□□ DataComponents.FOOD
// 2. □ItemStack NBT □□□□□□ getOrCreateTag()□□ stack.get/set(ComponentType)
// 3. □Food □□□□□FoodComponent -> Food□net.minecraft.world.food.Food□
// 4. CreativeModeTab.builder() □□□□□ 1.20.1 □□□
// 5. Item.Properties □□□□□□□□□.stacksTo()/.durability() □□
// □□ NeoForge □□□□□□
// 6. Forge □ ForgeRegistries.ITEMS□NeoForge □ DeferredRegister.createItems()
// 7. Forge □ RegistryObject<Item>□NeoForge □ DeferredItem<Item>
// □□ Fabric □□□□□□
// 8. Fabric □ Registry.register(Registries.ITEM) □□□□
//    Forge □ DeferredRegister.create(ForgeRegistries.ITEMS) □□□□
// ================================================================================

package com.example.examplemod.item;
```

```java
import com.example.examplemod.ExampleMod;
import net.minecraft.core.component.DataComponentType;
import net.minecraft.core.component.DataComponents;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.food.Food;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Rarity;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;
import com.mojang.serialization.Codec;

/**
 * □□□□□□ - Forge 1.21.1 Data Components □□
 */
public class ModItems {

    // □□□□□□
    // Forge 1.21.1 □□ ForgeRegistries.ITEMS
    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // □□□ Data Component □□□□□
    // □□□1.21+ □□□□□□□□ DataComponentType
    public static final DeferredRegister<DataComponentType<?>> COMPONENTS =
            DeferredRegister.create(Registries.DATA_COMPONENT_TYPE, ExampleMod.MODID);

    // ===== □□□□□□ Data Component =====
    // □□□1.21+ □□□□□□□□□□□□
    // □□□□□ ItemStack NBT
    public static final RegistryObject<DataComponentType<Integer>> CHARGE =
            COMPONENTS.register("charge",
                    () -> DataComponentType.<Integer>builder()
                            .persistent(Codec.INT)  // □□□ Codec
                            .build());

    // ===== □□□□□□□ =====
    public static final RegistryObject<Item> EXAMPLE_ITEM = ITEMS.register("example_item",
            () -> new Item(new Item.Properties()
                    .stacksTo(64)
                    // .rarity(Rarity.RARE)     // -> DataComponents.RARITY
                    // .fireResistant()          // -> DataComponents.FIRE_RESISTANT
                    // □□□□□ Item.Properties □□□□□□□□□□□□
                    // .component(CHARGE.get(), 0)
            ));

    // ===== □□□□□□□ =====
    // □□□□FoodComponent -> Food□1.21+□
    public static final RegistryObject<Item> EXAMPLE_FOOD = ITEMS.register("example_food",
            () -> new Item(new Item.Properties()
                    .food(new Food.Builder()
                            .nutrition(6)           // □□□□□ hunger()□
                            .saturationModifier(0.6f)
                            .meat()
                            .build()
                    )
            ));

    // ===== □□□□□□□□□ =====
    public static final RegistryObject<Item> EXAMPLE_DURABLE = ITEMS.register("example_durable",
            () -> new Item(new Item.Properties()
                    .durability(250)       // -> DataComponents.MAX_DAMAGE
                    .rarity(Rarity.UNCOMMON)
            ));

    // ===== □□□□□□□ =====
    // public static final RegistryObject<Item> EXAMPLE_BLOCK_ITEM = ITEMS.register("example_block",
    //         () -> new net.minecraft.world.item.BlockItem(
    //                 com.example.examplemod.block.ModBlocks.EXAMPLE_BLOCK.get(),
    //                 new Item.Properties()));

    // ===== □□□□□□□□□□□□□□□ =====
```

```
    // CreativeModeTab.builder()□□ 1.20.1 □□□
    public static final DeferredRegister<CreativeModeTab> CREATIVE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MODID);

    public static final RegistryObject<CreativeModeTab> EXAMPLE_TAB = CREATIVE_TABS.register("example_tab",
            () -> CreativeModeTab.builder()
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    .icon(() -> new ItemStack(EXAMPLE_ITEM.get()))
                    .displayItems((parameters, output) -> {
                        output.accept(EXAMPLE_ITEM.get());
                        output.accept(EXAMPLE_FOOD.get());
                        output.accept(EXAMPLE_DURABLE.get());
                    })
                    .build());

    public static void register(IEventBus modEventBus) {
        ITEMS.register(modEventBus);
        COMPONENTS.register(modEventBus);
        CREATIVE_TABS.register(modEventBus);
    }

    // ============================================================================
    // □1.21+ Data Components □□□□□
    //
    // □□□1.20.1□NBT □□□
    //   stack.getOrCreateTag().putInt("charge", 100);
    //   int charge = stack.getOrCreateTag().getInt("charge");
    //
    // □□□1.21.1□Data Components□
    //   stack.set(CHARGE.get(), 100);
    //   int charge = stack.get(CHARGE.get());  // □□ null □□□□□
    //   boolean has = stack.contains(CHARGE.get());
    //
    // □□□□Fabric □□□□□□ Data Components API□MC □□□
    // ============================================================================
}
```

item/tool_item.java

```
// ============================================================================
// Forge 1.21.1 □□/□□□□□□ - tool_item.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ============================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□!□
// 1. □Tier □□□□□□1.21+ □□□ getLevel()□int □□□□□
//    □□ getIncorrectDrops() □□ TagKey<Block>□□□□□□
//    □□□□□□□□□□□□□□□□□□□□□□□□
// 2. □□□□□□□□□□□1.21+ □□□□□□□□□□□□□□□□□□□□
//    □□ Item.Properties.attributes(XxxItem.createAttributes(tier, damage, speed))
// 3. 1.21+ □□□□□□□Item -> TieredItem -> DiggerItem -> {Axe, Hoe, Pickaxe, Shovel}
//                                          SwordItem□□□ TieredItem□
// 4. □□ DiggerItem □□□ createAttributes() □□□□
// □□ NeoForge □□□□□□
// 5. Forge □ ForgeRegistries.ITEMS□NeoForge □ DeferredRegister.createItems()
// □□ Fabric □□□□□□
// 6. Forge □ Tier □□□Fabric 1.21.1 □□ Tier □□□□□□□□
// 7. Forge □ DeferredRegister □□□Fabric □ Registry.register
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.tags.BlockTags;
import net.minecraft.tags.TagKey;
import net.minecraft.world.item.*;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraft.world.level.block.Block;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.registries.DeferredRegister;
import net.minecraftforge.registries.ForgeRegistries;
import net.minecraftforge.registries.RegistryObject;

/**
 * □□/□□□□□ - Forge 1.21.1 Tier □□
```

```
 */
public class ModTools {

    public static final DeferredRegister<Item> ITEMS =
            DeferredRegister.create(ForgeRegistries.ITEMS, ExampleMod.MODID);

    // ====================================================================
    // □□□ Tier□□□□□□
    //
    // □□□□□□□1.21+ □□□ getLevel()□int □□□□□
    // □□ getIncorrectDrops() □□ TagKey<Block>
    // □□□□□□□□□□□"□□□□□□□"□□□
    //
    // □□ 1.20.1:
    //   @Override
    //   public int getLevel() { return 3; }  // 1.21+ □□□!
    //
    // 1.21.1:
    //   @Override
    //   public TagKey<Block> getIncorrectDrops() {
    //       return TagKey.create(Registries.BLOCK,
    //           ResourceLocation.fromNamespaceAndPath(MODID, "incorrect_for_example_tool"));
    //   }
    // ====================================================================
    public static final Tier EXAMPLE_TIER = new Tier() {

        // □□□□□ TagKey<Block> □□ getLevel()
        // □□□□□□□□□BlockTags.INCORRECT_FOR_STONE_TOOL □
        // □□□□□□□□□□
        @Override
        public TagKey<Block> getIncorrectDrops() {
            return TagKey.create(
                    Registries.BLOCK,
                    ResourceLocation.fromNamespaceAndPath(
                            ExampleMod.MODID, "incorrect_for_example_tool"
                    )
            );
        }

        @Override
        public int getUses() { return 1561; }          // □□□

        @Override
        public float getSpeed() { return 8.0F; }       // □□□□

        @Override
        public float getAttackDamageBonus() { return 3.0F; } // □□□□□□

        @Override
        public int getEnchantmentValue() { return 10; } // □□□

        @Override
        public Ingredient getRepairIngredient() {
            return Ingredient.of(net.minecraft.world.item.Items.DIAMOND);
        }
    };

    // ====================================================================
    // □□□□□□□□□□□□□□ - □□□□□□□□□□□□□□□!
    //
    // 1.20.1 □□□:
    //   new SwordItem(tier, 3, -2.4f, new Item.Properties())
    //   (attackDamage □ attackSpeed □□□□□□□□□□)
    //
    // 1.21.1 □□□:
    //   new SwordItem(tier, new Item.Properties().attributes(
    //       SwordItem.createAttributes(tier, 3, -2.4f)
    //   ))
    //   (□□□□□□□ Item.Properties.attributes() □□)
    // ====================================================================

    // □□□ - □□□□□ 3, □□□□□ -2.4f
    public static final RegistryObject<SwordItem> EXAMPLE_SWORD = ITEMS.register("example_sword",
            () -> new SwordItem(EXAMPLE_TIER,
                    new Item.Properties().attributes(
                            // SwordItem.createAttributes(tier, attackDamageBonus, attackSpeedModifier)
```

```
                        SwordItem.createAttributes(EXAMPLE_TIER, 3, -2.4F)
                )
        ));

    // □□□□ - □□□□□ 1, □□□□□ -2.8f
    public static final RegistryObject<PickaxeItem> EXAMPLE_PICKAXE = ITEMS.register("example_pickaxe",
        () -> new PickaxeItem(EXAMPLE_TIER,
                new Item.Properties().attributes(
                        PickaxeItem.createAttributes(EXAMPLE_TIER, 1, -2.8F)
                )
        ));

    // □□□□ - □□□□□ 6, □□□□□ -3.2f
    public static final RegistryObject<AxeItem> EXAMPLE_AXE = ITEMS.register("example_axe",
        () -> new AxeItem(EXAMPLE_TIER,
                new Item.Properties().attributes(
                        AxeItem.createAttributes(EXAMPLE_TIER, 6, -3.2F)
                )
        ));

    // □□□□ - □□□□□ 1.5, □□□□□ -3.0f
    public static final RegistryObject<ShovelItem> EXAMPLE_SHOVEL = ITEMS.register("example_shovel",
        () -> new ShovelItem(EXAMPLE_TIER,
                new Item.Properties().attributes(
                        ShovelItem.createAttributes(EXAMPLE_TIER, 1.5F, -3.0F)
                )
        ));

    // □□□□ - □□□□□ -2, □□□□□ -1.0f
    public static final RegistryObject<HoeItem> EXAMPLE_HOE = ITEMS.register("example_hoe",
        () -> new HoeItem(EXAMPLE_TIER,
                new Item.Properties().attributes(
                        HoeItem.createAttributes(EXAMPLE_TIER, -2, -1.0F)
                )
        ));

    // ===== □□□□□□□□ =====
    // □□□□□□□□□□
    // data/examplemod/tags/block/needs_example_tool.json□□□□□□□□□□□□□
    // data/examplemod/tags/block/incorrect_for_example_tool.json□□□□□□□□□□□□□
    //
    // □□□□1.21+ □□□□□□□□□□block/□□□ blocks/□
    //         1.20.1 □□□□blocks/
    public static final TagKey<Block> NEEDS_EXAMPLE_TOOL = TagKey.create(
        Registries.BLOCK,
        ResourceLocation.fromNamespaceAndPath(ExampleMod.MODID, "needs_example_tool")
    );

    public static final TagKey<Block> INCORRECT_FOR_EXAMPLE_TOOL = TagKey.create(
        Registries.BLOCK,
        ResourceLocation.fromNamespaceAndPath(ExampleMod.MODID, "incorrect_for_example_tool")
    );

    public static void register(IEventBus modEventBus) {
       ITEMS.register(modEventBus);
    }
}
```

mixin/mixin_config.json

```
{
  "_comment_mixin_config": "================================================================================",
  "_comment_version": "Forge 1.21.1 Mixin □□ - mixin_config.json",
  "_comment_mc": "MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□",
  "_comment_diff_1": "1. compatibilityLevel □□ JAVA_21□1.20.1 □ JAVA_17□",
  "_comment_diff_2": "2. □□□□□□□□□",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□",
  "_comment_diff_3": "3. Forge □□ Mixin□□□ MANIFEST.MF □□ MixinConfigs",
  "   _comment_manifest": "   build.gradle: attributes([\"MixinConfigs\": \"examplemod.mixins.json\"])",
  "_comment_diff_4": "4. Fabric □ fabric.mod.json □ mixins □□□□",
  "_comment_diff_5": "5. Forge □ official □□□Fabric □ yarn/parchment",
  "_comment_location": "□□□□: src/main/resources/examplemod.mixins.json",
  "_comment_end": "================================================================================",

  "required": true,
```

```
  "minVersion": "0.8",
  "package": "com.example.examplemod.mixin",
  "compatibilityLevel": "JAVA_21",
  "_comment_compatibilityLevel": "□□□□1.21+ □□ JAVA_21□1.20.1 □ JAVA_17□",

  "refmap": "examplemod.refmap.json",
  "verbose": false,
  "showMessageTypes": false,

  "mixins": [
    {
      "name": "MinecraftServerMixin"
    }
  ],

  "client": [
    {
      "name": "ClientLevelMixin"
    },
    {
      "name": "GuiMixin"
    }
  ],

  "server": [
    {
      "name": "DedicatedServerMixin"
    }
  ],

  "injectors": {
    "defaultRequire": 1
  }
}
```

project/ModMain.java

```
// ================================================================================
// Forge 1.21.1 □□□□□□□□ - ModMain.java
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□!□
// 1. □Java 21!□□□□□ Java 21 □□□□□ Java 21 □□□□record□switch □□□□□□
// 2. Forge 1.21.1 □□□ net.minecraftforge.* □□□NeoForge □□ net.neoforged.*□
// 3. Forge 1.21.1 □□□ FMLJavaModLoadingContext□NeoForge □□□□□!□
// 4. 1.21+ □ Data Components □□□□□□□□□□□□□□
// 5. 1.21+ □□□□□□□□□□□□□□□□□□□□ Enchantment □□
// □□ NeoForge □□□□□□□
// 6. Forge □□ net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext
//    NeoForge □□□□□□□□□□□□□□□□□□ IEventBus
// 7. Forge □ MinecraftForge.EVENT_BUS□NeoForge □ NeoForge.EVENT_BUS
// □□ Fabric □□□□□□□
// 8. Fabric □□ ModInitializer □□□Forge □ @Mod □□ + DeferredRegister
// 9. Forge □ IEventBus □□□□□□□□Fabric □□□□□
// ================================================================================

package com.example.examplemod;

import net.minecraftforge.common.MinecraftForge;
import net.minecraftforge.eventbus.api.IEventBus;
import net.minecraftforge.fml.common.Mod;
import net.minecraftforge.fml.javafmlmod.FMLJavaModLoadingContext;

// □□□□□□□□□ @Mod □□
// □□□□Forge: net.minecraftforge.fml.common.Mod
//         NeoForge: net.neoforged.fml.common.Mod
//         Fabric: □□□□□□□ fabric.mod.json □ entrypoint □□
@Mod(ExampleMod.MODID)
public class ExampleMod {

    public static final String MODID = "examplemod";

    // □□□□□□□
    // □□□□Forge 1.21.1 □□ FMLJavaModLoadingContext.get().getModEventBus()
    //         NeoForge 1.21.1 □□□□□□□□□□□ IEventBus:
    //         public ExampleMod(IEventBus modEventBus, ModContainer modContainer)
```

```
    public ExampleMod() {
        IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();

        // ===== □□ DeferredRegister =====
        // ModBlocks.BLOCKS.register(modEventBus);
        // ModItems.ITEMS.register(modEventBus);
        // ModItems.CREATIVE_TABS.register(modEventBus);
        // ModEntities.ENTITIES.register(modEventBus);
        // ModMenus.MENU_TYPES.register(modEventBus);
        // ModEnchantments.ENCHANTMENT_COMPONENT_TYPES.register(modEventBus);

        modEventBus.addListener(this::commonSetup);

        // □□□ FORGE □□□□
        // □□□□Forge: MinecraftForge.EVENT_BUS
        //         NeoForge: NeoForge.EVENT_BUS
        MinecraftForge.EVENT_BUS.register(this);
    }

    private void commonSetup(final net.minecraftforge.fml.event.lifecycle.FMLCommonSetupEvent event) {
        // event.enqueueWork(() -> { ... });
    }
}
```

project/build.gradle

```
// ================================================================================
// Forge 1.21.1 □□□□□□□ - build.gradle
// MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
// ================================================================================
// □□ Forge 1.20.1 □□□□□□ - □□□□!□
// 1. □Java 21!□1.21+ □□□□ Java 21□1.20.1 □ Java 17□
// 2. Forge □□: 1.21.1 □ 51.x□1.20.1 □ 47.x□
// 3. ForgeGradle □□ 6.0.x □□□NeoForge □ ModDevGradle□
// 4. 1.21+ □□□ Data Components □□□□□□□□□□□□
// 5. 1.21+ □□□□□□□□□□□□□□□□□□ Enchantment □□
// 6. 1.21+ Tier □□□□□□□ TagKey<Block> □□ getLevel()□
// 7. 1.21+ ResourceLocation □□□□□□□□□□ fromNamespaceAndPath()
// □□ NeoForge □□□□□□□
// 8. Forge □ ForgeGradle □□□NeoForge □ ModDevGradle
// 9. Forge □□ mods.toml□NeoForge □ neoforge.mods.toml
// 10. Forge □□ net.minecraftforge.* □□□NeoForge □ net.neoforged.*
// □□ Fabric □□□□□□□
// 11. Forge □ ForgeGradle□□ Loom□□□□□□□□□
// ================================================================================

plugins {
    id 'eclipse'
    id 'idea'
    id 'maven-publish'
    id 'net.minecraftforge.gradle' version '6.0.24'
}

// --------------------------------------------------------------------------
// □□□□□□□□
// --------------------------------------------------------------------------
version = '1.0.0'
group = 'com.example.examplemod'

// □□□□□□□1.21+ □□□□□ Java 21!
// Forge 1.20.1 □ Java 17□1.21.1 □□□ Java 21
java.toolchain.languageVersion = JavaLanguageVersion.of(21)

// --------------------------------------------------------------------------
// Minecraft / Forge □□□□
// --------------------------------------------------------------------------
minecraft {
    // 1.21.1 □□□
    mappings channel: 'official', version: '1.21.1'
    // □□□□□ Parchment□
    // mappings channel: 'parchment', version: '2024.07.28-1.21.1'

    runs {
        client {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
```

```
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod { source sourceSets.main }
            }
        }

        server {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            mods {
                examplemod { source sourceSets.main }
            }
        }

        data {
            workingDirectory project.file('run')
            property 'forge.logging.markers', 'REGISTRIES'
            property 'forge.logging.console.level', 'debug'
            args '--mod', 'examplemod', '--all', '--output', file('src/generated/resources/'),
                    '--existing', file('src/main/resources/')
            mods {
                examplemod { source sourceSets.main }
            }
        }
    }
}

repositories {
    maven { url = 'https://maven.minecraftforge.net/' }
    maven { url = 'https://maven.parchmentmc.org' }
    maven { url = 'https://maven.blamejared.com' }
    mavenCentral()
}

dependencies {
    // Minecraft + Forge □□□□□1.21.1-51.0.x
    minecraft 'net.minecraftforge:forge:1.21.1-51.0.24'

    // JEI□1.21.1 □□□
    // compileOnly(fg.deobf("mezz.jei:jei-1.21.1-common-api:19.x.x.x:api"))
    // runtimeOnly(fg.deobf("mezz.jei:jei-1.21.1-forge:19.x.x.x"))
}

sourceSets.main.resources { srcDir 'src/generated/resources' }

jar {
    manifest {
        attributes([
            "Specification-Title"     : "examplemod",
            "Specification-Vendor"     : "examplemod",
            "Specification-Version"   : "1",
            "Implementation-Title"    : project.name,
            "Implementation-Version"  : project.jar.archiveVersion,
            "Implementation-Vendor"    : "examplemod",
            "MixinConfigs"            : "examplemod.mixins.json"
        ])
    }
}

tasks.named('processResources', ProcessResources).configure {
    var replaceProperties = [
            minecraft_version   : "1.21.1",
            minecraft_version_range: "[1.21.1,1.21.2)",
            forge_version       : "51.0.24",
            forge_version_range : "[51,)",
            loader_version_range: "[51,)",
            mod_id              : "examplemod",
            mod_name            : "Example Mod",
            mod_license         : "MIT",
            mod_version         : "1.0.0",
            mod_authors         : "Author",
            mod_description     : "Example mod for Forge 1.21.1"
    ]
    inputs.properties replaceProperties
    filesMatching(['META-INF/mods.toml', 'pack.mcmeta']) {
```

```
        expand replaceProperties + [project: project]
    }
}

// Java 21 □□□□
tasks.withType(JavaCompile).configureEach {
    options.encoding = 'UTF-8'
    options.release = 21
}
```

project/mods.toml

```
# ======================================================================================
# Forge 1.21.1 □□□□□□□□ - mods.toml
# MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x
# ======================================================================================
# □□ Forge 1.20.1 □□□□□□□
# 1. Forge 1.21.1 □□□ mods.toml□NeoForge □□ neoforge.mods.toml□
# 2. □□□□□□□□□1.21.1 □ [1.21.1,1.21.2)
# 3. 1.21.1 □ Forge □□□□□ [51,)
# 4. □□□ 1.20.1 □□□□
# □□ NeoForge □□□□□□□
# 5. Forge □ mods.toml□NeoForge □ neoforge.mods.toml
# 6. Forge □□□ modId □ "forge"□NeoForge □ "neoforge"
# □□ Fabric □□□□□□□
# 7. Forge □ mods.toml□TOML□□Fabric □ fabric.mod.json□JSON□
# ======================================================================================

modLoader="javafmloader"
loaderVersion="${loader_version_range}"
license="${mod_license}"

# --------------------------------------------------------------------------------------
# □□□□□
# --------------------------------------------------------------------------------------
[[mods]]
modId="${mod_id}"
version="${mod_version}"
displayName="${mod_name}"
authors="${mod_authors}"
description='''
${mod_description}
'''

# --------------------------------------------------------------------------------------
# Mixin □□□□□□Forge □ mods.toml □□□□
# □□□□□NeoForge □ [[mixins]] □□Forge □ MixinConfigs manifest □□
# --------------------------------------------------------------------------------------
#[[mixins]]
#config="${mod_id}.mixins.json"

# --------------------------------------------------------------------------------------
# □□□□□□
# --------------------------------------------------------------------------------------
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

[[dependencies.${mod_id}]]
    modId="forge"
    type="required"
    versionRange="${forge_version_range}"
    ordering="NONE"
    side="BOTH"

# □□□□□□□
# [[dependencies.${mod_id}]]
#     modId="jei"
#     type="optional"
#     versionRange="[19.0.0,)"
#     ordering="NONE"
#     side="BOTH"
```

recipe/shaped_recipe.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.21.1 □□□□□□ - shaped_recipe.json",
  "_comment_mc": "MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□",
  "_comment_diff_1": "1. □□□□□□□□1.21+ □□□□□□□□□□□□□□",
  "_comment_diff_2": "   data/<modid>/recipes/ -> data/<modid>/recipe/",
  "_comment_diff_3": "   □□□tags/blocks/ -> tags/block/, loot_tables/ -> loot_table/",
  "_comment_diff_4": "2. □result □□□□□result □□ \"item\" □□□□ \"id\"",
  "_comment_diff_5": "   1.20.1: \"result\": {\"item\": \"examplemod:example_item\", \"count\": 1}",
  "_comment_diff_6": "   1.21.1: \"result\": {\"id\": \"examplemod:example_item\", \"count\": 1}",
  "_comment_diff_7": "3. □Data Components□1.21+ □□ result □□□ components □□",
  "_comment_fabric_diff": "□□ Fabric □□□□Fabric □□□□□□□□□MC □□□□□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/recipe/example_shaped.json",
  "_comment_end": "================================================================================",

  "type": "minecraft:crafting_shaped",

  "pattern": [
    "III",
    "I I",
    "III"
  ],

  "key": {
    "I": {
      "tag": "forge:ingots/iron"
    }
  },

  "result": {
    "id": "examplemod:example_item",
    "count": 1,
    "_comment_components": "□□□1.21+ □□□□□□□□ Data Components",
    "components": {
      "_comment_example": "□□□□□□□□□□",
      "examplemod:charge": 10
    }
  },

  "conditions": [
    {
      "type": "forge:mod_loaded",
      "modid": "examplemod"
    }
  ]
}
```

tag/block_tag.json

```
{
  "_comment": "================================================================================",
  "_comment_version": "Forge 1.21.1 □□□□□□ - block_tag.json",
  "_comment_mc": "MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□",
  "_comment_diff_1": "1. □□□□□□□□1.21+ □□□□□□□□□□□□□□",
  "_comment_diff_2": "   data/<modid>/tags/blocks/ -> data/<modid>/tags/block/",
  "_comment_diff_3": "2. JSON □□□□□□",
  "_comment_diff_4": "3. Forge □□ forge: □□□□□NeoForge □ c: □ neoforge:□",
  "_comment_fabric_diff": "□□ Fabric □□□□Fabric □□□□□□□□□□ forge: □□□□",
  "_comment_location": "□□□□: src/main/resources/data/<tag_namespace>/tags/block/<tag_name>.json",
  "_comment_end": "================================================================================",

  "replace": false,

  "values": [
    "examplemod:example_block",
    "minecraft:stone",
    {
      "#": "minecraft:mineable/pickaxe",
      "required": false
    },
    {
      "id": "othermod:other_block",
```

```
      "required": false
    }
  ]
}
```

worldgen/biome_modifier.json

```
{
  "_comment_biome_modifier": "================================================================================",
  "_comment_version": "Forge 1.21.1 □□□□□□□□□□ - biome_modifier.json",
  "_comment_mc": "MC□□: 1.21.1 | Java□□: 21 | Forge□□: 51.0.x",
  "_comment_diff": "□□ Forge 1.20.1 □□□□□□",
  "_comment_diff_1": "1. Forge 1.21.1 □□ forge: □□□NeoForge □□ neoforge:□",
  "_comment_diff_2": "2. JSON □□□ 1.20.1 □□",
  "_comment_diff_3": "3. □□□□□□ data/<modid>/forge/biome_modifier/",
  "_comment_forge_vs_fabric": "□□ Fabric □□□□",
  "_comment_diff_4": "4. Forge □ biome_modifier JSON□□□□□□",
  "_comment_diff_5": "5. Fabric □ BiomeModification API□□□□□□",
  "_comment_location": "□□□□: src/main/resources/data/examplemod/forge/biome_modifier/example_modifier.json",
  "_comment_end": "================================================================================",

  "type": "forge:add_features",
  "_comment_type": "Forge 1.21.1 □□ forge: □□□NeoForge □ neoforge:□",

  "biomes": "#minecraft:is_overworld",

  "features": [
    "examplemod:example_ore_placed"
  ],

  "step": "underground_ores"
}
```

NeoForge 模组模板

NeoForge 1.20.4

block/simple_block.java

```
// ========================================================================
// NeoForge 1.20.4 □□□□□□□ (simple_block.java)
// NeoForge 1.20.4, Java 17
// ========================================================================
//
// □□ NeoForge 1.21.1 □□□□□□□
// 1. □BlockEntity □□/□□□□□□□□!□
//    NeoForge 1.20.4: saveAdditional(CompoundTag tag) - □ HolderLookup.Provider □□
//                     load(CompoundTag tag) - □□□□ load, □□ loadAdditional!
//    NeoForge 1.21.1: saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
//                     loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
//                     getUpdateTag(HolderLookup.Provider registries)
//
// 2. □ResourceLocation □□□□□□!□
//    NeoForge 1.20.4: new ResourceLocation("modid", "path")  (□□□□□□)
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//                     (□□□□□□□□, □□□□□□□)
//
// 3. □Block □□□□□□□□!□
//    NeoForge 1.20.4: □ MapCodec BlockType □□, Block □□□□□ codec()
//    NeoForge 1.21.1: □□□ MapCodec □□□□□□□, □□ codec() □□
//
// 4. □DeferredRegister.Blocks □□□□□
//    NeoForge 1.20.4: register() □□ DeferredBlock<T> (□□ DeferredHolder<Block, T>)
//                     createBlocks(MODID) □□□
//    NeoForge 1.21.1: □□, □ DeferredBlock □□□□□□□□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □DeferredRegister.Blocks □□□□
//    Forge 1.20.1:  □□□□, □□ DeferredRegister.create(ForgeRegistries.BLOCKS, MODID)
//    NeoForge 1.20.4: □□ DeferredRegister.Blocks, □□ DeferredRegister.createBlocks(MODID)
//                     □□ DeferredBlock<T> (□□ RegistryObject<Block>)
//
// 3. □□□□□□
//    Forge 1.20.1:  □□□□□□ ForgeRegistries □ (□ BLOCK_ENTITY_TYPES)
//    NeoForge 1.20.4: □□□□ net.minecraft.core.registries.Registries
//                     □ net.minecraft.core.registries.BuiltInRegistries
//
// 4. □RegistryObject vs DeferredHolder□
//    Forge 1.20.1:  □□ RegistryObject<T>
//    NeoForge 1.20.4: □□ DeferredHolder<R, T> □ DeferredBlock<T>
//
// 5. □BlockBehaviour.Properties□
//    1.20.x □□□□□□ AbstractBlock.Properties □□□□ BlockBehaviour.Properties
//    (□□□□□ 1.20.x □□□□□□, NeoForge □ Forge □□)
// ========================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.registries.BuiltInRegistries;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.protocol.Packet;
import net.minecraft.network.protocol.game.ClientGamePacketListener;
import net.minecraft.network.protocol.game.ClientboundBlockEntityDataPacket;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.world.level.Level;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.EntityBlock;
import net.minecraft.world.level.block.entity.BlockEntity;
```

```
import net.minecraft.world.level.block.entity.BlockEntityTicker;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.block.state.BlockState;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.MapColor;
import net.minecraft.sounds.SoundType;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;
import net.neoforged.neoforge.registries.DeferredBlock;

import javax.annotation.Nullable;

// ============================================================================
// □□□□□
// ============================================================================
public class ModBlocks {

    // ========================================================================
    // □□□□□□□ DeferredRegister.Blocks □□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCKS, MODID)
    //                □□ DeferredRegister<Block>, register() □□ RegistryObject<Block>
    // NeoForge 1.20.4: DeferredRegister.createBlocks(MODID)
    //                  □□ DeferredRegister.Blocks, register() □□ DeferredBlock<T>
    //
    // DeferredBlock<T> □□□ DeferredHolder<Block, T>, □□□□□□□□□□□
    // (□ NeoForge 1.21.1 □□□□□)
    // ========================================================================
    public static final DeferredRegister.Blocks BLOCKS =
            DeferredRegister.createBlocks(ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, MODID)
    // NeoForge 1.20.4: DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, MODID)
    //
    // □□□□□□□□□□:
    //   Forge: ForgeRegistries (Forge □□□□□□□)
    //   NeoForge: Registries / BuiltInRegistries (□□□□□□)
    //
    // □□□□□□□□□□: BLOCK_ENTITY_TYPE (□□ BLOCK_ENTITIES)
    // ========================================================================
    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITY_TYPES =
            DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□
    // □□□□□□□□□:
    //   Forge 1.20.1: RegistryObject<Block>
    //   NeoForge 1.20.4: DeferredBlock<Block>
    //
    // BlockBehaviour.Properties.of() □□□□□□□ (1.20.x □□□□□□)
    // - .mapColor() □□□□□□□ (1.20.x □□□□□□□ .material □□ .mapColor)
    // - .destroyTime() □□□□
    // - .explosionResistance() □□□□
    // - .sound() □□□□
    // - .lightLevel() □□□□ (□□□)
    // ========================================================================
    public static final DeferredBlock<Block> EXAMPLE_BLOCK = BLOCKS.register(
            "example_block",
            () -> new Block(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.STONE)
                    .destroyTime(2.0f)
                    .explosionResistance(10.0f)
                    .sound(SoundType.GRAVEL)
                    .lightLevel(state -> 7)
            )
    );

    // ========================================================================
    // □□□□□□□□□□□□□ (EntityBlock)
    // ========================================================================
    public static final DeferredBlock<ExampleEntityBlock> EXAMPLE_ENTITY_BLOCK = BLOCKS.register(
            "example_entity_block",
            () -> new ExampleEntityBlock(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.METAL)
                    .destroyTime(3.0f)
```

```
                    .explosionResistance(15.0f)
                    .sound(SoundType.METAL)
            )
    );

    // ========================================================================
    // 方块实体类型注册
    // 使用 BlockEntityType.Builder.of(supplier, blocks...).build(null)
    //
    // 注意：这是 Forge 1.20.1 的写法。
    //       在 NeoForge 中使用 BlockEntityType.BuilderDelegating 代替。
    //
    // build(null) - 传入 null 表示不使用 DataFixer
    // ========================================================================
    public static final DeferredHolder<BlockEntityType<?>, BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
            BLOCK_ENTITY_TYPES.register(
                    "example_block_entity",
                    () -> BlockEntityType.Builder.of(
                            ExampleBlockEntity::new,
                            EXAMPLE_ENTITY_BLOCK.get()
                    )
                    .build(null)
            );
}

// ========================================================================
// 方块实体类示例
// 实现 EntityBlock 接口, 提供 newBlockEntity() 方法
// ========================================================================
class ExampleEntityBlock extends Block implements EntityBlock {

    public ExampleEntityBlock(BlockBehaviour.Properties properties) {
        super(properties);
    }

    // 创建方块实体实例
    @Override
    public BlockEntity newBlockEntity(BlockPos pos, BlockState state) {
        return new ExampleBlockEntity(pos, state);
    }

    // ========================================================================
    // 获取 Ticker (用于 tick 处理器)
    // ========================================================================
    @Nullable
    @Override
    public <T extends BlockEntity> BlockEntityTicker<T> getTicker(
            Level level, BlockState state, BlockEntityType<T> type) {
        return type == ModBlocks.EXAMPLE_BLOCK_ENTITY.get()
                ? (lvl, pos, st, be) -> ExampleBlockEntity.tick(lvl, pos, st, (ExampleBlockEntity) be)
                : null;
    }

    // ========================================================================
    // 编解码器（1.20.4 的 Block 必须有!
    //
    // NeoForge 1.21.1 中必须有 codec() 方法返回 BlockType:
    //   @Override
    //   public MapCodec<? extends Block> codec() {
    //       return CODEC.value();
    //   }
    //
    // NeoForge 1.20.4 不需要! Block 类有 codec() 方法
    // ========================================================================
}

// ========================================================================
// 使用说明
// 创建方块实体（1.20.4 版本/数据驱动）!
//
// NeoForge 1.20.4 (与 Forge 1.20.1 相同):
//   saveAdditional(CompoundTag tag) - 用 HolderLookup.Provider 参数
//   load(CompoundTag tag) - 不存在 load (使用 loadAdditional!)
//   getUpdateTag() - 不变
//
// NeoForge 1.21.1:
```

```
//   saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
//   loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
//   getUpdateTag(HolderLookup.Provider registries)
// ============================================================================
class ExampleBlockEntity extends BlockEntity {

    private int value = 0;

    public ExampleBlockEntity(BlockPos pos, BlockState state) {
        super(ModBlocks.EXAMPLE_BLOCK_ENTITY.get(), pos, state);
    }

    // ========================================================================
    // □□□□□ (□ NBT)
    // □□□□□1.20.4: □□□□□ load, □ HolderLookup.Provider □□
    //         1.21.1: □□□□□ loadAdditional, □ HolderLookup.Provider □□
    //
    // Forge 1.20.1: public void load(CompoundTag tag)
    // NeoForge 1.20.4: public void load(CompoundTag tag) - □□!
    // NeoForge 1.21.1: public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
    // ========================================================================
    @Override
    public void load(CompoundTag tag) {
        super.load(tag);
        this.value = tag.getInt("value");
    }

    // ========================================================================
    // □□□□□ (□ NBT)
    // □□□□□1.20.4: □ HolderLookup.Provider □□
    //         1.21.1: □ HolderLookup.Provider □□
    //
    // Forge 1.20.1: public void saveAdditional(CompoundTag tag)
    // NeoForge 1.20.4: public void saveAdditional(CompoundTag tag) - □□!
    // NeoForge 1.21.1: public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
    // ========================================================================
    @Override
    public void saveAdditional(CompoundTag tag) {
        super.saveAdditional(tag);
        tag.putInt("value", this.value);
    }

    // ========================================================================
    // □□□□□ - □□□□□□
    // □□□□□1.20.4: getUpdateTag() □□□
    //         1.21.1: getUpdateTag(HolderLookup.Provider registries) □□□
    // ========================================================================
    @Override
    public CompoundTag getUpdateTag() {
        CompoundTag tag = new CompoundTag();
        saveAdditional(tag);
        return tag;
    }

    // □□□□□ - □□□□□□
    @Override
    public Packet<ClientGamePacketListener> getUpdatePacket() {
        return ClientboundBlockEntityDataPacket.create(this);
    }

    // ========================================================================
    // □ tick □□□□□
    // ========================================================================
    public static void tick(Level level, BlockPos pos, BlockState state, ExampleBlockEntity blockEntity) {
        if (!level.isClientSide) {
            // □□□□□□ (□□: □□□□□□□)
            // blockEntity.value++;
            // blockEntity.setChanged(); // □□□□□□□, □□□□□
        }
    }
}
```

command/simple_command.java

```
// ============================================================================
// NeoForge 1.20.4 □□□□□□□ (simple_command.java)
```

```
// NeoForge 1.20.4, Java 17
// ================================================================================
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□
// 1. □□□□□□□□□□
//    NeoForge 1.20.4: RegisterCommandsEvent (net.neoforged.neoforge.event.RegisterCommandsEvent)
//    NeoForge 1.21.1: RegisterCommandsEvent (net.neoforged.neoforge.event.RegisterCommandsEvent)
//    (□□□□□□□□□, 1.20.4 □ 1.21.1 □□□□□□)
//
// 2. □@Mod.EventBusSubscriber □ bus □□□
//    NeoForge 1.20.4: bus = Mod.EventBusSubscriber.Bus.FORGE (□□□□□□□)
//                     □□□□1.20.4 □□□ Bus.FORGE!
//    NeoForge 1.21.1: bus = Mod.EventBusSubscriber.Bus.GAME (□□□□□□□)
//                      Bus.FORGE □□□□□□ Bus.GAME!
//
// 3. □Brigadier API□
//    1.20.4 □ 1.21.1 □ Brigadier API □□□□□□□□□□
//    □□□□ Commands.literal(), Commands.argument() □□
//    CommandSourceStack □□□□□□□□□□□□
//
// 4. □sendSuccess □□□
//    1.20.4 □ 1.21.1 □□□ Supplier<Component> □ boolean □□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□□□
//    Forge 1.20.1:   net.minecraftforge.event.RegisterCommandsEvent
//    NeoForge 1.20.4: net.neoforged.neoforge.event.RegisterCommandsEvent
//    (□□□□□□□□□□□)
//
// 2. □@Mod □□□ @SubscribeEvent □□□
//    Forge 1.20.1:   net.minecraftforge.fml.common.Mod
//                    net.minecraftforge.eventbus.api.SubscribeEvent
//    NeoForge 1.20.4: net.neoforged.fml.common.Mod
//                     net.neoforged.bus.api.SubscribeEvent
//
// 3. □□□□□□□□□
//    Forge 1.20.1:   MinecraftForge.EVENT_BUS
//    NeoForge 1.20.4: NeoForge.EVENT_BUS (net.neoforged.neoforge.NeoForge)
//
// 4. □@Mod.EventBusSubscriber □ bus □□□
//    Forge 1.20.1:   Bus.FORGE
//    NeoForge 1.20.4: Bus.FORGE (□□!)
//    NeoForge 1.21.1: Bus.GAME (□□□!)
// ================================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.arguments.StringArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.commands.arguments.coordinates.BlockPosArgument;
import net.minecraft.core.BlockPos;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.Entity;
import net.neoforged.bus.api.SubscribeEvent;
import net.neoforged.fml.common.Mod;
import net.neoforged.neoforge.event.RegisterCommandsEvent;

// ================================================================================
// □□□□□
// ================================================================================
// □□□□RegisterCommandsEvent □ NeoForge □□□□□□□□□□
// □□ @Mod.EventBusSubscriber □□□□
//
// □□□□NeoForge 1.20.4: bus = Mod.EventBusSubscriber.Bus.FORGE (□□ FORGE!)
//         NeoForge 1.21.1: bus = Mod.EventBusSubscriber.Bus.GAME (□□□ GAME!)
//         Forge 1.20.1:    bus = Mod.EventBusSubscriber.Bus.FORGE (□□)
@Mod.EventBusSubscriber(modid = ExampleMod.MOD_ID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {
    // ================================================================================
```

```
// □□□□□ - □□ RegisterCommandsEvent
//
// □□□□□RegisterCommandsEvent □ NeoForge □□□□□□□□□□
// (NeoForge.EVENT_BUS), □□□□□□□□□□ (modEventBus)
//
// □□□□□:
//   @Mod.EventBusSubscriber(modid = MODID, bus = Bus.FORGE)
//   + @SubscribeEvent □□□□□
//   □□□□□□□□□□□□: NeoForge.EVENT_BUS.register(new ModCommands());
//
// □□□□□□□□□□:
//   Forge 1.20.1: MinecraftForge.EVENT_BUS
//   NeoForge 1.20.4: NeoForge.EVENT_BUS (net.neoforged.neoforge.NeoForge)
// ================================================================================
@SubscribeEvent
public static void register(RegisterCommandsEvent event) {
    // ============================================================================
    // □□1: /examplemod hello
    // □□□□□, □□□□, □□□□□ 0 (□□□□□)
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(0))
                    .then(Commands.literal("hello")
                            .executes(ModCommands::helloCommand)
                    )
                    .executes(context -> {
                        context.getSource().sendSuccess(
                                () -> Component.literal("ExampleMod command!"),
                                false
                        );
                        return Command.SINGLE_SUCCESS;
                    })
    );

    // ============================================================================
    // □□2: /examplemod setvalue <player> <value>
    // □□□□□□□□, □□□□□ 2 (OP)
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(2))
                    .then(Commands.literal("setvalue")
                            .then(Commands.argument("target", EntityArgument.player())
                                    .then(Commands.argument("value", IntegerArgumentType.integer(0, 100))
                                            .executes(ModCommands::setValueCommand)
                                    )
                            )
                    )
    );

    // ============================================================================
    // □□3: /examplemod message <message...>
    // □□□□□□□□ (□□□□□)
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(0))
                    .then(Commands.literal("message")
                            .then(Commands.argument("message", StringArgumentType.greedyString())
                                    .executes(context -> {
                                        String msg = StringArgumentType.getString(context, "message");
                                        context.getSource().sendSuccess(
                                                () -> Component.literal("Message: " + msg),
                                                false
                                        );
                                        return Command.SINGLE_SUCCESS;
                                    })
                            )
                    )
    );

    // ============================================================================
    // □□4: /examplemod teleport <position>
    // □□□□□□□, □□□□□ 2 (OP)
    // ============================================================================
```

```
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        .requires(src -> src.hasPermission(2))
                        .then(Commands.literal("teleport")
                                .then(Commands.argument("pos", BlockPosArgument.blockPos())
                                        .executes(context -> {
                                            BlockPos pos = BlockPosArgument.getBlockPos(context, "pos");
                                            Entity executor = context.getSource().getEntityOrPlayer();
                                            if (executor != null) {
                                                executor.teleportTo(
                                                        pos.getX() + 0.5,
                                                        pos.getY(),
                                                        pos.getZ() + 0.5
                                                );
                                                context.getSource().sendSuccess(
                                                        () -> Component.literal(
                                                                "Teleported to " + pos.toShortString()
                                                        ),
                                                        true
                                                );
                                            }
                                            return Command.SINGLE_SUCCESS;
                                        })
                                )
                        )
        );
    }

    // ===========================================================================
    // hello □□□□□□□□
    // ===========================================================================
    private static int helloCommand(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();

        // sendSuccess □□ Supplier<Component> (□□□□, □□□□)
        // 1.20.4 □ 1.21.1 □□□□
        source.sendSuccess(
                () -> Component.literal("Hello from ExampleMod!"),
                false  // false = □□□□□□ OP □□□
        );

        // □□□□□ (□□□□□□□□□□□)
        Entity entity = source.getEntity();
        if (entity != null) {
            source.sendSuccess(
                    () -> Component.literal("Executed by: " + entity.getName().getString()),
                    false
            );
        }

        return Command.SINGLE_SUCCESS;
    }

    // ===========================================================================
    // setvalue □□□□□□□□
    // ===========================================================================
    private static int setValueCommand(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        CommandSourceStack source = context.getSource();

        Entity target = EntityArgument.getPlayer(context, "target");
        int value = IntegerArgumentType.getInteger(context, "value");

        source.sendSuccess(
                () -> Component.literal(
                        "Set " + target.getName().getString() + "'s value to " + value
                ),
                true  // true = □□□ OP □□
        );

        return value;
    }
}
```

enchantment/custom_enchantment.java

```
// ================================================================================
// NeoForge 1.20.4 □□□□□□□□ (custom_enchantment.java)
// NeoForge 1.20.4, Java 17
// ================================================================================
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□□□□□!□
//
// 1. □□□□□□□□□□□□□□!□
//    NeoForge 1.20.4: □□ DeferredRegister □□□□□ (□□□□□!)
//                     DeferredRegister.create(Registries.ENCHANTMENT, MODID)
//                     □□□□□ Enchantment □□□□□□□□□□
//                     □ Forge 1.20.1 □□□□□!
//    NeoForge 1.21.1: □□□□□□□□□□□ (datapack registry)!
//                     □□□□ DeferredRegister □□!
//                     □□ RegistrySetBuilder □□□□□□□□□□□□
//                     □□□□□□□□□□, □□□□□□□□□□ (DataComponentType) □□
//
// 2. □□□□□□□□□□□□□□!□
//    NeoForge 1.20.4: □□□□□ Enchantment □□□□□□□□□□□□□:
//                     - getMaxLevel()
//                     - getMinCost() / getMaxCost()
//                     - checkCompatibility()
//                     - canEnchant()
//                     - doPostAttack() □
//                     □ Forge 1.20.1 □□□□□!
//    NeoForge 1.21.1: □□□□□□□□□□□□ (Enchantment Effect Components) □□:
//                     - □□□□□□□□, □□□□□ DataComponentMap □□□□□
//                     - □□ EnchantmentEffectComponents (□ DAMAGE, KNOCKBACK □)
//                     - □□□□□□□□□□□□□□□□□
//
// 3. □□□□□□□□□□□!□
//    NeoForge 1.20.4: □□□□□□ ItemStack NBT □ (Enchantments tag)
//                     □ Forge 1.20.1 □□!
//    NeoForge 1.21.1: □□□□□□ DataComponents.ENCHANTMENTS □□□□□□
//                     (1.20.5+ □□□ Data Components)
//
// 4. □Enchantments □□□□□□□□□□!□
//    NeoForge 1.20.4: Enchantments.SHARPNESS □□ Enchantment □□
//    NeoForge 1.21.1: Enchantments.SHARPNESS □□ ResourceKey<Enchantment>
//
// 5. □EnchantmentHelper □□□
//    NeoForge 1.20.4: EnchantmentHelper.getItemEnchantmentLevel(enchantment, stack)
//                     □□□□□□□□□□
//    NeoForge 1.21.1: EnchantmentHelper.runIterationOnItem(stack, visitor)
//                     □□□□□□□□□□, □□□□□□□□□□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □□□□□□□□
//    Forge 1.20.1:  ForgeRegistries.ENCHANTMENTS
//    NeoForge 1.20.4: Registries.ENCHANTMENT (□□□□□□)
//                     □ BuiltInRegistries.ENCHANTMENT
//
// 3. □□□□□□□
//    Forge 1.20.1:  RegistryObject<Enchantment>
//    NeoForge 1.20.4: DeferredHolder<Enchantment, Enchantment> □ Supplier<Enchantment>
//
// 4. □□□ API □□□□□!□
//    Forge 1.20.1 □ NeoForge 1.20.4 □□□ API □□□□:
//    - Enchantment □: Rarity, EnchantmentCategory, EquipmentSlot[]
//    - □□□□□: getMaxLevel, getMinCost, getMaxCost, checkCompatibility, etc.
//    - □□□□□: doPostAttack, doPostHurt □
// ================================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.entity.LivingEntity;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
```

```java
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

import java.util.function.Supplier;

// ============================================================================
// □□□□□□
// ============================================================================
public class ModEnchantments {

    // ========================================================================
    // □□□□□□□□□□□□ - □□□□□!
    //
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID)
    // NeoForge 1.20.4: DeferredRegister.create(Registries.ENCHANTMENT, MODID)
    //
    // □□□□□NeoForge 1.21.1:
    //   □□□□□□□□□□□□, □□□□ DeferredRegister □□!
    //   □□□□□ RegistrySetBuilder + □□□□□ (□□ JSON □□)
    //   □□ ResourceKey<Enchantment> □□□□□
    //
    // □□□□□□□□□□□□:
    //   Forge: ForgeRegistries.ENCHANTMENTS (Forge □□□□□□□□□)
    //   NeoForge: Registries.ENCHANTMENT (□□□□□□□□)
    // ========================================================================
    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(Registries.ENCHANTMENT, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□
    //
    // Enchantment □□□□□ (1.20.4 □□ API, □ Forge 1.20.1 □□):
    //   - Rarity rarity: □□□□ (COMMON=10 / UNCOMMON=5 / RARE=2 / VERY_RARE=1)
    //   - EnchantmentCategory category: □□□□□□□□□
    //   - EquipmentSlot[] slots: □□□□□□□□
    //
    // □□□□□NeoForge 1.21.1 □:
    //   □□□□□□□□□□□□□□□□!
    //   □□□□□ new Enchantment(description, definition, exclusiveSet, effects)
    //   □□ EnchantmentDefinition □ DataComponentMap □□
    //
    // EnchantmentCategory □□□:
    //   - ALL: □□□□□
    //   - ARMOR: □□/□□/□□/□□
    //   - ARMOR_FEET / ARMOR_LEGS / ARMOR_CHEST / ARMOR_HEAD: □□□□□
    //   - WEAPON: □ (□□□□□)
    //   - FISHING_ROD: □□□
    //   - BREAKABLE: □□□□□ (□□)
    //   - DIGGER: □/□/□/□
    //   - BOW: □
    //   - CROSSBOW: □
    //   - TRIDENT: □□□
    //   - VANISHABLE: □□□□
    // ========================================================================
    public static final Supplier<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND
                    )
            );

    // ========================================================================
    // □□□□□□□□ (□□□)
    // ========================================================================
    public static final Supplier<Enchantment> EXAMPLE_PROTECTION =
            ENCHANTMENTS.register("example_protection",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.COMMON,
                            EnchantmentCategory.ARMOR,
                            EquipmentSlot.values()  // □□□□□□□
                    )
            );
}
```

```
// ========================================================================
// □□□□□□□
// □□ Enchantment □□□□□□□□□□□□□
//
// □□□□□□□□ 1.20.4 □□□□□□□□□!
//          □ Forge 1.20.1 □□□□□!
//          1.21.1 □□□□□□□□□□□ (□□□□□□□□)
// ========================================================================
class ExampleEnchantment extends Enchantment {

    public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                              EquipmentSlot... slots) {
        super(rarity, category, slots);
    }

    // ====================================================================
    // □□□□□□□□□□ (□□□□□□□□□)
    // ====================================================================
    @Override
    public int getMinCost(int level) {
        return 10 * level;
    }

    // ====================================================================
    // □□□□□□□□□□
    // ====================================================================
    @Override
    public int getMaxCost(int level) {
        return getMinCost(level) + 15;
    }

    // ====================================================================
    // □□□□□□□
    // ====================================================================
    @Override
    public int getMaxLevel() {
        return 3;  // 3□□□
    }

    // ====================================================================
    // □□□□□□□□□□□□
    // ====================================================================
    @Override
    protected boolean checkCompatibility(Enchantment other) {
        // □□□□□
        return super.checkCompatibility(other)
                && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
    }

    // ====================================================================
    // □□□□□□□□□□
    // ====================================================================
    @Override
    public boolean canEnchant(ItemStack stack) {
        return super.canEnchant(stack);
    }

    // ====================================================================
    // □□□□□□□□ (□□□□)
    // ====================================================================
    @Override
    public boolean isCurse() {
        return false;
    }

    // ====================================================================
    // □□□□□□□□□□□
    // ====================================================================
    @Override
    public boolean isTradeable() {
        return true;
    }

    // ====================================================================
    // □□□□□□□□□□
    // ====================================================================
```

```
    @Override
    public boolean isDiscoverable() {
        return true;
    }

    // ========================================================================
    // □□□□□□□□□□
    // ========================================================================
    @Override
    public boolean isAllowedOnBooks() {
        return true;
    }

    // ========================================================================
    // □□□□□□ (□□□□□□)
    //
    // □□□□□NeoForge 1.20.4: □□□□□ doPostAttack() □□□□□
    //         NeoForge 1.21.1: □□□□□□□□□□ (EnchantmentEffectComponents) □□
    //                          □□□□□□□!
    // ========================================================================
    @Override
    public void doPostAttack(LivingEntity attacker, net.minecraft.world.entity.Entity target, int level) {
        // □□: □□□□□□□□□□□□
        if (target instanceof LivingEntity livingTarget) {
            livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                    net.minecraft.world.effect.MobEffects.POISON,
                    60,              // □□□□□ (tick, 60 = 3□)
                    level - 1         // □□□□□ (□□□□□ - 1)
            ));
        }
        super.doPostAttack(attacker, target, level);
    }

    // ========================================================================
    // □□□□□□□
    //
    // □□□□□□□□□□□□□□□□□□□:
    //
    // NeoForge 1.20.4 (□ Forge 1.20.1 □□):
    //   int level = EnchantmentHelper.getItemEnchantmentLevel(
    //       ModEnchantments.EXAMPLE_ENCHANTMENT.get(), stack);
    //   □□□□□□□□□□
    //
    // NeoForge 1.21.1:
    //   □□□□□□□□□□, □□□□□□□□□□:
    //   EnchantmentHelper.runIterationOnItem(stack, (holder, level) -> {
    //       // holder.value().effects().get(component)
    //   });
    //
    // □□□□□□□□□□:
    //   1.20.4: ItemStack NBT (Enchantments tag)
    //     stack.getEnchantments() -> Map<Enchantment, Integer>
    //     stack.enchant(enchantment, level)
    //   1.21.1: DataComponents.ENCHANTMENTS
    //     stack.get(DataComponents.ENCHANTMENTS) -> ItemEnchantments
    //     stack.set(DataComponents.ENCHANTMENTS, new ItemEnchantments(...))
    // ========================================================================
}

// ============================================================================
// □□□□□□ - NeoForge 1.20.4
// ============================================================================
// NeoForge 1.20.4 □□□□□□□□□□□□:
//
// @Mod.EventBusSubscriber(modid = ExampleMod.MOD_ID, bus = Mod.EventBusSubscriber.Bus.FORGE)
// public static class GameEvents {
//     @SubscribeEvent
//     public static void onEnchantLevel(net.neoforged.neoforge.event.enchantment.EnchantmentLevelSetEvent event) {
//         // event.getLevel() □□□□□□□
//         // event.getEnchantments() □□□□□ Map
//         // event.getPower() □□□□□□□□□□
//
//         // □□□□□□□□□□□□
//         // event.getEnchantments().put(ModEnchantments.EXAMPLE_ENCHANTMENT.get(), 1);
//     }
// }
```

```
//
// □□□□NeoForge 1.21.1 □□□□□□□□□□□□□
//         □□□□□□□□□□□□□□□□, □□□□□□□□□□□□□
// ================================================================================
```

entity/simple_entity.java

```
// ================================================================================
// NeoForge 1.20.4 □□□□□□□ (simple_entity.java)
// NeoForge 1.20.4, Java 17
// ================================================================================
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□
// 1. □□□□□□□□□□!□
//    NeoForge 1.20.4: hurt(DamageSource, float) - □□□□
//                     □ Forge 1.20.1 □□□□!
//    NeoForge 1.21.1: hurt() □□□□ final, □□□□ hurtServer() □ hurtClient()
//                     - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□
//                     - hurtClient(DamageSource, float): □□□□□□□□
//
// 2. □EntityType.Builder.build() □□□□!□
//    NeoForge 1.20.4: .build() □□□□□ String (datafixer type id)
//    NeoForge 1.21.1: .build(ResourceKey.create(Registries.ENTITY_TYPE, ResourceLocation))
//                     □□□□ ResourceKey!
//
// 3. □ResourceLocation □□□□□□!□
//    NeoForge 1.20.4: new ResourceLocation("modid", "path")
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//
// 4. □DeferredRegister.Entities □□□□
//    NeoForge 1.20.4: □□ DeferredRegister.create(Registries.ENTITY_TYPE, MODID)
//                     (1.20.4 □□□□ DeferredRegister.Entities □□□)
//    NeoForge 1.21.1: □□ DeferredRegister.createEntities(MODID) (□□□)
//                     □□ registerEntityType() □□□□□
//
// 5. □□□□□/□□□□□
//    NeoForge 1.20.4: readAdditionalSaveData(CompoundTag) / addAdditionalSaveData(CompoundTag)
//                     □ Forge 1.20.1 □□
//    NeoForge 1.21.1: □□ (1.21.2+ □□□ ValueInput/ValueOutput)
//
// 6. □defineSynchedData()□
//    NeoForge 1.20.4: defineSynchedData() - □□□ (□ Forge 1.20.1 □□)
//    NeoForge 1.21.1: □□ (1.21.5+ □□□ defineSynchedData(SynchedEntityData.Builder))
//
// □□ Forge 1.20.1 □□□□□□
// 1. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □□□□□□□□
//    Forge 1.20.1: ForgeRegistries.ENTITY_TYPES
//    NeoForge 1.20.4: Registries.ENTITY_TYPE (□□□□□□)
//
// 3. □EntityType.Builder □□□□
//    Forge 1.20.1: EntityType.Builder.create(factory, category)
//    NeoForge 1.20.4: EntityType.Builder.of(factory, category)
//                  (1.20.x □□□□□□□ create □□ of)
//
// 4. □□□□□□□
//    Forge 1.20.1: RegistryObject<EntityType<T>>
//    NeoForge 1.20.4: DeferredHolder<EntityType<?>, EntityType<T>> □ Supplier<EntityType<T>>
// ================================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.syncher.EntityDataAccessor;
import net.minecraft.network.syncher.EntityDataSerializers;
import net.minecraft.network.syncher.SynchedEntityData;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.world.damagesource.DamageSource;
import net.minecraft.world.entity.Entity;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
```

```java
import net.minecraft.world.level.Level;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

// ============================================================================
// □□□□□□
// ============================================================================
public class ModEntities {

    // ========================================================================
    // □□□□□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, MODID)
    // NeoForge 1.20.4: DeferredRegister.create(Registries.ENTITY_TYPE, MODID)
    //
    // □□□□□NeoForge 1.21.1:
    //   DeferredRegister.createEntities(MODID) (□□□□□□□)
    //   □□ registerEntityType() □□□□□□
    //
    // □□□□□□□□□□□□:
    //   Forge: ForgeRegistries (Forge □□□□□□□□)
    //   NeoForge: Registries (□□□□□□□□)
    // ========================================================================
    public static final DeferredRegister<EntityType<?>> ENTITY_TYPES =
            DeferredRegister.create(Registries.ENTITY_TYPE, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□
    //
    // □□□□□EntityType.Builder □□□:
    //   Forge 1.20.1: EntityType.Builder.create(factory, category)
    //   NeoForge 1.20.4: EntityType.Builder.of(factory, category)
    //   (1.20.x □□□□□□□ create □□ of)
    //
    // □□□□□build() □□:
    //   NeoForge 1.20.4: .build() □□□□ (□□ String □□ datafixer type id)
    //   NeoForge 1.21.1: .build(ResourceKey.create(
    //       Registries.ENTITY_TYPE, ResourceLocation.fromNamespaceAndPath(MODID, "name")))
    //   (1.21.x □□□□□□ ResourceKey!)
    //
    // □□□□□ResourceLocation □□:
    //   1.20.4: new ResourceLocation("modid", "path")
    //   1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
    // ========================================================================
    public static final DeferredHolder<EntityType<?>, EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITY_TYPES.register(
                    "example_entity",
                    () -> EntityType.Builder.<ExampleEntity>of(
                            ExampleEntity::new,
                            MobCategory.MISC
                    )
                    .sized(1.0f, 1.0f)              // □□□□□ (□ x □)
                    .clientTrackingRange(8)          // □□□□□□□□□ (□□□□)
                    .updateInterval(10)              // □□□□□ (tick)
                    .fireImmune()                    // □□□□□
                    // .canSpawnFarFromPlayer()       // □□□□□□□□□□
                    // .noSummon()                    // □□ /summon □□
                    // .noSave()                      // □□□□□□□□
                    // ================================================================================
                    // □□□□□□□build() □□!
                    // NeoForge 1.20.4: .build() □□□□ (□□ String)
                    // NeoForge 1.21.1: .build(ResourceKey.create(...)) □□□□ ResourceKey
                    // ================================================================================
                    .build("examplemod:example_entity")  // 1.20.4 □ String (datafixer type)
            );
}

// ============================================================================
// □□□□□□□
// □□ Entity (□□□□□□ LivingEntity, Mob □)
//
// Entity □□□□□, □□□□□□□□□□:
// 1. readAdditionalSaveData(CompoundTag) - □ NBT □□□□
// 2. addAdditionalSaveData(CompoundTag) - □□□□□□ NBT
// 3. defineSynchedData() - □□□□□□-□□□□□□□□
// 4. hurt(DamageSource, float) - □□□□□ (1.20.4 □□□□□)
// ============================================================================
```

```
class ExampleEntity extends Entity {

    // ========================================================================
    // □□□□□□□□ (□□□□□□-□□□□□□□□)
    // ========================================================================
    private static final EntityDataAccessor<Integer> DATA_VALUE =
            SynchedEntityData.defineId(ExampleEntity.class, EntityDataSerializers.INT);

    // ========================================================================
    // □□□□
    // □□□ (EntityType<? extends T>, Level) □□□□□□□□,
    // □□□□□□□ EntityFactory □□ (MyEntity::new)
    // ========================================================================
    public ExampleEntity(EntityType<? extends ExampleEntity> type, Level level) {
        super(type, level);
    }

    // ========================================================================
    // □□□□□□□□□□□□□□□□
    //
    // NeoForge 1.20.4 (□ Forge 1.20.1 □□):
    //   @Override
    //   public boolean hurt(DamageSource source, float amount) { ... }
    //   (□□□□□, □□□□□ hurt)
    //
    // NeoForge 1.21.1 (Damage Pipeline Rework):
    //   hurt() □□□□□ final! □□□□□ hurtServer() □/□ hurtClient()
    //   - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□□
    //   - hurtClient(DamageSource, float): □□□□□□ (□□□□□□□□/□□)
    // ========================================================================
    @Override
    public boolean hurt(DamageSource source, float amount) {
        if (this.isInvulnerableTo(source)) {
            return false;
        }
        // □□□□□□□□□□
        // □□: □□□□□□□□□□
        // if (source.is(net.minecraft.tags.DamageTypeTags.IS_FIRE)) {
        //     return false;
        // }
        // amount *= 2; // □□□□□□□
        return super.hurt(source, amount);
    }

    // ========================================================================
    // □ NBT □□□□□□□
    // □□□□1.20.4 □ 1.21.1 □□□□ CompoundTag (□□)
    //         (1.21.2+ □□□□ ValueInput)
    // ========================================================================
    @Override
    public void readAdditionalSaveData(CompoundTag tag) {
        super.readAdditionalSaveData(tag);
        int value = tag.getInt("value");
        this.entityData.set(DATA_VALUE, value);
    }

    // ========================================================================
    // □□□□□□□□□ NBT
    // ========================================================================
    @Override
    public void addAdditionalSaveData(CompoundTag tag) {
        super.addAdditionalSaveData(tag);
        tag.putInt("value", this.entityData.get(DATA_VALUE));
    }

    // ========================================================================
    // □□□□□□□□ (□□□-□□□□□□)
    //
    // 1.20.4: defineSynchedData() - □□□
    // (□ Forge 1.20.1 □□)
    //
    // □□□□1.21.5+ □□ defineSynchedData(SynchedEntityData.Builder builder)
    //         □ 1.20.4 □ 1.21.1 □□□□□□□□
    // ========================================================================
    @Override
    protected void defineSynchedData() {
```

```
        this.entityData.define(DATA_VALUE, 0);
    }

    // ========================================================================
    // □ tick □□□□□
    // ========================================================================
    @Override
    public void tick() {
        super.tick();
        if (!this.level().isClientSide) {
            // □□□□□
        }
    }

    // ========================================================================
    // □□/□□□□□□
    // ========================================================================
    public int getValue() {
        return this.entityData.get(DATA_VALUE);
    }

    public void setValue(int value) {
        this.entityData.set(DATA_VALUE, value);
    }

    // ========================================================================
    // □□□□□□□□ (□□□)
    // □□□□□□□□□□□□ FMLClientSetupEvent:
    //
    // □□□□1.20.4 □□ EntityRenderers.register() □ FMLClientSetupEvent □
    //         1.21.1 □□ RegisterRenderersEvent □□□□□□□□
    //
    // private void clientSetup(final FMLClientSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         EntityRenderers.register(ModEntities.EXAMPLE_ENTITY.get(),
    //             ExampleEntityRenderer::new);
    //     });
    // }
    //
    // □□□□ EntityRendererProvider.Context □□□□□□□□
    // ========================================================================
}
```

gui/container_screen.java

```
// ========================================================================
// NeoForge 1.20.4 GUI □□□□□□ (container_screen.java)
// NeoForge 1.20.4, Java 17
// ========================================================================
//
// □□ NeoForge 1.21.1 □□□□□ - GUI/□□□□□
// 1. □□□□□□□□□□□□!□
//    NeoForge 1.20.4: □□ IForgeMenuType.create(IContainerFactory) □□□□□□□□□□
//                     □□□□□□□: FriendlyByteBuf (□□ RegistryFriendlyByteBuf!)
//    NeoForge 1.21.1: □□ IMenuTypeExtension.create(IContainerFactory) □□
//                     □□□□□□□: RegistryFriendlyByteBuf
//                     (IForgeMenuType □□□ IMenuTypeExtension)
//
// 2. □Screen □□□□□□!□
//    NeoForge 1.20.4: □□ MenuScreens.register() □ FMLClientSetupEvent □□□
//                     (□ Forge 1.20.1 □□!)
//    NeoForge 1.21.1: □□ RegisterMenuScreensEvent □□□□□□□□□□
//                     (□□□, □□□□ FMLClientSetupEvent)
//
// 3. □□□□□□□□□!□
//    NeoForge 1.20.4: NetworkHooks.openScreen(serverPlayer, provider, buf)
//                     (□ Forge 1.20.1 □□!)
//    NeoForge 1.21.1: □□ ((ServerPlayer) player).openMenu(provider, buf)
//                     (NetworkHooks.openScreen □□□□□/□□)
//
// 4. □□□□□□□□
//    NeoForge 1.20.4: Registries.MENU (□ 1.21.1 □□)
//    Forge 1.20.1: ForgeRegistries.CONTAINERS
//
// 5. □BlockBehaviour#use □□□□
//    NeoForge 1.20.4: use(BlockState, Level, BlockPos, Player, InteractionHand, BlockHitResult)
```

```
//    NeoForge 1.21.1: useWithoutItem(BlockState, Level, BlockPos, Player, BlockHitResult)
//                     (use 无物品 useWithoutItem 而 useItemOn)
//
// 关于 Forge 1.20.1 的重要说明：
// 1. 菜单注册表名称
//    Forge 1.20.1: ForgeRegistries.CONTAINERS
//    NeoForge 1.20.4: Registries.MENU
//    (这是最大的 CONTAINERS 变为 MENU)
//
// 2. 的DeferredRegister 包不同
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 3. 的IForgeMenuType 包不同
//    Forge:     net.minecraftforge.common.extensions.IForgeMenuType
//    NeoForge:  net.neoforged.neoforge.common.extensions.IForgeMenuType
//    (这个类的位置不同)
//
// 4. 的NetworkHooks 包不同
//    Forge:     net.minecraftforge.network.NetworkHooks
//    NeoForge:  net.neoforged.neoforge.network.NetworkHooks
//
// 5. 菜单类型不同
//    Forge 1.20.1: RegistryObject<MenuType<T>>
//    NeoForge 1.20.4: DeferredHolder<MenuType<?>, MenuType<T>> 或 Supplier<MenuType<T>>
// ============================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.FriendlyByteBuf;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.ContainerLevelAccess;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.neoforge.common.extensions.IForgeMenuType;
import net.neoforged.neoforge.items.ItemStackHandler;
import net.neoforged.neoforge.items.SlotItemHandler;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

import java.util.function.Supplier;

// ============================================================================
// 菜单类型注册
// ============================================================================
class ModMenuTypes {

    // ========================================================================
    // 菜单类型延迟注册器
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID)
    //                使用: ForgeRegistries.CONTAINERS
    // NeoForge 1.20.4: DeferredRegister.create(Registries.MENU, MODID)
    //                  使用: Registries.MENU
    // 这是最大的变化！从 CONTAINERS 变为 MENU
    //
    // 这里是NeoForge 1.21.1: 使用 Registries.MENU (不变)
    // ========================================================================
    public static final DeferredRegister<MenuType<?>> MENU_TYPES =
            DeferredRegister.create(Registries.MENU, ExampleMod.MOD_ID);

    // ========================================================================
    // 注册菜单类型 (简单版本)
    // 使用 MenuType 构造器, 传入 MenuSupplier 和 FeatureFlags
    // ========================================================================
    public static final Supplier<MenuType<ExampleMenu>> EXAMPLE_MENU =
            MENU_TYPES.register("example_menu", () -> new MenuType<>(
                    ExampleMenu::new,
                    net.minecraft.world.flag.FeatureFlags.DEFAULT_FLAGS
```

```
            ));

    // ================================================================================
    // □□□□□□□□□□□□ (□□ IForgeMenuType)
    // □□□□□□□□□□□□□□□□□□ (□□□□□□), □□□□□□
    //
    // □□□□□□□□
    // NeoForge 1.20.4: IForgeMenuType.create(IContainerFactory)
    //                  □□□□□□□□: FriendlyByteBuf
    // NeoForge 1.21.1: IMenuTypeExtension.create(IContainerFactory)
    //                  □□□□□□□□: RegistryFriendlyByteBuf
    //                  (IForgeMenuType □□□□ IMenuTypeExtension)
    //
    // Forge 1.20.1: IForgeMenuType.create(IContainerFactory)
    //               (□□□□□□□□□□□□)
    // ================================================================================
    public static final Supplier<MenuType<ExampleMenuExtra>> EXAMPLE_MENU_EXTRA =
            MENU_TYPES.register("example_menu_extra", () ->
                    IForgeMenuType.create(ExampleMenuExtra::new)
            );
}

// ================================================================================
// □□□□□ (□□□□□□)
// □□ AbstractContainerMenu
// ================================================================================
class ExampleMenu extends AbstractContainerMenu {

    // ================================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□)
    // ================================================================================
    public ExampleMenu(int containerId, Inventory playerInventory) {
        this(containerId, playerInventory, new ItemStackHandler(9));
    }

    // ================================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□)
    // ================================================================================
    public ExampleMenu(int containerId, Inventory playerInventory, ItemStackHandler handler) {
        super(ModMenuTypes.EXAMPLE_MENU.get(), containerId);

        // □□□□□□□□□□□ (□□ SlotItemHandler, □□ IItemHandler □□)
        // NeoForge □□□□□ SlotItemHandler (□□ IItemHandler) □□□□ vanilla Slot
        for (int i = 0; i < 9; i++) {
            int x = 62 + (i % 3) * 18;
            int y = 18 + (i / 3) * 18;
            this.addSlot(new SlotItemHandler(handler, i, x, y));
        }

        // □□□□□□□□□□□ (27 □)
        for (int row = 0; row < 3; row++) {
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory,
                        col + row * 9 + 9,
                        8 + col * 18,
                        84 + row * 18
                ));
            }
        }

        // □□□□□□□□□□□ (9 □)
        for (int col = 0; col < 9; col++) {
            this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
        }
    }

    // ================================================================================
    // stillValid - □□□□□□□□□□□□
    // ================================================================================
    @Override
    public boolean stillValid(Player player) {
        return true;
    }

    // ================================================================================
    // quickMoveStack - Shift+□□□□□□□□
```

```
    // ========================================================================
    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasItem()) {
            return ItemStack.EMPTY;
        }

        ItemStack originalStack = slot.getItem();
        ItemStack copyStack = originalStack.copy();

        if (index < 9) {
            if (!this.moveItemStackTo(originalStack, 9, 45, false)) {
                return ItemStack.EMPTY;
            }
        } else {
            if (!this.moveItemStackTo(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        if (originalStack.isEmpty()) {
            slot.set(ItemStack.EMPTY);
        } else {
            slot.setChanged();
        }

        return copyStack;
    }
}

// ========================================================================
// □□□□□□□□□ (□□ IContainerFactory / IForgeMenuType)
// □□□□□□□□□□□□□□□□□□□□ (□□□□□□)
// ========================================================================
class ExampleMenuExtra extends AbstractContainerMenu {

    private final ContainerLevelAccess access;

    // ========================================================================
    // □□□□□□□□ (□□ FriendlyByteBuf □□□□)
    //
    // □□□□□□□□□□□□□□□□:
    //   NeoForge 1.20.4: FriendlyByteBuf (□ Forge 1.20.1 □□!)
    //   NeoForge 1.21.1: RegistryFriendlyByteBuf (□□□□□, □□□□□□□□)
    // ========================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, FriendlyByteBuf extraData) {
        this(containerId, playerInventory, extraData.readBlockPos());
    }

    // ========================================================================
    // □□□□□□□□
    // ========================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, BlockPos pos) {
        super(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), containerId);
        this.access = ContainerLevelAccess.create(playerInventory.player.level(), pos);
        // □□□□□ (□□, □□□□□)
    }

    @Override
    public boolean stillValid(Player player) {
        return AbstractContainerMenu.stillValid(this.access, player, Blocks.CRAFTING_TABLE);
    }

    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        return ItemStack.EMPTY;
    }

    // ========================================================================
    // □□□□□□□□□□□□□:
    //
    // □□□□□NeoForge 1.20.4: NetworkHooks.openScreen(serverPlayer, provider, buf)
    //         (□ Forge 1.20.1 □□!)
    //         NeoForge 1.21.1: serverPlayer.openMenu(provider, buf)
    //         (□□□□□ NetworkHooks)
```

```
    //
    // 在 BlockBehaviour#use 中调用 (1.20.4 官方):
    //   @Override
    //   public InteractionResult use(BlockState state, Level level, BlockPos pos,
    //           Player player, InteractionHand hand, BlockHitResult hitResult) {
    //       if (level.isClientSide) {
    //           return InteractionResult.SUCCESS;
    //       }
    //       NetworkHooks.openScreen((ServerPlayer) player, new SimpleMenuProvider(
    //           (containerId, playerInventory, p) -> new ExampleMenuExtra(containerId, playerInventory, pos),
    //           Component.translatable("menu.examplemod.example_menu")
    //       ), buf -> buf.writeBlockPos(pos));
    //       return InteractionResult.CONSUME;
    //   }
    //
    // 或者在1.21.1 的新版本 use 被拆为 useWithoutItem:
    //   public InteractionResult useWithoutItem(BlockState state, Level level, BlockPos pos,
    //           Player player, BlockHitResult hitResult) { ... }
    // ================================================================================
}

// ================================================================================
// 客户端屏幕 (Screen) - 负责渲染界面
// 继承 AbstractContainerScreen
// ================================================================================
class ExampleScreen extends net.minecraft.client.gui.screens.inventory.AbstractContainerScreen<ExampleMenu> {

    public ExampleScreen(ExampleMenu menu, Inventory playerInventory, Component title) {
        super(menu, playerInventory, title);
        // this.imageWidth = 176;
        // this.imageHeight = 166;
    }

    // ================================================================================
    // 渲染背景纹理
    // 1.20.4 使用 GuiGraphics (在 1.21.1 中)
    // ================================================================================
    @Override
    protected void renderBg(net.minecraft.client.gui.GuiGraphics guiGraphics,
                            float partialTick, int mouseX, int mouseY) {
        // 绘制背景纹理
        // guiGraphics.blit(
        //     new ResourceLocation(ExampleMod.MOD_ID, "textures/gui/example_container.png"),
        //     this.leftPos, this.topPos,
        //     0, 0,
        //     this.imageWidth, this.imageHeight
        // );
    }

    @Override
    public void render(net.minecraft.client.gui.GuiGraphics guiGraphics,
                       int mouseX, int mouseY, float partialTick) {
        this.renderBackground(guiGraphics, mouseX, mouseY, partialTick);
        super.render(guiGraphics, mouseX, mouseY, partialTick);
        this.renderTooltip(guiGraphics, mouseX, mouseY);
    }

    @Override
    protected void init() {
        super.init();
        this.titleLabelY = 6;
        this.inventoryLabelY = this.imageHeight - 94;
    }

    // ================================================================================
    // 注册 Screen (在客户端设置事件中):
    //
    // 别忘了注册Screen 哦！！!
    //
    // NeoForge 1.20.4 (与 Forge 1.20.1 相同):
    //   使用 MenuScreens.register() 在 FMLClientSetupEvent 中调用:
    //
    //   private void clientSetup(final FMLClientSetupEvent event) {
    //       event.enqueueWork(() -> {
    //           MenuScreens.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
    //           MenuScreens.register(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), ExampleScreenExtra::new);
```

```
    //       });
    //   }
    //   (□□□□ enqueueWork □□□□□□□)
    //
    // NeoForge 1.21.1:
    //   □□ RegisterMenuScreensEvent □□□□□□□□□:
    //
    //   @SubscribeEvent
    //   public static void registerScreens(RegisterMenuScreensEvent event) {
    //       event.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
    //   }
    //   (□□□□□ FMLClientSetupEvent □ enqueueWork)
    // ================================================================================
}
```

item/simple_item.java

```
// ================================================================================
// NeoForge 1.20.4 □□□□□□□ (simple_item.java)
// NeoForge 1.20.4, Java 17
// ================================================================================
//
// □□ NeoForge 1.21.1 □□□□□□□
// 1. □NBT □□ vs Data Components □□ (□□□□!)□
//    NeoForge 1.20.4: □□□ ItemStack NBT □□!
//                     itemStack.getTag(), itemStack.getOrCreateTag() □□□□□
//    NeoForge 1.21.1: □□ Data Components □□, NBT □□□!
//                     itemStack.get(DataComponents.XXX), itemStack.set() □
//                     (1.20.5+ □□□ Data Components)
//
// 2. □DeferredRegister.Items □□□□□
//    NeoForge 1.20.4: registerItem() □□ Supplier<Item> (□□ DeferredItem<T>)
//    NeoForge 1.21.1: registerItem() □□ DeferredItem<T> (□□□□)
//
// 3. □Item.Properties □□□
//    1.20.4 □ 1.21.1 □ Item.Properties □□□□□□:
//    .stacksTo(), .durability(), .rarity(), .food(), .fireResistant()
//    □ 1.21.1 □□□□□□ DataComponents, □ 1.20.4 □□□□
//
// 4. □CreativeModeTab □□□
//    1.20.4 □ 1.21.1 □□□ CreativeModeTab.builder() □□□
//    □□ DeferredRegister<CreativeModeTab> □□
//    □□□□□ BuildCreativeModeTabContentsEvent □□□□□□□□□□□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister.Items □□□□
//    Forge 1.20.1:  □□□□, □□ DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
//                   □□ RegistryObject<Item>
//    NeoForge 1.20.4: □□ DeferredRegister.Items, □□ DeferredRegister.createItems(MODID)
//                     □□ registerItem(), registerSimpleItem() □□□□□
//                     □□ Supplier<Item> (□□ RegistryObject<Item>)
//
// 2. □□□□□□□
//    Forge 1.20.1:  ForgeRegistries.ITEMS
//    NeoForge 1.20.4: BuiltInRegistries.ITEM (□□□□□□)
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.food.FoodProperties;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Rarity;
import net.neoforged.neoforge.registries.DeferredRegister;

import java.util.function.Supplier;

// ================================================================================
// □□□□□□
// ================================================================================
public class ModItems {
    // ============================================================================
```

```
// □□□□□□□ DeferredRegister.Items □□□□□□
// Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
//                □□ DeferredRegister<Item>, register() □□ RegistryObject<Item>
// NeoForge 1.20.4: DeferredRegister.createItems(MODID)
//                  □□ DeferredRegister.Items
//                  registerItem() □□ Supplier<Item>
//
// □□□□NeoForge 1.21.1: registerItem() □□ DeferredItem<T> (□□□□)
//         NeoForge 1.20.4: registerItem() □□ Supplier<Item>
// ============================================================================
public static final DeferredRegister.Items ITEMS =
        DeferredRegister.createItems(ExampleMod.MOD_ID);

// ============================================================================
// □□□□□□□□□
// □□□□□□□□□□:
//   Forge 1.20.1: RegistryObject<Item>
//   NeoForge 1.20.4: Supplier<Item>
//   NeoForge 1.21.1: DeferredItem<Item>
//
// □□: □□ registerItem □□□□ (□□)
// ============================================================================
public static final Supplier<Item> EXAMPLE_ITEM =
        ITEMS.registerItem("example_item", Item::new, new Item.Properties());

// ============================================================================
// □□□□□□□□□□□□□□□
// ============================================================================
public static final Supplier<Item> RARE_ITEM =
        ITEMS.registerItem("rare_item", Item::new, new Item.Properties()
                .rarity(Rarity.RARE)           // □□□: COMMON, UNCOMMON, RARE, EPIC
                .stacksTo(16)                   // □□□□□ (□□ 64)
                .fireResistant()                // □□□□ (□□□□□□□□□)
        );

// ============================================================================
// □□□□□□□□□
// FoodProperties □□ Item.Properties.food() □□
// □□□□1.20.4 □□□□ FoodProperties (□□□ DataComponents)
//         1.21.1 □□□□□ DataComponents.FOOD □□□□
// ============================================================================
public static final Supplier<Item> EXAMPLE_FOOD =
        ITEMS.registerItem("example_food", Item::new, new Item.Properties()
                .food(new FoodProperties.Builder()
                        .nutrition(6)                   // □□□ (□□□□)
                        .saturationMod(0.6f)             // □□□□□
                        .meat()                          // □□□□□
                        .build()
                )
        );

// ============================================================================
// □□□□□□□□□□□□□
// □□ durability □□□□□□□□□□□□□ 1
// ============================================================================
public static final Supplier<Item> EXAMPLE_DURABLE =
        ITEMS.registerItem("example_durable", Item::new, new Item.Properties()
                .durability(250)               // □□□
                .rarity(Rarity.UNCOMMON)
        );

// ============================================================================
// □□□□□□□ (□□□□□□□□)
// □□□□□□□□□□□□□□ Item □
// □□□□□□□□□□□□ (□□□□□), □□□□ Item □□□
// ============================================================================
public static final Supplier<Item> CUSTOM_ITEM =
        ITEMS.registerItem("custom_item", CustomItem::new, new Item.Properties()
                .rarity(Rarity.EPIC)
        );

// ============================================================================
// □□□□□□□□□□ (BlockItem)
// □□□□□□□registerSimpleBlockItem □□□□ BlockItem
// ============================================================================
// public static final Supplier<Item> EXAMPLE_BLOCK_ITEM =
```

```
    //         ITEMS.registerSimpleBlockItem(ModBlocks.EXAMPLE_BLOCK);
}

// ============================================================================
// □□□□□□□
// □□ Item □□□□□□□□□□□□□
// ============================================================================
class CustomItem extends Item {

    public CustomItem(Properties properties) {
        super(properties);
    }

    // ========================================================================
    // □□□□□□□NBT □□!
    //
    // NeoForge 1.20.4 (□ Forge 1.20.1 □□):
    //   □□ ItemStack NBT □□□□□□□□□:
    //   stack.getOrCreateTag().putInt("custom_value", 42);
    //   stack.getTag().getInt("custom_value");
    //
    // NeoForge 1.21.1:
    //   □□ Data Components □□ (NBT □□□):
    //   stack.set(ModDataComponents.CUSTOM_VALUE.get(), 42);
    //   stack.get(ModDataComponents.CUSTOM_VALUE.get());
    //   (1.20.5+ □□□□ Data Components)
    // ========================================================================

    // ========================================================================
    // ItemStack NBT □□□□□ (1.20.4 □□)
    //
    // public InteractionResult use(Level level, Player player, InteractionHand hand) {
    //     ItemStack stack = player.getItemInHand(hand);
    //
    //     // 1.20.4: □□ NBT □□□□□□□□
    //     CompoundTag tag = stack.getOrCreateTag();
    //     int currentValue = tag.getInt("custom_value");
    //     tag.putInt("custom_value", currentValue + 1);
    //
    //     // 1.21.1: □□ Data Components (□□□!)
    //     // stack.set(ModDataComponents.CUSTOM_VALUE.get(), currentValue + 1);
    //
    //     return InteractionResult.sidedSuccess(level.isClientSide);
    // }
    // ========================================================================
}

// ============================================================================
// □□□□□□□□□□□□□□□
// □□□□□1.20.4 □ 1.21.1 □□□□ CreativeModeTab.builder() □□□□
//         □□ DeferredRegister<CreativeModeTab> □□
// ============================================================================
class ModCreativeModeTabs {

    public static final DeferredRegister<CreativeModeTab> CREATIVE_MODE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□□□□□□□
    // □□ builder() □□□□□, □□□□□□□□□□□□□□□
    // ========================================================================
    public static final Supplier<CreativeModeTab> EXAMPLE_TAB =
            CREATIVE_MODE_TABS.register("example_tab", () -> CreativeModeTab.builder()
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    .icon(() -> new ItemStack(ModItems.EXAMPLE_ITEM.get()))
                    .displayItems((params, output) -> {
                        output.accept(ModItems.EXAMPLE_ITEM.get());
                        output.accept(ModItems.RARE_ITEM.get());
                        output.accept(ModItems.EXAMPLE_FOOD.get());
                        output.accept(ModItems.EXAMPLE_DURABLE.get());
                        output.accept(ModItems.CUSTOM_ITEM.get());
                    })
                    .build()
            );

    // ========================================================================
```

```
    // □□□□□□: □□ BuildCreativeModeTabContentsEvent □□□□□□□□□□□□
    // □□□□□□□□□□□□□□□□:
    //
    // @SubscribeEvent  // □□□□□□□□□ (bus = Bus.MOD)
    // public static void buildContents(BuildCreativeModeTabContentsEvent event) {
    //     if (event.getTabKey() == CreativeModeTabs.INGREDIENTS) {
    //         event.accept(ModItems.EXAMPLE_ITEM.get());
    //     }
    // }
    // ============================================================================
}
```

item/tool_item.java

```
// ============================================================================
// NeoForge 1.20.4 □□□□□□□□□ (tool_item.java)
// NeoForge 1.20.4, Java 17
// ============================================================================
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□
// 1. □Tier □□□□□□!□
//    NeoForge 1.20.4: □□ Tier □□, □□ getLevel() (int □□□□□□)
//                     □ Forge 1.20.1 □□□□!
//    NeoForge 1.21.1: □□□□ Tier □□, □□ getIncorrectDrops() (TagKey<Block>)
//                     □□□ getLevel() □□!
//                     □□ SimpleTier □□□□□□□□ Tier
//
// 2. □□□□□□□□□□□□!□
//    NeoForge 1.20.4: new SwordItem(tier, attackDamage, attackSpeed, properties)
//                     attackDamage □ attackSpeed □□□□□□□□□□
//    NeoForge 1.21.1: new SwordItem(tier, properties)
//                     attackDamage □ attackSpeed □□ Item.Properties.attributes() □□
//                     □□□□□□□□□□□□□□□□□□!
//
// 3. □ResourceLocation □□□□□□!□
//    NeoForge 1.20.4: new ResourceLocation("modid", "path")
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//
// 4. □ItemAbility / ToolAction □□□
//    NeoForge 1.20.4: □□□□ ToolAction / ToolActions (□ Forge 1.20.1 □□)
//    NeoForge 1.21.1: □□□□ ItemAbility / ItemAbilities
//
// □□ Forge 1.20.1 □□□□□□
// 1. □DeferredRegister.Items □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □Tier □□□□□
//    Forge 1.20.1 □ NeoForge 1.20.4 □ Tier □□□□□□!
//    □□□: getUses(), getSpeed(), getAttackDamageBonus(),
//            getLevel(), getEnchantmentValue(), getRepairIngredient()
//
// 3. □□□□□□□□□
//    Forge 1.20.1 □ NeoForge 1.20.4 □□□□□□□□□□□□!
//    new SwordItem(tier, attackDamage, attackSpeed, properties)
//
// 4. □□□□□□
//    Forge 1.20.1: RegistryObject<SwordItem>
//    NeoForge 1.20.4: Supplier<SwordItem>
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.tags.BlockTags;
import net.minecraft.tags.TagKey;
import net.minecraft.world.item.AxeItem;
import net.minecraft.world.item.HoeItem;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.ShovelItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
```

```
import net.minecraft.world.level.block.Block;
import net.neoforged.neoforge.registries.DeferredRegister;

import java.util.function.Supplier;

// ============================================================================
// □□□□□□□□
// ============================================================================
public class ModTools {

    public static final DeferredRegister.Items ITEMS =
            DeferredRegister.createItems(ExampleMod.MOD_ID);

    // ============================================================================
    // □□□□□□□□ Tier (□□□□) - □□□□!
    //
    // NeoForge 1.20.4 □ Forge 1.20.1 □ Tier □□□□□□:
    //   - getUses(): □□□
    //   - getSpeed(): □□□□
    //   - getAttackDamageBonus(): □□□□□□
    //   - getLevel(): □□□□ (int, □ wood=0, stone=1, iron=2, diamond=3, netherite=4)
    //   - getEnchantmentValue(): □□□
    //   - getRepairIngredient(): □□□□
    //
    // □□□□NeoForge 1.21.1 □ Tier □□:
    //   - getIncorrectDrops(): TagKey<Block> (□□ getLevel()!)
    //   - getUses(): int (□□)
    //   - getSpeed(): float (□□)
    //   - getAttackDamageBonus(): float (□□)
    //   - getEnchantmentValue(): int (□□)
    //   - getRepairIngredient(): Ingredient (□□)
    //   □□□□ SimpleTier □□□
    //
    // Forge 1.20.1 □□□ (□ 1.20.4 □□□□):
    //   public static final Tier COPPER_TIER = new Tier() {
    //       @Override public int getUses() { return 200; }
    //       @Override public float getSpeed() { return 5f; }
    //       @Override public float getAttackDamageBonus() { return 1.5f; }
    //       @Override public int getLevel() { return 2; }
    //       @Override public int getEnchantmentValue() { return 20; }
    //       @Override public Ingredient getRepairIngredient() { return ...; }
    //   };
    // ============================================================================
    public static final Tier COPPER_TIER = new Tier() {

        // □□□□□□□1.20.4 □□ getLevel() (int □□□□□□□)
        // 1.21.1 □□□□□□□□□, □□ getIncorrectDrops() □□ TagKey<Block>!
        @Override
        public int getLevel() {
            return 2;  // stone=1, iron=2, diamond=3
        }

        @Override
        public int getUses() {
            return 200;  // □□□□ (stone=131, iron=250, diamond=1561)
        }

        @Override
        public float getSpeed() {
            return 5f;  // □□□□□ (stone=4, iron=6, diamond=8)
        }

        @Override
        public float getAttackDamageBonus() {
            return 1.5f;  // □□□□□□□□ (stone=1, iron=2, diamond=3)
        }

        @Override
        public int getEnchantmentValue() {
            return 20;  // □□□□ (gold=22, iron=14, diamond=10)
        }

        @Override
        public Ingredient getRepairIngredient() {
            // □□□□□□ (□□ lazy supplier □□□□□□□□□)
            // □□□□□□ NeoForge □□: Tags.Items.INGOTS_COPPER
```

```
        return Ingredient.of(net.minecraft.world.level.block.Blocks.COPPER_BLOCK);
    }
};

// ============================================================================
// □□□□□□□□□□□□ - □□□□□□□□□□□!
//
// NeoForge 1.20.4 (□ Forge 1.20.1 □□):
//   new SwordItem(tier, attackDamage, attackSpeed, new Item.Properties())
//   (attackDamage □ attackSpeed □□□□□□□□□□□□)
//
// NeoForge 1.21.1:
//   new SwordItem(tier, new Item.Properties().attributes(
//       SwordItem.createAttributes(tier, attackDamage, attackSpeed)))
//   (□□ Item.Properties.attributes() □□, □□□□□□□□!)
//
// Forge 1.20.1 □□□□ (□ 1.20.4 □□□□□):
//   RegistryObject<SwordItem> COPPER_SWORD = ITEMS.register(
//       "copper_sword", () -> new SwordItem(COPPER_TIER, 3, -2.4f, new Item.Properties()));
//
// NeoForge 1.20.4 □□□:
//   Supplier<SwordItem> COPPER_SWORD = ITEMS.register(
//       "copper_sword", () -> new SwordItem(COPPER_TIER, 3, -2.4f, new Item.Properties()));
//
// □□□□□□□□□□□□□: RegistryObject -> Supplier
// ============================================================================

// □□ - □□□□□ 3, □□□□□ -2.4f
public static final Supplier<SwordItem> COPPER_SWORD =
        ITEMS.register("copper_sword", () -> new SwordItem(
                COPPER_TIER,
                3,          // □□□□□□□
                -2.4f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 1, □□□□□ -2.8f
public static final Supplier<PickaxeItem> COPPER_PICKAXE =
        ITEMS.register("copper_pickaxe", () -> new PickaxeItem(
                COPPER_TIER,
                1,          // □□□□□□□
                -2.8f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 6, □□□□□ -3.2f
public static final Supplier<AxeItem> COPPER_AXE =
        ITEMS.register("copper_axe", () -> new AxeItem(
                COPPER_TIER,
                6,          // □□□□□□□□ (□□□□□□□□)
                -3.2f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 1.5, □□□□□ -3.0f
public static final Supplier<ShovelItem> COPPER_SHOVEL =
        ITEMS.register("copper_shovel", () -> new ShovelItem(
                COPPER_TIER,
                1.5f,       // □□□□□□□
                -3.0f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 0, □□□□□ -1.0f (1.20.x □ HoeItem □□□□□□)
public static final Supplier<HoeItem> COPPER_HOE =
        ITEMS.register("copper_hoe", () -> new HoeItem(
                COPPER_TIER,
                0,          // □□□□□□□
                -1.0f,      // □□□□□□□
                new Item.Properties()
        ));

// ============================================================================
// □□□□□□□□ (□□□□□□□)
//
// 1.20.4 □□□□□□□□□□□□□:
```

```
    // data/examplemod/tags/blocks/needs_copper_tool.json
    //   - □□□□□□□□□□□□□□□
    //
    // □□□□1.21.1 □□□□□□□ incorrect_for_copper_tool.json □□
    //         (□□□□□□□□□□□□□□□□□□□□)
    //         1.20.4 □□□□□ incorrect □□ (□□□□□ getLevel() int □□)
    //
    // □□□□□ResourceLocation □□□□□:
    //   1.20.4: new ResourceLocation("modid", "path")
    //   1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
    // ============================================================================
    public static final TagKey<Block> NEEDS_COPPER_TOOL = TagKey.create(
            net.minecraft.core.registries.Registries.BLOCK,
            new ResourceLocation(ExampleMod.MOD_ID, "needs_copper_tool")
    );
}
```

mixin/mixin_config.json

```
{
    "_comment": "============================================================================",
    "_comment_2": "NeoForge 1.20.4 Mixin □□□□□ (mixin_config.json)",
    "_comment_3": "NeoForge 1.20.4, Java 17",
    "_comment_4": "============================================================================",
    "_comment_5": "Mixin □□□□□ - □□□□□ Mixin □□□□□□□",
    "_comment_6": "",
    "_comment_7": "□□□□□: src/main/resources/examplemod.mixins.json",
    "_comment_8": "□□□□□□ mods.toml □ [[mixins]] □□ config □□□□",
    "_comment_9": "",
    "_comment_10": "□□ NeoForge 1.21.1 □□□□□",
    "_comment_11": "1. Mixin □□□□□□ 1.20.4 □ 1.21.1 □□□□□□□",
    "_comment_12": "2. □□□□□□ SpongePowered Mixin □□",
    "_comment_13": "3. □ mods.toml □□□□ [[mixins]] □□□□ (1.20.4 □ mods.toml, 1.21.1 □ neoforge.mods.toml)",
    "_comment_14": "",
    "_comment_15": "□□ Forge 1.20.1 □□□□□",
    "_comment_16": "1. Forge 1.20.1 □ mods.toml □□ [[mixins]] □□□□ (□□)",
    "_comment_17": "2. Mixin □□ JSON □□□□□□□",
    "_comment_18": "3. □□□□□ mods.toml □□□□□□: Forge □ modId=forge, NeoForge □ modId=neoforge",
    "_comment_19": "4. NeoForge □ Mixin □□□□ Forge □□ (□□□□ SpongePowered Mixin)",
    "_comment_20": "",
    "_comment_21": "□□□□□□ mods.toml □□□□ Mixin □□:",
    "_comment_22": "  [[mixins]]",
    "_comment_23": "  config=\"${mod_id}.mixins.json\"",
    "_comment_24": "============================================================================",

    "required": true,
    "minVersion": "0.8.5",
    "package": "com.example.examplemod.mixin",
    "compatibilityLevel": "JAVA_17",
    "refmap": "examplemod.refmap.json",
    "mixins": [
        "example.MixinExampleClass",
        "example.MixinAnotherClass"
    ],
    "client": [
        "client.MixinClientClass"
    ],
    "server": [
        "server.MixinServerClass"
    ],
    "injectors": {
        "defaultRequire": 1
    }
}
```

project/ModMain.java

```
// ============================================================================
// NeoForge 1.20.4 □□□□□□□ (ModMain.java)
// NeoForge 1.20.4, Java 17
// ============================================================================
//
// □□ NeoForge 1.21.1 □□□□□□□
// 1. □Java □□□
//    NeoForge 1.20.4: Java 17 (□□ 21!)
//    NeoForge 1.21.1: Java 21
//
```

```
// 2. 获取IEventBus 的方式不同
//    NeoForge 1.20.4: 可以通过多种方式获取 IEventBus (推荐方式)
//                     FMLJavaModLoadingContext 在 1.20.4 中已弃用,
//                     但仍可以通过它获取事件总线
//    NeoForge 1.21.1: 只能通过构造函数注入 IEventBus 或 ModContainer
//                     FMLJavaModLoadingContext 已移除!
//
// 3. 在@Mod.EventBusSubscriber 的 bus 参数
//    NeoForge 1.20.4: Bus.FORGE (事件总线类型) / Bus.MOD (模组事件总线)
//                     但是在1.20.4 中已弃用 Bus.FORGE!
//    NeoForge 1.21.1: Bus.GAME (事件总线类型) / Bus.MOD (模组事件总线)
//                     Bus.FORGE 已经弃用 Bus.GAME
//
// 4. 获取ModContainer 的方式不同
//    NeoForge 1.20.4: 可以通过多种方式获取 ModContainer
//                     推荐使用 ModLoadingContext.get().getActiveContainer() (推荐)
//    NeoForge 1.21.1: 只能通过构造函数注入 ModContainer
//                     ModLoadingContext 已移除
//
// 5. 注册配置文件的方式
//    NeoForge 1.20.4: modContainer.registerConfig(ModConfig.Type.COMMON, SPEC)
//                     也可以 ModLoadingContext.get().registerConfig(...) (已弃用)
//    NeoForge 1.21.1: modContainer.registerConfig(ModConfig.Type.COMMON, SPEC)
//
// 如果 Forge 1.20.1 的话会有以下
// 1. 模组类的导入
//    Forge:     net.minecraftforge.fml.common.Mod
//    NeoForge:  net.neoforged.fml.common.Mod
//
// 2. 事件总线类的导入
//    Forge:     net.minecraftforge.eventbus.api.IEventBus
//    NeoForge:  net.neoforged.bus.api.IEventBus
//
// 3. 获取IEventBus 的方式不同
//    Forge 1.20.1:     FMLJavaModLoadingContext.get().getModEventBus()
//    NeoForge 1.20.4:  多种方式可以获取 (推荐)
//
// 4. 注册事件总线的方式
//    Forge:     MinecraftForge.EVENT_BUS
//    NeoForge:  NeoForge.EVENT_BUS (net.neoforged.neoforge.NeoForge)
//
// 5. 在DeferredRegister 的不同
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
// ============================================================================

package com.example.examplemod;

import com.example.examplemod.block.ModBlocks;
import com.example.examplemod.item.ModItems;
import com.example.examplemod.item.ModCreativeModeTabs;
import com.mojang.logging.LogUtils;
import net.neoforged.bus.api.IEventBus;
import net.neoforged.fml.ModContainer;
import net.neoforged.fml.common.Mod;
import net.neoforged.fml.config.ModConfig;
import org.slf4j.Logger;

// ============================================================================
// @Mod 注解 - 必须与 ID
// 如果你使用: net.neoforged.fml.common.Mod
//         Forge: net.minecraftforge.fml.common.Mod
// 必须匹配: @Mod("examplemod")
// ============================================================================
@Mod(ExampleMod.MOD_ID)
public class ExampleMod {

    // 模组 ID - 必须与 mods.toml 中的 modId 匹配
    // 如果你在1.20.4 使用 mods.toml (不是 neoforge.mods.toml!)
    public static final String MOD_ID = "examplemod";

    // 日志记录器
    private static final Logger LOGGER = LogUtils.getLogger();

    // ============================================================================
    // 构造函数
```

```
// □□□□NeoForge 1.20.4 □□□□□□□□□□□□□□ IEventBus □ ModContainer
//
// Forge 1.20.1 □□□□ (□□□□):
//   public ExampleMod() {
//       IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();
//       modEventBus.addListener(this::commonSetup);
//   }
//
// NeoForge 1.20.4 □□□□ (□□):
//   public ExampleMod(IEventBus modEventBus) {
//       modEventBus.addListener(this::commonSetup);
//   }
//
// NeoForge 1.21.1 □□□□:
//   public ExampleMod(IEventBus modEventBus, ModContainer modContainer) { ... }
//   (□ 1.20.4 □□□□□, □ ModContainer □□□□□□□)
//
// □□□□FMLJavaModLoadingContext □ NeoForge 1.20.4 □□□□□□□,
//         □□□□□□□□□□□□□□□□□□□□□□□□□□□□
//         □ NeoForge 1.20.5+ □ FMLJavaModLoadingContext □□□!
// ============================================================================
public ExampleMod(IEventBus modEventBus, ModContainer modContainer) {
    LOGGER.info("Loading Example Mod for NeoForge 1.20.4 (Java 17)");

    // □□ DeferredRegister □□□□□□□□
    // □□□□NeoForge □ DeferredRegister □□:
    //         net.neoforged.neoforge.registries.DeferredRegister
    //         Forge □ DeferredRegister □□:
    //         net.minecraftforge.registries.DeferredRegister
    // □□□□□□□□: deferredRegister.register(modEventBus)

    // □□□□□
    ModBlocks.BLOCKS.register(modEventBus);

    // □□□□□□□□□□□
    ModBlocks.BLOCK_ENTITY_TYPES.register(modEventBus);

    // □□□□□
    ModItems.ITEMS.register(modEventBus);

    // □□□□□□□□□□□□□□□□
    ModCreativeModeTabs.CREATIVE_MODE_TABS.register(modEventBus);

    // □□□□□□□□ (GUI)
    // ModMenuTypes.MENU_TYPES.register(modEventBus);

    // □□□□□□□□
    // ModEntities.ENTITY_TYPES.register(modEventBus);

    // □□□□□□ (1.20.4 □□□□□□□□□□□□)
    // ModEnchantments.ENCHANTMENTS.register(modEventBus);

    // □□□□□□ (□□□□□□□□□□□)
    // modEventBus.addListener(this::commonSetup);

    // □□□□□□□□
    // □□□□□Forge 1.20.1: ModLoadingContext.get().registerConfig(...)
    //         NeoForge 1.20.4: modContainer.registerConfig(...) (□□)
    //         NeoForge 1.21.1: modContainer.registerConfig(...) (ModLoadingContext □□□)
    modContainer.registerConfig(ModConfig.Type.COMMON, Config.SPEC);
}

// ============================================================================
// □□□□□□□□□□
// ============================================================================
// private void commonSetup(final FMLCommonSetupEvent event) {
//     event.enqueueWork(() -> {
//         LOGGER.info("Common setup complete");
//     });
// }

// ============================================================================
// NeoForge □□□□□□□□ (□□□□□□□□□□□□)
// □□□□□□□□ (modEventBus): □□□□□□□□□□□□ (DeferredRegister, □□□□□□)
// □□□□□□□□ (NeoForge.EVENT_BUS): □□□□□□□□□□□□ (□□□□□, □□□□□□)
//
```

```
    // □□□□Forge: MinecraftForge.EVENT_BUS
    //         NeoForge: NeoForge.EVENT_BUS (□□: net.neoforged.neoforge.NeoForge)
    //
    // □□□□□:
    // NeoForge.EVENT_BUS.register(new EventHandler());
    // □□□ @Mod.EventBusSubscriber □□ (□□□□□□)
    //
    // □□□□□□□@Mod.EventBusSubscriber □ bus □□:
    //   NeoForge 1.20.4: bus = Mod.EventBusSubscriber.Bus.FORGE (□□□□□□□)
    //                    bus = Mod.EventBusSubscriber.Bus.MOD  (□□□□□□□)
    //   NeoForge 1.21.1: bus = Mod.EventBusSubscriber.Bus.GAME  (□□□□□□□, FORGE □□□ GAME)
    //                    bus = Mod.EventBusSubscriber.Bus.MOD   (□□□□□□□)
    //
    // □□:
    // @Mod.EventBusSubscriber(modid = ExampleMod.MOD_ID, bus = Mod.EventBusSubscriber.Bus.FORGE)
    // public static class GameEventHandler {
    //     @SubscribeEvent
    //     public static void onPlayerInteract(PlayerInteractEvent.RightClickBlock event) {
    //         // □□□□□□□□□□□□
    //     }
    // }
    // ================================================================================
}
```

project/build.gradle

```
// ================================================================================
// NeoForge 1.20.4 □□□□□ (build.gradle)
// NeoForge 1.20.4, Java 17
// ================================================================================
//
// □□ NeoForge 1.21.1 □□□□□□ - Gradle □□□
// 1. □Gradle □□□□!□
//    NeoForge 1.20.4: □□ NeoGradle □□ (net.neoforged.gradle.userdev)
//                     □□ 7.1.x (□□ Userdev □□)
//    NeoForge 1.21.1: □□ ModDevGradle □□ (net.neoforged.moddev)
//                     □□ 1.0.x (□□□□□, □□□ NeoForge 21+)
//    □□□□ModDevGradle □□□□□□ 1.20.4!
//             1.20.4 □□□□□ NeoGradle (net.neoforged.gradle.userdev)!
//
// 2. □Java □□□□!□
//    NeoForge 1.20.4: Java 17 (Mojang □ 1.20.4 □□□□ Java 17)
//    NeoForge 1.21.1: Java 21 (1.20.5+ □□□□ Java 21)
//
// 3. □NeoForge □□□□□□
//    1.20.4: 20.4.xxx (□ 20.4.251)
//    1.21.1: 21.1.xxx
//
// 4. □□□□□□□□□
//    NeoGradle (1.20.4): implementation "net.neoforged:neoforge:${neo_version}"
//    ModDevGradle (1.21.1): neoForge { neoForgeVersion = ... }
//
// □□ Forge 1.20.1 □□□□□□
// 1. Forge 1.20.1 □□ ForgeGradle □□ (net.minecraftforge.gradleuserdev)
// 2. NeoForge 1.20.4 □□ NeoGradle □□ (net.neoforged.gradle.userdev)
// 3. □□: Forge: net.minecraftforge:forge  vs  NeoForge: net.neoforged:neoforge
// 4. □□□□: NeoGradle □□ runs { } □ (□ ForgeGradle □□□□□□□□□)
// ================================================================================

plugins {
    id 'java-library'
    id 'maven-publish'
    // □□□□NeoGradle □□ - NeoForge 1.20.4 □□□ Gradle □□
    // □□ 7.1.x □□□ MC 1.20.4 / NeoForge 20.4.x
    // □□□□□ ModDevGradle (net.neoforged.moddev), □□□□ NeoForge 21+!
    // □□□□□ ForgeGradle (net.minecraftforge.gradle), □□ Forge □□!
    id 'net.neoforged.gradle.userdev' version '7.1.38'
}

tasks.named('wrapper', Wrapper).configure {
    distributionType = Wrapper.DistributionType.BIN
}

version = mod_version
group = mod_group_id
sourceSets.main.resources {
```

```
    // □□□□□□□□□□□□□□□□
    srcDir('src/generated/resources')
    // □□□□□□□□□□□□□□□□
    exclude("**/*.bbmodels")
    exclude("src/generated/**/.cache")
}

repositories {
    // □□□□□□□□□□□□□□□□
}

base {
    archivesName = mod_id
}

// ============================================================================
// □□□□Java 17 - □□ 21!
// Mojang □ MC 1.20.4 □□□□□□□□□ Java 17
// □□□□NeoForge 1.21.1 □□ Java 21 (1.20.5+ □□□)
// ============================================================================
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// ============================================================================
// □□□□□ (Access Transformer) - □□
// NeoGradle □□□□□□□ ForgeGradle □□
// ============================================================================
//minecraft.accessTransformers.file rootProject.file('src/main/resources/META-INF/accesstransformer.cfg')
//minecraft.accessTransformers.entry public net.minecraft.client.Minecraft textureManager # textureManager

// ============================================================================
// □□□□□ - □□□□□□□□□□□□ Minecraft
// □□□□NeoGradle □□ runs { } □□□□, ForgeGradle □□ minecraft { runs { } }
// ============================================================================
runs {
    // □□□□□□□□□□□□
    configureEach {
        // □□□□□ (REGISTRIES □□□□□□□□□□□□)
        systemProperty 'forge.logging.markers', 'REGISTRIES'
        // □□□□□□□□
        systemProperty 'forge.logging.console.level', 'debug'
        // □□□□□
        workingDirectory project.layout.projectDirectory.dir('run').dir(name)
        // □□□□□
        modSource project.sourceSets.main
    }
    // □□□□□□□□
    client {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
    }
    // □□□□□□□□
    server {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
        argument '--nogui'
    }
    // □□□□□□□□
    gameTestServer {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
    }
    // □□□□□□□□□□
    data {
        arguments.addAll '--mod', project.mod_id, '--all', '--output',
                file('src/generated/resources/').getAbsolutePath(), '--existing',
                file('src/main/resources/').getAbsolutePath()
    }
}

// ============================================================================
// □□□□□□□□□□ (□□□□□□)
// ============================================================================
configurations {
    runtimeClasspath.extendsFrom localRuntime
}

// ============================================================================
// □□
// ============================================================================
```

```
dependencies {
    // □□□□□□□NeoGradle □□□□□□□□
    // NeoForge 1.20.4: implementation "net.neoforged:neoforge:${neo_version}"
    // Forge 1.20.1:    implementation "net.minecraftforge:forge:${minecraft_version}-${forge_version}"
    // ModDevGradle (1.21.1): □□ neoForge { } □□□, □□□□□□
    implementation "net.neoforged:neoforge:${neo_version}"

    // JEI □□□□□ (□□)
    // compileOnly "mezz.jei:jei-${mc_version}-common-api:${jei_version}"
    // compileOnly "mezz.jei:jei-${mc_version}-neoforge-api:${jei_version}"
    // localRuntime "mezz.jei:jei-${mc_version}-neoforge:${jei_version}"
}

// ============================================================================
// □□□□□□□□□
// □ gradle.properties □□□□□□□ mods.toml □□ ${} □□□
// ============================================================================
tasks.withType(ProcessResources).configureEach {
    var replaceProperties = [
            minecraft_version      : minecraft_version,
            minecraft_version_range: minecraft_version_range,
            neo_version            : neo_version,
            loader_version_range   : loader_version_range,
            mod_id                 : mod_id,
            mod_name               : mod_name,
            mod_license            : mod_license,
            mod_version            : mod_version,
    ]
    inputs.properties replaceProperties
    // □□□□□1.20.4 □□□□□ mods.toml (□□ neoforge.mods.toml!)
    filesMatching(['META-INF/mods.toml']) {
        expand replaceProperties
    }
}

// Maven □□□□□ (□□)
publishing {
    publications {
        register('mavenJava', MavenPublication) {
            from components.java
        }
    }
    repositories {
        maven {
            url "file://${project.projectDir}/repo"
        }
    }
}

// □□□□□
tasks.withType(JavaCompile).configureEach {
    options.encoding = 'UTF-8'
}

// IDEA □□□□□□□□
idea {
    module {
        downloadSources = true
        downloadJavadoc = true
    }
}
```

project/mods.toml

```
# ============================================================================
# NeoForge 1.20.4 □□□□□□□□ (mods.toml)
# NeoForge 1.20.4, Java 17
# ============================================================================
#
# □□ NeoForge 1.21.1 □□□□□□ - □□□□!□
# 1. □□□□□□□□!□
#    NeoForge 1.20.4: mods.toml  (□□ src/main/resources/META-INF/mods.toml)
#    NeoForge 1.21.1: neoforge.mods.toml  (□□ src/main/resources/META-INF/neoforge.mods.toml)
#    □□□□□□ NeoForge 20.5 (MC 1.20.5) □, □□□□□□ mods.toml □□ neoforge.mods.toml!
#             1.20.4 □□□□□ mods.toml!
#
```

```
# 2. 对loaderVersion 的说明：
#    NeoForge 1.20.4: loaderVersion="[1,)" (FML 主版本 1)
#    NeoForge 1.21.1: loaderVersion="[4,)" (FML 主版本 4, 从 1.20.5+ 开始)
#
# 3. 关于 modId：
#    NeoForge 1.20.4: modId="neoforge" (全小写 neoforge)
#    NeoForge 1.21.1: modId="neoforge" (全小)
#
# 如果 Forge 1.20.1 的模组迁移：
# 1. Forge 1.20.1: 使用旧的 mods.toml, 不是这个文件
# 2. 模组加载器标识符: Forge 用 modId="forge", NeoForge 用 modId="neoforge"
# 3. NeoForge 的 mods.toml 支持 [[mixins]] 和 [[accessTransformers]] 块 (在 mods.toml 中定义)
# ============================================================================

# 模组加载器 - NeoForge 使用 javafml (Java FML 加载器)
modLoader="javafml"

# 加载器版本范围 - NeoForge 1.20.4 用 FML 主版本 1
# 如果是1.21.1 用 loaderVersion="[4,)" (主版本 4)
loaderVersion="${loader_version_range}"

# 许可证
license="${mod_license}"

# 问题跟踪 URL (可选)
#issueTrackerURL="https://change.me.to.your.issue.tracker.example.invalid/"

# ============================================================================
# 模组属性 - 可以定义多个 [[mods]] 块
# ============================================================================
[[mods]]
# 模组 ID - 必须与 @Mod 注解的值匹配
modId="${mod_id}"

# 模组版本 - 来自 gradle.properties
version="${mod_version}"

# 显示名称
displayName="${mod_name}"

# 更新检查 JSON URL (可选)
#updateJSONURL="https://change.me.example.invalid/updates.json"

# 模组主页 URL (可选)
#displayURL="https://change.me.to.your.mods.homepage.example.invalid/"

# 模组徽标 (可选, 位于此 JAR 的根目录)
#logoFile="examplemod.png"

# 模组致谢 (可选)
#credits=""

# 作者 (可选)
#authors="ExampleModder"

# 模组描述 (支持多行)
description='''
Example mod for NeoForge 1.20.4.
'''

# ============================================================================
# Mixin 配置文件 (可选)
# 在 mods.toml 中定义 mixin 配置文件, NeoForge 会自动加载
# 这是相对于 Forge 1.20.1 的新增功能！
# ============================================================================
#[[mixins]]
#config="${mod_id}.mixins.json"

# ============================================================================
# 访问转换器配置 (可选)
# ============================================================================
#[[accessTransformers]]
#file="META-INF/accesstransformer.cfg"

# ============================================================================
# 依赖项
```

```
# ============================================================================
# □□ NeoForge □□
[[dependencies.${mod_id}]]
    modId="neoforge"
    type="required"
    # □□□□NeoForge □□□□
    # 1.20.4: "[20.4,)" (NeoForge 20.4.x)
    # 1.21.1: "[21.1,)" (NeoForge 21.1.x)
    versionRange="[${neo_version},)"
    ordering="NONE"
    side="BOTH"

# □□ Minecraft □□
[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

# ============================================================================
# □□□□□ (□□)
# □□□□□□□□ Java □□□OpenGL □□□
# ============================================================================
#[features.${mod_id}]
#javaVersion="[17,)"  # NeoForge 1.20.4 □□ Java 17
#openGLVersion="[3.2,)"
```

recipe/shaped_recipe.json

```
{
    "_comment": "============================================================================",
    "_comment_2": "NeoForge 1.20.4 □□□□□□□□□ (shaped_recipe.json)",
    "_comment_3": "NeoForge 1.20.4, Java 17",
    "_comment_4": "============================================================================",
    "_comment_5": "□□□□□□□ JSON □□ - NeoForge/Forge/Fabric □□!",
    "_comment_6": "",
    "_comment_7": "□□ NeoForge 1.21.1 □□□□□",
    "_comment_8": "1. □□□□□□□: data/<modid>/recipes/<name>.json",
    "_comment_9": "2. □□□□□□□ - □□□□□□□□□□ 1.20.x □ 1.21.x □□□□□□□",
    "_comment_10": "3. □□: 1.21.x □□□□□□□□□□ recipe book □□, □ JSON □□□□",
    "_comment_11": "",
    "_comment_12": "□□ Forge 1.20.1 □□□□□",
    "_comment_13": "1. Forge 1.20.1 □□□□□□□□□□□□□",
    "_comment_14": "2. □□□□ Forge □□□□□□□□ (conditions), NeoForge □□□□□□□",
    "_comment_15": "3. NeoForge □□ recipe_conditions (□ mod_loaded, item_exists □)",
    "_comment_16": "============================================================================",

    "type": "minecraft:crafting_shaped",
    "pattern": [
        "SSS",
        "S S",
        "SSS"
    ],
    "key": {
        "S": {
            "item": "examplemod:example_item"
        }
    },
    "result": {
        "item": "examplemod:example_block",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    "_comment": "============================================================================",
    "_comment_2": "NeoForge 1.20.4 □□□□□□□ (block_tag.json)",
    "_comment_3": "NeoForge 1.20.4, Java 17",
    "_comment_4": "============================================================================",
    "_comment_5": "□□□□□□□ JSON □□ - NeoForge/Forge/Fabric □□!",
    "_comment_6": "",
    "_comment_7": "□□: data/<modid>/tags/blocks/<tag_name>.json",
    "_comment_8": "(□□: □□□□□□ blocks, □□ block_tags)",
    "_comment_9": "",
```

```
    "_comment_10": "□□ NeoForge 1.21.1 □□□□□",
    "_comment_11": "1. □□□□□□□□ - □□□□□□ 1.20.x □ 1.21.x □□□□□□□",
    "_comment_12": "2. 1.21.x □□□□□□ remove □□□□□□□□□, □ 1.20.4 □□□□",
    "_comment_13": "",
    "_comment_14": "□□ Forge 1.20.1 □□□□□",
    "_comment_15": "1. Forge 1.20.1 □□□□□□□□□□□□□□",
    "_comment_16": "2. □□: Forge □□ Tags.Blocks / Tags.Items (□ ORES, INGOTS □)",
    "_comment_17": "   NeoForge □□ Tags.Blocks / Tags.Items (□□□□□, □□□□□)",
    "_comment_18": "   Forge:     net.minecraftforge.common.Tags",
    "_comment_19": "   NeoForge:  net.neoforged.neoforge.common.Tags",
    "_comment_20": "3. NeoForge □ Forge □□□□□□□ replace □ remove □□",
    "_comment_21": "================================================================================",

    "replace": false,
    "values": [
        "minecraft:stone",
        "minecraft:cobblestone",
        "minecraft:granite",
        "minecraft:diorite",
        "minecraft:andesite",
        "#minecraft:base_stone_overworld",
        {
            "id": "examplemod:example_block",
            "required": false
        },
        {
            "id": "#minecraft:mineable/pickaxe",
            "required": true
        }
    ]
}
```

worldgen/biome_modifier.json

```
{
    "_comment": "================================================================================",
    "_comment_2": "NeoForge 1.20.4 □□□□□□□□□□ (biome_modifier.json)",
    "_comment_3": "NeoForge 1.20.4, Java 17",
    "_comment_4": "================================================================================",
    "_comment_5": "□□□□□□□□ - □□□□□□□□□□□□/□□□□□□□□□□□□□□",
    "_comment_6": "",
    "_comment_7": "□□□□: data/<modid>/neoforge/biome_modifier/<name>.json",
    "_comment_8": "",
    "_comment_9": "□□ NeoForge 1.21.1 □□□□□□ - □□!□",
    "_comment_10": "1. □□□□□□!□",
    "_comment_11": "   NeoForge 1.20.4: type □□ neoforge: □□ (□ neoforge:add_features)",
    "_comment_12": "   NeoForge 1.21.1: type □□ neoforge: □□ (□□)",
    "_comment_13": "   □□□□□□ neoforge: □□, □□□ forge: □□!",
    "_comment_14": "",
    "_comment_15": "2. □□□□□□□□!□",
    "_comment_16": "   NeoForge 1.20.4: data/<modid>/neoforge/biome_modifier/<name>.json",
    "_comment_17": "   NeoForge 1.21.1: data/<modid>/neoforge/biome_modifier/<name>.json",
    "_comment_18": "",
    "_comment_19": "□□ Forge 1.20.1 □□□□□□ - □□□□□□!□",
    "_comment_20": "1. □□□□□□!□",
    "_comment_21": "   Forge 1.20.1: type □□ forge: □□ (□ forge:add_features)",
    "_comment_22": "   NeoForge 1.20.4: type □□ neoforge: □□ (□ neoforge:add_features)",
    "_comment_23": "   □□□□Forge □ forge: □□, NeoForge □ neoforge: □□!",
    "_comment_24": "",
    "_comment_25": "2. □□□□□□□□!□",
    "_comment_26": "   Forge 1.20.1: data/<modid>/forge/biome_modifier/<name>.json",
    "_comment_27": "   NeoForge 1.20.4: data/<modid>/neoforge/biome_modifier/<name>.json",
    "_comment_28": "   □□□□Forge □ forge/ □□, NeoForge □ neoforge/ □□!",
    "_comment_29": "",
    "_comment_30": "3. □□□□□□□□□□□□",
    "_comment_31": "   Forge 1.20.1 □ NeoForge 1.20.4 □□□□□□□:",
    "_comment_32": "   - forge:add_features / neoforge:add_features: □□□□",
    "_comment_33": "   - forge:add_spawns / neoforge:add_spawns: □□□□□□",
    "_comment_34": "   - forge:remove_features / neoforge:remove_features: □□□□",
    "_comment_35": "   - forge:remove_spawns / neoforge:remove_spawns: □□□□□□",
    "_comment_36": "   - forge:replace_biome / neoforge:replace_biome: □□□□□□ (□□□)",
    "_comment_37": "================================================================================",

    "type": "neoforge:add_features",
    "biomes": "#minecraft:is_overworld",
    "features": "examplemod:ore_example",
```

```
    "step": "underground_ores"
}
```

NeoForge 1.20.6

block/simple_block.java

```
// ================================================================================
// NeoForge 1.20.6 □□□□□□□ (simple_block.java)
// NeoForge 1.20.6, Java 21
// ================================================================================
//
// □□ NeoForge 1.20.4 □□□□□□□
// 1. □Java □□□□!□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21 (1.20.5+ □□□□ Java 21!)
//
// 2. □BlockEntity □□/□□□□□□□□!□
//    NeoForge 1.20.4: saveAdditional(CompoundTag tag) - □ HolderLookup.Provider □□
//                     load(CompoundTag tag) - □□□□ load, □□ loadAdditional!
//                     getUpdateTag() - □□□
//    NeoForge 1.20.6: saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
//                     loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
//                     getUpdateTag(HolderLookup.Provider registries)
//    □□□□1.20.5+ □□□□□□ HolderLookup.Provider □□, □□□□□□ load □□ loadAdditional!
//
// 3. □ItemStack □□□□□□!□
//    NeoForge 1.20.4: ItemStack NBT □□ (getOrCreateTag() □)
//    NeoForge 1.20.6: Data Components □□ (1.20.5+ □□, NBT □□□!)
//
// 4. □ResourceLocation □□□□ - □□□□□□
//    NeoForge 1.20.4: new ResourceLocation("modid", "path") (□□□□□□)
//    NeoForge 1.20.6: new ResourceLocation("modid", "path") □□□□
//                     (1.21.x □□□□□□□□□□, □□□□□□)
//                     □□□□□□□□ ResourceLocation.fromNamespaceAndPath()
//
// 5. □Block □□□□□□□!□
//    NeoForge 1.20.6: □ MapCodec BlockType □□, Block □□□□□ codec()
//    NeoForge 1.21.1: □□□ MapCodec □□□□□□□, □□ codec() □□
//
// □□ NeoForge 1.21.1 □□□□□□□
// 1. □Block □□□□□ - 1.21.x □□□
//    NeoForge 1.20.6: Block □□□ codec() □□
//    NeoForge 1.21.1: □□□□□ codec() □□□ BlockType
//
// 2. □ResourceLocation □□□□□
//    NeoForge 1.20.6: new ResourceLocation() □□□□ (□□□□)
//    NeoForge 1.21.1: □□□□□□□□, □□□□□□□□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □DeferredRegister.Blocks □□□□
//    Forge 1.20.1:  □□□□, □□ DeferredRegister.create(ForgeRegistries.BLOCKS, MODID)
//    NeoForge 1.20.6: □□ DeferredRegister.Blocks, □□ DeferredRegister.createBlocks(MODID)
//                     □□ DeferredBlock<T> (□□ RegistryObject<Block>)
//
// 3. □□□□□□
//    Forge 1.20.1:  □□□□□□ ForgeRegistries □ (□ BLOCK_ENTITY_TYPES)
//    NeoForge 1.20.6: □□□□ net.minecraft.core.registries.Registries
//
// 4. □RegistryObject vs DeferredHolder□
//    Forge 1.20.1:  □□ RegistryObject<T>
//    NeoForge 1.20.6: □□ DeferredHolder<R, T> □ DeferredBlock<T>
// ================================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.HolderLookup;
import net.minecraft.core.registries.Registries;
```

```java
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.protocol.Packet;
import net.minecraft.network.protocol.game.ClientGamePacketListener;
import net.minecraft.network.protocol.game.ClientboundBlockEntityDataPacket;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.world.level.Level;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.EntityBlock;
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraft.world.level.block.entity.BlockEntityTicker;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.block.state.BlockState;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.MapColor;
import net.minecraft.sounds.SoundType;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;
import net.neoforged.neoforge.registries.DeferredBlock;

import javax.annotation.Nullable;

// ============================================================================
// □□□□□
// ============================================================================
public class ModBlocks {

    // ========================================================================
    // □□□□□□□ DeferredRegister.Blocks □□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCKS, MODID)
    //                □□ DeferredRegister<Block>, register() □□ RegistryObject<Block>
    // NeoForge 1.20.6: DeferredRegister.createBlocks(MODID)
    //                  □□ DeferredRegister.Blocks, register() □□ DeferredBlock<T>
    //
    // DeferredBlock<T> □□□ DeferredHolder<Block, T>, □□□□□□□□□□□
    // (□ NeoForge 1.21.1 □□□□□)
    // ========================================================================
    public static final DeferredRegister.Blocks BLOCKS =
            DeferredRegister.createBlocks(ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, MODID)
    // NeoForge 1.20.6: DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, MODID)
    //
    // □□□□□□□□□□:
    //   Forge: ForgeRegistries (Forge □□□□□□□)
    //   NeoForge: Registries (□□□□□□)
    //
    // □□□□□□□□□□: BLOCK_ENTITY_TYPE (□□ BLOCK_ENTITIES)
    // ========================================================================
    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITY_TYPES =
            DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□
    // □□□□□□□□□:
    //   Forge 1.20.1: RegistryObject<Block>
    //   NeoForge 1.20.6: DeferredBlock<Block>
    //
    // BlockBehaviour.Properties.of() □□□□□□□
    // - .mapColor() □□□□□□□
    // - .destroyTime() □□□□
    // - .explosionResistance() □□□□
    // - .sound() □□□□
    // - .lightLevel() □□□□ (□□□)
    // ========================================================================
    public static final DeferredBlock<Block> EXAMPLE_BLOCK = BLOCKS.register(
            "example_block",
            () -> new Block(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.STONE)
                    .destroyTime(2.0f)
                    .explosionResistance(10.0f)
                    .sound(SoundType.GRAVEL)
                    .lightLevel(state -> 7)
            )
    );
```

```
    // ========================================================================
    // □□□□□□□□□□□□□□□ (EntityBlock)
    // ========================================================================
    public static final DeferredBlock<ExampleEntityBlock> EXAMPLE_ENTITY_BLOCK = BLOCKS.register(
            "example_entity_block",
            () -> new ExampleEntityBlock(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.METAL)
                    .destroyTime(3.0f)
                    .explosionResistance(15.0f)
                    .sound(SoundType.METAL)
            )
    );

    // ========================================================================
    // □□□□□□□□□□
    // □□ BlockEntityType.Builder.of(supplier, blocks...).build(null)
    //
    // build(null) - □□ null □□□□□□□□ DataFixer
    // ========================================================================
    public static final DeferredHolder<BlockEntityType<?>, BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
            BLOCK_ENTITY_TYPES.register(
                    "example_block_entity",
                    () -> BlockEntityType.Builder.of(
                            ExampleBlockEntity::new,
                            EXAMPLE_ENTITY_BLOCK.get()
                    )
                    .build(null)
            );
}

// ========================================================================
// □□□□□□□□□□□
// □□ EntityBlock □□, □□ newBlockEntity() □□
// ========================================================================
class ExampleEntityBlock extends Block implements EntityBlock {

    public ExampleEntityBlock(BlockBehaviour.Properties properties) {
        super(properties);
    }

    // □□□□□□□□□□
    @Override
    public BlockEntity newBlockEntity(BlockPos pos, BlockState state) {
        return new ExampleBlockEntity(pos, state);
    }

    // ========================================================================
    // □□ Ticker (□ tick □□□□)
    // ========================================================================
    @Nullable
    @Override
    public <T extends BlockEntity> BlockEntityTicker<T> getTicker(
            Level level, BlockState state, BlockEntityType<T> type) {
        return type == ModBlocks.EXAMPLE_BLOCK_ENTITY.get()
                ? (lvl, pos, st, be) -> ExampleBlockEntity.tick(lvl, pos, st, (ExampleBlockEntity) be)
                : null;
    }

    // ========================================================================
    // □□□□1.20.6 □ Block □□□□!
    //
    // NeoForge 1.21.1 □□□□□ codec() □□□□□ BlockType:
    //   @Override
    //   public MapCodec<? extends Block> codec() {
    //       return CODEC.value();
    //   }
    //
    // NeoForge 1.20.6 □□□□! Block □□□ codec() □□□
    // ========================================================================
}

// ========================================================================
// □□□□□□
// □□□□□□□1.20.6 □□□/□□□□□□!
//
// NeoForge 1.20.4:
```

```
//   saveAdditional(CompoundTag tag) - 带 HolderLookup.Provider 参数
//   load(CompoundTag tag) - 现在叫 load (不是 loadAdditional!)
//   getUpdateTag() - 无参数
//
// NeoForge 1.20.6 (1.20.5+ 过渡):
//   saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
//   loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
//   getUpdateTag(HolderLookup.Provider registries)
//   方法名和旧版一样了, 但现在是 load 不是 loadAdditional!
//
// NeoForge 1.21.1:
//   和 1.20.6 相同 (saveAdditional/loadAdditional + HolderLookup.Provider)
// ============================================================================
class ExampleBlockEntity extends BlockEntity {

    private int value = 0;

    public ExampleBlockEntity(BlockPos pos, BlockState state) {
        super(ModBlocks.EXAMPLE_BLOCK_ENTITY.get(), pos, state);
    }

    // ========================================================================
    // 读取数据 (从 NBT)
    // 旧版本1.20.6: 方法名 loadAdditional, 带 HolderLookup.Provider 参数
    //         1.20.4: 方法名 load, 无 HolderLookup.Provider 参数
    //
    // 这个方法在 1.20.5+ 被重命名!
    // ========================================================================
    @Override
    public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.loadAdditional(tag, registries);
        this.value = tag.getInt("value");
    }

    // ========================================================================
    // 写入数据 (到 NBT)
    // 旧版本1.20.6: 带 HolderLookup.Provider 参数
    //         1.20.4: 无 HolderLookup.Provider 参数
    // ========================================================================
    @Override
    public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.saveAdditional(tag, registries);
        tag.putInt("value", this.value);
    }

    // ========================================================================
    // 客户端同步 - 获取更新标签
    // 旧版本1.20.6: getUpdateTag(HolderLookup.Provider registries) 带参数
    //         1.20.4: getUpdateTag() 无参数
    // ========================================================================
    @Override
    public CompoundTag getUpdateTag(HolderLookup.Provider registries) {
        CompoundTag tag = new CompoundTag();
        saveAdditional(tag, registries);
        return tag;
    }

    // 客户端同步 - 更新数据包
    @Override
    public Packet<ClientGamePacketListener> getUpdatePacket() {
        return ClientboundBlockEntityDataPacket.create(this);
    }

    // ========================================================================
    // 可 tick 的方法示例
    // ========================================================================
    public static void tick(Level level, BlockPos pos, BlockState state, ExampleBlockEntity blockEntity) {
        if (!level.isClientSide) {
            // 服务端逻辑 (例如: 生产物品等)
            // blockEntity.value++;
            // blockEntity.setChanged(); // 标记需要保存, 谨慎使用
        }
    }
}
```

command/simple_command.java

```java
// ========================================================================
// NeoForge 1.20.6 □□□□□□□ (simple_command.java)
// NeoForge 1.20.6, Java 21
// ========================================================================
//
// □□ NeoForge 1.20.4 □□□□□□ - □□□□□□
// 1. □Java □□□□□□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 2. □□□□□□□□□□!□
//    NeoForge 1.20.4 □ 1.20.6 □□□ API □□□□:
//    - □□□ RegisterCommandsEvent
//    - □□□ Bus.FORGE (□□□□□□□)
//    - Brigadier API □□
//
// □□ NeoForge 1.21.1 □□□□□□□
// 1. □□□□□□□□□□!□
//    NeoForge 1.20.6: bus = Mod.EventBusSubscriber.Bus.FORGE (□□□□□□□)
//                     □□□□1.20.6 □□□ Bus.FORGE!
//    NeoForge 1.21.1: bus = Mod.EventBusSubscriber.Bus.GAME (□□□□□□□)
//                      Bus.FORGE □□□□□□ Bus.GAME!
//
// 2. □□□□□□□□□ - □□□
//    NeoForge 1.20.6: RegisterCommandsEvent (net.neoforged.neoforge.event.RegisterCommandsEvent)
//    NeoForge 1.21.1: RegisterCommandsEvent (net.neoforged.neoforge.event.RegisterCommandsEvent)
//    (□□□□□□□□)
//
// 3. □Brigadier API - □□□□□
//    □□□□ Commands.literal(), Commands.argument() □□
//    CommandSourceStack □□□□□□□□□□□
//
// 4. □sendSuccess □□ - □□□□□
//    □□□ Supplier<Component> □ boolean □□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□□□
//    Forge 1.20.1:   net.minecraftforge.event.RegisterCommandsEvent
//    NeoForge 1.20.6: net.neoforged.neoforge.event.RegisterCommandsEvent
//    (□□□□□□□□□□)
//
// 2. □@Mod □□□ @SubscribeEvent □□□
//    Forge 1.20.1:   net.minecraftforge.fml.common.Mod
//                    net.minecraftforge.eventbus.api.SubscribeEvent
//    NeoForge 1.20.6: net.neoforged.fml.common.Mod
//                     net.neoforged.bus.api.SubscribeEvent
//
// 3. □□□□□□□□□
//    Forge 1.20.1:   MinecraftForge.EVENT_BUS
//    NeoForge 1.20.6: NeoForge.EVENT_BUS (net.neoforged.neoforge.NeoForge)
//
// 4. □@Mod.EventBusSubscriber □ bus □□□
//    Forge 1.20.1:   Bus.FORGE
//    NeoForge 1.20.6: Bus.FORGE (□□!)
//    NeoForge 1.21.1: Bus.GAME (□□□!)
// ========================================================================

package com.example.examplemod.command;

import com.example.examplemod.ExampleMod;
import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.arguments.StringArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.commands.arguments.coordinates.BlockPosArgument;
import net.minecraft.core.BlockPos;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.Entity;
import net.neoforged.bus.api.SubscribeEvent;
import net.neoforged.fml.common.Mod;
```

```java
import net.neoforged.neoforge.event.RegisterCommandsEvent;

// ==========================================================================
// □□□□□□
// ==========================================================================
// □□□□□RegisterCommandsEvent □ NeoForge □□□□□□□□□□□
// □□ @Mod.EventBusSubscriber □□□□□
//
// □□□□□NeoForge 1.20.6: bus = Mod.EventBusSubscriber.Bus.FORGE (□□ FORGE!)
//         NeoForge 1.21.1: bus = Mod.EventBusSubscriber.Bus.GAME (□□□□ GAME!)
//         Forge 1.20.1:    bus = Mod.EventBusSubscriber.Bus.FORGE (□□)
@Mod.EventBusSubscriber(modid = ExampleMod.MOD_ID, bus = Mod.EventBusSubscriber.Bus.FORGE)
public class ModCommands {

    // ==========================================================================
    // □□□□□ - □□ RegisterCommandsEvent
    //
    // □□□□□RegisterCommandsEvent □ NeoForge □□□□□□□□□□□
    // (NeoForge.EVENT_BUS), □□□□□□□□□□ (modEventBus)
    //
    // □□□□□:
    //   @Mod.EventBusSubscriber(modid = MODID, bus = Bus.FORGE)
    //   + @SubscribeEvent □□□□□
    //   □□□□□□□□□□□□□: NeoForge.EVENT_BUS.register(new ModCommands());
    //
    // □□□□□□□□□□□:
    //   Forge 1.20.1: MinecraftForge.EVENT_BUS
    //   NeoForge 1.20.6: NeoForge.EVENT_BUS (net.neoforged.neoforge.NeoForge)
    // ==========================================================================
    @SubscribeEvent
    public static void register(RegisterCommandsEvent event) {
        // ======================================================================
        // □□1: /examplemod hello
        // □□□□□, □□□□, □□□□□ 0 (□□□□□)
        // ======================================================================
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        .requires(src -> src.hasPermission(0))
                        .then(Commands.literal("hello")
                                .executes(ModCommands::helloCommand)
                        )
                        .executes(context -> {
                            context.getSource().sendSuccess(
                                    () -> Component.literal("ExampleMod command!"),
                                    false
                            );
                            return Command.SINGLE_SUCCESS;
                        })
        );

        // ======================================================================
        // □□2: /examplemod setvalue <player> <value>
        // □□□□□□□, □□□□□ 2 (OP)
        // ======================================================================
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        .requires(src -> src.hasPermission(2))
                        .then(Commands.literal("setvalue")
                                .then(Commands.argument("target", EntityArgument.player())
                                        .then(Commands.argument("value", IntegerArgumentType.integer(0, 100))
                                                .executes(ModCommands::setValueCommand)
                                        )
                                )
                        )
        );

        // ======================================================================
        // □□3: /examplemod message <message...>
        // □□□□□□□□ (□□□□□)
        // ======================================================================
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        .requires(src -> src.hasPermission(0))
                        .then(Commands.literal("message")
                                .then(Commands.argument("message", StringArgumentType.greedyString())
                                        .executes(context -> {
```

```
                                    String msg = StringArgumentType.getString(context, "message");
                                    context.getSource().sendSuccess(
                                            () -> Component.literal("Message: " + msg),
                                            false
                                    );
                                    return Command.SINGLE_SUCCESS;
                                })
                        )
                )
        );

        // ========================================================================
        // □□4: /examplemod teleport <position>
        // □□□□□□□, □□□□□ 2 (OP)
        // ========================================================================
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        .requires(src -> src.hasPermission(2))
                        .then(Commands.literal("teleport")
                                .then(Commands.argument("pos", BlockPosArgument.blockPos())
                                        .executes(context -> {
                                            BlockPos pos = BlockPosArgument.getBlockPos(context, "pos");
                                            Entity executor = context.getSource().getEntityOrPlayer();
                                            if (executor != null) {
                                                executor.teleportTo(
                                                        pos.getX() + 0.5,
                                                        pos.getY(),
                                                        pos.getZ() + 0.5
                                                );
                                                context.getSource().sendSuccess(
                                                        () -> Component.literal(
                                                                "Teleported to " + pos.toShortString()
                                                        ),
                                                        true
                                                );
                                            }
                                            return Command.SINGLE_SUCCESS;
                                        })
                                )
                        )
        );
    }

    // ========================================================================
    // hello □□□□□□□□
    // ========================================================================
    private static int helloCommand(CommandContext<CommandSourceStack> context) {
        CommandSourceStack source = context.getSource();

        // sendSuccess □□ Supplier<Component> (□□□□□, □□□□□)
        // 1.20.6 □ 1.21.1 □□□□
        source.sendSuccess(
                () -> Component.literal("Hello from ExampleMod!"),
                false  // false = □□□□□□□ OP □□□
        );

        // □□□□□□ (□□□□□□□□□□□□□)
        Entity entity = source.getEntity();
        if (entity != null) {
            source.sendSuccess(
                    () -> Component.literal("Executed by: " + entity.getName().getString()),
                    false
            );
        }

        return Command.SINGLE_SUCCESS;
    }

    // ========================================================================
    // setvalue □□□□□□□□
    // ========================================================================
    private static int setValueCommand(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        CommandSourceStack source = context.getSource();

        Entity target = EntityArgument.getPlayer(context, "target");
```

```
        int value = IntegerArgumentType.getInteger(context, "value");

        source.sendSuccess(
                () -> Component.literal(
                        "Set " + target.getName().getString() + "'s value to " + value
                ),
                true  // true = □□□ OP □□
        );

        return value;
    }
}
```

enchantment/custom_enchantment.java

```
// ========================================================================
// NeoForge 1.20.6 □□□□□□□□ (custom_enchantment.java)
// NeoForge 1.20.6, Java 21
// ========================================================================
//
// □□ NeoForge 1.20.4 □□□□□□ - □□□□□□
// 1. □Java □□□□□□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 2. □□□□□□□□ - □□□□!□
//    NeoForge 1.20.4: □□ DeferredRegister □□□□ (□□□□)
//                     DeferredRegister.create(Registries.ENCHANTMENT, MODID)
//                     □□□□ Enchantment □□□□□□□□□
//    NeoForge 1.20.6: □□! □□□□ DeferredRegister + □□ Enchantment □
//
// 3. □□□□□□□□ - □□□□!□
//    □□□□□ Enchantment □□□□□□□□□□:
//    - getMaxLevel()
//    - getMinCost() / getMaxCost()
//    - checkCompatibility()
//    - canEnchant()
//    - doPostAttack() □
//
// 4. □□□□□□□□ - □□!□
//    NeoForge 1.20.4: □□□□□ ItemStack NBT □ (Enchantments tag)
//    NeoForge 1.20.6: □□□□□ DataComponents.ENCHANTMENTS □□□□□
//                    (1.20.5+ □□ Data Components, NBT □□□!)
//    □□□□□□□□□□□□□□□, □□□□□□□□□□□□□□□!
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□□□□!□
//
// 1. □□□□□□□□□□□□□!□
//    NeoForge 1.20.6: □□ DeferredRegister □□□□ (□□□□!)
//                     DeferredRegister.create(Registries.ENCHANTMENT, MODID)
//                     □□□□ Enchantment □□□□□□□□□
//                     □ Forge 1.20.1 / 1.20.4 □□□□!
//    NeoForge 1.21.1: □□□□□□□□□□ (datapack registry)!
//                     □□□ DeferredRegister □□!
//                     □□ RegistrySetBuilder □□□□□□□□□□□
//                     □□□□□□□□□, □□□□□□□□□□ (DataComponentType) □□
//
// 2. □□□□□□□□□□□□□!□
//    NeoForge 1.20.6: □□□□ Enchantment □□□□□□□□□□:
//                     - getMaxLevel()
//                     - getMinCost() / getMaxCost()
//                     - checkCompatibility()
//                     - canEnchant()
//                     - doPostAttack() □
//                     □ Forge 1.20.1 □□□□!
//    NeoForge 1.21.1: □□□□□□□□□□ (Enchantment Effect Components) □□:
//                     - □□□□□□□, □□□□ DataComponentMap □□□□
//                     - □□ EnchantmentEffectComponents (□ DAMAGE, KNOCKBACK □)
//                     - □□□□□□□□□□□□□□□
//
// 3. □EnchantmentTarget □□ - 1.20.6 □□□!□
//    NeoForge 1.20.6: EnchantmentCategory □□□□□□ (WEAPON, ARMOR, DIGGER □)
//    NeoForge 1.21.1: EnchantmentCategory □□□□□, □□□□□□□□□□□□□□□
//
// 4. □Enchantments □□□□□□□□□!□
//    NeoForge 1.20.6: Enchantments.SHARPNESS □□ Enchantment □□
```

```
//    NeoForge 1.21.1: Enchantments.SHARPNESS □□ ResourceKey<Enchantment>
//
// 5. □EnchantmentHelper □□□
//    NeoForge 1.20.6: EnchantmentHelper.getItemEnchantmentLevel(enchantment, stack)
//                     □□□□□□□□□
//    NeoForge 1.21.1: EnchantmentHelper.runIterationOnItem(stack, visitor)
//                     □□□□□□□□□, □□□□□□□□□
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □□□□□□□□
//    Forge 1.20.1:  ForgeRegistries.ENCHANTMENTS
//    NeoForge 1.20.6: Registries.ENCHANTMENT (□□□□□□)
//
// 3. □□□□□□□
//    Forge 1.20.1:  RegistryObject<Enchantment>
//    NeoForge 1.20.6: DeferredHolder<Enchantment, Enchantment> □ Supplier<Enchantment>
//
// 4. □□□ API - □□□□□!□
//    Forge 1.20.1 □ NeoForge 1.20.6 □□□ API □□□□:
//    - Enchantment □: Rarity, EnchantmentCategory, EquipmentSlot[]
//    - □□□□: getMaxLevel, getMinCost, getMaxCost, checkCompatibility, etc.
//    - □□□□: doPostAttack, doPostHurt □
// ============================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.world.entity.EquipmentSlot;
import net.minecraft.world.entity.LivingEntity;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.world.item.enchantment.EnchantmentCategory;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

import java.util.function.Supplier;

// ============================================================================
// □□□□□
// ============================================================================
public class ModEnchantments {

    // ========================================================================
    // □□□□□□□□□□ - □□□□!
    //
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID)
    // NeoForge 1.20.6: DeferredRegister.create(Registries.ENCHANTMENT, MODID)
    //
    // □□□□□□NeoForge 1.21.1:
    //   □□□□□□□□□□, □□□ DeferredRegister □□!
    //   □□□□ RegistrySetBuilder + □□□□ (□□ JSON □□)
    //   □□ ResourceKey<Enchantment> □□□□
    //
    // □□□□1.20.6 □□□□ DeferredRegister, □□□□ 1.21.x □□□□□□□!
    // ========================================================================
    public static final DeferredRegister<Enchantment> ENCHANTMENTS =
            DeferredRegister.create(Registries.ENCHANTMENT, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□
    //
    // Enchantment □□□□ (1.20.6 □□ API, □ Forge 1.20.1 □□):
    //   - Rarity rarity: □□□ (COMMON=10 / UNCOMMON=5 / RARE=2 / VERY_RARE=1)
    //   - EnchantmentCategory category: □□□□□□□□
    //   - EquipmentSlot[] slots: □□□□□□□□
    //
    // □□□□NeoForge 1.21.1 □:
    //   □□□□□□□□□□□□□□□!
    //   □□□□ new Enchantment(description, definition, exclusiveSet, effects)
    //   □□ EnchantmentDefinition □ DataComponentMap □□
    //
```

```
    // EnchantmentCategory □□□□ (1.20.6 □□□□!):
    //   - ALL: □□□□□
    //   - ARMOR: □□/□□/□□/□□
    //   - ARMOR_FEET / ARMOR_LEGS / ARMOR_CHEST / ARMOR_HEAD: □□□□□□
    //   - WEAPON: □ (□□□□□)
    //   - FISHING_ROD: □□□
    //   - BREAKABLE: □□□□□ (□□)
    //   - DIGGER: □/□/□/□
    //   - BOW: □
    //   - CROSSBOW: □
    //   - TRIDENT: □□□
    //   - VANISHABLE: □□□□□
    //
    // □□□□1.21.x □ EnchantmentCategory □□□□□□!
    // ============================================================================
    public static final Supplier<Enchantment> EXAMPLE_ENCHANTMENT =
            ENCHANTMENTS.register("example_enchantment",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.RARE,
                            EnchantmentCategory.WEAPON,
                            EquipmentSlot.MAINHAND,
                            EquipmentSlot.OFFHAND
                    )
            );

    // ============================================================================
    // □□□□□□□□□ (□□□)
    // ============================================================================
    public static final Supplier<Enchantment> EXAMPLE_PROTECTION =
            ENCHANTMENTS.register("example_protection",
                    () -> new ExampleEnchantment(
                            Enchantment.Rarity.COMMON,
                            EnchantmentCategory.ARMOR,
                            EquipmentSlot.values()  // □□□□□□□□
                    )
            );
}

// ============================================================================
// □□□□□□□□
// □□ Enchantment □□□□□□□□□□□□□□□
//
// □□□□□□□□ 1.20.6 □□□□□□□□□!
//          □ Forge 1.20.1 / 1.20.4 □□□□□!
//          1.21.1 □□□□□□□□□□□□ (□□□□□□□□ + □□□□□□)
// ============================================================================
class ExampleEnchantment extends Enchantment {

    public ExampleEnchantment(Rarity rarity, EnchantmentCategory category,
                              EquipmentSlot... slots) {
        super(rarity, category, slots);
    }

    // ============================================================================
    // □□□□□□□□□□ (□□□□□□□□□□)
    // ============================================================================
    @Override
    public int getMinCost(int level) {
        return 10 * level;
    }

    // ============================================================================
    // □□□□□□□□□□
    // ============================================================================
    @Override
    public int getMaxCost(int level) {
        return getMinCost(level) + 15;
    }

    // ============================================================================
    // □□□□□□□□
    // ============================================================================
    @Override
    public int getMaxLevel() {
        return 3;  // 3□□□
    }
```

```
// ========================================================================
// □□□□□□□□□□□□
// ========================================================================
@Override
protected boolean checkCompatibility(Enchantment other) {
    // □□□□□□
    return super.checkCompatibility(other)
            && other != net.minecraft.world.item.enchantment.Enchantments.SHARPNESS;
}

// ========================================================================
// □□□□□□□□□□□
// ========================================================================
@Override
public boolean canEnchant(ItemStack stack) {
    return super.canEnchant(stack);
}

// ========================================================================
// □□□□□□□□□ (□□□□□)
// ========================================================================
@Override
public boolean isCurse() {
    return false;
}

// ========================================================================
// □□□□□□□□□□□□
// ========================================================================
@Override
public boolean isTradeable() {
    return true;
}

// ========================================================================
// □□□□□□□□□□□
// ========================================================================
@Override
public boolean isDiscoverable() {
    return true;
}

// ========================================================================
// □□□□□□□□□□
// ========================================================================
@Override
public boolean isAllowedOnBooks() {
    return true;
}

// ========================================================================
// □□□□□□ (□□□□□□)
//
// □□□□□NeoForge 1.20.6: □□□□□ doPostAttack() □□□□□
//         NeoForge 1.21.1: □□□□□□□□□□ (EnchantmentEffectComponents) □□
//                          □□□□□□□!
// ========================================================================
@Override
public void doPostAttack(LivingEntity attacker, net.minecraft.world.entity.Entity target, int level) {
    // □□: □□□□□□□□□□
    if (target instanceof LivingEntity livingTarget) {
        livingTarget.addEffect(new net.minecraft.world.effect.MobEffectInstance(
                net.minecraft.world.effect.MobEffects.POISON,
                60,              // □□□□ (tick, 60 = 3□)
                level - 1         // □□□□ (□□□□ - 1)
        ));
    }
    super.doPostAttack(attacker, target, level);
}

// ========================================================================
// □□□□□□□
//
// □□□□□□□□□□□□□□□□□□:
//
// NeoForge 1.20.6 (□ Forge 1.20.1 □□):
```

```
    //   int level = EnchantmentHelper.getItemEnchantmentLevel(
    //       ModEnchantments.EXAMPLE_ENCHANTMENT.get(), stack);
    //   □□□□□□□□□□
    //
    // NeoForge 1.21.1:
    //   □□□□□□□□□□, □□□□□□□□□□:
    //   EnchantmentHelper.runIterationOnItem(stack, (holder, level) -> {
    //       // holder.value().effects().get(component)
    //   });
    //
    // □□□□□□□□□□:
    //   1.20.6: DataComponents.ENCHANTMENTS (1.20.5+ □□ Data Components)
    //     stack.get(DataComponents.ENCHANTMENTS) -> ItemEnchantments
    //   1.21.1: DataComponents.ENCHANTMENTS (□□)
    //
    // □□□□□□□ 1.20.6 □□□□ DataComponents □□□□□,
    //         □ EnchantmentHelper □ API □□□□□□□□□!
    // ================================================================================
}

// ================================================================================
// □□□□□□ - NeoForge 1.20.6
// ================================================================================
// NeoForge 1.20.6 □□□□□□□□□□□□□□:
//
// @Mod.EventBusSubscriber(modid = ExampleMod.MOD_ID, bus = Mod.EventBusSubscriber.Bus.FORGE)
// public static class GameEvents {
//     @SubscribeEvent
//     public static void onEnchantLevel(net.neoforged.neoforge.event.enchantment.EnchantmentLevelSetEvent event) {
//         // event.getLevel() □□□□□□□
//         // event.getEnchantments() □□□□□ Map
//         // event.getPower() □□□□□□□□□□
//
//         // □□□□□□□□□□□□
//         // event.getEnchantments().put(ModEnchantments.EXAMPLE_ENCHANTMENT.get(), 1);
//     }
// }
//
// □□□□□NeoForge 1.21.1 □□□□□□□□□□□□□
//         □□□□□□□□□□□□□□□□, □□□□□□□□□□□□□
// ================================================================================
```

entity/simple_entity.java

```
// ================================================================================
// NeoForge 1.20.6 □□□□□□□ (simple_entity.java)
// NeoForge 1.20.6, Java 21
// ================================================================================
//
// □□ NeoForge 1.20.4 □□□□□□□
// 1. □Java □□□□□□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 2. □□□□□□□□□□□!□
//    NeoForge 1.20.4 □ 1.20.6 □□□□ API □□□□□:
//    - hurt() □□□□□□□□□ (1.21 □□□□□ final)
//    - EntityType.Builder.build() □□ String □□
//    - defineSynchedData() □□□□□
//    - □□/□□□□□ CompoundTag
//
// 3. □□□□□□□□□ - Data Components□
//    □□□□□□□ API □□, □□□□□□□ NBT □□ Data Components
//    □□□□□□□ ItemStack, □□□□□ Data Components □□
//
// □□ NeoForge 1.21.1 □□□□□□ - □□□□□□
// 1. □□□□□□□□□□□!□
//    NeoForge 1.20.6: hurt(DamageSource, float) - □□□□□, □□□□
//                     □ Forge 1.20.1 □□□□□!
//    NeoForge 1.21.1: hurt() □□□□□ final, □□□□□ hurtServer() □ hurtClient()
//                     - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□
//                     - hurtClient(DamageSource, float): □□□□□□□□
//
// 2. □EntityType.Builder.build() □□□□!□
//    NeoForge 1.20.6: .build() □ String (datafixer type id)
//    NeoForge 1.21.1: .build(ResourceKey.create(Registries.ENTITY_TYPE, ResourceLocation))
```

```
//                    □□□□□ ResourceKey!
//
// 3. 【ResourceLocation □□□□□□□!】
//    NeoForge 1.20.6: new ResourceLocation("modid", "path") □□□□□
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//
// 4. 【DeferredRegister.Entities □□□□】
//    NeoForge 1.20.6: □□ DeferredRegister.create(Registries.ENTITY_TYPE, MODID)
//    NeoForge 1.21.1: □□ DeferredRegister.createEntities(MODID) (□□□)
//                     □□ registerEntityType() □□□□□□
//
// 5. □□□□□□/□□□□□□
//    NeoForge 1.20.6: readAdditionalSaveData(CompoundTag) / addAdditionalSaveData(CompoundTag)
//    NeoForge 1.21.1: □□ (1.21.2+ □□□ ValueInput/ValueOutput)
//
// 6. 【defineSynchedData()】
//    NeoForge 1.20.6: defineSynchedData() - □□□
//    NeoForge 1.21.1: □□ (1.21.5+ □□□ defineSynchedData(SynchedEntityData.Builder))
//
// □□ Forge 1.20.1 □□□□□□□
// 1. 【DeferredRegister □□□】
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □□□□□□□□
//    Forge 1.20.1: ForgeRegistries.ENTITY_TYPES
//    NeoForge 1.20.6: Registries.ENTITY_TYPE (□□□□□□)
//
// 3. 【EntityType.Builder □□□□】
//    Forge 1.20.1: EntityType.Builder.create(factory, category)
//    NeoForge 1.20.6: EntityType.Builder.of(factory, category)
//                  (1.20.x □□□□□□□ create □□ of)
//
// 4. □□□□□□□
//    Forge 1.20.1: RegistryObject<EntityType<T>>
//    NeoForge 1.20.6: DeferredHolder<EntityType<?>, EntityType<T>>
// ============================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.syncher.EntityDataAccessor;
import net.minecraft.network.syncher.EntityDataSerializers;
import net.minecraft.network.syncher.SynchedEntityData;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.world.damagesource.DamageSource;
import net.minecraft.world.entity.Entity;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.level.Level;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

// ============================================================================
// □□□□□□
// ============================================================================
public class ModEntities {

    // ========================================================================
    // □□□□□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, MODID)
    // NeoForge 1.20.6: DeferredRegister.create(Registries.ENTITY_TYPE, MODID)
    //
    // □□□□NeoForge 1.21.1:
    //   DeferredRegister.createEntities(MODID) (□□□□□□)
    //   □□ registerEntityType() □□□□□□
    //
    // □□□□□□□□□□□:
    //   Forge: ForgeRegistries (Forge □□□□□□□□)
    //   NeoForge: Registries (□□□□□□□)
    // ========================================================================
    public static final DeferredRegister<EntityType<?>> ENTITY_TYPES =
            DeferredRegister.create(Registries.ENTITY_TYPE, ExampleMod.MOD_ID);
    // ========================================================================
```

```
    // □□□□□□□□
    //
    // □□□□□EntityType.Builder □□□□:
    //   Forge 1.20.1: EntityType.Builder.create(factory, category)
    //   NeoForge 1.20.6: EntityType.Builder.of(factory, category)
    //   (1.20.x □□□□□□□□ create □□ of)
    //
    // □□□□□build() □□:
    //   NeoForge 1.20.6: .build(String) □ String □□ datafixer type id
    //   NeoForge 1.21.1: .build(ResourceKey.create(...)) □□□□ ResourceKey
    //
    // □□□□□1.20.6 □□□□ String, □□□□□□ 1.21.x □ ResourceKey □□!
    // ================================================================================
    public static final DeferredHolder<EntityType<?>, EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITY_TYPES.register(
                    "example_entity",
                    () -> EntityType.Builder.<ExampleEntity>of(
                            ExampleEntity::new,
                            MobCategory.MISC
                    )
                    .sized(1.0f, 1.0f)              // □□□□□□ (□ x □)
                    .clientTrackingRange(8)          // □□□□□□□□□□ (□□□□)
                    .updateInterval(10)              // □□□□□□ (tick)
                    .fireImmune()                    // □□□□□□
                    // .canSpawnFarFromPlayer()       // □□□□□□□□□□□□
                    // .noSummon()                    // □□ /summon □□
                    // .noSave()                      // □□□□□□□□□□
                    // ================================================================================
                    // □□□□□□□□□build() □□!
                    // NeoForge 1.20.6: .build(String) □ String (datafixer type)
                    // NeoForge 1.21.1: .build(ResourceKey.create(...)) □□□□ ResourceKey
                    // ================================================================================
                    .build("examplemod:example_entity")
            );
}

// ============================================================================
// □□□□□□□□
// □□ Entity (□□□□□□ LivingEntity, Mob □)
//
// Entity □□□□□, □□□□□□□□□□:
// 1. readAdditionalSaveData(CompoundTag) - □ NBT □□□□□
// 2. addAdditionalSaveData(CompoundTag) - □□□□□□ NBT
// 3. defineSynchedData() - □□□□□□□-□□□□□□□□□
// 4. hurt(DamageSource, float) - □□□□□ (1.20.6 □□□□□, □□□□)
// ============================================================================
class ExampleEntity extends Entity {

    // ============================================================================
    // □□□□□□□□□ (□□□□□□-□□□□□□□□□)
    // ============================================================================
    private static final EntityDataAccessor<Integer> DATA_VALUE =
            SynchedEntityData.defineId(ExampleEntity.class, EntityDataSerializers.INT);

    // ============================================================================
    // □□□□□
    // □□□□ (EntityType<? extends T>, Level) □□□□□□□□□,
    // □□□□□□□□ EntityFactory □□ (MyEntity::new)
    // ============================================================================
    public ExampleEntity(EntityType<? extends ExampleEntity> type, Level level) {
        super(type, level);
    }

    // ============================================================================
    // □□□□□□□□□□□□□□□□□□
    //
    // NeoForge 1.20.6 (□ Forge 1.20.1 / 1.20.4 □□):
    //   @Override
    //   public boolean hurt(DamageSource source, float amount) { ... }
    //   (□□□□□, □□□□□ hurt)
    //
    // NeoForge 1.21.1 (Damage Pipeline Rework):
    //   hurt() □□□□□ final! □□□□□ hurtServer() □/□ hurtClient()
    //   - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□□
    //   - hurtClient(DamageSource, float): □□□□□□ (□□□□□□□□/□□)
    //
```

```
// □□□□□1.20.6 □□□□□ hurt(), □□□□□ 1.21.x □ hurtServer()/hurtClient()!
// ============================================================================
@Override
public boolean hurt(DamageSource source, float amount) {
    if (this.isInvulnerableTo(source)) {
        return false;
    }
    // □□□□□□□□□□
    // □□: □□□□□□□□□
    // if (source.is(net.minecraft.tags.DamageTypeTags.IS_FIRE)) {
    //     return false;
    // }
    // amount *= 2; // □□□□□□□
    return super.hurt(source, amount);
}

// ============================================================================
// □ NBT □□□□□□□
// □□□□□1.20.6 □ 1.21.1 □□□□ CompoundTag (□□)
//          (1.21.2+ □□□□ ValueInput)
// ============================================================================
@Override
public void readAdditionalSaveData(CompoundTag tag) {
    super.readAdditionalSaveData(tag);
    int value = tag.getInt("value");
    this.entityData.set(DATA_VALUE, value);
}

// ============================================================================
// □□□□□□□□□ NBT
// ============================================================================
@Override
public void addAdditionalSaveData(CompoundTag tag) {
    super.addAdditionalSaveData(tag);
    tag.putInt("value", this.entityData.get(DATA_VALUE));
}

// ============================================================================
// □□□□□□□□ (□□□□-□□□□□□)
//
// 1.20.6: defineSynchedData() - □□□
// (□ Forge 1.20.1 □□)
//
// □□□□□1.21.5+ □□ defineSynchedData(SynchedEntityData.Builder builder)
//          □ 1.20.6 □ 1.21.1 □□□□□□□□
// ============================================================================
@Override
protected void defineSynchedData() {
    this.entityData.define(DATA_VALUE, 0);
}

// ============================================================================
// □ tick □□□□□
// ============================================================================
@Override
public void tick() {
    super.tick();
    if (!this.level().isClientSide) {
        // □□□□□
    }
}

// ============================================================================
// □□/□□□□□□
// ============================================================================
public int getValue() {
    return this.entityData.get(DATA_VALUE);
}

public void setValue(int value) {
    this.entityData.set(DATA_VALUE, value);
}

// ============================================================================
// □□□□□□□□ (□□□)
// □□□□□□□□□□□□□ FMLClientSetupEvent:
```

```
    //
    // □□□□1.20.6 □□ EntityRenderers.register() □ FMLClientSetupEvent □
    //         1.21.1 □□ RegisterRenderersEvent □□□□□□□□□
    //
    // private void clientSetup(final FMLClientSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         EntityRenderers.register(ModEntities.EXAMPLE_ENTITY.get(),
    //             ExampleEntityRenderer::new);
    //     });
    // }
    // ============================================================================
}
```

gui/container_screen.java

```
// ============================================================================
// NeoForge 1.20.6 GUI □□□□□□□ (container_screen.java)
// NeoForge 1.20.6, Java 21
// ============================================================================
//
// □□ NeoForge 1.20.4 □□□□□□ - GUI/□□□□□□
// 1. □Java □□□□□□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 2. □GUI/□□□□□□□□□!□
//    NeoForge 1.20.4 □ 1.20.6 □□□□ API □□□□□:
//    - □□□□ IForgeMenuType
//    - □□□□ FriendlyByteBuf □□□□□□□□□
//    - □□□□ MenuScreens.register() □ FMLClientSetupEvent □□□□ Screen
//    - □□□□ NetworkHooks.openScreen() □□□□□
//    - BlockBehaviour#use □□□□□□
//
// □□ NeoForge 1.21.1 □□□□□□ - GUI/□□□□□□
// 1. □□□□□□□□□□□□□□□!□
//    NeoForge 1.20.6: □□ IForgeMenuType.create(IContainerFactory) □□□□□□□□□□□□
//                     □□□□□□□□: FriendlyByteBuf
//    NeoForge 1.21.1: □□ IMenuTypeExtension.create(IContainerFactory) □□
//                     □□□□□□□□: RegistryFriendlyByteBuf
//                     (IForgeMenuType □□□□ IMenuTypeExtension)
//
// 2. □Screen □□□□□□□!□
//    NeoForge 1.20.6: □□ MenuScreens.register() □ FMLClientSetupEvent □□□□
//                     (□ Forge 1.20.1 □□!)
//    NeoForge 1.21.1: □□ RegisterMenuScreensEvent □□□□□□□□□□□□
//                     (□□□□, □□□□□ FMLClientSetupEvent)
//
// 3. □□□□□□□□□□□!□
//    NeoForge 1.20.6: NetworkHooks.openScreen(serverPlayer, provider, buf)
//    NeoForge 1.21.1: □□ ((ServerPlayer) player).openMenu(provider, buf)
//                     (NetworkHooks.openScreen □□□□□□/□□)
//
// 4. □BlockBehaviour □□□□□□!□
//    NeoForge 1.20.6: use(BlockState, Level, BlockPos, Player, InteractionHand, BlockHitResult)
//    NeoForge 1.21.1: useWithoutItem(BlockState, Level, BlockPos, Player, BlockHitResult)
//                     (use □□□□ useWithoutItem □ useItemOn)
//
// 5. □□□□□□□ - □□□
//    NeoForge 1.20.6 □ 1.21.1 □□□□ Registries.MENU
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□□□□
//    Forge 1.20.1: ForgeRegistries.CONTAINERS
//    NeoForge 1.20.6: Registries.MENU
//    (□□□□□□□□ CONTAINERS □□ MENU)
//
// 2. □DeferredRegister □□□
//    Forge:     net.minecraftforge.registries.DeferredRegister
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 3. □IForgeMenuType □□□
//    Forge:     net.minecraftforge.common.extensions.IForgeMenuType
//    NeoForge:  net.neoforged.neoforge.common.extensions.IForgeMenuType
//
// 4. □NetworkHooks □□□
//    Forge:     net.minecraftforge.network.NetworkHooks
```

```
//    NeoForge:  net.neoforged.neoforge.network.NetworkHooks
//
// 5. □□□□□□□
//    Forge 1.20.1: RegistryObject<MenuType<T>>
//    NeoForge 1.20.6: DeferredHolder<MenuType<?>, MenuType<T>> □ Supplier<MenuType<T>>
// ============================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.FriendlyByteBuf;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.ContainerLevelAccess;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.neoforge.common.extensions.IForgeMenuType;
import net.neoforged.neoforge.items.ItemStackHandler;
import net.neoforged.neoforge.items.SlotItemHandler;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

import java.util.function.Supplier;

// ============================================================================
// □□□□□□□
// ============================================================================
class ModMenuTypes {

    // ========================================================================
    // □□□□□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID)
    //                □□□: ForgeRegistries.CONTAINERS
    // NeoForge 1.20.6: DeferredRegister.create(Registries.MENU, MODID)
    //                  □□□: Registries.MENU
    // □□□□□□□□□□□□ CONTAINERS □□ MENU
    //
    // □□□□NeoForge 1.21.1: □□ Registries.MENU (□□)
    // ========================================================================
    public static final DeferredRegister<MenuType<?>> MENU_TYPES =
            DeferredRegister.create(Registries.MENU, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□ (□□□□□□)
    // □□ MenuType □□□□, □□ MenuSupplier □ FeatureFlags
    // ========================================================================
    public static final Supplier<MenuType<ExampleMenu>> EXAMPLE_MENU =
            MENU_TYPES.register("example_menu", () -> new MenuType<>(
                    ExampleMenu::new,
                    net.minecraft.world.flag.FeatureFlags.DEFAULT_FLAGS
            ));

    // ========================================================================
    // □□□□□□□□□□□□ (□□ IForgeMenuType)
    // □□□□□□□□□□□□□□□□ (□□□□□□), □□□□□□
    //
    // □□□□□□□
    // NeoForge 1.20.6: IForgeMenuType.create(IContainerFactory)
    //                  □□□□□□□: FriendlyByteBuf
    // NeoForge 1.21.1: IMenuTypeExtension.create(IContainerFactory)
    //                  □□□□□□□: RegistryFriendlyByteBuf
    //                  (IForgeMenuType □□□ IMenuTypeExtension)
    //
    // Forge 1.20.1: IForgeMenuType.create(IContainerFactory)
    //               (□□□□□□□□□□)
    // ========================================================================
    public static final Supplier<MenuType<ExampleMenuExtra>> EXAMPLE_MENU_EXTRA =
            MENU_TYPES.register("example_menu_extra", () ->
                    IForgeMenuType.create(ExampleMenuExtra::new)
            );
```

```
}

// ============================================================================
// □□□□□ (□□□□□□□)
// □□ AbstractContainerMenu
// ============================================================================
class ExampleMenu extends AbstractContainerMenu {

    // ========================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□)
    // ========================================================================
    public ExampleMenu(int containerId, Inventory playerInventory) {
        this(containerId, playerInventory, new ItemStackHandler(9));
    }

    // ========================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□)
    // ========================================================================
    public ExampleMenu(int containerId, Inventory playerInventory, ItemStackHandler handler) {
        super(ModMenuTypes.EXAMPLE_MENU.get(), containerId);

        // □□□□□□□□□□□ (□□ SlotItemHandler, □□ IItemHandler □□)
        // NeoForge □□□□□ SlotItemHandler (□□ IItemHandler) □□□□ vanilla Slot
        for (int i = 0; i < 9; i++) {
            int x = 62 + (i % 3) * 18;
            int y = 18 + (i / 3) * 18;
            this.addSlot(new SlotItemHandler(handler, i, x, y));
        }

        // □□□□□□□□□□□ (27 □)
        for (int row = 0; row < 3; row++) {
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory,
                        col + row * 9 + 9,
                        8 + col * 18,
                        84 + row * 18
                ));
            }
        }

        // □□□□□□□□□□□ (9 □)
        for (int col = 0; col < 9; col++) {
            this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
        }
    }

    // ========================================================================
    // stillValid - □□□□□□□□□□□
    // ========================================================================
    @Override
    public boolean stillValid(Player player) {
        return true;
    }

    // ========================================================================
    // quickMoveStack - Shift+□□□□□□□□
    // ========================================================================
    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasItem()) {
            return ItemStack.EMPTY;
        }

        ItemStack originalStack = slot.getItem();
        ItemStack copyStack = originalStack.copy();

        if (index < 9) {
            if (!this.moveItemStackTo(originalStack, 9, 45, false)) {
                return ItemStack.EMPTY;
            }
        } else {
            if (!this.moveItemStackTo(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }
```

```
        if (originalStack.isEmpty()) {
            slot.set(ItemStack.EMPTY);
        } else {
            slot.setChanged();
        }

        return copyStack;
    }
}

// ============================================================================
// □□□□□□□□□ (□□ IContainerFactory / IForgeMenuType)
// □□□□□□□□□□□□□□□□□□□□ (□□□□□□)
// ============================================================================
class ExampleMenuExtra extends AbstractContainerMenu {

    private final ContainerLevelAccess access;

    // ========================================================================
    // □□□□□□□□ (□□ FriendlyByteBuf □□□□)
    //
    // □□□□□□□□□□□□□□□□:
    //   NeoForge 1.20.6: FriendlyByteBuf (□ Forge 1.20.1 □□!)
    //   NeoForge 1.21.1: RegistryFriendlyByteBuf (□□□□, □□□□□□□□)
    // ========================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, FriendlyByteBuf extraData) {
        this(containerId, playerInventory, extraData.readBlockPos());
    }

    // ========================================================================
    // □□□□□□□□
    // ========================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, BlockPos pos) {
        super(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), containerId);
        this.access = ContainerLevelAccess.create(playerInventory.player.level(), pos);
        // □□□□□ (□□, □□□□□)
    }

    @Override
    public boolean stillValid(Player player) {
        return AbstractContainerMenu.stillValid(this.access, player, Blocks.CRAFTING_TABLE);
    }

    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        return ItemStack.EMPTY;
    }

    // ========================================================================
    // □□□□□□□□□□□□□:
    //
    // □□□□□NeoForge 1.20.6: NetworkHooks.openScreen(serverPlayer, provider, buf)
    //         (□ Forge 1.20.1 □□!)
    //         NeoForge 1.21.1: serverPlayer.openMenu(provider, buf)
    //         (□□□□□ NetworkHooks)
    //
    // □ BlockBehaviour#use □□□□ (1.20.6 □□):
    //   @Override
    //   public InteractionResult use(BlockState state, Level level, BlockPos pos,
    //           Player player, InteractionHand hand, BlockHitResult hitResult) {
    //       if (level.isClientSide) {
    //           return InteractionResult.SUCCESS;
    //       }
    //       NetworkHooks.openScreen((ServerPlayer) player, new SimpleMenuProvider(
    //           (containerId, playerInventory, p) -> new ExampleMenuExtra(containerId, playerInventory, pos),
    //           Component.translatable("menu.examplemod.example_menu")
    //       ), buf -> buf.writeBlockPos(pos));
    //       return InteractionResult.CONSUME;
    //   }
    //
    // □□□□□1.21.1 □□□□□□ use □□ useWithoutItem:
    //   public InteractionResult useWithoutItem(BlockState state, Level level, BlockPos pos,
    //           Player player, BlockHitResult hitResult) { ... }
    // ========================================================================
}
// ============================================================================
```

```
// □□□□□ (Screen) - □□□□□□
// □□ AbstractContainerScreen
// ============================================================================
class ExampleScreen extends net.minecraft.client.gui.screens.inventory.AbstractContainerScreen<ExampleMenu> {

    public ExampleScreen(ExampleMenu menu, Inventory playerInventory, Component title) {
        super(menu, playerInventory, title);
        // this.imageWidth = 176;
        // this.imageHeight = 166;
    }

    // ============================================================================
    // □□□□□□□
    // 1.20.6 □□ GuiGraphics (□ 1.21.1 □□)
    // ============================================================================
    @Override
    protected void renderBg(net.minecraft.client.gui.GuiGraphics guiGraphics,
                            float partialTick, int mouseX, int mouseY) {
        // □□□□□□□
        // guiGraphics.blit(
        //     new ResourceLocation(ExampleMod.MOD_ID, "textures/gui/example_container.png"),
        //     this.leftPos, this.topPos,
        //     0, 0,
        //     this.imageWidth, this.imageHeight
        // );
    }

    @Override
    public void render(net.minecraft.client.gui.GuiGraphics guiGraphics,
                       int mouseX, int mouseY, float partialTick) {
        this.renderBackground(guiGraphics, mouseX, mouseY, partialTick);
        super.render(guiGraphics, mouseX, mouseY, partialTick);
        this.renderTooltip(guiGraphics, mouseX, mouseY);
    }

    @Override
    protected void init() {
        super.init();
        this.titleLabelY = 6;
        this.inventoryLabelY = this.imageHeight - 94;
    }

    // ============================================================================
    // □□ Screen (□□□□□□□□□□):
    //
    // □□□□□□□Screen □□□□□!
    //
    // NeoForge 1.20.6 (□ Forge 1.20.1 □□):
    //   □□ MenuScreens.register() □ FMLClientSetupEvent □□□:
    //
    //   private void clientSetup(final FMLClientSetupEvent event) {
    //       event.enqueueWork(() -> {
    //           MenuScreens.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
    //           MenuScreens.register(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), ExampleScreenExtra::new);
    //       });
    //   }
    //   (□□□ enqueueWork □□□□□□)
    //
    // NeoForge 1.21.1:
    //   □□ RegisterMenuScreensEvent □□□□□□□□:
    //
    //   @SubscribeEvent
    //   public static void registerScreens(RegisterMenuScreensEvent event) {
    //       event.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
    //   }
    //   (□□□□ FMLClientSetupEvent □ enqueueWork)
    // ============================================================================
}
```

item/simple_item.java

```
// ============================================================================
// NeoForge 1.20.6 □□□□□□ (simple_item.java)
// NeoForge 1.20.6, Java 21
// ============================================================================
//
```

```
// ⚠ NeoForge 1.20.4 □□□□□□ - □□□□!□
// 1. □NBT □□ vs Data Components □□ (□□□□!)□
//    NeoForge 1.20.4: □□□ ItemStack NBT □□!
//                     itemStack.getTag(), itemStack.getOrCreateTag() □□□□□
//    NeoForge 1.20.6: □□ Data Components □□, NBT □□□!
//                     itemStack.get(DataComponentTypes.XXX), itemStack.set() □
//                     □□□□1.20.5+ □□ Data Components, □□ NBT!
//
// 2. □Java □□□□!□
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 3. □□□□□□□□□□□□
//    NeoForge 1.20.4: FoodProperties □□□□□□ Item □
//    NeoForge 1.20.6: FoodProperties □□ DataComponents.FOOD □□□□□□□
//                    (Item.Properties.food() □□□□□□□□, □□□□□□ DataComponents)
//
// 4. □DeferredRegister.Items □□□□□□
//    NeoForge 1.20.4: registerItem() □□ Supplier<Item>
//    NeoForge 1.20.6: registerItem() □□ Supplier<Item> (□ 1.20.4 □□)
//                    (DeferredItem<T> □□□□□□ 1.21.x □□□□□□)
//
// ⚠ NeoForge 1.21.1 □□□□□□□□
// 1. □DeferredRegister.Items □□□□□□
//    NeoForge 1.20.6: registerItem() □□ Supplier<Item>
//    NeoForge 1.21.1: registerItem() □□ DeferredItem<T> (□□□□□)
//
// 2. □ResourceLocation □□□□□□
//    NeoForge 1.20.6: new ResourceLocation() □□□□
//    NeoForge 1.21.1: □□□□□□□□□, □□□□□□□□□
//
// 3. □□□□□□ - Tier □□□
//    NeoForge 1.20.6: □□ Tier □□, □□ getLevel() (int)
//    NeoForge 1.21.1: □□□□□ Tier □□, □□ getIncorrectDrops() (TagKey<Block>)
//
// ⚠ Forge 1.20.1 □□□□□□□□
// 1. □DeferredRegister.Items □□□□□
//    Forge 1.20.1:  □□□□□, □□ DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
//                   □□ RegistryObject<Item>
//    NeoForge 1.20.6: □□ DeferredRegister.Items, □□ DeferredRegister.createItems(MODID)
//                     □□ registerItem(), registerSimpleItem() □□□□□□
//
// 2. □□□□□□□□
//    Forge 1.20.1:  ForgeRegistries.ITEMS
//    NeoForge 1.20.6: Registries.ITEM (□□□□□□)
//
// 3. □□□□□□□□□□□
//    Forge 1.20.1: ItemStack NBT
//    NeoForge 1.20.6: Data Components (1.20.5+ □□)
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.food.FoodProperties;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Rarity;
import net.neoforged.neoforge.registries.DeferredRegister;

import java.util.function.Supplier;

// ============================================================================
// □□□□□□
// ============================================================================
public class ModItems {

    // ========================================================================
    // □□□□□□□ DeferredRegister.Items □□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
    //                □□ DeferredRegister<Item>, register() □□ RegistryObject<Item>
    // NeoForge 1.20.6: DeferredRegister.createItems(MODID)
    //                  □□ DeferredRegister.Items
```

```
//                  registerItem() 返回 Supplier<Item>
//
// 注意：NeoForge 1.21.1: registerItem() 返回 DeferredItem<T> (不是直接)
//         NeoForge 1.20.6: registerItem() 返回 Supplier<Item> (与 1.20.4 相同)
// ============================================================================
public static final DeferredRegister.Items ITEMS =
        DeferredRegister.createItems(ExampleMod.MOD_ID);

// ============================================================================
// 示例：简单物品
// 返回类型在不同版本:
//   Forge 1.20.1: RegistryObject<Item>
//   NeoForge 1.20.6: Supplier<Item>
//   NeoForge 1.21.1: DeferredItem<Item>
//
// 提示: 使用 registerItem 方法引用 (推荐)
// ============================================================================
public static final Supplier<Item> EXAMPLE_ITEM =
        ITEMS.registerItem("example_item", Item::new, new Item.Properties());

// ============================================================================
// 示例：带属性的物品（稀有度、堆叠数量）
// Item.Properties 方法:
// - .rarity(): 稀有度 (COMMON, UNCOMMON, RARE, EPIC)
// - .stacksTo(): 最大堆叠数量 (不能与耐久 DataComponents.MAX_STACK_SIZE)
// - .fireResistant(): 防火物品 (添加组件 DataComponents.FIRE_RESISTANT)
// - .durability(): 耐久度 (添加组件 DataComponents.MAX_DAMAGE)
// - .food(): 食物属性 (添加组件 DataComponents.FOOD)
// ============================================================================
public static final Supplier<Item> RARE_ITEM =
        ITEMS.registerItem("rare_item", Item::new, new Item.Properties()
                .rarity(Rarity.RARE)           // 稀有的
                .stacksTo(16)                   // 最大堆叠数量 (默认 64)
                .fireResistant()                // 防火物品 (掉落物不会燃烧)
        );

// ============================================================================
// 示例：食物物品
// FoodProperties 来自 Item.Properties.food() 方法
// 注意：1.20.4 及以前 FoodProperties
//         1.20.6 及以后使用 DataComponents.FOOD 组件系统
//         Item.Properties.food() 方法设置营养, 饱和度和特效
// ============================================================================
public static final Supplier<Item> EXAMPLE_FOOD =
        ITEMS.registerItem("example_food", Item::new, new Item.Properties()
                .food(new FoodProperties.Builder()
                        .nutrition(6)                    // 营养值 (半个鸡腿)
                        .saturationMod(0.6f)              // 饱和度修饰
                        .meat()                           // 肉类食物
                        .build()
                )
        );

// ============================================================================
// 示例：耐久物品（工具、武器）
// 设置 durability 会自动将最大堆叠数量设为 1
// 注意：1.20.6 及以后使用 DataComponents.MAX_DAMAGE 组件系统
// ============================================================================
public static final Supplier<Item> EXAMPLE_DURABLE =
        ITEMS.registerItem("example_durable", Item::new, new Item.Properties()
                .durability(250)               // 耐久度
                .rarity(Rarity.UNCOMMON)
        );

// ============================================================================
// 示例：自定义物品 (重写物品行为方法)
// 如果需要自定义物品行为，创建 Item 的
// 子类并重写相关方法 (如右键使用), 然后使用 Item 构造
// ============================================================================
public static final Supplier<Item> CUSTOM_ITEM =
        ITEMS.registerItem("custom_item", CustomItem::new, new Item.Properties()
                .rarity(Rarity.EPIC)
        );

// ============================================================================
// 示例：方块物品 (BlockItem)
```

```
    // 方块物品：registerSimpleBlockItem 自动创建 BlockItem
    // ========================================================================
    // public static final Supplier<Item> EXAMPLE_BLOCK_ITEM =
    //         ITEMS.registerSimpleBlockItem(ModBlocks.EXAMPLE_BLOCK);
}

// ========================================================================
// 自定义物品类
// 继承 Item 并可以添加自定义行为
// ========================================================================
class CustomItem extends Item {

    public CustomItem(Properties properties) {
        super(properties);
    }

    // ========================================================================
    // 使用物品时：Data Components 方式! (1.20.5+ 推荐)
    //
    // NeoForge 1.20.4 (及 Forge 1.20.1 等):
    //   使用 ItemStack NBT 存储自定义数据:
    //   stack.getOrCreateTag().putInt("custom_value", 42);
    //   stack.getTag().getInt("custom_value");
    //
    // NeoForge 1.20.6 (1.20.5+ 引入):
    //   使用 Data Components 系统 (NBT 已弃用!):
    //   stack.set(ModDataComponents.CUSTOM_VALUE.get(), 42);
    //   stack.get(ModDataComponents.CUSTOM_VALUE.get());
    //
    // 常用 Data Component 类型 (net.minecraft.core.component.DataComponentTypes):
    //   DataComponentTypes.MAX_STACK_SIZE   // 最大堆叠数
    //   DataComponentTypes.MAX_DAMAGE         // 最大耐久
    //   DataComponentTypes.DAMAGE             // 当前耐久
    //   DataComponentTypes.RARITY             // 稀有度
    //   DataComponentTypes.FOOD               // 食物属性
    //   DataComponentTypes.CUSTOM_NAME        // 自定义名称
    //   DataComponentTypes.LORE               // 描述文本
    //   DataComponentTypes.ENCHANTMENTS       // 附魔
    //   DataComponentTypes.REPAIR_COST        // 修复成本
    // ========================================================================

    // ========================================================================
    // Data Components 使用示例 (1.20.6 及以上)
    //
    // import net.minecraft.core.component.DataComponentTypes;
    //
    // public InteractionResult use(Level level, Player player, InteractionHand hand) {
    //     ItemStack stack = player.getItemInHand(hand);
    //
    //     // 1.20.6: 使用 Data Components 存储自定义数据
    //     // 需要先注册自己的 DataComponentType
    //     // Integer currentValue = stack.get(ModComponents.CUSTOM_VALUE);
    //     // if (currentValue == null) currentValue = 0;
    //     // stack.set(ModComponents.CUSTOM_VALUE, currentValue + 1);
    //
    //     // 读取耐久信息
    //     // int maxDamage = stack.get(DataComponentTypes.MAX_DAMAGE);
    //     // int damage = stack.get(DataComponentTypes.DAMAGE);
    //
    //     return InteractionResult.sidedSuccess(level.isClientSide);
    // }
    // ========================================================================
}

// ========================================================================
// 创造模式物品栏标签页
// 注意：1.20.6 使用 CreativeModeTab.builder() 模式
//         需要 DeferredRegister<CreativeModeTab> 注册
// ========================================================================
class ModCreativeModeTabs {

    public static final DeferredRegister<CreativeModeTab> CREATIVE_MODE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MOD_ID);

    // ========================================================================
    // 创造模式物品栏标签页注册
```

```
    // 使用 builder() 方法创建, 并注册到创造模式标签页
    // ================================================================================
    public static final Supplier<CreativeModeTab> EXAMPLE_TAB =
            CREATIVE_MODE_TABS.register("example_tab", () -> CreativeModeTab.builder()
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    .icon(() -> new ItemStack(ModItems.EXAMPLE_ITEM.get()))
                    .displayItems((params, output) -> {
                        output.accept(ModItems.EXAMPLE_ITEM.get());
                        output.accept(ModItems.RARE_ITEM.get());
                        output.accept(ModItems.EXAMPLE_FOOD.get());
                        output.accept(ModItems.EXAMPLE_DURABLE.get());
                        output.accept(ModItems.CUSTOM_ITEM.get());
                    })
                    .build()
            );

    // ================================================================================
    // 添加物品到现有标签页: 监听 BuildCreativeModeTabContentsEvent 事件并在其中添加物品
    // 取消下面的注释并修改以使用:
    //
    // @SubscribeEvent  // 确保已订阅到 MOD 总线 (bus = Bus.MOD)
    // public static void buildContents(BuildCreativeModeTabContentsEvent event) {
    //     if (event.getTabKey() == CreativeModeTabs.INGREDIENTS) {
    //         event.accept(ModItems.EXAMPLE_ITEM.get());
    //     }
    // }
    // ================================================================================
}
```

item/tool_item.java

```
// ================================================================================
// NeoForge 1.20.6 工具物品模板 (tool_item.java)
// NeoForge 1.20.6, Java 21
// ================================================================================
//
// 适用 NeoForge 1.20.4 及更早版本 - 重要说明
// 1. 需Java 版本升级
//    NeoForge 1.20.4: Java 17
//    NeoForge 1.20.6: Java 21
//
// 2. 工具等级系统变化!!!
//    NeoForge 1.20.4 及 1.20.6 的 Tier 接口方法与更新版本不同!
//    如果你使用的是 1.21.x, 请看 1.20.6。
//
// 3. 数据组件系统 - Data Components。
//    新版本使用组件系统, 而不是直接的 NBT 操作 Data Components
//    工具属性 (攻击伤害、速度) 现在由 Tier 决定而不是, 直接设
//
// 适用 NeoForge 1.21.1 及更新版本 - 最大的变化!!!
// 1. 【Tier 接口变化】!!!
//    NeoForge 1.20.6: 实现 Tier 接口, 包含 getLevel() (int 等级数值)
//                     和 Forge 1.20.1 / 1.20.4 相同!
//    NeoForge 1.21.1: 不再实现 Tier 接口, 改为 getIncorrectDrops() (TagKey<Block>)
//                     不再有 getLevel() 方法!
//                     改用 SimpleTier 构造函数创建 Tier
//
// 2. 【工具物品构造函数】!!!
//    NeoForge 1.20.6: new SwordItem(tier, attackDamage, attackSpeed, properties)
//                     attackDamage 和 attackSpeed 是构造函数参数
//    NeoForge 1.21.1: new SwordItem(tier, properties)
//                     attackDamage 和 attackSpeed 通过 Item.Properties.attributes() 设置
//                     攻击伤害和速度现在是属性修饰符!
//
// 3. 【ResourceLocation 变化】!!!
//    NeoForge 1.20.6: new ResourceLocation("modid", "path") 公开的
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//                     (更明确的命名, 防止错误)
//
// 4. 【ItemAbility / ToolAction 变化】
//    NeoForge 1.20.6: ToolAction / ToolActions (和 Forge 1.20.1 相同)
//    NeoForge 1.21.1: 改名为 ItemAbility / ItemAbilities
//
// 适用 Forge 1.20.1 及更早版本
// 1. 【DeferredRegister.Items 变化】
//    Forge:     net.minecraftforge.registries.DeferredRegister
```

```
//    NeoForge:  net.neoforged.neoforge.registries.DeferredRegister
//
// 2. □Tier □□□□ - □□□□!□
//    Forge 1.20.1 □ NeoForge 1.20.6 □ Tier □□□□□□□!
//    □□□: getUses(), getSpeed(), getAttackDamageBonus(),
//            getLevel(), getEnchantmentValue(), getRepairIngredient()
//
// 3. □□□□□□□□ - □□□□!□
//    new SwordItem(tier, attackDamage, attackSpeed, properties)
//
// 4. □□□□□□□□□
//    Forge 1.20.1: RegistryObject<SwordItem>
//    NeoForge 1.20.6: Supplier<SwordItem>
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.tags.BlockTags;
import net.minecraft.tags.TagKey;
import net.minecraft.world.item.AxeItem;
import net.minecraft.world.item.HoeItem;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.ShovelItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraft.world.level.block.Block;
import net.neoforged.neoforge.registries.DeferredRegister;

import java.util.function.Supplier;

// ============================================================================
// □□□□□□□□
// ============================================================================
public class ModTools {

    public static final DeferredRegister.Items ITEMS =
            DeferredRegister.createItems(ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□ Tier (□□□□□) - □□□□!
    //
    // NeoForge 1.20.6 □ Forge 1.20.1 □ Tier □□□□□□:
    //   - getUses(): □□□
    //   - getSpeed(): □□□□
    //   - getAttackDamageBonus(): □□□□□□
    //   - getLevel(): □□□□ (int, □ wood=0, stone=1, iron=2, diamond=3, netherite=4)
    //   - getEnchantmentValue(): □□□
    //   - getRepairIngredient(): □□□□
    //
    // □□□□□□□NeoForge 1.21.1 □ Tier □□:
    //   - getIncorrectDrops(): TagKey<Block> (□□ getLevel()!)
    //   - getUses(): int (□□)
    //   - getSpeed(): float (□□)
    //   - getAttackDamageBonus(): float (□□)
    //   - getEnchantmentValue(): int (□□)
    //   - getRepairIngredient(): Ingredient (□□)
    //
    // □□□□1.20.6 □□□□ getLevel() (int), □□□□ 1.21.x □ getIncorrectDrops()!
    // ========================================================================
    public static final Tier COPPER_TIER = new Tier() {

        // □□□□1.20.6 □□ getLevel() (int □□□□□□)
        // 1.21.1 □□□□□□□□, □□ getIncorrectDrops() □□ TagKey<Block>!
        @Override
        public int getLevel() {
            return 2;  // stone=1, iron=2, diamond=3
        }

        @Override
        public int getUses() {
            return 200;  // □□□ (stone=131, iron=250, diamond=1561)
```

```
    }

    @Override
    public float getSpeed() {
        return 5f;  // □□□□□ (stone=4, iron=6, diamond=8)
    }

    @Override
    public float getAttackDamageBonus() {
        return 1.5f;  // □□□□□□□ (stone=1, iron=2, diamond=3)
    }

    @Override
    public int getEnchantmentValue() {
        return 20;  // □□□ (gold=22, iron=14, diamond=10)
    }

    @Override
    public Ingredient getRepairIngredient() {
        // □□□□□
        // □□□□□ NeoForge □□: Tags.Items.INGOTS_COPPER
        return Ingredient.of(net.minecraft.world.level.block.Blocks.COPPER_BLOCK);
    }
};

// ============================================================================
// □□□□□□□□□□□□ - □□□□□□□□!
//
// NeoForge 1.20.6 (□ Forge 1.20.1 / 1.20.4 □□):
//   new SwordItem(tier, attackDamage, attackSpeed, new Item.Properties())
//   (attackDamage □ attackSpeed □□□□□□□□□□□□)
//
// NeoForge 1.21.1 (□□□!):
//   new SwordItem(tier, new Item.Properties().attributes(
//       SwordItem.createAttributes(tier, attackDamage, attackSpeed)))
//   (□□ Item.Properties.attributes() □□, □□□□□□□□!)
//
// □□□□□1.20.6 □□□□□□□□□□□□, □□□□□ 1.21.x □□□□□□!
// ============================================================================

// □□ - □□□□□ 3, □□□□□ -2.4f
public static final Supplier<SwordItem> COPPER_SWORD =
        ITEMS.register("copper_sword", () -> new SwordItem(
                COPPER_TIER,
                3,          // □□□□□□□
                -2.4f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 1, □□□□□ -2.8f
public static final Supplier<PickaxeItem> COPPER_PICKAXE =
        ITEMS.register("copper_pickaxe", () -> new PickaxeItem(
                COPPER_TIER,
                1,          // □□□□□□□
                -2.8f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 6, □□□□□ -3.2f
public static final Supplier<AxeItem> COPPER_AXE =
        ITEMS.register("copper_axe", () -> new AxeItem(
                COPPER_TIER,
                6,          // □□□□□□□ (□□□□□□□)
                -3.2f,      // □□□□□□□
                new Item.Properties()
        ));

// □□ - □□□□□ 1.5, □□□□□ -3.0f
public static final Supplier<ShovelItem> COPPER_SHOVEL =
        ITEMS.register("copper_shovel", () -> new ShovelItem(
                COPPER_TIER,
                1.5f,       // □□□□□□□
                -3.0f,      // □□□□□□□
                new Item.Properties()
        ));
// □□ - □□□□□ 0, □□□□□ -1.0f
```

```
    public static final Supplier<HoeItem> COPPER_HOE =
            ITEMS.register("copper_hoe", () -> new HoeItem(
                    COPPER_TIER,
                    0,          // □□□□□□□
                    -1.0f,      // □□□□□□□
                    new Item.Properties()
            ));

    // ============================================================================
    // □□□□□□□□ (□□□□□□□)
    //
    // 1.20.6 □□□□□□□□□□:
    // data/examplemod/tags/blocks/needs_copper_tool.json
    //   - □□□□□□□□□□□□□□□
    //
    // □□□□1.21.1 □□□□□□□ incorrect_for_copper_tool.json □□
    //         (□□□□□□□□□□□□□□□□□□□□□□□□)
    //         1.20.6 □□□□□ incorrect □□ (□□□□□ getLevel() int □□)
    //
    // □□□□□ResourceLocation □□□□□:
    //   1.20.6: new ResourceLocation("modid", "path") □□□□
    //   1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
    // ============================================================================
    public static final TagKey<Block> NEEDS_COPPER_TOOL = TagKey.create(
            net.minecraft.core.registries.Registries.BLOCK,
            new ResourceLocation(ExampleMod.MOD_ID, "needs_copper_tool")
    );
}
```

mixin/mixin_config.json

```
{
    "_comment": "============================================================================",
    "_comment_2": "NeoForge 1.20.6 Mixin □□□□□ (mixin_config.json)",
    "_comment_3": "NeoForge 1.20.6, Java 21",
    "_comment_4": "============================================================================",
    "_comment_5": "Mixin □□□□□ - □□□□□ Mixin □□□□□□□",
    "_comment_6": "",
    "_comment_7": "□□□□□: src/main/resources/examplemod.mixins.json",
    "_comment_8": "□□□□□□ neoforge.mods.toml □ [[mixins]] □□ config □□□",
    "_comment_9": "",
    "_comment_10": "□□ NeoForge 1.20.4 □□□□□□□",
    "_comment_11": "1. □Java □□□□!□",
    "_comment_12": "   NeoForge 1.20.4: compatibilityLevel = \"JAVA_17\"",
    "_comment_13": "   NeoForge 1.20.6: compatibilityLevel = \"JAVA_21\"",
    "_comment_14": "   □□□□1.20.5+ □□□□ Java 21!",
    "_comment_15": "",
    "_comment_16": "2. □□□□□□□□□!□",
    "_comment_17": "   NeoForge 1.20.4: □ mods.toml □□□ [[mixins]] □□",
    "_comment_18": "   NeoForge 1.20.6: □ neoforge.mods.toml □□□ [[mixins]] □□",
    "_comment_19": "   (1.20.5+ □□□□ mods.toml □□ neoforge.mods.toml)",
    "_comment_20": "",
    "_comment_21": "3. Mixin □□ JSON □□□□□□□",
    "_comment_22": "",
    "_comment_23": "□□ NeoForge 1.21.1 □□□□",
    "_comment_24": "1. Mixin □□□□□ 1.20.6 □ 1.21.1 □□□□□□",
    "_comment_25": "2. □□□ JAVA_21 □ compatibilityLevel",
    "_comment_26": "3. □□ neoforge.mods.toml □□□ [[mixins]] □□",
    "_comment_27": "",
    "_comment_28": "□□ Forge 1.20.1 □□□□",
    "_comment_29": "1. Forge 1.20.1 □ mods.toml □□ [[mixins]] □□□ (□□□□)",
    "_comment_30": "2. Mixin □□ JSON □□□□□□□",
    "_comment_31": "3. □□□□ mods.toml □□□□□: Forge □ modId=forge, NeoForge □ modId=neoforge",
    "_comment_32": "",
    "_comment_33": "□□□□□ neoforge.mods.toml □□□ Mixin □□:",
    "_comment_34": "  [[mixins]]",
    "_comment_35": "  config=\"${mod_id}.mixins.json\"",
    "_comment_36": "============================================================================",

    "required": true,
    "minVersion": "0.8.5",
    "package": "com.example.examplemod.mixin",
    "compatibilityLevel": "JAVA_21",
    "refmap": "examplemod.refmap.json",
    "mixins": [
        "example.MixinExampleClass",
```

```
        "example.MixinAnotherClass"
    ],
    "client": [
        "client.MixinClientClass"
    ],
    "server": [
        "server.MixinServerClass"
    ],
    "injectors": {
        "defaultRequire": 1
    }
}
```

project/build.gradle

```
// ============================================================================
// NeoForge 1.20.6 □□□□□ (build.gradle)
// NeoForge 1.20.6, Java 21
// ============================================================================
//
// □□ NeoForge 1.20.4 □□□□□□□
// 1. □Java □□□□□!□
//    NeoForge 1.20.4: Java 17 (Mojang □ 1.20.4 □□□□□ Java 17)
//    NeoForge 1.20.6: Java 21 (1.20.5+ □□□□□ Java 21!)
//    □□□□□□□1.20.5+ □□□□□ Java 21
//
// 2. □NeoForge □□□□□□□
//    1.20.4: 20.4.xxx (□ 20.4.251)
//    1.20.6: 20.6.xxx (□ 20.6.62)
//
// 3. □mods.toml → neoforge.mods.toml!□
//    NeoForge 1.20.4: □□ mods.toml (□□ META-INF/mods.toml)
//    NeoForge 1.20.6: □□ neoforge.mods.toml (□□ META-INF/neoforge.mods.toml)
//    □□□□□□ NeoForge 20.5 (MC 1.20.5) □, □□□□□ mods.toml □□ neoforge.mods.toml!
//
// 4. □loaderVersion □□□□□□□
//    NeoForge 1.20.4: loaderVersion="[1,)" (FML □□□□ 1)
//    NeoForge 1.20.6: loaderVersion="[4,)" (FML □□□□ 4, □ 1.20.5+ □□□□)
//
// 5. □Data Components □□□□□□
//    NeoForge 1.20.5+ □□□□ Data Components □□, □□ ItemStack NBT
//
// □□ NeoForge 1.21.1 □□□□□□ - Gradle □□□
// 6. NeoForge 1.20.6: □□□□□ NeoGradle □□ (net.neoforged.gradle.userdev)
//    NeoForge 1.21.1: □□ ModDevGradle □□ (net.neoforged.moddev)
//    □□□□□ModDevGradle □□□□ NeoForge 21.0+ (MC 1.21+)
//             1.20.6 □□□□□ NeoGradle!
// ============================================================================

plugins {
    id 'java-library'
    id 'maven-publish'
    // □□□□□NeoGradle □□ - NeoForge 1.20.6 □ Gradle □□
    // □□□□□NeoForge 1.21.1 □□ ModDevGradle (net.neoforged.moddev)
    //         NeoForge 1.20.6 □□ NeoGradle (net.neoforged.gradle.userdev)
    //         ModDevGradle □□□□ NeoForge 21+ (MC 1.21+)□□□□□□□□ 1.20.6!
    // □□□□□ ModDevGradle (net.neoforged.moddev), □□□□□ NeoForge 21+!
    id 'net.neoforged.gradle.userdev' version '7.1.+'
}

tasks.named('wrapper', Wrapper).configure {
    distributionType = Wrapper.DistributionType.BIN
}

version = mod_version
group = mod_group_id

sourceSets.main.resources {
    // □□□□□□□□□□□□□□□□
    srcDir('src/generated/resources')
    // □□□□□□□□□□□□□□□□
    exclude("**/*.bbmodels")
    exclude("src/generated/**/.cache")
}

repositories {
```

```
    // □□□□□□□□□□□□□□
}

base {
    archivesName = mod_id
}

// ============================================================================
// □□□□Java 21 - □□ 17!
// □□□□□□□1.20.5+ □□□□□ Java 21 (1.20.4 □ Java 17)
// Mojang □ MC 1.20.5 □□□□□□□□□□□□ Java 21
// □□□□□ JDK 21□□ Temurin 21 / Microsoft OpenJDK 21□
// ============================================================================
java.toolchain.languageVersion = JavaLanguageVersion.of(21)

// ============================================================================
// □□□□□□ (Access Transformer) - □□
// NeoGradle □□□□□□
// ============================================================================
//minecraft.accessTransformers.file rootProject.file('src/main/resources/META-INF/accesstransformer.cfg')
//minecraft.accessTransformers.entry public net.minecraft.client.Minecraft textureManager # textureManager

// ============================================================================
// □□□□□ - □□□□□□□□□□□□ Minecraft
// ============================================================================
runs {
    // □□□□□□□□□□
    configureEach {
        // □□□□□ (REGISTRIES □□□□□□□□□□□□)
        systemProperty 'forge.logging.markers', 'REGISTRIES'
        // □□□□□□□□
        systemProperty 'forge.logging.console.level', 'debug'
        // □□□□□
        workingDirectory project.layout.projectDirectory.dir('run').dir(name)
        // □□□□□
        modSource project.sourceSets.main
    }
    // □□□□□□□□
    client {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
    }
    // □□□□□□□□
    server {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
        argument '--nogui'
    }
    // □□□□□□□□
    gameTestServer {
        systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
    }
    // □□□□□□□□□
    data {
        arguments.addAll '--mod', project.mod_id, '--all', '--output',
                file('src/generated/resources/').getAbsolutePath(), '--existing',
                file('src/main/resources/').getAbsolutePath()
    }
}

// ============================================================================
// □□□□□□□□□□ (□□□□□□)
// ============================================================================
configurations {
    runtimeClasspath.extendsFrom localRuntime
}

// ============================================================================
// □□
// ============================================================================
dependencies {
    // □□□□□□□NeoGradle □□□□□□□□
    // NeoForge 1.20.6: implementation "net.neoforged:neoforge:${neo_version}"
    // neo_version □ gradle.properties □□□ 20.6.62□□□□□ 20.6.x□
    implementation "net.neoforged:neoforge:${neo_version}"

    // JEI □□□□ (□□)
    // compileOnly "mezz.jei:jei-${mc_version}-common-api:${jei_version}"
```

```
    // compileOnly "mezz.jei:jei-${mc_version}-neoforge-api:${jei_version}"
    // localRuntime "mezz.jei:jei-${mc_version}-neoforge:${jei_version}"
}

// ============================================================================
// □□□□□□□□□□
// □ gradle.properties □□□□□□□ neoforge.mods.toml □□ ${} □□□
// □□□□□□□□1.20.6 □□□□□ neoforge.mods.toml (□□ mods.toml!)
//             1.20.4 □□□□□ mods.toml
// ============================================================================
tasks.withType(ProcessResources).configureEach {
    var replaceProperties = [
            minecraft_version      : minecraft_version,
            minecraft_version_range: minecraft_version_range,
            neo_version            : neo_version,
            loader_version_range   : loader_version_range,
            mod_id                 : mod_id,
            mod_name               : mod_name,
            mod_license            : mod_license,
            mod_version            : mod_version,
    ]
    inputs.properties replaceProperties
    // □□□□□1.20.6 □□ neoforge.mods.toml (1.20.4 □□□□□ mods.toml!)
    filesMatching(['META-INF/neoforge.mods.toml']) {
        expand replaceProperties
    }
}

// Maven □□□□□ (□□)
publishing {
    publications {
        register('mavenJava', MavenPublication) {
            from components.java
        }
    }
    repositories {
        maven {
            url "file://${project.projectDir}/repo"
        }
    }
}

// □□□□□
tasks.withType(JavaCompile).configureEach {
    options.encoding = 'UTF-8'
}

// IDEA □□□□□□□□
idea {
    module {
        downloadSources = true
        downloadJavadoc = true
    }
}
```

project/version_diff.md

```
# NeoForge 1.20.6 □ 1.20.4 □□□□□□□

> □□□□□□ NeoForge 1.20.6 □□□□ 1.20.4 □□□□ API □□□□
> 1.20.5/1.20.6 □□□□□□□□□□□□Java □□□□ 17 □□□□ 21□
> □□□□ Data Components □□□mods.toml □□□□ neoforge.mods.toml□
> □□□□□□□□□□□□□□□□□□□□□□□□
> AI □□□□ 1.20.6 □ NeoForge □□□□□□□□□□□□□□□□□□

---

## □□□□□

1.20.6 □□ 1.20.x □□□□□□□□□□1.20.4 → 1.20.5 → 1.20.6□□
□□ MC □□□□□□□□□□□□ 1.20.5 □□□□□□□□□□□□□

| □□□□ | 1.20.4 | 1.20.6 |
|--------|--------|--------|
| **Java □□** | **17** | **21□□□□□□** |
| NeoForge □□ | 20.4.x□□ 20.4.251□ | 20.6.x□□ 20.6.62□ |
```

Gradle □□	NeoGradle (userdev 7.1.x)	NeoGradle (userdev 7.1.x□□□)
□□□□□□□	**mods.toml**	**neoforge.mods.toml**
loaderVersion	`[1,)` (FML □□□ 1)	`[4,)` (FML □□□ 4)
□□□□□□□□	NBT (`ItemStack.getOrCreateTag()`)	**Data Components** (`stack.get/set()`)
□□□□□	□□ `Enchantment` □	□□ `Enchantment` □□□□□□1.21 □□□□□□□□
EnchantmentTarget □□	□□	□□□1.21 □□□□□
FoodComponent	`net.minecraft.world.food.FoodProperties`	□□□□□□□□□□□□
DeferredRegister	□□	□□□□□□□

□□Java □□□□□□□□□□□

2.1 Java 21 □□□□

1.20.5+ **□□□□ Java 21**□1.20.4 □□□ Java 17□

```gradle
// 1.20.4 build.gradle
java.toolchain.languageVersion = JavaLanguageVersion.of(17)
```

```gradle
// 1.20.6 build.gradle
java.toolchain.languageVersion = JavaLanguageVersion.of(21)
```

□□□□□□□□□ JDK 21□□□ Microsoft OpenJDK 21 □ Eclipse Temurin 21□□

2.2 □□□ Java 21 □□

1.20.6 □□□□□□□□ Java 21 □□□□□□
- Record □□Java 16 □□□
- Switch □□□□□□Java 21 □□□
- □□□ `"""..."""`□Java 15+ □□□15 □□□
- □□□□□□Virtual Threads□Java 21□
- `sealed` □□□□Java 17 □□□

□□□□□□□□□□□□□□□□□□□□□□

3.1 mods.toml → neoforge.mods.toml

□ NeoForge 20.5□MC 1.20.5□□□□□□□□□□□□ `mods.toml` □□ `neoforge.mods.toml`□

□□□	1.20.4	1.20.6
□□□	`mods.toml`	`neoforge.mods.toml`
□□□□	`META-INF/mods.toml`	`META-INF/neoforge.mods.toml`

build.gradle □□□□□□□□□□□□□□

```gradle
// 1.20.4
filesMatching(['META-INF/mods.toml']) {
    expand replaceProperties
}

// 1.20.6
filesMatching(['META-INF/neoforge.mods.toml']) {
    expand replaceProperties
}
```

3.2 neoforge.mods.toml □□□□

1.20.6 □ `neoforge.mods.toml` □□□ 1.20.4 □ `mods.toml` □□□□□
□□□□□□□□□ `loaderVersion` □□□□

```toml
# 1.20.4 mods.toml
loaderVersion="${loader_version_range}"  # [1,)

# 1.20.6 neoforge.mods.toml
loaderVersion="${loader_version_range}"  # [4,)
```

```

### 3.3 FML □□□□□□

1.20.5+ □ NeoForge □□□□□ FML □□□□□ 1 □□ 4□

| □□ | FML □□□ | loaderVersion |
|---|---|---|
| 1.20.4 | 1 | `[1,)` |
| 1.20.6 | 4 | `[4,)` |
| 1.21.1 | 4 | `[4,)` |

---

## □□Data Components □□□□□□□□

### 4.1 □□

1.20.5+ □□□ **Data Components □□**□□□□□□□ `ItemStack` NBT □□□□□□□□
□□□□□□□□□□□□□□□ Data Components □□□

| □□□ | 1.20.4 (NBT) | 1.20.6 (Data Components) |
|---|---|---|
| □□□□ | `stack.getOrCreateTag().getInt("value")` | `stack.get(MY_COMPONENT)` |
| □□□□ | `stack.getOrCreateTag().putInt("value", 100)` | `stack.set(MY_COMPONENT, 100)` |
| □□□□ | `stack.getOrCreateTag().contains("value")` | `stack.contains(MY_COMPONENT)` |
| □□□□ | NBT tag □□ | `DataComponentTypes.XXX` |

### 4.2 □□ Data Component □□

```java
import net.minecraft.core.component.DataComponentTypes;

DataComponentTypes.MAX_STACK_SIZE   // □□□□□
DataComponentTypes.MAX_DAMAGE         // □□□□□
DataComponentTypes.DAMAGE             // □□□□□
DataComponentTypes.RARITY             // □□□
DataComponentTypes.FOOD               // □□□□
DataComponentTypes.CUSTOM_NAME        // □□□□□
DataComponentTypes.LORE               // □□□□
DataComponentTypes.ENCHANTMENTS       // □□
DataComponentTypes.REPAIR_COST        // □□□□
DataComponentTypes.CREATIVE_SLOT_LOCK // □□□□□□
```

> □□□NeoForge □□ Mojang □□□□□□□□□□ `net.minecraft.core.component`□
> □□ Fabric □ Yarn □□ `net.minecraft.component`□

4.3 □□ ItemStack □□□□

```java
import net.minecraft.core.component.DataComponentTypes;
import net.minecraft.world.item.ItemStack;

// □□
int maxDamage = stack.get(DataComponentTypes.MAX_DAMAGE);

// □□
stack.set(DataComponentTypes.MAX_DAMAGE, 500);

// □□
boolean hasFood = stack.contains(DataComponentTypes.FOOD);
```

4.4 □□□□□ Data Component

```java
import net.minecraft.core.component.DataComponentType;
import net.minecraft.core.registries.Registries;
import net.neoforged.neoforge.registries.DeferredRegister;
import com.mojang.serialization.Codec;

public class ModComponents {
    public static final DeferredRegister<DataComponentType<?>> COMPONENTS =
        DeferredRegister.create(Registries.DATA_COMPONENT_TYPE, "examplemod");
    public static final DataComponentType<Integer> CHARGE =
```
```

```
        COMPONENTS.register("charge", () -> DataComponentType.<Integer>builder()
            .codec(Codec.INT)
            .build());
}
```

□□ItemStack NBT □□□□

1.20.5+ □□□□□□ `ItemStack.getOrCreateTag()` □ NBT □□□
□□□□□□□□□ Data Components □□□

```java
// 1.20.4 —— NBT □□
CompoundTag tag = stack.getOrCreateTag();
tag.putInt("value", 100);
int value = tag.getInt("value");

// 1.20.6 —— Data Components □□
stack.set(ModComponents.CHARGE, 100);
Integer charge = stack.get(ModComponents.CHARGE);
```

□□□□□□□□□

6.1 FoodProperties □□

NeoForge □□□□□□□□□□□□□□□□ `FoodProperties`□
1.20.5+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□ Data Components □□□

```java
// 1.20.6 —— □□ Data Components □□□□□□□
boolean hasFood = stack.contains(DataComponentTypes.FOOD);
```

6.2 Item.Properties □□□□

```java
// 1.20.6
new Item(new Item.Properties()
    .food(new FoodProperties.Builder()
        .nutrition(6)
        .saturationMod(0.6f)
        .meat()
        .build())
);
```

□□□□□□□□□□□□□□□□

7.1 □□□□

1.20.6 □□□□□□□□□□ `Enchantment` □□□ 1.20.4 □□□□
□ 1.20.5+ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```java
// 1.20.6 —— □□□□□□ Enchantment
public class MyEnchantment extends Enchantment {
    public MyEnchantment(Rarity rarity, EnchantmentTarget target, EquipmentSlot... slots) {
        super(rarity, target, slots);
    }
    @Override
    public int getMinCost(int level) { return 10 + (level - 1) * 15; }
    @Override
    public int getMaxLevel() { return 3; }
}
```

> □□□1.20.6 □□□□□□□□□□ 1.20.4 □□□□□□□□□□□□□□□□ 1.21+ □□□□□
> 1.20.6 □□□□□□□□□□□□ `EnchantmentTarget` □□□1.21 □□□□□□□□□□□
> □□□□□□□□□□□□□ `getMinPower` → `getMinCost`□□□□□□□□□□□□

□□BlockEntity □□/□□□□

1.20.5+ □ `BlockEntity` □ `saveAdditional` □ `loadAdditional` □□□□□ `HolderLookup.Provider` □□□

```java
// 1.20.6
@Override
public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.saveAdditional(tag, registries);
    tag.putInt("value", this.value);
}

@Override
public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
    super.loadAdditional(tag, registries);
    this.value = tag.getInt("value");
}
```

□□□□□□□□□

9.1 DeferredRegister □□□□

1.20.6 □ `DeferredRegister` □□□ 1.20.4 □□□□□

```java
// 1.20.6
import net.neoforged.neoforge.registries.DeferredRegister;
import net.minecraft.core.registries.Registries;

public static final DeferredRegister<Item> ITEMS =
    DeferredRegister.create(Registries.ITEM, "examplemod");

public static final RegistryObject<Item> EXAMPLE_ITEM =
    ITEMS.register("example_item", () -> new Item(new Item.Properties()));
```

9.2 □□ DATA_COMPONENT_TYPE □□□

```java
// 1.20.6 □□
Registries.DATA_COMPONENT_TYPE  // Data Component □□□□□
```

□□□□□□□□1.20.4 → 1.20.6□

1. **□□ JDK □ 21**□□□ build.gradle □□ Java □□□17 → 21□□
2. **□□□□□□□□□**□`mods.toml` → `neoforge.mods.toml`□
3. □□ build.gradle □ `filesMatching` □□□□□□□
4. `loaderVersion` □ `[1,)` □□ `[4,)`□
5. ItemStack NBT □□ → Data Components □□□□□□□□□
6. `BlockEntity.saveAdditional(CompoundTag)` → `saveAdditional(CompoundTag, HolderLookup.Provider)`□
7. □□ `gradle.properties` □□□□□□
   - NeoForge□`20.4.251` → `20.6.62`
   - Minecraft□`1.20.4` → `1.20.6`
   - loaderVersion□`[1,)` → `[4,)`

□□□□□□□□

11.1 □ 1.20.6 □□□ Java 17

```gradle
// □ □□□1.20.6 □□ Java 21
java.toolchain.languageVersion = JavaLanguageVersion.of(17)

// □ □□□□□ Java 21
java.toolchain.languageVersion = JavaLanguageVersion.of(21)
```

11.2 □ 1.20.6 □□□□□□ mods.toml □□□

```
// □ □□□1.20.6 □□ neoforge.mods.toml□□□□□ mods.toml
META-INF/mods.toml

// □ □□□1.20.6 □□ neoforge.mods.toml
META-INF/neoforge.mods.toml
```

11.3 □ 1.20.6 □□□□□ NBT API

```java
// □ □□□1.20.6 □□□□ getOrCreateTag()
CompoundTag tag = stack.getOrCreateTag();

// □ □□□□□□ Data Components
stack.set(ModComponents.CHARGE, 100);
```

11.4 □ 1.20.6 □□□□ ModDevGradle □□

```gradle
// □ □□□□ModDevGradle □□□□ NeoForge 21+□MC 1.21+□□□□□□ 1.20.6!
id 'net.neoforged.moddev' version '2.0.146'

// □ □□□1.20.6 □□ NeoGradle
id 'net.neoforged.gradle.userdev' version '7.1.+'
```

11.5 □ 1.20.6 □□□□□□ loaderVersion

```toml
// □ □□□1.20.6 □ FML □□□□□ 4
loaderVersion="[1,)"

// □ □□□1.20.6 □□□□□□ 4
loaderVersion="[4,)"
```

recipe/shaped_recipe.json

```
{
    "_comment": "================================================================================",
    "_comment_2": "NeoForge 1.20.6 □□□□□□□□□□ (shaped_recipe.json)",
    "_comment_3": "NeoForge 1.20.6, Java 21",
    "_comment_4": "================================================================================",
    "_comment_5": "□□□□□□□ JSON □□ - NeoForge/Forge/Fabric □□!",
    "_comment_6": "",
    "_comment_7": "□□□□□: data/<modid>/recipes/<name>.json",
    "_comment_8": "(□□: □□□□□□ recipes □□, 1.21.x □□□□□□ recipe)",
    "_comment_9": "",
    "_comment_10": "□□ NeoForge 1.20.4 □□□□□",
    "_comment_11": "1. □□□□□□□□□□ - 1.20.4 □ 1.20.6 □□□□□□",
    "_comment_12": "2. □□ recipes □□□□□□",
    "_comment_13": "3. result □□□ \"item\" □□",
    "_comment_14": "",
    "_comment_15": "□□ NeoForge 1.21.1 □□□□□□!□",
    "_comment_16": "1. □□□□□□□□□□1.21+ □□□□□□□□□□□□□□□□□□□□:",
    "_comment_17": "   NeoForge 1.20.6: data/<modid>/recipes/ (□□)",
    "_comment_18": "   NeoForge 1.21.1: data/<modid>/recipe/ (□□)",
    "_comment_19": "",
    "_comment_20": "2. □result □□□□!□",
    "_comment_21": "   NeoForge 1.20.6: \"result\": {\"item\": \"examplemod:example_item\"}",
    "_comment_22": "   NeoForge 1.21.1: \"result\": {\"id\": \"examplemod:example_item\"}",
    "_comment_23": "   □□□□□□□□ \"item\" □□ \"id\"!",
    "_comment_24": "",
    "_comment_25": "3. □Data Components□",
    "_comment_26": "   NeoForge 1.20.6: result □□□□□□ Data Components",
    "_comment_27": "   NeoForge 1.21.1: result □□□□□□ components □□",
    "_comment_28": "",
    "_comment_29": "□□ Forge 1.20.1 □□□□",
    "_comment_30": "1. Forge 1.20.1 □□□□□□□□□□□□",
    "_comment_31": "2. NeoForge □□ neoforge:conditions □□□□",
    "_comment_32": "   (Forge □□ forge:conditions)",
    "_comment_33": "================================================================================",
```

```
    "type": "minecraft:crafting_shaped",
    "pattern": [
        "SSS",
        "S S",
        "SSS"
    ],
    "key": {
        "S": {
            "item": "examplemod:example_item"
        }
    },
    "result": {
        "item": "examplemod:example_block",
        "count": 1
    }
}
```

tag/block_tag.json

```
{
    "_comment": "================================================================================",
    "_comment_2": "NeoForge 1.20.6 □□□□□□□ (block_tag.json)",
    "_comment_3": "NeoForge 1.20.6, Java 21",
    "_comment_4": "================================================================================",
    "_comment_5": "□□□□□□□ JSON □□ - NeoForge/Forge/Fabric □□!",
    "_comment_6": "",
    "_comment_7": "□□□□: data/<modid>/tags/blocks/<tag_name>.json",
    "_comment_8": "(□□: □□□□□□ blocks □□, 1.21.x □□□□□□ block)",
    "_comment_9": "",
    "_comment_10": "□□ NeoForge 1.20.4 □□□□",
    "_comment_11": "1. □□□□□□□□□ - 1.20.4 □ 1.20.6 □□□□□□",
    "_comment_12": "2. □□ tags/blocks □□□□□□",
    "_comment_13": "",
    "_comment_14": "□□ NeoForge 1.21.1 □□□□□□!□",
    "_comment_15": "1. □□□□□□□□□1.21+ □□□□□□□□□□□□□□□□□□□□:",
    "_comment_16": "   NeoForge 1.20.6: data/<modid>/tags/blocks/ (□□)",
    "_comment_17": "   NeoForge 1.21.1: data/<modid>/tags/block/ (□□)",
    "_comment_18": "",
    "_comment_19": "2. □□ JSON □□□□□□□: values, replace, remove □□□□□□□",
    "_comment_20": "",
    "_comment_21": "□□ Forge 1.20.1 □□□□□",
    "_comment_22": "1. Forge 1.20.1 □□□□□□□□□□□□□",
    "_comment_23": "2. □□: Forge □□ Tags.Blocks / Tags.Items (□□□□)",
    "_comment_24": "   Forge:     net.minecraftforge.common.Tags",
    "_comment_25": "   NeoForge:  net.neoforged.neoforge.common.Tags",
    "_comment_26": "3. NeoForge □ Forge □□□□□□□ replace □ remove □□",
    "_comment_27": "================================================================================",

    "replace": false,
    "values": [
        "minecraft:stone",
        "minecraft:cobblestone",
        "minecraft:granite",
        "minecraft:diorite",
        "minecraft:andesite",
        "#minecraft:base_stone_overworld",
        {
            "id": "examplemod:example_block",
            "required": false
        },
        {
            "id": "#minecraft:mineable/pickaxe",
            "required": true
        }
    ]
}
```

worldgen/biome_modifier.json

```
{
    "_comment": "================================================================================",
    "_comment_2": "NeoForge 1.20.6 □□□□□□□□□□ (biome_modifier.json)",
    "_comment_3": "NeoForge 1.20.6, Java 21",
    "_comment_4": "================================================================================",
    "_comment_5": "□□□□□□□□ - □□□□□□□□□□□□/□□□□□□□□□□□□□□",
    "_comment_6": "",
    "_comment_7": "□□□□: data/<modid>/neoforge/biome_modifier/<name>.json",
```

```
    "_comment_8": "(□□: □□□□□□ biome_modifier □□)",
    "_comment_9": "",
    "_comment_10": "□□ NeoForge 1.20.4 □□□□□",
    "_comment_11": "1. □□□□□: □□□□ neoforge: □□",
    "_comment_12": "2. □□□□□: □□ neoforge/biome_modifier/ □□",
    "_comment_3": "3. □□□□□□□□ - □□□□□□□□□□ 1.20.4 □ 1.20.6 □□□□□□",
    "_comment_14": "",
    "_comment_15": "□□ NeoForge 1.21.1 □□□□□",
    "_comment_16": "1. □□□□□□!□□□□ neoforge: □□",
    "_comment_17": "",
    "_comment_18": "2. □□□□□□□□□□□□",
    "_comment_19": "   NeoForge 1.20.6: neoforge/biome_modifier/ (□□)",
    "_comment_20": "   NeoForge 1.21.1: neoforge/biome_modifiers/ (□□)",
    "_comment_21": "   (1.21.x □□□□□□□□□□/□□□□□□)",
    "_comment_22": "",
    "_comment_23": "□□ Forge 1.20.1 □□□□□□ - □□□□□□!□",
    "_comment_24": "1. □□□□□□!□",
    "_comment_25": "   Forge 1.20.1: type □□ forge: □□ (□ forge:add_features)",
    "_comment_26": "   NeoForge 1.20.6: type □□ neoforge: □□ (□ neoforge:add_features)",
    "_comment_27": "   □□□□Forge □ forge: □□, NeoForge □ neoforge: □□!",
    "_comment_28": "",
    "_comment_29": "2. □□□□□□□□!□",
    "_comment_30": "   Forge 1.20.1: data/<modid>/forge/biome_modifier/<name>.json",
    "_comment_31": "   NeoForge 1.20.6: data/<modid>/neoforge/biome_modifier/<name>.json",
    "_comment_32": "   □□□□Forge □ forge/ □□, NeoForge □ neoforge/ □□!",
    "_comment_33": "",
    "_comment_34": "3. □□□□□□□□□□□ - □□□□□□",
    "_comment_35": "   - neoforge:add_features / forge:add_features: □□□□",
    "_comment_36": "   - neoforge:add_spawns / forge:add_spawns: □□□□□□",
    "_comment_37": "   - neoforge:remove_features / forge:remove_features: □□□□",
    "_comment_38": "   - neoforge:remove_spawns / forge:remove_spawns: □□□□□□",
    "_comment_39": "============================================================================",

    "type": "neoforge:add_features",
    "biomes": "#minecraft:is_overworld",
    "features": "examplemod:ore_example",
    "step": "underground_ores"
}
```

NeoForge 1.21.1

block/simple_block.java

```
// ==========================================================================
// NeoForge 1.21.1 □□□□□□□ (simple_block.java)
// ==========================================================================
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □DeferredRegister □□□
//    Forge:   net.minecraftforge.registries.DeferredRegister
//    NeoForge: net.neoforged.neoforge.registries.DeferredRegister
// 2. □DeferredRegister.Blocks □□□□
//    NeoForge □□□ DeferredRegister.Blocks □□□, □□ DeferredBlock<T>
//    (□□□ DeferredHolder<Block, T>), □□□□□□□□□□□□□□□
//    □□ DeferredRegister.createBlocks(MODID) □□□
// 3. □Registries □□
//    Forge 1.20.1 □□□□□□□□ ForgeRegistries □ (□ BlockEntityType)□
//    NeoForge 1.21.1 □□□□□□ net.minecraft.core.registries.Registries
//    □ net.minecraft.core.registries.BuiltInRegistries□
// 4. □BlockBehaviour.Properties□
//    □□□□ AbstractBlock.Properties □□□□ BlockBehaviour.Properties□
//    (□□□□□ 1.20.x □□□□□□□)
// 5. □BlockEntityType.Builder□
//    □□ BlockEntityType.Builder.of(supplier, blocks...).build(null) □□□
//    NeoForge □□□□ BlockEntityType.BuilderDelegating □□□□
// 6. □EntityBlock □□□
//    □□ EntityBlock □□□□□□ newBlockEntity() □□, □□□ Forge □□□
// 7. □Block □□□□ (□)□
//    1.21+ □□□ MapCodec □□□□□□□, □□□□ BlockType □□□ codec() □□□
// ==========================================================================

package com.example.examplemod.block;

import com.example.examplemod.ExampleMod;
```

```
import com.mojang.serialization.MapCodec;
import net.minecraft.core.BlockPos;
import net.minecraft.core.HolderLookup;
import net.minecraft.core.registries.BuiltInRegistries;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.protocol.Packet;
import net.minecraft.network.protocol.game.ClientGamePacketListener;
import net.minecraft.network.protocol.game.ClientboundBlockEntityDataPacket;
import net.minecraft.world.level.Level;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.Blocks;
import net.minecraft.world.level.block.EntityBlock;
import net.minecraft.world.level.block.entity.BlockEntity;
import net.minecraft.world.level.block.entity.BlockEntityTicker;
import net.minecraft.world.level.block.entity.BlockEntityType;
import net.minecraft.world.level.block.state.BlockState;
import net.minecraft.world.level.block.state.BlockBehaviour;
import net.minecraft.world.level.material.MapColor;
import net.minecraft.sounds.SoundType;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;
import net.neoforged.neoforge.registries.DeferredBlock;

import javax.annotation.Nullable;

// ================================================================================
// □□□□□□
// ================================================================================
public class ModBlocks {

    // ================================================================================
    // □□□□□□□ DeferredRegister.Blocks □□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCKS, MODID)
    // NeoForge 1.21.1: DeferredRegister.createBlocks(MODID)
    //
    // createBlocks □□ DeferredRegister.Blocks, □:
    // - register() □□□□□ DeferredBlock<T> (□□□□ DeferredHolder<Block, T>)
    // - □□□ registerBlock()□registerSimpleBlock() □□□□□□
    // ================================================================================
    public static final DeferredRegister.Blocks BLOCKS =
            DeferredRegister.createBlocks(ExampleMod.MOD_ID);

    // ================================================================================
    // □□□□□□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.BLOCK_ENTITY_TYPES, MODID)
    // NeoForge 1.21.1: DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, MODID)
    //
    // □□□□□□□□□□□□: BLOCK_ENTITY_TYPE (□□ BLOCK_ENTITIES)
    //         □□□□□: Registries (□□ ForgeRegistries)
    // ================================================================================
    public static final DeferredRegister<BlockEntityType<?>> BLOCK_ENTITY_TYPES =
            DeferredRegister.create(Registries.BLOCK_ENTITY_TYPE, ExampleMod.MOD_ID);

    // ================================================================================
    // □□□□□□□□□□
    // □□□□□□□□□□: DeferredBlock<Block> (Forge □□ RegistryObject<Block>)
    // ================================================================================
    public static final DeferredBlock<Block> EXAMPLE_BLOCK = BLOCKS.register(
            "example_block",
            () -> new Block(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.STONE)      // □□□□□ (1.20.x □□ .material)
                    .destroyTime(2.0f)             // □□□□□
                    .explosionResistance(10.0f)     // □□□□□
                    .sound(SoundType.GRAVEL)        // □□□□□
                    .lightLevel(state -> 7)         // □□□□□ (□□□□)
            )
    );

    // ================================================================================
    // □□□□□□□□□□□□□□□□ (EntityBlock)
    // ================================================================================
    public static final DeferredBlock<ExampleEntityBlock> EXAMPLE_ENTITY_BLOCK = BLOCKS.register(
            "example_entity_block",
            () -> new ExampleEntityBlock(BlockBehaviour.Properties.of()
                    .mapColor(MapColor.METAL)
```

```
                    .destroyTime(3.0f)
                    .explosionResistance(15.0f)
                    .sound(SoundType.METAL)
            )
    );

    // ==========================================================================
    // □□□□□□□□□□
    // □□□□□□□□ BlockEntityType.Builder.of(supplier, blocks...).build(null)
    //          Forge 1.20.1 □□□□□, □ supplier □□□□□□□□□□□□
    //
    // NeoForge □□□□□ BlockEntityType.BuilderDelegating □□□,
    // □□□□□ Builder, □□□□□□□□□□□□□□□□□□□□□ Builder□
    // ==========================================================================
    public static final DeferredHolder<BlockEntityType<?>, BlockEntityType<ExampleBlockEntity>> EXAMPLE_BLOCK_ENTITY =
            BLOCK_ENTITY_TYPES.register(
                    "example_block_entity",
                    () -> BlockEntityType.Builder.of(
                            ExampleBlockEntity::new,
                            EXAMPLE_ENTITY_BLOCK.get()
                    )
                    // build(null) - □□ null □□□□□□□□ DataFixer
                    .build(null)
            );
}

// ==========================================================================
// □□□□□□□□□□□□
// □□ EntityBlock □□, □□ newBlockEntity() □□
// ==========================================================================
class ExampleEntityBlock extends Block implements EntityBlock {

    public ExampleEntityBlock(BlockBehaviour.Properties properties) {
        super(properties);
    }

    // □□□□□□□□□□
    @Override
    public BlockEntity newBlockEntity(BlockPos pos, BlockState state) {
        return new ExampleBlockEntity(pos, state);
    }

    // ==========================================================================
    // □□ Ticker (□ tick □□□□)
    // ==========================================================================
    @Nullable
    @Override
    public <T extends BlockEntity> BlockEntityTicker<T> getTicker(
            Level level, BlockState state, BlockEntityType<T> type) {
        // □□ createTickerHelper □□□□□□□□
        return type == ModBlocks.EXAMPLE_BLOCK_ENTITY.get()
                ? (lvl, pos, st, be) -> ExampleBlockEntity.tick(lvl, pos, st, (ExampleBlockEntity) be)
                : null;
    }

    // ==========================================================================
    // □□□Block □□□□□ (1.21+)
    // 1.21+ □□□□ MapCodec □□□□□□□□□□□ Block □□□□□□□ codec() □□,
    // □□ MapCodec □□□□ BuiltInRegistries.BLOCK_TYPE □□□□□
    // □□□□□□□□ (□□□□ Properties □□), □□□□□ BlockBehaviour#simpleCodec□
    // ==========================================================================
    // @Override
    // public MapCodec<? extends Block> codec() {
    //     return CODEC.value();
    // }
}

// ==========================================================================
// □□□□□□□
// □□□□□□1.21.x □ saveAdditional □ loadAdditional □□□□□□
//         HolderLookup.Provider registries □□□
//         Forge 1.20.1: saveAdditional(CompoundTag tag)
//         NeoForge 1.21.1: saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
// ==========================================================================
class ExampleBlockEntity extends BlockEntity {
    private int value = 0;
```

```
    public ExampleBlockEntity(BlockPos pos, BlockState state) {
        super(ModBlocks.EXAMPLE_BLOCK_ENTITY.get(), pos, state);
    }

    // ========================================================================
    // □□□□□ (□ NBT)
    // □□□□loadAdditional □□□ HolderLookup.Provider □□
    // Forge 1.20.1: public void load(CompoundTag tag)
    // NeoForge 1.21.1: public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries)
    // ========================================================================
    @Override
    public void loadAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.loadAdditional(tag, registries);
        this.value = tag.getInt("value");
    }

    // ========================================================================
    // □□□□□ (□ NBT)
    // □□□□saveAdditional □□□ HolderLookup.Provider □□
    // Forge 1.20.1: public void saveAdditional(CompoundTag tag)
    // NeoForge 1.21.1: public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries)
    // ========================================================================
    @Override
    public void saveAdditional(CompoundTag tag, HolderLookup.Provider registries) {
        super.saveAdditional(tag, registries);
        tag.putInt("value", this.value);
    }

    // ========================================================================
    // □□□□□ - □□□□□□
    // ========================================================================
    @Override
    public CompoundTag getUpdateTag(HolderLookup.Provider registries) {
        CompoundTag tag = new CompoundTag();
        saveAdditional(tag, registries);
        return tag;
    }

    // □□□□□ - □□□□□□
    @Override
    public Packet<ClientGamePacketListener> getUpdatePacket() {
        return ClientboundBlockEntityDataPacket.create(this);
    }

    // ========================================================================
    // □ tick □□□□□□
    // ========================================================================
    public static void tick(Level level, BlockPos pos, BlockState state, ExampleBlockEntity blockEntity) {
        if (!level.isClientSide) {
            // □□□□□□ (□□: □□□□□□□)
            // blockEntity.value++;
            // blockEntity.setChanged(); // □□□□□□□, □□□□
        }
    }
}
```

command/simple_command.java

```
// ========================================================================
// NeoForge 1.21.1 □□□□□□□ (simple_command.java)
// ========================================================================
//
// □□ Forge 1.20.1 □□□□□□ - □□□□□□
// 1. □□□□□□□□□□
//    Forge 1.20.1: RegisterCommandsEvent (net.minecraftforge.event.RegisterCommandsEvent)
//    NeoForge 1.21.1: RegisterCommandsEvent (net.neoforged.neoforge.event.RegisterCommandsEvent)
//    □□□□□□□□□□□□□□□□ (net.minecraftforge -> net.neoforged)
//
// 2. □Brigadier API□
//    1.21.x □ Brigadier API □□□□□□□□□□
//    □□□□ Commands.literal(), Commands.argument() □□
//    CommandSourceStack □□□□□□□□□□□□
//
// 3. □□□□□□□
//    RegisterCommandsEvent □ NeoForge □□□□□□□□□□□□:
//    NeoForge.EVENT_BUS (□□□□□□□□□ modEventBus!)
```

```
//
//    Forge 1.20.1: MinecraftForge.EVENT_BUS.register(...)
//    NeoForge 1.21.1: NeoForge.EVENT_BUS.register(...)
//
// 4. □□□□□□□
//    Commands.literal("name").requires(source -> source.hasPermission(level))
//    level 0: □□□□□, level 2+: OP □□, level 4: □□□
// ============================================================================

package com.example.examplemod.command;

import com.mojang.brigadier.Command;
import com.mojang.brigadier.arguments.IntegerArgumentType;
import com.mojang.brigadier.arguments.StringArgumentType;
import com.mojang.brigadier.context.CommandContext;
import com.mojang.brigadier.exceptions.CommandSyntaxException;
import net.minecraft.commands.CommandSourceStack;
import net.minecraft.commands.Commands;
import net.minecraft.commands.arguments.EntityArgument;
import net.minecraft.commands.arguments.coordinates.BlockPosArgument;
import net.minecraft.core.BlockPos;
import net.minecraft.network.chat.Component;
import net.minecraft.world.entity.Entity;
import net.neoforged.bus.api.SubscribeEvent;
import net.neoforged.neoforge.event.RegisterCommandsEvent;
import net.neoforged.neoforge.event.CommandEvent;

import java.util.Collection;

// ============================================================================
// □□□□□□
// ============================================================================
public class ModCommands {

    // ========================================================================
    // □□□□□ - □□ RegisterCommandsEvent
    //
    // □□□□□RegisterCommandsEvent □ NeoForge □□□□□□□□□□□
    // (NeoForge.EVENT_BUS), □□□□□□□□□ (modEventBus)
    //
    // □□□□□:
    //   □□□□□□□□□: NeoForge.EVENT_BUS.register(new ModCommands());
    //   □□□□ @SubscribeEvent □□□□□ + □□□□□ NeoForge.EVENT_BUS
    //
    // Forge 1.20.1: MinecraftForge.EVENT_BUS.register(new ModCommands());
    // NeoForge 1.21.1: NeoForge.EVENT_BUS.register(new ModCommands());
    // ========================================================================
    @SubscribeEvent
    public static void registerCommands(RegisterCommandsEvent event) {
        CommandSourceStack source = event.getDispatcher().getSource();

        // ====================================================================
        // □□1: /examplemod hello
        // □□□□□, □□□
        // ====================================================================
        event.getDispatcher().register(
                Commands.literal("examplemod")
                        // □□□□□□□□ (0 = □□□□□, 2 = OP, 4 = □□□)
                        .requires(src -> src.hasPermission(0))
                        .then(Commands.literal("hello")
                                // □□□□□□□
                                .executes(ModCommands::helloCommand)
                        )
                        // □□□□□ (□□□□□): /examplemod
                        .executes(context -> {
                            context.getSource().sendSuccess(
                                    () -> Component.literal("ExampleMod command!"),
                                    false
                            );
                            return Command.SINGLE_SUCCESS;
                        })
        );

        // ====================================================================
        // □□2: /examplemod setvalue <player> <value>
        // □□□□□□□
```

```
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(2))  // OP □□
                    .then(Commands.literal("setvalue")
                            .then(Commands.argument("target", EntityArgument.player())
                                    .then(Commands.argument("value", IntegerArgumentType.integer(0, 100))
                                            .executes(ModCommands::setValueCommand)
                                    )
                            )
                    )
    );

    // ============================================================================
    // □□3: /examplemod message <message...>
    // □□□□□□□□ (□□□□□)
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(0))
                    .then(Commands.literal("message")
                            .then(Commands.argument("message", StringArgumentType.greedyString())
                                    .executes(context -> {
                                        String msg = StringArgumentType.getString(context, "message");
                                        context.getSource().sendSuccess(
                                                () -> Component.literal("Message: " + msg),
                                                false
                                        );
                                        return Command.SINGLE_SUCCESS;
                                    })
                            )
                    )
    );

    // ============================================================================
    // □□4: /examplemod teleport <position>
    // □□□□□□□
    // ============================================================================
    event.getDispatcher().register(
            Commands.literal("examplemod")
                    .requires(src -> src.hasPermission(2))
                    .then(Commands.literal("teleport")
                            .then(Commands.argument("pos", BlockPosArgument.blockPos())
                                    .executes(context -> {
                                        BlockPos pos = BlockPosArgument.getBlockPos(context, "pos");
                                        Entity executor = context.getSource().getEntityOrPlayer();
                                        if (executor != null) {
                                            executor.teleportTo(
                                                    pos.getX() + 0.5,
                                                    pos.getY(),
                                                    pos.getZ() + 0.5
                                            );
                                            context.getSource().sendSuccess(
                                                    () -> Component.literal(
                                                            "Teleported to " + pos.toShortString()
                                                    ),
                                                    true  // true = □□□ OP □□
                                            );
                                        }
                                        return Command.SINGLE_SUCCESS;
                                    })
                            )
                    )
    );
}

// ============================================================================
// hello □□□□□□□□
// ============================================================================
private static int helloCommand(CommandContext<CommandSourceStack> context) {
    // □□□□□□
    CommandSourceStack source = context.getSource();

    // □□□□□□□ (□□ Supplier<Component> □□□□□ Component)
    // □□□□1.21.x □ sendSuccess □□ Supplier<Component>
    source.sendSuccess(
```

```
                () -> Component.literal("Hello from ExampleMod!"),
                false  // false = □□□□□□□ OP □□□
        );

        // □□□□□□□□
        // source.sendFailure(Component.literal("Command failed!"));

        // □□□□□□□ (□□□□□□□□□□□□□□)
        Entity entity = source.getEntity();
        if (entity != null) {
            source.sendSuccess(
                    () -> Component.literal("Executed by: " + entity.getName().getString()),
                    false
            );
        }

        return Command.SINGLE_SUCCESS;  // □□ 1 □□□□□
    }

    // ============================================================================
    // setvalue □□□□□□□□
    // □□ EntityArgument □ IntegerArgumentType □□□□
    // ============================================================================
    private static int setValueCommand(CommandContext<CommandSourceStack> context)
            throws CommandSyntaxException {
        CommandSourceStack source = context.getSource();

        // □□□□□□□□□□
        Entity target = EntityArgument.getPlayer(context, "target");

        // □□□□□□□□
        int value = IntegerArgumentType.getInteger(context, "value");

        // □□□□□□□ (□□□□□□□□□□□□□□□)
        source.sendSuccess(
                () -> Component.literal(
                        "Set " + target.getName().getString() + "'s value to " + value
                ),
                true  // true = □□□ OP □□
        );

        return value;  // □□□□□□□ (□□□ /execute store)
    }

    // ============================================================================
    // □□□□ Brigadier □□□□:
    // ============================================================================
    // 1. □□□□: EntityArgument.entity(), EntityArgument.player(),
    //          EntityArgument.entities(), EntityArgument.players()
    // 2. □□□□: BlockPosArgument.blockPos(), CoordinatesArgument
    // 3. □□□□: ItemArgument.item(), ItemStackArgument
    // 4. □□/□□□□: StringArgumentType.string(), StringArgumentType.word(),
    //                  StringArgumentType.greedyString()
    // 5. □□□□: IntegerArgumentType.integer(min, max),
    //              FloatArgumentType.floatArg(min, max),
    //              LongArgumentType.longArg()
    // 6. □□□□: BoolArgumentType.bool()
    //
    // □□□□□□□:
    // 1. .executes() - □□□□
    // 2. .then() - □□□□□□/□□
    // 3. .requires() - □□□□□□□
    // 4. .suggests() - □□□□□□□□□
    //
    // □□□□□□:
    // event.getDispatcher() - □□□□□□□□□
    // event.getBuildContext() - □□□□□□□□ (1.21.x)
    // ============================================================================
}
```

enchantment/custom_enchantment.java

```
// ============================================================================
// NeoForge 1.21.1 □□□□□□□□ (custom_enchantment.java)
// ============================================================================
//
```

```
// ═══ Forge 1.20.1 ═══
//
// 1. ═══
//    Forge 1.20.1: ═══ (BuiltInRegistries/ForgeRegistries),
//                  ═══ DeferredRegister.create(ForgeRegistries.ENCHANTMENTS, MODID) ═══,
//                  ═══ Enchantment ═══
//    NeoForge 1.21.1: ═══ (datapack registry)!
//                     ═══ JSON ═══, ═══ Java ═══
//                     ═══: Registries.ENCHANTMENT
//                     ═══ Enchantment ═══ Holder<Enchantment>
//
// 2. ═══ Enchantment ═══!
//    Forge 1.20.1: ═══ extends Enchantment, ═══:
//                  - getMaxLevel()
//                  - getMinCost() / getMaxCost()
//                  - checkCompatibility()
//                  - canEnchant()
//                  - doGetDamageProtection() ═
//    NeoForge 1.21.1: ═══ Enchantment ═══, ═══:
//                     - description (Component): ═══
//                     - definition (EnchantmentDefinition): ═══
//                     - exclusiveSet (HolderSet<Enchantment>): ═══
//                     - effects (DataComponentMap): ═══
//
// 3. ═══ (Enchantment Effect Components)═
//    1.21+ ═══
//    ═══, ═══
//    ═══ DataComponentType<?>, ═══
//    BuiltInRegistries.ENCHANTMENT_EFFECT_COMPONENT_TYPE═
//    ═══ (═ minecraft:damage, minecraft:knockback ═)═
//
// 4. ═══
//    Forge 1.20.1: ═══ ItemStack NBT ═ (Enchantments tag)
//    NeoForge 1.21.1: ═══ DataComponents.ENCHANTMENTS ═══
//                     ═══ ItemEnchantments
//
// 5. ═Enchantments ═══
//    Forge 1.20.1: Enchantments.SHARPNESS ═ Enchantment ═
//    NeoForge 1.21.1: Enchantments.SHARPNESS ═ ResourceKey<Enchantment>
//                     ═══ RegistryAccess ═ Holder<Enchantment>
//
// 6. ═EnchantmentHelper ═══
//    ═══, ═══ EnchantmentEffectComponents ═══
//    ═ EnchantmentHelper.runIterationOnItem() ═══
//    ═ Enchantment.applyEffects() ═══
// ================================================================================

package com.example.examplemod.enchantment;

import com.example.examplemod.ExampleMod;
import com.mojang.serialization.Codec;
import com.mojang.serialization.codecs.RecordCodecBuilder;
import net.minecraft.core.HolderSet;
import net.minecraft.core.RegistrySetBuilder;
import net.minecraft.core.registries.BuiltInRegistries;
import net.minecraft.core.registries.Registries;
import net.minecraft.core.Holder;
import net.minecraft.data.worldgen.BootstrapContext;
import net.minecraft.network.chat.Component;
import net.minecraft.resources.ResourceKey;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.enchantment.Enchantment;
import net.minecraft.core.component.DataComponentType;
import net.minecraft.core.component.DataComponents;
import net.minecraft.world.item.enchantment.Enchantment.EnchantmentDefinition;
import net.minecraft.world.item.enchantment.Enchantment.Cost;
import net.minecraft.world.entity.EquipmentSlotGroup;
import net.minecraft.world.item.enchantment.EnchantmentEffectComponents;
import net.minecraft.world.item.enchantment.ItemEnchantments;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

import java.util.List;
import java.util.Optional;
import java.util.concurrent.atomic.AtomicInteger;
```

```java
import java.util.function.Consumer;

// ============================================================================
// □□□□□□□□□□□□
// ============================================================================
public class ModEnchantments {

    // ========================================================================
    // □□□□□□□□□□□□□□□□□□□□
    //
    // □□□□□□□□□□□□□□□□□□□□
    // BuiltInRegistries.ENCHANTMENT_EFFECT_COMPONENT_TYPE □□□,
    // □□□ DataComponentType<?>□
    //
    // Forge 1.20.1 □□□□□□□□ - □□□□□□□□□□□□□□□□
    // ========================================================================

    // □□□□□□□□□□□□
    public static final DeferredRegister<DataComponentType<?>> ENCHANTMENT_COMPONENT_TYPES =
            DeferredRegister.create(BuiltInRegistries.ENCHANTMENT_EFFECT_COMPONENT_TYPE, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□□□□□: □□□□□□
    //
    // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    // ========================================================================
    public record Increment(int value) {
        // □□□□□□□□ Codec (□□□□□□/□□□□□)
        public static final Codec<Increment> CODEC = RecordCodecBuilder.create(instance ->
                instance.group(
                        Codec.INT.fieldOf("value").forGetter(Increment::value)
                ).apply(instance, Increment::new)
        );

        public int add(int x) {
            return value() + x;
        }
    }

    // □□□□□□□□□□□□□□□□□□
    public static final DeferredHolder<DataComponentType<?>, DataComponentType<Increment>> INCREMENT =
            ENCHANTMENT_COMPONENT_TYPES.register("increment",
                    () -> DataComponentType.<Increment>builder()
                            .persistent(Increment.CODEC)   // □□□□ Codec
                            .build()
            );

    // ========================================================================
    // □□□□□□□□ (ConditionalEffect)
    // □□□□ ConditionalEffect □□□□□□ LootContext □□□□□□□□□□
    // ========================================================================
    public static final DeferredHolder<DataComponentType<?>,
            DataComponentType<net.minecraft.world.level.storage.loot.predicates.ConditionalEffect<Increment>>>
            CONDITIONAL_INCREMENT = ENCHANTMENT_COMPONENT_TYPES.register("conditional_increment",
                    () -> DataComponentType.<net.minecraft.world.level.storage.loot.predicates.ConditionalEffect<Increment>>builder()
                            // ConditionalEffect.codec □□: □□□□□□ Codec □ LootContextParamSet
                            .persistent(
                                    net.minecraft.world.level.storage.loot.predicates.ConditionalEffect.codec(
                                            Increment.CODEC,
                                            net.minecraft.world.level.storage.loot.parameters.LootContextParamSets.ENCHANTED_DAMAGE
                                    )
                            )
                            .build()
            );

    // ========================================================================
    // □□ ResourceKey □□ (□□ DeferredRegister!)
    //
    // □□□□□□□□Forge 1.20.1 □□ DeferredRegister □□□□□:
    //   public static final RegistryObject<Enchantment> MY_ENCHANT =
    //       ENCHANTMENTS.register("my_enchant", () -> new MyEnchantment(...));
    //
    // NeoForge 1.21.1 □□ ResourceKey + □□□□□ (RegistrySetBuilder):
    //   □□□□□□□□□□□□, □□□□ DeferredRegister □□!
    //   □□ RegistrySetBuilder □□□□□□□□□□□□□□
    // ========================================================================
```

```
public static final ResourceKey<Enchantment> EXAMPLE_ENCHANTMENT = ResourceKey.create(
        Registries.ENCHANTMENT,
        ResourceLocation.fromNamespaceAndPath(ExampleMod.MOD_ID, "example_enchantment")
);

// ================================================================================
// □□□□□ - □□ RegistrySetBuilder □□□□□
//
// □□ builder □□□□□□ DatapackBuiltinEntriesProvider
// (□ GatherDataEvent □□□□□□)
//
// Forge 1.20.1 □□□□□□□□ - □□□□□□ Java □□□□□□□
// NeoForge 1.21.1 □□□□□□□□□□□□□□□ (□□ JSON □□)□
// ================================================================================
public static final RegistrySetBuilder BUILDER = new RegistrySetBuilder();

static {
    BUILDER.add(Registries.ENCHANTMENT, ModEnchantments::bootstrapEnchantments);
}

// ================================================================================
// □□□□□□□□ (□□□□□□□□□□□)
// □□□□□□□ Enchantment □□□□□□□□□□
// ================================================================================
private static void bootstrapEnchantments(BootstrapContext<Enchantment> context) {
    // □□□□□□□□□
    context.register(EXAMPLE_ENCHANTMENT, new Enchantment(
            // 1. □□□□□ (□□□□□□□□□)
            Component.translatable("enchantment.examplemod.example_enchantment"),

            // 2. □□□□□ (EnchantmentDefinition)
            new EnchantmentDefinition(
                    // supportedItems: □□□□□□ (HolderSet)
                    // □□□□□□□□□□, □ #examplemod:enchantable/example
                    HolderSet.direct(),

                    // primaryItems: □□□□□ (Optional, □□□□□□□□)
                    // □□□□□□□, □ supportedItems □□
                    Optional.empty(),

                    // weight: □□□□□□□□ [1, 1024]
                    10,

                    // maxLevel: □□□□□ [1, 255]
                    3,

                    // minCost: □□□□□□□
                    new Cost(
                            5,   // base: □□□□
                            8    // perLevelAboveFirst: □□□□□□
                    ),

                    // maxCost: □□□□□□□
                    new Cost(
                            25,  // base
                            9    // perLevelAboveFirst
                    ),

                    // anvilCost: □□□□□ [1, inf)
                    2,

                    // slots: □□□□□□□□□□□□
                    List.of(EquipmentSlotGroup.MAINHAND)
            ),

            // 3. exclusiveSet: □□□□□ (HolderSet<Enchantment>)
            // □□□□□□□□□□□□□□□□□□□□□□□□
            HolderSet.empty(),

            // 4. effects: □□□□□□□ (DataComponentMap)
            // ================================================================================
            // □□□□□□□□□□ DataComponentMap □□, □□□□□□□□□□!
            //
            // Forge 1.20.1 □□□□□□□□□□:
            //   @Override
            //   public float getDamageBonus(int level, ...) { return level * 2.5f; }
```

```
                //
                // NeoForge 1.21.1 □□□□□□□□:
                //   □□ DataComponentMap.builder()
                //   .set(EnchantmentEffectComponents.DAMAGE, ...)
                //   .set(MY_CUSTOM_COMPONENT, new MyData())
                //   .build()
                // ============================================================================
                net.minecraft.util.DataComponentMap.builder()
                        // □□□□□□□□□□□□ (□□□□□□)
                        // .set(EnchantmentEffectComponents.DAMAGE, ...)
                        // □□□□□□□□□□□
                        .set(INCREMENT.get(), new Increment(2))
                        .build()
        ));
    }

    // ============================================================================
    // □ GatherDataEvent □□□□□□□:
    //
    // @SubscribeEvent
    // public static void gatherData(GatherDataEvent event) {
    //     DataGenerator generator = event.getGenerator();
    //     PackOutput packOutput = generator.getPackOutput();
    //     CompletableFuture<HolderLookup.Provider> lookupProvider = event.getLookupProvider();
    //
    //     // □□□□□□□□□□
    //     generator.addProvider(event.includeServer(),
    //         new DatapackBuiltinEntriesProvider(packOutput, lookupProvider, BUILDER,
    //             Set.of(ExampleMod.MOD_ID))
    //     );
    // }
    //
    // □□□ src/generated/data/examplemod/enchantment/example_enchantment.json □□ JSON □□
    // ============================================================================

    // ============================================================================
    // □□ JSON □□ (□ data/examplemod/enchantment/example_enchantment.json □)
    //
    // □ Java □□□□:
    // {
    //     "description": { "translate": "enchantment.examplemod.example_enchantment" },
    //     "supported_items": "#examplemod:enchantable/example",
    //     "primary_items": ["examplemod:item_a"],
    //     "exclusive_set": "#examplemod:exclusive_set",
    //     "weight": 10,
    //     "max_level": 3,
    //     "min_cost": { "base": 5, "per_level_above_first": 8 },
    //     "max_cost": { "base": 25, "per_level_above_first": 9 },
    //     "anvil_cost": 2,
    //     "slots": ["mainhand"],
    //     "effects": {
    //         "examplemod:increment": [
    //             { "effect": { "value": 2 } }
    //         ]
    //     }
    // }
    // ============================================================================
}

// ============================================================================
// □□□□□□□□□□
// ============================================================================
class EnchantmentEffectUsage {

    // ============================================================================
    // □□ EnchantmentHelper □□□□□□□□□
    //
    // □□□□Forge 1.20.1 □□□:
    //   EnchantmentHelper.getEnchantmentLevel(Enchantments.SHARPNESS, stack)
    //   EnchantmentHelper.getItemEnchantmentLevel(enchantment, stack)
    //
    // NeoForge 1.21.1 □□□:
    //   EnchantmentHelper.runIterationOnItem(stack, (enchantmentHolder, level) -> { ... })
    //   □□□□□□□□□, □□□□□□□□□
    // ============================================================================
    public static int applyEnchantmentEffect(ItemStack stack, int baseValue) {
```

```
    AtomicInteger result = new AtomicInteger(baseValue);

    // □□□□□□□□□□□□
    // □□: (ItemStack, EnchantmentHelper.EnchantmentVisitor)
    // EnchantmentVisitor □□□□□□□□□: (Holder<Enchantment>, int level) -> void
    net.minecraft.world.item.enchantment.EnchantmentHelper.runIterationOnItem(
            stack,
            (enchantmentHolder, enchantLevel) -> {
                // □□□ Holder □□□□□□□
                // enchantmentHolder.value() □□ Enchantment □□
                // .effects() □□ DataComponentMap
                Increment increment = enchantmentHolder.value()
                        .effects()
                        .get(ModEnchantments.INCREMENT.get());

                // □□□□□□□ Increment □□□□, □□□□□□
                if (increment != null) {
                    result.set(increment.add(result.get()));
                }
            }
    );

    return result.get();
}

// ============================================================================
// □□ Enchantment.applyEffects() □□□□□□□
//
// □□ ConditionalEffect □□□□□□□□:
// ============================================================================
public static void applyConditionalEffects(
        Enchantment enchant,
        net.minecraft.world.level.storage.loot.LootContext lootContext
) {
    // □□ Enchantment.applyEffects() □□□□□□□
    // □□: (List<ConditionalEffect<T>>, LootContext, Consumer<T>)
    Enchantment.applyEffects(
            enchant.getEffects(EnchantmentEffectComponents.KNOCKBACK),  // □□□□□□□□
            lootContext,
            (effectData) -> {
                // □□□□□□□□□□□□□□□□
                // effectData □□□□□□□□□□□□
            }
    );
}

// ============================================================================
// □□□□□□□□□□□□ (□□ ResourceKey)
//
// □□□□Forge 1.20.1:
//   EnchantmentHelper.getItemEnchantmentLevel(MyEnchantments.MY_ENCHANT.get(), stack)
//
// NeoForge 1.21.1:
//   □□□□ RegistryAccess □□ Holder<Enchantment>,
//   □□□□ ItemEnchantments □□
// ============================================================================
public static int getEnchantmentLevel(
        ItemStack stack,
        ResourceKey<Enchantment> enchantmentKey,
        net.minecraft.core.HolderLookup.Provider registryAccess
) {
    // □□□□□□□□□□
    ItemEnchantments enchantments = stack.get(DataComponents.ENCHANTMENTS);
    if (enchantments == null) {
        return 0;
    }

    // □□ RegistryAccess □□□□ Holder
    var enchantmentRegistry = registryAccess.lookupOrThrow(Registries.ENCHANTMENT);
    Optional<Holder<Enchantment>> enchantmentHolder = enchantmentRegistry.get(enchantmentKey);

    return enchantmentHolder.map(holder -> enchantments.getLevel(holder))
            .orElse(0);
}

// ============================================================================
```

```
    // □□□□□□□□□□ 1.20.1 vs 1.21.1 □□
    // ============================================================================
    //
    // ┌──────────────────┬──────────────────────────┬──────────────────────────────┐
    // │ □□               │ Forge 1.20.1             │ NeoForge 1.21.1              │
    // ├──────────────────┼──────────────────────────┼──────────────────────────────┤
    // │ □□□□□            │ □□□□□□□ (DeferredRegister) │ □□□□□□□ (RegistrySetBuilder)│
    // │ □□□              │ □□ Enchantment           │ □□ Enchantment □□ (□□□□□) │
    // │ □□□□□            │ □□□□□                    │ □□□□□ (DataComponentType)    │
    // │ □□□□□            │ ItemStack NBT            │ DataComponents.ENCHANTMENTS  │
    // │ □□□□□            │ Enchantment              │ Holder<Enchantment>          │
    // │ Enchantments.X   │ Enchantment □□           │ ResourceKey<Enchantment>     │
    // │ □□□□□            │ getEnchantmentLevel()    │ ItemEnchantments.getLevel()  │
    // │ □□□□□            │ □□□□                     │ EnchantmentHelper.runIteration │
    // └──────────────────┴──────────────────────────┴──────────────────────────────┘
    // ============================================================================
}
```

entity/simple_entity.java

```
// ============================================================================
// NeoForge 1.21.1 □□□□□□□ (simple_entity.java)
// ============================================================================
//
// □□ Forge 1.20.1 □□□□□□ - □□□□□□
// 1. □DeferredRegister.Entities □□□□□
//    NeoForge □□□ DeferredRegister.Entities □□□□,
//    □□□ registerEntityType() □□□□□□□
//    □□ DeferredRegister.createEntities(MODID) □□□
//    Forge 1.20.1: DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, MODID)
//
// 2. □EntityType.Builder.build() □□□□□□
//    Forge 1.20.1: EntityType.Builder.create(factory, category).build()
//                  build() □□□□□□ (□□ ResourceLocation)
//    NeoForge 1.21.1: EntityType.Builder.of(factory, category)
//                     .build(ResourceKey.create(Registries.ENTITY_TYPE, ResourceLocation))
//    □□□□build() □□□□□ create □□ of, □ build() □□□□□ ResourceKey!
//
// 3. □□□□□□/□□□□□□
//    Forge 1.20.1:
//      readAdditionalSaveData(CompoundTag tag)
//      addAdditionalSaveData(CompoundTag tag)
//    NeoForge 1.21.1 (1.21.x):
//      readAdditionalSaveData(CompoundTag tag)   - □□□□□ CompoundTag
//      addAdditionalSaveData(CompoundTag tag)    - □□□□□ CompoundTag
//    (□□: 1.21.2+ □□ ValueInput/ValueOutput, □ 1.21.1 □□ CompoundTag)
//
// 4. □defineSynchedData()□
//    1.21.1: defineSynchedData() - □□□, □□ this.entityData.define()
//    (□□: 1.21.5+ □□ defineSynchedData(SynchedEntityData.Builder), □ 1.21.1 □□□□)
//
// 5. □□□□□□ (NeoForge □□)□
//    Forge 1.20.1: □□ hurt(DamageSource, float)
//    NeoForge 1.21.1: hurt() □□□□□ final, □□□□□ hurtServer() □ hurtClient()
//    - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□
//    - hurtClient(DamageSource, float): □□□□□□□□□ (□□□□□□□□/□□)
//
// 6. □ResourceLocation □□□□□□□□□
//    Forge 1.20.1: new ResourceLocation("modid", "path")
//    NeoForge 1.21.1: ResourceLocation.fromNamespaceAndPath("modid", "path")
//    □ ResourceLocation.parse("modid:path") □ ResourceLocation.withDefaultNamespace("path")
//
// 7. □MobCategory□
//    MobCategory □□□□□□□ (NeoForge □□)□
//    □□□: MONSTER, CREATURE, AMBIENT, MISC □□
// ============================================================================

package com.example.examplemod.entity;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.nbt.CompoundTag;
import net.minecraft.network.syncher.EntityDataAccessor;
import net.minecraft.network.syncher.EntityDataSerializers;
import net.minecraft.network.syncher.SynchedEntityData;
import net.minecraft.resources.ResourceKey;
```

```java
import net.minecraft.resources.ResourceLocation;
import net.minecraft.server.level.ServerLevel;
import net.minecraft.world.damagesource.DamageSource;
import net.minecraft.world.entity.Entity;
import net.minecraft.world.entity.EntityType;
import net.minecraft.world.entity.MobCategory;
import net.minecraft.world.level.Level;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

// ================================================================================
// □□□□□□
// ================================================================================
public class ModEntities {

    // ============================================================================
    // □□□□□□□□ DeferredRegister.Entities □□□□□□
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ENTITY_TYPES, MODID)
    // NeoForge 1.21.1: DeferredRegister.createEntities(MODID)
    // ============================================================================
    public static final DeferredRegister.Entities ENTITY_TYPES =
            DeferredRegister.createEntities(ExampleMod.MOD_ID);

    // ============================================================================
    // □□□□□□□□
    //
    // □□□□□EntityType.Builder □□□□:
    //   Forge 1.20.1: EntityType.Builder.create(factory, category)
    //   NeoForge 1.21.1: EntityType.Builder.of(factory, category)
    //
    // □□□□□build() □□:
    //   Forge 1.20.1: .build() (□□□□□□□ ResourceLocation □□ datafixer type)
    //   NeoForge 1.21.1: .build(ResourceKey.create(
    //       Registries.ENTITY_TYPE, ResourceLocation.fromNamespaceAndPath(MODID, "name")))
    //
    // □□□□□ (□□):
    //   registerEntityType("name", factory, category)
    //   registerEntityType("name", factory, category, builderConfigurer)
    // ============================================================================
    public static final DeferredHolder<EntityType<?>, EntityType<ExampleEntity>> EXAMPLE_ENTITY =
            ENTITY_TYPES.register(
                    "example_entity",
                    () -> EntityType.Builder.<ExampleEntity>of(
                            ExampleEntity::new,      // EntityFactory (EntityType, Level) -> T
                            MobCategory.MISC          // □□□□□
                    )
                    // □□□□□□□□ (□ x □), □□□□□□□□□□□□□□□
                    // □□: 0.6f x 1.8f
                    .sized(1.0f, 1.0f)
                    // □□□□□□□□ (□□: height * 0.85), □□□□ sized() □□□□□
                    .eyeHeight(0.85f)
                    // □□□□□□□□□ (□□□□□□□□), □□ 5
                    .clientTrackingRange(8)
                    // □□□□□ (□ x tick □□□□□□□□), □□ 3
                    .updateInterval(10)
                    // □□□□□□□□□
                    .fireImmune()
                    // □□□□□□□□□□ (□□□□□□□□□□)
                    // .canSpawnFarFromPlayer()
                    // □□□□□ /summon □□
                    // .noSummon()
                    // □□□□□□□□□
                    // .noSave()
                    // ========================================================================
                    // □□□□□build() □□□□□ ResourceKey!
                    // Forge 1.20.1: .build("examplemod:example_entity") □ .build()
                    // NeoForge 1.21.1: .build(ResourceKey.create(...))
                    // ========================================================================
                    .build(ResourceKey.create(
                            Registries.ENTITY_TYPE,
                            ResourceLocation.fromNamespaceAndPath(
                                    ExampleMod.MOD_ID, "example_entity"
                            )
                    ))
            );
    // ============================================================================
```

```
    // □□□□□ (□□□□□):
    // public static final DeferredHolder<EntityType<?>, EntityType<ExampleEntity>> EXAMPLE_ENTITY =
    //     ENTITY_TYPES.registerEntityType("example_entity", ExampleEntity::new, MobCategory.MISC);
    //
    // □□□□□□□□□□□□□:
    // public static final DeferredHolder<EntityType<?>, EntityType<ExampleEntity>> EXAMPLE_ENTITY =
    //     ENTITY_TYPES.registerEntityType("example_entity", ExampleEntity::new, MobCategory.MISC,
    //         builder -> builder.sized(2.0f, 2.0f).eyeHeight(1.5f).updateInterval(5));
    // ============================================================================
}

// ============================================================================
// □□□□□□□□
// □□ Entity (□□□□□□ LivingEntity, Mob □)
//
// Entity □□□□□, □□□□□□□□□□:
// 1. readAdditionalSaveData(CompoundTag) - □ NBT □□□□□
// 2. addAdditionalSaveData(CompoundTag) - □□□□□□ NBT
// 3. defineSynchedData() - □□□□□□-□□□□□□□□
// 4. hurt(DamageSource, float) □ hurtServer(ServerLevel, DamageSource, float) - □□□□□
// ============================================================================
class ExampleEntity extends Entity {

    // ============================================================================
    // □□□□□□□□□ (□□□□□□-□□□□□□□□)
    // □□□□□□□□□□□□□□□□□□□□□□
    // ============================================================================
    private static final EntityDataAccessor<Integer> DATA_VALUE =
            SynchedEntityData.defineId(ExampleEntity.class, EntityDataSerializers.INT);

    // ============================================================================
    // □□□□□
    // □□□□ (EntityType<? extends T>, Level) □□□□□□□□,
    // □□□□□□□□ EntityFactory □□ (MyEntity::new)
    // ============================================================================
    public ExampleEntity(EntityType<? extends ExampleEntity> type, Level level) {
        super(type, level);
    }

    // □□: □□□□□□□□□ (□□□□□□□□□□□□□□)
    public ExampleEntity(EntityType<? extends ExampleEntity> type, Level level, double x, double y, double z) {
        this(type, level);
        this.setPos(x, y, z);
    }

    // ============================================================================
    // □□□□□□□□□□□□□□□□
    //
    // Forge 1.20.1:
    //   @Override
    //   public boolean hurt(DamageSource source, float amount) { ... }
    //
    // NeoForge 1.21.1 (Damage Pipeline Rework):
    //   hurt() □□□□□ final! □□□□□ hurtServer() □/□ hurtClient()
    //
    // - hurtServer(ServerLevel, DamageSource, float): □□□□□□□□
    //   □□ true □□□□□□□□□□
    // - hurtClient(DamageSource, float): □□□□□□ (□□□□□□□□/□□)
    // ============================================================================
    @Override
    public boolean hurtServer(ServerLevel level, DamageSource source, float amount) {
        if (this.isInvulnerableTo(source)) {
            return false;
        }
        // □□□□□□□□□□
        // □□: □□□□□□□□□, □□□□□□□□□□
        // if (source.is(net.minecraft.tags.DamageTypeTags.IS_FIRE)) {
        //     return false;
        // }
        // amount *= 2; // □□□□□□□
        // return super.hurtServer(level, source, amount);
        return super.hurtServer(level, source, amount);
    }

    // ============================================================================
    // □ NBT □□□□□□□
```

```
    // 1.21.1 □□□□□□ CompoundTag (□□ ValueInput, ValueInput □ 1.21.2+)
    // ========================================================================
    @Override
    public void readAdditionalSaveData(CompoundTag tag) {
        super.readAdditionalSaveData(tag);
        // □□□□□□□□
        int value = tag.getInt("value");
        this.entityData.set(DATA_VALUE, value);
    }

    // ========================================================================
    // □□□□□□□□ NBT
    // 1.21.1 □□□□□□ CompoundTag (□□ ValueOutput, ValueOutput □ 1.21.2+)
    // ========================================================================
    @Override
    public void addAdditionalSaveData(CompoundTag tag) {
        super.addAdditionalSaveData(tag);
        // □□□□□□□□
        tag.putInt("value", this.entityData.get(DATA_VALUE));
    }

    // ========================================================================
    // □□□□□□□□ (□□□-□□□□□□)
    //
    // 1.21.1: defineSynchedData() - □□□□, □□ this.entityData.define()
    // (□□: 1.21.5+ □□ defineSynchedData(SynchedEntityData.Builder builder),
    //  □ 1.21.1 □□□□□□□□□□)
    //
    // Forge 1.20.1 □□□□ defineSynchedData() □□□□, □□□□□
    // ========================================================================
    @Override
    protected void defineSynchedData() {
        // □□ entityData.define() □□□□□□□□□□□
        // □□1: EntityDataAccessor (□□ SynchedEntityData.defineId □□)
        // □□2: □□□□
        this.entityData.define(DATA_VALUE, 0);
    }

    // ========================================================================
    // □ tick □□□□□□
    // ========================================================================
    @Override
    public void tick() {
        super.tick();
        // □□□□ tick □□□□
        // □□: □□□□□□□□□□□
        if (!this.level().isClientSide) {
            // □□□□□□
            // this.entityData.set(DATA_VALUE, this.entityData.get(DATA_VALUE) + 1);
        }
    }

    // ========================================================================
    // □□□□□□□□ (□□□□□□□)
    // ========================================================================
    public int getValue() {
        return this.entityData.get(DATA_VALUE);
    }

    public void setValue(int value) {
        this.entityData.set(DATA_VALUE, value);
    }

    // ========================================================================
    // □□□□□□□□□□ (□□□□)
    // □□□□□□□□□□□□□□□□ RegisterRenderersEvent:
    //
    // @SubscribeEvent  // □□□□□□□□□□□ (□□□□□)
    // public static void registerRenderers(RegisterRenderersEvent event) {
    //     event.registerEntityRenderer(ModEntities.EXAMPLE_ENTITY.get(),
    //         ExampleEntityRenderer::new);
    // }
    //
    // □□□□□ EntityRendererProvider.Context □□□□□□□□□□
    // ========================================================================
}
```

gui/container_screen.java

```
// ========================================================================
// NeoForge 1.21.1 GUI □□□□□□□ (container_screen.java)
// ========================================================================
//
// □□ Forge 1.20.1 □□□□□□ - GUI/□□□□□□
// 1. □MenuType □□□
//    Forge 1.20.1: DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID)
//                  □□□□□: ForgeRegistries.CONTAINERS
//    NeoForge 1.21.1: DeferredRegister.create(Registries.MENU, MODID)
//                     □□□□□: Registries.MENU (□□ CONTAINER □ MENU_TYPE)
//    □□□□□□□□□□□□□□: CONTAINERS -> MENU
//
// 2. □AbstractContainerMenu □□□
//    □□□□□□□: (MenuType, int containerId)
//    □□□□□□□□:
//    - stillValid(Player): □□□□□□□□□□□□
//    - quickMoveStack(Player, int): Shift+□□□□□□
//    1.21.x □□□□□□□□□□ Forge 1.20.1 □□□
//
// 3. □Screen □□□□□
//    Forge 1.20.1: MenuScreens.register() □□□□□□□□□□□□
//    NeoForge 1.21.1: □□ RegisterMenuScreensEvent (□□□□□□□□□)
//    @SubscribeEvent
//    public static void registerScreens(RegisterMenuScreensEvent event) {
//        event.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
//    }
//
// 4. □SlotItemHandler□
//    NeoForge □□□□ IItemHandler □□□□ (□□ vanilla Container)□
//    □□ SlotItemHandler □□ Slot □□□□□□□□
//    (SlotItemHandler □□ IItemHandler □□□ Container)
//
// 5. □IContainerFactory / IMenuTypeExtension□
//    □□□□□□□□□□□□□□□□ (□□□□□), □□ IContainerFactory□
//    □□ IMenuTypeExtension.create() □□ MenuType□
//    (Forge 1.20.1 □□ IContainerFactory, NeoForge □□ IMenuTypeExtension)
//
// 6. □RegistryFriendlyByteBuf□
//    1.21.x □□□□□□□ RegistryFriendlyByteBuf (□□ FriendlyByteBuf)□
//    □□□□□□□□□□□□□□□□□□□□□□□□
// ========================================================================

package com.example.examplemod.gui;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.BlockPos;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.FriendlyByteBuf;
import net.minecraft.network.chat.Component;
import net.minecraft.network.codec.ByteBufCodecs;
import net.minecraft.network.RegistryFriendlyByteBuf;
import net.minecraft.world.SimpleContainerData;
import net.minecraft.world.entity.player.Inventory;
import net.minecraft.world.entity.player.Player;
import net.minecraft.world.inventory.AbstractContainerMenu;
import net.minecraft.world.inventory.ContainerLevelAccess;
import net.minecraft.world.inventory.MenuType;
import net.minecraft.world.inventory.Slot;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.neoforge.items.ItemStackHandler;
import net.neoforged.neoforge.items.SlotItemHandler;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredHolder;

// ========================================================================
// □□□□□□□
// ========================================================================
class ModMenuTypes {

    // ====================================================================
    // □□□□□□□□□□□
    // Forge 1.20.1: DeferredRegister.create(ForgeRegistries.CONTAINERS, MODID)
    //               □□□□: ForgeRegistries.CONTAINERS
```

```
    // NeoForge 1.21.1: DeferredRegister.create(Registries.MENU, MODID)
    //                  □□□□: Registries.MENU
    // □□□□□□□□□□□□ CONTAINERS □□ MENU
    // ========================================================================
    public static final DeferredRegister<MenuType<?>> MENU_TYPES =
            DeferredRegister.create(Registries.MENU, ExampleMod.MOD_ID);

    // ========================================================================
    // □□□□□□□□ (□□□□□□□)
    // □□ MenuType □□□□□, □□ MenuSupplier
    // ========================================================================
    public static final DeferredHolder<MenuType<?>, MenuType<ExampleMenu>> EXAMPLE_MENU =
            MENU_TYPES.register("example_menu", () -> new MenuType<>(
                    ExampleMenu::new,   // MenuSupplier: (containerId, playerInv) -> Menu
                    net.minecraft.world.flag.FeatureFlags.DEFAULT_FLAGS
            ));

    // ========================================================================
    // □□□□□□□□□□□□ (□□ IMenuTypeExtension)
    // □□□□□□□□□□□□□□□□□□□□ (□□□□□□□), □□□□□□
    //
    // □□□□□Forge 1.20.1: IContainerFactory + new MenuType<>(IContainerFactory::create)
    //         NeoForge 1.21.1: IMenuTypeExtension.create(IContainerFactory)
    //
    // ========================================================================
    public static final DeferredHolder<MenuType<?>, MenuType<ExampleMenuExtra>> EXAMPLE_MENU_EXTRA =
            MENU_TYPES.register("example_menu_extra", () ->
                    // IMenuTypeExtension.create() □□□□□□□□□□□□□□□□□□□□□□□□
                    net.neoforged.neoforge.client.extensions.IMenuTypeExtension.create(
                            ExampleMenuExtra::new  // (containerId, playerInv, extraData) -> Menu
                    )
            );
}

// ============================================================================
// □□□□□ (□□□□□□□)
// □□ AbstractContainerMenu
// ============================================================================
class ExampleMenu extends AbstractContainerMenu {

    // ========================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□□)
    // □□: containerId (□□□□□□□), playerInventory (□□□□□□□)
    // □□□□□□□□□ MenuSupplier □□, □□□□□□□□□
    // ========================================================================
    public ExampleMenu(int containerId, Inventory playerInventory) {
        this(containerId, playerInventory, new ItemStackHandler(9));
    }

    // ========================================================================
    // □□□□□□□□□ (□□□□□□□□□□□□□□)
    // □□□□□□□□□□
    // ========================================================================
    public ExampleMenu(int containerId, Inventory playerInventory, ItemStackHandler handler) {
        super(ModMenuTypes.EXAMPLE_MENU.get(), containerId);

        // □□□□□□□□□□□ (□□ SlotItemHandler, □□ IItemHandler □□)
        // □□□□□NeoForge □□□□□ SlotItemHandler (□□ IItemHandler)
        //         □□□□ vanilla Slot (□□ Container)
        // □□: (IItemHandler inventory, int index, int x, int y)
        for (int i = 0; i < 9; i++) {
            // □□□□□: 3x3 □□, □ (62, 18) □□
            int x = 62 + (i % 3) * 18;
            int y = 18 + (i / 3) * 18;
            this.addSlot(new SlotItemHandler(handler, i, x, y));
        }

        // □□□□□□□□□□□ (27 □)
        for (int row = 0; row < 3; row++) {
            for (int col = 0; col < 9; col++) {
                this.addSlot(new Slot(playerInventory,
                        col + row * 9 + 9,   // □□□□□
                        8 + col * 18,          // X □□
                        84 + row * 18          // Y □□
                ));
            }
```

```
        }

        // □□□□□□□□□□□ (9 □)
        for (int col = 0; col < 9; col++) {
            this.addSlot(new Slot(playerInventory, col, 8 + col * 18, 142));
        }
    }

    // ========================================================================
    // stillValid - □□□□□□□□□□□□
    // □□□□□□□□ true; □□□□□□□□□□□□□□□□
    // ========================================================================
    @Override
    public boolean stillValid(Player player) {
        // □□□□□□□, □□□□□□□ true
        return true;

        // □□□□□□□□□□□□, □□ ContainerLevelAccess:
        // return AbstractContainerMenu.stillValid(
        //     ContainerLevelAccess.create(player.level(), blockPos),
        //     player,
        //     Blocks.CRAFTING_TABLE  // □□□□□
        // );
    }

    // ========================================================================
    // quickMoveStack - Shift+□□□□□□□□
    // □□: (Player, int slotIndex)
    // □□: □□□□□□ ItemStack
    // ========================================================================
    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        Slot slot = this.slots.get(index);
        if (!slot.hasItem()) {
            return ItemStack.EMPTY;
        }

        ItemStack originalStack = slot.getItem();
        ItemStack copyStack = originalStack.copy();

        // □□□□□□□:
        // 0-8: □□□□□□□□
        // 9-35: □□□□□□
        // 36-44: □□□□□□

        if (index < 9) {
            // □□□□□□□□□□□□□□□□/□□□
            if (!this.moveItemStackTo(originalStack, 9, 45, false)) {
                return ItemStack.EMPTY;
            }
        } else {
            // □□□□□□□/□□□□□□□□□□□
            if (!this.moveItemStackTo(originalStack, 0, 9, false)) {
                return ItemStack.EMPTY;
            }
        }

        if (originalStack.isEmpty()) {
            slot.set(ItemStack.EMPTY);
        } else {
            slot.setChanged();
        }

        return copyStack;
    }
}

// ============================================================================
// □□□□□□□□□ (□□ IContainerFactory / IMenuTypeExtension)
// □□□□□□□□□□□□□□□□□□□□ (□□□□□)
// ============================================================================
class ExampleMenuExtra extends AbstractContainerMenu {

    private final ContainerLevelAccess access;

    // ========================================================================
```

```
    // □□□□□□□□ (□□ RegistryFriendlyByteBuf □□□□)
    // □□ IMenuTypeExtension.create □□□□□
    // ============================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, RegistryFriendlyByteBuf extraData) {
        this(containerId, playerInventory, extraData.readBlockPos());
    }

    // ============================================================================
    // □□□□□□□□
    // ============================================================================
    public ExampleMenuExtra(int containerId, Inventory playerInventory, BlockPos pos) {
        super(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), containerId);
        this.access = ContainerLevelAccess.create(playerInventory.player.level(), pos);

        // □□□□□ (□□, □□□□□)
        // ...
    }

    @Override
    public boolean stillValid(Player player) {
        // □□□□□□□□□□□□□ (8 □□□□□)
        return AbstractContainerMenu.stillValid(this.access, player, Blocks.CRAFTING_TABLE);
    }

    @Override
    public ItemStack quickMoveStack(Player player, int index) {
        // □□□□□□□ (□□)
        return ItemStack.EMPTY;
    }

    // ============================================================================
    // □□□□□□□□□□□□□□:
    //
    // // □ BlockBehaviour#useWithoutItem □□□□□□□:
    // @Override
    // public InteractionResult useWithoutItem(BlockState state, Level level, BlockPos pos,
    //         Player player, BlockHitResult hitResult) {
    //     if (level.isClientSide) {
    //         return InteractionResult.SUCCESS;  // □□□□□□ SUCCESS
    //     }
    //     // □□□: □□□□
    //     ((ServerPlayer) player).openMenu(new SimpleMenuProvider(
    //         (containerId, playerInventory, p) -> new ExampleMenuExtra(containerId, playerInventory, pos),
    //         Component.translatable("menu.examplemod.example_menu")
    //     ), buf -> buf.writeBlockPos(pos));  // □□□□□□□□□□□□
    //     return InteractionResult.CONSUME;
    // }
    // ============================================================================
}

// ============================================================================
// □□□□ (Screen) - □□□□□
// □□ AbstractContainerScreen
// ============================================================================
class ExampleScreen extends net.minecraft.client.gui.screens.inventory.AbstractContainerScreen<ExampleMenu> {

    // ============================================================================
    // Screen □□□□
    // □□: (Menu menu, Inventory playerInventory, Component title)
    // ============================================================================
    public ExampleScreen(ExampleMenu menu, Inventory playerInventory, Component title) {
        super(menu, playerInventory, title);

        // □□□□□□□□□ (□□□ 176x166)
        // this.imageWidth = 176;
        // this.imageHeight = 166;
    }

    // ============================================================================
    // □□□□□ (1.21.x □□□ RenderSystem □ GuiGraphics)
    // 1.21.x □□□□□□ GuiGraphics □□□□□□□□ PoseStack
    // ============================================================================
    @Override
    protected void renderBg(net.minecraft.client.gui.GuiGraphics guiGraphics,
                            float partialTick, int mouseX, int mouseY) {
        // □□□□□□□□
```

```
        // 1.21.x □□□ GuiGraphics.blit() □□
        // guiGraphics.blit(
        //     TEXTURE,                        // ResourceLocation □□
        //     this.leftPos, this.topPos,       // □□□□
        //     0, 0,                           // □□ UV □□
        //     this.imageWidth, this.imageHeight // □□□□
        // );
    }

    // ========================================================================
    // □□□□□ (□□□)
    // ========================================================================
    @Override
    public void render(net.minecraft.client.gui.GuiGraphics guiGraphics,
                       int mouseX, int mouseY, float partialTick) {
        // 1.21.x □□□ GuiGraphics □□
        this.renderBackground(guiGraphics, mouseX, mouseY, partialTick);
        super.render(guiGraphics, mouseX, mouseY, partialTick);
        // □□□□□□□
        this.renderTooltip(guiGraphics, mouseX, mouseY);
    }

    // ========================================================================
    // □□□□□□ (□□□□□□)
    // ========================================================================
    @Override
    protected void init() {
        super.init();
        // □□□□□□□ (□□□□□□□□□□)
        this.titleLabelY = 6;
        this.inventoryLabelY = this.imageHeight - 94;
    }

    // ========================================================================
    // □□ Screen (□□□□□□□□□□):
    //
    // @SubscribeEvent  // □□□□□□□□□ (□□□□)
    // public static void registerScreens(RegisterMenuScreensEvent event) {
    //     event.register(ModMenuTypes.EXAMPLE_MENU.get(), ExampleScreen::new);
    //     event.register(ModMenuTypes.EXAMPLE_MENU_EXTRA.get(), ExampleScreenExtra::new);
    // }
    //
    // □□□□Forge 1.20.1 □□ MenuScreens.register() □ FMLClientSetupEvent □
    //         NeoForge 1.21.1 □□ RegisterMenuScreensEvent □□□□□□□□
    // ========================================================================
}
```

item/simple_item.java

```
// ========================================================================
// NeoForge 1.21.1 □□□□□□ (simple_item.java)
// ========================================================================
//
// □□ Forge 1.20.1 □□□□□□
// 1. □DeferredRegister.Items □□□□
//    NeoForge □□□ DeferredRegister.Items □□□ (□□ DeferredRegister<Item>),
//    □□□ registerItem()□registerSimpleItem()□registerSimpleBlockItem() □□□□□□
//    □□ DeferredRegister.createItems(MODID) □□□
// 2. □□□□□□ - Data Components □□ (□□□□)□
//    1.20.5+ □□□ Data Components □□, □□□□□□□□□□□□ setter □□,
//    □□□□ DataComponents □□□□□□□□□□
//    Forge 1.20.1: item.getMaxDamage() □ setter □□
//    NeoForge 1.21.1: □□ Item.Properties □□□□□ (□□□□□ DataComponents)
//    - stacksTo -> DataComponents.MAX_STACK_SIZE
//    - durability -> DataComponents.MAX_DAMAGE
//    - rarity -> DataComponents.RARITY
//    - food -> DataComponents.FOOD
//    □□□
// 3. □NBT □□□
//    1.20.5+ □□□ ItemStack NBT □□, □□□□□□ DataComponents □□□
//    Forge 1.20.1 □ ItemStack □ getOrCreateTag() □□□□
//    NeoForge 1.21.1 □□□ stack.get(DataComponents.XXX) □ stack.set() □□□
// 4. □CreativeModeTab □□□□□
//    1.21.x □□□ CreativeModeTab.builder() □□□□□□□□□
//    □□ DeferredRegister<CreativeModeTab> □□□
//    □□□□□ BuildCreativeModeTabContentsEvent □□□□□□□□□□□
```

```
// 5. 在Item.Properties 中配置属性
//    - .rarity() 设置稀有度
//    - .food() 设置食物属性 (FoodProperties)
//    - .durability() 设置耐久
//    - .stacksTo() 设置堆叠
//    - .fireResistant() 设置防火
// ============================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.network.chat.Component;
import net.minecraft.world.food.FoodProperties;
import net.minecraft.world.item.CreativeModeTab;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.ItemStack;
import net.minecraft.world.item.Rarity;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredItem;

import java.util.function.Supplier;

// ============================================================================
// 物品注册类
// ============================================================================
public class ModItems {

    // ========================================================================
    // 创建物品注册器 DeferredRegister.Items 用于注册物品
    // Forge 1.20.1:  DeferredRegister.create(ForgeRegistries.ITEMS, MODID)
    // NeoForge 1.21.1: DeferredRegister.createItems(MODID)
    //
    // createItems 返回 DeferredRegister.Items, 用于:
    // - register() / registerItem() 返回 DeferredItem<T> (实际是 DeferredHolder<Item, T>)
    // - 支持 registerItem()、registerSimpleItem()、registerSimpleBlockItem() 等便捷方法
    // ========================================================================
    public static final DeferredRegister.Items ITEMS =
            DeferredRegister.createItems(ExampleMod.MOD_ID);

    // ========================================================================
    // 示例物品：基础物品
    // 注册后获得的对象: DeferredItem<Item> (Forge 中是 RegistryObject<Item>)
    //
    // 参数1: 物品 registerItem 的名称 (小写)
    // ========================================================================
    public static final DeferredItem<Item> EXAMPLE_ITEM =
            ITEMS.registerItem("example_item", Item::new, new Item.Properties());

    // ========================================================================
    // 示例物品：带稀有度和堆叠限制
    // ========================================================================
    public static final DeferredItem<Item> RARE_ITEM =
            ITEMS.registerItem("rare_item", Item::new, new Item.Properties()
                    .rarity(Rarity.RARE)           // 稀有度: COMMON, UNCOMMON, RARE, EPIC
                    .stacksTo(16)                   // 最大堆叠数 (默认 64)
                    .fireResistant()                // 防火属性 (等同 DataComponents.FIRE_RESISTANT)
            );

    // ========================================================================
    // 示例食物：苹果
    // 注意：1.21.x 中 FoodProperties 通过 Item.Properties.food() 设置
    //         食物组件为 DataComponents.FOOD 组件类型
    // ========================================================================
    public static final DeferredItem<Item> EXAMPLE_FOOD =
            ITEMS.registerItem("example_food", Item::new, new Item.Properties()
                    .food(new FoodProperties.Builder()
                            .nutrition(6)                   // 饥饿值 (半个鸡腿)
                            .saturationMod(0.6f)             // 饱和度修正
                            .meat()                          // 肉类食物
                            .build()
                    )
            );

    // ========================================================================
    // 示例物品：带耐久的工具
```

```
    // 设置 durability 使其可以被损坏（默认为 1）
    // ============================================================================
    public static final DeferredItem<Item> EXAMPLE_DURABLE =
            ITEMS.registerItem("example_durable", Item::new, new Item.Properties()
                    .durability(250)               // 耐久值
                    .rarity(Rarity.UNCOMMON)
            );

    // ============================================================================
    // 自定义物品类 (需要创建对应的类)
    // 如果你想要使用自己的 Item 类
    // 首先创建一个类继承 (例如下面), 然后将 Item 改为
    // ============================================================================
    public static final DeferredItem<CustomItem> CUSTOM_ITEM =
            ITEMS.registerItem("custom_item", CustomItem::new, new Item.Properties()
                    .rarity(Rarity.EPIC)
            );
}

// ============================================================================
// 自定义物品类
// 用于 Item 的自定义行为和属性
// ============================================================================
class CustomItem extends Item {

    public CustomItem(Properties properties) {
        super(properties);
    }

    // 1.21.x 中可以重写使用 ItemStack 的方法来自定义 Item
    // 例如: use, useOn, getUseDuration, getUseAnimation 等
}

// ============================================================================
// 创造模式物品栏注册类
// 注意：1.21.x 使用新的 CreativeModeTab.builder() 模式
//         需要 DeferredRegister<CreativeModeTab> 来
//         (Forge 1.20.1 及之前版本不同, 但 1.21.x 是 API 变化)
// ============================================================================
class ModCreativeModeTabs {

    public static final DeferredRegister<CreativeModeTab> CREATIVE_MODE_TABS =
            DeferredRegister.create(Registries.CREATIVE_MODE_TAB, ExampleMod.MOD_ID);

    // ============================================================================
    // 示例创造模式物品栏标签
    // 使用 builder() 模式创建, 可以自定义图标和显示顺序
    // ============================================================================
    public static final Supplier<CreativeModeTab> EXAMPLE_TAB =
            CREATIVE_MODE_TABS.register("example_tab", () -> CreativeModeTab.builder()
                    // 设置物品栏标题 (使用翻译键)
                    .title(Component.translatable("itemGroup.examplemod.example_tab"))
                    // 设置物品栏图标 (Supplier<ItemStack>)
                    .icon(() -> new ItemStack(ModItems.EXAMPLE_ITEM.get()))
                    // 添加要显示的物品
                    .displayItems((params, output) -> {
                        // 添加物品 (ItemLike 或 Supplier<Item>)
                        output.accept(ModItems.EXAMPLE_ITEM.get());
                        output.accept(ModItems.RARE_ITEM.get());
                        output.accept(ModItems.EXAMPLE_FOOD.get());
                        output.accept(ModItems.EXAMPLE_DURABLE.get());
                        output.accept(ModItems.CUSTOM_ITEM.get());
                    })
                    .build()
            );

    // ============================================================================
    // 事件监听: 需要 BuildCreativeModeTabContentsEvent 来添加物品到现有标签
    // 在主模组类或事件总线中:
    //
    // @SubscribeEvent  // 不是静态方法
    // public static void buildContents(BuildCreativeModeTabContentsEvent event) {
    //     if (event.getTabKey() == CreativeModeTabs.INGREDIENTS) {
    //         event.accept(ModItems.EXAMPLE_ITEM.get());
    //     }
    // }
```

```
    // ================================================================================
}
```

item/tool_item.java

```
// ================================================================================
// NeoForge 1.21.1 □□□□□□□□□□ (tool_item.java)
// ================================================================================
//
// □□ Forge 1.20.1 □□□□□□ - □□□□□□
// 1. □Tier □□□
//    Forge 1.20.1: Tier □□ + Tier □□□□□ (□ IItemTier)
//    NeoForge 1.21.1: □□□□□ Tier □□, □□□ SimpleTier □□□
//    SimpleTier □□□□□□□□□□□□□ Tier □□□□□□□□
//
// 2. □Tier □□□□□□
//    Forge 1.20.1 □ Tier □□□□□: getUses()□getSpeed()□getAttackDamageBonus()□
//    getLevel()□getEnchantmentValue()□getRepairIngredient()□
//    NeoForge 1.21.1 □ SimpleTier □□□□□:
//    - incorrectBlocksDrops: TagKey<Block> (□□ getLevel())
//      □□□□□□□□□□□ int level □□□□□□□, □□□□□□□□!
//    - durability: int (□□□, □□ getUses())
//    - speed: float (□□□□□, □□ getSpeed())
//    - attackDamageBonus: float (□□□□□□□, □□ getAttackDamageBonus())
//    - enchantmentValue: int (□□□, □□ getEnchantmentValue())
//    - repairIngredient: Supplier<Ingredient> (□□□□□, □□ getRepairIngredient())
//
// 3. □□□□□□□□□□□□
//    Forge 1.20.1: new SwordItem(tier, attackDamage, attackSpeed, properties)
//                  (attackDamage □□□□□□□□□□□)
//    NeoForge 1.21.1: new SwordItem(tier, properties)
//                     (attackDamage □□ Item.Properties.attributes() □□)
//    □□□□1.21.x □□□□□□□□□□□□□□□□□□□□□□□□□□□□,
//             □□□□ Item.Properties.attributes() □□□□!
//
// 4. □□□□□□□□
//    Item -> TieredItem -> DiggerItem -> {AxeItem, HoeItem, PickaxeItem, ShovelItem}
//                                                  SwordItem (□□□ TieredItem)
//    TieredItem: □□ Tier □□□□
//    DiggerItem: □□□□□□□□□, □ createAttributes() □□□□
//
// 5. □createAttributes □□□
//    □□□□□ (SwordItem, PickaxeItem □) □□ createAttributes() □□□□,
//    □□□□□□□□□□□□□□□ (AttributeModifiers)□
//    □□: (tier, attackDamageBonus, attackSpeedModifier)
//
// 6. □ItemAbility □□ (□)□
//    Forge 1.20.1 □ ToolAction □□□ NeoForge □□□□□□ ItemAbility□
//    (ItemAbilities □□ ToolActions, ItemAbility □□ ToolAction)
//    □□ IItemExtension#canPerformAction() □□□
//
// 7. □1.21.2+ □□□
//    □ 1.21.2+ □, Tier □ ToolMaterial □□, □□□□□□
//    □ 1.21.1 □□□□ Tier + SimpleTier□
// ================================================================================

package com.example.examplemod.item;

import com.example.examplemod.ExampleMod;
import net.minecraft.core.registries.Registries;
import net.minecraft.resources.ResourceLocation;
import net.minecraft.tags.BlockTags;
import net.minecraft.tags.TagKey;
import net.minecraft.world.item.AxeItem;
import net.minecraft.world.item.HoeItem;
import net.minecraft.world.item.Item;
import net.minecraft.world.item.PickaxeItem;
import net.minecraft.world.item.ShovelItem;
import net.minecraft.world.item.SwordItem;
import net.minecraft.world.item.Tier;
import net.minecraft.world.item.Tiers;
import net.minecraft.world.item.crafting.Ingredient;
import net.minecraft.world.level.block.Block;
import net.minecraft.world.level.block.Blocks;
import net.neoforged.neoforge.registries.DeferredRegister;
import net.neoforged.neoforge.registries.DeferredItem;
```

```
import java.util.function.Supplier;

// ================================================================================
// □□□□□□□□
// ================================================================================
public class ModTools {

    public static final DeferredRegister.Items ITEMS =
            DeferredRegister.createItems(ExampleMod.MOD_ID);

    // ================================================================================
    // □□□□□□□□ Tier (□□□□)
    //
    // Forge 1.20.1 □□□□□□ IItemTier □□□□□□□□□
    // NeoForge 1.21.1 □□□ SimpleTier, □□□□□□□□□□□□
    //
    // □□□□□□□□□□ int level □□□□□□□!
    // □□□ TagKey<Block> (incorrectBlocksDrops) □□□□□□□□□□□□□□□□
    // □□: minecraft:incorrect_for_stone_tool □□□□□□□□□□□□□□□□□□□□□□□□□□□□
    //
    // □□ Forge 1.20.1:
    //   public static final Tier COPPER_TIER = new Tier() {
    //       @Override public int getUses() { return 200; }
    //       @Override public float getSpeed() { return 5f; }
    //       @Override public float getAttackDamageBonus() { return 1.5f; }
    //       @Override public int getLevel() { return 2; }  // 1.21.x □□□□□!
    //       @Override public int getEnchantmentValue() { return 20; }
    //       @Override public Ingredient getRepairIngredient() { return ...; }
    //   };
    // ================================================================================
    public static final Tier COPPER_TIER = new net.minecraft.world.item.Tier() {

        // ============================================================================
        // □□□□□□□□□□□ getLevel() (int □□□□□□□□),
        // □□□ getIncorrectDrops() □□□□□ TagKey<Block>,
        // □□□□□□□□□□□□□□□□□□□□□□□□□□□□
        //
        // □□□□□□□□□□□: BlockTags.INCORRECT_FOR_STONE_TOOL □
        // □□□□□□□□□□□□□
        // ============================================================================
        @Override
        public TagKey<Block> getIncorrectDrops() {
            // □□□□□□□□□ (□□□ data/<modid>/tags/block/ □□□)
            // □□□□□□□□□□□: BlockTags.INCORRECT_FOR_STONE_TOOL
            return TagKey.create(
                    Registries.BLOCK,
                    ResourceLocation.fromNamespaceAndPath(
                            ExampleMod.MOD_ID, "incorrect_for_copper_tool"
                    )
            );
        }

        @Override
        public int getUses() { return 200; }            // □□□ (stone=131, iron=250)

        @Override
        public float getSpeed() { return 5f; }           // □□□□ (stone=4, iron=6)

        @Override
        public float getAttackDamageBonus() { return 1.5f; } // □□□□□□□ (stone=1, iron=2)

        @Override
        public int getEnchantmentValue() { return 20; }  // □□□ (gold=22)

        @Override
        public Ingredient getRepairIngredient() {
            // □□□□□ (□□ lazy supplier □□□□□□□□)
            // □□□□□ NeoForge □□: Tags.Items.INGOTS_COPPER
            return Ingredient.of(Blocks.COPPER_BLOCK);
        }
    };

    // ================================================================================
    // □□: □□□□□□□□ Tier □□□□□□□
    // NeoForge □□□□□ SimpleTier □□□ (□□□□□), □□□□□□□□ Tier □□□
    //
```

```
// □□□□□ SimpleTier:
// public static final Tier COPPER_TIER = new SimpleTier(
//     TagKey.create(Registries.BLOCK,
//         ResourceLocation.fromNamespaceAndPath(MOD_ID, "incorrect_for_copper_tool")),
//     200,        // durability
//     5f,         // speed
//     1.5f,       // attackDamageBonus
//     20,         // enchantmentValue
//     () -> Ingredient.of(Tags.Items.INGOTS_COPPER)  // repairIngredient
// );
// ============================================================================

// ============================================================================
// □□□□□□□□□□□□ - □□□□□□□□□□□□□□□□□□□□!
//
// Forge 1.20.1 □□□:
//   public static final RegistryObject<SwordItem> COPPER_SWORD = ITEMS.register(
//       "copper_sword", () -> new SwordItem(COPPER_TIER, 3, -2.4f, new Item.Properties()));
//   (attackDamage=3, attackSpeed=-2.4f □□□□□□□□□□□□)
//
// NeoForge 1.21.1 □□□:
//   public static final DeferredItem<SwordItem> COPPER_SWORD = ITEMS.register(
//       "copper_sword", () -> new SwordItem(COPPER_TIER,
//           new Item.Properties().attributes(
//               SwordItem.createAttributes(COPPER_TIER, 3, -2.4f)
//           )
//       ));
//   (□□□□□□□ Item.Properties.attributes() □□, □□□□□□□□!)
// ============================================================================

// □□ - □□□□□ 3, □□□□□ -2.4f
public static final DeferredItem<SwordItem> COPPER_SWORD =
        ITEMS.register("copper_sword", () -> new SwordItem(
                COPPER_TIER,
                new Item.Properties().attributes(
                        // □□ SwordItem.createAttributes() □□□□□□□□
                        // □□1: Tier
                        // □□2: □□□□□□□□□□□□ (sword=3, shovel=1.5, pickaxe=1)
                        // □□3: □□□□□□□□□□□□ (sword=-2.4f, shovel=-3f, pickaxe=-2.8f)
                        SwordItem.createAttributes(COPPER_TIER, 3, -2.4f)
                )
        ));

// □□ - □□□□□ 1, □□□□□ -2.8f
public static final DeferredItem<PickaxeItem> COPPER_PICKAXE =
        ITEMS.register("copper_pickaxe", () -> new PickaxeItem(
                COPPER_TIER,
                new Item.Properties().attributes(
                        PickaxeItem.createAttributes(COPPER_TIER, 1, -2.8f)
                )
        ));

// □□
public static final DeferredItem<AxeItem> COPPER_AXE =
        ITEMS.register("copper_axe", () -> new AxeItem(
                COPPER_TIER,
                new Item.Properties().attributes(
                        AxeItem.createAttributes(COPPER_TIER, 6, -3.2f)
                )
        ));

// □□
public static final DeferredItem<ShovelItem> COPPER_SHOVEL =
        ITEMS.register("copper_shovel", () -> new ShovelItem(
                COPPER_TIER,
                new Item.Properties().attributes(
                        ShovelItem.createAttributes(COPPER_TIER, 1.5f, -3.0f)
                )
        ));

// □□
public static final DeferredItem<HoeItem> COPPER_HOE =
        ITEMS.register("copper_hoe", () -> new HoeItem(
                COPPER_TIER,
                new Item.Properties().attributes(
                        HoeItem.createAttributes(COPPER_TIER, -2, -1.0f)
```

```
                )
        ));

    // ========================================================================
    // □□□□□□□□ (□□□□□□□)
    //
    // □□□□□□□□□□:
    // data/examplemod/tags/block/needs_copper_tool.json
    //   - □□□□□□□□□□□□□□□
    //
    // data/examplemod/tags/block/incorrect_for_copper_tool.json
    //   - □□□□□□□□□□□□□□□ (□□□ incorrect_for_stone_tool □□□□□□ needs_copper_tool)
    //
    // □□ incorrect_for_copper_tool.json:
    // {
    //     "values": ["#minecraft:incorrect_for_stone_tool"],
    //     "remove": ["#examplemod:needs_copper_tool"]
    // }
    //
    // □□ needs_copper_tool.json:
    // {
    //     "values": [
    //         "minecraft:gold_block",
    //         "minecraft:gold_ore",
    //         "minecraft:deepslate_gold_ore"
    //     ]
    // }
    // ========================================================================
    public static final TagKey<Block> NEEDS_COPPER_TOOL = TagKey.create(
            Registries.BLOCK,
            ResourceLocation.fromNamespaceAndPath(ExampleMod.MOD_ID, "needs_copper_tool")
    );

    public static final TagKey<Block> INCORRECT_FOR_COPPER_TOOL = TagKey.create(
            Registries.BLOCK,
            ResourceLocation.fromNamespaceAndPath(ExampleMod.MOD_ID, "incorrect_for_copper_tool")
    );

    // ========================================================================
    // □□□□ItemAbility □□ (NeoForge □□□)
    //
    // Forge 1.20.1: ToolAction / ToolActions
    // NeoForge 1.21.1: ItemAbility / ItemAbilities (□□□)
    //
    // □□: □□□□□□□□□□□□□□□□ (□□□□□□□□□□□)
    //
    // □□□□:
    //   stack.canPerformAction(ItemAbilities.SWORD_SWEEP)  // □□□□□
    //   stack.canPerformAction(ItemAbilities.AXE_DIG)       // □□□□□
    //
    // □□□□□ ItemAbility:
    //   public static final ItemAbility MY_ACTION = ItemAbility.get("examplemod:my_action");
    //
    // □□□□□□□□□:
    //   @Override
    //   public boolean canPerformAction(ItemStack stack, ItemAbility action) {
    //       return action == MY_ACTION || super.canPerformAction(stack, action);
    //   }
    // ========================================================================
}
```

mixin/mixin_config.json

```
{
    // ========================================================================
    // NeoForge 1.21.1 Mixin □□□□ (mixin_config.json)
    // ========================================================================
    //
    // □□ Forge 1.20.1 □□□□□□ - Mixin □□□
    // 1. □Mixin □□□□□□
    //    Forge 1.20.1 □□ Forge □□□□ Mixin □□ (□□ SpongePowered Mixin)□
    //    NeoForge 1.20.4+ □□□□□ Fabric-like Mixin □□ (□□ Fabric □ Mixin □□□)□
    //    □□□□□□□□□ (JSON) □□□□□□
    //
    // 2. □□□□□□□
    //    Forge 1.20.1: □ mods.toml □□□□ mixins □□□
```

```
    //                  (□□□□□□ mods.toml)
    //    NeoForge 1.21.1: □ neoforge.mods.toml □□□ [[mixins]] □□□
    //                     [[mixins]]
    //                     config="${mod_id}.mixins.json"
    //
    // 3. □□□□□□□□□
    //    □□□□□: src/main/resources/examplemod.mixins.json
    //    (□ src/main/resources/examplemod.mixins.client.json □□□□□□ Mixin)
    //
    // 4. □Refmap□
    //    NeoForge □□□□ Mixin refmap, □□□□□□□□□
    // ============================================================================

    // □□: □□□□□□□ Mixin □□□□□□□□
    "package": "com.example.examplemod.mixin",

    // □□: □□□□□□□□□□
    "mixins": [
        // □□□/□□ Mixin □□□ (□□□□□□□□□□□□□□)
        // "ExampleMixin",
        // "block.BlockMixin"
    ],

    // □□: □□□□□□□□ Mixin □
    "client": [
        // "client.ClientExampleMixin",
        // "client.rendering.RenderMixin"
    ],

    // □□: □□□□□□□□ Mixin □
    "server": [
        // "server.ServerExampleMixin"
    ],

    // Mixin □□□□□□□□□ (□□□□□□□□□)
    "priority": 0,

    // □□□□ (Refmap) - NeoForge □□□□
    // □□ refmap JSON □□□□□ (NeoForge □□□□)
    "refmap": "examplemod.refmap.json",

    // □□□□□□ (□□)
    "minVersion": "0.8",

    // □□□□□□□ (□□, □□ false)
    "compatibilityLevel": "JAVA_21",

    // ============================================================================
    // Mixin □□□□□□□□:
    //
    // □ mixin □□□□□□:
    //   @Mixin(TargetClass.class)
    //   public class ExampleMixin {
    //       @Inject(method = "targetMethod", at = @At("HEAD"))
    //       private void onTargetMethod(CallbackInfo ci) {
    //           // □□□□
    //       }
    //
    //       @Redirect(method = "targetMethod", at = @At(value = "INVOKE", target = "..."))
    //       private void redirectMethod() { ... }
    //
    //       @ModifyVariable(method = "targetMethod", at = @At("STORE"))
    //       private int modifyVar(int original) { return original + 1; }
    //   }
    // ============================================================================
}
```

project/ModMain.java

```
// ============================================================================
// NeoForge 1.21.1 □□□□□□□ (ModMain.java)
// ============================================================================
//
// □□ Forge 1.20.1 □□□□□□□
// 1. □□□□□□□Forge: net.minecraftforge.*  ->  NeoForge: net.neoforged.*
//    - @Mod □□: net.minecraftforge.fml @Mod  ->  net.neoforged.fml @Mod
```

```
//    - □□□□□: net.minecraftforge.eventbus.api.IEventBus
//               -> net.neoforged.bus.api.IEventBus
//    - DeferredRegister: net.minecraftforge.registries.DeferredRegister
//                      -> net.neoforged.neoforge.registries.DeferredRegister
//    - □□□: BuiltInRegistries (1.19.3+ □ Forge □□□ Vanilla,
//             1.20.1 □□□□□ Forge □□□)
// 2. □□□□□□□□□□□
//    - Forge 1.20.1: □□ FMLJavaModLoadingContext.get().getModEventBus() □□
//    - NeoForge 1.21.1: □□ ModContainer □□□□□□□□□□ IEventBus
//      (FMLJavaModLoadingContext □□□□!)
// 3. □□□□□□□□□NeoForge □□□□□□□□□□□ ModContainer □ IEventBus
// 4. □@Mod □□□NeoForge □ @Mod □□□□□□□□□□□□□□□
// ============================================================================

package com.example.examplemod;

import com.example.examplemod.block.ModBlocks;
import com.example.examplemod.item.ModItems;
import com.example.examplemod.item.ModCreativeModeTabs;
import com.mojang.logging.LogUtils;
import net.neoforged.bus.api.IEventBus;
import net.neoforged.fml.ModContainer;
import net.neoforged.fml.common.Mod;
import net.neoforged.fml.config.ModConfig;
import net.neoforged.neoforge.common.NeoForgeMod;
import org.slf4j.Logger;

// ============================================================================
// @Mod □□ - □□□□□ ID
// □□□□□□□: net.neoforged.fml.common.Mod (Forge: net.minecraftforge.fml.Mod)
// □□□□ Forge 1.20.1 □□: @Mod("examplemod")
// ============================================================================
@Mod(ExampleMod.MOD_ID)
public class ExampleMod {

    // □□ ID - □□□□ neoforge.mods.toml □□ modId □□
    public static final String MOD_ID = "examplemod";

    // □□□□□□
    private static final Logger LOGGER = LogUtils.getLogger();

    // ============================================================================
    // □□□□□
    // □□□□□□□NeoForge 1.21.1 □□□□□□□□□□□□□□□ IEventBus □ ModContainer
    //
    // Forge 1.20.1 □□□:
    //   public ExampleMod() {
    //       IEventBus modEventBus = FMLJavaModLoadingContext.get().getModEventBus();
    //       modEventBus.addListener(this::commonSetup);
    //   }
    //
    // NeoForge 1.21.1 □□□:
    //   public ExampleMod(IEventBus modEventBus, ModContainer modContainer) {
    //       modEventBus.addListener(this::commonSetup);
    //   }
    //
    // □□□□□FMLJavaModLoadingContext □ NeoForge 1.20.5+ □□□□□□!
    //         □□□□□□ FMLJavaModLoadingContext.get()
    // ============================================================================
    public ExampleMod(IEventBus modEventBus, ModContainer modContainer) {
        LOGGER.info("Loading Example Mod for NeoForge 1.21.1");

        // □□ DeferredRegister □□□□□□□□
        // □□□□□NeoForge □ DeferredRegister □□: net.neoforged.neoforge.registries.DeferredRegister
        //         Forge □ DeferredRegister □□: net.minecraftforge.registries.DeferredRegister
        // □□□□□□□□: deferredRegister.register(modEventBus)

        // □□□□□
        ModBlocks.BLOCKS.register(modEventBus);

        // □□□□□□□□□□□
        ModBlocks.BLOCK_ENTITY_TYPES.register(modEventBus);

        // □□□□□
        ModItems.ITEMS.register(modEventBus);
        // □□□□□□□□□□□□□□□
```

```
        ModCreativeModeTabs.CREATIVE_MODE_TABS.register(modEventBus);

        // □□□□□□□□□□□□ (1.21+ □□□□)
        // ModEnchantments.ENCHANTMENT_COMPONENT_TYPES.register(modEventBus);

        // □□□□□□□ (GUI)
        // ModMenuTypes.MENU_TYPES.register(modEventBus);

        // □□□□□□□
        // ModEntities.ENTITY_TYPES.register(modEventBus);

        // □□□□ (□□□□□□□□□□)
        // modEventBus.addListener(this::commonSetup);

        // □□□□□□□
        // □□□□Forge: ModLoadingContext.get().registerConfig(...)
        //         NeoForge: modContainer.registerConfig(ModConfig.Type.COMMON, SPEC)
        //         ModLoadingContext □ NeoForge □□□□□!
        modContainer.registerConfig(ModConfig.Type.COMMON, Config.SPEC);
    }

    // ===========================================================================
    // □□□□□□□□□
    // ===========================================================================
    // private void commonSetup(final FMLCommonSetupEvent event) {
    //     event.enqueueWork(() -> {
    //         LOGGER.info("Common setup complete");
    //     });
    // }

    // ===========================================================================
    // NeoForge □□□□□□□ (□□□□□□□□□□)
    // □□□□□□□ (modEventBus): □□□□□□□□□□ (DeferredRegister, □□□□□)
    // □□□□□□□ (NeoForge.EVENT_BUS): □□□□□□□□□□ (□□□□, □□□□□)
    //
    // □□□□Forge: MinecraftForge.EVENT_BUS
    //         NeoForge: NeoForge.EVENT_BUS (□□: net.neoforged.neoforge.NeoForge)
    //
    // □□□□:
    // NeoForge.EVENT_BUS.register(new EventHandler());
    // □□□□□□□□ @SubscribeEvent □□
    // ===========================================================================
}
```

project/build.gradle

```
// ===========================================================================
// NeoForge 1.21.1 □□□□ (build.gradle)
// ===========================================================================
//
// □□ Forge 1.20.1 □□□□□□
// 1. Forge 1.20.1 □□ net.minecraftforge.gradle □□ (ForgeGradle);
//    NeoForge 1.21.1 □□ net.neoforged.moddev □□ (ModDevGradle)□
// 2. Forge 1.20.1 □□ "minecraft" + "forge" □□□□□;
//    NeoForge 1.21.1 □□ neoForge {} □□□□□□□□
// 3. Forge 1.20.1 □□ userdev □ official □□;
//    NeoForge 1.21.1 □□ Parchment □□ (parchment {} □)□
// 4. Forge 1.20.1 □ reobf □□□ NeoForge □□□□□□□
// 5. □□□□1.21+ □□ Java 21 (□□ Java 17!)
//    Forge 1.20.1 □□ Java 17, NeoForge 1.21.1 □□□□ Java 21□
// 6. NeoForge 1.21.1 □□ ModDevGradle □□, □□□□ NeoGradle (□□ NeoGradle □□□□)□
// ===========================================================================

plugins {
    id 'java-library'
    id 'maven-publish'
    // NeoForge ModDevGradle □□
    // □□□□□□□□: https://projects.neoforged.net/neoforged/ModDevGradle
    // □□□□Forge □□: id 'net.minecraftforge.gradle' version '6.0.x'
    //         NeoForge □□: id 'net.neoforged.moddev' version '2.0.146'
    id 'net.neoforged.moddev' version '2.0.146'
    id 'idea'
}

// Gradle Wrapper □□
tasks.named('wrapper', Wrapper).configure {
```

```
    distributionType = Wrapper.DistributionType.BIN
}

version = mod_version
group = mod_group_id

// □□□□□□□□□□□□□□□□□□□□
sourceSets.main.resources {
    srcDir('src/generated/resources')
    exclude("**/*.bbmodel")
    exclude("src/generated/**/.cache")
}

repositories {
    // □□□□□□□□ Maven □□ (□□□□□)
    // NeoForge □□□□□□□□□□□□□□
}

base {
    archivesName = mod_id
}

// ============================================================================
// □□□□□1.21+ □□□□□ Java 21!
// Forge 1.20.1 □□ Java 17, □ Minecraft 1.21 □□□□□ Java 21 □□□,
// □□□□□□□□□□□□ Java 21 □□□□
// ============================================================================
java.toolchain.languageVersion = JavaLanguageVersion.of(21)

// ============================================================================
// NeoForge □□□□
// □□□□□Forge □□ minecraft { } □ dependencies { forge ... } □□
//         NeoForge □□ neoForge { } □□□□□□□□
// ============================================================================
neoForge {
    // □□ NeoForge □□
    // □□□□□1.21.1 □□□□ NeoForge □□□□ 21.1.x (□ 21.1.77)
    // Forge 1.20.1 □□ Forge □□ 47.x.x
    version = project.neo_version

    // Parchment □□ (□□□□□□□□□□□ Javadoc)
    // □□□□□Forge □□ official □□□□ MCP □□
    //         NeoForge □□□□□ Parchment □□
    parchment {
        mappingsVersion = project.parchment_mappings_version
        minecraftVersion = project.parchment_minecraft_version
    }

    // Access Transformers (□□, □□□□□)
    // accessTransformers = project.files('src/main/resources/META-INF/accesstransformer.cfg')

    // □□□□□
    runs {
        client {
            client()
            systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
        }
        server {
            server()
            programArgument '--nogui'
            systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
        }
        // □□□□□□□□□
        gameTestServer {
            type = "gameTestServer"
            systemProperty 'neoforge.enabledGameTestNamespaces', project.mod_id
        }
        // □□□□□□□□
        data {
            data()
            programArguments.addAll '--mod', project.mod_id, '--all',
                '--output', file('src/generated/resources/').getAbsolutePath(),
                '--existing', file('src/main/resources/').getAbsolutePath()
        }
        // □□□□□□□□□□□□
        configureEach {
```

```
            // □□□□□: SCAN, REGISTRIES, REGISTRYDUMP
            systemProperty 'forge.logging.markers', 'REGISTRIES'
            logLevel = org.slf4j.event.Level.DEBUG
        }
    }

    // □□ <-> □□□□□□
    mods {
        "${mod_id}" {
            sourceSet(sourceSets.main)
        }
    }
}

// localRuntime □□: □□□□□□□□□□□□□□□□□□□□
configurations {
    runtimeClasspath.extendsFrom localRuntime
}

dependencies {
    // □□: JEI □□
    // compileOnly "mezz.jei:jei-${mc_version}-common-api:${jei_version}"
    // compileOnly "mezz.jei:jei-${mc_version}-neoforge-api:${jei_version}"
    // localRuntime "mezz.jei:jei-${mc_version}-neoforge:${jei_version}"
}

// ================================================================================
// □□□□□□□□ (□□ neoforge.mods.toml □□ ${} □□□□)
// ================================================================================
var generateModMetadata = tasks.register("generateModMetadata", ProcessResources) {
    var replaceProperties = [
            minecraft_version      : minecraft_version,
            minecraft_version_range: minecraft_version_range,
            neo_version            : neo_version,
            loader_version_range   : loader_version_range,
            mod_id                 : mod_id,
            mod_name               : mod_name,
            mod_license            : mod_license,
            mod_version            : mod_version,
    ]
    inputs.properties replaceProperties
    expand replaceProperties
    from "src/main/templates"
    into "build/generated/sources/modMetadata"
}
sourceSets.main.resources.srcDir generateModMetadata
neoForge.ideSyncTask generateModMetadata

// Maven □□□□
publishing {
    publications {
        register('mavenJava', MavenPublication) {
            from components.java
        }
    }
    repositories {
        maven {
            url "file://${project.projectDir}/repo"
        }
    }
}

tasks.withType(JavaCompile).configureEach {
    options.encoding = 'UTF-8'
}

idea {
    module {
        downloadSources = true
        downloadJavadoc = true
    }
}
```

project/neoforge.mods.toml

```
# ================================================================================
```

```
# NeoForge 1.21.1 □□□□□□□□ (neoforge.mods.toml)
# ============================================================================
#
# □□ Forge 1.20.1 □□□□□□□
# 1. □□□□□□□□Forge □□ "mods.toml", NeoForge 1.20.5+ □□ "neoforge.mods.toml"□
#    □□□□□: src/main/resources/META-INF/neoforge.mods.toml
#    (□□□ MDK □, □□ src/main/templates/META-INF/ □□□□□□□□□)
# 2. □modLoader □□NeoForge □□□□□ "javafml" □□ modLoader □
#    (□□: □□ "javafmloader", □□ MDK □□□□□□ "javafml")□
# 3. □□□□□□□NeoForge □□□ modId □ "neoforge" (□□ Forge □ "forge")□
# 4. □Mixin □□□Mixin □□□□□□□□□□ [[mixins]] □□□□
# 5. □AccessTransformer□AT □□□□ [[accessTransformers]] □□□□
# ============================================================================

# □□□□□□□□ - NeoForge □□ "javafml" (□ Forge □□)
# □□□□□modLoader □□□□ "javafml", □□ "javafmloader"
modLoader="javafml"

# □□□□□□□□□ - FML □□□
loaderVersion="${loader_version_range}"

# □□□□□□
license="${mod_license}"

# □□□□□ URL (□□)
#issueTrackerURL="https://change.me.to.your.issue.tracker.example.invalid/"

# ============================================================================
# □□□□□
# ============================================================================
[[mods]]
modId="${mod_id}"
version="${mod_version}"
displayName="${mod_name}"
#updateJSONURL="https://change.me.example.invalid/updates.json"
#displayURL="https://change.me.to.your.mods.homepage.example.invalid/"
#logoFile="examplemod.png"
#credits=""
#authors=""
description='''
Example mod description.
'''

# ============================================================================
# Mixin □□□□□
# □□□□□NeoForge 1.20.5+ □ neoforge.mods.toml □□□□ Mixin □□
#         Forge □ mods.toml □□□□ mixins □□
# ============================================================================
#[[mixins]]
#config="${mod_id}.mixins.json"

# ============================================================================
# Access Transformer □□
# ============================================================================
#[[accessTransformers]]
#file="META-INF/accesstransformer.cfg"

# ============================================================================
# □□□□□
# □□□□□□□□NeoForge □□□□ modId □ "neoforge" (Forge □□ "forge")
# ============================================================================
[[dependencies.${mod_id}]]
    modId="neoforge"  # □□□□□Forge: modId="forge", NeoForge: modId="neoforge"
    type="required"
    versionRange="[${neo_version},)"
    ordering="NONE"
    side="BOTH"

[[dependencies.${mod_id}]]
    modId="minecraft"
    type="required"
    versionRange="${minecraft_version_range}"
    ordering="NONE"
    side="BOTH"

# ============================================================================
```

```
# □□□□□ (□□)
# ================================================================================
#[features.${mod_id}]
#openGLVersion="[3.2,)"
```

recipe/shaped_recipe.json

```
{
    // ================================================================================
    // NeoForge 1.21.1 □□□□□□□ (shaped_recipe.json)
    // ================================================================================
    //
    // □□ Forge 1.20.1 □□□□□□ - □□□□□□
    // 1. □□□□□□□□□1.21+ □□□□□□□□□□□□□□□□:
    //    Forge 1.20.1: data/<modid>/recipes/ (□□)
    //    NeoForge 1.21.1: data/<modid>/recipe/ (□□)
    //    □□□□□□: tags/blocks -> tags/block, loot_tables -> loot_table □
    //
    // 2. □□□□□□□□□JSON □□□ Forge 1.20.1 □□□□□:
    //    type, pattern, key, result
    //
    // 3. □result □□ (1.21.x)□
    //    Forge 1.20.1: "result": {"item": "examplemod:example_item"}
    //    NeoForge 1.21.1: "result": {"id": "examplemod:example_item", "count": 1}
    //    □□□□□□□□ "item" □□ "id"! □□□□□ "components" □□□
    //
    // 4. □Data Components□1.21.x □□□□ result □□□ Data Components:
    //    "result": {
    //        "id": "examplemod:example_item",
    //        "count": 1,
    //        "components": {
    //            "minecraft:enchantment_glint_override": true
    //        }
    //    }
    //
    // 5. □□□□□□□
    //    - minecraft:crafting_shaped (□□□□)
    //    - minecraft:crafting_shapeless (□□□□)
    //    - minecraft:smelting, minecraft:blasting, minecraft:smoking □
    //
    // □□□□: data/examplemod/recipe/example_shaped.json
    // ================================================================================

    "type": "minecraft:crafting_shaped",

    // □□□□ NeoForge □□□□□□ (□□)
    // "neoforge:conditions": {
    //     "type": "neoforge:mod_loaded",
    //     "modid": "examplemod"
    // },

    // □□□□
    // □□□□□□□□□□□□□□□□
    "pattern": [
        "SSS",
        " S ",
        " S "
    ],

    // □□□
    // □□□□□□□□□ ID □ NeoForge □□
    "key": {
        "S": {
            "item": "minecraft:stick"
        },
        // NeoForge □□□□:
        // "I": {
        //     "tag": "c:ingots/iron"
        // }
        // 1.21.x □□□□□□:
        // "I": {
        //     "tag": "minecraft:planks"
        // }
    },

    // ================================================================================
```

```
    // □□□□□□□result □□□□□
    // Forge 1.20.1: "result": {"item": "examplemod:example_item", "count": 4}
    // NeoForge 1.21.1: "result": {"id": "examplemod:example_item", "count": 4}
    //                  "item" □□□□□ "id"!
    // ============================================================================
    "result": {
        "id": "examplemod:example_item",
        "count": 4,
        // 1.21.x □□□□□□□□□ Data Components (□□)
        "components": {
            // "minecraft:enchantment_glint_override": true,
            // "minecraft:rarity": "rare"
        }
    },

    // □□ (□□) - □□□□□□□□□□□□□□□□□□□□□□□□
    "group": "examplemod:example_item"
}
```

tag/block_tag.json

```
{
    // ============================================================================
    // NeoForge 1.21.1 □□□□□□□ (block_tag.json)
    // ============================================================================
    //
    // □□ Forge 1.20.1 □□□□□□ - □□□□□□
    // 1. □□□□□□□□□1.21+ □□□□□□□□□□□□□□□□□□□□:
    //    Forge 1.20.1: data/<modid>/tags/blocks/ (□□)
    //    NeoForge 1.21.1: data/<modid>/tags/block/ (□□)
    //
    // 2. □NeoForge □□□□□□□□□
    //    NeoForge 1.21.x □□□□ "remove" □□, □□□□□□□□□□□□□□□□□
    //    Forge 1.20.1 □□□□□□□□
    //
    // 3. □□□□□□□□□□
    //    - minecraft: □□□□□ (□ minecraft:mineable/pickaxe)
    //    - c: □□□□□□□ (□ c:ores, c:ingots) - NeoForge □□□□
    //    - <modid>: □□□□□□□□
    //
    // □□□□: data/examplemod/tags/block/needs_copper_tool.json
    // ============================================================================

    // □□□□□□□□□□□□
    "values": [
        // □□□□□□□□ ID
        "minecraft:gold_block",
        "minecraft:raw_gold_block",
        "minecraft:gold_ore",
        "minecraft:deepslate_gold_ore",
        "minecraft:redstone_ore",
        "minecraft:deepslate_redstone_ore",

        // □□□□□□□□□□ (# □□)
        "#minecraft:gold_ores",

        // □□□□□□□□□□□□□□
        "examplemod:example_block",
        "examplemod:example_entity_block",

        // □□: □□□□□□ (□□□□□□□□□/□□□□□□□□□)
        // "required": false  // □□ true, □□ false □□□□□□□□□□□□□□
    ],

    // ============================================================================
    // □NeoForge □□□remove □□ - □□□□□□□□□
    // □□ NeoForge □□□□□□, Forge 1.20.1 □□□□□□□
    //
    // □□: □□□□ incorrect_for_copper_tool.json □,
    //   □□ incorrect_for_stone_tool □□□□□□, □□□□□ copper □□□□□□□□□
    //
    // □□ (incorrect_for_copper_tool.json):
    // {
    //     "values": ["#minecraft:incorrect_for_stone_tool"],
    //     "remove": ["#examplemod:needs_copper_tool"]
    // }
```

```
    // ============================================================================
    // "remove": [
    //     "minecraft:bedrock",
    //     "#some:other_tag"
    // ]
}
```

worldgen/biome_modifier.json

```
{
    // ============================================================================
    // NeoForge 1.21.1 □□□□□□□□□□ (biome_modifier.json)
    // ============================================================================
    //
    // □□ Forge 1.20.1 □□□□□□ - □□□□□□
    // 1. □□□□□□□□□□□□
    //    Forge 1.20.1 □□ BiomeModifier □□ (□□ Forge □ datapack □□)□
    //    NeoForge 1.21.1 □□□□□□□□ BiomeModifier □□, □□□□□□ "forge:" □□ "neoforge:"□
    //
    // 2. □□□□□□□□□□
    //    Forge 1.20.1: "type": "forge:add_spawns"
    //    NeoForge 1.21.1: "type": "neoforge:add_spawns"
    //
    // 3. □□□□□□□□□□□□
    //    - neoforge:add_spawns: □□□□□□□
    //    - neoforge:remove_spawns: □□□□□□□
    //    - neoforge:add_features: □□□□
    //    - neoforge:remove_features: □□□□
    //    - neoforge:add_carvers: □□□□□□□□
    //    - neoforge:remove_carvers: □□□□□□□□
    //    - neoforge:add_mob_settings: □□□□□□□□□
    //    - neoforge:modify_water_spring: □□□□
    //    - neoforge:modify_lava_spring: □□□□□
    //
    // □□□□: data/examplemod/neoforge/biome_modifiers/example_add_spawns.json
    //
    // 4. □1.21.x □□□□□
    //    □□□□□□□□ 1.21.x □□□□□□□□□□□□□□,
    //    □□□□□ API □□□□□□□ JSON □□□□□□□□
    // ============================================================================

    // ============================================================================
    // □□1: □□□□□□□ (add_spawns)
    // ============================================================================
    "type": "neoforge:add_spawns",

    // □□□□□□□
    // □□□: □□ biome ID□biome ID □□□□ biome □□ (# □□)
    "biomes": "#minecraft:is_jungle",

    // □□□□
    // weight: □□□□ (□□□□□□)
    // minCount: □□□□□□□
    // maxCount: □□□□□□□
    "spawners": [
        {
            "type": "minecraft:cow",
            "weight": 8,
            "minCount": 4,
            "maxCount": 4
        }
    ]

    // ============================================================================
    // □□2: □□□□ (add_features) - □□□□□□□
    // ============================================================================
    // "type": "neoforge:add_features",
    // "biomes": "#minecraft:is_overworld",
    // "features": [
    //     "examplemod:example_ore",
    //     "examplemod:example_flower"
    // ],
    // // □□□□: □□□□□□□□□□□□□□□
    // // □□□: raw_generation, lakes, local_modifications, underground_structures,
    // //         surface_structures, strongholds, underground_ores, underground_decoration,
    // //         fluid_springs, vegetal_decoration, top_layer_modification
```

```
    // "step": "underground_ores"

    // ================================================================================
    // □□3: □□□□□□□□ (remove_carvers)
    // ================================================================================
    // "type": "neoforge:remove_carvers",
    // "biomes": ["minecraft:plains", "minecraft:forest"],
    // "carvers": "minecraft:cave"
}
```

KunYuEcosystem

M7 AI 助手接线方案

参考包集成设计 / 四问题修复 / TypeScript Context Builder / Manifest 机器可读索引 / Scope 控制 / 映射声明

维度	数值
模板文件总数	243
M7 active 模板数	41 (Fabric 1.20.1)
M7 inactive 模板数	202 (其余 16 个版本组合)
错误模式条数	50
API 索引条目数	40
覆盖加载器数	3 (Fabric / Forge / NeoForge)
覆盖 MC 版本	7 (1.16.5 -> 1.21.3)
加载器 x 版本组合	17

一、发现的四个问题及修复方案

问题	严重性	修复方案	状态
1. context_builder.py 与技术栈不匹配	最要紧	TypeScript 在主进程重写	已完成
2. 模板缺乏编译验证元数据	高	manifest 中 verified/build 标记 pending	待执行
3. 缺机器可读检索索引	高	新增 manifest.json 三层数据	已完成
4. 覆盖范围与当前 MVP 错位	中	scopeControl 区分 active/inactive	已完成

1.1 问题一: context_builder.py 与技术栈不匹配

第 7 章写的是「由 prompts/context_builder.py 自动完成」，但工具是 Electron + Vue + TypeScript，渲染进程里跑 Python 不现实。参考包保持文件形态不变，但 context builder 用 TypeScript 在主进程重写，逻辑很简单: 读 manifest -> 按 (loader, mcVersion, contentType) 检索 -> 拼接 system prompt。参考包本身就是纯文本文件，TS 读写没有障碍。

TypeScript ContextBuilder 核心 API:

方法	用途	M7 任务
buildSystemPrompt(config, userQuery?)	构建完整 system prompt	T7.1
detectContentTypes(userQuery)	从用户查询检测内容类型	T7.2
getTemplateContents(loader, ver, types[])	检索模板内容 (限 5 个)	T7.2
matchErrorPatterns(log, loader, ver)	匹配编译错误模式	T7.3
getErrorFixContext(log, config)	获取错误修复上下文	T7.3
buildSystemPromptWithError(cfg, log, query?)	构建带错误修复的 prompt	T7.3
isVersionActive(loader, version)	Scope 控制检查	全局

关键代码片段 (context_builder.ts):

```
export class ContextBuilder {
  private manifest: Manifest | null = null;

  // □□□□: □□□ active=true □□□□□
  searchTemplates(loader, mcVersion, contentTypes) {
    const manifest = this.loadManifest();
    return manifest.templates.filter(t => {
      if (t.loader !== loader) return false;
      if (t.version !== mcVersion) return false;
      if (!t.active) return false;  // scope control
      if (contentTypes && !contentTypes.includes(t.type)) return false;
      return true;
    });
  }

  // □□□□ system prompt
  buildSystemPrompt(config, userQuery?, contentTypesFilter?) {
    // 1. □□□□□□
    // 2. □□□□ (□ MAX_INJECTION_TEMPLATES=5)
    // 3. □□ API □□
    // 4. □□□□□□
    // 5. □□□□□ -> □□□□ system prompt
  }

  // □□□□ (T7.3)
  matchErrorPatterns(errorLog, loader, mcVersion) {
    // □□ 50 □□□□□□ regex
    // □□: □□□□□ loader + mcVersion
    // □□ ErrorMatchResult[]
  }
}
```

原 context_builder.py 已标注 [DEPRECATED]，仅作参考实现保留，不随 M7 更新。

1.2 问题二: 模板编译验证状态不明

文档 9.5.2 要求片段顶部加注释: 验证日期、验证结果，但这版参考包的模板里没有这个元数据。243 个模板文件如果未经批量编译验证，它们本身就是新的幻觉来源。manifest.json 中所有模板的 verified 字段为 null，build 字段为 pending。后续需要对 14 个全类型版本组合各组装一个最小工程，批量 gradlew build，把结果回写。

优先级	版本组合	模板数	理由
P0	Fabric 1.20.1	41	M7 active 路径，必须先验证
P1	Forge 1.20.1	12	下一个适配版本
P2	NeoForge 1.21.1	12	差异化优势版本

P3	其余 11 个组合	~178	储备资产，增量维护

manifest.json 中的验证字段示例:

```
{
  "loader": "fabric",
  "version": "1.20.1",
  "type": "block",
  "file": "templates/fabric/1.20.1/block/simple_block.java",
  "keywords": ["□□", "block", "□□□□"],
  "verified": null,          // <- □□□ ISO □□
  "build": "pending",        // <- □□□: pass / fail
  "active": true              // M7 □□□□
}
```

增量维护: 新增模板时 build 设为 pending; 修改模板时重置为 pending; 删除模板时从 manifest.json 移除条目。

1.3 问题三: 缺机器可读的检索索引

模板的组织是人读的 (目录结构)，但 M7 的注入逻辑需要机读索引。新增 manifest.json 包含三层数据: templates (243 条)、errorPatterns (50 条)、apiIndex (40 条)。检索键就是项目配置里的 loader + MC 版本，精确匹配 + 关键词兜底足够。

层	字段	条数	说明
模板索引	templates[]	243	loader/version/type/file/keywords/verified/build/active
错误模式	errorPatterns[]	50	regex pattern + cause + solution + fix_code
API 索引	apiIndex[]	40	class_path + field + type + example + common_error
Scope 控制	scopeControl	1	m7ActiveVersions + registeredButInactive
映射类型	mappingTypes	3	各加载器的映射体系声明

1.4 问题四: 覆盖范围和当前 MVP 错位

243 个模板对应 17 个版本组合，但 M7 跑起来时产品只支持 Fabric 1.20.1 (41 个模板 + 工程模板)。这不是错，它是对的资产储备，Multi-loader 是对 AutoMods 的差异化优势。但要防止 M7 scope 蔓延: M7 的注入逻辑只接通 1.20.1 一条路径，其余版本模板只在 manifest 里登记、不接检索。等做对应版本适配时再启用。

字段	值	说明
m7ActiveVersions	[{fabric, 1.20.1}]	M7 检索只接通这条路径 (41 模板)
registeredButInactive	16 个版本组合	在 manifest 中登记但不暴露给用户 (202 模板)
启用方式	更新 m7ActiveVersions 数组	无需改代码，改 manifest.json 即可

二、M7 接线方案

2.1 数据流总览

把参考包接进 M7 后，三个任务的形态变成以下流程:

任务	描述	ContextBuilder 方法
T7.1 对话面板	system prompt 常驻环境声明	buildSystemPrompt()
T7.2 上下文注入	根据提问选 3-5 个模板注入	detectContentTypes() + getTemplateContents()
T7.3 修复建议	错误日志 -> 正则匹配 -> 修复方案注入	matchErrorPatterns() + getErrorFixContext()

完整数据流:

```
M2 □□□□ (loader, mcVersion, modId, packageName)
  |
  v
ContextBuilder.constructor(refPackageDir)
  |-> □□□ manifest.json + system_prompt_base.md
  |
  v
T7.1: buildSystemPrompt(config)
  |-> □□ active □□ (□ fabric/1.20.1, □ 5 □)
  |-> □□ {{MAPPING_DECLARATION}} (□□□□□□)
  |-> □□ {{REFERENCE_TEMPLATES}} (□□□□)
  |-> □□ {{REFERENCE_API}} (API □□)
  |-> □□□□ system prompt
  |
  v
T7.2: detectContentTypes(userQuery) -> getTemplateContents()
  |-> □□□□□: "□□" -> ["block"]
  |-> □□ 3-5 □□□□□□□
  |-> □□□ system prompt
  |
  v
T7.3: matchErrorPatterns(errorLog) -> getErrorFixContext()
  |-> □□ 50 □□□□□□ regex
  |-> □□: □□□□□ loader + mcVersion
  |-> □□: □□□□ + □□□□ + □□□□
  |-> buildSystemPromptWithError() □□□□□□□
  |-> AI □□□ -> □□ -> □□□□ (3 □□□)
```

2.2 T7.1 对话面板 - 环境声明常驻

system prompt v2.0 常驻一段「当前项目环境」声明，所有回答受它约束。包含: loader/MC 版本/Java/映射类型/模组 ID/包名/已有文件。Scope 控制: context_builder.ts 的 searchTemplates() 只返回 active=true 的模板。

段落	内容	占位符
一、角色定义	锁定 {{LOADER}} {{MC_VERSION}}	LOADER, MC_VERSION
二、项目上下文	环境变量 + 映射声明	MOD_ID, JAVA_VERSION, MAPPING_TYPE, MAPPING_DECLARATION
2.1 映射声明 (新增)	显式声明 Yarn/Mojang 映射	MAPPING_DECLARATION
三、参考包注入	模板内容 + API 索引	REFERENCE_TEMPLATES, REFERENCE_API
四、约束规则	21 条反幻觉规则	(静态文本)
五-七、输出/不确定/类型映射	格式与检索表	PACKAGE_NAME_PATH 等

2.3 T7.2 上下文注入 - 参考包检索注入

收窄为「参考包检索注入」: 根据用户提问和当前打开的文件类型选模板，单次注入控制在 3-5 个文件 (MAX_INJECTION_TEMPLATES = 5)，别把 243 个全塞进去。

用户需求关键词	对应内容类型	参考包目录
方块, block	block	templates/{loader}/{ver}/block/
物品, item, 工具, 盔甲	item	templates/{loader}/{ver}/item/
实体, entity, 生物	entity	templates/{loader}/{ver}/entity/
配方, recipe, 合成	recipe	templates/{loader}/{ver}/recipe/
界面, gui, screen	gui	templates/{loader}/{ver}/gui/
混入, mixin, 注入	mixin	templates/{loader}/{ver}/mixin/

用户需求关键词	对应内容类型	参考包目录
世界生成, worldgen, biome	worldgen	templates/{loader}/{ver}/worldgen/

2.4 T7.3 修复建议 - 错误模式匹配 + 修复循环

M5 的 T5.2 解析出错误日志 -> 正则匹配错误模式库 (50 条的 regex 字段现成) -> 命中则把「错误 + 修复方案 + 相关模板」一起注入 -> AI 出补丁 -> 应用 -> 重新构建，3 轮上限，对应文档已有的降级策略。

修复循环流程:

```
M5 □□□□ -> □□□□□□□
  |
  v
matchErrorPatterns(errorLog, loader, mcVersion)
  |-> □□ 50 □□□□□□□ regex pattern
  |-> □□: □□□□□□ loader + mcVersion □□□□□□
  |-> □□ ErrorMatchResult[] (□ cause + solution + fix_code)
  |
  v
getErrorFixContext(errorLog, config)
  |-> □□□□□□□□□□□□□□□□□□□□
  |-> □□□□□□□ (3-5 □)
  |-> □□ { matches, relatedTemplates }
  |
  v
buildSystemPromptWithError(config, errorLog, userQuery)
  |-> □□□ system prompt □□□ "□□□□□□□□□□□□"
  |-> □□: □□□□□ + □□□□□ + □□□□□ + □□□□□
  |
  v
AI □□□□□□□□□□□□□□
  |-> □□□□□ -> □□ gradlew build
  |-> □□□□□ -> □□□ (□□ 3 □)
  |-> 3 □□□□□□ -> □□: □□□□□□□□□ + □□□□□□□□
```

错误模式匹配示例:

```
□□□□: "error: cannot find symbol / symbol: class RegistryMethods"
  |
  v
□□ ERR_001 (pattern: "cannot find symbol[\s\S]*RegistryMethods")
  |-> □□: AI□□□RegistryMethods.getRegistry()□□
  |-> □□: □□Registry.register(Registries.BLOCK, new Identifier(...), block)
  |-> □□□□: Registry.register(Registries.BLOCK,
       new Identifier("examplemod", "example_block"), block);
```

三、映射声明防幻觉机制

模板全是 Yarn 映射的写法，而 Forge/NeoForge 用的是官方 Mojang 映射。给 AI 的 system prompt 里要显式声明「本项目使用 Yarn 映射」或对应版本，否则 AI 容易混用方法名 -- 这是跨加载器时最隐蔽的幻觉类型。

加载器	映射类型	标识符类	注册表类
Fabric	Yarn	net.minecraft.util.Identifier	net.minecraft.registry.Registries
Forge	Mojang 官方	net.minecraft.resources.ResourceLocation	net.minecraftforge.registries.ForgeRegistries
NeoForge	Mojang 官方	net.minecraft.resources.ResourceLocation	net.neoforged.neoforge.registries.DeferredRegister

system prompt v2.0 新增的映射声明占位符:

```
> **□□□□□□□**: □□□□□□ **Yarn □□**
>   (net.minecraft.util.Identifier, net.minecraft.registry.Registries)□
> □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
> □□□□□□□□□□ (□ Yarn □ Mojang □□□□□□□□□□□□□□□□)□
> □□□: fabric | □□: yarn

**□□□□□□□□□□□□□□□□**:

| □□□□□ | □□□□□□□ | □□□□□□□ | □□□ |
|---|---|---|---|
| Yarn | Identifier | Registries | Fabric |
| Mojang □□ | ResourceLocation | ForgeRegistries | Forge |

**□□□□□□□□□□**
```

这直接防止了 ERR_050 (Forge 中使用 Identifier) 和 ERR_049 (Fabric 中使用 @Mod 注解) 等跨映射幻觉。manifest.json 的 mappingTypes 字段声明每个加载器的映射类型，context_builder.ts 的 buildMappingDeclaration() 方法生成映射声明文本。

四、文件变更清单

4.1 新增文件

文件路径	用途

manifest.json	机器可读索引 (templates + errorPatterns + apiIndex + scopeControl)
prompts/context_builder.ts	TypeScript context builder (Electron 主进程，生产用)
M7_INTEGRATION_DESIGN.md	接线方案设计文档

4.2 修改文件

文件路径	变更内容
prompts/system_prompt_base.md	v2.0: 新增映射声明/Java 版本占位符/映射混用禁止规则
prompts/context_builder.py	标注 [DEPRECATED]，仅作参考保留
README.md	v3.1: 更新目录结构/manifest 说明/版本历史/统计信息

五、Manifest.json 数据结构

5.1 顶层结构

```
{
  "version": "3.0.0",
  "generatedAt": "2026-09-13T...",
  "description": "KunYuEcosystem AI □□□□□□□□□□",

  "scopeControl": {
    "m7ActiveVersions": [{"loader":"fabric","version":"1.20.1"}],
    "registeredButInactive": [/* 16 □□□□□ */],
    "rule": "M7 □□□□□□□ active=true □□□□□..."
  },

  "mappingTypes": {
    "fabric":   {"type":"yarn", "description":"..."},
    "forge":    {"type":"mojang_official", "description":"..."},
    "neoforge": {"type":"mojang_official", "description":"..."}
  },

  "templates": [/* 243 □ */],
  "errorPatterns": [/* 50 □ */],
  "apiIndex": [/* 40 □ */],

  "stats": {
    "templateCount": 243,
    "activeTemplateCount": 41,
    "inactiveTemplateCount": 202,
    "errorPatternCount": 50,
    "apiIndexCount": 40
  }
}
```

5.2 Template 条目结构

```
{
  "loader": "fabric",
  "version": "1.20.1",
  "type": "block",
  "file": "templates/fabric/1.20.1/block/simple_block.java",
  "filename": "simple_block.java",
  "keywords": ["□□","block","blockstate","□□□□"],
  "verified": null,       // ISO □□ □ null
  "build": "pending",    // pending / pass / fail
  "active": true         // M7 □□□□□□□
}
```

5.3 ErrorPattern 条目结构

```
{
  "id": "ERR_001",
  "category": "API□□□□□/□□",
  "error_type": "□□□□□",
  "pattern": "cannot find symbol[\s\S]*RegistryMethods",
  "severity": "critical",
  "mc_versions": ["1.20.1","1.20.4"],
  "loaders": ["fabric"],
  "cause": "AI□□□□□□□RegistryMethods.getRegistry()□□",
  "solution": "□□Registry.register(Registries.BLOCK, ...)",
  "fix_code": "Registry.register(Registries.BLOCK, ...);",
  "example_error": "error: cannot find symbol..."
}
```

5.4 M7 Active 模板分布 (Fabric 1.20.1)

内容类型	模板数	文件示例
advancement	1	advancement.json
block	5	block_entity.java
command	1	simple_command.java
enchantment	1	custom_enchantment.java
entity	4	entity_attribute.java
gui	2	screen_with_menu.java
item	5	armor_item.java
mixin	3	inject_mixin.java
project	9	LICENSE
recipe	5	shaped_recipe.json
tag	3	block_tag.json
worldgen	2	biome_modifier.json

六、 Scope 控制: Active vs Inactive 版本

M7 的注入逻辑只接通 active=true 的版本路径 (Fabric 1.20.1, 41 个模板)。其余 16 个版本组合 (202 个模板) 在 manifest 里登记但不接检索。等做对应版本适配时再启用，别让未验证的路径提前暴露给用户。

状态	加载器	版本	模板数	说明
Active	fabric	1.20.1	41	M7 接通检索
Inactive	fabric	1.16.5	1	legacy, project-only
Inactive	fabric	1.18.2	12	not yet adapted for M7
Inactive	fabric	1.19.2	12	not yet adapted for M7
Inactive	fabric	1.20.4	12	not yet adapted for M7
Inactive	fabric	1.20.6	12	not yet adapted for M7
Inactive	fabric	1.21.1	12	not yet adapted for M7
Inactive	forge	1.16.5	2	legacy, project-only
Inactive	forge	1.18.2	12	not yet adapted for M7
Inactive	forge	1.19.2	11	not yet adapted for M7
Inactive	forge	1.20.1	12	not yet adapted for M7
Inactive	forge	1.20.4	12	not yet adapted for M7
Inactive	forge	1.21.1	12	not yet adapted for M7
Inactive	neoforge	1.20.4	12	not yet adapted for M7
Inactive	neoforge	1.20.6	12	not yet adapted for M7
Inactive	neoforge	1.21.1	12	not yet adapted for M7
Inactive	neoforge	1.21.3	1	transition version, diff only

七、待执行事项

事项	优先级	状态	说明
context_builder.ts 集成到 Electron 主进程	P0	待集成	TS 文件已就位，需接入 IPC
Fabric 1.20.1 模板批量编译验证	P0	待执行	41 个模板 gradlew build 回写
T7.1 对话面板接通 system prompt	P0	待开发	调用 buildSystemPrompt()
T7.2 上下文注入逻辑	P1	待开发	detectContentTypes + getTemplateContents
T7.3 错误修复循环	P1	待开发	matchErrorPatterns + 3 轮重试
Forge 1.20.1 模板编译验证	P1	待执行	12 个模板
NeoForge 1.21.1 模板编译验证	P2	待执行	12 个模板
其余 11 个版本组合验证	P3	待执行	~178 个模板，增量维护

KunYuEcosystem

版本漂移监测与 AI 参考包集成

MC 版本号冻结但运行环境不冻结 / 版本指纹化 / 四条漂移路径 / 17 个版本组合健康状态 / AI 漂移意识约束

核心判断: MC 的版本号是冻结的，但让一个旧版本能编译、能跑起来的整条供应链不是。所谓全版本覆盖，覆盖的不是一个版本号，而是当时那串加载器 build、Mappings、JDK 与在线服务的组合。KunYuEcosystem 必须把版本组合指纹化，并对老版本组合做定期健康回测，否则覆盖 1.20.1 这句话半年后可能名不副实。

维度	数值
模板文件总数	243
M7 active 模板数	41 (Fabric 1.20.1)
M7 inactive 模板数	202
错误模式条数	50
API 索引条目数	40
版本组合指纹数	17
漂移路径数	4
健康: green	0
健康: yellow	0
健康: red	0

健康: unknown	17

一、核心认知: 版本号冻结但运行环境不冻结

MC 各版本之间确实相互独立 -- 但这个独立性只成立在游戏内容与存档格式这一层: 1.20.1 的方块行为、NBT 结构、合成配方不会因为 Mojang 发布了 26.2 而改变。问题在于，开发者实际要交付的能跑的 mod，依赖的远不止那一个版本号。它是一长串同时成立的条件: MC 的 jar、Mojang 托管的 libraries 与资产索引、当时兼容的加载器 build、当时那版 Mappings、指定小版本的 JDK，以及在线登录服务的协议。这串条件里任何一环动了，看似冻结的旧版本就会重新出问题。

二、四条让旧版本被动摇的路径

按发生频率从高到低排列。每一条都不需要 MC 本身发新版，就足以让一份曾经编译通过的旧模板重新报错。

漂移路径	频率	机制	对模板的影响	AI 行为
加载器对老版本滚动发版	高	MC 版本不变，但 Forge 持续出新的小版本 build、Fabric loader 不停更新，改 Mixin、改事件注册方式、动 API 签名。	模板里写的 DeferredRegister、事件总线调用，在新 build 上可能签名已变；用户用最新稳定加载器构建旧模板就编译失败。	当 health=yellow (加载器已出新 build 待重测) 时，AI 应在生成代码前提示: 此版本组合的加载器...
Mappings 持续修正	中	同一个 MC 版本，Yarn / Official 映射会被持续改名、改字段、改参数；mod 代码用的是映射名而非混淆名。	模板里引用的某个字段名在新映射里被改名或移除，旧代码直接编译不过；这与 MC 版本号无关。	AI 应在 system prompt 的映射声明中注明映射版本和验证日期。如果 mappings 健康状态非 gree...
外部环境演进	中	JDK 小版本升级带来强封装与 TLS 证书要求，微软登录 / session server 协议升级，启动器被迫跟进。	以前能跑的模板在新 JDK 下起不来、或登录失败；问题不在 mod 代码，而在运行环境，却同样表现为交付物不可用。	AI 应在环境声明中包含 JDK 版本。如果 JDK 小版本已超出验证时的版本，AI 应提示可能的兼容性问题。
Mojang 回头改托管文件	低但最难复现	旧版本指向的 libraries URL、资产索引、甚至 server jar，Mojang 理论上可以替换；在线服务端点升级后旧客户端连不上。	Gradle 拉依赖 404、资产加载失败、旧版无法登录；这类问题最难复现，因为它不在模板仓库里。	当 Gradle 构建失败且错误信息包含 404 或 connection refused 时，AI 应优先检查是否为 ...

2026-09-13 实际观测数据:

维度	值
MC 最新正式版	26.2
Fabric loader	0.19.5
NeoForge 最新	26.2.0.88
Forge 主线停在	1.19.4

备注	连加载器对各版本的支持力度本身都在随时间重排

三、版本组合指纹库 (17 个组合)

版本组合要记成带时间戳的指纹。覆盖 1.20.1 = 在这套指纹下、于某日编译通过，而不是一个含糊的版本号。用户换了更新的加载器 build 时，系统应能提示此模板在 X 版加载器上验证过，当前为 Y，可能需要重测。

ID	MC 版本	加载器	Loader 版本	映射	JDK	健康	漂移风险	漂移说明
fabric-1.16.5	1.16.5	fabric	0.19.5	yarn	8	unknown	high	JDK 8 已过时; legacy 版本, 加载器可能不再维护此版本
fabric-1.18.2	1.18.2	fabric	0.19.5	yarn	17	unknown	medium	Fabric loader 持续滚动发版, 老版本可能签名漂移
fabric-1.19.2	1.19.2	fabric	0.19.5	yarn	17	unknown	medium	Material 类在 1.19.3 被移除; 1.19.2 的 loader build 可能变化
fabric-1.20.1	1.20.1	fabric	0.19.5	yarn	17	unknown	low	M7 active 路径; Fabric loader 滚动发版需定期回测
fabric-1.20.4	1.20.4	fabric	0.19.5	yarn	17	unknown	medium	Registry API 变更: Registries.ITEM -> Registry.ITEM;...
fabric-1.20.6	1.20.6	fabric	0.19.5	yarn	21	unknown	medium	JDK 21 要求; Fabric API 签名可能漂移
fabric-1.21.1	1.21.1	fabric	0.19.5	yarn	21	unknown	medium	JDK 21 + 新 Registry API; loader 持续更新
forge-1.16.5	1.16.5	forge	36.2.42	mojang_official	8	unknown	high	JDK 8 过时; Forge 可能不再维护 1.16.5; Gradle 依赖 URL 可能 40...
forge-1.18.2	1.18.2	forge	40.2.0	mojang_official	17	unknown	medium	Forge 持续出小版本 build; DeferredRegister 签名可能变化
forge-1.19.2	1.19.2	forge	43.4.0	mojang_official	17	unknown	medium	Forge 主线停在 1.19.4; 1.19.2 的 build 更新频率低但漂移风险仍在
forge-1.20.1	1.20.1	forge	47.4.23	mojang_official	17	unknown	low	Forge 47.x 持续滚动发版; 事件总线 API 可能微调
forge-1.20.4	1.20.4	forge	45.0.66	mojang_official	17	unknown	medium	Forge 1.20.4 是过渡版本; NeoForge 从此分裂出去
forge-1.21.1	1.21.1	forge	51.0.98	mojang_official	21	unknown	medium	JDK 21; Forge 新版 build 频繁更新
neoforge-1.20.4	1.20.4	neoforge	20.4.240	mojang_official	17	unknown	medium	NeoForge 首个稳定版; API 仍在演进; DeferredRegister 签名变化频繁

ID	MC 版本	加载器	Loader 版本	映射	JDK	健康	漂移风险	漂移说明
neoforge-1.20.6	1.20.6	neoforge	20.6.132	mojang_official	21	unknown	medium	JDK 21; NeoForge 持续滚动发版
neoforge-1.21.1	1.21.1	neoforge	21.1.250	mojang_official	21	unknown	medium	JDK 21; NeoForge 21.1.x 持续更新; 事件 API 可能微调
neoforge-1.21.3	1.21.3	neoforge	21.3.0-beta	mojang_official	21	unknown	high	过渡版本; beta 状态; API 不稳定; 仅做差异参考

健康状态图例:

状态	颜色	含义
green	绿色	最近回测通过 -- 模板在真实加载器 + 真实 JDK 下仍能编译通过
yellow	黄色	加载器已出新 build 待重测 -- 模板可能基于旧版加载器，签名可能已漂移
red	红色	回测编译失败 -- 模板可能已失效，建议等待重测后再使用
unknown	灰色	从未真实回测 -- 模板基于文档和 API 参考编写，未在真实 Gradle 环境中验证过

四、监测层: 三个脚本的职责

监测层把发现事实与决定覆盖严格分离: 监测脚本只标状态，编译探针才是唯一签字方。未接入真实 Gradle 环境时，所有组合返回 skipped，health 保持 unknown，yellow 不会自动恢复为 green -- 这是故意的，防止没测过被当成测过了。

脚本	盯什么	输出与动作
monitor.py	有没有新 MC 版本、Fabric/Forge/NeoForge 全局最新版	report.md; 首次建基线，之后报新增
legacy_watch.py	每个已覆盖组合的加载器是否出了新 build	更新 version_matrix.json 的 loader_version; 有新 build 则 health 置为 yellow
health_check.py	老组合是否需要重测; 调度编译探针	对 yellow/unknown/超过阈值天数未回测的组合跑探针; pass->green, fail->red

每日监测工作流:

```
# □□□□□□□□ (□□□□□□□)
python3 monitor.py         # 1. □ MC □□ / □□□□□□□□
python3 legacy_watch.py    # 2. □□□□□□□□□ build (□□□ yellow)
python3 health_check.py --max-age-days 7  # 3. □□□□□□□□□
```

接入真实编译探针:

```
# □□□□□□□□□□□□□□□ Gradle □□
# □□□ 0 = pass -> health □□□ green
export BUILD_PROBE_CMD="./gradlew -Pmc={mc_version} -Ploader={loader} -PloaderVer={loader_version} build"
python3 health_check.py

# □□□□□: {id} {mc_version} {loader} {loader_version} {jdk}
```

五、AI 漂移意识与行为约束

system_prompt_base.md v3.0 新增三个占位符，让 AI 在生成代码前感知版本漂移: {{VERSION_FINGERPRINT}} 注入当前版本组合的完整指纹， {{HEALTH_STATUS}} 注入健康状态描述， {{DRIFT_WARNING}} 根据 health 生成不同级别的警告。

health	AI 行为约束	生成代码前的提示
green	正常生成，无额外约束	此版本组合最近回测通过，模板在当前环境下可信赖

health	AI 行为约束	生成代码前的提示
yellow	生成代码前提示漂移风险	此版本组合的加载器可能已漂移。生成的代码基于上次验证时的加载器版本。如遇编译失败，请检查加载器版本差异。
red	警告模板可能已失效	此版本组合最近回测失败，模板可能已失效。建议等待重测后再使用，或手动验证。
unknown	提示未经验证	此版本组合尚未经过编译验证回测。模板基于文档和 API 参考编写，但未在真实 Gradle 环境中验证过。

context_builder.ts 新增 API:

方法	用途
getVersionFingerprint(loader, mcVersion)	获取版本组合指纹 (loader_version/mappings/jdk/health/verified_at)
getHealthStatus(loader, mcVersion)	获取健康状态 (green/yellow/red/unknown)
getDriftPaths()	获取四条漂移路径信息
getMonitoringConfig()	获取监测层配置 (cron/max-age/build_probe_cmd)
getAllFingerprintsSummary()	获取全部 17 个版本指纹概要
buildVersionFingerprint()	构建指纹文本注入 {{VERSION_FINGERPRINT}}
buildHealthStatus()	构建健康状态文本注入 {{HEALTH_STATUS}}
buildDriftWarning()	构建漂移警告注入 {{DRIFT_WARNING}} (分级警告)

system prompt v3.0 中 AI 看到的漂移警告示例 (health=yellow):

```
> [WARNING] □□□□: □□□□□□□□□□□□□□□ build (health=yellow)□
> □□□□□□□□□□□□□□□□□□□□□ (0.19.5)□
> □□□□□□□□□□□□□:
>   1. □□□□□□□□□ build (□□ changelog)
>   2. □□□□ / DeferredRegister □□□□□□□□□
>   3. Mappings □□□□□□□□□□
```

六、结论

MC 的版本号是冻结的，但运行它的世界不是。对 KunYuEcosystem 而言，全版本矩阵不是一张建好就静止的资产清单，而是一张需要持续回测的活矩阵: 用指纹记录每一次覆盖成立的条件，用监测发现加载器与环境的漂移，用编译探针周期性确认老组合今天还活着。三件事做齐，检索优先于生成、编译通过才交付这条护城河才不会因为上游悄悄变动而自己漏水。

AI 参考包通过 manifest.json v4.0 的版本指纹化、system prompt v3.0 的漂移意识注入、context_builder.ts 的健康状态暴露，将这一认知落地到 AI 生成代码的每一个环节。AI 不再只是被动引用模板，而是主动感知模板的健康状态，在 health 非 green 时主动提示漂移风险，引导用户检查加载器版本差异。